

# Choupo

The Open Source Chemical Process Simulator

# Theory Guide

Choupo-dev

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# Contents

**Preface — how to read this manual** 1

**Part I — Foundations** 2

**1 The sequential modular architecture** 3
1.1 Why this chapter exists . . . 3
1.2 Westerberg's classification of flowsheeting strategies . . . 3
1.3 Ports vs parameters — the CAPE-OPEN contract . . . 3
1.4 What lives in `operation`: hardware and operating setpoints, both as parameters . . . 4
1.5 Common practice . . . 5
1.6 The unit-op interface in Choupo . . . 5
1.7 The flowsheet executive . . . 5
1.8 The outer layer — DesignSpec, SweepDriver, OptimizationDriver . . . 6
1.9 Recycles and tear streams . . . 6
1.10 Choupo's compliance scorecard . . . 6

**2 What a process simulator predicts, and how** 8
2.1 The question . . . 8
2.2 Pillar 1 — conservation laws . . . 8
2.3 Pillar 2 — thermodynamic laws . . . 8
2.4 Pillar 3 — physical-chemistry models . . . 9
2.5 Power and limit, the running theme . . . 9
2.6 What this manual does . . . 9

**3 What a unit accepts: modelled, a declared approximation, or refused** 11
3.1 The question a connection does not answer . . . 11
3.2 Three states, never two . . . 11
3.3 A declared approximation is identified twice . . . 11
3.4 Where the contract lives . . . 12
3.5 Reading a refusal . . . 14
3.6 What the contract does not do . . . 14

**4 One equation, many declarations: the map** 15

**5 Notation, units, and basis** 16
5.1 Units . . . 16
5.2 Vectors and components . . . 16
5.3 Vapour fraction . . . 16

**Part II — Pure-component thermodynamic models** 17

**1 Critical properties and corresponding states** 18
1.1 The pure-component $P$-$T$ diagram . . . 18
1.2 The law of corresponding states . . . 18
1.3 The acentric factor $\omega$ . . . 19
1.4 The compressibility factor $Z$ . . . 19
1.5 Where critical properties land in the stack . . . 20

**2 Vapour pressure: from Clausius-Clapeyron to Antoine** 21
2.1 The thermodynamic starting point . . . 21
2.2 Two physical approximations . . . 21
2.3 Integrating with constant $\Delta H_{\text{vap}}$: the rigid form . . . 21
2.4 The empirical fix: Antoine . . . 22
2.5 Failure modes of Antoine . . . 22
2.6 Watson for $\Delta H_{\text{vap}}(T)$ . . . 23
2.7 Worked example: water from $\ln P$ vs $1/T$ to Antoine . . . 23
2.8 Ambrose-Walton: $P^{\text{sat}}$ when there are no Antoine data . . . 23

**3 Sublimation: the solid-vapour curve and the triple point** 26
3.1 Why a solid sublimes: the same identity, a different phase . . . 26
3.2 The triple point and $\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$ . . . 26
3.3 The melting line: the full Clapeyron slope . . . 26

3.4 What the algorithm needs — and what it draws . . . 27

**4 Saturated-liquid molar volume: the Rackett equation 28**
4.1 Why a special correlation for the liquid? . . . 28
4.2 From the critical point to the Rackett equation . . . 28
4.3 The Spencer-Danner / Yamada-Gunn substitution: $Z_c \to Z_{RA}$ . . . 28
4.4 Mixtures: mole-weighted molar volume . . . 29

**5 Heat capacity 30**

**6 From component data to integral properties 31**
6.1 Anatomy of a component data file . . . 31
6.2 Pure-component enthalpy . . . 32
6.3 The reference state, and why a reaction needs the elements . . . 33
6.4 Which rung does the data file pin? The `referenceState` keyword . . . 34
6.5 Where every number comes from — the three questions a black-box tool won't answer . . . 36
6.6 The Gibbs side: $s^\circ_{298}$, third-law entropy, and the absolute ideal-gas free energy . . . 37
6.7 Where the two numbers come from: the NASA-CEA import . . . 38
6.8 Latent heat at any $T$: Watson . . . 39
6.9 Pure-component entropy and Gibbs energy . . . 39
6.10 Mixing: ideal versus real . . . 40
6.11 Gibbs-Duhem: the consistency check . . . 40
6.12 Putting it together: an example energy balance . . . 41

**7 What is fittable, what is not 42**

**Part III — Mixtures: from ideal to real 43**

**1 Ideal mixing 44**
1.1 What "ideal mixing" means . . . 44
1.2 The combinatorial entropy of mixing . . . 44
1.3 The ideal-mixture chemical potential . . . 44
1.4 Ideal-gas mixtures and Dalton's law . . . 45
1.5 Ideal solutions and Raoult's law . . . 45
1.6 The driving force for mixing . . . 45
1.7 Where the ideal limit goes wrong . . . 45

**2 Activity coefficients 47**
2.1 The excess Gibbs energy bridge . . . 47
2.2 Ideal solution . . . 47
2.3 NRTL (Renon and Prausnitz, 1968) . . . 47
2.4 Wilson (1964) . . . 49
2.5 UNIQUAC (Abrams and Prausnitz, 1975) . . . 51
2.6 Worked example . . . 53
2.7 UNIFAC: predicting $\gamma_i$ from molecular structure . . . 53

**3 Electrolyte solutions: ionic strength, the osmotic coefficient, and Pitzer 56**
3.1 Why an electrolyte needs its own model . . . 56
3.2 How Choupo carries the split: one salt, model ions, reference rung . . . 56
3.3 Osmotic pressure from the solvent chemical potential . . . 56
3.4 The osmotic coefficient $\phi$ . . . 57
3.5 The van't Hoff baseline: $\phi = 1$ . . . 57
3.6 Ionic strength . . . 57
3.7 Pitzer's osmotic coefficient for a 1:1 salt . . . 57
3.8 Why this matters for NF/RO — and where it is worst . . . 58

**4 Electrolyte-NRTL: the activity coefficient of the *ion* 60**
4.1 Why a second electrolyte model? . . . 60
4.2 The three-layer excess Gibbs energy . . . 60
4.3 Long range: the Pitzer–Debye–Hückel term . . . 61
4.4 Short range: the local-composition (NRTL) term . . . 61
4.5 Temperature dependence . . . 61
4.6 The mixed solvent: drowning-out and the Born term . . . 62
4.7 Validation and the power-and-limit . . . 62
4.8 Multi-salt eNRTL: the symmetric multicomponent formulation . . . 63

**5 Cohesive energy density and the solubility parameter 65**
5.1 The definition . . . 65
5.2 What Eq. (103) cannot do . . . 65
5.3 The Hansen split, and the Teas triangle . . . 65
5.4 Choosing a solvent, or a solvent mixture: $R_a$ and RED . . . 66
5.5 How Choupo evaluates it . . . 67

**6 Multi-ion Pitzer: the Harvie–Møller–Weare model 68**
6.1 From one salt to many ions . . . 68
6.2 The per-ion working equations . . . 68
6.3 The higher-order electrostatic term $E_\theta(I)$ . . . 69
6.4 The multi-ion osmotic coefficient and the water activity . . . 69
6.5 The single-salt collapse — the skeptic's gate . . . 70

**7 Molecular solutes in a Pitzer expansion: the Edwards truncation 72**
7.1 Why $\gamma = 1$ for a neutral is a structural silence . . . 72
7.2 The three-term working equation . . . 72
7.3 One measured family, four estimation rules . . . 72
7.4 What it is worth on a real feed . . . 73

**8 Aqueous speciation: mass action, charge balance, and the pH closure 74**
8.1 Master ions and the mass-action network . . . 74
8.2 The conservation equations: mole balances . . . 74
8.3 Two closures for $H^+$: pH given vs pH solved . . . 74
8.4 The PHREEQC percent-error convention . . . 76
8.5 Distribution diagrams: the Bjerrum plot, and the titrant nobody declared . . . 76
8.6 The solution algorithm: Newton in log-molality . . . 78
8.7 The Davies equation: a parameter-free single-ion $\gamma_i$ . . . 78
8.8 Activity model and water activity . . . 80
8.9 Temperature dependence of $K$ . . . 80
8.10 Open systems: the Henry gas pin . . . 80
8.11 From speciation to saturation: the scaling link . . . 81

**9 Fugacity and the vapour phase 83**
9.1 Why we need fugacity at all . . . 83
9.2 Fugacity of a pure liquid . . . 83
9.3 Partial fugacity in mixtures . . . 83
9.4 Phase-equilibrium criterion in fugacity form . . . 84
9.5 When $\varphi_i = 1$ fails . . . 84
9.6 Where $\varphi_i$ comes from when not 1 . . . 85
9.7 The $\gamma$-$\varphi$ implementation . . . 85
9.8 Henry's law: the dilute-solute K-value . . . 85
9.9 Why the simulator uses two solver layers, not one . . . 87

**10 The SRK cubic equation of state 88**
10.1 From van der Waals to Soave . . . 88
10.2 Pure-component parameters . . . 88
10.3 Mixing rules . . . 88
10.4 The cubic in $Z$ and its Cardano root . . . 89
10.5 The fugacity coefficient . . . 89
10.6 Departure (residual) functions for energy balances . . . 89
10.7 Peng-Robinson: the same machine, two constants . . . 89

**11 COSMO-SAC: an activity coefficient from a charge surface 92**

**12 PC-SAFT: a molecular equation of state 93**
12.1 Association: Wertheim's theory, visible . . . 93

**13 $K$-values and Rachford-Rice 95**
13.1 Defining the $K$-value . . . 95
13.2 Material balance and Rachford-Rice . . . 95
13.3 Why exactly one root, in $(0, 1)$, when there is two-phase . . . 95
13.4 Initialisation: the Wilson $K$-value . . . 96
13.5 Worked example . . . 96

**Part IV — Numerical methods** **98**

**1 Numerical methods, gently** **99**
1.1 One problem, four strategies . . . 99
1.2 Strategy 1 — guess and adjust . . . 99
1.3 Strategy 2 — bisection . . . 99
1.4 Strategy 3 — secant . . . 100
1.5 Strategy 4 — Newton-Raphson, in one breath . . . 100
1.6 The four strategies side by side . . . 101
1.7 Why the rest of Part III matters . . . 101

**2 Newton-Raphson, one dimension** **103**
2.1 Derivation . . . 103
2.2 Convergence . . . 103
2.3 Damping . . . 104
2.4 Worked example . . . 104

**3 Newton-Raphson, $n$ dimensions** **105**
3.1 Why we need a multi-dimensional Newton . . . 105
3.2 Generalising Taylor: the Jacobian matrix . . . 105
3.3 The Newton step in $n$ dimensions . . . 105
3.4 Finite-difference Jacobian . . . 106
3.5 Solving the linear system . . . 107
3.6 Damping by backtracking line search . . . 107
3.7 Worked example: a $2 \times 2$ system . . . 108

**4 Parameter estimation: Levenberg-Marquardt** **110**
4.1 A different question . . . 110
4.2 Problem statement . . . 110
4.3 The gradient of $\chi^2$: deriving $\mathbf{J}^\top \mathbf{r}$ . . . 110
4.4 The Hessian splits in two: keep one piece, drop the other . . . 111
4.5 Levenberg's idea: a knob between two extremes . . . 111
4.6 Marquardt's invariance fix . . . 112
4.7 The trust-region update: 3 down, 5 up . . . 112
4.8 Finite-difference Jacobian . . . 112
4.9 Stopping criteria . . . 113
4.10 The simulator's implementation . . . 113

**5 Wegstein acceleration** **115**
5.1 The setting: fixed-point problems . . . 115
5.2 Direct substitution and why it can be slow . . . 115
5.3 The Wegstein idea: model $f$ as a straight line . . . 115
5.4 A worked iteration . . . 116
5.5 When the linear model misleads: the $q$-clip . . . 116
5.6 Where this matters in the simulator: the $\gamma$-loop . . . 117

**6 Closing a recycle: Newton on the tear stream** **119**
6.1 The recycle as a fixed-point in the tear stream . . . 119
6.2 What goes in the tear vector . . . 120
6.3 Newton on the tear residual . . . 120
6.4 Why the residual must be relative — a bug story . . . 120
6.5 Wegstein or Newton? Choosing the recycle solver . . . 121
6.6 This is not equation-oriented — and that is the point . . . 121

**7 Direct search: Nelder-Mead** **123**

**8 Constrained optimisation: line-search SQP** **127**
8.1 The problem and the KKT conditions . . . 127
8.2 Shadow prices: what the multiplier is worth . . . 127
8.3 From KKT to a quadratic program . . . 128
8.4 Keeping the QP convex: damped-BFGS on the Lagrangian . . . 128
8.5 The $\ell_1$ merit function and the Armijo line search . . . 129
8.6 Termination: the five-residual KKT certificate . . . 130
8.7 The inner active-set QP . . . 130
8.8 A worked self-check: Hock–Schittkowski . . . 132

**9 Runge-Kutta 4 for the PFR** **134**
9.1 The reactor as an ODE . . . 134
9.2 RK4 step . . . 134

**Part V — Transport properties and mixing rules** **135**

**1 Why transport: where pressure drop, heat transfer, and HETP come from** **136**
1.1 Three coefficients, three modes of irreversibility . . . 136
1.2 Where each property is consumed . . . 136
1.3 Status note . . . 137

**2 Viscosity** **138**
2.1 Pure liquid: Andrade . . . 138
2.2 Pure gas: Sutherland . . . 138
2.3 Either phase: Chapman-Enskog with Lennard-Jones . . . 139
2.4 The Chung correlation — what Choupo actually uses . . . 139
2.5 Declaring the models — the grammar the engine reads . . . 140

**3 Thermal conductivity** **141**
3.1 Pure liquid: Sato-Riedel . . . 141
3.2 Per-component fits, and how far corresponding states gets you . . . 141
3.3 Pure gas: Eucken . . . 141
3.4 Declaring the models . . . 142

**4 Diffusivity** **143**
4.1 Gas binary: Fuller-Schettler-Giddings . . . 143
4.2 Liquid binary, dilute: Wilke-Chang . . . 143
4.3 Composition dependence: Vignes . . . 143
4.4 Where the ingredient data live . . . 143

**5 Mixture rules** **144**
5.1 Gas viscosity: Wilke . . . 144
5.2 Liquid viscosity: log-mole-fraction average . . . 144
5.3 Liquid thermal conductivity: mass-fraction average . . . 144
5.4 Gas thermal conductivity: Wassiljewa with Mason-Saxena . . . 144
5.5 Diffusivity in multicomponent: Vignes plus Maxwell-Stefan . . . 144

**6 Where transport feeds the downstream chapters** **145**
6.1 Pressure drop . . . 145
6.2 Heat-transfer coefficients . . . 145
6.3 Mass-transfer coefficients, NTU and HETP . . . 145

**Part VI — Selected unit operations** **146**

**1 A unit operation is a physical equipment** **147**
1.1 The friction we want to remove . . . 147
1.2 Where this idea comes from . . . 147
1.3 What the student sees . . . 148

**2 A stream is a thermodynamic state, not five numbers** **149**
2.1 Degrees of freedom for a single-phase pure-component stream . . . 149
2.2 The Choupo stream record carries six fields . . . 149
2.3 Specification + consistency check on import . . . 149
2.4 `vaporFraction`: the general two-spec rule (why $\beta$ is often essential) . . . 150
2.5 The flow $F$: extensive, source-driven vs. demand-driven . . . 150
2.6 Where in the code this happens . . . 151

**3 Balances: mass, elements, energy** **152**
3.1 Why moles are not conserved, and atoms are . . . 152
3.2 The seal tests every element, not the total . . . 152
3.3 Honesty states: FULL, PARTIAL, UNAVAILABLE . . . 152
3.4 The three regimes . . . 153

**4 Isothermal flash, vapour–liquid** **154**
4.1 Problem statement . . . 154
4.2 Degrees of freedom: what you specify, what the flash computes . . . 154

4.3 Algorithm . . . 155
4.4 Diagnostics in the log . . . 156

**5 Phase changer: the dome-crossing boiler / condenser 157**
5.1 Where it sits between a heater and a flash . . . 157
5.2 Degrees of freedom: one outlet coordinate . . . 157
5.3 The saturation temperature . . . 157
5.4 Dome-aware enthalpy and the vf closure . . . 158
5.5 The duty is always a result; latent / sensible split . . . 158
5.6 Boiler and condenser are emergent labels . . . 158

**6 Continuous stirred-tank reactor (CSTR) 160**
6.1 Material balance with reaction extent . . . 160
6.2 First-order kinetics — analytic check . . . 160
6.3 Worked example . . . 160

**7 The rate law: from mass action to Langmuir–Hinshelwood 161**
7.1 Where mass action stops . . . 161
7.2 The one formula Choupo evaluates . . . 161
7.3 Per gram of catalyst . . . 162
7.4 A worked, cited example, and a trap . . . 162

**8 Plug-flow reactor (PFR) 163**
8.1 The differential material balance . . . 163
8.2 Algorithm: RK4 integration . . . 163
8.3 First-order kinetics: the analytic check . . . 163
8.4 Worked example . . . 163

**9 The catalyst pellet: the Thiele modulus and the effectiveness factor 165**
9.1 The balance inside one pellet . . . 165
9.2 The modulus is a ratio of rates . . . 165
9.3 The three closed forms . . . 165
9.4 One length that almost erases the shape . . . 166
9.5 The two asymptotes, and what they say about design . . . 166
9.6 $\eta$ multiplies the rate — and the size of the error . . . 167
9.7 The Weisz–Prater criterion: deciding without knowing $k$ . . . 167
9.8 What is deliberately not modelled . . . 167

**10 Single-effect evaporator: the credo case study 168**
10.1 What the evaporator solves . . . 168
10.2 The Mode-2 equations . . . 168
10.3 What *you* write vs what the simulator gives back . . . 170
10.4 Why mode-switching inside the unit op was wrong . . . 170
10.5 The three canonical design questions, one unit op . . . 170

**11 Liquid-liquid equilibrium by Gibbs minimisation 171**
11.1 The phase-stability pre-check: Michelsen's tangent plane . . . 171
11.2 The common tangent: where the two liquids come from . . . 174
11.3 Ternary phase diagrams: sweeping the simplex . . . 177
11.4 The tie-triangle and the Hunter–Nash construction . . . 177

**12 Counter-current liquid–liquid extraction 179**
12.1 The cascade and its variables . . . 179
12.2 One stage is one Gibbs-min flash . . . 179
12.3 Tearing the cascade: under-relaxed Gauss–Seidel . . . 180
12.4 What the column reports . . . 180

**13 Stoichiometric conversion reactor 182**
13.1 The extent of reaction, made primary . . . 182
13.2 Honesty guards in the algorithm . . . 182
13.3 The isothermal duty . . . 183
13.4 Why a *finite* conversion is the whole point of a recycle . . . 183
13.5 Conversion vs. Gibbs: two opposite idealisations . . . 183

**14 Gibbs reactor: equilibrium by free-energy minimisation 185**

14.1 The problem . . . 185
14.2 The Lagrangian . . . 186
14.3 The residual system . . . 186
14.4 The initial guess that makes this work . . . 186
14.5 Worked example: water-gas shift at 800 K . . . 187
14.6 When the ideal-gas Gibbs reactor stops being right . . . 188
14.7 Equilibrium maps and the temperature approach . . . 188

**15 Distillation column, Wang-Henke bubble-point 189**
15.1 Stage model . . . 189
15.2 Constant molar overflow . . . 189
15.3 Wang-Henke bubble-point method . . . 190
15.4 When this method struggles . . . 190

**16 Gas absorption and stripping: the coupled stage cascade 192**
16.1 The counter-current cascade and the absorption factor . . . 192
16.2 Why the energy balance matters . . . 192
16.3 The stripper: the same cascade, mirrored . . . 193

**17 Crystallisation: the population balance and the method of moments 194**
17.1 The driving force: solubility and supersaturation . . . 194
17.2 Where the saturation comes from for a salt: the solubility product . . . 194
17.3 The two-layer $K_{sp}(T)$: which heat of solution drives van't Hoff? . . . 195
17.4 The population balance: a balance over size . . . 196
17.5 The MSMPR steady state: the famous straight line . . . 196
17.6 The method of moments . . . 197
17.7 Kinetics and closing for the supersaturation . . . 198
17.8 The batch crystalliser: integrating the moments in time . . . 198
17.9 When the moments do not close: the discretised PBE . . . 199

**18 Spiral-wound membrane: solution-diffusion and concentration polarisation 201**
18.1 Solution-diffusion: two independent fluxes . . . 201
18.2 Where the coefficients come from — and where the model ends . . . 201
18.3 Concentration polarisation: the film model . . . 201
18.4 Polarisation of a multi-ionic solution: one field at the interface . . . 202
18.5 The coupling, and why it needs an inner Newton . . . 203
18.6 Marching down the channel . . . 204
18.7 Modules as records: a spiral element and a laboratory cell . . . 204
18.8 Stages, arrays and the feed spacer as a record . . . 205
18.9 Batch diafiltration: the same flux law, once per instant . . . 206

**19 Nanofiltration with charge: the Donnan-Steric Pore Model with Dielectric Exclusion (DSPM-DE) 208**
19.1 The pore picture and the three exclusions . . . 208
19.2 Hindered transport inside the pore . . . 209
19.3 The model: the whole boundary-value problem . . . 209
19.4 Dimensionless form and the Péclet numbers . . . 211
19.5 The numerical method: every node at once . . . 211
19.6 The mesh and its observed order . . . 215
19.7 Verification: is the equation solved right? . . . 215
19.8 Practical guidance: when the pore is stiff, and how many nodes . . . 216
19.9 Closing the water flux and marching the channel . . . 217

**20 Nanofiltration without a fixed charge: solution-diffusion with one electric field (SDEM) 219**
20.1 The silence of the uncoupled law . . . 219
20.2 One field, zero current: the equations . . . 219
20.3 The single-salt limit: an ambipolar permeance, derived . . . 219
20.4 A trace ion in a dominant salt: negative rejection . . . 220
20.5 Numerics: shoot from the face where the unknown is . . . 220

**21 Electrodialysis: Faraday transfer, the Nernst stack, and the limiting current 222**
21.1 The stack: where charge meets selectivity . . . 222
21.2 Faraday's law: charge in, moles out . . . 222
21.3 The membrane potential: Nernst on real activities . . . 223
21.4 The ohmic drop: membranes and solutions in series . . . 223

21.5 Assembling the stack voltage and power . . . 223
21.6 The limiting current: what Cowan and Brown actually gave us . . . 223
21.7 The multi-ionic limiting current, predicted . . . 224
21.8 Where the mass-transfer coefficient comes from — and the velocity . . . 225
21.9 The stack as a record . . . 225
21.10 The limiting current in TIME: the batch rig, and in SPACE: the train . . . 226

**22 Ion-exchange softening: Gaines–Thomas mass action 228**
22.1 The exchange half-reaction and the selectivity constant . . . 228
22.2 The Gaines–Thomas convention: activity = equivalent fraction . . . 228
22.3 One extra unknown, one extra balance — folded into the speciation Newton . . . 229
22.4 Hardness, leakage, and the salt penalty . . . 229

**23 Adsorption: isotherms, LDF uptake, and the fixed bed 231**
23.1 The equilibrium loading: Henry and Langmuir . . . 231
23.2 Competition: the extended Langmuir, honestly . . . 231
23.3 The rate: linear driving force, and whose datum $k$ is . . . 231
23.4 The 1-D isothermal fixed bed . . . 232
23.5 The stoichiometric time: where the front must arrive . . . 232
23.6 Reading a "steady" swing unit: the twin-bed time average . . . 232

**24 Rotating equipment: compressors, turbines, pumps 234**
24.1 The isentropic reference and the efficiency . . . 234
24.2 Why a nested Newton (compressor / turbine) . . . 234
24.3 The pump: incompressible, no EoS, no Newton . . . 234
24.4 The pressure exchanger: pressure moved from one liquid to another . . . 235

**25 Pipe pressure drop: Darcy–Weisbach and the friction factor 236**
25.1 The mechanical-energy balance for an incompressible liquid . . . 236
25.2 Darcy–Weisbach: the friction head . . . 236
25.3 The friction factor in each regime . . . 236
25.4 Minor losses and the assembled $\Delta P$ . . . 237
25.5 Gas-liquid two-phase flow: Lockhart–Martinelli . . . 238

**26 Pneumatic conveying: carrying solids in a gas 240**
26.1 The additive pressure budget . . . 240
26.2 Particle velocity: a force balance, not a fit . . . 240
26.3 Solids friction: Yang (1974) . . . 240
26.4 The saltation floor — will the line plug? . . . 240
26.5 Bends: the re-acceleration that dominates . . . 241
26.6 Scope: dilute phase . . . 241

**27 Gas-solid separation: cyclone, bag filter, ideal splitter 242**
27.1 Grade efficiency: the unifying idea . . . 242
27.2 Cyclone: cheap centrifugal classification . . . 242
27.3 Bag filter: the cake is the filter . . . 243
27.4 Ideal splitter: when you just want a number . . . 243

**28 Drying: spray dryer and solid dryer 245**
28.1 The four coupled pieces of a spray dryer . . . 245
28.2 The residual moisture: equilibrium vs kinetics . . . 245
28.3 The solid dryer: finishing an already-solid powder . . . 246

**29 Two-stream heat exchanger: the $\varepsilon$-NTU rating method 247**

**30 Heat exchanger DESIGN: geometry, pressure drop, and the sizing workflow 248**
30.1 Two questions: rating vs design . . . 248
30.2 The overall coefficient, built . . . 248
30.3 Pressure drop: the other half of design . . . 248
30.4 The design loop . . . 249
30.5 The design workflow . . . 249
30.6 The deliverable: the datasheet . . . 249

**31 The water cooling tower: Merkel's enthalpy driving force 250**
31.1 The construction . . . 250

31.2 What the model assumes, and says so . . . 250
31.3 The two temperatures that matter . . . 250

**32 Pinch analysis: the target before the network 251**
32.1 Streams, CP, and why a duty becomes a line . . . 251
32.2 The shift: why $\Delta T_{\min}$ moves the axes . . . 251
32.3 The problem table, cascade by cascade . . . 252
32.4 Composite curves: the same result, drawn . . . 252
32.5 Candidate matches: what the physics admits . . . 253
32.6 The violation diagnostics, and why they add up . . . 253
32.7 What this pass does not do . . . 253

**33 Shortcut distillation: Fenske–Underwood–Gilliland 254**
33.1 Reading $N(R)$: two limits and a fit between them . . . 254
33.2 What the shortcut returns — and what it does not . . . 256
33.3 Shortcut against rigorous: the benzene/toluene numbers . . . 256

**34 Rigorous distillation by simultaneous MESH Newton 258**
34.1 The model ladder (*all models are wrong, some useful*) . . . 258
34.2 Wang–Henke vs the simultaneous CMO Newton . . . 258
34.3 The full MESH (Naphtali–Sandholm): the energy balance . . . 258
34.4 The warm-started ladder (switching model mid-solve) . . . 259
34.5 Multiple feeds and side draws . . . 259
34.6 Reactive distillation . . . 259
34.7 A column over a chemistry: the effective stage $K$ . . . 259

**35 The enthalpy-concentration diagram: putting energy inside the geometry 262**
35.1 Two curves and the ties between them . . . 262
35.2 Why three balances become one straight line . . . 262
35.3 What constant molar overflow actually assumes . . . 263
35.4 What Choupo publishes, and what it does not . . . 263

**36 Tray hydraulics: can the column be built? 265**
36.1 First, the tray is not ideal: Murphree efficiency . . . 265
36.2 Entrainment flooding: Souders–Brown and Fair . . . 265
36.3 Pressure drop, as heads of liquid . . . 266
36.4 Downcomer backup . . . 266
36.5 Weeping . . . 267
36.6 The two directions . . . 267

**37 Multi-stream heat exchanger (MHeatX): a checker, not a solver 268**
37.1 The degrees of freedom: why verify, not solve . . . 268
37.2 Per-stream duty and the first-law audit . . . 268
37.3 The internal pinch: pooled composite curves . . . 268
37.4 Worked mini-example . . . 269
37.5 Power and limit . . . 269

**38 Drying of solids and spray drying 270**
38.1 Moisture content and the equilibrium . . . 270
38.2 The drying-rate curve: constant and falling rate . . . 270
38.3 Spray drying . . . 270
38.4 Atomisation — the drop-size correlations . . . 270

**39 The simple unit operations 273**

**Part VII — Equipment design and economics 274**

**1 Vessel sizing: the stirred tank 275**
1.1 Volume to geometry . . . 275
1.2 Wall thickness: ASME stress design . . . 275
1.3 Weight . . . 275

**2 Heat-exchanger sizing: shell-and-tube 276**

**3 Discounted cash flow: from a cost to a decision 277**

**4 Turton bare-module costing** **278**
4.1 Purchased cost from size . . . 278
4.2 Bare-module factor . . . 278
4.3 Total-module cost . . . 278
4.4 Tray stacks: bought by the tray . . . 278
4.5 Year and currency scaling . . . 278

**Appendix A — Symbol glossary** **280**

**Appendix B — References** **281**

**Appendix C — Historical context: Choupo and FLOWTRAN** **289**

**Part VIII — Dynamic simulation and process control** **292**

**1 Time-stepping with a packed state vector** **293**

**2 Adiabatic batch reactor: coupled mass + energy balance** **294**

**3 Rayleigh batch distillation** **296**

**4 Continuous dynamic CSTR with finite hold-up** **301**

**5 The liquid tank with a level: integrating and self-regulating inventory** **303**

**6 Feedback control: the ideal PID, derivative-on-PV, anti-windup** **304**

**7 The process reaction curve: identifying an FOPDT model, and tuning from it** **307**

**8 Choosing a column control structure, and where the instruments go** **311**

**9 Numerical integration of stiff systems** **314**
9.1 The initial-value problem and explicit stepping . . . 314
9.2 What stiffness is . . . 315
9.3 The implicit cure . . . 315
9.4 Rosenbrock23: the scheme Choupo uses . . . 316
9.5 The Jacobian, and two correctness traps . . . 317

**10 Chemical kinetics in the gas phase** **319**
10.1 Mass action and the Arrhenius law . . . 319
10.2 Third-body and pressure fall-off reactions . . . 319
10.3 Reversible reactions: the reverse rate from thermodynamics . . . 319

**11 Polymerisation: chain statistics from conversion** **321**
11.1 Carothers: degree of polymerisation from conversion . . . 321
11.2 The distribution and the polydispersity index . . . 321

# Preface — how to read this manual

*What I cannot create, I do not understand.*
Richard P. Feynman, on his blackboard, February 1988

Every equation in this manual maps to lines of source code you can grep for, and to runnable cases in `tutorials/`. The goal is that nothing in the simulator is unaccounted for in writing. If you find a mismatch — a line of code without a justification here, or a derivation here that does not match what the simulator prints — that is a bug.

The structure is the obvious one. Foundations first — notation, equilibrium thermodynamics, $K$-values and Rachford-Rice. Then the families of models the simulator carries (vapour pressure, activity, EoS, heat capacity). Then the numerical machinery the models are solved with. Finally, the unit operations that compose them into a flowsheet.

Each chapter follows the same rhythm:

1. Motivation — why this object exists.
2. Hypotheses — what we assume, in plain language.
3. Derivation — the equations, step by step.
4. Worked example — a small numerical case carried by hand.
5. Runnable case — the tutorial that exercises the same model.

Read it linearly the first time. After that, treat it as a reference: the worked examples are written so they can be re-read as standalone exercises.

**Key idea.** The simulator is a glass box. This manual is the lens.

## A note on units

All equations in this manual are written in SI: pressure in Pa, molar flow in kmol/s, temperature in K, length in m, area in $m^2$, volume in $m^3$, energy in J, mass in kg, amount in kmol. Choupo's parser accepts named units in case files (`P 1 bar;`, `F 100 kmol/h;`) and converts to canonical SI internally, so the case files you read in the tutorials are written in the industrial-convention units the engineer expects while the equations on the page stay in the SI form they take in textbooks. Every dict-loaded scalar carries a five-int dimension tag $[M\,L\,T\,\Theta\,N]$ for mass, length, time, temperature and amount; a mismatch between a value's declared dimensions and the expected ones is caught with an explicit error before the solver runs. Throughout the manual the intermediate expressions and the numerical worked examples are SI; the final result is reported in whichever unit is most natural (kJ/kmol, bar, kmol/h,...) with the conversion written out.

# Part I — Foundations

*A map is not the territory.*
Alfred Korzybski, *Science and Sanity* (1933)

# 1 The sequential modular architecture

*What you'll learn.* The architectural credo Choupo follows, set out by Westerberg *et al.* in 1979 [1], codified by the CAPE-OPEN consortium in the late 1990s [2], and embodied in process flowsheeting ever since. Memorise five sentences and the rest of this manual reads as a footnote.

1. **The flowsheet is a directed graph.** Nodes are *unit operations*; edges are *streams* (material, energy, or information). Topology comes first; equations come second.
2. **Every unit operation is a black box with two signatures.** *Ports* for streams (inputs / outputs); *parameters* for hardware (area, volume, tray count, ...). Nothing else.
3. **Every stream that enters a unit must be fully specified.** The simulator computes the outputs; the engineer specifies the inputs (or seeds them and the outer driver iterates).
4. **Solve in the declared order, and check it.** Each unit's `solve()` runs in the order the case declares the units, once its inputs are known. Choupo never re-sorts that order behind the author's back: it checks it (`Flowsheet::validateSequentialPlan`) — every material input must be a plant inlet, an EARLIER unit's output, or a declared tear — and refuses an order that breaks the rule, naming a valid one. Recycles are broken by the *tear streams* the author declares, which the outer Wegstein iteration converges [3, 1].
5. **Design targets live on top of the simulation, not inside it.** An outer *DesignSpec* manipulates the engineer's free parameters until KPI constraints are satisfied. The simulator itself never solves the design problem; it solves the analysis problem.

## 1.1 Why this chapter exists

The honest reason: the project's first version of an evaporator unit op accepted a duty $Q$ directly in its operation block, even though there was no heating-medium stream attached. Students reading the case asked, reasonably, "where does this $Q$ come from?". The answer involved a long story about energy balances that the unit op was computing internally. After several rounds of confusion, we went back and read the textbooks. The architecture was already fully worked out forty years before Choupo existed; what we needed was discipline, not invention.

This chapter writes the discipline down so that no future author (or AI assistant) can reinvent it differently. Anything in the manual that contradicts the five rules above is a bug.

## 1.2 Westerberg's classification of flowsheeting strategies

Following [1] Chapter 1, a process simulator can attack the global flowsheet equations in three ways:

1. **Sequential modular (SM).** Each unit operation is a self-contained *module* with its own internal `solve()`. The flowsheet executive (a) orders the units topologically, (b) calls each module in order, (c) loops Wegstein over tear streams to close recycles. Each module is self-contained; the executive only routes streams. This is the classic sequential-modular architecture, the one Choupo uses — with one deliberate difference in (a): Choupo takes the order from the case and VALIDATES it rather than computing it (rule 4 above), so the order a student reads in the flowsheetDict is the order the engine runs.
2. **Equation-oriented (EO).** Strip every module of its `solve`() and dump all the algebraic equations from every unit into one big system. Solve the system with a sparse Newton. Pros: faster on highly recycled flowsheets; can be wrapped in a non-linear programming solver without iteration overhead. Cons: unit-op authors lose locality (an error in one module infects the whole Jacobian); debug becomes harder.
3. **Simultaneous modular (SMM).** A hybrid: modules keep their internal solvers, but the executive linearises them at each outer iteration and solves a sparse system of the linearised modules. Used in optimisation-heavy applications.

For pedagogical and open-source purity reasons Choupo adopts SM throughout: students see one module's equations at a time, can breakpoint inside any `solve()`, and the recycle loop is the only place where multiple modules interact. EO is on the v1.0+ roadmap as an alternative back-end behind the same module interface.

## 1.3 Ports vs parameters — the CAPE-OPEN contract

The CAPE-OPEN Unit Operation Interface Specification [2] states the contract precisely:

> *A Unit Operation may have several **Ports** that allow it to be connected to other Unit Operations and to exchange material, energy or information with them. Unit operations also have sets of **Parameters**. These represent information that is not associated with the Ports and include design specifications and computed results.*

The credo translated:

| | Port | Parameter |
|---|---|---|
| What it carries | A stream (material / energy / info) | A scalar (or list, or sub-dict) |
| In the dict | Under `inputs (...)` / `outputs (...)` | Under `operation { ...}` |
| Examples | `feed`, `liveSteam`, `Q_in` | area, $V_R$, refluxRatio, $N_{\text{stages}}$ |
| Who writes the value | Upstream unit OR the engineer | The engineer (or outer DesignSpec) |
| Who reads it | The unit op via the port handle | The unit op via `operation.lookupScalar(...)` |
| Physical meaning | The actual fluid / energy crossing the boundary | A property of the equipment itself |

Three kinds of port that CAPE-OPEN sanctions:

- **Material port** — carries a fluid; the standard Choupo stream record (`ProcessStream`). Choupo supports this.
- **Energy port** — carries a heat duty $Q$ in W. Used to model a heating utility *without* representing the carrier fluid explicitly (a so-called "energy stream"). Choupo does not yet expose this as a first-class type; cases that need a duty source pass it through a material stream carrying the utility fluid explicitly. Energy streams are on the roadmap.
- **Information port** — carries a scalar or vector of values for control signals, set-points, sensor readings. Used in dynamic flowsheets (the `choupoCtrl` Controller layer); the steady-state simulator (`choupoSolve`) ignores them.

## 1.4 What lives in `operation`: hardware and operating setpoints, both as parameters

A subtlety students invariably trip over: the `operation { ...}` block of a unit op mixes two physically different things, both classified as *parameters* (not ports) by the CAPE-OPEN contract.

**Hardware parameters** are intrinsic to the equipment: the heat-exchange surface area $A$, the overall heat-transfer coefficient $U$, the reactor volume $V_R$, the number of trays $N$, the tube length $L$. They describe *geometry* or material properties. The engineer chooses them at design time and they typically do not change during operation.

**Operating setpoints** are control choices: the operating pressure of an evaporator vessel ($P_{\text{op}}$), the jacket-fluid temperature of a CSTR ($T_{\text{jacket}}$), the reflux ratio of a column ($R$). They describe the *set-point of an external controller* (valve, vacuum pump, DCS loop). An operator can change them in real time without touching the equipment.

**The dict treats them identically.** Both appear under `operation { ...}` as scalar parameters; the dict parser does not distinguish them. This is deliberate — the unit op needs both to solve, and the engineer specifies both at design time, so there is no reason to fork the syntax.

**The output streams inherit operating setpoints.** An operating pressure declared as `operation { P 14 kPa; }` on a flash drum becomes the pressure of the vapour and liquid outputs: $P_{V,\,\text{out}} = P_{L,\,\text{out}} = P_{\text{op}}$ by mechanical equilibrium within the vessel. Students often read this as "the simulator computed the output pressure", which is technically correct but misleading — the simulator copied the operating-setpoint parameter to the output stream. The non-trivial computation is on the conjugate variable: in this example, the boiling temperature $T_{\text{boil}} = T_{\text{sat}}(P_{\text{op}}) + \Delta T_{\text{BPE}}$, found by inverting the saturation curve.

**Where does $P_{\text{op}}$ physically come from?** From equipment *outside* the unit op model: a vacuum pump on the vapour line, a back-pressure valve, a flare header. The unit op does not model the controller — it takes $P_{\text{op}}$ as given, the same way it takes $A$ as given. In a multi-effect cascade where only the terminal $P_N$ is set by external equipment, the intermediate $P_i$ are *design unknowns*: the equal-area design problem is typically solved by a `DesignSpec` that manipulates them until the cascade closes (see Choupo's `evaporator03_co_current` tutorial for the worked case).

**The conjugate-spec flavour.** Most P-spec operations can be written as T-spec equivalently. A flash block typically accepts $\{T, P\}$, $\{T, V/F\}$, $\{P, V/F\}$, $\{P, Q\}$, $\{T, Q\}$ — any two of the five conjugate variables ($T, P, V/F, Q$, plus the duty constraint). Choupo evaporator defaults to P-spec for tutorial alignment with the textbook problems; a T-spec flavour is a one-line addition behind the same module interface.

> *Intuition.* The pedagogical heuristic from §1 sharpens: if you can imagine *walking out to the plant* and adjusting the setting with a wrench (resizing a pipe, replacing a tube bundle), it is a hardware parameter.

> If you can adjust it from the DCS in five minutes (changing a valve position, nudging a set-point), it is an operating setpoint. Both live in `operation`; the simulator treats them as data, not as unknowns.

## 1.5 Common practice

A common rule of thumb, in one sentence:

> *Inlet streams must be defined by the user as process variables, while outlet streams can be calculated.*

And it gives a two-tier hierarchy of heat-transfer unit ops that Choupo now mirrors:

| Block | Ports | Parameters |
|---|---|---|
| **Heater** | 1 material in + (optional) 1 energy in → 1 material out | $T_{\text{out}}$ *or* $Q$ |
| **HeatX** | 2 material in (hot + cold) → 2 material out | area, $U$, geometry |

The Heater is a deliberate *abstraction*: when the utility-side fluid is not pedagogically interesting, you provide a duty number or an energy stream and the block does the math. The HeatX is the full physical-equipment model. Choupo will ship both (`heater` and `heatExchanger`); the single-block hybrid is being split.

## 1.6 The unit-op interface in Choupo

In code (`src/unitOperations/UnitOperation.H`), the credo materialises as four virtual methods:

```
class UnitOperation {
public:
    // Streams in / streams out / parameters all arrive in ‘dict‘.
    virtual int solve(const DictPtr& dict,
                      const ThermoPackage& thermo,
                      int verbosity) = 0;

    // Streams produced by this unit’s last solve().  Order matches
    // the user’s ‘outputs (...)‘ declaration in flowsheetDict.
    virtual std::vector<ProcessStream> producedStreams() const = 0;

    // Scalar results exposed by name (‘A_needed‘, ‘duty‘, ‘economy‘,
    //...) for the outer DesignSpec / SweepDriver / postProcessor.
    virtual std::map<std::string,scalar> kpis() const { return {}; }

    // Optional 1-D internal profile (axial sweep for PFR, stage
    // table for a column,...).  Read by the GUI’s profile plot.
    virtual std::optional<UnitProfile> profile() const
                                          { return std::nullopt; }
};
```

Notice what is *absent*: there is no method for the unit op to write back into its inputs, no method for the unit op to demand a stream that was not declared, no method for the unit op to inject parameters into the outer dict. The interface is deliberately one-way: *streams in → solve → streams out + KPIs*. The credo enforced by the type system.

## 1.7 The flowsheet executive

The dispatcher (`src/unitOperations/flowsheet/Flowsheet.cpp`) runs the SM loop:

1. **Load streams.** `readStateDir()` parses the case’s `0/` directory — one file per stream, the complete initial state — into `ProcessStreams`, applying flash-on-import to complete the thermodynamic state (§2). (A `streams {}` block inside `flowsheetDict` is refused: topology lives in the dict, numbers live in `0/`.)
2. **Topological order.** Read `units (...)` in the order written. For acyclic flowsheets (the DAG case) this order *is* the calculation order. For recycles, the user declares `tearStreams (...)` and the executive solves the tear variables — Newton on the tears by default, Wegstein selectable (`recycleSolver`).
3. **Per-unit dispatch.** `runUnit()` builds the augmented unit dict (injects each input stream’s $T, P, F, \mathbf{z}$ as sub-dicts; merges `solverDict` defaults; resolves named reactions), then calls `UnitOperation::New(type)->solve()`.

4. **Register outputs.** `producedStreams()` are bound to the names in `outputs (...)` and put into the stream registry, available to downstream units and to the result emitter.

## 1.8 The outer layer — DesignSpec, SweepDriver, OptimizationDriver

When a case has a `system/outerDict`, `main.cpp` instantiates an outer driver that *wraps the simulator functor*. The driver:

- declares its own *manipulated variables* (typically case-level `$variables`),
- declares its own *targets* (a list of either "KPI/stream-field equals a literal" or "two computed quantities are equal"),
- runs Newton-ND (or Nelder-Mead, or a sweep) over the manipulated variables, calling the simulator functor at each evaluation,
- ends with a final replay at the converged design point.

This outer layer — *Design Spec*, *Sensitivity*, *Optimization* — Choupo provides the same names. The pedagogical point: the simulator never solves a design problem; it solves an analysis problem (forward simulation of a given hardware). The design problem belongs to the outer driver.

## 1.9 Recycles and tear streams

Westerberg [1] Chapter 6 frames the recycle problem cleanly. In any SM flowsheet whose directed graph has a cycle, at least one stream on the cycle must be *torn* — that is, declared with an initial guess and updated each pass until the new value matches the old to a tolerance. Choupo exposes this directly in `flowsheetDict`:

```
tearStreams            (recycle );
recycleTol             1.0e-5;
recycleMaxIter         60;
recycleWegsteinQmin   -0.5;
recycleWegsteinQmax    0.0;
```

The Wegstein [3] accelerator is the default; $q_{\min} = q_{\max} = 0$ degenerates to plain direct substitution for cases where acceleration is unsafe (flash regime switches, strongly non-linear recycles).

> *Intuition.* Why does a flowsheet of hundreds of unit operations, each resting on its own assumptions, end up *consistent*? A student once put this very question to Prof. Luiz Alves (1920–1990) — professor of Chemical Engineering, Director of Instituto Superior Técnico and later head of its Chemical Engineering Department, and a director of CUF/Quimigal, then Portugal's largest chemical company. His answer, from a man who had actually built and run chemical plants: *"Don't worry — in the end it all checks out."* This is not faith. It checks out because a process engineer carries a physical sense of the *bounds* of every stream (a temperature in roughly 300–500 K, a vapour fraction in $[0, 1]$, a flow of the order of the feed) and of the operating *limits* of every unit. Numbers that wander outside those ranges are caught — by the engineer, and, in a glass-box simulator, by the bounds the author writes down and that the solver reports when they bind. The consistency is *earned*, by judgement made explicit — never conjured by a hidden crutch.

## 1.10 Choupo's compliance scorecard

Where the project stands today against the credo:

| Rule | Status | Notes |
|---|---|---|
| 1. Topology first, equations second | ✓ | `flowsheetDict.units (...)` is the DAG |
| 2. Ports + parameters separation | ✓ | `inputs/outputs` vs `operation {...}` |
| 3. Inlets fully specified | ✓ | dropped the demand-driven F shortcut |
| 4. Topological solve + Wegstein | ✓ | Both DAG and recycle paths in place |
| 5. Outer DesignSpec for targets | ✓ | `designSpec` driver |
| Material ports | ✓ | `ProcessStream` |
| Energy ports | × | on roadmap; today via state-tagged material |
| Information ports | partial | in `choupoCtrl` (`Controller` layer) only |

The remaining gap (energy ports) is the only piece of the standard architecture not yet implemented. Everything else in this manual — thermodynamics, numerical methods, unit-op algorithms — is the *content* that fills the architecture.

*Intuition.* If at any point the manual seems to describe a unit op doing something *outside* the streams-in / parameters / streams-out contract, treat that as a bug report. The architecture is the load-bearing pedagogical commitment: it is what makes the source code a glass box. Anything else is a leak.

# 2 What a process simulator predicts, and how

*What you'll learn.* Before any equation, a clear-eyed view of what a simulator like Choupo actually does: it takes a handful of measured numbers about pure components and binary mixtures, applies the three universal pillars of process engineering (conservation laws, thermodynamic laws, physical-chemistry models), and predicts the behaviour of a system that has never been built or measured. This chapter sets the scope, the power, and the limits that the rest of the manual will fill in.

## 2.1 The question

A chemical engineer is handed an assignment:

> *"You have a stream of 50/50 mol ethanol/water at 300 K and 1.01 bar leaving a fermenter, 100 kmol/h. The next step in the process needs this stream at 360 K, still liquid. Size the heater: tell me $Q$, the heat-exchange area, and the cooling-water cost."*

No one has ever measured this exact stream at 360 K — it is a *design* question, not a *reporting* one. The data sheet on your desk lists pure ethanol properties, pure water properties, a table of NRTL binary parameters fitted from VLE measurements at 1 atm, and the price of cooling water. How do you get from those numbers to a $Q$ you can defend in a design review?

A process simulator answers the question. Internally it stacks three layers of reasoning, each grounded in a different kind of knowledge:

| pillar | what it asserts | where it fails |
|---|---|---|
| conservation laws | mass, energy, momentum balance, always | never (an exact identity) |
| thermodynamic laws | equilibrium criteria, $\Delta G$ direction | never (an exact identity) |
| physical-chemistry models | how $\gamma$, $P^{\mathrm{sat}}$, $c_p$, $k$ depend on state | outside the calibration range |

The first two pillars are **universal**: a balance is a balance, the second law is the second law, no engineer ever has to fit them. The third pillar is **empirical**: every model has a domain of validity bounded by the data it was fitted to, and pushing it past those bounds is where engineering judgement begins. The rest of this manual is in effect a tour of the third pillar — what each model means, what its parameters mean, where they come from, and when the simulator will quietly lie to you because they were not quite the right model for your problem.

## 2.2 Pillar 1 — conservation laws

The mass and energy balances are the simulator's only commandments that admit no exception. For a steady-state control volume around the heater of the opening question:

$$\dot{n}_{\mathrm{in}} = \dot{n}_{\mathrm{out}} \qquad \text{(mole balance, per component if no reaction)}, \tag{1}$$

$$\dot{n}_{\mathrm{in}}\, h_{\mathrm{in}} + \dot{Q} = \dot{n}_{\mathrm{out}}\, h_{\mathrm{out}} \qquad \text{(energy balance, no shaft work)}. \tag{2}$$

With one unknown ($\dot{Q}$) and one equation, the duty falls out *algebraically* once the inlet and outlet enthalpies are known. That is the value of pillar 1: it converts a design question into a property-evaluation question. In the balances are written by hand inside each unit operation (`src/unitOperations/...`); a future equation-oriented version will hand them all to a single sparse Newton solver (roadmap, post-v1).

*Intuition.* Balances are *exact* by construction. Every error a simulator ever commits has to come from the property models that feed them. That is why the next pillar's models, and the third pillar's parameter fits, deserve so much of this manual's attention.

## 2.3 Pillar 2 — thermodynamic laws

The first law of thermodynamics is the energy balance dressed up; the second law gives us *direction* and *equilibrium*. The simulator uses it in three concrete shapes:

- **Phase-equilibrium criterion.** At a flash, two phases coexist when the chemical potential of every component is equal across them: $\mu_i^{\mathrm{L}} = \mu_i^{\mathrm{V}}$. Rewritten in fugacities and divided through by $RT$, this becomes $\gamma_i x_i P_i^{\mathrm{sat}} \varphi_i^{\mathrm{sat}} = y_i \varphi_i P$ — the $\gamma$-$\varphi$ formulation we will derive piece by piece in Part III. The shape is universal; the specific functions $\gamma_i$, $\varphi_i$ are pillar-3 models.
- **Equilibrium constant of a reaction.** $K_{\mathrm{eq}}(T) = \exp(-\Delta G_r(T)/RT)$ comes from the second law's minimum-Gibbs criterion; we use it to bound conversions even when kinetics are slow. (Not yet consumed by the reactor models, which are kinetic only.)

- **Phase-stability test.** A trial flash that satisfies $\mu^{\mathrm{L}} = \mu^{\mathrm{V}}$ may still be unstable against splitting into three phases. Michelsen's tangent-plane distance criterion screens for that; ships the detector but not yet the full LL flash (§9 of CLAUDE.md).

> *Intuition.* Pillars 1 and 2 together tell you *where the system wants to sit* but say nothing about *how the parameters look in practice.* A spreadsheet that knew only these two pillars could not answer the opening question — it would need to be told, for every $T$ on the path from 300 K to 360 K, what the enthalpy is. That information lives in pillar 3.

## 2.4 Pillar 3 — physical-chemistry models

Each property the simulator needs (vapour pressure, latent heat, heat capacity, activity coefficient, fugacity coefficient, viscosity, ...) is represented by a *model* — a closed-form functional dependence on $T$, $P$, composition — with two or three fitted parameters. Antoine for $P^{\mathrm{sat}}(T)$; Watson for $\Delta h^{\mathrm{vap}}(T)$; NRTL or Wilson for $\gamma_i$ in a non-ideal liquid; ideal-gas $\varphi_i = 1$ for the vapour (currently); Arrhenius for reaction rate constants. Each model is a *compression* of experimental data: it remembers the shape of the data with few enough numbers that the simulator carries them cheaply, and forgets the rest.

That compression is where the power comes from. Once Antoine constants are fitted to four or five $(T, P^{\mathrm{sat}})$ points, $P^{\mathrm{sat}}(T)$ becomes evaluable at any $T$ in the validity range — including $T$ values nobody measured. Multiply this by ten components and the simulator can evaluate $10 \times 10$ binary flash diagrams from a few hundred numbers. The same compression is also where the failure comes from: a model fitted between 280 and 380 K may extrapolate badly to 500 K; a model fitted on Raoult mixtures may misbehave on a hydrogen-bonded one; a model fitted at 1 bar may need a Poynting correction at 100 bar.

The simulator does **not** hide this from the user: every model records its declared validity range in the data file (`Trange` in the Antoine block, $T_b < T < 0.95\,T_c$ for Watson,...) and warns when it is asked to extrapolate. Chapter 7 will return to this and ask, model by model, what is genuinely fittable and what is hard-coded.

## 2.5 Power and limit, the running theme

A process simulator is at heart **compression plus extrapolation**. A short list of measured numbers (component properties, binary pair parameters) is compressed into a stack of models. The models are then extrapolated to a vast space of process conditions and topologies the original measurements never covered.

The compression is what makes the simulator useful: it would be absurd to measure the duty of every conceivable ethanol/water heater to a precision of a percent. The extrapolation is what makes the simulator dangerous: every prediction sits at some distance from the measurements that fed the models, and the engineer's job is to keep that distance reasonable.

> *Intuition.* The right relationship with a process simulator is the same as the right relationship with any instrument: trust the readings inside the calibration range, treat them as estimates outside it, and always ask which underlying measurement contributed to which output number. The glass-box mission of Choupo exists precisely so that question can be answered by reading the source, not by trust.

## 2.6 What this manual does

The rest of the manual is a tour of pillar 3 — the models the simulator uses, the parameters they need, and the boundaries within which they can be trusted — followed by the numerical methods that glue them together into a working flowsheet.

| Part | content |
|---|---|
| Part I | this chapter; the notation and units the simulator uses internally |
| Part II | thermodynamic models for *pure* components: Antoine, Watson, Cp, plus integrals into $h$, $s$, $g$ |
| Part III | thermodynamic models for *mixtures*: ideal mixing, activity coefficients, fugacity, $\gamma$-$\varphi$, K-values |
| Part IV | numerical methods: Newton-1D, Newton-ND, Levenberg-Marquardt, Wegstein, RK4 |
| Part V | transport properties: viscosity (Chung), thermal conductivity (Eucken), diffusivity (Fuller), mixture rules – shipped |
| Part VI | selected unit operations as they are wired |
| Part VII | equipment sizing and Guthrie/Turton costing |

The order is roughly: *what does the simulator know about a single pure substance?* then *what changes when substances mix?* then *how does the code actually solve the resulting equations?* then *how does it apply them to a unit operation and to design?* By Part VII you can read any line of the simulator's run log and know which equation produced it, which parameter is upstream of that equation, and where in the data file the parameter came from.

**Runnable case.** **The part numbering above is the target layout.** of this manual currently presents Parts II, III, IV (mixtures and numerical methods) in a slightly different order; reorganisation into the table above is the active editorial work. See the session log in CLAUDE.md §14 for the current state.

# 3 What a unit accepts: modelled, a declared approximation, or refused

*What you'll learn.* A stream is not accepted by a unit just because the ports connect. Every unit operation models a *subset* of what a stream can carry, and the rest of this manual's honesty depends on the engine saying, for every component that arrives, which of three things happens to it: it is **modelled**, it **crosses the unit as a declared approximation**, or the run is **refused** — by name, with the remedy. This section explains why a unit declares what it accepts, how a declared approximation is identified on the input and on the output, and how to read a refusal.

## 3.1 The question a connection does not answer

Two feeds, one question. A wine is water with a few per cent of ethanol; a brine going to an electrodialysis stack is water with its ions and, by accident, a trace of benzene in a second liquid phase. Should a unit that was written for aqueous solutions take them?

- The **ethanol in the wine** is, for a reverse-osmosis module, an ordinary solute: dealcoholisation by membrane is industrial practice, and the transport model prices ethanol's activity inside its *declared domain*. Refusing it because "ethanol is not an ion" would refuse a legitimate process.
- The **benzene in the stack** enters none of the stack's equations: the stack moves ions across membranes, and benzene is neither an ion nor priced by anything the stack computes. It is not *wrong* for it to cross the stack — but nobody should be able to miss that it did.

A simulator with only two answers ("accept" and "refuse") must either refuse the wine or silently carry the benzene. Before this contract existed Choupo did something worse than either, in two units: the reverse-osmosis module re-counted a volatile solute's moles as water (the outlets then held 0.16327 kmol/s against a 0.16111 kmol/s feed, exit 0), and the batch diafilter *retained* a permeable solute whole through 5.43 diavolumes that washed the salt to 50 %. Neither run said anything. A unit that has not decided what it does with a component decides it by accident.

## 3.2 Three states, never two

For a unit and the stream it is handed, every *present* component is in exactly one state:

1. **Modelled.** The component enters the unit's equations, because the declared models can price it inside the model's own validity domain. The domain is a statement of the *model* (the Davies activity coefficient states its ionic-strength range; the aqueous speciation needs water as the dominant solvent) — never a universal threshold the engine invents.
2. **A declared approximation (it passes through).** The component crosses the unit without entering its equations, and the *case* says so: a `passThrough` block in the unit's entry names the component, the outlet it leaves by (its *fate*) and the reason. The engine never chooses a fate: where a component goes in a separator *is* a model choice.
3. **Refused.** The run stops before the unit computes anything, naming the unit, the stream, the component and the fact, and offering both remedies: a model that prices it, or the `passThrough` declaration.

## 3.3 A declared approximation is identified twice

The benzene case is the one to study (`tutorials/steady/electrodialysis/ed10_passthrough_benzene`). The case adds, to the stack's entry,

```
passThrough
{
    benzene { fate diluate; }
    reason  "trace organic in a second liquid phase; the stack's ion
             transport does not act on it";
}
```

and the engine does three things with it.

- **On the input** it judges the declaration: a component the unit *models* cannot be listed (a listing nobody needs reads like a decision), a missing `fate` or `reason` refuses, and a block on a unit that offers no pass-through refuses. Authorisation is the block's presence: an unlisted component that the unit does not model is still refused.
- **In the run** the component is carried unchanged in amount to the named outlet and counted in *every* balance — mass, elements and energy — so conservation stays checkable. In `ed10` the stack's mass row closes at 100.0000 % on 653.6157 kg/h and the carbon row carries benzene's 0.06 kmol C/h in and out.
- **On the output** it is recorded as a *declared approximation* on the `problemDivergence` channel: in the run banner, in the result JSON and in `converged/problemDivergence`, above the caveats. The answer therefore says, on its face, that the question it answered was "the stack, with benzene passing through".

Every stack result in `ed10` equals the benzene-free twin's to the last printed digit, which is the point: the approximation claims that benzene does not act on the stack, and the run shows that nothing in the stack moved. What the stack *cannot* show is whether the claim is physically true of the real brine — that remains the author's, which is why the `reason` is mandatory.

## 3.4 Where the contract lives

Each unit class answers one function, `acceptance()`, and the answer is data: the inlet phases the model is written for (for the whole unit, or *per port* — an absorber's gas feed must be a vapour and its wash a liquid), whether the declared aqueous solvent must be present, a closed rule naming which components the equations carry, the package capabilities it needs, the pass-through fates it offers, and any further refusal condition with its sentence. The default accepts every phase and models every component, so a unit that has not declared a contract behaves as it always did.

One function writes every acceptance refusal. It is called before any unit runs, on every *authored* inlet (so `choupo-lint` reports it too and the message points at the stream the student wrote), and again at the unit's own solve on the stream it is actually handed (a *produced* stream, and a recycle, exist only then). The documentation is generated from the same declaration: `choupoSolve -describe-units` prints every contract as JSON, the *Model* tab of the GUI and `docs/ai/unit-ops.md` draw the "models / may pass through / refuses / needs" lists from it, and the table below is written by the same generator. Nothing about what a unit accepts is written twice.

| Unit | Models | May pass | Refuses (clause names) |
|---|---|---|---|
| `FUG`, `shortcutColumn` | Every component of the feed, at the constant relative volatilities of the feed bubble point (Fenske-Underwood-Gilliland); no azeotrope is representable, and the model is a declared approximation, recorded on every run. | none | nothing |
| `absorber` | The declared solvent (held in the liquid, never evaporated), every `role solute` component that has a Henry pair against it (moved between the phases by the Kremser cascade), and every other gas component as an insoluble carrier that leaves in the gas. | none | phase of the gas feed: vapour only; phase of the liquid wash: liquid only; noSolvent; noSolute; soluteWithoutHenryPair; nonSoluteInTheLiquid |
| `adiabaticFlash` | Every component of the feed, split among the phases the package's equilibrium finds at the outlet pressure and the inlet's enthalpy (a vapour, one or two liquids, and any solid phase the case declares). | none | nothing |
| `batchCrystalliser`, `crystalliser` | Every component the stream carries. | none | crystallisableSolute; volatileCarrier |
| `batchDiafilter`, `membraneSW`, `spiralWoundModule` | The aqueous solvent (water unless the case declares another) and the solutes the case declares `role nonvolatile;`, each with its own permeance. | none | no aqueous solvent; component outside the model |
| `batchElectrodialysis` | The aqueous solvent and the IONS – every other component must cross its declared component -> species bridge (`aqueousMapping`) to ONE aqueous species with a record in the species catalogue. | none | no aqueous solvent; component outside the model; needs aqueousActivityModel |
| `column`, `distillationColumn` | Every component of every feed, equilibrated stage by stage on the package's K-values under constant molar overflow (a nonvolatile, K = 0, goes to the bottoms); the stage energy balance is NOT solved by the CMO method (the full MESH route solves it), and a Murphree efficiency only where declared. | none | nothing |
| `compressor` | Every component of the suction, as one vapour mixture priced by the package's enthalpy and entropy. | none | phase: vapour only |

| Unit | Models | May pass | Refuses (clause names) |
|---|---|---|---|
| `electrodialysisStack` | The aqueous solvent and the IONS – every other component must cross its declared component -> species bridge (`aqueousMapping`) to ONE aqueous species with a record in the species catalogue. | fate diluate \| concentrate | no aqueous solvent; component outside the model; needs aqueousActivityModel |
| `evaporator` | One volatile solvent (the single feed component whose role is `volatile` or `solute`), evaporated, and every other component as a nonvolatile solute that stays in the concentrate and raises its boiling point. | none | oneVolatileSolvent |
| `extract`, `extractor` | Every component of both streams, split between the extract and the raffinate by the package's liquid-liquid equilibrium at the column's one temperature. | none | phase of the feed: liquid only; phase of the solvent: liquid only |
| `flash`, `isothermalFlash` | Every component of the feed, split among the phases the package's equilibrium finds at the specified conditions (a vapour, one or two liquids, and any solid phase the case declares); a nonvolatile stays in the liquid (K = 0). | none | nothing |
| `heatExchanger` | Every component of both streams, by the effectiveness-NTU method on a constant heat capacity per stream (re-evaluated at its mean temperature); a phase change inside the exchanger is not modelled by the eps-NTU relation – the outlet is re-resolved at the package's equilibrium and the change is announced. | none | nothing |
| `ionExchanger` | The aqueous solvent and the IONS of a water analysis (Ca, Mg, Na, K, Cl, SO4, HCO3, ...) – every other component must cross its declared component -> species bridge (`aqueousMapping`) to ONE aqueous species with a record in the species catalogue. | none | no aqueous solvent; component outside the model; needs aqueousActivityModel |
| `mixer` | Every component of every inlet, summed; the outlet is the package's equilibrium at the outlet pressure, at the temperature that closes the adiabatic balance or at the declared one (then the duty it implies is published and the outlet carries the dominant phase's label). | none | nothing |
| `pressureExchanger` | Every component the two streams carry, each with a liquid molar volume or an aqueous bridge (a dissolved solute, whose volume is then the solvent's). | none | phase: liquid only; no aqueous solvent |
| `pump` | Every component the stream carries. | none | phase: liquid only |
| `splitter` | Every component of the inlet, divided by ONE fraction per outlet: it separates nothing, so every outlet has the inlet's composition, temperature and pressure. | none | nothing |
| `stripper` | The declared solvent (held in the liquid, never evaporated), every `role solute` component that has a Henry pair against it (stripped out of the liquid by the Kremser cascade), and every other gas component as an inert carrier that leaves in the gas. | none | phase of the rich liquid: liquid only; phase of the stripping gas: vapour only; noSolvent; noSolute; soluteWithoutHenryPair; nonSoluteInTheLiquid; solventInTheStrippingGas |
| `turbine` | Every component of the inlet, as one vapour mixture priced by the package's enthalpy and entropy. | none | phase: vapour only |

| Unit | Models | May pass | Refuses (clause names) |
|---|---|---|---|
| `valve` | Every component of the feed, throttled isenthalpically: the outlet is the package's equilibrium at the lower pressure carrying the inlet's enthalpy, in whatever phase the feed is. | none | nothing |

## 3.5 Reading a refusal

A refusal has the same anatomy everywhere.

```
absorber 'scrubber' (type absorber): the feed (stream 'gasFeed') carries
component(s) declared solutes that this unit cannot transfer:
      'air' (5 mol%) is ‘role solute‘ but has no Henry pair against 'water'.
  A component is transferred only when its record declares ‘role solute;‘
  AND a Henry pair against the solvent exists.  A solute with no pair is not
  transferred -- the cascade would carry it through as an inert, in silence,
  and report a gas it never washed ... at exit 0.
  REMEDY: curate the Henry pair <component>-<solvent> (a curation act, cited
  to its primary source), or change the component's role if it is meant to be
  an inert carrier.
```

Read it in four parts: *who* (the unit and its type), *what it was handed* (the stream and the component, with its amount), *why the unit cannot* (the sentence of the clause, which is the same sentence the generated documentation prints), and the *remedy*. Two habits follow. First, the remedy is a choice, not an order: pricing the component with a model that covers it and declaring it a pass-through are different scientific claims, and only the author can make either. Second, a refusal is not a failure of the simulator; it is the simulator declining to produce a number whose meaning it cannot state.

> **Common pitfall. A silent inert is an undeclared approximation.** The Kremser absorber transfers a component only when its record says `role solute` *and* a Henry pair against the solvent exists. A solute without a pair used to fall through to the inert branch: the gas passed on unwashed and the run exited 0. It is a contract clause now. When you write a unit, or a case, ask of every component that crosses it: is it in the equations, is it declared, or is it silently somewhere in between?

## 3.6 What the contract does not do

It does not decide physics. It refuses or announces only what a unit already cannot do, using the unit's own classification; it invents no threshold and no default fate. It also does not yet cover every unit: the table lists those that declare a contract, and an undeclared unit accepts what it always accepted. The remaining units are migrated as each is read for what it silently ignores.

# 4 One equation, many declarations: the map

*What you'll learn.* Every phase-equilibrium computation in this simulator is ONE equation — $\hat{f}_i^V = \hat{f}_i^L$ for each species — dressed in different, explicitly declared standard states. This section is the map between the vocabulary of the thermodynamics course (conventions, standard states, activity, fugacity) and the declarations a case file carries. Read it once and every formulation name in the engine becomes an instance of something you already know.

Phase equilibrium demands equal fugacities. The vapour side is $\hat{f}_i^V = y_i \hat{\varphi}_i P$ (ideal gas: $\hat{\varphi}_i = 1$). For the liquid the course teaches two *conventions*, distinguished only by the reference (standard) state chosen per species:

**Raoult / Lewis–Randall convention** (pure-liquid reference, $\gamma_i \to 1$ as $x_i \to 1$): $\hat{f}_i^L = x_i \gamma_i P_i^{\mathrm{sat}}(T)$ — the natural choice for condensable components.

**Henry convention** (infinite-dilution reference, $\gamma_i^* \to 1$ as $x_i \to 0$): $\hat{f}_i^L = x_i \gamma_i^* H_i$, or in the molal form used by the aqueous network, $a_i = m_i \gamma_i^{(m)}$ with $a_i = K(T)\, p_i$ — the natural choice for dissolved solutes.

The *unsymmetric convention* uses both at once in one liquid: the solvent on Raoult, the solutes on Henry. That is not a trick — it is the textbook treatment of electrolyte and dilute-solute systems, and it is exactly what the reactive flash implements: water through its activity ($a_w$), ammonia and acetic acid through molal gas–liquid constants tied to the speciation network (mass action + charge balance, pH solved), a molecular co-volatile through $\gamma\, x\, P^{\mathrm{sat}}$.

| course concept | engine declaration | tutorial |
|---|---|---|
| Raoult + $\gamma(x)$ | `gammaPhi` (NRTL, Wilson, …) | flash01–03 |
| $\hat{\varphi}$ from an EoS | `phiPhi` (SRK, PC-SAFT) | flash09_n2ch4_stryjek |
| Henry, dilute solute | `diluteSolution`; Henry pairs | flash08 |
| $\gamma_\pm$ on molality | `electrolyteGammaPhi` | crystallisers |
| law of mass action, pH | the speciation network | flash09–13 |
| unsymmetric convention | the reactive shape | flash09–13 |
| ideal-solution approximation | `approximations{}` (delimited) | flash12 |
| composite mixed-solvent v1 | `ionic davies; molecular NRTL;` | flash13 |

Two reading rules make every run header self-explaining. When an announcement names a *convention*, it is telling you which standard state a $\gamma$ or a $K$ refers to. When it names an *approximation*, it is telling you which physics was deliberately, visibly neglected — and which named slice will replace it. Nothing in the engine selects a model silently: the case declares, the run announces, and the limits of every composite treatment are proven by automated tests, not assumed.

# 5 Notation, units, and basis

*What you'll learn.* The conventions used throughout this manual and inside the simulator, so you can read the run-time log without translation.

## 5.1 Units

The simulator works in SI-derived process units, not strict SI:

| Quantity | Symbol | Unit |
|---|---|---|
| Molar flow rate | $F, L, V$ | kmol/h |
| Temperature | $T$ | K |
| Pressure | $P$ | bar |
| Volume | $V_R$ | $\mathrm{m}^3$ |
| Heat duty | $Q$ | kJ/kmol of feed |
| Specific heat capacity | $C_p$ | J/(mol K) |
| Heat of vaporisation | $\Delta H_{\mathrm{vap}}$ | J/mol |
| Saturation pressure | $P^{\mathrm{sat}}$ | bar |

*Intuition.* Plant data sheets use these units. SI purists wince at bar and `kmol/h`, but these are what every plant data sheet uses and the arithmetic is friendlier. The simulator does no hidden conversions — what you read in the log is in the unit above.

## 5.2 Vectors and components

For a mixture with $N_c$ components labelled $i = 1, \dots, N_c$:

- $z_i$ — feed mole fraction of component $i$; vector $\mathbf{z}$.
- $x_i$ — liquid-phase mole fraction; vector $\mathbf{x}$.
- $y_i$ — vapour-phase mole fraction; vector $\mathbf{y}$.
- $K_i$ — equilibrium $K$-value, $K_i \equiv y_i / x_i$; vector $\mathbf{K}$.

Mole fractions always sum to unity:

$$\sum_{i=1}^{N_c} z_i = 1, \qquad \sum_{i=1}^{N_c} x_i = 1, \qquad \sum_{i=1}^{N_c} y_i = 1.$$

## 5.3 Vapour fraction

For a feed of total molar flow $F$ partitioning into a vapour stream $V$ and a liquid stream $L$:

$$\beta \equiv \frac{V}{F}, \qquad 1 - \beta = \frac{L}{F}, \qquad \beta \in [0, 1]. \tag{3}$$

$\beta = 0$ is a saturated liquid (bubble point); $\beta = 1$ is a saturated vapour (dew point). All flash problems are, in the end, problems for $\beta$.

# Part II — Pure-component thermodynamic models

*What I do not know I do not think I know either.*
Socrates, in Plato's *Apology* 21d

# 1 Critical properties and corresponding states

*What you'll learn.* Every data file ships three numbers — $T_c$, $P_c$ and the acentric factor $\omega$ — that look mysterious until you place them on a $P$-$T$ phase diagram and ask what they actually mark. This chapter unpacks the geometry: $T_c$ and $P_c$ are the coordinates where the vapour-pressure line stops because liquid and vapour cease to be distinguishable; reduced coordinates collapse the behaviour of all simple fluids onto a single master curve (the law of corresponding states); $\omega$ is the third parameter that corrects for non-spherical, polar or hydrogen-bonded molecules. Three downstream correlations in the simulator (Watson, Sato-Riedel, the planned cubic EOS) consume nothing more than these three numbers, plus $T_b$.

## 1.1 The pure-component $P$-$T$ diagram

Pure substances exist as solid, liquid, or vapour depending on the $(T, P)$ chosen. The boundaries on a $P$-$T$ diagram are three coexistence curves — sublimation (S/V), melting (S/L), and vapourisation (L/V) — meeting at the *triple point*. Only the vapourisation curve is relevant to this manual, because every unit op deals with liquid + vapour fluid streams (no solids yet).

Tracing the L/V curve from the triple point upward, both $T$ and $P$ rise. But at a specific $(T_c, P_c)$, the curve simply **ends**: above that point, no matter how high $P$, no separate liquid and vapour exist — you have a single supercritical fluid phase. These two coordinates are the *critical temperature* and *critical pressure*, the values listed in every component's `.dat` file:

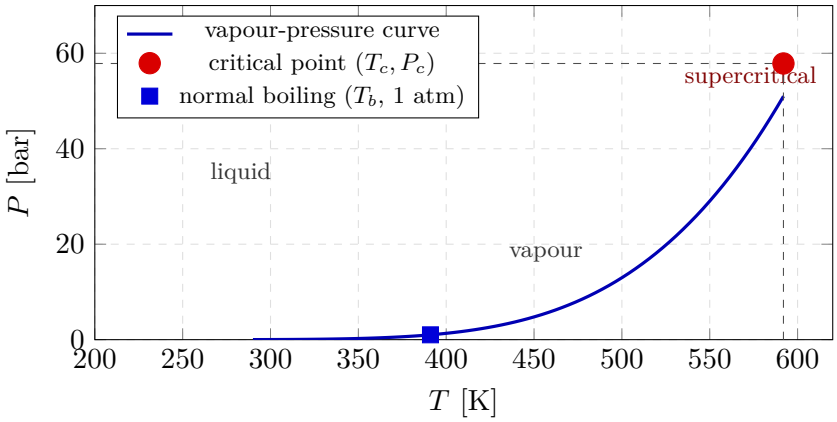



**Figure 1:** Vapour-pressure (saturation) curve of acetic acid in the $P$-$T$ plane, computed from the Antoine constants in `data/standards/components/aceticAcid.dat`. The square marks the normal boiling point ($T_b = 391$ K at 1 atm). The curve terminates at the critical point ($T_c = 591.95$ K, $P_c = 57.86$ bar): beyond it there is no liquid/vapour distinction.

Two physical readings of the critical point are worth carrying:

- **It is where liquid and vapour become indistinguishable.** Both phases approach a common density as $T \to T_c$; the latent heat $\Delta h^{\text{vap}}$ smoothly decays to zero (this is exactly what Watson (39) encodes).
- **It is the upper limit of the saturation curve.** Any Antoine fit, valid by definition only along the saturation line, ceases to make sense beyond $T_c$. That is why every `vaporPressure` block in the data files carries a `Trange` that always lies below $T_c$.

## 1.2 The law of corresponding states

van der Waals' great insight (1873) was that if you express $T$ and $P$ in units of their critical values, all simple fluids collapse onto the same curves. Define the *reduced* coordinates

$$T_r \equiv \frac{T}{T_c}, \qquad P_r \equiv \frac{P}{P_c}, \qquad V_r \equiv \frac{V}{V_c}, \tag{4}$$

and the two-parameter law of corresponding states asserts: any property expressed in reduced form is the *same function* of $(T_r, P_r)$ for every fluid. Argon and methane and benzene, plotted as $P_r$ versus $T_r$, give one universal saturation curve to within a few percent.

The simulator exploits this collapse repeatedly without naming it:

- **Watson** (39) writes $\Delta h^{\text{vap}}(T)/\Delta h^{\text{vap}}(T_b) = [(1 - T_r)/(1 - T_{br})]^{0.38}$ — a function of the single reduced variable $1 - T_r$.

- **Sato-Riedel** (258) for the liquid thermal conductivity is a function of $T_r$ and $T_{br}$ only.
- **The planned cubic EOS** (SRK/PR, v1.x) writes the attractive parameter $a$ as $a(T) = a_c \cdot \alpha(T_r, \omega)$, with $\alpha$ a universal function.

> *Intuition.* Corresponding states is what makes a thermophysical-property data sheet *short*. Without it, every fluid would need a custom correlation for every property. With it, we measure $T_c$, $P_c$ (and one extra parameter — next subsection) for every component and reuse universal forms. This is the same engineering compression mentioned in Chapter 2: a few numbers per component, many predictions.

## 1.3 The acentric factor $\omega$

Two-parameter corresponding states works for simple, near-spherical molecules (argon ∼0, methane ∼0.01). Real molecules deviate: longer chains, polar groups, hydrogen bonds all add **shape and stickiness** that two scalars cannot capture. Pitzer's 1955 fix was a third reduced parameter, the *acentric factor*,

$$\boxed{\omega \equiv -\log_{10}\left[P_r^{\text{sat}}(T_r = 0.7)\right] - 1.} \tag{5}$$

The definition takes the reduced saturation pressure at $T_r = 0.7$ (a representative point on the curve) and measures how far below 0.1 it lies. Simple fluids sit at $\omega \approx 0$ by construction (argon, oxygen). Hydrocarbons grow $\omega$ with chain length (*n*-hexane: 0.30; *n*-decane: 0.49). Hydrogen-bonded and dimerising species reach $\omega \approx 0.4$–$0.7$ (water: 0.34; ethanol: 0.65; acetic acid: 0.47 — moderated by vapour-phase dimerisation).

With $\omega$ in hand, the corresponding-states recipe becomes **three**-parameter: every dimensionless property is some universal function $f(T_r, P_r, \omega)$. This is the empirical backbone of the SRK and PR cubic EOS that will land in v1.x.

**Worked example 1.1.** Computing $\omega$ from Antoine for acetic acid At $T_r = 0.7$ we need $T = 0.7 \times 591.95 = 414.36$ K. From the Antoine constants in `aceticAcid.dat` ($A = 4.68206$, $B = 1642.54$, $C = -39.764$):

$$\log_{10} P^{\text{sat}}_{[\text{bar}]}(414.36) = 4.68206 - \frac{1642.54}{414.36 - 39.764} = 4.68206 - 4.3863 = 0.2957,$$

so $P^{\text{sat}}(414.36) = 1.977$ bar. In reduced form, $P_r^{\text{sat}} = 1.977/57.86 = 0.0342$, hence

$$\omega = -\log_{10}(0.0342) - 1 = 1.466 - 1 = 0.466,$$

which agrees with the tabulated $\omega = 0.4665$ in the data file to three significant figures. Notice that $\omega$ is not an independent measurement: it is derivable from $T_c$, $P_c$ and a single $(T, P^{\text{sat}})$ point at $T_r = 0.7$.

## 1.4 The compressibility factor $Z$

The compressibility factor measures how far a real fluid departs from the ideal-gas law:

$$Z \equiv \frac{PV}{nRT}, \qquad Z_{\text{idealgas}} = 1. \tag{6}$$

For low pressures, $Z \approx 1$ for any gas. As $P \to P_c$, $Z$ drops below 1 (attraction-dominated, vapour denser than ideal) or rises above 1 (repulsion-dominated, vapour less dense than ideal) depending on the molecule. Corresponding states tells us that $Z(T_r, P_r, \omega)$ is a *universal* function — generalised compressibility charts (Pitzer-Curl, Lee-Kesler) tabulate it once for all fluids. takes the ideal-gas limit $Z = 1$ throughout: justified for the vapour-pressure ranges of our 9 stock components at ambient operating pressures, breaks down above ∼10 bar. When a cubic EOS lands (post-v1) it will compute $Z(T_r, P_r, \omega)$ via the roots of the appropriate cubic. The data files already carry all the inputs.

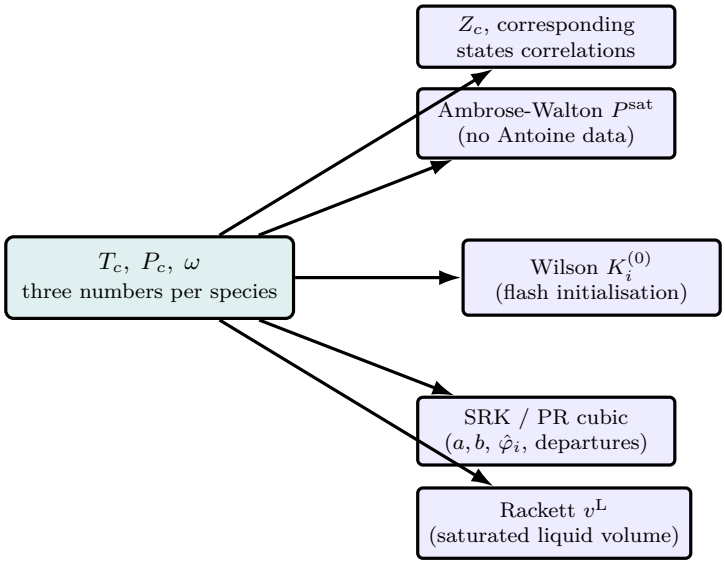



**Figure 2:** The corresponding-states payoff: three numbers per species — the critical temperature $T_c$, critical pressure $P_c$, and acentric factor $\omega$ — seed an entire thermodynamic stack. The same triple supplies the Ambrose-Walton vapour pressure when no Antoine constants exist, the Wilson $K$-value that initialises every flash, the $a(T)$ and $b$ of the SRK/PR cubic (and through it $\hat{\varphi}_i$ and the enthalpy/entropy departures), the Rackett saturated-liquid volume, and the generalised $Z_c$ correlations. Carrying $T_c, P_c, \omega$ in every component data file is what keeps the stack forward-compatible: select `model SRK;` and those three numbers *become* the equation of state, no new data required.

## 1.5 Where critical properties land in the stack

| which property uses it | how | in source code |
|---|---|---|
| $\Delta h^{\mathrm{vap}}(T)$ via Watson | $T_c$, $T_b$, $\Delta h^{\mathrm{vap}}_{T_b}$ | `Component::Hvap_latent` |
| $\kappa^{\mathrm{liq}}(T)$ via Sato-Riedel | $T_c$, $T_b$, $M$ | `SatoRiedel` (ships; `transport.liquid.thermalConductivity`) |
| $\varphi_i$ via cubic EOS | $T_c$, $P_c$, $\omega$ | post-v1 |
| $\mu^{\mathrm{gas}}(T)$ via Chapman-Enskog | $\sigma$, $\epsilon/k$ (Lennard-Jones) | `ChapmanEnskog` (ships since 2026-09-06; reads the record's `lennardJones{}` block, drafted from Svehla 1962 by `bin/curate/propose_lennard_jones.py`; `Chung` needs no such block) |
| $P^{\mathrm{sat}}(T)$ validity ceiling | $T_c$ as the upper bound on Antoine `Trange` | data-file metadata |

*Intuition.* A practical sanity check when staring at the numbers in a fresh `.dat` file: $T_b/T_c$ should sit in roughly the range 0.55–0.70 for non-polar molecules (acetic acid: $391/592 = 0.66$; benzene: $353/562 = 0.63$). If you see $T_b/T_c$ near 1 or below 0.5, you almost certainly have a data-entry error. The ratio $T_b/T_c$ is what Watson, Sato-Riedel and the corresponding- states machinery rely on; getting it wrong propagates everywhere.

# 2 Vapour pressure: from Clausius-Clapeyron to Antoine

*What you'll learn.* Antoine is not magic. It is the engineering refinement of a thermodynamic identity — the Clausius-Clapeyron equation — after two physical approximations and one empirical fix. This chapter walks through that derivation so the three constants $A, B, C$ have names, the $\log_{10}$ has a reason, and the failure modes of the correlation are visible by construction. Watson then enters as the $\gamma$-$\varphi$-free way to get $\Delta H_{\text{vap}}$ at any $T$ from a single calorimetric measurement at $T_b$.

## 2.1 The thermodynamic starting point

Along the liquid-vapour coexistence curve, the chemical potentials of liquid and vapour are equal: $\mu^{\text{L}}(T, P) = \mu^{\text{V}}(T, P)$. Take the differential along the curve,

$$\mathrm{d}\mu^{\text{L}} \;=\; \mathrm{d}\mu^{\text{V}} \;\Longrightarrow\; -s^{\text{L}}\,\mathrm{d}T + v^{\text{L}}\,\mathrm{d}P \;=\; -s^{\text{V}}\,\mathrm{d}T + v^{\text{V}}\,\mathrm{d}P,$$

where $s$ and $v$ are the molar entropy and molar volume of each phase. Solving for $\mathrm{d}P/\mathrm{d}T$ and using $\Delta s_{\text{vap}} = \Delta H_{\text{vap}}/T$ (from the Gibbs relation $\Delta g = 0$ at coexistence) gives the **Clausius-Clapeyron equation**:

$$\boxed{\frac{\mathrm{d}P^{\text{sat}}}{\mathrm{d}T} \;=\; \frac{\Delta H_{\text{vap}}}{T\,\Delta v_{\text{vap}}},} \tag{7}$$

where $\Delta v_{\text{vap}} = v^{\text{V}} - v^{\text{L}}$. Equation (7) is exact: no approximation has been made. It is a consequence of the second law (pillar 2 of Chapter 2) and no fitted constants enter.

## 2.2 Two physical approximations

Equation (7) is not yet usable because $v^{\text{V}}$ is itself a function of $T$ and $P$. Two approximations make it algebraic:

1. **The vapour behaves as an ideal gas:** $v^{\text{V}} = RT/P^{\text{sat}}$. Valid well below the critical point where vapour density is low.
2. **The liquid volume is negligible against the vapour volume:** $v^{\text{L}} \ll v^{\text{V}}$. At 1 bar, $T = T_b$, water has $v^{\text{L}} \approx 1.8 \times 10^{-5}$ m³/mol and $v^{\text{V}} \approx 3.1 \times 10^{-2}$ m³/mol — three orders of magnitude apart. The approximation worsens as $T \to T_c$ where the two volumes converge.

Under both approximations, $\Delta v_{\text{vap}} \approx RT/P^{\text{sat}}$, and (7) reduces to

$$\frac{\mathrm{d}P^{\text{sat}}}{\mathrm{d}T} \;=\; \frac{\Delta H_{\text{vap}}\,P^{\text{sat}}}{RT^2}, \tag{8}$$

or, dividing through by $P^{\text{sat}}$,

$$\boxed{\frac{\mathrm{d}\ln P^{\text{sat}}}{\mathrm{d}T} \;=\; \frac{\Delta H_{\text{vap}}}{RT^2}.} \tag{9}$$

This is the form usually called "simple Clausius-Clapeyron". It relates the slope of the log-saturation pressure to the latent heat divided by $RT^2$.

## 2.3 Integrating with constant $\Delta H_{\text{vap}}$: the rigid form

The simplest closure is to treat $\Delta H_{\text{vap}}$ as temperature-independent. Then (9) integrates immediately:

$$\ln P^{\text{sat}}(T) \;=\; -\frac{\Delta H_{\text{vap}}}{R\,T} \;+\; C_0, \tag{10}$$

where $C_0$ is a constant of integration fixed by one reference point (e.g. the normal boiling point: at $T_b$, $P^{\text{sat}} = 1$ atm). The rigid form (10) is a straight line on a $\ln P$-vs-$1/T$ plot, with slope $-\Delta H_{\text{vap}}/R$.

It is also pedagogically powerful: *measuring* $P^{\text{sat}}$ at two temperatures gives you $\Delta H_{\text{vap}}$ from the slope, with no calorimetry needed.

*Intuition.* The rigid form is the simplest closed expression a thermodynamic identity can be turned into. Its weakness is the constancy assumption on $\Delta H_{\text{vap}}$. Watson (39) already told us $\Delta H_{\text{vap}}$ varies with $T$ — decaying to zero at $T_c$ — so the rigid form must drift from the truth as $T$ moves away from the reference point. Across a 100 K range it is typically wrong by 5–10% in $P^{\text{sat}}$, ample reason to want a better fit.

## 2.4 The empirical fix: Antoine

What if $\Delta H_{\text{vap}}$ were not constant but *weakly varied* with $T$? Then (9) would integrate to something like $-F(T)/R+C_0$ with $F(T)$ a slowly-varying function. Fitting a flexible-enough empirical form to thirty years of measurement across the laboratory range, Antoine (1888) proposed

$$\boxed{\log_{10} P_i^{\text{sat}}(T)\,[\text{bar}] \;=\; A_i \;-\; \frac{B_i}{T\,[\text{K}] + C_i}.} \tag{11}$$

The modern statement of (11), together with the constant tables the component data files follow, is Poling, Prausnitz and O'Connell, Chapter 6 [4]. Antoine's own 1888 note is not held anywhere in this project — not in the bibliography, not in `src/`, not in a data record — so it is named above and deliberately not cited: a reference this project cannot check is a reference it does not make.

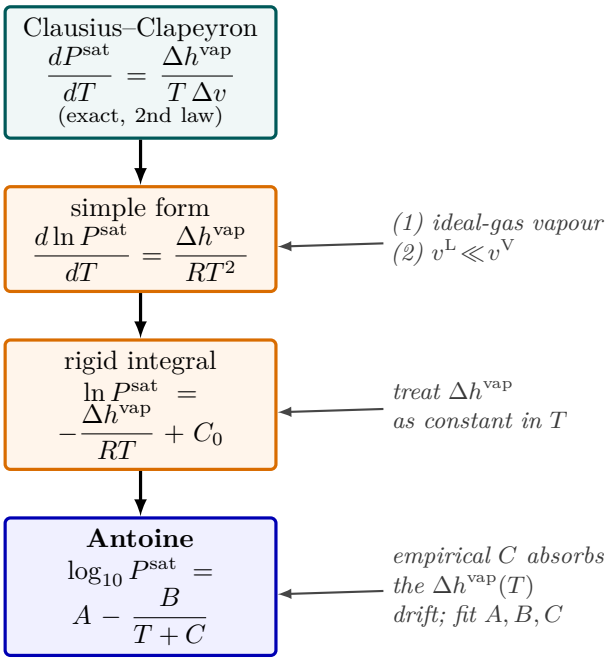



**Figure 3:** From a thermodynamic identity to the Antoine fit, in three honest steps. The exact Clausius–Clapeyron equation (teal, no fitted constants) becomes algebraic only after *two physical approximations* (orange): an ideal-gas vapour and a negligible liquid volume. Integrating with a constant latent heat gives the rigid straight-line-on-ln $P$-vs-$1/T$ form; the empirical Antoine constant $C$ (blue) then absorbs the temperature drift of $\Delta h^{\text{vap}}$ that the rigid form ignores. Each downward arrow is a named assumption, so the student sees exactly what was traded for usability.

Recognise the shape: it is the rigid Clausius-Clapeyron form (10) with one cosmetic and one substantive modification.

- $\log_{10}$ **vs** ln **(cosmetic).** Antoine historically used base-10 logs because engineers read tables with them. $A \log 10$ and $B \log 10$ differ from the natural-log constants by a factor of $\ln 10 \approx 2.303$; the physics is unchanged.
- $T + C$ **in the denominator (substantive).** Instead of $1/T$ the denominator is $1/(T + C)$. The constant $C$ absorbs the leading-order temperature dependence of $\Delta H_{\text{vap}}$. In the rigid form $B$ was identified with $\Delta H_{\text{vap}}/(R \ln 10)$; in Antoine $B$ no longer has that clean meaning, but the three-parameter fit gains an extra order of magnitude in accuracy over the same range.

> *Intuition.* The Antoine constants are *not three independent measurements*; they are the three numbers that best track $\ln P^{\text{sat}}$ across the range in which the fit was done. Reusing them outside that range is asking the formula to extrapolate. Asking a fit to extrapolate is asking the fit to invent.

## 2.5 Failure modes of Antoine

Three ways Antoine breaks down, all visible from the structure of (11):

- **Beyond the validity range.** Antoine is calibrated on $[T_{\min}, T_{\max}]$, usually straddling $T_b$. Push far below $T_{\min}$ and the rigid Clausius-Clapeyron approximation (constant $\Delta H_{\text{vap}}$) starts to lie; push past $T_{\max}$ approaching $T_c$ and *both* approximations (constant $\Delta H_{\text{vap}}$ *and* ideal-gas vapour) collapse.

- $T + C$ **flips sign.** Antoine fits with $C$ negative produce undefined logarithms when $T = |C|$. Typically $|C| < T_{\min}$ so this never bites inside the range; if you extrapolate below $|C|$ you get a domain error.
- **Near the critical point.** Above $\sim 0.95\,T_c$ the vapour is no longer dilute. No three-parameter form can recover the critical exponents; Pitzer/Watson-style corresponding-states relations take over.

Component data files in `data/standards/components/` declare a `Trange` explicitly:

```
vaporPressure
{
    model         Antoine;
    coefficients  (4.68206  1642.540  -39.764);   // aceticAcid
    Trange        (290  392);                     // K
}
```

does not yet *enforce* the range at run time; this is on the bug-watch. But the data are there and the reader of a `.dat` is expected to honour them.

## 2.6 Watson for $\Delta H_{\text{vap}}(T)$

The latent heat is already derived in Chapter 6 (Eq. 39). Repeated here for completeness:

$$\Delta H_{\text{vap}}(T) \;=\; \Delta H_{\text{vap}}(T_b)\left(\frac{T_c - T}{T_c - T_b}\right)^{0.38}. \tag{12}$$

Watson is a corresponding-states result: it depends only on $T_r$ and $T_{br}$, and the 0.38 exponent is the universal value Watson extracted from a broad survey of vapour-pressure curves in 1943[5]. The simulator computes it in `Component::Hvap_latent`.

## 2.7 Worked example: water from $\ln P$ vs $1/T$ to Antoine

**Worked example 2.1.** Two ways to compute $P^{\text{sat}}$ of water at 320 KData from `data/standards/components/water.dat` (DIPPR-style, valid 273–373 K): $A = 5.40221$, $B = 1838.675$, $C = -31.737$, $T_b = 373.15$ K, $T_c = 647.10$ K, $\Delta H_{\text{vap}}(T_b) = 40\,650$ J/mol.
**Antoine, direct.**

$$P^{\text{sat}}(320) = 10^{\,5.40221 - 1838.675/(320-31.737)} = 10^{\,5.40221 - 6.3781} \approx 10^{-0.9759} \approx 0.1058 \text{ bar}.$$

**Rigid Clausius-Clapeyron, for comparison.** Anchored at $T_b = 373.15$ K, $P^{\text{sat}} = 1.01325$ bar, using $\Delta H_{\text{vap}}(T_b) = 40\,650$ J/mol and the rigid form (10),

$$\ln\frac{P^{\text{sat}}(320)}{1.01325} = -\frac{40\,650}{8.314}\left(\frac{1}{320} - \frac{1}{373.15}\right) \approx -2.171,$$

giving $P^{\text{sat}} \approx 1.01325 \cdot e^{-2.171} \approx 0.116$ bar.
**Difference.** Antoine: 0.106 bar. Rigid form: 0.116 bar. A ∼10% gap, exactly the order of magnitude we predicted in §Integrating with constant $\Delta H_{\text{vap}}$. Antoine wins because its extra fitted constant $C$ corrects for the weak temperature dependence of $\Delta H_{\text{vap}}$ that the rigid form ignored.
**Watson at 320 K, for the energy balance.**

$$\Delta H_{\text{vap}}(320) = 40\,650\left(\frac{647.10 - 320}{647.10 - 373.15}\right)^{0.38} \approx 40\,650 \cdot 1.070 \approx 43\,480 \text{ J/mol}.$$

Water below its boiling point is harder to evaporate than at $T_b$ itself, by about $\sim 7\%$ — consistent with $\Delta H_{\text{vap}} \to 0$ as $T \to T_c$ being a slow approach.

*Intuition.* Antoine and Watson serve two different masters. Antoine asks "what is the saturation pressure?" — the answer is needed *every time* we compute a $K$-value in the flash. Watson asks "how much heat does this phase change move?" — the answer is needed every time the energy balance crosses the L/V boundary (adiabatic flash, distillation reboiler, condenser duty). The two correlations are independent in the simulator; you can use one without the other.

## 2.8 Ambrose-Walton: $P^{\text{sat}}$ when there are no Antoine data

Antoine needs three numbers *fitted to measured vapour pressures*. A component you have just *estimated* from its molecular structure (Joback groups → $T_c$, $P_c$; Lee-Kesler → $\omega$; Chapter 1) has no such measurements — so Antoine cannot be filled, and without $P^{\text{sat}}$ the species is not flashable. The Ambrose-Walton correlation closes exactly that

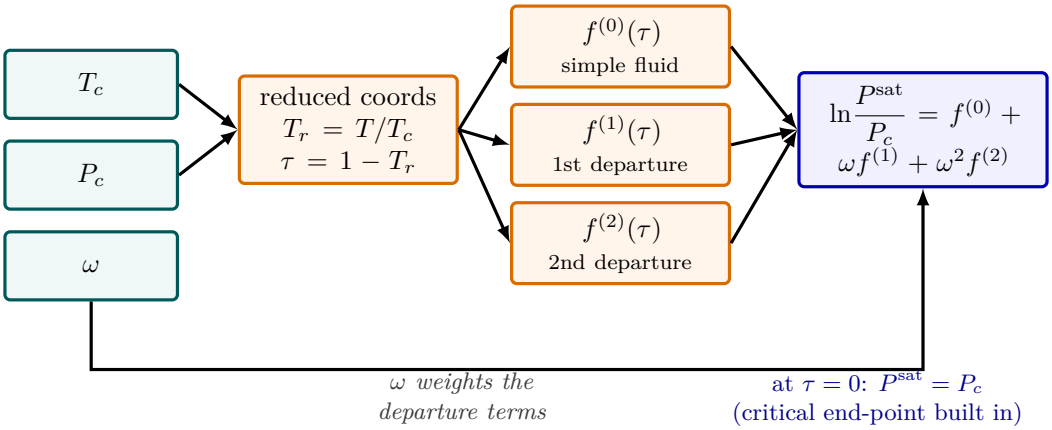



**Figure 4:** Ambrose–Walton turns the *three* corresponding-states constants an estimate already produces ($T_c, P_c, \omega$, teal) into the *whole* saturation curve, with no measured vapour pressure. The reduced temperature feeds three *universal* shape functions $f^{(0)}, f^{(1)}, f^{(2)}$ (orange, the same for every fluid); the acentric factor $\omega$ weights the two departure terms and the sum returns $\ln(P^{\mathrm{sat}}/P_c)$ (blue). Because every $f^{(i)}$ vanishes at $\tau = 1 - T_r = 0$, the curve passes through $P^{\mathrm{sat}} = P_c$ at the critical point *by construction*—which is exactly what makes a freshly estimated component flashable.

gap. It is a *three-parameter corresponding-states* form: it asks for only $T_c$, $P_c$ and $\omega$ — precisely the three constants the estimate already produced — and returns the whole saturation curve.

The idea continues the corresponding-states reasoning of §1. Pitzer's three-parameter principle says the reduced vapour pressure $P^{\mathrm{sat}}/P_c$ is a *universal* function of the reduced temperature $T_r = T/T_c$ once the acentric factor $\omega$ is admitted as a third coordinate. Expanding $\ln(P^{\mathrm{sat}}/P_c)$ as a power series in $\omega$ and truncating at $\omega^2$,

$$\ln\frac{P^{\mathrm{sat}}}{P_c} \;=\; f^{(0)}(T_r) \;+\; \omega\, f^{(1)}(T_r) \;+\; \omega^2 f^{(2)}(T_r), \tag{13}$$

where $f^{(0)}$ is the simple-fluid reference (argon/krypton/xenon, $\omega \approx 0$) and $f^{(1)}, f^{(2)}$ are the first- and second-order departures. Ambrose and Walton (1989) fitted the three shape functions (transcribed here, as in `AmbroseWalton.H`, from [4] Eq. 7-2.7; the 1989 original was not re-read) to high-quality data using a Wagner-style basis in

$$\tau \;\equiv\; 1 - T_r,$$

which vanishes at the critical point and grows toward the triple point. Their result, the form the simulator uses verbatim (`AmbroseWalton::psat`), is

$$\begin{aligned} f^{(0)} &= \left(-5.97616\,\tau + 1.29874\,\tau^{1.5} - 0.60394\,\tau^{2.5} - 1.06841\,\tau^{5}\right)/T_r, \\ f^{(1)} &= \left(-5.03365\,\tau + 1.11505\,\tau^{1.5} - 5.41217\,\tau^{2.5} - 7.46628\,\tau^{5}\right)/T_r, \\ f^{(2)} &= \left(-0.64771\,\tau + 2.41539\,\tau^{1.5} - 4.26979\,\tau^{2.5} + 3.25259\,\tau^{5}\right)/T_r. \end{aligned} \tag{14}$$

The fractional powers $\tau^{1.5}$ and $\tau^{2.5}$ are exactly the Wagner exponents that let a smooth analytic form track the steep curvature of $\ln P^{\mathrm{sat}}$ down toward the triple point — the same curvature Antoine captures with its single empirical constant $C$. At the critical point $\tau = 0$, every term vanishes, so $\ln(P^{\mathrm{sat}}/P_c) = 0$, i.e. $P^{\mathrm{sat}} = P_c$ at $T = T_c$ *by construction*. The simulator caps the output at $P_c$ for any $T \geq T_c$ (there is no saturation above the critical point).

**What the code does.** `AmbroseWalton::psat` is a pure function of $(T, T_c, P_c, \omega)$ — it carries no fitted state, so the estimate op (`estimateComponent`) calls the *same* function to fill a `.dat` and the flash calls it at run time. Note one bookkeeping detail visible in the source: $P_c$ is stored in **bar** everywhere in Choupo (as the equations of state also assume), and the constructor converts it to Pa once so that `Psat_Pa` returns pure SI. The three constants are *not* re-declared inside the `vaporPressure` block — they already live at component level, and `Component::Component` injects them, so a case author writes only

```
vaporPressure { model AmbroseWalton; }   // Tc, Pc, omega taken from the component
```

**Worked example 2.2.** Ambrose-Walton $P^{\text{sat}}$ of benzeneBenzene is a near-simple fluid: $T_c = 562.05$ K, $P_c = 48.95$ bar, $\omega = 0.210$ (Poling et al., 5th ed., App. A). Evaluate at the normal boiling point $T_b = 353.25$ K — a self-consistency check, since the answer should land on 1 atm.
First the reduced coordinates,

$$T_r = \frac{353.25}{562.05} = 0.6285, \qquad \tau = 1 - T_r = 0.3715.$$

The $\tau$ powers: $\tau^{1.5} = 0.22643$, $\tau^{2.5} = 0.08412$, $\tau^5 = 0.00708$. The shape functions from (14) (each already divided by $T_r$):

$$f^{(0)} = -3.1574, \quad f^{(1)} = -3.3820, \quad f^{(2)} = -0.0475.$$

Assemble (13) with $\omega = 0.210$,

$$\ln\frac{P^{\text{sat}}}{P_c} = -3.1574 + (0.210)(-3.3820) + (0.210)^2(-0.0475) = -3.8697,$$

so

$$P^{\text{sat}} = 48.95\,\text{bar} \times e^{-3.8697} = 48.95 \times 0.020862 = 1.021\ \text{bar},$$

within 0.8% of the 1.013 bar it *should* return at $T_b$. The residual gap is honest: $\omega$ here came from a Lee-Kesler relation, while the saturation curve came from the Ambrose-Walton fit — two different correlations, so they disagree by a fraction of a percent even on a textbook fluid. At 25 °C (298.15 K) the same evaluation gives 0.132 bar against a measured ≈ 0.127 bar, about 4% high — the expected accuracy band for a constants-only estimate.

**Power and limit.** The power is that Ambrose-Walton needs

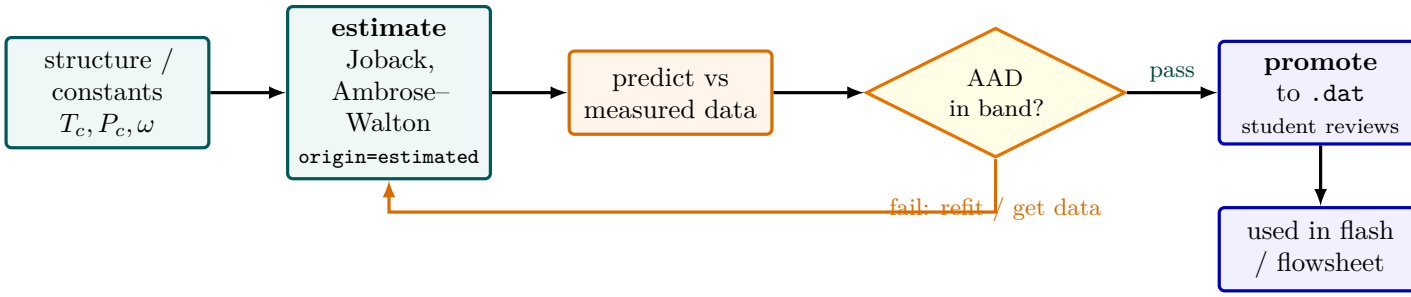



**Figure 5:** The glass-box property pipeline: **estimate**, then *validate*, then promote. From a molecular structure or three critical constants a group-contribution or corresponding-states method produces a value tagged `origin=estimated` (teal). Before it is trusted it is compared against measured data and scored by the average-absolute-deviation (AAD); the gate (yellow) is the project's no-silent-crutch rule applied to data quality. Only a value that clears its accuracy band is *promoted* into the curated `.dat` the student reviews (blue) and the solver then reads. A failing estimate loops back to be refitted or replaced by laboratory data—it is never silently used.

*nothing measured about the vapour pressure itself*: feed it the three critical/acentric constants — which corresponding states gives you even for a brand-new molecule — and it returns the entire curve from triple point to critical point, with the correct critical end-point built in. This is what makes an *estimated* component flashable without a single laboratory point. The limit is that it is an *estimate*, not data. It is good to a few percent for non-polar and weakly polar species; it *degrades* for strongly polar, hydrogen-bonding and associating compounds — water, alcohols, carboxylic acids — where the universal $f^{(i)}$ shapes simply do not describe the real curve. The glass-box rule therefore stands: the estimate op overlays the predicted Ambrose-Walton curve on any measured points in the component's `reference{}` / `experimental{}` blocks, and you look before you trust it for design. When you *do* have measured vapour pressures, fit Antoine (§2) and use that — Ambrose-Walton is the fallback that keeps the simulator running until the data arrive, not a replacement for them.

*Intuition.* Antoine and Ambrose-Walton answer the same question — "what is $P^{\text{sat}}(T)$?" — from opposite directions. Antoine is *data-first*: three constants regressed from measurements, accurate inside the fitted window, silent about everything outside it. Ambrose-Walton is *theory-first*: a universal corresponding-states shape with no compound-specific tuning beyond $(T_c, P_c, \omega)$, defined everywhere from triple to critical point but only as accurate as the universality assumption holds. A polar molecule breaks the universality; a molecule outside its Antoine `Trange` breaks the fit. The honest workflow is to use Antoine where the data live and fall back to Ambrose-Walton where they do not — and to plot both whenever you can.

# 3 Sublimation: the solid-vapour curve and the triple point

*What you'll learn.* The same Clausius-Clapeyron identity that produced Antoine (§2) for the liquid-vapour curve produces the *solid-vapour* (sublimation) curve when the latent heat is the heat of sublimation rather than of vapourisation. This section derives the sublimation line, shows why the three coexistence curves must meet at one point with $\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$, and documents the exact algorithm `PurePhaseDiagram` uses to draw the solid/liquid/vapour $P$-$T$ diagram — sublimation by Clausius-Clapeyron, melting by the full Clapeyron slope — so the triple-point picture of §1 is finally closed on both ends.

## 3.1 Why a solid sublimes: the same identity, a different phase

The derivation of §2 never used the fact that the condensed phase was a liquid. It used only that two phases coexist, so their chemical potentials are equal along the boundary, and that the Gibbs relation gives $\Delta s = \Delta h / T$ at coexistence. Replace "liquid" by "solid" everywhere and the *exact* Clapeyron equation (7) reads, for the solid-vapour boundary,

$$\frac{\mathrm{d}P^{\text{sub}}}{\mathrm{d}T} \;=\; \frac{\Delta H_{\text{sub}}}{T\,\Delta v_{\text{sub}}}, \qquad \Delta v_{\text{sub}} = v^{\text{V}} - v^{\text{S}}, \tag{15}$$

where $\Delta H_{\text{sub}}$ is the molar enthalpy of sublimation (solid → vapour, directly, with no liquid in between). Apply the *same* two approximations as before — the vapour is ideal, $v^{\text{V}} = RT/P$, and the solid volume is negligible against the vapour, $v^{\text{S}} \ll v^{\text{V}}$ (a solid is even denser than the liquid, so this is *safer* than for the L/V curve) — and (15) collapses to the simple form

$$\frac{\mathrm{d}\ln P^{\text{sub}}}{\mathrm{d}T} \;=\; \frac{\Delta H_{\text{sub}}}{R\,T^2}. \tag{16}$$

This is (9) with $\Delta H_{\text{vap}}$ swapped for $\Delta H_{\text{sub}}$ — "Antoine extended to the solid", except that we do not bother with a three-constant Antoine fit here. Sublimation pressures are tiny and rarely measured across a wide range, so the simulator integrates (16) in its *rigid* (constant-$\Delta H_{\text{sub}}$) form, anchored at the triple point $(T_t, P_t)$:

$$\boxed{P^{\text{sub}}(T) \;=\; P_t\,\exp\!\Big[-\frac{\Delta H_{\text{sub}}}{R}\Big(\frac{1}{T}-\frac{1}{T_t}\Big)\Big].} \tag{17}$$

This is exactly the line in `PurePhaseDiagram::run`,

```
const scalar P = Pt * std::exp(-(Hsub / R) * (1.0 / T - 1.0 / Tt));
```

evaluated on a grid of $m = \max(6, n/2)$ points from $0.78\,T_t$ up to $T_t$ (the lower bound keeps $P^{\text{sub}}$ within about three decades of $P_t$ so it does not vanish off the log axis).

## 3.2 The triple point and $\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$

The three coexistence curves — sublimation (S/V), melting (S/L) and vapourisation (L/V) — are not independent. At the unique *triple point* $(T_t, P_t)$ all three phases coexist, so the solid, liquid and vapour chemical potentials are equal there. Enthalpy is a state function, so the energy to go solid → vapour must equal solid → liquid → vapour around the triple point:

$$\boxed{\Delta H_{\text{sub}} \;=\; \Delta H_{\text{fus}} \;+\; \Delta H_{\text{vap}}} \qquad \text{(at the triple point)}. \tag{18}$$

Because $\Delta H_{\text{sub}} > \Delta H_{\text{vap}}$, the sublimation line is *steeper* than the vapourisation line on a $\ln P$ vs $1/T$ plot (slope $-\Delta H_{\text{sub}}/R$ against $-\Delta H_{\text{vap}}/R$, both negative): the two curves leave the triple point at different slopes, with the sublimation curve below and to the left. This kink at $T_t$ is the qualitative signature of the triple point, and (18) is the consistency check the curator applies before writing a `sublimation{}` block.

## 3.3 The melting line: the full Clapeyron slope

The solid-liquid boundary is the one curve where neither phase is a gas, so the ideal-gas trick of §2 is *unavailable*: $\Delta v_{\text{fus}} = v^{\text{L}} - v^{\text{S}}$ is a small difference of two condensed-phase volumes, not an $RT/P$ that dominates. We must keep the full Clapeyron equation (7):

$$\frac{\mathrm{d}P^{\text{fus}}}{\mathrm{d}T} \;=\; \frac{\Delta H_{\text{fus}}}{T\,\Delta v_{\text{fus}}}. \tag{19}$$

Over the modest temperature span of a melting line, $T$ barely moves from $T_t$ and $\Delta H_{\text{fus}}$, $\Delta v_{\text{fus}}$ are nearly constant, so (19) integrates to a near-straight, very steep line. The simulator linearises it directly, anchored at the triple point:

$$T^{\text{fus}}(P) \;=\; T_t \;+\; \frac{T_t\,\Delta v_{\text{fus}}}{\Delta H_{\text{fus}}}\,(P - P_t), \tag{20}$$

the line `const scalar T = Tt + (Tt * dVfus / Hfus) * (P - Pt)` in `PurePhaseDiagram`, swept from $P_t$ up to $\max(P_c, 10^8\ \text{Pa})$. The sign of $\Delta v_{\text{fus}}$ is what makes water famous: for almost every substance the solid is denser than

the liquid, so $\Delta v_{\text{fus}} > 0$ and the melting line leans *right* (higher pressure raises the melting point). Water is the anomaly — ice floats, so $\Delta v_{\text{fus}} < 0$ and the line leans *left* (pressure *lowers* the melting point). In `phase01_water_full` the curator enters `deltaVfus -1.63e-6` ($\text{m}^3$/mol, *negative*), and the melting line tilts the right way by construction.

## 3.4 What the algorithm needs — and what it draws

`PurePhaseDiagram` reads the solid anchor from two places, in precedence order (this matches the code exactly):

1. the component's curated `sublimation{}` block — `tripleT` $T_t$, `tripleP` $P_t$, `Hfus`, `Hsub` — the intrinsic pure-compound constants that live *with* the species (`Component::subTripleT()` etc.);
2. the op-dict `solid{}` block, which fills any field the component left blank *and* is the sole source of `deltaVfus` $\Delta v_{\text{fus}}$ — the fusion-line slope, deliberately kept out of the component block because it is a sample/measurement quantity, not a clean intrinsic constant.

The drawing logic then branches on what is available:

- **L/V only** (no solid anchor): the saturation curve runs from $0.45\,T_c$ to $T_c$ and the critical point is marked — the historical behaviour, every existing `.dat` unaffected.
- **+ sublimation** ($T_t, P_t, \Delta H_{\text{sub}}$ all present, flag `hasSub`): the saturation curve now starts at $T_t$, the sublimation curve (17) is drawn, and the triple point is marked.
- **+ fusion** (additionally $\Delta H_{\text{fus}} > 0$ and $\Delta v_{\text{fus}} \neq 0$, flag `hasFusion`): the melting line (20) is added.

Each row of the output CSV is tagged `saturation`, `sublimation`, `fusion`, `triple` or `critical`, so the GUI can colour the three curves and place the two singular points without re-deriving anything.

**Worked example 3.1.** Triple-point closure for `watertutorials/props/scan/phase01_water_full` draws the full water diagram with the op-dict anchor

$$T_t = 273.16\ \text{K}, \quad P_t = 611.657\ \text{Pa}, \quad \Delta H_{\text{fus}} = 6010, \quad \Delta H_{\text{sub}} = 51059\ \text{J/mol}.$$

**Consistency by (18).** The implied heat of vapourisation at the triple point is $\Delta H_{\text{vap}} = \Delta H_{\text{sub}} - \Delta H_{\text{fus}} = 51059 - 6010 = 45049$ J/mol, against the calorimetric $\approx 45\,054$ J/mol for water at 0 °C — the block is internally consistent to better than 0.02%.
**A point on the sublimation curve.** Evaluate (17) at $T = 260$ K (ice subliming below freezing):

$$P^{\text{sub}}(260) = 611.657\,\exp\!\left[-\frac{51059}{8.31446}\left(\tfrac{1}{260} - \tfrac{1}{273.16}\right)\right] = 611.657\,e^{-1.1383} = 195.9\ \text{Pa},$$

about 1.96 mbar — the vapour pressure of ice at −13 °C, within a few percent of the tabulated $\approx 196$ Pa, and the basis of freeze-drying (lyophilisation): pull the chamber below $P_t$ and ice leaves as vapour without ever melting.
**The melting tilt.** With $\Delta v_{\text{fus}} = -1.63 \times 10^{-6}$ $\text{m}^3$/mol, the slope (20) is

$$\frac{\mathrm{d}T}{\mathrm{d}P} = \frac{T_t\,\Delta v_{\text{fus}}}{\Delta H_{\text{fus}}} = \frac{273.16 \times (-1.63 \times 10^{-6})}{6010} = -7.4 \times 10^{-8}\ \text{K/Pa},$$

i.e. raising the pressure by 100 bar ($10^7$ Pa) lowers the melting point by about 0.74 K — the classic regelation number, and negative, as only water and a handful of other anomalies are.

*Intuition.* Dry ice and water ice are the two halves of the same picture. $CO_2$ has its triple point *above* 1 atm ($T_t = 216.6$ K, $P_t = 5.18$ bar), so at atmospheric pressure the liquid field is never crossed: warm a block of solid $CO_2$ and it goes straight to gas along the sublimation curve — that is why it is "dry". Water's triple point is *below* 1 atm (611 Pa), so at room pressure you cross the melting line first and ice becomes a puddle. Same three curves, same Clausius-Clapeyron slope $-\Delta H_{\text{sub}}/R$; the only thing that decides "dry" versus "wet" is whether the ambient pressure sits above or below $P_t$.

**Power and limit.** The power is reuse: one identity (Clausius-Clapeyron) plus one bookkeeping law ($\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$) gives the entire solid region from four curated numbers, with the triple point and critical point as the two anchors that make the diagram close. The limit is the same constant-$\Delta H_{\text{sub}}$ rigidity that makes the rigid L/V form (10) only approximate: (17) treats $\Delta H_{\text{sub}}$ as temperature-independent, exact only near $T_t$ and drifting as $T$ falls (which is why the curve is drawn only down to $0.78\,T_t$); and the melting line (20) is linearised, fine over its narrow span but never extrapolate it to the exotic high-pressure ice polymorphs. As always in Choupo, the diagram is *drawn, not asserted*: the CSV carries every $(T, P)$ point and its curve tag, so a student sees the kink at the triple point and the direction of the melting tilt rather than taking them on faith.

# 4 Saturated-liquid molar volume: the Rackett equation

Ambrose-Walton (§2.8) gave us $P^{\text{sat}}(T)$ from the same three constants $(T_c, P_c, \omega)$ an estimate produces. Its sibling on the *liquid* side answers the companion question: how much volume does one mole of saturated liquid occupy? The simulator needs this in three independent places — the liquid **density** reported for any stream (`ThermoPackage::density`), the pure-component liquid molar volumes $V_i^L$ that Wilson's $\Lambda_{ij} = \frac{V_j^L}{V_i^L}\exp(\cdots)$ demands ((69)), and the `Vliq` field an `estimateComponent` writes into a fresh `.dat`. All three route through one closure, `closures::rackettVliq`, so that what the estimate stores and what the flash later reads are the *same* number from the *same* formula — the glass-box invariant we hold for every estimated property.

## 4.1 Why a special correlation for the liquid?

A cubic equation of state (SRK, PR; Chapter 10) will happily return a liquid root $v^L$, but that root is the EoS's notorious weak spot: SRK over-predicts saturated-liquid volumes by 10–30%, and even the volume-translated forms drift with temperature. The liquid is nearly incompressible, so its molar volume is governed almost entirely by how close $T$ sits to $T_c$ — a one-dimensional problem along the saturation line — and a purpose-built corresponding-states correlation beats a general EoS by an order of magnitude here. That correlation is Rackett's.

## 4.2 From the critical point to the Rackett equation

Start at the critical point, where by definition

$$Z_c \;=\; \frac{P_c V_c}{R_{\text{g}} T_c}, \qquad \text{so} \qquad V_c \;=\; \frac{R_{\text{g}} T_c}{P_c}\, Z_c. \tag{21}$$

Rackett (1970) observed that if you plot $\ln V^{\text{sat}}$ against $(1-T_r)^{2/7}$ for a saturated liquid, the data fall on a remarkably straight line that, extrapolated to $T_r \to 1$, passes through $V_c$. Anchoring the line at the critical point and letting its slope be set by $Z_c$ gives the one-constant Rackett equation,

$$V^{\text{sat}}(T) \;=\; V_c\, Z_c^{(1-T_r)^{2/7}} \;=\; \frac{R_{\text{g}} T_c}{P_c}\, Z_c^{\,1+(1-T_r)^{2/7}}, \qquad T_r = \frac{T}{T_c}. \tag{22}$$

The two forms are identical: substituting $V_c = (R_{\text{g}}T_c/P_c)Z_c$ folds the leading $Z_c$ into the exponent, raising the 1. The peculiar exponent $2/7 \approx 0.2857$ is purely empirical — it is the power that linearises the most species at once — and the same $2/7$ you will see in COSTALD and other liquid-volume correlations descends from Rackett. At $T = T_c$ the bracket $(1-T_r)^{2/7} = 0$, the exponent is 1, and (22) returns exactly $V_c$, as a correlation seeded at the critical point must.

## 4.3 The Spencer-Danner / Yamada-Gunn substitution: $Z_c \to Z_{RA}$

Equation (22) has a flaw: the *true* $Z_c$ is itself only known to a percent or two, and using it makes the correlation miss the actual measured volume at lower $T_r$ where you care most. Spencer and Danner (1972) fixed this by replacing $Z_c$ with an *adjustable* Rackett parameter $Z_{RA}$ — a number fitted per compound so that (22) hits one known liquid density. $Z_{RA}$ is numerically close to $Z_c$ but not equal to it. When a fitted $Z_{RA}$ is *not* available — exactly the situation for an estimated component — Yamada and Gunn (1973) supplied a corresponding-states correlation for it in terms of the acentric factor,

$$Z_{RA} \;=\; 0.29056 \;-\; 0.08775\,\omega. \tag{23}$$

This is the form the simulator hard-codes. The constant 0.29056 is the generic small-molecule $Z_c \approx 0.29$; the $-0.08775\,\omega$ term shrinks it as molecules become more elongated and acentric, mirroring how $Z_c$ falls along a homologous series. Putting (23) into (22) is the operational formula:

$$\boxed{V^L(T) \;=\; \frac{R_{\text{g}} T_c}{P_c}\, Z_{RA}^{\,1+(1-T_r)^{2/7}}, \qquad Z_{RA} = 0.29056 - 0.08775\,\omega.} \tag{24}$$

**What the code does.** `closures::rackettVliq(T, Tc, Pc_Pa, omega)` is (24) verbatim: it forms $T_r = T/T_c$, computes $Z_{RA} = 0.29056 - 0.08775\,\omega$, raises it to $1 + (1-T_r)^{2/7}$, and multiplies by $R_{\text{g}}T_c/P_c$ (note $P_c$ passed in **Pa** here, so the result is SI, $\text{m}^3/\text{mol}$). Like the Ambrose-Walton closure it carries no fitted state — the estimate op and the run-time density call the identical function. This correlation is variously labelled "Rackett", "modified Rackett (Spencer-Danner)" or "Rackett/Yamada-Gunn" depending on whether $Z_{RA}$ is fitted or predicted; Choupo predicts it from $\omega$, so the diagnostic JSON tags the value `Rackett / Yamada-Gunn`.

## 4.4 Mixtures: mole-weighted molar volume

For a liquid mixture the simulator does *not* mix the constants; it computes each pure-component saturated volume at the mixture $T$ and mole-averages them (`ThermoPackage::density`, branch (3)),

$$V^L_{\text{mix}}(T,\mathbf{x}) \;=\; \sum_i x_i\, V^L_i(T) \;=\; \sum_i x_i\, \frac{R_g T_c^{(i)}}{P_c^{(i)}}\, Z_{RA,i}^{\,1+(1-T/T_c^{(i)})^{2/7}}, \tag{25}$$

and the reported liquid density is then $\rho = \overline{M}/V^L_{\text{mix}}$ with $\overline{M} = \sum_i x_i M_i$ the mole-averaged molar mass. This is the simplest defensible mixing rule: it assumes an *ideal liquid mixture* (no excess volume, $V^E = 0$), so the partial molar volume of each species equals its pure-component value. It is the same Amagat-additivity assumption that underlies treating the liquid as volumetrically ideal, and it is exactly the assumption Wilson's $V_j^L/V_i^L$ ratio also makes. Real mixtures show a few-percent $V^E$ (ethanol-water famously contracts), which this rule does not capture — the honest limit flagged below.

**Worked example 4.1.** Rackett liquid volume and density of benzeneBenzene: $T_c = 562.05$ K, $P_c = 48.95$ bar $= 4.895 \times 10^6$ Pa, $\omega = 0.210$, $M = 78.11$ g/mol. Evaluate at 25 °C ($T = 298.15$ K), the temperature `estimateComponent` pins.
Reduced temperature and the Rackett parameter,

$$T_r = \frac{298.15}{562.05} = 0.5305, \qquad Z_{RA} = 0.29056 - 0.08775(0.210) = 0.27213.$$

The exponent,

$$1 + (1 - T_r)^{2/7} = 1 + (0.4695)^{0.2857} = 1 + 0.8084 = 1.8084.$$

The prefactor,

$$\frac{R_g T_c}{P_c} = \frac{8.314 \times 562.05}{4.895 \times 10^6} = 9.547 \times 10^{-4}\ \text{m}^3/\text{mol}.$$

Assemble (24),

$$V^L = 9.547 \times 10^{-4} \times (0.27213)^{1.8084} = 9.547 \times 10^{-4} \times 0.09367 = 8.94 \times 10^{-5}\ \text{m}^3/\text{mol} = 89.4\ \text{cm}^3/\text{mol}.$$

The measured saturated-liquid volume of benzene at 25 °C is $\approx 89.4$ cm$^3$/mol — agreement to better than 1% for a non-polar, near-spherical molecule. The density follows,

$$\rho = \frac{M}{V^L} = \frac{78.11 \times 10^{-3}\ \text{kg/mol}}{8.94 \times 10^{-5}\ \text{m}^3/\text{mol}} = 874\ \text{kg/m}^3,$$

against the handbook 874 kg/m$^3$. Rackett is at its best precisely here.

**Power and limit.** The power mirrors Ambrose-Walton's: from only $(T_c, P_c, \omega)$ — the three constants corresponding states hands you even for a freshly estimated molecule — Rackett returns the whole saturated-liquid volume curve, correct to 1–2% for normal fluids and seeded exactly at the critical point. It is far better than reading the liquid root of a cubic EoS, and it costs one `pow`. The limits are three. (i) It is a *saturated*-liquid correlation: it carries *no* pressure dependence, so it ignores the (small) compression of a sub-cooled liquid above its saturation pressure — a good approximation because liquids are nearly incompressible, but not exact for high-pressure work. (ii) Accuracy *degrades* for the same polar, hydrogen-bonding species that defeat Ambrose-Walton — water, alcohols, acids — because the universal $Z_{RA}(\omega)$ shape no longer describes them; with a measured density you should fit $Z_{RA}$ (Spencer-Danner) rather than predict it from $\omega$. (iii) The mole-weighting (25) assumes $V^E = 0$, so it misses the volume change on mixing of strongly non-ideal solutions. For pedagogy and for the density figure on a flowsheet stream these are acceptable; for a custody-transfer liquid density they are not, and the glass-box rule applies — the estimate op overlays the Rackett curve on any measured liquid densities in the component file so you look before you trust it.

> *Intuition.* Ambrose-Walton and Rackett are the two halves of one corresponding-states bargain. Give up three constants $(T_c, P_c, \omega)$ and you get back *both* legs of the saturation envelope — the pressure $P^{\text{sat}}(T)$ *and* the liquid volume $V^L(T)$ — each anchored at the critical point and each accurate to a few percent for normal fluids. Neither needs a single measurement of the property it predicts; both fail on the same polar molecules; both are fallbacks that keep an estimated component fully simulatable until real data arrive, and both are overlaid on data, never substituted for it.

# 5 Heat capacity

The molar heat capacity is parametrised as a polynomial in temperature:

$$C_p^{\mathrm{ig}}(T) \;=\; a_0 + a_1 T + a_2 T^2 + a_3 T^3 + a_4 T^4 \quad [\mathrm{J/(mol\,K)}]. \tag{26}$$

Two such polynomials per component: one for the ideal gas ($C_p^{\mathrm{ig}}$, used in vapour-phase energy balances) and one for the liquid ($C_p^{L}$, used in liquid-phase energy balances).

For an enthalpy difference between $T_{\mathrm{ref}}$ and $T$:

$$h(T) - h(T_{\mathrm{ref}}) \;=\; \int_{T_{\mathrm{ref}}}^{T} C_p(T')\,dT'. \tag{27}$$

With the polynomial form, this integral is closed-form term by term — the simulator does it analytically, not by quadrature.

# 6 From component data to integral properties

*What you'll learn.* Each `data/standards/components/*.dat` file holds a short list of numbers: $T_c$, $P_c$, $\omega$, $T_b$, $\Delta h^{\mathrm{vap}}_{T_b}$, $V^{\mathrm{liq}}$, Antoine constants, $c_p$ polynomials. This chapter shows what each one *means*, where the simulator reads it, and how those tiles are assembled into the integral quantities that drive every energy balance and every Gibbs-energy minimisation: pure-component enthalpy and entropy, the latent heat at an arbitrary $T$ via Watson, mixture enthalpy and mixture Gibbs energy, and Gibbs-Duhem as the consistency check on activity-coefficient fits.

## 6.1 Anatomy of a component data file

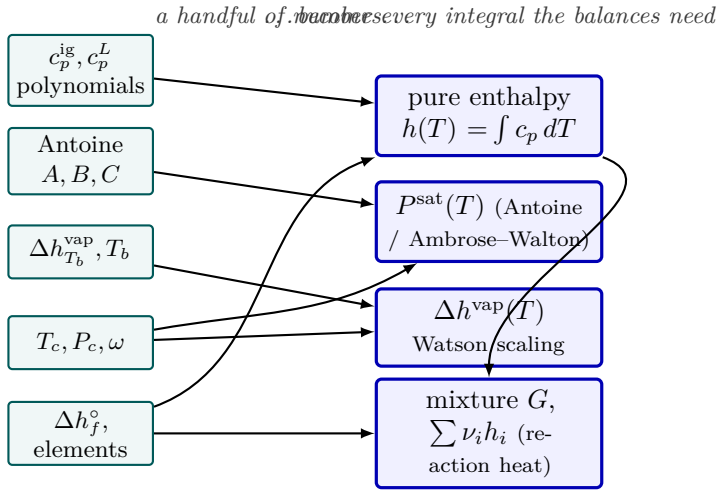



**Figure 6:** What a component `.dat` file is *for*. Each entry is a small tile (teal): a few $c_p$ coefficients, Antoine or critical constants, a single latent heat at $T_b$, the formation enthalpy and its element basis. The simulator assembles those tiles—analytically, term by term—into the integral quantities (blue) that every energy balance and every Gibbs-energy minimisation actually consumes: pure-component enthalpy, the saturation curve, the temperature-corrected latent heat via Watson, and the mixture Gibbs energy whose $\sum_i \nu_i h_i$ returns the heat of reaction. One number can feed several integrals (note the shared $T_c$ and $\Delta h^\circ_f$ arrows).

A typical entry, slightly trimmed for the page:

```
name        aceticAcid;
formula     C2H4O2;
CAS         64-19-7;

MW          60.052;         // kg/kmol
Tc          591.95;         // K
Pc          57.86;          // bar
omega       0.4665;         // [-]
Tb          391.05;         // K
HvapTb      24390;          // J/mol
Vliq        5.752e-5;       // m^3/mol  (57.52 cm^3/mol)

vaporPressure
{
    model         Antoine;
    coefficients  (4.68206  1642.540  -39.764);
    Trange        (290  392);
}

idealGasHeatCapacity {... }
liquidHeatCapacity   {... }
```

Each scalar plays a specific role. The right-hand column below points to where the simulator reads it:

| field | unit | meaning, and where it is consumed |
|---|---|---|
| `MW` | kg/kmol | Molar mass. Carries through mass-flow conversions and the sizing chapters; entered in every stoichiometry / yield calculation. |
| `Tc` | K | Critical temperature. Sets the upper bound of every saturation curve (vapour and liquid become indistinguishable at $T_c$) and is the explicit input to the Watson correlation (39). Also feeds the SRK/PR $\alpha(T,\omega)$ (Chapter 10) and the Wilson $K$-value seed (185). |
| `Pc` | bar | Critical pressure. Feeds the SRK/PR $a, b$ parameters, the Wilson $K$-value seed (185), and the Chung viscosity estimate (Chapter 2). |
| `omega` | – | Pitzer's *acentric factor*, defined as $\omega = -\log_{10} P_r^{\mathrm{sat}}(T_r = 0.7) - 1$. Zero for spherical molecules (argon, methane), positive for elongated and hydrogen-bonded ones (acetic acid is on the high side at 0.47 because of its dimerisation in the vapour). Used by SRK/PR's $\alpha(T,\omega)$, the Wilson $K$-value seed, and the Chung viscosity estimate. |
| `Tb` | K | Normal boiling point ($P^{\mathrm{sat}}(T_b) = 1$ atm). Anchors the Watson correlation: the temperature where the latent heat is tabulated. Also a sanity-check ruler for Antoine fits. |
| `HvapTb` | J/mol | Latent heat of vaporisation *at* $T_b$, the reference value Watson extends to other temperatures. |
| `Vliq` | m³/mol | Liquid molar volume at 25 °C. The PFR reads it to convert molar feed to a volumetric flow $Q = F \cdot \bar{V}^{\mathrm{liq}}$ for the concentration $C_i = F_i/Q$; the stirred-tank sizing chain uses it for liquid hold-up. Treated as constant; pressure-volume corrections would be a ten-line edit. |
| `vaporPressure` | Antoine + Trange | $\log_{10} P^{\mathrm{sat}}_{[\mathrm{bar}]} = A - B/(T_{[\mathrm{K}]} + C)$. Used everywhere VLE shows up (flash, bubble-T, dew-T, distillation). |
| `idealGasHeatCapacity` | polynomial | $c_p^{\mathrm{ig}}(T) = a_0 + a_1 T + a_2 T^2 + a_3 T^3$, J/(mol · K). Drives the gas-phase part of every energy balance. |
| `liquidHeatCapacity` | polynomial | Same shape, for the saturated liquid. Drives the liquid-phase part of energy balances and the entropy integral below. |

*Intuition.* The data file looks like a flat list of numbers because it *is* one: a thermodynamic ground truth the simulator's models do not re-derive, only consult. Every entry is either **measured** (Tc, Pc from Reid/Prausnitz/Poling tables[4]; Tb, HvapTb from boiling-point calorimetry; Vliq from density at 25 °C; Antoine from regression of vapour-pressure tables; $c_p$ from calorimetric measurements) or **empirical** ($\omega$ from a defined regression formula). Nothing is invented; nothing is fitted by the simulator at run time. Chapter 7 maps out which numbers in the whole stack *would* be fittable in principle, and which the simulator currently takes as fixed.

## 6.2 Pure-component enthalpy

The thermodynamic ground level is fixed by convention: the simulator sets

$$h_i^{\mathrm{liq}}(T_{\mathrm{ref}}) \equiv 0, \qquad T_{\mathrm{ref}} = 298.15\ \mathrm{K}, \tag{28}$$

for every component in the liquid phase. Departures from $T_{\mathrm{ref}}$ in the liquid follow from $c_p^{\mathrm{liq}}$:

$$h_i^{\mathrm{liq}}(T) = \int_{T_{\mathrm{ref}}}^{T} c_{p,i}^{\mathrm{liq}}(T')\,\mathrm{d}T'. \tag{29}$$

With the polynomial $c_p^{\mathrm{liq}} = a_0 + a_1 T + a_2 T^2 + \dots$ shipped in the data file, the integral is term-by-term elementary; the simulator evaluates it in `src/thermo/heatCapacity/PolynomialCp.cpp`, called via `Component::Hliq_pure(T, Tref)`.

The vapour enthalpy of a pure component sits a latent heat above the liquid:

$$\boxed{h_i^{\mathrm{vap}}(T) \;=\; h_i^{\mathrm{liq}}(T) \;+\; \Delta h_i^{\mathrm{vap}}(T).} \tag{30}$$

The two terms are computed in `Component::Hvap_pure` / `Component::Hliq_pure` / `Component::Hvap_latent`; the next subsection is about how the latent heat at an arbitrary $T$ is obtained from the single tabulated value $\Delta h_{T_b}^{\mathrm{vap}}$.

> *Intuition.* The choice $h_i^{\mathrm{liq}}(T_{\mathrm{ref}}) = 0$ is a reference zero, like sea level in altitude. For a unit with *no reaction* only differences enter the energy balance ($h_{\mathrm{out}} - h_{\mathrm{in}}$, $Q = \dot{n}\,\Delta h_{\mathrm{stage}}$), and the per-component zero cancels — a flash, a column, a heater, an evaporator all balance correctly on this datum. If you compare these $h$ values against a table that picked $h_i^{\mathrm{ig}}(298) = 0$ for an *ideal gas* reference, they differ by exactly $\Delta h_i^{\mathrm{vap}}(T_{\mathrm{ref}})$.
> **But beware: this datum does NOT survive a chemical reaction.** With a separate zero *for each component*, the difference $h_B(T) - h_A(T)$ for a reaction $A \to B$ at fixed $T$ is essentially zero (both are zeroed at $T_{\mathrm{ref}}$ plus a near-equal sensible term) — the heat of reaction has vanished. An energy balance written across a reactor on this datum is simply wrong. The next subsection fixes this.

## 6.3 The reference state, and why a reaction needs the elements

The sensible+latent datum of the previous subsection sets $h_i^{\mathrm{liq}}(T_{\mathrm{ref}}) = 0$ *per component* — every species gets its own private zero. That is harmless as long as the set of species is conserved (no reaction): in $h_{\mathrm{out}} - h_{\mathrm{in}}$ each species' zero appears once on each side and cancels.

A reaction breaks that bookkeeping. Consider the isomerization neo$C_5 \to n$-$C_5$ (used in the `process05_isomerization_recycle` tutorial). On the per-component datum, $h_{n\mathrm{C_5}}(T) - h_{\mathrm{neoC_5}}(T) \approx 0$ at equal $T$, so an energy balance over the reactor reports $\Delta H \approx 0$. The *true* heat of reaction, however, is the difference of formation enthalpies,

$$\Delta h_{\mathrm{rxn}}(T_{\mathrm{ref}}) \;=\; \sum_i \nu_i\, \Delta h_{f,i}^{\circ} \;=\; \Delta h_{f,n\mathrm{C_5}}^{\circ} - \Delta h_{f,\mathrm{neoC_5}}^{\circ} \;=\; -146{,}760 - (-168{,}050) \;=\; +21{,}290\ \mathrm{J/mol}, \tag{31}$$

where the index $i$ runs over the *species* taking part in the reaction and $\nu_i$ is the **stoichiometric coefficient** of species $i$ — *negative for reactants, positive for products.* For neo$C_5 \to n$-$C_5$ that is $\nu_{\mathrm{neoC_5}} = -1$ and $\nu_{n\mathrm{C_5}} = +1$, so the sum is $(+1)\,\Delta h_{f,n\mathrm{C_5}}^{\circ} + (-1)\,\Delta h_{f,\mathrm{neoC_5}}^{\circ}$ — "products minus reactants", each weighted by how many moles the balanced equation consumes or produces. (The same $i$ indexes a component everywhere in this guide: $x_i$ its mole fraction, $h_i$ its molar enthalpy, $\Delta h_{f,i}^{\circ}$ its formation enthalpy.) The result is +21,290 J/mol, i.e. *endothermic* — the isothermal reactor must be *heated*. The per-component datum simply cannot see this number.

The cure is to measure every substance's enthalpy from the same floor: the **chemical elements it is made of**, each in its naturally stable form at 298.15 K, 1 bar.

**What "the elements datum" actually means.** By universal convention, each chemical element is assigned enthalpy *zero* in its standard state — its stable form at 298.15 K, 1 bar:

$$h^{\circ}(\mathrm{C,\ graphite}) = 0, \quad h^{\circ}(\mathrm{H_2, gas}) = 0, \quad h^{\circ}(\mathrm{O_2, gas}) = 0, \quad h^{\circ}(\mathrm{N_2, gas}) = 0, \ \ldots$$

The **standard enthalpy of formation** $\Delta h_{f,i}^{\circ}$ of a compound is then the heat of *building one mole of it from those elements.* For methane,

$$\mathrm{C(graphite)} + 2\,\mathrm{H_2(gas)} \;\longrightarrow\; \mathrm{CH_4(gas)}, \qquad \Delta h_{f,\mathrm{CH_4}}^{\circ} = -74{,}520\ \mathrm{J/mol}.$$

Since the elements on the left are worth zero, the compound's *absolute* enthalpy on this datum simply *is* its formation enthalpy: $h_{\mathrm{CH_4}}^{\circ}(298) = \Delta h_{f,\mathrm{CH_4}}^{\circ} = -74{,}520$ J/mol. Negative means methane sits *below* its elements — energy was released when it formed.

> *Intuition.* Think of altitude. Measuring each city's height from its own local sea level (the per-component datum) is useless for comparing a Portuguese city with a Spanish one. Pick *one* global sea level and all height differences become meaningful. The chemical elements are that global sea level for enthalpy: because *every* compound is measured from the same atomic floor, the atoms cancel in "products minus reactants" (mass is conserved in a reaction), and what is left over is exactly the heat of reaction. That is why this is the only datum that survives a reactor.

**Worked example 6.1.** Burning methane — the elements never appear in the sum, yet they make it workEveryone knows methane burns and releases a lot of heat. Let us *compute* how much, using only formation enthalpies, and watch the elements datum do its job. The balanced combustion is

$$\mathrm{CH_4} + 2\,\mathrm{O_2} \;\longrightarrow\; \mathrm{CO_2} + 2\,\mathrm{H_2O(g)}.$$

Standard formation enthalpies (kJ/mol), straight from the data files:

| species | $\Delta h_f^\circ$ [kJ/mol] | note |
|---|---|---|
| $CH_4$ | $-74.5$ | a compound |
| $O_2$ | $0$ | **an element — zero by definition** |
| $CO_2$ | $-393.5$ | a compound |
| $H_2O$(g) | $-241.8$ | a compound |

Apply "products minus reactants", each weighted by its stoichiometric $\nu$:

$$\Delta h_{\text{rxn}} = \big[\,1\cdot(-393.5) + 2\cdot(-241.8)\,\big] \;-\; \big[\,1\cdot(-74.5) + 2\cdot 0\,\big]$$
$$= (-877.1) - (-74.5) \;=\; \boxed{-802.6\ \text{kJ/mol}}.$$

Negative and large — strongly *exothermic*, exactly as the blue flame on a stove tells you. Two things to notice. First, $O_2$ contributed nothing (it is an element, its enthalpy is the zero of the scale) — yet had we measured each compound from its *own* private zero, the three formation numbers would have been on three incompatible scales and this subtraction would have been meaningless. The common elemental floor is what lets us add them. Second, no separate "heat of reaction" had to be supplied to the simulator: it is simply $h_{\text{out}} - h_{\text{in}}$ once every stream carries its $\Delta h_f^\circ$.

This is the datum behind $\Delta h_{f,i}^\circ$ and the one `Component::h_pure_ig` already uses (introduced for the Gibbs reactor, Chapter 14). An ideal gas at temperature $T$ is its formation enthalpy plus the sensible heat from 298.15 K up to $T$:

$$h_i^{\text{ig}}(T) \;=\; \underbrace{\Delta h_{f,i}^\circ}_{\text{build it from elements}} \;+\; \underbrace{\int_{298.15}^{T} c_{p,i}^{\text{ig}}(T')\,\mathrm{d}T'}_{\text{then heat it to } T}. \tag{32}$$

On this datum the liquid sits a latent heat *below* the ideal gas, so the elements-referenced liquid enthalpy is

$$\boxed{\,h_i^{\text{liq}}(T) \;=\; h_i^{\text{ig}}(T) \;-\; \Delta h_i^{\text{vap}}(T),\,} \tag{33}$$

and now *both* phases share the same zero. Because each species carries its $\Delta h_{f,i}^\circ$, the sum $\sum_i \nu_i h_i$ over a reaction returns the heat of reaction automatically — no separate $\Delta h_{\text{rxn}}$ term is needed, it falls out of $h_{\text{out}} - h_{\text{in}}$. This is the datum every process simulator uses internally, for exactly this reason.

> *Intuition.* [The standard enthalpy-reference convention] The standard convention defines the enthalpy reference state as *the elements at 298.15 K*, with the ideal-gas enthalpy pinned to the standard heat of formation, $h_i^{\text{ig}}(T_{\text{ref}}) = \Delta h_{f,i}^\circ$ — the internal calculation path is elements → ideal-gas compounds at 298 K → ideal gas at $T$ → real fluid at $T, P$, the same ladder as Figure 7. It is why a stream of pure ethanol at 298 K reports a *negative* enthalpy: that number is its (negative) heat of formation, not a bug. Choupo follows this convention, with the reference *pressure* pinned at 1 bar (the post-1982 NIST/JANAF convention against which the `S_ig` method was validated). The gap is *zero* for enthalpy (ideal-gas $h$ does not depend on pressure) and a flat $R\ln(1.01325) \approx 0.11$ J/(mol · K) for entropy — negligible, and never accumulating.

> *Intuition.* Two datums, one rule. *Sensible+latent* (per-component zero) is fine for any non-reacting unit and is what the flash / column / evaporator unit ops use internally for their local duties. The *elements* datum (formation-enthalpy zero) is the only one that survives a reactor, so it is what the `energyBalance` report uses when every component carries a `standardThermochemistry` block; the report falls back to the sensible datum (and flags the row) only when some species lacks $\Delta h_f^\circ$. The two agree exactly on any unit where nothing reacts — which is why `flash01` reports the same 267 kW duty on either datum, while `process05`'s reactor only shows its +4.5 kW of reaction heat on the elements datum.

The whole picture fits on one ladder. Figure 7 stacks the states of a pure substance on a vertical enthalpy axis: the two candidate zeros, the two latent jumps ($\Delta h^{\text{fus}}$, $\Delta h^{\text{vap}}$), and the sensible ramps ($\int c_p\,dT$) that connect them. Read it once and the reference-state question stops being mysterious: an enthalpy value is just a *height* on this ladder, and a "datum" is just the choice of where to nail $h = 0$.

## 6.4 Which rung does the data file pin? The `referenceState` keyword

The ladder of Figure 7 shows that the formation enthalpy $\Delta h_{f,i}^\circ$ is a *distance*: from the elements baseline (the green dashed line) up to one of the rungs of the species $i$. But to which rung exactly? Different tables answer this question differently:

- NIST and JANAF tabulate $\Delta h_{f,i}^\circ$ in the **ideal-gas state at** 298.15 **K,** 1 bar for compounds that can be brought to the gas phase — water, ethanol, methane, and the bulk of the volatiles and radicals shipped with Choupo. This is the standard convention used internally for every species.

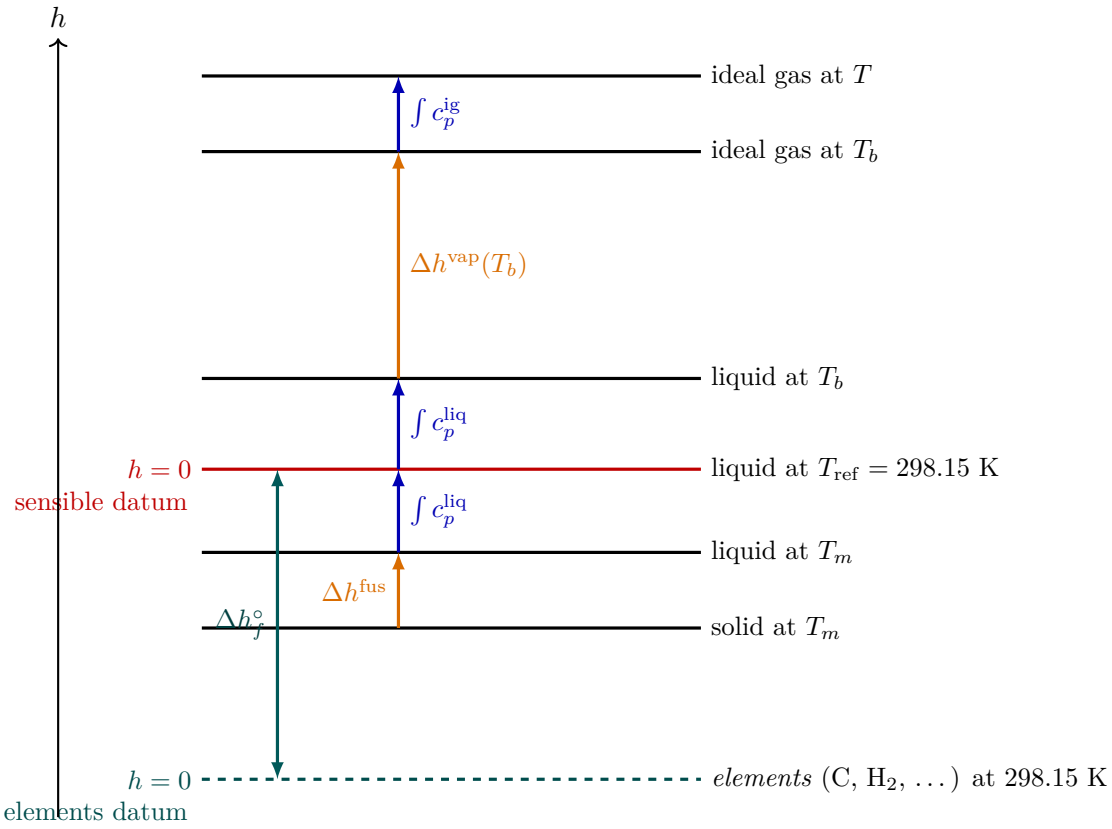



**Figure 7:** The enthalpy ladder of a pure substance (heights qualitative, not to scale). Each state is a rung; latent transitions ($\Delta h^{\text{fus}}$, $\Delta h^{\text{vap}}$) and sensible ramps ($\int c_p\,dT$) connect them. The **sensible datum** nails $h = 0$ at the liquid at $T_{\text{ref}}$, *per component* — convenient, but every species has its own zero, so it loses the heat of reaction. The **elements datum** nails $h = 0$ at the chemical elements; the rung-to-rung distance up to the substance is its formation enthalpy $\Delta h_f^\circ$, and because all species are measured from the same elemental floor, $\sum_i \nu_i h_i$ over a reaction returns the reaction enthalpy automatically. Choupo's vapour enthalpy is $h^{\text{vap}} = h^{\text{liq}} + \Delta h^{\text{vap}}$ (one rung up); on the elements datum the liquid is $h^{\text{ig}} - \Delta h^{\text{vap}}$ (one rung down from the gas).

- For **crystalline non-volatiles** that decompose before they vaporise (sucrose, glucose, the bulk of solid solutes), NIST tabulates $\Delta h_f^\circ$ in the **solid** state — there is no honest ideal-gas value to give.
- For a small group of compounds (mostly experimentally-measured condensed-phase enthalpies of formation), the tabulated number is the **liquid** datum.

Choupo records the chosen rung as a one-word `referenceState` field inside the component's `standardThermochemistry` block — the thermodynamic *standard state* the values are on, not merely a physical phase:

```
standardThermochemistry
{
    dHf_298    -241826.0;        // J/mol  (ideal-gas reference)
    s_298       188.834;         // J/(mol*K)  (third-law absolute)
    referenceState  idealGas;    // <-- tells the solver which rung
}                                //     (the default when absent)
```

```
standardThermochemistry                  // sucrose, from NIST WebBook
{
    dHf_298   -2226100.0;        // J/mol  (crystalline)
    s_298         392.4;         // J/(mol*K)
    referenceState  pureSolid;   // <-- never goes through a gas leg
}
```

The reason the field matters is operational. When the energy balance asks for $h_i^{\mathrm{liq}}(T)$ on the elements datum, the solver builds a path through the ladder:

$$\texttt{referenceState = idealGas}: \quad h_i^{\mathrm{liq}}(T) = \Delta h_{f,i}^{\circ} + \int_{298.15}^{T} c_p^{\mathrm{ig}}\, dT - \Delta h^{\mathrm{vap}}(T)$$

$$\texttt{referenceState = pureLiquid}: \quad h_i^{\mathrm{liq}}(T) = \Delta h_{f,i}^{\circ} + \int_{298.15}^{T} c_p^{\mathrm{liq}}\, dT$$

$$\texttt{referenceState = pureSolid}: \quad h_i^{\mathrm{liq}}(T) \approx \Delta h_{f,i}^{\circ} + \int_{298.15}^{T} c_p^{\mathrm{liq}}\, dT \qquad (c_p^{\mathrm{solid}} \approx c_p^{\mathrm{liq}},\ \Delta h^{\mathrm{fus}} \approx 0)$$

That is why a sucrose-bearing fermentation stream **does not need an ideal-gas $c_p$ for sucrose**: the solver never integrates through a sucrose vapour leg, because sucrose's natural rung is on the solid line of the ladder and `referenceState pureSolid;` tells it so. Forcing every species through a strict-IG path — as a naive strict-IG implementation would — would require the case author to invent an artificial $c_p^{\mathrm{ig}}$ for crystalline sucrose just so the equations work. Choupo refuses that compromise: the natural-phase convention keeps the per-species data *honest*, and the energy-balance report still closes through the reactor because every species carries its own $\Delta h_f^{\circ}$.

> *Intuition.* Think of `referenceState` as the answer to "starting from the elements, where on the ladder does the data sheet stop?" For gases the table stops at the top rung (ideal gas at 298 K). For sucrose it stops at the bottom rung (the crystal). When the energy balance later asks the liquid enthalpy at some other temperature, the solver knows how to walk from the recorded rung to the one the balance needs — and how to refuse the walk when the latent jump it would need does not exist (sucrose → vapour throws a clear error rather than fabricate numbers).

**Default for backward compatibility.** If `referenceState` is absent the solver assumes `idealGas` (the NIST / JANAF convention, correct for the overwhelming majority of volatile species); the standard catalogue's condensed-phase records (sucrose, glucose, and the other non-volatiles) declare `pureSolid` / `pureLiquid` explicitly. Per-case overlays under `<case>/constant/components/<name>.dat` follow the same rule: a non-gas datum must declare its `referenceState` so the datum is unambiguous at the source of truth.

## 6.5 Where every number comes from — the three questions a black-box tool won't answer

The `referenceState` keyword of the previous subsection is one instance of a larger discipline. The elements datum (§6.3) tells you *which floor* a formation enthalpy is measured from; it does not tell you *where the number on the data sheet came from*, *which model consumed it*, or *at which level of the case it was overridden*. A black-box simulator answers none of these: you pick "Property Method: NRTL-RK" from a dropdown and a thousand numbers are sourced, substituted, and defaulted silently behind it. Choupo's governing rule is the opposite — a student points at any number on the screen and asks *"where did THIS come from?"*, and the run has already answered it, at the point of use. Three questions, three rules.

**(1) An "in-water" property is pair data wearing a convenience hat.** The formation enthalpy of crystalline sucrose, $\Delta h_f^{\circ} = -2226.1$ kJ/mol, is *intrinsic*: its definition names sucrose and its elements, nothing else. Its enthalpy of *solution*, $\Delta h_{\mathrm{soln}} = +5.40$ kJ/mol (crystal → aqueous), is a different animal: its definition names *two* species, sucrose *and water*. That is *arity-2* data, and it does not live in `sucrose.dat` — it lives in a by-name catalogue tier, `data/standards/parameters/solution/sucrose-water.dat`, exactly as the infinite-dilution ion enthalpies live in `data/standards/species/<ion>.dat` on the $h_f^{\circ}(\mathrm{H}^+_{aq}) = 0$ convention. Water is ubiquitous enough to earn *one canonical, named aqueous reference tier* — never an *implied component slot*. When a solution property is needed and no solvent is named, the resolver applies the package default and *announces it*:

```
[thermo] dHsoln(sucrose): solvent not named -> DEFAULT solvent = water
         source: solution/sucrose-water.dat  [Putnam & Kilday 1986]
```

Off-default, it refuses with a remedy rather than silently substitute the water value. The student therefore always sees that a second species was assumed, which one, from which file, and from whose paper — the four things a dropdown hides.

**(2) A model's parameters live in the model's own block.** A cubic equation of state (SRK, PR) consumes only the corresponding-states triad $\{T_c, P_c, \omega\}$ and derives its own $a_c, b, \kappa$ *in source* — adding PR after SRK touched no data file. A model that needs parameters it *cannot* derive — PC-SAFT's segment number

$m$, segment diameter $\sigma$, well depth $\varepsilon/k$ — carries them in a *model-keyed* block on the component, `pcsaft{ m; sigma; epsilonK; }`, with its own provenance. One molecule then carries SRK, PR, PC-SAFT and the next EOS simultaneously, never colliding, and the run announces *per component* whether its numbers were derived from corresponding states or read from a fitted block, and from whose paper:

```
[thermo] EOS = PR     -- benzene: a_c,b,kappa derived from Tc,Pc,omega (intrinsic)
[thermo] EOS = PCSAFT -- benzene: m,sigma,epsilon/k from the pcsaft{} block
                         [Gross & Sadowski 2001]  origin: regressed
```

A model missing a required block fails with a remedy — never a silent corresponding-states fallback.

**(3) A molecule is whole at every level.** The case tree is a plant and its sectors (case ⊃ sector ⊃ unit), and a lower node may *overlay* a refined number when its scope makes it more true — a particular powder's measured sorption isotherm, say. But the overlay is a *patch, never a fork*: the molecule's identity $(\mathrm{MW}, T_c, \omega)$ is born once, at the standard tier, and is the same molecule at every level. The overlay merges *block-by-block* — override the whole `solid{}` reference-state block, never a lone scalar inside it (a lone-scalar overlay would silently splice a sample's $\rho_p$ onto the catalogue's $k_v$, a hidden hybrid). And a number that is really a *rate* (crystallisation kinetics) or a *geometry* (a particle-size distribution) is not component data at all — it belongs to the equipment, in the operation's `constant/`, and the PSD travels as a *stream* attribute. The cascade line tells the student which level won.

> *Intuition.* *All models are wrong, some are useful* — and Choupo tells you exactly *which* one you used and *where its numbers came from*, so you judge the usefulness yourself. A datum's home is decided by **arity** (does its definition name a second species?), then by **scope** (at what level is it true?), then by **kind** (is it the molecule, or the molecule-in-a-machine?). Nothing — no solvent, no model, no level — is ever implicit on disk. The curator's full decision procedure is `docs/ai/data-doctrine.md`.

## 6.6 The Gibbs side: $s^\circ_{298}$, third-law entropy, and the absolute ideal-gas free energy

The previous subsections settled the *enthalpy* half of the formation datum: $\Delta h^\circ_{f,i}$ pins the substance's height above the elements floor. But the Gibbs reactor of Chapter 14 minimises the *Gibbs* energy, and Eq. (326) needs the absolute standard-state free energy $g^\circ_i(T)$, which carries an entropy term. This subsection derives the second half of the `standardThermochemistry` block — the field $s^\circ_{298}$ — and the chain `Component::g_pure_ig` walks to assemble $g^\circ_i(T)$.

**Why entropy needs a *different* reference from enthalpy.** Enthalpy has no natural zero, so we *chose* one (the elements). Entropy does have a natural zero: the third law of thermodynamics fixes $s = 0$ for a perfect crystal at $T = 0$ K. The number tabulated in $s^\circ_{298}$ is therefore the **absolute (third-law) entropy** of the substance at 298.15 K, 1 bar — the integral of $c_p/T$ all the way up from 0 K, plus the entropy of every phase change crossed on the way:

$$s^\circ_i(298) \;=\; \int_0^{298.15} \frac{c_{p,i}(T')}{T'}\,\mathrm{d}T' \;+\; \sum_{\text{transitions}} \frac{\Delta h^{\text{trs}}}{T^{\text{trs}}}. \tag{34}$$

This is *not* an "entropy of formation" — it is an absolute value, and the elements carry a non-zero $s^\circ_{298}$ of their own. That distinction is why the data file stores $\Delta h^\circ_f$ (a *difference* from the elements) but a plain absolute $s^\circ_{298}$ (a *third-law* value). The reaction's entropy change reconstructs automatically as $\sum_i \nu_i\, s^\circ_i$, because the atoms cancel exactly as they did for enthalpy.

**From the two stored numbers to $g^\circ_i(T)$.** Both stored values are pinned at 298.15 K; the simulator transports them to the reactor temperature $T$ by integrating the ideal-gas heat capacity. Enthalpy uses $\int c_p\,dT$, entropy uses $\int c_p/T\,dT$:

$$h^\circ_i(T) = \Delta h^\circ_{f,i} \;+\; \int_{298.15}^{T} c^{\text{ig}}_{p,i}(T')\,\mathrm{d}T', \tag{35}$$

$$s^\circ_i(T) = s^\circ_i(298) \;+\; \int_{298.15}^{T} \frac{c^{\text{ig}}_{p,i}(T')}{T'}\,\mathrm{d}T', \tag{36}$$

$$\boxed{\,g^\circ_i(T) \;=\; h^\circ_i(T) \;-\; T\,s^\circ_i(T).\,} \tag{37}$$

These are exactly `Component::h_pure_ig(T)`, `Component::s_pure_ig(T)` and `Component::g_pure_ig(T)` in `src/thermo/Component.`. The two integrals are the $H$ and $S$ methods of the heat-capacity model (`src/thermo/heatCapacity/`); for the

polynomial form $c_p^{\mathrm{ig}} = a_0 + a_1T + a_2T^2 + a_3T^3$ both are elementary:

$$\int_{298.15}^{T} c_p\,dT' = \sum_k \frac{a_k}{k+1}\big(T^{k+1} - 298.15^{k+1}\big), \quad \int_{298.15}^{T} \frac{c_p}{T'}\,dT' = a_0 \ln\frac{T}{298.15} + \sum_{k\geq 1}\frac{a_k}{k}\big(T^k - 298.15^k\big).$$

The *element-potential* solver of the Gibbs reactor reads only the combination $g_i^\circ(T)/RT$ (it appears as `g_pure_ig(T)/RT` in `gibbsMethod/ElementPotential.cpp`), so the 1 bar standard reference cancels into the $\ln(y_iP)$ term of Eq. (326) and never appears as a separate offset.

**Worked example 6.2.** Reforming: the entropy term decides the equilibriumThe water-gas-shift reaction $\mathrm{CO} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO_2} + \mathrm{H_2}$ has almost no entropy change (two moles to two moles), so it is dominated by $\Delta h$; but the steam-reforming reaction $\mathrm{CH_4} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO} + 3\,\mathrm{H_2}$ goes from two moles of gas to four, and $\Delta s^\circ$ is large and positive. Below $\sim 900$ K the $-T\Delta s$ term in $\Delta g^\circ = \Delta h^\circ - T\Delta s^\circ$ is too small to overcome the endothermic $\Delta h^\circ \approx +206$ kJ/mol, and equilibrium sits on the methane side; raise $T$ and the entropy term flips the sign of $\Delta g^\circ$, driving conversion to syngas. A Gibbs reactor *cannot* reproduce this temperature swing from $\Delta h_f^\circ$ alone — it needs each species' $s_{298}^\circ$. That is why the `standardThermochemistry` block carries *both* numbers, and why a component lacking $s_{298}^\circ$ cannot be placed in a Gibbs reactor.

## 6.7 Where the two numbers come from: the NASA-CEA import

A component's $\Delta h_f^\circ$ and $s_{298}^\circ$ are *curated*, not computed at run time — they are measured thermochemistry the simulator only consults. Choupo's gas-phase formation data are imported from the **NASA Glenn** thermodynamic database (NASA-7 polynomials), via the curation tool `bin/curate/import_gibbs_nasa.py`. The primary source is McBride, Gordon & Reno, NASA TM-4513 (1993) [1], a public-domain US-government work.

The connection between a NASA-7 polynomial and the two fields Choupo needs is direct and exact. The NASA polynomials are themselves referenced to the elements, so the standard-state enthalpy they return at 298.15 K *is* the enthalpy of formation, and the entropy they return is the third-law absolute entropy:

$$\Delta h_{f,i}^\circ \;=\; H_i^0(298.15), \qquad s_i^\circ(298) \;=\; S_i^0(298.15), \tag{38}$$

which is precisely what the import reads off the polynomial (`standard_enthalpies_RT` and `standard_entropies_R` evaluated at 298.15 K, 1 atm).

The combustion radicals of the catalogue (CH, HCO, CN, NCO, ...) take the same two numbers, and their heat capacity, from the report's successor, McBride, Zehe & Gordon, NASA/TP-2002-211556 (2002) [2]: $\Delta h_f^\circ$ from the record line of its appendix D, $s_{298}^\circ$ from its Table B1, and the nine-coefficient polynomial $C_p/R = a_1T^{-2} + a_2T^{-1} + a_3 + a_4T + a_5T^2 + a_6T^3 + a_7T^4$ read as printed (`model NASA9`) — no refit, so a reader can hold each record to its page (`bin/curate/nasa_tp2002_to_choupo.py` prints the blocks and checks $H$ and $S$ at 298.15 K against the report before it does).

**Two glass-box guards (why the import is trustworthy).** The tool refuses to write a number it cannot defend:

- **Anchor verify.** Before any file is touched, the tool re-derives $\Delta h_f^\circ$ for five well-known species ($\mathrm{CO_2}, \mathrm{H_2O}, \mathrm{NH_3}$, benzene, CO) and aborts if any deviates from the textbook value by more than 0.5 kJ/mol — a unit-conversion or transcription slip is caught at the door.
- **Isomer guard.** NASA names many species by formula only, and the trap is real: "C3H6O" in TM-4513 is *propylene oxide* ($\Delta h_f^\circ \approx -93.7$ kJ/mol), not acetone ($-217.1$). Every name-to-NASA mapping therefore carries an *expected* $\Delta h_f^\circ$, and the import **refuses** any species whose NASA value deviates by more than 5 kJ/mol. A wrong-isomer match is a transcription error to catch, never a number to trust.

A real, curated measured value already in `data/standards/components/` is never overwritten; the import only *fills* an absent block or *refits* one tagged as an estimate (Joback, `origin=estimated`). Species with no clean, isomer-unambiguous NASA match (acetone, the longer $n$-alkanes, the xylenes, diethyl ether) are deliberately left carrying their honestly-tagged Joback estimate rather than a guessed value — the gap is kept visible, not papered over.

[1] B. J. McBride, S. Gordon, M. A. Reno, *Coefficients for Calculating Thermodynamic and Transport Properties of Individual Species*, NASA Technical Memorandum 4513 (1993). A US-government work, in the public domain; also the data behind the NASA CEA program (Gordon & McBride, NASA RP-1311, 1994).

[2] B. J. McBride, M. J. Zehe, S. Gordon, *NASA Glenn Coefficients for Calculating Thermodynamic Properties of Individual Species*, NASA/TP-2002-211556 (2002). A US-government work, in the public domain.

*Intuition.* [Power and limit] **Power:** two small numbers per species — $\Delta h_f^\circ$ and $s_{298}^\circ$ — plus a $c_p^{\mathrm{ig}}(T)$ polynomial are enough to give the absolute Gibbs energy of *any* ideal-gas mixture at *any* temperature, which is all the element-potential equilibrium solver needs. The data are public-domain, element-referenced, and audited. **Limit:** the chain is strictly *ideal-gas*. $g_i^\circ(T)$ carries no pressure correction beyond the ideal $RT\ln(y_iP)$, no real-gas fugacity, and no liquid- or solid-phase free energy — so a Choupo Gibbs reactor is a gas-phase equilibrium tool. And the values are only as good as the NASA fit: TM-4513 $c_p$ polynomials are reliable to $\sim$ 1000–6000 K depending on species, and an estimated (Joback) entry can be off by several kJ/mol, which the per-value provenance tag makes the student check before trusting an equilibrium prediction.

## 6.8 Latent heat at any $T$: Watson

Each data file lists $\Delta h_i^{\mathrm{vap}}(T_b)$ at the normal boiling point only. The simulator transports it to any other temperature with Watson's correlation [5],

$$\boxed{\Delta h_i^{\mathrm{vap}}(T) \;=\; \Delta h_i^{\mathrm{vap}}(T_b)\left(\frac{T_c - T}{T_c - T_b}\right)^{0.38}.} \tag{39}$$

This is not a fit; it is a dimensionally motivated power law that collapses to $\Delta h^{\mathrm{vap}} \to 0$ as $T \to T_c$ (vapour and liquid become indistinguishable) and reproduces the tabulated $\Delta h^{\mathrm{vap}}(T_b)$ at $T = T_b$ by construction. The exponent 0.38 is the empirical Watson value; Pitzer's later acentric-factor formulation refines it but the simple form is good to a few percent across the practically interesting range $T_b < T < 0.95\,T_c$.

Implementation: `Component::Hvap_latent(T)` in `src/thermo/Component.cpp`, 6 lines.

**Worked example 6.3.** Acetic acid latent heat at $T = 350$ KWith $T_c = 591.95$ K, $T_b = 391.05$ K, $\Delta h_{T_b}^{\mathrm{vap}} = 24{,}390$ J/mol from `data/standards/components/aceticAcid.dat`,

$$\Delta h^{\mathrm{vap}}(350) \;=\; 24390\cdot\left(\frac{591.95-350}{591.95-391.05}\right)^{0.38} \;\approx\; 24390\cdot(1.2042)^{0.38} \;\approx\; 26{,}180\ \mathrm{J/mol}.$$

Acetic acid is harder to evaporate at 350 K than at its boiling point (by $\sim 7\%$). The trend is universal: $\Delta h^{\mathrm{vap}}$ *decreases* as $T$ rises toward $T_c$.

## 6.9 Pure-component entropy and Gibbs energy

Entropy is the analogous integral against $c_p/T$ — but with one structural difference from everything enthalpy did in this chapter. Enthalpy ran on a *chosen* reference (the liquid-referenced convention above, or the elements datum of §6.6); entropy has no second convention anywhere in the engine. There is *one* chain, and it is absolute: the measured third-law anchor $s_{298}^\circ$ from the data file, plus $\int c_p/T$,

$$s_i^\circ(T) \;=\; s_{i,298}^\circ \;+\; \int_{298.15}^{T} \frac{c_{p,i}(T')}{T'}\,\mathrm{d}T'. \tag{40}$$

Each $c_p$ correlation supplies this integral in closed form (for the polynomial, $a_0\ln(T/298.15) + \sum_{k\geq 1} a_k(T^k - 298.15^k)/k$), so no quadrature is ever run.

**Crossing a phase is an explicit leg — and the obvious latent step is wrong at the datum.** *Along the saturation curve* the two phases are in equilibrium, $\Delta g = 0$, so the latent entropy step is exactly the familiar

$$\Delta s_i^{\mathrm{vap}}\big(T^{\mathrm{sat}}\big) \;=\; \frac{\Delta h_i^{\mathrm{vap}}\big(T^{\mathrm{sat}}\big)}{T^{\mathrm{sat}}}. \tag{41}$$

But the data file's anchors live at the *standard states* — liquid at 298.15 K and 1 bar, ideal gas at 298.15 K and 1 bar — and those two states are *not* in equilibrium with each other: water's $P^{\mathrm{sat}}$ at 298 K is about 3.17 kPa, not 1 bar. The transfer between them carries a non-zero Gibbs change, and dividing $\Delta h^{\mathrm{vap}}$ by $T$ alone would silently drop it. What the engine implements is the general identity $\Delta s = (\Delta h - \Delta g)/T$:

$$\Delta s_i^{\mathrm{vap}}(298.15) \;=\; \frac{\Delta h_i^{\mathrm{vap}}(298.15) - \Delta g}{298.15}, \qquad \Delta g \;=\; -RT\ln\frac{P_i^{\mathrm{sat}}(298.15)}{P^\circ} \tag{42}$$

(`Component::s_formation` in `src/thermo/Component.cpp`, with this reason stated in its comment). The correction is not small: for water, $R\ln(P^{\mathrm{sat}}/P^\circ) \approx -28.7$ J/(mol · K) — about a fifth of the whole step. The published tables confirm the identity among themselves: the third-law difference $s_{298}^\circ(\mathrm{H_2O,g}) - s_{298}^\circ(\mathrm{H_2O,l}) = 188.8 - 70.0 = 118.9$ J/(mol · K), while the calorimetric $\Delta h^{\mathrm{vap}}(298)/298.15 \approx 44.0\times10^3/298.15 \approx 147.5$ J/(mol · K) — and the gap between those two published numbers *is* the Gibbs term, not an error in either. The same construction handles a solid datum's melting leg (opt-in: it needs a solid $c_p$ *and* $\Delta h^{\mathrm{fus}}$); a crossing the data cannot price refuses by name rather than guessing.

With $h$ and $s$ in hand, the pure-component Gibbs energy is the Legendre transform that defines it,

$$g_i^\circ(T) \;=\; h_i^\circ(T) - T\, s_i^\circ(T) \tag{43}$$

(`Component::g_pure_ig`) — the enthalpy on the elements datum, the entropy absolute, and the pairing correct *only* across a balanced stoichiometry, as §6.6 explains.

> *Intuition.* [Where an entropy value actually decides something] Two places, and they use it very differently. The **Gibbs machinery** consumes $g_i^\circ(T)$: the equilibrium reactors evaluate $K_p = \exp(-\sum_i \nu_i\, g_i^\circ/RT)$, and the Gibbs reactor minimises total $G$ directly — there the entropy of mixing never appears by name, because it is implicit in every $\ln y_i$ term of the chemical potentials. The **isentropic machines** (compressor, turbine) hold $s$ itself constant: an inner Newton finds the temperature at which $s(T, P_{\text{out}}, y) = s_{\text{in}}$, and each machine publishes the entropy *generated* by its declared irreversibility, $\Delta S_{\text{gen}} = s_{\text{out}} - s_{\text{in}} \geq 0$, as a KPI. Note what cancels: with the composition held fixed across the machine, the $s_{298}^\circ$ anchors, the mixing term and the reference pressure all drop out of the *difference* — the isentropic answer hangs only on $\int c_p/T$, $-R\ln(P_2/P_1)$ and the model's departure. That cancellation is why the ledger can be audited one line at a time: the "What is entropy?" EduTool page re-adds the lines from a live run of `tutorials/props/molecular/entropy01_air_ledger`, and `docs/design/entropy-glass-box-trace.md` carries the file-by-file trace of the whole chain.

## 6.10 Mixing: ideal versus real

The simulator's mixture enthalpy is the ideal-mixing combination of the pure values, weighted by phase fractions:

$$h_{\text{mix}}(T) \;=\; (1-V)\sum_i x_i\, h_i^{\text{liq}}(T) \;+\; V\sum_i y_i\, h_i^{\text{vap}}(T), \tag{44}$$

exactly as implemented in `ThermoPackage::Hmixture`. This assumes the *excess enthalpy* $h^E$ of mixing is negligible. For Raoult systems (benzene/toluene, $n$-hexane/$n$-heptane) and moderately non-ideal systems with weak hydrogen bonding, $h^E$ is indeed tens of J/mol against pure $h$'s of tens of kJ/mol — a sub-percent correction. For strongly hydrogen-bonded mixtures (ethanol/water, acetic acid/water) $h^E$ can reach $\sim 500$ J/mol and the energy balance will under-predict, e.g., the duty of a condenser by a fraction of a percent up to a few percent.

The Gibbs energy of mixing is where activity coefficients enter. For a single-phase mixture

$$\boxed{\Delta G_{\text{mix}} \;=\; RT\sum_i x_i \ln\!\big(x_i\gamma_i\big) \;=\; \underbrace{RT\sum_i x_i \ln x_i}_{\text{ideal mixing, always} \leq 0} \;+\; \underbrace{RT\sum_i x_i \ln\gamma_i}_{\equiv\, G^E,\ \text{excess Gibbs}}\ .} \tag{45}$$

The ideal part is the universal $-T\,\Delta s_{\text{mix}}^{\text{ideal}} = RT\sum x_i \ln x_i$ that always drives mixing forward at finite $T$ for components that mix at all. The excess part $G^E$ is what NRTL, Wilson and friends actually parameterise, via $\ln\gamma_i = \partial(n\,G^E/RT)/\partial n_i$ at constant $T, P, n_{j\neq i}$. A consistency check is built in: an activity model has to satisfy the integrability constraint that follows next.

## 6.11 Gibbs-Duhem: the consistency check

At constant $T$ and $P$, partial molar quantities of a mixture are not independent. The Gibbs-Duhem identity for the activity coefficients reads

$$\boxed{\sum_i x_i\, \mathrm{d}(\ln\gamma_i) \;=\; 0 \quad (\text{constant } T, P).} \tag{46}$$

It says: you cannot fit $\ln\gamma_1$ and $\ln\gamma_2$ to data independently and freely — once $\ln\gamma_1(x_1)$ is chosen, $\ln\gamma_2(x_1)$ is determined up to a constant of integration fixed by the pure-limit $\gamma_i(x_i \to 1) = 1$.

For a binary system the constraint linearises to a useful test on experimental data: at a series of $(x_1, \gamma_1, \gamma_2)$ measurements, the area under $\ln(\gamma_1/\gamma_2)$ over the full composition range must integrate to zero:

$$\int_0^1 \ln\frac{\gamma_1(x_1)}{\gamma_2(x_1)}\,\mathrm{d}x_1 \;=\; 0. \tag{47}$$

This is the celebrated Redlich-Kister consistency test. When the `fitParameters` regression fits NRTL or Wilson parameters to a $(x_1, T_{\text{bubble}})$ dataset (the `tutorials/steady/optimisation/fitNRTL01_ethanol_water` case), the chosen functional form for $G^E$ *automatically* satisfies Eq. (46) by construction — NRTL and Wilson are both derived from a $G^E$ model, and any pair of $\ln\gamma_i$ derived from a single $G^E$ obeys Gibbs-Duhem identically. The fit quality therefore measures how well the data themselves are thermodynamically consistent: a poor LM convergence is sometimes the data's fault, not the optimiser's.

**Common pitfall.** Gibbs-Duhem is exact for a binary mixture as written, but its appearance changes in $n$-component systems — there it becomes $\sum_i x_i \, \mathrm{d}\ln\gamma_i = 0$ along any infinitesimal path in composition space. Most published binary $\gamma$-data sets satisfy GD to within experimental noise; ternary data sets often do not (a hint that one of the three binaries is missing the cross-term). If you ever fit a ternary by combining three independent binary NRTL pairs, you have implicitly trusted the GD consistency of each pair.

## 6.12 Putting it together: an example energy balance

Take an isobaric heater that brings 100 kmol/h of equimolar ethanol/water from $T_{\mathrm{in}} = 300$ K to $T_{\mathrm{out}} = 360$ K, no phase change ($V = 0$ throughout, liquid only). The duty per kmol of feed is

$$\begin{aligned} q &= h_{\mathrm{liq}}(360) - h_{\mathrm{liq}}(300) \\ &= \sum_i x_i \left[ h_i^{\mathrm{liq}}(360) - h_i^{\mathrm{liq}}(300) \right] \\ &= \sum_i x_i \int_{300}^{360} c_{p,i}^{\mathrm{liq}}(T) \, \mathrm{d}T. \end{aligned} \tag{48}$$

Using the constant-$c_p$ approximations shipped (ethanol: $c_p^{\mathrm{liq}} \approx 165$ J/(mol · K); water: $\approx 75.4$ J/(mol · K)), this evaluates to

$$q \;\approx\; 0.5 \cdot 165 \cdot 60 + 0.5 \cdot 75.4 \cdot 60 \;\approx\; 7{,}212 \text{ J/mol feed.}$$

At 100 kmol/h that is a duty of about $7{,}212 \cdot 10^5/3600 \approx 200$ kW — which is exactly what `tutorials/steady/heat/hx01_heater` reports when you run it. Every step uses only Eqs. (28)–(44) and the polynomial $c_p$ from the data file.

**Runnable case.** `runCase tutorials/steady/heat/hx01_heater` executes the duty calculation above end-to-end and writes the result to the run log. Modify `tutorials/steady/heat/hx01_heater/system/flowsheetDict` to change $T_{\mathrm{out}}$ and watch how $Q$ scales linearly with $\Delta T$ — a direct consequence of constant $c_p$.

# 7 What is fittable, what is not

Every model in this Part has parameters that ultimately come from *somewhere*. The simulator distinguishes three sources, in descending order of trust:

1. **Literature curation.** Antoine constants $(A_i, B_i, C_i)$, critical properties $(T_c, P_c, \omega)$, normal boiling points, latent heats, heat-capacity polynomials, liquid molar volumes — these live in `data/standards/components/<name>.dat` and are cited per-file at the top of each file. They are *not* estimated by the simulator. To add a new component, copy an existing `.dat` file and populate it from DIPPR / Reid-Prausnitz-Poling [4] / NIST.
2. **Literature pair coefficients.** NRTL $(a_{ij}, b_{ij}, a_{ji}, b_{ji}, \alpha_{ij})$ and Wilson $\Lambda_{ij}$ live in `data/standards/parameters/<model>/<c1>-<c2>.dat`, also with provenance. The same DECHEMA / RPP sources apply.
3. **In-simulator regression.** For binary interaction parameters *only*, the simulator can refit $(a, b)$ against an experimental $T_b$-vs-$x$ dataset by the Levenberg-Marquardt `fitParameters` property operation of Chapter 4 (a `choupoProps` run; the earlier outer driver is retired and its factory refuses by name). Pure-component properties cannot be refitted by the simulator — if your candidate fluid is missing Antoine constants, you must either find them in the literature or estimate them by a group-contribution method (Lydersen, Joback, UNIFAC) outside the simulator, then write them into a `.dat` file.

**Common pitfall.** Pure-component property gaps will silently bend any flowsheet result. If `<name>.dat` has Antoine constants fitted only between 273 and 373 K, do not run a simulation that requires $P^{\text{sat}}$ at $T = 600$ K — the extrapolation can give negative pressures. The simulator does not yet enforce `validityRange`; that is on the bug-watch.

# Part III — Mixtures: from ideal to real

*Essentially, all models are wrong, but some are useful.*

George E. P. Box & Norman R. Draper, *Empirical Model-Building and Response Surfaces* (1987), p. 424

# 1 Ideal mixing

*What you'll learn.* Before any activity coefficient, every multi-component thermodynamic model is anchored in an *ideal* reference: a hypothetical mixture where molecules of different species feel no preference for who their neighbours are. This chapter builds the ideal-mixture chemical potential $\mu_i^{\mathrm{id}} = \mu_i^* + RT\ln x_i$ piece by piece (the $\ln x_i$ term is pure combinatorial entropy, no model), shows how it gives the ideal-gas mixture (Dalton's law) and the ideal-solution mixture (Raoult's law) as special cases, and ends with the binary $\Delta G_{\mathrm{mix}}^{\mathrm{id}}$ curve that explains why mixing is thermodynamically favourable at any finite temperature. The next chapter promotes $\gamma_i$ to the role of the excess factor that captures *real* non-ideality on top of this ideal baseline.

## 1.1 What "ideal mixing" means

Take $N_A$ molecules of species $A$ and $N_B$ molecules of species $B$, mix them at constant $T$ and $P$. An *ideal* mixture is the limit in which:

- the volume of mixing is zero ($\Delta V_{\mathrm{mix}} = 0$);
- the enthalpy of mixing is zero ($\Delta H_{\mathrm{mix}} = 0$);
- the molecules retain their identity — $A$ feels the same molecular field whether surrounded by $A$ or by $B$.

These three statements are equivalent under mild regularity assumptions, and together they fix *every* mixture property once the pure-component properties are known. The only thing that changes on mixing is the **configurational entropy**: there are many more ways to arrange $N_A$ $A$'s and $N_B$ $B$'s among a fixed set of sites than there are ways to arrange the pure components separately. This is the only "mixing" effect.

## 1.2 The combinatorial entropy of mixing

Place $N = N_A + N_B$ indistinguishable-within-species molecules on a lattice. The number of distinct arrangements is the binomial coefficient

$$W \;=\; \binom{N}{N_A} \;=\; \frac{N!}{N_A!\,N_B!}. \tag{49}$$

Boltzmann's statistical entropy is $S = k_B \ln W$. For molar amounts ($N_i = x_i N_{\mathrm{Avogadro}}$), Stirling's approximation gives, per mole of mixture,

$$\boxed{\Delta s_{\mathrm{mix}}^{\mathrm{id}} \;=\; -R\sum_i x_i \ln x_i.} \tag{50}$$

This is a *pure* consequence of how many ways the molecules can be arranged — there is no fitted constant, no model assumption about intermolecular forces. It is positive (since $x_i < 1$ and $\ln x_i < 0$), maximised at the equimolar composition where the combinatorial freedom is greatest, and zero at the pure-component limits.

*Intuition.* The $-R\sum x_i \ln x_i$ term shows up in every mixing calculation that follows — ideal or not, gas or liquid, in chemical-equilibrium expressions, in distillation $\Delta G$ profiles. It is the same function that appears in Shannon's information entropy of a discrete distribution, for the same combinatorial reason. Engineers and information theorists rediscovered each other's result decades apart.

## 1.3 The ideal-mixture chemical potential

Combining $\Delta s_{\mathrm{mix}}^{\mathrm{id}} = -R\sum x_i \ln x_i$ with $\Delta h_{\mathrm{mix}}^{\mathrm{id}} = 0$ gives the molar Gibbs energy of mixing

$$\Delta g_{\mathrm{mix}}^{\mathrm{id}} \;=\; \Delta h_{\mathrm{mix}}^{\mathrm{id}} - T\Delta s_{\mathrm{mix}}^{\mathrm{id}} \;=\; RT\sum_i x_i \ln x_i. \tag{51}$$

The full molar Gibbs energy of the mixture is then the sum of the pure-component values, weighted by mole fraction, plus this mixing correction:

$$g^{\mathrm{id}}(T,P,\mathbf{x}) \;=\; \sum_i x_i\, g_i^*(T,P) \;+\; RT\sum_i x_i \ln x_i, \tag{52}$$

where $g_i^*$ is the pure-component molar Gibbs energy. Take the partial derivative with respect to $n_i$ at constant $T, P, n_{j\neq i}$ to recover the **ideal-mixture chemical potential**:

$$\boxed{\mu_i^{\mathrm{id}}(T,P,\mathbf{x}) \;=\; \mu_i^*(T,P) \;+\; RT\ln x_i.} \tag{53}$$

Equation (53) is the single most important formula in mixture thermodynamics. The pure-component term $\mu_i^*$ carries all the physical chemistry of species $i$ alone (saturation pressure, latent heat, $c_p$); the $RT\ln x_i$ term is the combinatorial correction — universal across substances and models.

*Intuition.* Notice what (53) does *not* contain: any parameter that depends on which other species $i$ is mixed with. In an ideal mixture the chemical potential of $A$ at $x_A = 0.3$ is the same whether $A$ is dissolved in $B$, $C$, or in $A + B + C$. That is exactly the simplification that breaks for real mixtures, where neighbour identity matters. The activity coefficient $\gamma_i$ of the next chapter is the multiplicative correction (53) needs — $\mu_i = \mu_i^* + RT\ln(x_i\gamma_i)$ — to absorb that effect.

## 1.4 Ideal-gas mixtures and Dalton's law

Specialise (53) to a gas mixture in which every component obeys the ideal-gas law. The pure-component chemical potential is

$$\mu_i^{*,\mathrm{ig}}(T,P) \;=\; \mu_i^{0,\mathrm{ig}}(T) \;+\; RT\ln(P/P_0), \tag{54}$$

with $P_0$ a reference pressure (1 bar). Substituting into (53) and using $P_i = y_i P$ (the partial pressure):

$$\mu_i^{\mathrm{ig,mix}}(T,P,\mathbf{y}) \;=\; \mu_i^{0,\mathrm{ig}}(T) \;+\; RT\ln(y_iP/P_0) \;=\; \mu_i^{0,\mathrm{ig}}(T) \;+\; RT\ln(P_i/P_0). \tag{55}$$

The mixture chemical potential is therefore identical to that of pure $i$ at its partial pressure $P_i$. Going back through the ideal-gas equation of state, this is exactly Dalton's law:

$$P \;=\; \sum_i P_i, \qquad P_i \;=\; y_iP. \tag{56}$$

For the simulator: the fugacity coefficient of every species in an ideal-gas mixture is $\varphi_i = 1$ *by construction*. This is the vapour-phase model.

## 1.5 Ideal solutions and Raoult's law

Apply (53) to a liquid mixture in which the intermolecular forces between unlike molecules are the same as between like ones — the classical *ideal solution*. The pure-component liquid chemical potential at saturation is

$$\mu_i^{*,\mathrm{L}}(T,P) \;=\; \mu_i^{*,\mathrm{V,sat}}(T,P_i^{\mathrm{sat}}) \;\approx\; \mu_i^{0,\mathrm{ig}}(T) \;+\; RT\ln(P_i^{\mathrm{sat}}/P_0), \tag{57}$$

where the second equality uses ideal-gas vapour fugacity. Phase equilibrium $\mu_i^{\mathrm{L,mix}} = \mu_i^{\mathrm{V,mix}}$ then gives, after cancelling the reference,

$$\boxed{\;y_iP \;=\; x_i\,P_i^{\mathrm{sat}}(T).\;} \tag{58}$$

This is **Raoult's law**: the partial pressure of species $i$ above an ideal solution is its pure-component saturation pressure weighted by its liquid mole fraction. In the simulator, the activity coefficient $\gamma_i = 1$ *by construction*.

Raoult's law is exact for ideal solutions; for many real systems (chemically similar species, e.g. benzene/toluene, $n$-hexane/$n$-heptane) it is accurate to a percent or so. Where it breaks — ethanol/water, acetic acid/water, ethyl acetate/ethanol — is the territory of activity coefficients in the next chapter.

## 1.6 The driving force for mixing

For a binary at any composition, (51) reads

$$\Delta g_{\mathrm{mix}}^{\mathrm{id}}(x_1) \;=\; RT\big[x_1\ln x_1 + (1-x_1)\ln(1-x_1)\big], \tag{59}$$

which is concave-up and negative, with its minimum at $x_1 = 0.5$ where $\Delta g_{\mathrm{mix}}^{\mathrm{id}} = -RT\ln 2 \approx -1.72$ kJ/mol at 300 K. Because $\Delta g_{\mathrm{mix}}^{\mathrm{id}} < 0$ for every $x_1 \in (0,1)$, **ideal mixing is always spontaneous**. This is why miscibility is the rule rather than the exception for chemically similar liquids; breaking it requires real intermolecular forces that the ideal model cannot represent.

*Intuition.* The cusps at $x_1 = 0$ and $x_1 = 1$ are characteristic of combinatorial mixing: an infinitesimal amount of impurity in a pure liquid produces a finite drop in $g$, because the configurational entropy gain $-Rx\ln x \to +\infty$ as $x \to 0^+$. This is the thermodynamic root of why ultra-purification is hard: every additional decade of purity costs an infinite Gibbs energy.

## 1.7 Where the ideal limit goes wrong

Ideal mixing is the right *reference*, not the right *description*, for most real liquids:

- **Chemically similar species, weak interactions (alkanes, aromatics):** Raoult holds to within 1–5%. Activity coefficients are close to 1; the ideal model is useful directly.
- **Polar / non-polar pairs (water with hydrocarbons):** $\gamma_i \gg 1$, often $> 10$. Ideal mixing under-predicts the escape tendency of one component in the other by orders of magnitude; liquid-liquid splits become possible.
- **Hydrogen-bonded systems (ethanol/water, acid/water):** $\gamma_i$ varies smoothly but non-trivially across composition; can hit azeotropes (minimum or maximum on the bubble curve) that the ideal model cannot represent at all.

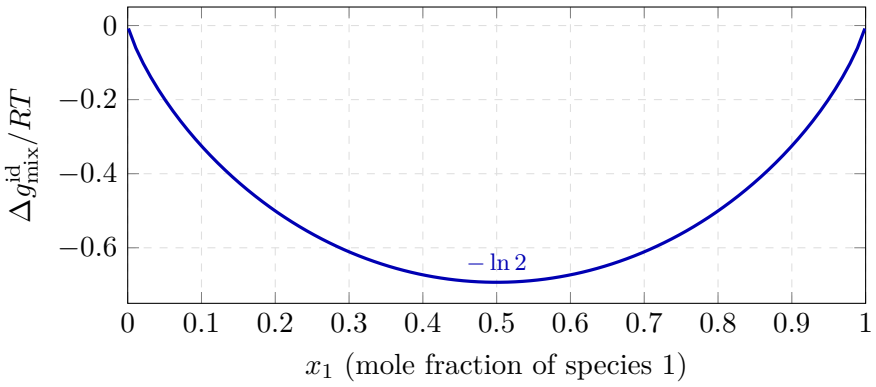



**Figure 8:** Ideal-mixture Gibbs energy of mixing for a binary at fixed $T$. Negative everywhere except at the pure-component endpoints ($x_1 = 0$ or 1, where the curve is tangent to zero with a vertical slope from the ln singularity). The minimum at $x_1 = 0.5$ is exactly $-RT \ln 2$.

- **Strong specific interactions (acid/base, hydrogen- bond donors with acceptors):** $\gamma_i < 1$ (negative deviations), maximum-boiling azeotropes; again outside the ideal-mixture description.

The activity coefficient $\gamma_i$ is the multiplicative correction that puts (53) back on top of the experimental data:

$$\mu_i(T, P, \mathbf{x}) \;=\; \mu_i^*(T, P) \;+\; RT \ln(x_i\, \gamma_i) \;=\; \mu_i^{\mathrm{id}}(T, P, \mathbf{x}) \;+\; RT \ln \gamma_i. \tag{60}$$

The next chapter is about how that correction is parametrised (NRTL, Wilson) and how the excess Gibbs energy $G^E$ generates $\gamma_i$ through a partial-derivative recipe.

# 2 Activity coefficients

*What you'll learn.* Three models cover the simulator's needs. They share a common derivation: an expression for excess Gibbs energy $G^E$ from which $\gamma_i$ falls out by differentiation.

## 2.1 The excess Gibbs energy bridge

For a real liquid mixture, the molar Gibbs energy is split as

$$G^L(T,P,\mathbf{x}) \;=\; G^{L,\mathrm{ideal}}(T,P,\mathbf{x}) \;+\; G^E(T,\mathbf{x}). \tag{61}$$

The activity coefficient is the composition derivative:

$$\boxed{\ln\gamma_i \;=\; \frac{1}{R_\mathrm{g}T}\left(\frac{\partial(n\,G^E)}{\partial n_i}\right)_{T,P,n_{j\neq i}}} \tag{62}$$

Pick a model for $G^E$, do the derivative, and you have an activity model. That is what NRTL, Wilson, UNIQUAC and friends are: each is one choice of $G^E(\mathbf{x},T)$.

## 2.2 Ideal solution

If $G^E = 0$, then $\gamma_i = 1$ for every component. Raoult's law: $y_i P = x_i P_i^{\mathrm{sat}}(T)$. Useful as a baseline, a sanity check, and a starting point for the Wegstein outer loop.

## 2.3 NRTL (Renon and Prausnitz, 1968)

The model that earns its keep in this simulator is NRTL — the *Non-Random Two-Liquid* model. It is the default for non-ideal VLE and the engine behind every liquid–liquid split: NRTL is the only $G^E$ model in Choupo that can represent demixing, because (unlike Wilson, §2.4) its denominators admit the doubly-tangent $G^E$ shape that two coexisting liquids require. Everything below is implemented in `src/thermo/activityCoefficient/NRTL.cpp`.

**The local-composition idea.** Renon and Prausnitz's [6] starting point — the same postulate that underlies Wilson (67) — is that a real liquid is not randomly mixed. Around a central molecule of type $i$, the local mole fraction of $j$-neighbours, $x_{ji}$, is biased

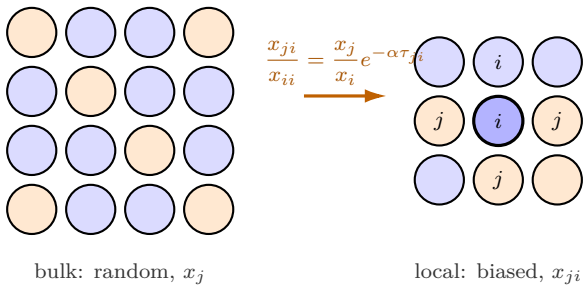



**Figure 9:** The local-composition postulate behind NRTL and Wilson. In a *randomly* mixed liquid (left) every molecule's neighbour shell reflects the bulk fractions $x_j$. Real mixtures are not random: around a central molecule $i$, the local fraction of $j$-neighbours $x_{ji}$ is re-weighted by a Boltzmann factor in the pair-energy difference (right). NRTL's twist is the extra non-randomness constant $\alpha$ in the exponent — $\alpha \to 0$ recovers the random regular-solution lattice, larger $\alpha$ sharpens the local ordering. Every $\tau_{ij}$ and $\Lambda_{ij}$ in the activity-coefficient formulas is just this biased shell, counted.

away from the bulk $x_j$ by the difference in pair energy. NRTL's distinctive twist is a *two-liquid* Boltzmann weighting that carries an extra non-randomness factor $\alpha$:

$$\frac{x_{ji}}{x_{ii}} \;=\; \frac{x_j}{x_i}\,\exp\!\Big[-\alpha_{ji}\,\frac{g_{ji}-g_{ii}}{R_\mathrm{g}T}\Big], \tag{63}$$

where $g_{ji}$ is the energy of a $j$–$i$ pair and $\alpha_{ji}$ *quenches* the non-randomness: $\alpha \to \infty$ forces complete

local order, $\alpha = 0$ recovers a random (regular-solution) mixture. In practice $\alpha$ is an adjustable but *transferable* constant, taken symmetric ($\alpha_{ij} = \alpha_{ji}$) and in the narrow band $\alpha \in [0.20,\ 0.47]$: $\sim$0.2 near ideality, 0.3 for typical polar mixtures (the simulator's default when a pair file omits it), $\sim$0.47 for strongly self-associating or phase-splitting systems.

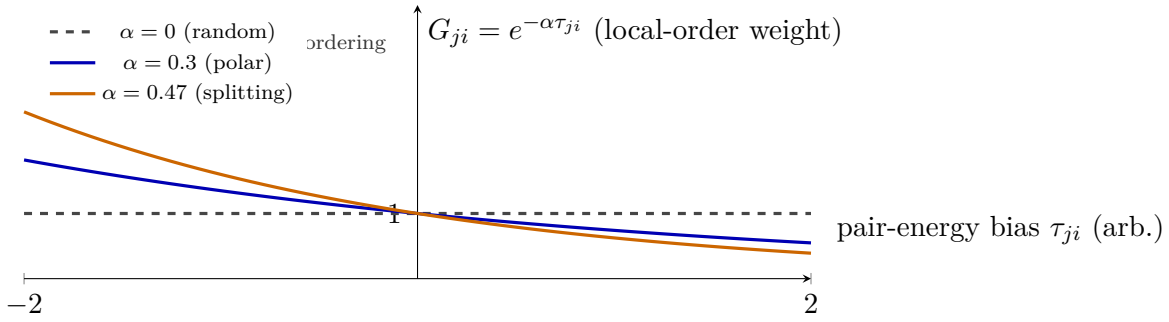



**Figure 10:** What the NRTL non-randomness constant $\alpha$ actually does. The factor $G_{ji} = e^{-\alpha\tau_{ji}}$ is the weight a $j$-neighbour gets in the shell around an $i$-molecule. At $\alpha = 0$ (flat dashed line) $G = 1$ for every bias — the shell is random and NRTL collapses to a regular solution. Cranking $\alpha$ up steepens the curve, so the same pair-energy bias $\tau$ produces a sharper deviation of the local shell from the bulk: $\alpha \approx 0.3$ for typical polar mixtures (Choupo's default when a pair file omits it), $\alpha \approx 0.47$ for the strongly self-associating, phase-splitting systems where the extra ordering is what carries the liquid–liquid split. Because $\alpha$ is held fixed during a fit, a poor choice cannot be repaired by the energy parameters.

**The grouped parameters.** Collecting the two quantities the model actually stores,

$$\boxed{\tau_{ij} \;=\; \frac{g_{ij} - g_{jj}}{R_{\mathrm{g}} T} \;=\; a_{ij} + \frac{b_{ij}}{T}, \qquad G_{ij} \;=\; \exp(-\alpha_{ij}\,\tau_{ij})} \tag{64}$$

with the diagonal conventions $\tau_{ii} = 0$, $G_{ii} = 1$, $\alpha_{ii} = 0$. Two facts matter. First, $\tau$ is *asymmetric*: $\tau_{ij} \neq \tau_{ji}$ in general, because the energy to place a $j$ next to an $i$ differs from the reverse once each is referenced to its own pure-fluid pair. Second, the temperature dependence is the affine form $\tau_{ij} = a_{ij} + b_{ij}/T$ — exactly the storage in the code: each ordered pair carries four energy parameters $(a_{ij}, b_{ij}, a_{ji}, b_{ji})$ and one shared $\alpha_{ij}$ (the `aMat_`, `bMat_`, `alphaMat_` matrices). Here $a_{ij}$ is the high-$T$ residual (entropic, temperature-independent) and $b_{ij}$ [K] is the enthalpic slope. A $b$-only fit ($a = 0$) forces $\tau \to 0$ as $T \to \infty$ — the mixture becomes ideal when thermal energy swamps the pair energy — whereas carrying $a \neq 0$ admits a residual that survives.

**Excess Gibbs energy.** Substituting the local compositions (63) into the two-fluid lattice energy and referencing to the pure components gives

$$\frac{G^E}{R_{\mathrm{g}} T} \;=\; \sum_{i=1}^{N_c} x_i \,\frac{\displaystyle\sum_{j=1}^{N_c} \tau_{ji} G_{ji} x_j}{\displaystyle\sum_{k=1}^{N_c} G_{ki} x_k}. \tag{65}$$

**Activity coefficient.** Apply the partial-molar derivative (62). Write $n_i = n x_i$ and abbreviate the denominators $S_j \equiv \sum_k G_{kj} x_k$ (these are `S[j]` in the code) and the numerators $D_j \equiv \sum_k \tau_{kj} G_{kj} x_k$ (`Tsum[j]`). Differentiating $nG^E/R_{\mathrm{g}}T = \sum_i n_i D_i / S_i$ with respect to $n_i$ at fixed $n_{m\neq i}$ — using $\partial S_j/\partial n_i = (G_{ij} - S_j)/n$ and $\partial D_j/\partial n_i = (\tau_{ij} G_{ij} - D_j)/n$ — the $1/n$ factors cancel and the result is the standard form

$$\boxed{\ln\gamma_i \;=\; \underbrace{\frac{\displaystyle\sum_j \tau_{ji} G_{ji} x_j}{\displaystyle\sum_k G_{ki} x_k}}_{D_i/S_i} \;+\; \sum_j \frac{x_j G_{ij}}{S_j}\left[\tau_{ij} - \frac{D_j}{S_j}\right]} \tag{66}$$

which is term-for-term the loop in `NRTL::gamma()`: `termA` is $D_i/S_i$, `termB` accumulates the $j$-sum, and the two are exponentiated. Pre-computing $S_j$ and $D_j$ once per temperature (rather than inside every $i, j$ pass) is the only optimisation; the algebra is otherwise the textbook formula. Check a hand derivation against Renon and Prausnitz [6] or Reid–Prausnitz–Poling [4], Eq. 8-10.

> *Intuition.* Read (66) as "the environment $i$ creates plus the environments $i$ disturbs." The first term is how strongly $i$ is held by its own surroundings; the sum is the back-reaction — $i$ entering each cell $j$ and changing its tally. At infinite dilution ($x_i \to 0$) the cell of the majority solvent dominates and $\ln\gamma_i^{\infty}$ collapses to a closed

expression in the pair parameters — the quantity most sensitive to a fit.

**Pairs and the ideal-default rule.** Binary parameter sets are tabulated in the DECHEMA Vapor–Liquid Equilibrium Data Collection or regressed from data; the tutorials carry literature values from the DECHEMA bank (each pair file's `provenance` block cites its primary). The simulator resolves each pair in a fixed precedence (`locatePairFile`): an inline `pairs(...)` block in the case's declared system, then a per-node `constant/parameters/NRTL/<a>-<b>.da` then the case root, then `data/standards/parameters/NRTL/`. For an $N_c$-component mixture there are $N_c(N_c-1)/2$ binaries; NRTL is a strictly *pairwise* model with *no* ternary parameters, so a multicomponent prediction is only as good as its constituent binaries.

A pair with no parameters anywhere does *not* abort the run: it **defaults to ideal** ($\tau = 0 \Rightarrow$ no $G^E$ contribution; $\alpha$ set to 0.3 harmlessly). This is deliberate — it lets a thermo package be built one pair at a time — but it is never silent: every ideal-defaulted pair is announced on `stdout` and in the advisory log, so the student *sees* exactly which interactions are unconstrained. To produce one's own fitted set from a $T_b$-vs-$x$ dataset, run the parameter-estimation driver of Chapter 4 (`tutorials/steady/optimisation/fitNRTL01_ethanol_water`).

**Power.** Two energy parameters per direction plus one non-randomness constant span ideality, strong positive deviation, and liquid–liquid immiscibility within a single algebraic form; the affine $\tau(T)$ extrapolates across a modest temperature range; and the model is analytically differentiable (used for the bubble-point Jacobian).

**Limit.** NRTL is a *fitting* model, not a predictive one: the parameters have no first-principles values and must come from data (contrast UNIFAC, which predicts them from group contributions). It carries no information about the vapour or the solid — it gives $\gamma$ for the liquid only; you still supply $P^{\text{sat}}$ from Antoine (§2) and any $K_{sp}$ separately. The affine $\tau(T)$ is calorimetrically uncalibrated unless each pair is refit against measured $H^E$ (the `calorimetricFit` gate that NRTL exposes), so excess-enthalpy predictions from a VLE-only fit are not to be trusted. And because $\alpha$ is held fixed during a fit, a poor $\alpha$ choice cannot be repaired by the energy parameters.

## 2.4 Wilson (1964)

Simpler in form than NRTL, faster numerically (no $\alpha$, no nested $G$-exponentials), and built on a more physical picture — but with one structural cost: it *cannot represent* a liquid-liquid split. That is a limitation of the functional form, not of the fit, and the reason is worth understanding rather than memorising (§2.4).

**The idea: local compositions.** Wilson's [7] starting point is that the composition *immediately around* a central molecule is not the bulk composition. If molecule 1 attracts molecule 2 more strongly than it attracts its own kind, then the local mole fraction of 2 near a 1 is enriched above $x_2$. Wilson postulated that these *local* mole fractions $x_{ji}$ (species $j$ around a central $i$) are related to the bulk ones through Boltzmann factors of the interaction energies $\lambda_{ji}$:

$$\frac{x_{ji}}{x_{ii}} \;=\; \frac{x_j\,\exp(-\lambda_{ji}/R_{\text{g}}T)}{x_i\,\exp(-\lambda_{ii}/R_{\text{g}}T)}. \tag{67}$$

Weighting each local environment by the molar volume of the central species and collecting the result into an excess Gibbs energy yields the one-line model below. The same local-composition postulate underlies NRTL and UNIQUAC; Wilson is its earliest and leanest realisation.

**Excess Gibbs energy.**

$$\boxed{\frac{G^E}{R_{\text{g}}T} \;=\; -\sum_{i=1}^{N_c} x_i \ln\!\left(\sum_{j=1}^{N_c} x_j \Lambda_{ij}\right)} \tag{68}$$

with the asymmetric, dimensionless interaction parameter

$$\Lambda_{ij} \;=\; \frac{V_j^L}{V_i^L}\,\exp\!\Big[-\underbrace{(\lambda_{ij}-\lambda_{ii})}_{A_{ij}}\Big/(R_{\text{g}}T)\Big], \qquad \Lambda_{ii}=1. \tag{69}$$

Here $V_i^L$ is the pure-component liquid molar volume and $\lambda_{ij}$ the energy of an $i$–$j$ interaction. Only the *difference* $A_{ij} \equiv \lambda_{ij}-\lambda_{ii}$ ever appears, so Wilson carries **two** energy parameters per binary, $A_{ij}$ and $A_{ji}$ (in J/mol), and these are *asymmetric*: $A_{ij} \neq A_{ji}$ in general, hence $\Lambda_{ij} \neq \Lambda_{ji}$. There is no third "non-randomness" factor as in NRTL.

**From $G^E$ to $\gamma_i$.** Apply the partial-molar recipe (62) to (68). Writing $n_i = nx_i$ and $nG^E/R_{\text{g}}T = -\sum_i n_i \ln\!\big(\sum_j n_j\Lambda_{ij}/n\big)$, the derivative $\partial/\partial n_i$ contributes one term from the explicit ln at $i$ and one from every other molecule's denominator

in which $i$ appears. After re-dividing by $n$ the result collapses to the classic two-term form:

$$\boxed{\ln\gamma_i \;=\; 1 - \ln\!\left(\sum_j x_j\Lambda_{ij}\right) \;-\; \sum_k \frac{x_k\,\Lambda_{ki}}{\sum_j x_j\,\Lambda_{kj}}} \tag{70}$$

This is exactly the loop in `src/thermo/activityCoefficient/Wilson.cpp`: the code first builds the $\Lambda$ matrix from (69), precomputes $S_k = \sum_j x_j\Lambda_{kj}$ once, then assembles (70) as $1 - \ln S_i - \sum_k x_k\Lambda_{ki}/S_k$.

**Built-in consistency.** Two checks fall out for free, and the code reproduces them. At infinite dilution of $i$ in pure $j$ ($x_j \to 1$) the second sum keeps only the $k = j$ term, giving the limiting activity coefficient

$$\ln\gamma_i^{\infty} \;=\; 1 - \ln\Lambda_{ij} - \Lambda_{ji}, \tag{71}$$

a convenient handle for fitting to a single $\gamma^{\infty}$ datum. And in the pure limit $x_i \to 1$ every $S_k \to \Lambda_{ki}$ and (70) returns $\ln\gamma_i = 1 - \ln 1 - 1 = 0$, i.e. $\gamma_i \to 1$ as it must.

**Temperature dependence — and an honest caveat.** The $T$-dependence in Choupo's Wilson is *implicit only*, through the Boltzmann factor $\exp(-A_{ij}/R_gT)$ and the (weakly $T$-varying) volume ratio in (69): the energy parameters $A_{ij}$ themselves are stored as single constants in J/mol, not as a $T$-expansion. This is deliberately *unlike* the NRTL implementation, whose $\tau_{ij} = a_{ij} + b_{ij}/T$ (65) carries an explicit linear-in-$1/T$ term. If a wider temperature range demands it, the physically standard refinement is to let $A_{ij} = A_{ij}^0 + A_{ij}^1\,T$ (a linear-in-$T$ energy, the form used by DECHEMA for Wilson); that extra slope is not yet wired into the loader, so over a broad $T$ window prefer a pair fitted near the operating temperature. The single-constant choice keeps the glass-box small: one number per direction, visible in the `.dat`.

**Wilson vs NRTL at a glance.**

| | **Wilson** | **NRTL** |
|---|---|---|
| Parameters / binary | $A_{ij}, A_{ji}$ (2) | $a_{ij}, b_{ij}, a_{ji}, b_{ji}, \alpha$ (5) |
| Needs $V_i^L$? | yes (liquid molar volume) | no |
| $T$-model in Choupo | $\exp(-A/R_gT)$ only | explicit $a + b/T$ in $\tau$ |
| Liquid-liquid split | **impossible** | yes (its main strength) |
| Cost per $\gamma$ eval | one ln, one nested sum | two nested sums + exp |

**Why Wilson cannot demix.** A liquid-liquid split requires $G^E$ to be non-convex in composition (a region where $\partial^2 G/\partial x^2 < 0$). For a binary, $G^E/R_gT = -x_1\ln(x_1 + x_2\Lambda_{12}) - x_2\ln(x_2 + x_1\Lambda_{21})$ is mathematically incapable of producing two minima for any positive

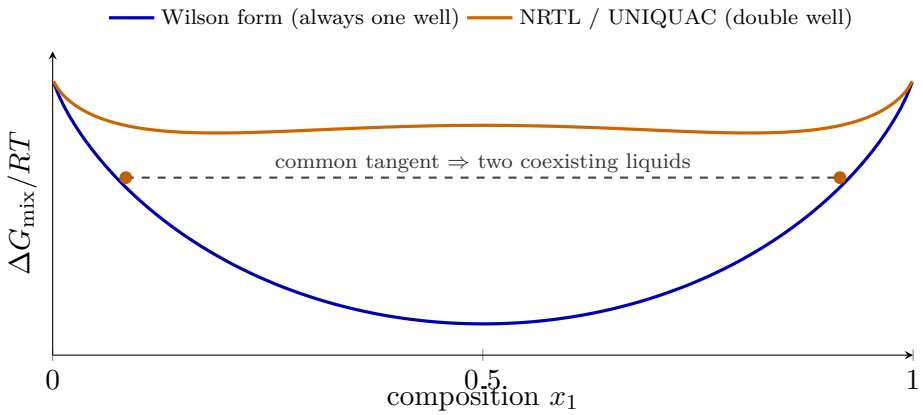



**Figure 11:** Why the model's *form* — not its fit — decides whether a liquid can split. A liquid–liquid split needs $\Delta G_{\mathrm{mix}}$ to be non-convex: a hump with two wells, so a common tangent touches two distinct compositions (the orange curve). Wilson's single logarithm per term, $G^E/RT = -x_1\ln(x_1 + x_2\Lambda_{12}) - x_2\ln(x_2 + x_1\Lambda_{21})$, is mathematically incapable of producing that hump for any positive $\Lambda$ — it stays convex (blue), so no parameter set makes Wilson demix. NRTL and UNIQUAC carry the extra freedom for a double well, which is exactly why Choupo's LLE Gibbs-minimisation accepts those two and refuses Wilson.

$\Lambda_{12}, \Lambda_{21}$ — the single logarithm per term keeps the curve convex. No choice of parameters fixes this; it is the price of the one-parameter-pair-per-direction simplicity. For partially miscible systems (e.g. the LLE Gibbs-minimisation of Chapter 11) the simulator falls back to NRTL or UNIQUAC, which carry the extra freedom needed for a double-well $G^E$.

**Parameter sources.** Binary $(A_{ij}, A_{ji})$ sets resolve from a case-local `constant/parameters/Wilson/<i>-<j>.dat`, then `data/standards/parameters/Wilson/` — though the standard tree currently ships no Wilson pairs, so the tutorials declare theirs inline in the case's thermophysical system; a pair with no parameters anywhere *defaults to ideal* for that binary ($\Lambda = 1$, no contribution) and the run announces it aloud rather than aborting — the same no-silent-crutch hybrid used by NRTL. Literature values are tabulated in the DECHEMA collection and Reid–Prausnitz–Poling [4]; fit your own with the parameter-estimation driver of Chapter 4.

**Worked example 2.1.** Wilson $\gamma$ for ethanol(1)/water(2) at 350 K, $x_1 = 0.20$Parameters $A_{12} = 1157.95$, $A_{21} = 4081.65$ J/mol (the values the `flash03_wilson_ethanol_water` tutorial declares inline) and liquid molar volumes $V_1^L = 5.87 \times 10^{-5}$, $V_2^L = 1.81 \times 10^{-5}$ m$^3$/mol. With $R_g T = 8.314 \times 350 = 2910$ J/mol:

$$\Lambda_{12} = \tfrac{V_2}{V_1} e^{-A_{12}/R_g T} = \tfrac{1.81}{5.87}\, e^{-0.398} = 0.207, \quad \Lambda_{21} = \tfrac{V_1}{V_2} e^{-A_{21}/R_g T} = \tfrac{5.87}{1.81}\, e^{-1.403} = 0.797.$$

At $x_1 = 0.20$: $S_1 = x_1 + x_2\Lambda_{12} = 0.20 + 0.80(0.207) = 0.366$ and $S_2 = x_2 + x_1\Lambda_{21} = 0.80 + 0.20(0.797) = 0.959$. Then from (70),

$$\ln\gamma_1 = 1 - \ln 0.366 - \left[\frac{x_1\Lambda_{11}}{S_1} + \frac{x_2\Lambda_{21}}{S_2}\right] = 1 - (-1.005) - \left[\tfrac{0.20}{0.366} + \tfrac{0.80\cdot 0.797}{0.959}\right] \approx 0.79,$$

so $\gamma_1 \approx 2.2$. Ethanol is strongly non-ideal here — the same qualitative verdict as the NRTL worked example below, with a different model and two fewer parameters.

## 2.5 UNIQUAC (Abrams and Prausnitz, 1975)

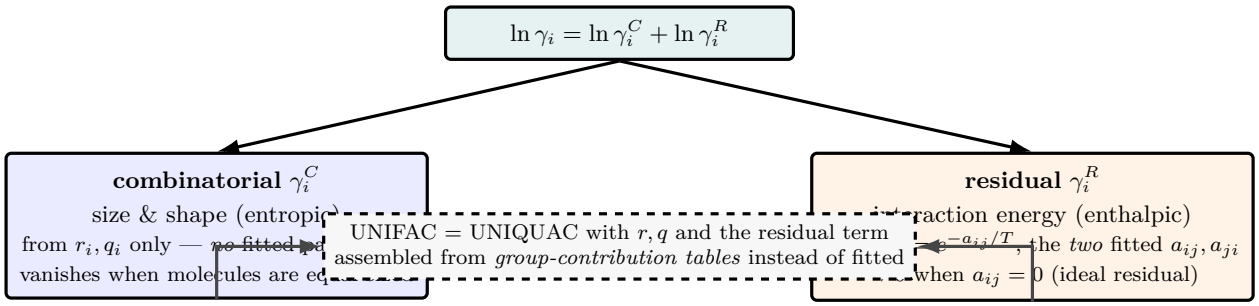



**Figure 12:** UNIQUAC's defining move: split the excess Gibbs energy into two physically distinct pieces. The **combinatorial** term (blue) carries the entropic penalty of packing molecules of unequal size and shape — it needs only the structural constants $r_i, q_i$ and has *zero* fitted parameters, vanishing entirely when all molecules are the same size. The **residual** term (orange) carries the interaction energy through $\tau_{ij} = e^{-a_{ij}/T}$ and holds the *two* binary energies $(a_{ij}, a_{ji})$ — one fewer than NRTL, which also needs $\alpha$. Because the size/shape physics lives in $r, q$, this is the exact structure UNIFAC reuses: keep the combinatorial term as is and build the residual from group-contribution tables instead of a binary fit.

NRTL and Wilson are *energy-only* models: they describe non-ideality through interaction parameters and quietly assume the molecules are the same size and shape. That assumption frays for mixtures of very unequal molecules — a polymer in a solvent, a sugar in water, an alkane in an alcohol — where much of the deviation from ideality is *entropic*, born of the difficulty of packing big and small molecules onto a common lattice. UNIQUAC (UNIversal QUAsi-Chemical) [8] splits the excess Gibbs energy into two physically distinct pieces and gives each its own term:

$$\frac{G^E}{R_g T} = \underbrace{\frac{G^E_{\text{comb}}}{R_g T}}_{\text{size/shape (entropic)}} + \underbrace{\frac{G^E_{\text{res}}}{R_g T}}_{\text{energy (enthalpic)}}, \qquad \Longrightarrow \qquad \ln\gamma_i = \ln\gamma_i^C + \ln\gamma_i^R. \tag{72}$$

The *combinatorial* part $\gamma_i^C$ depends only on *structural* constants; the *residual* part $\gamma_i^R$ carries the two adjustable binary energies. This is exactly the structure that lets UNIFAC *predict* $\gamma$ from group counts: UNIFAC is UNIQUAC with $r$, $q$ and the residual term assembled from group-contribution tables instead of being fitted.

**Structural constants.** Each component carries two pure-component numbers, computed from its van der Waals group volumes and surface areas (Bondi, 1968):

$$r_i \;=\; \text{relative molecular } \textit{volume}, \qquad q_i \;=\; \text{relative molecular } \textit{surface area}. \tag{73}$$

These are not fitted — they are tabulated constants the student *cites* (Poling–Prausnitz–O'Connell [4], Table 8E; original values in [8]). Each standard component carries its own `uniquac { r; q; }` block in its `.dat`, from which the engine fills the model's `rq { ...}` block automatically; an inline `rq { ...}` entry in the declared system always wins, and a component with $r, q$ nowhere refuses to run (no silent default for a structural property). From $r$ and $q$ the model builds, at a given composition, the *segment fraction* $\phi_i$ and *area fraction* $\theta_i$:

$$\phi_i \;=\; \frac{r_i x_i}{\sum_j r_j x_j}, \qquad \theta_i \;=\; \frac{q_i x_i}{\sum_j q_j x_j}. \tag{74}$$

Two limits aid intuition: if all $r_i$ are equal then $\phi_i = x_i$, and if additionally all $q_i$ are equal then $\theta_i = x_i$ and the combinatorial term vanishes — equal-sized molecules carry no packing penalty.

**Combinatorial (size/shape) term.** The entropic piece is a Staverman–Guggenheim lattice count, with activity coefficient

$$\ln\gamma_i^{C} \;=\; \ln\frac{\phi_i}{x_i} \;+\; \frac{z}{2}\,q_i \ln\frac{\theta_i}{\phi_i} \;+\; l_i \;-\; \frac{\phi_i}{x_i}\sum_j x_j l_j, \qquad l_i \;=\; \frac{z}{2}(r_i - q_i) - (r_i - 1), \tag{75}$$

where the *lattice coordination number* is fixed at $z = 10$ (each molecule is taken to have ten nearest neighbours — the canonical UNIQUAC value, hard-coded as `z_coord_` in the engine, not a tunable). This term contains *no* adjustable parameters: feed it the $r$'s and $q$'s and it is fully determined. It alone already predicts the bulk of athermal (zero-energy) non-ideality.

**Residual (energy) term.** The enthalpic piece uses the *quasi-chemical* idea — a molecule's surroundings are biased by interaction energies through Boltzmann factors

$$\tau_{ij} \;=\; \exp\!\Big(-\frac{u_{ij} - u_{jj}}{R_{\mathrm{g}} T}\Big) \;=\; \exp\!\Big(-\frac{a_{ij}}{T}\Big), \qquad a_{ij} \;=\; \frac{u_{ij} - u_{jj}}{R_{\mathrm{g}}} \;\;[\mathrm{K}], \tag{76}$$

with $u_{ij}$ the $i$–$j$ interaction energy. Note $\tau_{ii} = 1$ and the matrix is asymmetric ($\tau_{ij} \neq \tau_{ji}$); each binary contributes *two* energies $a_{ij}, a_{ji}$ — one fewer than NRTL, which also needs $\alpha$. The engine stores $a_{ij}$ in *kelvin* and supports an optional polynomial extension $a_{ij}^{\mathrm{eff}} = a_{ij} + b_{ij}\,T + c_{ij}\,T^2$ for sets fitted over a wide temperature range (most literature sets use $a$ alone, $b = c = 0$). The residual activity coefficient is

$$\ln\gamma_i^{R} \;=\; q_i \left[\, 1 - \ln\!\Big(\sum_j \theta_j \tau_{ji}\Big) - \sum_j \frac{\theta_j \tau_{ij}}{\sum_k \theta_k \tau_{kj}} \,\right]. \tag{77}$$

Equations (75) and (77) are exactly what `src/thermo/activityCoefficient/UNIQUAC.cpp` evaluates in `gamma()` — read the two side by side.

**Parameter sources and the ideal default.** The two energies $(a_{ij}, a_{ji})$ per binary come from VLE regression (DECHEMA, or the parameter-estimation driver of Chapter 4), supplied inline as `pairs (...)` or from a UNIQUAC pair catalogue (`constant/parameters/UNIQUAC/<a>-<b>.dat`, the same precedence NRTL and Wilson use). A binary with no parameters anywhere defaults to $a_{ij} = a_{ji} = 0$, i.e. $\tau = 1$ (ideal residual contribution), and the run *announces* the defaulted pairs aloud — the same no-silent-crutch hybrid as NRTL and Wilson.

**Power and limit.** UNIQUAC needs only *two* energy parameters per binary (vs three for NRTL) yet, because the size/shape physics is built in through $r$ and $q$, it extrapolates over composition and to multicomponent mixtures better than NRTL or Wilson for molecules of disparate size. Unlike Wilson it *can* describe liquid–liquid demixing (its $G^E$ admits a double well), so the LLE Gibbs-minimisation of Chapter 11 accepts it alongside NRTL. And it is the structural parent of UNIFAC, so a UNIQUAC *fit* and a UNIFAC *prediction* are directly comparable on the same components. The **limit**: the two-parameter residual term is less flexible than NRTL's three-parameter form for a single tightly-fitted binary — for a strongly hydrogen-bonded pair fitted on its own data NRTL often edges it. The combinatorial term is a mean-field lattice count that sees molecular shape only through the single number $q$, so it is imperfect for very flat or branched species. And, like every $G^E$ model, it is silent about the vapour ($\varphi^{\mathrm{sat}} = 1$ here) and about ions (use eNRTL/Pitzer, §3).

**Worked example 2.2.** UNIQUAC $\gamma$ for ethanol(1)/water(2) at 351 K, $x_1 = 0.20$The `tutorials/steady/flash/bubbleT02_uniquac_ethanol_water` system. Structural constants $r_1 = 2.1055,\ q_1 = 1.9720,\ r_2 = 0.9200,\ q_2 = 1.4000$; energies $a_{12} = -21.44\,\mathrm{K}$, $a_{21} = 222.47\,\mathrm{K}$.
*Step 1 — $\tau$ at $T = 351\,K$* (Eq. 76): $\tau_{12} = e^{21.44/351} = 1.063$, $\tau_{21} = e^{-222.47/351} = 0.531$ (and $\tau_{11} = \tau_{22} = 1$).
*Step 2 — fractions* (Eq. 74) at $x = (0.20, 0.80)$: $\sum r_j x_j = 1.157$, $\sum q_j x_j = 1.514$, hence $\phi_1 = 0.364,\ \theta_1 = 0.260$ (and $\phi_2 = 0.636,\ \theta_2 = 0.740$).
*Step 3 — combinatorial* (Eq. 75) with $z = 10$: $l_1 = 5(2.1055 - 1.9720) - (2.1055 - 1) = -0.438$, $l_2 = -2.320$, $\sum x_j l_j = -1.944$, giving $\ln \gamma_1^C = 0.398$.
*Step 4 — residual* (Eq. 77): $\sum_j \theta_j \tau_{j1} = 0.260 + 0.740(0.531) = 0.653$, the second sum evaluates to 1.172, so $\ln \gamma_1^R = 1.972\,[\,1 - \ln 0.653 - 1.172\,] = 0.501$.
*Result:* $\ln \gamma_1 = 0.398 + 0.501 = 0.899 \Rightarrow \gamma_1 \approx 2.46$ (and $\gamma_2 \approx 1.12$). Driving $x_1 \to 0$ gives the infinite-dilution value $\gamma_1^\infty \approx 7.7$ — the strongly-non-ideal range that makes ethanol/water a minimum-boiling azeotrope, and the very value the tutorial's declared system energies were fitted to reproduce.

**Runnable case.** `tutorials/steady/flash/bubbleT02_uniquac_ethanol_water` — the bubble temperature of 30/70 ethanol/water at 1 atm, landing on the same $T_{\mathrm{bubble}} \approx 354\,\mathrm{K}$ as its NRTL and Wilson twins: a clean three-way model comparison on identical components.

## 2.6 Worked example

**Worked example 2.3.** NRTL $\gamma$ for ethanol/water at 350 K, $x_1 = 0.20$Components 1 =ethanol, 2 =water. Literature parameters (the same set carried in `data/standards/parameters/NRTL/ethanol-water.dat`, from the DECHEMA bank its provenance cites): $a_{12} = -0.8009$, $b_{12} = 246.18\,\mathrm{K}$, $a_{21} = 3.4578$, $b_{21} = -586.0809\,\mathrm{K}$, $\alpha = 0.30$.
*Step 1 — $\tau$ and $G$ at $T = 350\,K$* from (64):

$$\tau_{12} = -0.8009 + \tfrac{246.18}{350} = -0.0975, \quad \tau_{21} = 3.4578 - \tfrac{586.08}{350} = \quad 1.7833,$$

$$G_{12} = e^{-0.3(-0.0975)} = 1.0297, \quad G_{21} = e^{-0.3(1.7833)} = 0.5857.$$

*Step 2 — the binary* $\ln \gamma_1$. For two components (66) reduces to a two-term sum ($\tau_{11} = \tau_{22} = 0$, $G_{11} = G_{22} = 1$). With $x_1 = 0.20$, $x_2 = 0.80$ the denominators are $S_1 = x_1 + G_{21} x_2 = 0.20 + 0.5857(0.80) = 0.6686$ and $S_2 = G_{12} x_1 + x_2 = 1.0297(0.20) + 0.80 = 1.0059$, so

$$\ln \gamma_1 = x_2^2 \left[ \tau_{21} \left( \frac{G_{21}}{S_1} \right)^2 + \frac{\tau_{12} G_{12}}{S_2^2} \right] = 0.64\,[\,1.368 - 0.099\,] = 0.812,$$

giving $\gamma_1 \approx 2.25$.
*Step 3 — read it.* Ethanol is strongly non-ideal at this dilute composition; Raoult's law ($\gamma_1 = 1$) would under-predict the ethanol partial pressure more than twofold. Pushing $x_1 \to 0$ in the same formula gives the infinite-dilution value $\gamma_1^\infty \approx 5.4$ — the steepest, most fit-sensitive point of the curve. At the other end the same parameters place the well-known minimum-boiling azeotrope near $x_1 \approx 0.894$, where $\gamma_1 \approx \gamma_2 P^{\mathrm{sat}}$ balance crosses (`tutorials/batch/still/still02_ethanol_water_azeo`).

**Runnable case.** `tutorials/steady/flash/flash02_ethanol_water` — watch the Wegstein outer loop converge in ∼7 iterations vs ∼29 by direct substitution.

## 2.7 UNIFAC: predicting $\gamma_i$ from molecular structure

NRTL and Wilson are *correlative*: they reproduce a binary's non-ideality only after someone has fitted $(a_{ij}, b_{ij}, \alpha)$ or $(A_{ij}, A_{ji})$ to measured VLE for *that pair*. Confront them with a binary no one has measured and they have nothing to say. UNIFAC (**UNI**QUAC **F**unctional-group **A**ctivity **C**oefficients) is *predictive*: it computes $\gamma_i(T, \mathbf{x})$ from the *functional groups* that make up each molecule, using one published parameter table shared across all of chemistry. No binary fit. The headline glass-box win is to predict the ethanol–water azeotrope from structure alone and then *see* the residual error against data. UNIFAC is the group-contribution realisation of UNIQUAC [9, 4].

**The central idea: a solution of groups.** A molecule is decomposed into a multiset of *subgroups* (e.g. ethanol = 1

`CH3` + 1 `CH2` + 1 `OH`). The mixture is then treated as a solution not of molecules but of those groups. A `CH2` in ethanol is assumed to interact with a `H2O` exactly as a `CH2` in any other molecule would — the *group-additivity hypothesis*. This is what lets one parameter matrix (group $m$ against group $n$) serve every conceivable molecule built from those groups.

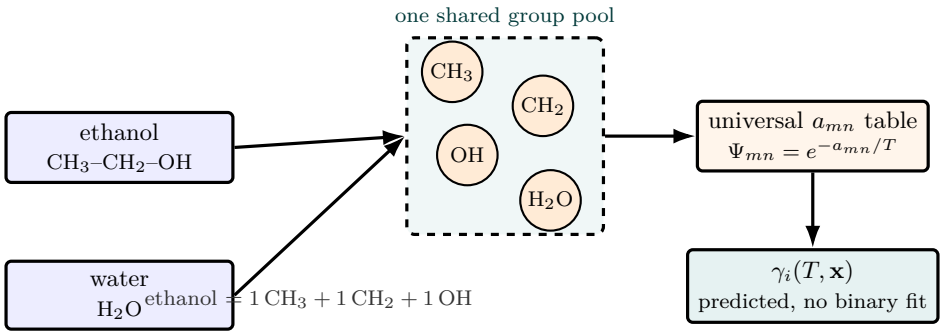



**Figure 13:** UNIFAC's central idea, drawn: a molecule is dissolved into a *multiset of functional groups* (ethanol = $CH_3$+$CH_2$+OH), and the mixture is treated as a solution of those groups rather than of molecules. A $CH_2$ is assumed to interact with $H_2O$ the same way in every molecule it appears in — the group-additivity hypothesis — so a *single universal* $a_{mn}$ table ($\Psi_{mn} = e^{-a_{mn}/T}$) serves all of chemistry. That is what turns UNIQUAC's *fitted* residual term into a *prediction*: feed in the group counts and out comes $\gamma_i(T, \mathbf{x})$ for a binary nobody ever measured. The price is proximity effects and isomers the group count cannot see (see the limits).

**Two-part structure.** As in UNIQUAC, the activity coefficient splits into a size/shape (entropic) part and an energetic part:

$$\ln \gamma_i \;=\; \underbrace{\ln \gamma_i^{C}}_{\text{combinatorial}} \;+\; \underbrace{\ln \gamma_i^{R}}_{\text{residual}} . \tag{78}$$

**Group constants.** Each subgroup $k$ carries a van der Waals volume $R_k$ and surface area $Q_k$ (tabulated, $T$-independent). Summing over the $\nu_k^{(i)}$ groups in molecule $i$ gives its molecular volume and surface parameters

$$r_i = \sum_k \nu_k^{(i)} R_k, \qquad q_i = \sum_k \nu_k^{(i)} Q_k. \tag{79}$$

These are exactly the UNIQUAC $r_i, q_i$, which is why UNIFAC and UNIQUAC share the combinatorial term. In the simulator $R_k, Q_k$ live in `data/standards/parameters/UNIFAC/groups.dat` (Hansen *et al.* [10], open-literature original-UNIFAC values).

**Combinatorial term (Staverman–Guggenheim).** Purely geometric — it depends on $r_i, q_i$ only, never on temperature:

$$\ln \gamma_i^{C} = \ln \frac{\phi_i}{x_i} + \frac{Z}{2}\, q_i \ln \frac{\theta_i}{\phi_i} + l_i - \frac{\phi_i}{x_i} \sum_j x_j l_j, \tag{80}$$

with the volume and surface-area fractions

$$\phi_i = \frac{r_i x_i}{\sum_j r_j x_j}, \qquad \theta_i = \frac{q_i x_i}{\sum_j q_j x_j}, \qquad l_i = \frac{Z}{2}(r_i - q_i) - (r_i - 1), \tag{81}$$

and lattice coordination number $Z = 10$. This term alone already captures the entropic non-ideality of mixing molecules of unequal size (it is what the Flory–Huggins correction does for polymers). In the code the quantity actually stored is $l_i$; $\phi_i/x_i$ and $\theta_i/\phi_i$ are formed directly from the running sums $\sum_j r_j x_j$ and $\sum_j q_j x_j$.

**Residual term: $\Gamma_k$ on two scales.** The energetic part is the difference between how each group "feels" in the mixture versus in the pure component:

$$\ln \gamma_i^{R} \;=\; \sum_k \nu_k^{(i)} \left[ \ln \Gamma_k - \ln \Gamma_k^{(i)} \right], \tag{82}$$

where $\Gamma_k$ is the residual activity coefficient of group $k$ in the actual mixture, and $\Gamma_k^{(i)}$ is the *same* group's residual activity in a reference solution containing only the groups of pure component $i$. The subtraction is what enforces $\gamma_i \to 1$ as $x_i \to 1$: a group surrounded by its own molecule has $\Gamma_k = \Gamma_k^{(i)}$, so the bracket vanishes. Both scales use the identical functional form:

$$\ln \Gamma_k = Q_k \left[ 1 - \ln\Big( \sum_m \Theta_m \Psi_{mk} \Big) - \sum_m \frac{\Theta_m \Psi_{km}}{\sum_n \Theta_n \Psi_{nm}} \right], \tag{83}$$

with the group surface fraction and the group-interaction Boltzmann factor

$$\Theta_m = \frac{Q_m X_m}{\sum_n Q_n X_n}, \qquad \Psi_{mn} = \exp\!\left(-\frac{a_{mn}}{T}\right). \tag{84}$$

$X_m$ is the mole fraction of group $m$ over all groups in the relevant solution. The *main-group* interaction parameter $a_{mn}$ [K] is asymmetric ($a_{mn} \neq a_{nm}$); groups in the same main group have $a_{mn} = 0$ (so $\Psi_{mn} = 1$). These are the only adjustable numbers in the whole model, and they are *universal*, not per-binary — `data/standards/parameters/UNIFAC/interactions.dat` (Hansen *et al.* [10]). The implementation in `src/thermo/activityCoeffic` uses this single-parameter $a_{mn}$ (original UNIFAC); the later temperature-extended form $\Psi_{mn} = \exp[-(a_{mn} + b_{mn}T + c_{mn}T^2)/T]$ of modified UNIFAC (Dortmund) is *not* implemented.

**Subgroups vs. main groups.** A subtle but load-bearing point: $\Psi_{mn}$ is indexed by *main group*, while $R_k, Q_k$ and the group counting use the *subgroup*. Subgroups sharing a main group (e.g. `CH3`, `CH2`, `CH` all in main group `CH2`) have distinct $R_k, Q_k$ but *share* every interaction parameter. That is how UNIFAC keeps the $a$-matrix small while still telling a methyl apart from a methylene by size.

**The algorithm (glass-box).** For each component $i$ the code: (1) builds $r_i, q_i, l_i$ from the declared groups (79); (2) evaluates the combinatorial term (80) from the mixture composition; (3) computes $\ln \Gamma_k$ once for the *mixture* group composition and once for each *pure*-component group composition $X^{(i)}$ via (83); (4) assembles the residual (82); (5) returns $\gamma_i = \exp(\ln \gamma_i^C + \ln \gamma_i^R)$. The pure-component $\Gamma_k^{(i)}$ are temperature dependent (through $\Psi$), so they are recomputed at every call rather than cached. Each component's decomposition lives in its own `.dat`: a `groups { unifac ( { group ...; count ...; } ... ); }` block, from which the engine assembles the model (a case-local component overlay serves for a pedagogical alternative; an inline `groups{}` block in the `activityModel` is refused). A participating component with no group decomposition is an *error* with a remedy, never a silent $\gamma_i = 1$ (no silent crutch).

> **Worked example 2.4.** Decomposing acetone + *n*-hexaneAcetone = 1 `CH3CO` + 1 `CH3`; *n*-hexane = 2 `CH3` + 4 `CH2`. With the tabulated $R_{\mathrm{CH3}} = 0.9011$, $Q_{\mathrm{CH3}} = 0.848$ and $R_{\mathrm{CH2}} = 0.6744$, $Q_{\mathrm{CH2}} = 0.540$, hexane gets $r = 2(0.9011) + 4(0.6744) = 4.500$ and $q = 2(0.848) + 4(0.540) = 3.856$ from (79) alone — no fitting, no data. The non-zero $a_{mn}$ between main groups `CH2CO` and `CH2` then drives the group $\Gamma_k$ apart in the residual term and predicts the strong positive deviations (and the homogeneous azeotrope) of this notoriously non-ideal pair. Compare the prediction against the measured Txy through the validation weapon (Chapter 4) — that overlay, prediction vs. data, is the whole pedagogical point.

**Power.** A genuine prediction — $\gamma_i$ for an *unmeasured* binary, even an azeotrope, from structure and a single universal table. Indispensable when no fitted NRTL/Wilson set exists, and the natural default when screening many candidate solvents at once.

**Limit.** (1) *Only as good as the group table.* A missing $a_{mn}$ pair is silently treated as $a_{mn} = 0$ (ideal between those groups) — the header of `interactions.dat` flags this as a known honesty gap, and a declared group with no $a$-rows yields an unreliable $\gamma$. (2) *Group-additivity breaks for proximity effects* — neighbouring polar groups, intramolecular hydrogen bonds, and isomers the group count cannot distinguish (UNIFAC sees no difference between two positional isomers with the same groups). (3) *Weak near a mixture's critical region and for strongly dilute or asymmetric systems*, and the original-UNIFAC $\Psi = \exp(-a/T)$ gives only a crude temperature trend (the reason modified UNIFAC was developed). (4) *No liquid–liquid split* from this code path; UNIFAC-LLE uses a separate parameter table not bundled here. The standing lesson: a prediction to be *checked against data*, never trusted blind — if you *have* measured VLE for the pair, a fitted NRTL will beat UNIFAC, and UNIFAC's value is precisely when you do not.

# 3 Electrolyte solutions: ionic strength, the osmotic coefficient, and Pitzer

*What you'll learn.* A dissolved salt does not obey Raoult's law and does not have a meaningful vapour pressure, yet it governs the single most important number in reverse osmosis: the *osmotic pressure* $\pi$, the hydraulic head a membrane must overcome before a drop of water will cross. This chapter builds $\pi$ from first principles — the equality of solvent chemical potential across the membrane — arrives at the ideal-dilute *van't Hoff* law $\pi = \nu R_g T c$ (the Raoult of electrolytes), then shows why it over-predicts at seawater concentration and how *Pitzer's osmotic coefficient* $\phi(I)$ corrects it. This is the electrolyte branch of the same "ideal → real" arc that took us from Raoult to NRTL.

## 3.1 Why an electrolyte needs its own model

The activity models of the previous chapter ($\gamma_i$ from $G^E$) describe *neutral* molecules whose non-ideality is short-range: hydrogen bonds, size differences, weak dipoles. An ion is a different animal. When NaCl dissolves it dissociates,

$$\mathrm{NaCl} \longrightarrow \mathrm{Na^+} + \mathrm{Cl^-},$$

into $\nu = 2$ particles, and those particles carry charge. The dominant non-ideality is now the *long-range* Coulomb interaction between ions — an energy that decays as $1/r$, far slower than the $1/r^6$ of a van der Waals force, and which therefore matters even in a dilute solution. No $G^E$ polynomial in mole fraction captures a $\sqrt{\text{concentration}}$ singularity at infinite dilution; the electrolyte needs its own functional form, and that form is what Debye–Hückel (1923) and later Pitzer (1973) [11] supply.

In Choupo this matters in exactly one place so far — the `spiralWoundModule` (NF/RO) — where $\pi$ is the back-pressure that opposes the permeate water flux. A component flagged `nonvolatile true; dissociation` $\nu$`;` in its `.dat` is a salt; it is invisible to the VLE machinery ($K_i = 0$) and visible to the osmotic model.

## 3.2 How Choupo carries the split: one salt, model ions, reference rung

The physics above is also the *data* architecture. The user writes one component, `NaCl`; the solver reasons in the two *model species* it dissociates into, $\mathrm{Na^+}$ and $\mathrm{Cl^-}$. Choupo records exactly that, without duplicating the salt: the component (`components/NaCl.dat`) carries the identity and the ion stoichiometry `dissociatesTo { Na 1; Cl 1; }` — a formula-like fact, like the molecular formula itself — while the *model species* (the records `Na` and `Cl`, one `recordType modelSpecies` file each — `species/Na.dat`, `species/Cl.dat`) carry the charge and the aqueous reference data. We deliberately do not call the ions "true" species: an ion is not more real than the salt — complete dissociation is itself a model, the $K \to \infty$ limit of an association equilibrium, and the moment a real $K$ appears (an ion pair, a hydrate) the split moves from the component's stoichiometry into `chemistry/` as an equilibrium.

The *reference state* matters as much as the model. A neutral activity model is anchored on pure-liquid Raoult ($\gamma_i \to 1$ as $x_i \to 1$); an ion has no pure liquid, so its datum is the *aqueous infinite-dilution* reference on the $\mathrm{H^+(aq)} = 0$ convention. Choupo does not hardcode "aqueous": the reference basis is a *consequence of the declared formulation* — declaring a Pitzer formulation settles the aqueous infinite-dilution rung for the ions and the ion-derived datum for the salt, where declaring NRTL settles pure-liquid Raoult; formulation, standard states and Henry treatment travel together (the one-knob rule). A case *declares* its property package inline in `constant/thermoPhysPropDict` — the formulation, the salt (whose `dissociatesTo` names the ions), and the Pitzer pairs; the Developer Guide (*the thermophysical property architecture*) gives the full picture. Here we keep to the physics: why $\phi(I)$, built next, is the electrolyte's correction to van't Hoff.

## 3.3 Osmotic pressure from the solvent chemical potential

Picture a membrane permeable to water but not to salt, with pure water on the left at pressure $P$ and a salt solution on the right. At equilibrium the *solvent* chemical potential must be equal across the membrane (the salt cannot equilibrate — it cannot cross). The dissolved salt lowers the water's chemical potential through its activity $a_w < 1$; the only way to raise it back to the pure-water value is to raise the pressure on the solution side by an amount $\pi$:

$$\mu_w^*(T, P) \;=\; \mu_w^*(T, P + \pi) \;+\; R_g T \ln a_w, \tag{85}$$

where the star denotes pure water. Expand the pressurised pure-water term with the (nearly pressure-independent) molar volume of pure water $\bar{V}_w$,

$$\mu_w^*(T, P + \pi) \;=\; \mu_w^*(T, P) + \int_P^{P+\pi} \bar{V}_w \, dP' \;\approx\; \mu_w^*(T, P) + \bar{V}_w \, \pi,$$

substitute into (85), and cancel $\mu_w^*(T, P)$:

$$\boxed{\pi \;=\; -\frac{R_g T}{\bar{V}_w} \ln a_w.} \tag{86}$$

This is exact apart from treating $\bar{V}_w$ as pressure-independent — an excellent approximation for liquid water. Everything that follows is a model for the one unknown it contains, the solvent activity $a_w$.

## 3.4 The osmotic coefficient $\phi$

For an electrolyte it is conventional to carry the deviation of $\ln a_w$ from ideality not in an activity coefficient but in the *osmotic coefficient* $\phi$, defined by

$$\ln a_w \;\equiv\; -\,\phi\, M_w \sum_i m_i \;=\; -\,\phi\, M_w\, \nu m, \tag{87}$$

where $m$ is the salt molality (mol per kg solvent), $\nu$ the number of ions it releases, and $M_w$ the solvent molar mass (kg/mol). The sum $\sum_i m_i = \nu m$ runs over all ionic species. By construction $\phi \to 1$ as $m \to 0$: at infinite dilution the solution is ideal.

Substituting (87) into (86) and using $M_w/\bar{V}_w = \rho_w$ (the solvent density) together with the dilute-solution identity $m\,\rho_w \approx c$ (molality times solvent density $\approx$ molarity):

$$\boxed{\;\pi \;=\; \phi\, \nu\, R_{\mathrm{g}} T\, c.\;} \tag{88}$$

This is the form the code evaluates (`OsmoticModel::osmoticPressure` returns $\phi\,\nu\,RTc$, with $R$ in J/(kmol K) and $c$ in kmol/m$^3$). The whole physics of the electrolyte has been distilled into a single dimensionless number $\phi$.

> *Intuition.* $\phi$ is to an electrolyte what $\gamma$ is to a neutral mixture: a multiplicative "reality factor" on an ideal law. $\phi = 1$ is the ideal (van't Hoff) baseline; $\phi < 1$ means the real solution exerts *less* osmotic pressure than $\nu$ free particles would — the ions are not free, they are partly paired and screened, so they push on the membrane less hard than their count suggests.

## 3.5 The van't Hoff baseline: $\phi = 1$

Setting $\phi = 1$ in (88) gives van't Hoff's law,

$$\pi_{\mathrm{vH}} \;=\; \nu\, R_{\mathrm{g}} T\, c, \tag{89}$$

the colligative analogue of the ideal-gas law (van't Hoff, 1887, drew the analogy explicitly: solute particles in a solvent behave, for $\pi$, like gas molecules in a vacuum). It is the simulator's default osmotic model (`model vanHoff;`) and the right first answer: exact in the dilute limit, requiring nothing but the dissociation number $\nu$. Its failure is quantitative, not qualitative. In the seawater regime ($m \lesssim 1$ mol/kg) it fails on the high side — $\phi < 1$, so van't Hoff over-predicts $\pi$, the cautious-but-wasteful side for a membrane designer (an over-estimated $\pi$ under-estimates the achievable flux and recovery). The error is *not* one-signed for all concentrations, though: for NaCl $\phi$ bottoms out near $m \approx 0.7$, climbs back through $\phi = 1$ around $m \approx 2.4$, and *exceeds* 1 in concentrated brine — where van't Hoff *under*-predicts. A high-recovery design, a brine concentrator, or simply the polarised wall can reach that regime, so even the sign of the correction is concentration-dependent.

## 3.6 Ionic strength

The natural concentration variable for ion–ion electrostatics is not molality but the *ionic strength*

$$I \;=\; \tfrac{1}{2} \sum_i m_i\, z_i^2, \tag{90}$$

which weights each ion by the *square* of its charge $z_i$ — the Coulomb energy scales as charge squared, so a divalent ion counts four times a monovalent one. For a 1:1 salt such as NaCl, $z_+ = +1$, $z_- = -1$, $m_+ = m_- = m$, so

$$I \;=\; \tfrac{1}{2}\,(m \cdot 1 + m \cdot 1) \;=\; m \qquad \text{(1:1 salt)}. \tag{91}$$

The single-electrolyte Pitzer model in Choupo assumes this 1:1 case (seawater RO); $I = m$ throughout.

## 3.7 Pitzer's osmotic coefficient for a 1:1 salt

Pitzer's ion-interaction model writes $\phi - 1$ as the sum of a *long-range* electrostatic term (a Debye–Hückel limiting law, softened to stay finite) and *short-range* ion-specific corrections. For a single 1:1 electrolyte it collapses to the compact form Choupo implements:

$$\boxed{\;\phi \;=\; 1 \;\underbrace{-\,A_\phi\, \frac{\sqrt{I}}{1 + b\sqrt{I}}}_{\text{Debye–Hückel}} \;+\, m\, B^{\phi} \;+\; m^2\, C^{\phi},\;} \tag{92}$$

with the second-virial coefficient itself a function of ionic strength,

$$B^{\phi} \;=\; \beta^{(0)} + \beta^{(1)}\, e^{-\alpha\sqrt{I}}. \tag{93}$$

Read term by term:

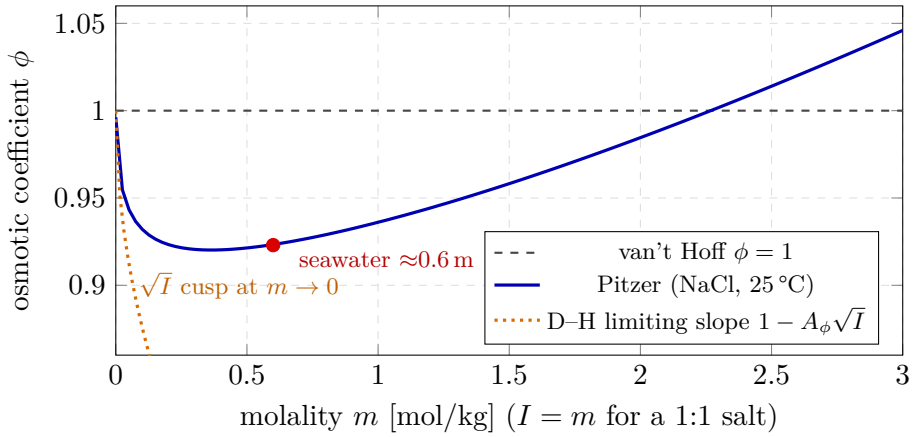



**Figure 14:** Pitzer's osmotic coefficient for NaCl, the electrolyte analogue of $\gamma$ correcting an ideal law. The dashed grey line is the van't Hoff ideal baseline $\phi = 1$. The dotted orange curve is the Debye–Hückel limiting law $\phi \to 1 - A_\phi\sqrt{I}$: note its infinite slope (the $\sqrt{I}$ cusp) at infinite dilution—a singularity no mole-fraction polynomial can reproduce. The full Pitzer curve (blue) follows this electrostatic dip, then the salt-specific pair term pulls it back up, crossing $\phi = 1$ near $m \approx 2.4$. At seawater concentration (red point, $\phi \approx 0.92$) the real osmotic pressure is about 8 % below van't Hoff—directly the back-pressure an RO membrane need not overcome.

- $-A_\phi\sqrt{I}/(1 + b\sqrt{I})$ — the Debye–Hückel term. $A_\phi \approx 0.391\,\mathrm{kg}^{1/2}\,\mathrm{mol}^{-1/2}$ at 25 °C is the (universal, solvent-only) limiting-law slope; $b = 1.2$ is the universal closest-approach parameter. At low $I$ this $\sim -A_\phi\sqrt{I}$ behaviour is the exact electrostatic limit — the famous square-root dependence that no mole-fraction polynomial can reproduce. It *lowers* $\phi$: ion screening reduces the effective particle count.
- $m\,B^\phi$ — the short-range, salt-specific pair interaction. $\beta^{(0)}, \beta^{(1)}$ are tabulated per salt and *raise* $\phi$ back up at higher $m$. The exponential in (93) lets the pair term decay smoothly from its low-$I$ value $\beta^{(0)} + \beta^{(1)}$ toward $\beta^{(0)}$; $\alpha = 2.0$ is the universal decay rate for 1:1 salts.
- $m^2\,C^\phi$ — a small triple-ion correction, important only in concentrated brine.

For NaCl at 25 °C the simulator ships the standard set

$$\beta^{(0)} = 0.0765, \quad \beta^{(1)} = 0.2664, \quad C^\phi = 0.00127, \quad A_\phi = 0.391, \quad b = 1.2, \quad \alpha = 2.0,$$

every one of which is a parameter the `osmotic` sub-block can override (and a future `FitParameters` run could calibrate to a measured $\phi(m)$ curve — exactly the spirit of Chapter 4).

**Worked example 3.1.** NaCl at seawater concentrationA 35 g/L NaCl feed ($M = 58.44\,\mathrm{g/mol}$) is $c \approx 0.60\,\mathrm{kmol/m^3}$, i.e. molality $m \approx 0.60\,\mathrm{mol/kg}$, so $I = 0.60$ and $\sqrt{I} = 0.775$.
*Debye–Hückel term:*

$$-\frac{0.391 \times 0.775}{1 + 1.2 \times 0.775} = -\frac{0.303}{1.930} = -0.157.$$

*Pair term:* $B^\phi = 0.0765 + 0.2664\,e^{-2.0\times0.775} = 0.0765 + 0.2664 \times 0.213 = 0.133$, so $mB^\phi = 0.60 \times 0.133 = 0.080$.
*Triple term:* $m^2C^\phi = 0.36 \times 0.00127 = 0.0005$.

$$\phi = 1 - 0.157 + 0.080 + 0.0005 \approx 0.923.$$

The osmotic pressure is therefore about **7.7 % lower** than van't Hoff predicts:

$$\pi_{\mathrm{vH}} = \nu R_\mathrm{g} T c = 2 \times 8.314 \times 298 \times 599 \approx 29.7 \text{ bar}, \qquad \pi_{\mathrm{Pitzer}} \approx 0.923 \times 29.7 \approx 27.4 \text{ bar}.$$

Against a 55 bar applied pressure that ∼2.3 bar relief in the back-pressure is not cosmetic: it raises the net driving pressure $\Delta P - \Delta\pi$ and therefore the water flux and the recovery.

## 3.8 Why this matters for NF/RO — and where it is worst

The osmotic correction looks modest at the feed (8 % here), but the quantity that actually throttles the flux is the osmotic pressure *at the membrane wall*, not in the bulk. Concentration polarisation (Chapter 1 and the membrane tutorials) piles solute up against the wall, $c_m = c_b\,e^{J_w/k}$, so the wall concentration can be several times the feed. The correction $(1 - \phi)\,\nu R_\mathrm{g} T c$ therefore reaches its largest magnitude — and, at brine concentrations, can even change sign — exactly at the wall, where the membrane is most starved; getting $\phi$ right matters most there. This is

the bread-and-butter regime of Vítor's NF/RO research, and the reason the Pitzer model is the first electrolyte refinement Choupo carries.

**Key idea.** The osmotic pressure is built in two independent layers, mirroring the whole simulator's "ideal baseline + correction" design. The *exact* thermodynamics is $\pi = -(R_{\mathrm{g}}T/\bar{V}_w)\ln a_w$; the *model* supplies $\ln a_w$ through one dimensionless number $\phi$. van't Hoff sets $\phi = 1$ (ideal); Pitzer makes $\phi$ a function of ionic strength, capturing the $\sqrt{I}$ electrostatic limit that no polynomial activity model can. The two are interchangeable `OsmoticModel` sub-models — a one-word change in the case.

**Runnable case.** `tutorials/steady/membranes/membrane06_pitzer` — the seawater RO case of `membrane01` with `osmotic { model Pitzer; }`; $\phi \approx 0.92$ lifts recovery from 15.6 % to 17.2 %. `tutorials/steady/membranes/membrane01_RO_NaCl_seawater` is the van't Hoff baseline for comparison; `membrane03` sweeps $k_{\mathrm{film}}$ to expose the concentration-polarisation curve where the osmotic model matters most.

# 4 Electrolyte-NRTL: the activity coefficient of the *ion*

*What you'll learn.* Pitzer's $\phi(I)$ (§3) gives the deviation of the *solvent* — the osmotic pressure that throttles an RO membrane. But an evaporator, a crystalliser, or a drowning-out precipitator needs the deviation of the *salt*: the mean ionic activity coefficient $\gamma_\pm$, which fixes the boiling-point rise and the supersaturation ratio. This chapter derives the *electrolyte-NRTL* (eNRTL) model of Chen, Song and Evans [12, 13], the natural extension of the molecular NRTL of §2 to a solution of ions. Its premise is a clean *three-layer* split of the excess Gibbs energy — a long-range Pitzer–Debye–Hückel (PDH) term for the ion atmosphere, a short-range local-composition (NRTL) term for the molecular neighbourhood, and (for a mixed solvent) a Born term for transferring an ion between dielectrics — each one a separately readable contribution to $\ln\gamma_\pm$. We write the closed form Choupo evaluates for one fully-dissociated 1:1 salt, give the $T$-dependence, and validate it against Hamer & Wu NaCl data to AAD 1.8 %.

## 4.1 Why a second electrolyte model?

The Pitzer osmotic model of §3 answers exactly one question — *what is the water activity $a_w$?* — and answers it beautifully (NaCl $\phi$ to $< 0.2\,\%$). That is all an RO designer needs, because the flux is set by $\Delta\pi \propto -\ln a_w$. But move to a process that *concentrates* or *precipitates* the salt and the governing quantity flips to the salt's own activity:

- an **evaporator** raises the boiling point by $\propto -\ln a_w$, but the duty and the achievable concentration are bounded by where the salt *saturates*, i.e. by $\gamma_\pm$;
- a **crystalliser** nucleates when the activity product $(\gamma_\pm m)^\nu$ exceeds $K_{sp}$ — the driving force is $\gamma_\pm$, not $m$;
- a **drowning-out** (antisolvent) precipitator works *because* adding ethanol drops the dielectric constant, which *raises* $\gamma_\pm$ and so the supersaturation — an effect with no counterpart in a single-solvent model.

eNRTL supplies $\gamma_\pm$ *and* $a_w$ from one consistent excess-Gibbs surface, and — unlike Pitzer, whose virial coefficients are tied to molality in pure water — it lives on the mole-fraction scale, so it extends naturally to the mixed solvent. In Choupo it is the `activity { model eNRTL; }` sub-model of the same electrolyte slot that carries Pitzer (`src/thermo/electrolyte/ENRTLSingleSalt.H`).

## 4.2 The three-layer excess Gibbs energy

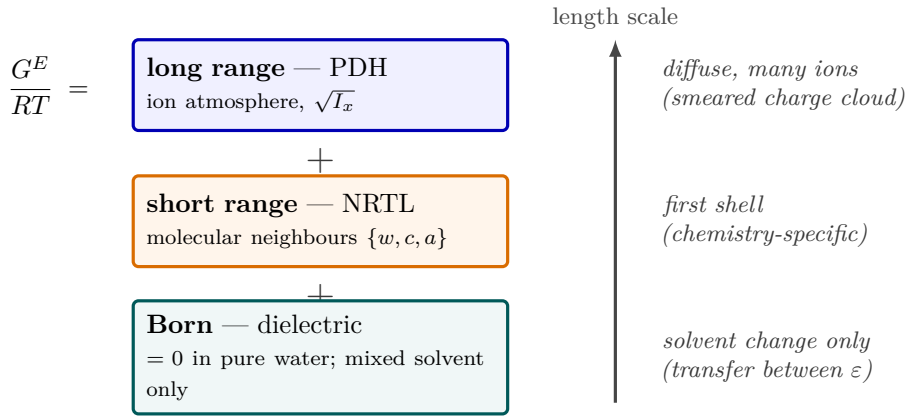



**Figure 15:** Electrolyte-NRTL splits the excess Gibbs energy into three separately-readable layers, each acting on a different length scale—the model's whole idea. The *long-range* Pitzer–Debye–Hückel term (blue) sees only the diffuse ion atmosphere and its $\sqrt{I_x}$ electrostatics; the *short-range* local-composition NRTL term (orange) sees an ion's immediate molecular neighbours $\{w, c, a\}$ and carries the chemistry-specific interactions; the Born term (teal) enters only when the solvent's dielectric constant changes, and is identically zero in pure water. Because they act on different scales they simply *add*, and $\ln\gamma_\pm$ for each species is the corresponding derivative—an evaporator and a crystalliser read the same surface that an RO membrane does.

eNRTL writes the molar excess Gibbs energy as a sum of physically distinct contributions,

$$\frac{G^E}{RT} = \underbrace{\frac{G^{E,\mathrm{PDH}}}{RT}}_{\text{long range}} + \underbrace{\frac{G^{E,\mathrm{lc}}}{RT}}_{\text{short range (NRTL)}} + \underbrace{\frac{G^{E,\mathrm{Born}}}{RT}}_{\text{dielectric transfer}}, \tag{94}$$

and the activity coefficient of every species is the corresponding derivative. The split is the model's whole idea: an ion feels the diffuse *ion atmosphere* (a smeared-out, charge-only effect that only the PDH term knows about) *and* its immediate *molecular neighbours* (a short-range, chemistry-specific effect that only the local-composition term knows about). The two act on different length scales, so they add. The Born term enters only when the solvent itself changes (the mixed-solvent case, §4.6); in pure water it is identically zero.

**Reference state — unsymmetric for ions.** A neutral component's $\gamma \to 1$ as $x \to 1$ (the pure liquid, §2). An ion has no pure liquid, so its reference is the *unsymmetric* (aqueous infinite-dilution) convention: $\gamma_i^* \to 1$ as the salt $\to 0$ in pure water. In the code this is enforced literally — every ionic local term is evaluated at the actual composition *minus* its value at $\mathbf{X}_{\text{ref}} = (x_w{=}1,\, x_c{=}0,\, x_a{=}0)$ — so the subtraction $lc(\mathbf{X}) - lc(\mathbf{X}_{\text{ref}})$ in `ENRTLSingleSalt.H` *is* the unsymmetric normalisation. The solvent (water) keeps the ordinary symmetric reference ($\gamma_w \to 1$ as $x_w \to 1$).

## 4.3 Long range: the Pitzer–Debye–Hückel term

For the long-range part eNRTL borrows Pitzer's mole-fraction-scale Debye–Hückel expression. Its contribution to the *unsymmetric* ionic $\ln\gamma$ is

$$\ln\gamma_i^{\text{PDH}} \;=\; -A_{\text{DH}}\left[\frac{2z_i^2}{\rho}\,\ln\!\big(1+\rho\sqrt{I_x}\big) \;+\; \frac{z_i^2\sqrt{I_x} - 2\,I_x^{3/2}}{1+\rho\sqrt{I_x}}\right], \tag{95}$$

where $I_x = \tfrac{1}{2}\sum_i x_i z_i^2$ is the *mole-fraction* ionic strength (note: $x_i$, not molality $m_i$ — eNRTL lives on the mole-fraction scale), $\rho = 14.9$ is the closest-approach parameter (Pitzer's universal value), and $A_{\text{DH}}$ is the mole-fraction-scale Debye–Hückel slope,

$$A_{\text{DH}} \;=\; \frac{1}{3}\sqrt{\frac{2\pi N_A}{\bar V_w}}\left(\frac{e^2}{4\pi\varepsilon_w\varepsilon_0 k_B T}\right)^{3/2}, \tag{96}$$

which evaluates to $A_{\text{DH}} = 2.9127$ for water at 25 °C ($\bar V_w = 1.807\times10^{-5}\,\text{m}^3/\text{mol}$, $\varepsilon_w = 78.45$). The solvent term gets the symmetric companion $\ln\gamma_w^{\text{PDH}} = 2A_{\text{DH}}\,I_x^{3/2}/(1+\rho\sqrt{I_x})$. This is the same $\sqrt{I}$ electrostatic limit as Pitzer's (92) — the part no polynomial activity model can reproduce — just written for $\ln\gamma$ instead of $\phi$ and on the mole-fraction basis.

## 4.4 Short range: the local-composition (NRTL) term

The short-range term is the NRTL local-composition idea of §2, now applied to a mixture of *three* species $\{w, c, a\}$ (water, cation, anion) with two rules borrowed from Chen's electrolyte theory:

- **like-ion repulsion**: no cation sits next to a cation (Coulomb forbids it), so a like-charged species is *excluded* from an ion's local cell;
- **local electroneutrality**: the immediate neighbourhood of a central molecule is charge-balanced.

With a single salt the energy landscape collapses to just *two* adjustable numbers — the molecule → electrolyte and electrolyte → molecule interaction energies $\tau_{m,ca}$ and $\tau_{ca,m}$ — plus the nonrandomness factor $\alpha$. The Boltzmann factors are the familiar NRTL form $G_{ij} = \exp(-\alpha\,\tau_{ij})$, and all ion–ion same-pair terms vanish ($\tau_{ca} = 0$, $G_{ca} = 1$). The resulting ionic local-composition $\ln\gamma$ is the Chen–Song–Evans three-term expression (as implemented in `ENRTLSingleSalt::lnGammaX`, following IDAES Eqns. 26–27),

$$\ln\gamma_s^{\text{lc}} \;=\; z_s\left[\frac{X_w G_{sw}}{\sum_k X_k G_{kw}}\left(\tau_{sw} - \frac{\sum_k X_k G_{kw}\tau_{kw}}{\sum_k X_k G_{kw}}\right) + \frac{\sum_k' X_k G_{ks}\tau_{ks}}{\sum_k' X_k G_{ks}} + \frac{X_{s'} G_{ss'}}{\sum_k'' X_k G_{ks'}}\left(\tau_{ss'} - \frac{\sum_k'' X_k G_{ks'}\tau_{ks'}}{\sum_k'' X_k G_{ks'}}\right)\right], \tag{97}$$

where $s'$ is the ion of *opposite* charge to $s$; the primed sums $\sum'$ exclude the species like-charged with $s$, and the double-primed sums $\sum''$ exclude the species like-charged with $s'$ (the repulsion rule). Read it as the three neighbourhoods an ion sees: term 1 the solvent cell, term 2 its own charge cell, term 3 the counter-ion cell. The mean ionic value follows from the salt's stoichiometry,

$$\ln\gamma_\pm^{(x)} \;=\; \frac{\nu_c\big(\ln\gamma_c^{\text{PDH}} + \ln\gamma_c^{\text{lc}}\big) + \nu_a\big(\ln\gamma_a^{\text{PDH}} + \ln\gamma_a^{\text{lc}}\big)}{\nu_c + \nu_a}, \qquad \gamma_\pm = \gamma_\pm^{(x)}\Big/\big(1 + 10^{-3} M_w\,\nu\,m\big), \tag{98}$$

the divisor converting the mole-fraction-scale $\gamma_\pm^{(x)}$ to the molality-scale $\gamma_\pm$ that lab data report ($M_w$ the solvent molar mass in g/mol, $\nu = \nu_c + \nu_a$).

## 4.5 Temperature dependence

Three places in the model carry $T$, and Choupo handles each by anchoring to the validated 25 °C kernel so that every existing golden master stays byte-identical at the reference and moves physically away from it:

1. **NRTL energies.** Like the molecular NRTL's $\tau = a + b/T$, the electrolyte $\tau$ is taken in the $\Delta g/RT$ form with a $T$-independent interaction energy,

$$\tau_{ij}(T) = \tau_{ij}^{\mathrm{ref}} \frac{T_{\mathrm{ref}}}{T}, \qquad T_{\mathrm{ref}} = 298.15\,\mathrm{K}, \tag{99}$$

   i.e. $a = 0$, $b = \tau^{\mathrm{ref}} T_{\mathrm{ref}}$ — the single published 25 °C value sets $b$ and the form reduces to $\tau^{\mathrm{ref}}$ at the reference.
2. **Debye–Hückel slope.** $A_{\mathrm{DH}}(T) = A_{\mathrm{DH}}(25)\, f(T)$ with the Born–Bjerrum factor

$$f(T) = \left(\frac{\rho_w(T)}{\rho_w(25)}\right)^{1/2} \left(\frac{\varepsilon_w(25)\, T_{25}}{\varepsilon_w(T)\, T}\right)^{3/2}, \tag{100}$$

   evaluated through the Malmberg–Maryott $\varepsilon_w(T)$ and Kell $\rho_w(T)$ correlations (`src/thermo/electrolyte/SolventProperties.H`); $f(25) = 1$.
3. **Born and PDH** are evaluated at the actual $T$, never a hard-coded 298.15.

> **Common pitfall.** The anchored $\tau(T) \propto 1/T$ is a *calorimetrically uncalibrated* guess: it reproduces $\gamma_\pm$ and $\phi$ at and near 25 °C, but its temperature *slope* (hence any heat of dilution derived from it) has not been refitted against d$H$ data. This is the same caveat that gates the Pitzer $L_\phi$ enthalpy branch; the eNRTL package deliberately does *not* inherit the `calorimetricFit` flag, so it stays in the honest, uncalibrated regime. Use it for $\gamma_\pm$, $a_w$, boiling-point rise and solubility; do not read a heat of dilution off its $T$-slope without a refit.

## 4.6 The mixed solvent: drowning-out and the Born term

Add an antisolvent (ethanol) at salt-free mole fraction $x_{\mathrm{anti}}$ and two things happen, both of which eNRTL captures from (94):

1. **Dielectric mixing.** The mixture permittivity $\varepsilon_{\mathrm{mix}}$ (a mass-fraction average of the two pure values), molar volume $\bar{V}_{\mathrm{mix}}$ and molar mass $M_{\mathrm{mix}}$ replace the pure-water values in (96). Because $\varepsilon_{\mathrm{mix}} < \varepsilon_w$, the slope $A_{\mathrm{DH}}$ *grows* (electrostatics is less screened in a less polar medium) — the PDH term strengthens.
2. **Born transfer.** Moving an ion from pure water ($\varepsilon_w$) into the lower-$\varepsilon$ mixture costs the Born self-energy difference,

$$\ln \gamma_i^{\mathrm{Born}} \;=\; \frac{e^2}{8\pi\varepsilon_0 k_B T} \left(\frac{1}{\varepsilon_{\mathrm{mix}}} - \frac{1}{\varepsilon_w}\right) \frac{1}{r_i}, \tag{101}$$

   with $r_i$ the Born radius (calibrated to KCl in water+ethanol). It is positive — the ion is *less* comfortable in the mixture — so $\gamma_\pm$ rises.

Both effects push $\gamma_\pm$ up, which lifts the activity product $(\gamma_\pm m)^\nu$ above $K_{sp}$ and *precipitates* the salt without removing any water — the thermodynamic engine of antisolvent crystallisation. In Choupo this is the `gammaPMMixed` entry point; $x_{\mathrm{anti}} \to 0$ recovers the aqueous $\gamma_\pm$ exactly, and $T = 298.15$ recovers the validated 25 °C kernel.

> **Worked example 4.1.** NaCl mean ionic activity coefficient at 25 °CTake $m = 1.0$ mol/kg NaCl ($\nu_c = \nu_a = 1$, $z_c = z_a = 1$, $M_w = 18.015$). The solvent amount is $n_w = 1000/18.015 = 55.51$ mol/kg, so $n_{\mathrm{tot}} = 55.51 + 2(1.0) = 57.51$ and the mole fractions are $X_w = 0.9652$, $X_c = X_a = 0.01739$, giving a mole-fraction ionic strength $I_x = \frac{1}{2}(0.01739 + 0.01739) = 0.01739$, $\sqrt{I_x} = 0.1319$.
> *PDH term* (Eq. 95, $z = 1$, $A_{\mathrm{DH}} = 2.9127$, $\rho = 14.9$):
>
> $$\ln \gamma_c^{\mathrm{PDH}} = -2.9127 \left[\frac{2}{14.9} \ln(1 + 14.9 \times 0.1319) + \frac{0.1319 - 2(0.01739)^{3/2}}{1 + 14.9 \times 0.1319}\right] \approx -0.550.$$
>
> *Local term* (Eq. 97, with $\tau_{m,ca} = 8.885$, $\tau_{ca,m} = -4.549$, $\alpha = 0.2$, evaluated unsymmetrically) adds a positive short-range contribution; summed and mole-fraction-to-molality converted via (98) the kernel returns
>
> $$\gamma_\pm \;\approx\; 0.648,$$
>
> against the Hamer & Wu tabulated value 0.657 — a 1.3 % error, typical of the model across the range.

## 4.7 Validation and the power-and-limit

The kernel was checked term-by-term against an independent Python prototype (`bin/curate/enrtl_ref/enrtl_validated.py`, agreeing to $5 \times 10^{-5}$) and against the classic NaCl reference data of Hamer & Wu [14] (and Robinson & Stokes) over 0.1–6 mol/kg at 25 °C:

| $m$ (mol/kg) | eNRTL $\gamma_\pm$ | lit. $\gamma_\pm$ | error |
|---|---|---|---|
| 0.1 | 0.773 | 0.778 | −0.6 % |
| 0.5 | 0.670 | 0.681 | −1.7 % |
| 1.0 | 0.648 | 0.657 | −1.3 % |
| 2.0 | 0.674 | 0.668 | +0.9 % |
| 3.0 | 0.732 | 0.714 | +2.5 % |
| 6.0 | 0.950 | 0.986 | −3.6 % |
| AAD $\gamma_\pm = 1.8$ %; | | osmotic $\phi$ AAD = 1.4 % | |

The eNRTL captures the entire shape: the dip near $m \approx 1$ (PDH screening dominates) and the strong rise past $m \approx 3$ (the local term takes over) — the same non-monotone curve van't Hoff cannot see.

**Key idea.** **Power.** One mole-fraction-scale excess-Gibbs surface delivers both $\gamma_\pm$ *and* $a_w$ from two energy parameters per salt, extends without re-parameterisation to mixed solvents (the Born term is parameter-light), and reproduces the non-monotone $\gamma_\pm(m)$ to ∼2 % — the engine behind boiling-point rise, solubility, and antisolvent crystallisation.
**Limit.** Choupo's kernel is a *single fully-dissociated salt*; it does not by itself do multi-salt common-ion mixtures, ion pairing, or weak-electrolyte speciation (those live on the multi-ion Pitzer speciation path of §6). Its $T$-slope is calorimetrically uncalibrated (`calorimetricFit` stays off), and for high-concentration *aqueous* accuracy a per-salt-fitted Pitzer (∼0.2 %) beats it (∼1.8 %) — eNRTL's edge is the mixed solvent and the unified $\gamma_\pm/a_w$ surface, not single-salt aqueous precision.

**Runnable case.** `tutorials/steady/evaporation/evaporator07_nacl_enrtl` — a NaCl evaporator with `activityModel { model eNRTL; }`, where $\gamma_\pm$ sets the boiling-point rise and the saturation limit. `tutorials/steady/crystallisation/crystalliser06_nacl_antisolvent` exercises the mixed-solvent Born term (ethanol drowning-out). Contrast with the Pitzer membrane cases of §3 (`membrane06_pitzer`), which solve the *solvent* side of the same physics.

## 4.8 Multi-salt eNRTL: the symmetric multicomponent formulation

When two salts share the water — NaCl + KCl in a crystalliser purge, a mixed brine — each ion's environment contains *foreign* ions, and the single-salt expressions no longer apply. The multicomponent formulation (Song & Chen 2009 [15], DOI `10.1021/ie9004578`) keeps the two eNRTL hypotheses (local electroneutrality, like-ion repulsion) and writes the local term species-wise, with every effective ion–molecule and ion–ion binary assembled from the *pair* parameters by charge-fraction mixing rules:

$$Y_c = \frac{X_c}{\sum_{c'} X_{c'}}, \qquad G_{am} = \sum_c Y_c\, G_{ca,m}, \qquad G_{ca} = \sum_{c'} Y_{c'}\, G_{ca,c'a}, \qquad \tau = -\ln G/\alpha, \tag{102}$$

so a chloride ion in a Na/K mixture sees a *composition-weighted* water–salt energy. The long-range term is the same PDH expression with the multicomponent ionic strength. Two structural results, both golden-locked in the tutorial:

1. **Exact single-salt reduction.** With one salt the charge fractions collapse to unity and the model IS the single-salt kernel of the previous sections (the tutorial pins the pure endpoints at $\sim 10^{-15}$ relative).
2. **Harned's rule emerges.** At constant total molality, $\ln\gamma_\pm$ of each salt is nearly *linear* in the salt fraction ($R^2 > 0.999$ at 4 mol/kg with zero electrolyte–electrolyte parameters) — the century-old empirical rule falls out of the local-composition structure.

*Intuition.* **A declared inconsistency — and its fix, side by side.** The published multicomponent activity-coefficient expressions differentiate the excess Gibbs energy *treating the mixing-rule binaries as constants*, yet those binaries depend on composition through the charge fractions $Y$. The result is not exactly Gibbs–Duhem consistent — the motivation for the *refined* eNRTL of Bollas, Chen & Barton (2008) [16]. Choupo ships **both formulations** (`formulation published;` vs `consistent;`): the consistent one keeps the chain rule through the $Y$-dependent mixing rules, so $\gamma$ *is* the exact derivative of $G^{ex}$ and the finite-difference oracle closes ($\sim 10^{-9}$) at *every* composition — while the published one prints its mid-sweep deviation (∼0.5) next to the pure-salt state where it, too, is exact. At 4 mol/kg NaCl–KCl the two differ by ∼4 % in mid-sweep $\gamma_\pm$. Robinson's 1961 isopiestic data judge them: the published form wins the dominant Harned coefficient ($\alpha_{\mathrm{NaCl}}$ +14 % vs −29 %), both overstate the small $\alpha_{\mathrm{KCl}}$, and no single $\tau_{EE}$ closes both — the residual is structural. Bollas et al.'s own Figure 1 (this very system) cross-checks both branches: their original-eNRTL panel matches our published curves to the pixel, and their refined panel shows the same signature our consistent form shows — the $\tau$ fan nearly collapses, because an exact derivative is almost insensitive to $\tau_{EE}$.

**Runnable case.** `tutorials/props/electrolyte/enrtl_multisalt_nacl_kcl` — the paper's own demonstration (NaCl–KCl–$H_2O$, 4 mol/kg Harned sweep), reproducing its Figure 1 $\tau_{ca,c'a}$ sensitivity fan (in $\log_{10}$ — the figure's axis label says ln but plots $\log_{10}$) and locking the reduction pin, the Harned slopes, and the

declared Gibbs–Duhem deviation.

# 5 Cohesive energy density and the solubility parameter

Before activity coefficients could be regressed from data, engineers needed some way to guess whether two liquids would mix. The answer that survived is one number per substance, and it survived because it is derived from quantities everybody already measures.

## 5.1 The definition

Vaporising a mole of liquid separates its molecules completely. The energy that costs, less the work $RT$ the vapour does pushing the atmosphere aside, is the *energy of vaporisation*; divided by the liquid's molar volume it is the *cohesive energy density*, and Hildebrand's parameter is its square root:

$$\delta(T) \;=\; \sqrt{\frac{\Delta H_{\mathrm{vap}}(T) - RT}{V_m}} \tag{103}$$

Its unit, $\mathrm{Pa}^{1/2}$, is not an accident of algebra: energy per volume is a pressure, so $\delta$ is the square root of an internal pressure. Compilations print $\mathrm{MPa}^{1/2}$, a factor of $10^3$ smaller.

The rule of thumb follows from regular-solution theory, in which the excess enthalpy of mixing two non-polar liquids is

$$\Delta H^{E} \;=\; V_m\,\phi_1\phi_2\,(\delta_1 - \delta_2)^2 \tag{104}$$

with $\phi_i$ the volume fractions. The difference enters *squared* and is never negative: on this model mixing is always endothermic, and the closer the two $\delta$ are, the smaller the penalty. "Like dissolves like" is Eq. (104) read backwards.

## 5.2 What Eq. (103) cannot do

The assumption behind Eq. (104) is that the only forces are dispersive — that a molecule's interaction with a neighbour does not depend on which neighbour. Where that fails, the theory fails with it, and it fails in a specific direction: $\Delta H^{E} \geq 0$ always, so a *negative* enthalpy of mixing cannot be represented at all. Hydrogen bonding between unlike molecules is exactly that case.

The consequence is not subtle. Water's $\delta$ is about 48 $\mathrm{MPa}^{1/2}$ and ethanol's about 26; the difference of 22 says they should be strongly immiscible, and they are miscible in all proportions. The theory has compressed dispersion, polarity and hydrogen bonding into a single scalar, and water is where that compression costs everything.

This is why Hansen's later three-parameter split is what formulation practice uses, and it is the subject of the next subsection.

## 5.3 The Hansen split, and the Teas triangle

Hansen's move is to stop pretending the cohesive energy is one thing. It is the sum of three contributions — London dispersion, permanent-dipole interaction, and hydrogen bonding — and because they add as energies, the parameters add in quadrature:

$$\boxed{\delta^2 \;=\; \delta_d^2 \;+\; \delta_p^2 \;+\; \delta_h^2} \tag{105}$$

The single Hildebrand $\delta$ is the *length* of that vector. Water and ethanol are far apart in length; what the length cannot show is that they point in nearly the same direction, both dominated by $\delta_h$, which is why they mix.

**The Teas triangle.** Three numbers per substance want a three-dimensional plot, which is awkward on paper and worse in a lecture. Teas's device [17] is to plot the *fractions*

$$f_d = \frac{\delta_d}{\delta_d + \delta_p + \delta_h}, \qquad f_p = \frac{\delta_p}{\delta_d + \delta_p + \delta_h}, \qquad f_h = \frac{\delta_h}{\delta_d + \delta_p + \delta_h}, \tag{106}$$

which sum to one by construction and therefore place every solvent as a single point in a ternary diagram whose corners are pure dispersion, pure polar and pure hydrogen bonding. A polymer's solubility region becomes an *area* on that triangle, and choosing a solvent becomes reading whether a point falls inside it — which is what the picture is for.

**What the triangle throws away**

Equation (106) normalises the magnitude out. Two solvents with the same *balance* of forces and very different total cohesive strength plot at the *same point*. The triangle shows the character of the interaction, never its strength, and it is not a miscibility map: reading it as one is the second mistake of this chapter, after reading the Hildebrand ladder as one. Choupo prints $|\delta|_{\text{trio}}$ in the column beside the fractions for exactly this reason.

**Note that the denominators differ.** Equation (105) adds in quadrature; Eq. (106) divides by the plain sum. That is not an inconsistency — the Teas fractions are a projection chosen for legibility, not a physical decomposition, and Teas said so. It is worth labouring because normalising by $|\delta|$ instead produces three numbers that still look like Teas coordinates, still order the solvents plausibly, and are wrong; they do not sum to one.

**What Choupo does and does not have.** The engine reads the split from a `hansen { dD; dP; dH; Tref; }` block on the component record, in $\mathrm{Pa}^{1/2}$ with the temperature the placement was made at, and prints the Teas fractions from it. *No component in the shipped catalogue carries one.* Hansen parameters are a separate curated dataset this project does not hold in its catalogue, and none is invented — a triple typed from memory would produce a Teas point that looks entirely reasonable and is fiction. What the tree does carry, since the solvent-selection lesson of §5.4, is a handful of *case-local* records transcribed from Hansen's 1967 thesis [18], one citation per value and marked interim. A run with no triple says so, naming the block it would read, so that the absence is legible as missing *data* and not as a missing capability.

What the engine does get for free is an audit. Because Choupo derives $\delta$ independently, from the latent heat and the molar volume, any triple that arrives can be tested against Eq. (105): a value mistyped, entered in $\mathrm{MPa}^{1/2}$ where $\mathrm{Pa}^{1/2}$ is meant, or transcribed off by a row cannot reproduce a number computed from an enthalpy of vaporisation. The limit of that audit is worth stating in the same breath: it constrains the *length* of the vector and says nothing about where it points, so a triple wrong in direction but right in magnitude passes it untouched.

## 5.4 Choosing a solvent, or a solvent mixture: $R_a$ and RED

*Declared anchor:* `ch:hansen-mixture`.

The question a formulator actually asks is not "what is this solvent's triple" but "which solvent, or which blend, for *this* substance". Hansen's answer [18] is geometric. A solute — a polymer, a resin, a dye — is placed in the same three-dimensional space as the solvents, at the centre $(\delta_{d0}, \delta_{p0}, \delta_{h0})$ of the cloud of solvents observed to dissolve it, and the cloud's radius $R_0$ is recorded with it. The thesis found those clouds to be spheres only when the dispersion axis is drawn at twice the scale of the other two, which is where the factor 4 in the distance comes from (its Eq. 8.2):

$$R_a^2 \;=\; 4\,(\delta_d - \delta_{d0})^2 \;+\; (\delta_p - \delta_{p0})^2 \;+\; (\delta_h - \delta_{h0})^2, \qquad \mathrm{RED} \;=\; \frac{R_a}{R_0}. \tag{107}$$

A solvent with RED $< 1$ lies inside the sphere; one with RED $> 1$ outside. The ratio is defined only where a record declares $R_0$: Choupo publishes $R_a$ alone otherwise, because a cut-off the record did not measure would be an invented one.

**The mixture rule.** A blend of solvents is placed by the volume-fraction average of each parameter,

$$\delta_{k,\mathrm{mix}} \;=\; \sum_i \phi_i\, \delta_{k,i}, \qquad k = d,\, p,\, h, \tag{108}$$

which assumes that each solvent keeps its own split of cohesive energy when mixed and that the mixture's volume is the sum of the parts. It is a rule of thumb, not a theorem — but it is the rule Hansen checked his placements against: two *non*-solvents lying on opposite sides of a solute's sphere give a blend that lies inside it, and "this has always been found to be the case, with the exception of a very few boundary region cases" (thesis, §2.5). A composition declared by mass or by mole is converted to volume fractions through each solvent's molar volume (the record's $V_{\mathrm{liq}}$ at 25 °C, which carries that approximation too), and Choupo prints the conversion beside every point.

**The temperature is not decorative.** Every `hansen {}` block carries `Tref`, the temperature its triple was placed at, and the engine applies *no* correction to move it elsewhere: the parameters are held constant at `Tref` as an approximation, and the run says so. The declared $T$ of a `hansenMixture` operation decides one thing only — the phase check below.

**The phase check: three outcomes.** The mixture rule describes *one* liquid. At every blend composition Choupo asks the case's own activity model whether the solvent pair is stable as one liquid, through Michelsen's tangent-plane test (§11; the same test the flash's `phaseSet auto` runs), and publishes a word per point:

- **homogeneous** — one liquid; the mixture rule applies, with its assumptions stated;
- **twoPhase** — a second liquid. Choupo *refuses* to represent the pair as one homogeneous solvent there: no averaged triple, no $R_a$, no RED is published for the point. A two-phase pair is not useless — it may serve an extraction — it is only not one solvent;
- **notVerified** — nothing in the case can decide. An `ideal` liquid ($\gamma = 1$) has no excess Gibbs energy and can never demix, so its "stable" would be said of every pair; a pair running on the authorised ideal default is in the same position. The rule is applied and the point is marked.

**Affinity is not solubility**

$R_a$ is a distance in a space fitted to dissolves / does-not-dissolve observations. A small $R_a$, or a RED below one, says the solvent lies in the region where the substance was *observed* to dissolve or swell — a compatibility indicator in this approach. No saturation concentration in g/L follows from it, and Choupo publishes none. The thesis also states the method's own residual: about 2.5 % of the solvents beyond $R_0 + 0.5$ dissolve, and about 2.5 % within $R_0 - 0.5$ do not (in its original $(\mathrm{cal/cm^3})^{1/2}$ units).

**Where the numbers come from.** Hansen's parameters are decades-old published placements, and Choupo takes them only from a primary source with the citation on each value. The witnesses of this section carry case-local records transcribed from the thesis's Tables 2.1 and 4.1 — thirteen solvents and polystyrene (its solute G) — in `tutorials/props/solvents/`: a synthetic pair whose minimum sits at 50:50 by construction (`solvent01`), hexane blended with acetone and with nitromethane against polystyrene (`solvent02`; UNIFAC finds the nitromethane pair two liquids exactly where the mixture rule would have liked it best), and a screen of the thirteen pure solvents beside the thesis's own dissolution ratings of Table 2.3 (`solvent03`). The EduTool *Choosing a solvent, or a solvent mixture* draws them; the `hansenMixture` operation computes them (its grammar is in the Properties Guide's operations reference).

## 5.5 How Choupo evaluates it

*Declared anchor:* `ch:solubility-parameter`.

$\delta$ is *derived*, never stored. Every input in Eq. (103) is already in the component record — $\Delta H_{\mathrm{vap}}$ at the normal boiling point, $T_c$, and $V_m$ — so a stored $\delta$ would be a second home for a fact the record already fixes. The latent heat at $T$ comes from the Watson correlation $\Delta H_{\mathrm{vap}}(T) = \Delta H_{\mathrm{vap}}(T_b)\,[(T_c - T)/(T_c - T_b)]^{0.38}$, the same one the enthalpy legs use.

Where a record carries a published $\delta$, it is stored as an *anchor*: never an input, only a check. The three ways Eq. (103) is got wrong — dropping $RT$, a wrong unit on $V_m$, and reporting $\mathrm{Pa}^{1/2}$ where the literature uses $\mathrm{MPa}^{1/2}$ — all produce a number that still reads as a solubility parameter, and only an independent determination beside it turns any of them into a visible disagreement.

Note finally that $\delta \to 0$ as $T \to T_c$: at the critical point there is no longer a liquid to pull apart. A table of solubility parameters without a temperature beside it is a table of numbers from unrelated states, which is why the evaluation temperature is announced on every run.

# 6 Multi-ion Pitzer: the Harvie–Møller–Weare model

*What you'll learn.* The single-salt osmotic coefficient of the previous section answers "how much does *this one* dissolved salt lower the water activity?" Real brines, seawater, scaling waters and the inside of an evaporator carry *many* ions at once — $Na^+$, $K^+$, $Ca^{2+}$, $Mg^{2+}$, $Cl^-$, $SO_4^{2-}$, $OH^-$, the carbonates — and an ion's activity then depends on *every* ion present, not just its own counter-ion. This section builds the Harvie–Møller–Weare (HMW) assembly of Pitzer's model, which gives a *per-ion* activity coefficient $\gamma_i$ from the same long-range Debye–Hückel term plus binary virials *plus* ternary mixing terms. We show exactly the sums Choupo evaluates (`PitzerHMW::evaluate`), the higher-order electrostatic term $E_\theta(I)$ that no other Pitzer code makes optional, and the self-test that proves the multi-ion model collapses to the single-salt kernel of §3.

## 6.1 From one salt to many ions

The compact form (92) is what you get when the general Pitzer expansion is specialised to a *single* fully dissociated salt. The general expansion does not work with salts at all — it works with *ions*. Its natural output is not a mean activity coefficient $\gamma_\pm$ but an individual $\gamma_i$ for each ionic species $i$, on the molality scale, referenced to infinite dilution in pure water. Two reasons force this move:

- **A speciation solver needs $\gamma_i$, not $\gamma_\pm$.** When the same water carries dissolved $CO_2$ that speciates into $HCO_3^-$ and $CO_3^{2-}$, or when you solve for pH, the chemical-equilibrium solver (`SpeciationSolver`) writes a mass-action law for each reaction and needs the activity of *each* ion individually. A lumped $\gamma_\pm$ cannot be unbundled.
- **Cross interactions are real.** In a mixed $Na^+/Mg^{2+}/Cl^-/SO_4^{2-}$ brine the activity of $Na^+$ is changed by the presence of $Mg^{2+}$ (a like-sign neighbour) and of $SO_4^{2-}$ (a second anion it never pairs with as a pure salt). These are the *mixing terms*; they are zero in any single-salt solution and are precisely what HMW adds.

Choupo implements the HMW assembly as the `pitzer AqueousActivity` model (`PitzerHMW`), selected by the same one word a single-salt case uses — `activityModel pitzer;` — but now reached through the speciation path, where it replaces the cheap Davies fallback with a full per-ion $\gamma_i$. All the arithmetic comes from the *same* `PitzerForm` header the single-salt kernel uses, so there is one Pitzer surface in the code, never two that can drift apart.

## 6.2 The per-ion working equations

For a cation $M$ (charge $z_M$) and an anion $X$ (charge $z_X$, here $z_X<0$), in a solution of cations $\{c\}$, anions $\{a\}$ and neutrals $\{n\}$ at molalities $m_i$, ionic strength $I=\tfrac12\sum_i z_i^2 m_i$, the HMW equations Choupo evaluates are (Pitzer 1991, ch. 3; [19])

$$\begin{aligned}\ln\gamma_M \;=\; & z_M^2\,F \;+\; \sum_a m_a\big(2B_{Ma}+Z\,C_{Ma}\big) \;+\; |z_M|\sum_c\sum_a m_c m_a\,C_{ca} \\ & +\sum_c m_c\Big(2\Phi_{Mc}+\sum_a m_a\,\psi_{Mca}\Big) \;+\; \sum_{a<a'} m_a m_{a'}\,\psi_{Maa'},\end{aligned} \tag{109}$$

$$\begin{aligned}\ln\gamma_X \;=\; & z_X^2\,F \;+\; \sum_c m_c\big(2B_{cX}+Z\,C_{cX}\big) \;+\; |z_X|\sum_c\sum_a m_c m_a\,C_{ca} \\ & +\sum_a m_a\Big(2\Phi_{Xa}+\sum_c m_c\,\psi_{Xac}\Big) \;+\; \sum_{c<c'} m_c m_{c'}\,\psi_{Xcc'},\end{aligned} \tag{110}$$

with the long-range function $F$ collecting the Debye–Hückel slope and the ionic-strength *derivatives* of the binary and mixing terms,

$$F \;=\; \underbrace{-A_\phi\left[\frac{\sqrt{I}}{1+b\sqrt{I}}+\frac{2}{b}\ln\!\big(1+b\sqrt{I}\big)\right]}_{\text{Debye–Hückel}} \;+\; \sum_c\sum_a m_c m_a\,B'_{ca} \;+\; \sum_{c<c'} m_c m_{c'}\Phi'_{cc'} \;+\; \sum_{a<a'} m_a m_{a'}\Phi'_{aa'}, \tag{111}$$

and the charge-weighted total molality

$$Z \;=\; \sum_i |z_i|\,m_i. \tag{112}$$

Reading these against the code (`PitzerHMW::evaluate`), every symbol is one of a handful of `PitzerForm` calls:

- $B_{ca}=\beta^{(0)}_{ca}+\beta^{(1)}_{ca}\,g(\alpha_1\sqrt{I})+\beta^{(2)}_{ca}\,g(\alpha_2\sqrt{I})$, with $g(x)=2\big[1-(1+x)e^{-x}\big]/x^2$ — the binary pair term, per (cation, anion) read from the shared catalogue `data/standards/parameters/Pitzer/pairs/`. The $\beta^{(2)}$ slot is the 2:2 ion-pairing term ($\alpha_1=1.4$, $\alpha_2=12$); it is zero for 1:1/2:1/1:2 salts.

- $B'_{ca} = \left[\beta^{(1)}_{ca}\, g'(\alpha_1\sqrt{I}) + \beta^{(2)}_{ca}\, g'(\alpha_2\sqrt{I})\right]/I$, with $g'(x) = -2\left[1 - (1 + x + \tfrac{1}{2}x^2)e^{-x}\right]/x^2$ — the $I$-derivative that lives in $F$.
- $C_{ca} = C^{\phi}_{ca}/\left(2\sqrt{|z_c z_a|}\right)$ — the triplet (concentrated-brine) coefficient. The $|z_i|\sum_c\sum_a m_c m_a C_{ca}$ term is common to every ion (only $|z_i|$ differs); the code computes the double sum once and reuses it.
- $\Phi_{ij} = \theta_{ij} + E_{\theta\,ij}(I)$ and $\Phi'_{ij} = E'_{\theta\,ij}(I)$ — the *like-sign* mixing parameter (§6.3). $\theta_{ij}$ and the triplet $\psi_{ijk}$ are read from `data/standards/parameters/Pitzer/theta/ and psi/`; an absent entry is 0.

**Neutral solutes.** A neutral species ($z = 0$, e.g. dissolved $CO_2$) is excluded from every charged sum and from $I$; it interacts only through the salting-out terms

$$\ln\gamma_n \;=\; 2\sum_{\text{ion}} m_{\text{ion}}\,\lambda_{n,\text{ion}} \;+\; \sum_c\sum_a m_c m_a\,\zeta_{n,c,a}, \tag{113}$$

with the symmetric back-reactions $+2m_n\lambda_{n,i}$ added to each ion's $\ln\gamma_i$ and $+m_n m_a\zeta_{n,c,a}$ to the cation/anion legs. A neutral with no tabulated $\lambda/\zeta$ gets $\gamma_n = 1$ exactly — there is no Debye–Hückel term for an uncharged species.

## 6.3 The higher-order electrostatic term $E_\theta(I)$

The like-sign mixing parameter $\Phi_{ij}$ is *not* simply the fitted constant $\theta_{ij}$. Pitzer (1975) [20] showed that two ions of the same sign but *different charge* (e.g. $Na^+$ and $Mg^{2+}$, or $Cl^-$ and $SO_4^{2-}$) feel an additional, purely electrostatic, ionic-strength-dependent interaction — the *unsymmetrical higher-order* term $E_\theta(I)$ — that exists even when $\theta_{ij} = 0$. It is built from a single integral $J(x)$:

$$E_{\theta\,ij}(I) \;=\; \frac{z_i z_j}{4I}\Big[\,J(x_{ij}) - \tfrac{1}{2}J(x_{ii}) - \tfrac{1}{2}J(x_{jj})\,\Big], \qquad x_{ij} = 6\,z_i z_j A_\phi\sqrt{I}, \tag{114}$$

$$E'_{\theta\,ij}(I) \;=\; -\frac{E_{\theta\,ij}}{I} + \frac{z_i z_j}{8I^2}\Big[\,x_{ij}J'(x_{ij}) - \tfrac{1}{2}x_{ii}J'(x_{ii}) - \tfrac{1}{2}x_{jj}J'(x_{jj})\,\Big]. \tag{115}$$

The single integral $J(x)$ Choupo evaluates is Pitzer's published closed-form approximation (the same one used by the USGS PHRQPITZ / Plummer et al. 1988),

$$J(x) \;=\; \frac{x}{4 + C_1\,x^{-C_2}\exp\!\left(C_3\,x^{C_4}\right)}, \qquad \begin{aligned} &C_1 = 4.581,\;\; C_2 = 0.7237,\\ &C_3 = 0.0120,\;\; C_4 = 0.528, \end{aligned} \tag{116}$$

with $J'(x)$ taken *analytically* (not by finite difference) from (116). Two limits make $E_\theta$ harmless where it should be: as $x \to 0$, $J \sim x^2$ so $E_\theta \to 0$ and the Debye–Hückel limiting law is untouched; as $x \to \infty$, $J \to x/4$. The crucial algebraic fact is that for two ions of the *same charge* ($z_i = z_j$) one has $x_{ij} = x_{ii} = x_{jj}$, so the bracket in (114) vanishes *identically*: $E_\theta = E'_\theta = 0$. Same-charge mixing (e.g. $Na^+/K^+$) therefore keeps the plain constant-$\theta$ form, and *every single salt* — which has one cation and one anion — has no like-sign pairs at all, so $E_\theta$ never fires.

> **Key idea.** $E_\theta(I)$ is the term most Pitzer tutorials wave away and most codes hard-wire on. Choupo computes it from first principles via the $J(x)$ kernel and *announces it*: at `verbosity` $\geq 3$ the model prints which $\theta$, $\psi$, $\lambda$, $\zeta$ binaries and triplets are active for the present ion set (`announceParameters`), so a student sees the *actual* parameter set in play — never a black box. The same kernel that carries $E_\theta$ is the difference between a correct $Mg^{2+}$-rich seawater $\gamma$ and a 10–20 % error.

## 6.4 The multi-ion osmotic coefficient and the water activity

The per-ion $\gamma_i$ answers "how non-ideal is each solute?" Its thermodynamic *sibling* answers the question that matters for the *solvent*: "how much do all these ions, together, lower the activity of the water?" That is the osmotic coefficient $\phi$ of the whole brine, and it is built from the *same* virial parameters as the $\gamma_i$ — the Gibbs–Duhem relation forces it. Skipping it (the $\phi = 1$ dilute approximation) is the single most common way a Pitzer code quietly cheats: the ions get the full treatment, the water is left ideal. In a real brine the water is *not* ideal — $a_w < x_w$ — and that departure *is* the osmotic pressure $\Pi = -(R_g T/\bar{V}_w)\ln a_w$ that drives every membrane process, so a $\phi = 1$ water activity is exactly the error that wrecks an osmosis or evaporator calculation at high concentration.

Choupo therefore evaluates the full Harvie–Møller–Weare osmotic coefficient ([11], Pitzer 1991 ch. 3; HMW 1984), the osmotic-form companion of the $\gamma_i$ sums above:

$$\begin{aligned}\Big(\phi - 1\Big)\sum_i m_i \;=\; 2\Bigg[ &-\frac{A_\phi\, I^{3/2}}{1 + b\sqrt{I}} \;+\; \sum_c\sum_a m_c m_a\left(B^{\phi}_{ca} + Z\,C_{ca}\right)\\ &+\sum_{c<c'} m_c m_{c'}\left(\Phi^{\phi}_{cc'} + \textstyle\sum_a m_a\,\psi_{cc'a}\right) + \sum_{a<a'} m_a m_{a'}\left(\Phi^{\phi}_{aa'} + \textstyle\sum_c m_c\,\psi_{aa'c}\right)\\ &+\sum_n\sum_i m_n m_i\,\lambda_{ni} \;+\; \sum_n\sum_c\sum_a m_n m_c m_a\,\zeta_{nca}\Bigg],\end{aligned} \tag{117}$$

with $Z = \sum_i |z_i| m_i$, the osmotic second virial $B^{\phi}_{ca} = \beta^{(0)}_{ca} + \beta^{(1)}_{ca} e^{-\alpha_1\sqrt{I}} + \beta^{(2)}_{ca} e^{-\alpha_2\sqrt{I}}$ (note: the bare exponentials, not the $g(x)$ of the $\gamma$ form), and $C_{ca} = C^{\phi}_{ca}/(2\sqrt{|z_c z_a|})$. Term by term this is the same parameter set the $\gamma_i$ used: the long-range Debye–Hückel head $-A_\phi I^{3/2}/(1 + b\sqrt{I})$ is $I\, f^{\phi}(I)$, the binaries reuse $\beta^{(0,1,2)}, C^{\phi}$, and the ternaries reuse $\theta, \psi$.

The one genuinely different object is the like-sign mixing term, which takes its *osmotic* form

$$\Phi^{\phi}_{ij} \;=\; \theta_{ij} \;+\; {}^{E}\theta_{ij}(I) \;+\; I\,{}^{E}\theta'_{ij}(I), \tag{118}$$

which carries the extra $I\,{}^{E}\theta'$ piece that the $\gamma$ mixing term $\Phi_{ij} = \theta_{ij} + {}^{E}\theta_{ij}$ does *not* — both built from the *same* $J(x)$ kernel (116). For two same-charge ions ${}^{E}\theta = {}^{E}\theta' = 0$, so (118) collapses to the plain $\theta_{ij}$ and every single salt is untouched. The water activity then follows from the $\phi$ definition (88),

$$\ln a_w \;=\; -\,M_w\,\phi \sum_i m_i, \tag{119}$$

the same shape as the dilute form (142) but with the *real* $\phi$ in front instead of 1. In code this is `PitzerHMW::evaluate` filling `ActivityResult::osmotic` from (117), which `SpeciationSolver` reads for (119); Davies leaves it null and keeps $\phi = 1$, announced as such.

**Key idea.** The same virials, two faces. Equation (117) shares *every* parameter with the $\gamma_i$ sums — as Gibbs–Duhem demands — so a brine's water activity is no longer a free guess but a *prediction*. Validated where the literature is sharpest: the NaCl osmotic coefficient from (117) tracks Pitzer–Peiper–Busey/Robinson–Stokes to better than 0.2 % from 0.1 to 6 mol/kg ($\phi = 0.936$ at 1 m, 1.045 at 3 m, 1.27 at 6 m), and standard seawater ($S = 35$, $I \approx 0.7$) returns $a_w = 0.981$ against the measured 0.980–0.981. Before this, Choupo (like many codes) used $\phi = 1$; that is now a *declared* fallback of the Davies model only, never a silent default under Pitzer.

## 6.5 The single-salt collapse — the skeptic's gate

The multi-ion sums *must* reduce to the closed single-salt kernel of §3 when only one salt is present, or there is a summation bug. This is not asserted by faith: `PitzerHMW::verify` loops over *every* binary in the `parameters/Pitzer/pairs/` catalogue, builds the single-salt solution at $m \in \{0.1, 0.5, 1, 3\}$ mol/kg, forms the mean

$$\ln \gamma^{\mathrm{HMW}}_{\pm} = \frac{\nu_c \ln \gamma_M + \nu_a \ln \gamma_X}{\nu_c + \nu_a}, \tag{120}$$

and compares it to `PitzerSingleSalt::lnGammaPM`. The max relative deviation across all binaries is $\sim 1.6 \times 10^{-14}$ — floating-point reassociation only. The same routine checks the Debye–Hückel limiting law (as $I \to 0$, $\ln \gamma_M \to -3A_\phi z_M^2\sqrt{I} = -A_\gamma z_M^2\sqrt{I}$, ratio $\to 1$) and the $J(x)$ kernel against the tabulated anchors $J(0.1) \approx 0.00353$, $J(1) \approx 0.1158$, $J(10) \approx 2.040$ (the $\sim 10^{-3}$ deviation there is the published accuracy of the approximation (116) itself, not an error).

**Worked example 6.1.** Why $E_\theta$ is not optional in seawaterStandard seawater is roughly $Na^+$ 0.49, $Mg^{2+}$ 0.055, $Cl^-$ 0.57, $SO_4^{2-}$ 0.029 mol/kg, giving $I \approx 0.72$ mol/kg. For the like-sign pair $Cl^-/SO_4^{2-}$ ($z_i = -1$, $z_j = -2$), the arguments are $x_{ij} = 6 \cdot 2 \cdot 0.391\sqrt{0.72} = 3.98$, $x_{ii} = 6 \cdot 1 \cdot 0.391\sqrt{0.72} = 1.99$, $x_{jj} = 6 \cdot 4 \cdot 0.391\sqrt{0.72} = 7.96$. From (116), $J(3.98) \approx 0.695$, $J(1.99) \approx 0.291$, $J(7.96) \approx 1.574$, so

$$E_\theta = \frac{(-1)(-2)}{4 \times 0.72}\left[0.695 - \tfrac{1}{2}(0.291) - \tfrac{1}{2}(1.574)\right] = 0.694 \times (-0.238) \approx -0.17.$$

This $-0.17$ *adds* to whatever constant $\theta_{\mathrm{Cl,SO_4}}$ the catalogue carries (typically $\sim 0.02$): the electrostatic part *dominates* the fitted constant here. Dropping $E_\theta$ — the "constant-theta" shortcut — would mis-state the $Cl^-/SO_4^{2-}$ interaction by an order of magnitude, and with it the $\gamma$ of every ion that sees both anions.

**Power.** One header of pure $J(x)$ arithmetic turns the single-salt osmotic coefficient into a full multi-component aqueous activity model: any mix of Na, K, Ca, Mg, H cations and Cl, $SO_4$, OH (plus the carbonate subsystem) anions, with the higher-order electrostatics that make 2:1 and 2:2 brines correct, all driven from the flat per-record catalogues under `parameters/Pitzer/` (`pairs/`, and the mixing kinds `theta/`, `psi/`, `lambda/`, `zeta/`) that cite a primary source per value and accept a per-case overlay. The same surface feeds the speciation solver, the osmotic pressure, and — through the Gibbs–Helmholtz identity $L_\phi = -RT^2\,\partial g^{ex}/\partial T$ — the heats of dilution.

**Limit.** The shipped parameters are the *25 °C* HMW base set; the full $\beta(T)$, $\theta(T)$ temperature surface is deferred (the dominant temperature dependence enters only through $A_\phi(T)$, scaled by the Born–Bjerrum factor, and through the single-salt $\partial\beta/\partial T$ calorimetric slots). The ion roster is the core seawater/brine system plus carbonates — borate,

$H_4SiO_4$, $HSO_4^-$ and the higher neutral system are intentionally out. And $J(x)$ is Pitzer's *approximation* to the defining integral (good to $\sim 10^{-3}$), not the integral itself — accurate enough that the limiting law and the single-salt collapse hold to machine precision, but a Chebyshev-grade $J(x)$ would be the next refinement if sub-0.1% $E_\theta$ ever mattered.

**Runnable case.** The HMW model is the `pitzer AqueousActivity` reached through the speciation path (`SpeciationSolver`); a single-salt case still uses the closed `PitzerSingleSalt` kernel and the osmotic-pressure membrane tutorials (`membrane06_pitzer`). Run any speciation case at `verbosity 4` to see `announceParameters` list the active binaries, $\theta/\psi$ ternaries and $E_\theta$ pairs.

# 7 Molecular solutes in a Pitzer expansion: the Edwards truncation

*What you'll learn.* Every model so far has priced *ions*. A sour water — the condensate off a crude unit, a refinery stripper feed, an ammonia scrubber blowdown — is difficult for the opposite reason: the species whose escape you care about, dissolved $NH_3$ and $CO_2$, are *neutral*. A charge-only correlation cannot say anything about them at all: put $z = 0$ into Debye–Hückel and you get $\gamma = 1$, exactly, at every ionic strength. This section builds the truncation of Edwards, Maurer, Newman and Prausnitz (1978) [21], which extends Pitzer's virial expansion to molecular solutes, and — just as importantly — shows how little of it is measured and how the rest is estimated.

## 7.1 Why $\gamma = 1$ for a neutral is a structural silence

It is worth being precise about what a charge-only model does and does not claim. The Davies equation [22] is

$$\log_{10}\gamma_i = -Az_i^2\left(\frac{\sqrt{I}}{1+\sqrt{I}} - 0.3\,I\right), \tag{121}$$

a function of $|z_i|$ and $I$ alone. For a neutral solute $z_i = 0$ and the whole right-hand side vanishes, so $\gamma_i = 1$ identically — not approximately, not as a small-correction limit, but as an algebraic consequence of the model having no term that could depend on the solute's identity. It is a *structural* silence, and a student who has only met Davies can easily read that 1 as "the correction is negligible" when what it means is "this model has nothing to say".

Physically the correction is not negligible. A neutral molecule in a concentrated brine is *salted out*: the ions organise the water around themselves, less free solvent is available to the neutral, and its escaping tendency rises. Pricing that is exactly what a stripper design turns on.

## 7.2 The three-term working equation

Edwards keeps Pitzer's expansion to the second virial coefficient and lets the sums run over *all* solute species, ions and molecules alike. With $m$ in mol per kg of *water* and the indices $i, j, k$ over solutes,

$$\begin{aligned}\ln\gamma_i^{*} = &-A_\phi z_i^2\left[\frac{\sqrt{I}}{1+1.2\sqrt{I}} + \frac{2}{1.2}\ln\!\left(1+1.2\sqrt{I}\right)\right]\\ &+2\sum_j m_j\left\{\beta_{ij}^{(0)} + \frac{\beta_{ij}^{(1)}}{2I}\left[1-\left(1+2\sqrt{I}\right)e^{-2\sqrt{I}}\right]\right\}\\ &-\frac{z_i^2}{4I^2}\sum_j\sum_k m_j m_k \beta_{jk}^{(1)}\left\{1-\left(1+2\sqrt{I}+2I\right)e^{-2\sqrt{I}}\right\},\end{aligned} \tag{122}$$

with $I = \tfrac{1}{2}\sum_j z_j^2 m_j$ as always, and $A_\phi = 2.303\,A_\gamma/3$ the osmotic Debye–Hückel slope. The water activity comes from the Gibbs–Duhem partner of (122),

$$\ln a_w = M_w\left[\frac{2A_\phi I^{3/2}}{1+1.2\sqrt{I}} - \sum_i\sum_j m_i m_j\left(\beta_{ij}^{(0)} + \beta_{ij}^{(1)}e^{-2\sqrt{I}}\right)\right] - M_w\sum_i m_i. \tag{123}$$

Read (122) one term at a time. The **first** is Debye–Hückel and carries $z_i^2$, so it vanishes for a molecule — a neutral gets *no* long-range term, correctly, because it has no charge to screen. The **second** is the pairwise virial, and it is the only term a molecule has: its $\gamma$ comes entirely from how it interacts with each species present, weighted by that species' molality. The **third** is a $z_i^2$-weighted double sum, so it too is silent for a molecule.

Three rules the paper states *around* the equation rather than in it, each load-bearing and none visible in the algebra:

- if either index is a molecular species, $\beta_{ij}^{(1)} = 0$ ("because experimental data are inadequate");
- for ions of *like* charge, $\beta_{ij}^{(0)} = \beta_{ij}^{(1)} = 0$ — Brønsted's assumption that like charges do not approach closely enough to interact specifically;
- $m$ must be *molal* for (122) as written.

## 7.3 One measured family, four estimation rules

This is the part a user must not skip. Edwards measured $\beta^{(0)}$ for *like molecule pairs* only, as a two-constant temperature fit

$$\beta_{aa}^{(0)}(T) = E + F/T, \tag{124}$$

tabulated for $NH_3$, $CO_2$, $H_2S$ and $SO_2$. Everything else in the matrix that (122) sums over is *estimated* from those by four rules:

$$\text{unlike molecules:} \qquad \beta_{ij}^{(0)} = \tfrac{1}{2}\left(\beta_{ii}^{(0)} + \beta_{jj}^{(0)}\right), \tag{125}$$

$$\text{ion–ion (Bromley):} \qquad \beta_{+-}^{(0)} = \beta_{+}^{(0)} + \beta_{-}^{(0)}, \tag{126}$$

$$\text{molecule–ion:} \qquad \beta_{m\text{-}i}^{(0)} = \beta_{m}^{(0)} + \beta_{i}^{(0)}, \tag{127}$$

$$\text{and, for every ion pair:} \qquad \beta^{(1)} = 0.018 + 3.06\,\beta^{(0)}. \tag{128}$$

Equation (128) is what makes a multisolute calculation tractable at all: $\beta^{(1)}$ is never an independent datum in this model, only a correlation of $\beta^{(0)}$. The paper calls both of its parameter tables "preliminary results subject to change as more and better data become available", and a case using them may not present them as measured constants.

Choupo takes that seriously enough to make it visible. The estimated pairs are *not stored*: only the measured tables live on disk, and each estimated $\beta$ is derived when the parameter set is assembled and printed with the rule that produced it. At `verbosity 3` a run lists every active pair as, for instance,

```
CO2aq -    CO2aq   beta0 =  0.008219   beta1 =  0.000000   [Eq 14, measured]
CO2aq -    NH3aq   beta0 =  0.011720   beta1 =  0.000000   [Eq 21, estimated]
 HCO3 -      NH4   beta0 = -0.077000   beta1 = -0.217620   [Eq 22 + Eq 24, estimated]
 HCO3 -     HCO3   beta0 =  0.000000   beta1 =  0.000000   [Bronsted, like signs -> 0]
```

so that the one measured number on the page is distinguishable at a glance from the three estimated ones beside it.

## 7.4 What it is worth on a real feed

Take an ammonia-rich sour water, total $NH_3$ 2.9 and total $CO_2$ 1.45 mol/kg, at 25 °C and pH 9. Under Davies the two neutral species come back at $\gamma = 1.000000$ and 1.000000. Under the Edwards truncation they come back at 1.0224 and 1.0101: both salted out, ammonia rather more, because both measured self-terms are positive at 25 °C and ammonia is the more abundant of the two by three orders of magnitude, so its own $\beta_{aa}^{(0)}$ dominates its sum in (122).

The larger difference, though, is not molecular at all. The two models disagree about the *ions* — Davies returns $\gamma_{NH_4^+} = 0.956$ at $I \approx 2$, because its bracket saturates and then turns back upward, where (122) returns 0.301 — and the $NH_4^+/NH_3$ equilibrium reads those coefficients. So the speciation itself moves: $I = 2.78$ against 1.78, with $m_{NH_3}$ 0.41 against 1.01. A dilute correlation used far outside its band does not merely lose accuracy; it changes the *shape* of the answer.

> **Runnable case.** `props/electrolyte/edwards01_sour_water_activity` runs both models on one feed so the comparison above is reproducible, and prints the full parameter table with its rule labels. The case is a *structural* witness: it validates the machinery, not the numbers. Measured validation requires the vapour side — Henry's constants on the paper's own convention and the two-phase equilibrium — which is a separate development.

# 8 Aqueous speciation: mass action, charge balance, and the pH closure

*What you'll learn.* A water analysis hands you a short list of *total* concentrations — so much calcium, so much carbonate, so much chloride — but the quantities that actually matter (the $CO_3^{2-}$ that scales a membrane, the free $H^+$ that sets the pH, the $CaSO_4$(aq) ion pair that hides calcium from a saturation index) are *distributed* among dozens of dissolved species by chemical equilibrium. This section builds the *speciation* calculation that does the distributing: a small set of mass-action equilibria written against a handful of *master* ions, closed either by a measured pH or, when none is reliable, by *electroneutrality*. We solve it the way the code does — one Newton iteration in $\ln m$ inside a frozen-$\gamma$ fixed point — and we meet the PHREEQC *percent-error* convention that tells you, before you trust a single number, whether the lab sheet even balances its charges. This is the engine behind the `speciate` and `scalingScan` property operations and behind every membrane-scaling and softener audit downstream.

## 8.1 Master ions and the mass-action network

The previous two chapters gave the *activity* of an ion that is already free in solution. Speciation answers the prior question: given the totals, how much of each ion *is* free? When a carbonate water sits at equilibrium, the dissolved inorganic carbon is split between $H_2CO_3$, $HCO_3^-$, $CO_3^{2-}$ and pairs like $CaHCO_3^+$; calcium is split between free $Ca^{2+}$, $CaHCO_3^+$, $CaCO_3$(aq), $CaSO_4$(aq), and so on. Each non-master *species* $s$ is the product of a *formation reaction* written against a fixed, minimal basis of *master* ions (Ca, Mg, Na, K, Cl, $SO_4$, $HCO_3$, …) plus $H^+$ and water. Its concentration is fixed by a mass-action law,

$$m_s\,\gamma_s \;=\; K_s(T)\,\prod_j a_j^{\,\nu_{sj}}, \qquad a_j = \gamma_j\,m_j, \tag{129}$$

where $m_j$ are the *free* master molalities (mol per kg water), $\nu_{sj}$ the stoichiometric coefficient of master $j$ in the formation of $s$, $K_s(T)$ the formation constant, and $\gamma$ the single-ion activity coefficients of the previous chapter. Water enters with its own activity $a_w$ raised to $\nu_{s,\mathrm{H_2O}}$ (so the autoprotolysis $H_2O = H^+ + OH^-$ writes $OH^-$ with $a_{\mathrm{OH}} = K_w\,a_w/a_{\mathrm{H}}$). The whole network — the species list, their stoichiometries, $\log_{10} K_s$ at 25 °C and an optional temperature law — is read from a `speciation.dat` catalogue derived from the public-domain USGS PHREEQC databases [23].

*Intuition.* Speciation is not a new kind of physics; it is "more $\partial/\partial n$ of the same Gibbs surface." Equation (129) is just $\Delta G_{\mathrm{rxn}} = 0$ for each formation reaction — the same condition that gives an equilibrium constant in any reactor. What is new is the *bookkeeping*: many simultaneous equilibria sharing the same free ions, closed by conservation of mass and charge.

## 8.2 The conservation equations: mole balances

There are far more species than masters, so (129) alone cannot close the system. The closure is conservation. For each master ion $j$ the analysis specifies a *total* molality $T_j$ (everything containing that element, free or bound), and the model must reproduce it:

$$T_j \;=\; m_j \;+\; \sum_s \nu_{sj}\,m_s \qquad (j = 1,\dots,n). \tag{130}$$

Substituting the mass-action expression (129) for each $m_s$ turns (130) into $n$ equations in the $n$ free master molalities — everything else is algebra. This is the core problem the solver faces. What remains is to decide what to do with $H^+$, the master that a water analysis almost never reports as a total.

## 8.3 Two closures for $H^+$: pH given vs pH solved

**pH given.** If a trustworthy pH is on the sheet, set

$$a_{\mathrm{H}} \;=\; 10^{-\mathrm{pH}}, \tag{131}$$

treat $H^+$ as *known*, and remove its mole balance from the system: $H^+$ (and the $OH^-$ that chains off it) are simply algebraic functions of the fixed $a_{\mathrm{H}}$. The unknowns stay the $n$ master molalities. This is the dict form `pH 7.8;`.

**pH solved (electroneutrality).** Often the reported pH is stale, drifted in the sample bottle, or simply missing. Then $H^+$ becomes an *extra unknown* and an extra equation must replace its missing total. That equation is *electroneutrality*: a real solution carries no net charge,

$$\boxed{\;\sum_i z_i\,m_i \;=\; 0\;} \tag{132}$$

summed over *all* species — masters, $H^+$, $OH^-$, and every complex. The pH is then an *output*: the free $H^+$ adjusts until the cations and anions balance. This is the dict form `pH solve;`.

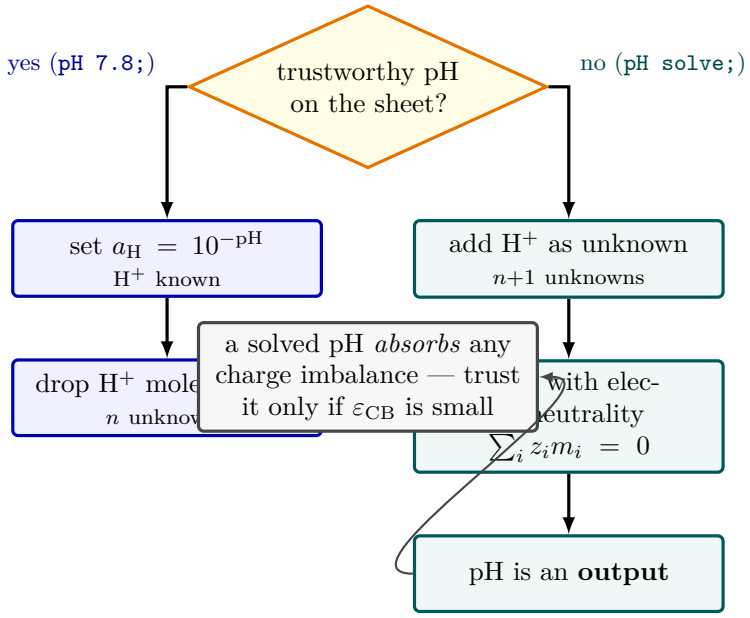



**Figure 16:** The two ways the speciation solver closes the proton balance. If a trustworthy pH is on the analysis (left, blue), $\mathrm{H}^+$ is *known*: fix $a_{\mathrm{H}} = 10^{-\mathrm{pH}}$ and drop its mole balance, leaving $n$ unknowns. If the pH is stale or missing (right, teal), $\mathrm{H}^+$ becomes an extra unknown and *electroneutrality* $\sum_i z_i m_i = 0$ supplies the missing equation—the pH then falls out as an output, exactly as a titration determines it physically. The caveat (grey): a solved pH silently absorbs any charge-balance error in the input, so it is only as good as $\varepsilon_{\mathrm{CB}}$.

**Common pitfall. A precipitating mineral does not belong in (132).** Calcite dissolves as $\mathrm{CaCO_3(s)} + \mathrm{H^+} = \mathrm{Ca^{2+}} + \mathrm{HCO_3^-}$, so precipitating it *releases* a proton and the solved pH must fall. The tempting move — add $\nu_{\mathrm{pH}}\, n_p$ to the charge row so the freed proton is "accounted for" — counts it *twice*. The crystal is neutral, so its dissolution reaction is charge-balanced by construction, $\sum_j z_j \nu_{pj} + \nu_{\mathrm{pH}} = 0$ ($+2 - 1 - 1$ here). Precipitating $n_p$ moles of it does not conserve charge *by accident*: it removes $\mathrm{Ca^{2+}}$ and $\mathrm{HCO_3^-}$ one for one, and one for one *in moles* is $+1$ *in charge* — so a proton must appear to make up the difference. That is the whole intuition, and the algebra only confirms it. With the mineral sink the mole balance (130) reads

$$T_j \;=\; m_j \;+\; \sum_s \nu_{sj}\, m_s \;+\; \sum_p \nu_{pj}\, n_p, \tag{133}$$

$n_p$ being the moles of solid $p$ per kilogram of water and $\nu_{pj}$ the moles of master $j$ its dissolution RELEASES. Substituting (133) into the charge sum and using charge conservation of the formation reactions ($z_s = \sum_j \nu_{sj} z_j + \nu_{s\mathrm{H}}$) gives

$$\sum_i z_i m_i \;=\; m_{\mathrm{H}} + \sum_s \nu_{s\mathrm{H}} m_s \;+\; \sum_j z_j T_j \;+\; \sum_p \nu_{\mathrm{pH}}\, n_p,$$

so setting it to zero *already* shifts the proton condition by $\nu_{\mathrm{pH}} n_p$: the sink pulls $\mathrm{Ca^{2+}}$ and $\mathrm{HCO_3^-}$ out in 1:1 and the only species left to restore the balance is $\mathrm{H^+}$. The acidification is a *consequence* of the mole balances, never a term to be added. The double count is silent — the solver still converges, on $\sum_i z_i m_i = +n_p$ — and it under-reads the calcite ceiling of an RO concentrate by 37 % (equivalently, the correct ceiling is 58 % higher: 12.1 → 19.2 kg/day at 85 % recovery in `precipitation_ro_brackish`). Which is why Choupo reports the converged charge residual next to the answer and gates it.

*Intuition.* Charge balance closing the pH is exactly what a titration does physically. When the cation and anion totals on a sheet do not add up to zero charge, something is wrong — a missing analyte, a unit slip, an alkalinity mis-reported — and the *solved* pH silently *absorbs* that error: it moves $\mathrm{H^+}$ (and hence $\mathrm{OH^-}$, $\mathrm{HCO_3^-}$, $\mathrm{CO_3^{2-}}$) to make the books balance. A solved pH is only as trustworthy as the charge balance of the analysis behind it. That is why the next subsection insists on measuring the imbalance *before* believing the answer.

## 8.4 The PHREEQC percent-error convention

A defensible speciation report must say how far the input was from charge balance. Following PHREEQC [23], Choupo reports the *charge-balance percent error*

$$\boxed{\varepsilon_{\mathrm{CB}} \;=\; 100 \cdot \frac{\left|\, S_+ - S_- \,\right|}{\frac{1}{2}\,(S_+ + S_-)} \;=\; 200 \cdot \frac{\left|\, S_+ - S_- \,\right|}{S_+ + S_-},} \tag{134}$$

where $S_+ = \sum_{z_j>0} |z_j|\, T_j$ and $S_- = \sum_{z_j<0} |z_j|\, T_j$ are the cation and anion *equivalent* sums over the *master totals* ($\mathrm{H^+/OH^-}$ are not part of a water analysis, so they are excluded). The factor 200 (not 100) normalises by the *average* of the two sums, so a $+5\,\%$ cation excess and a $-5\,\%$ anion deficit each register as the same $\approx 5\,\%$. Choupo prints $\varepsilon_{\mathrm{CB}}$ *before* it solves, and when it exceeds $\approx 5\,\%$ raises a loud advisory: the analysis is internally inconsistent and the solved pH is not trustworthy because it has absorbed the lab error. This is the simulator's no-silent-crutch rule applied to data quality — the user is told what the number is carrying.

## 8.5 Distribution diagrams: the Bjerrum plot, and the titrant nobody declared

The two closures of the previous subsections are usually met one at a time: a case declares one pH and reads one answer. Sweep the *given*-pH closure instead — solve the same network again and again, imposing a different pH at each point, and plot every species against it — and the result is the oldest picture in aqueous chemistry, the *distribution diagram* or *Bjerrum plot*. It is what the carbonate system, the ammonia system and every polyprotic acid *are*, in one figure: which species carries the material at which pH, where the hand-over happens, and how sharp it is.

**Where the crossings come from.** Nothing new is needed to read the shape — it is (129) with $a_{\mathrm{H}}$ walked along the axis. For a monoprotic acid $\mathrm{HA = H^+ + A^-}$ with $K_a = a_{\mathrm{H}} a_{\mathrm{A}}/a_{\mathrm{HA}}$, the ratio of the two bands is

$$\frac{a_{\mathrm{A^-}}}{a_{\mathrm{HA}}} \;=\; \frac{K_a}{a_{\mathrm{H}}} \qquad\Longrightarrow\qquad \log_{10}\frac{a_{\mathrm{A^-}}}{a_{\mathrm{HA}}} \;=\; \mathrm{pH} - \mathrm{p}K_a, \tag{135}$$

so the two curves *cross at* $\mathrm{pH} = \mathrm{p}K_a$ and separate by one decade per pH unit on either side. A diprotic acid gives two crossings, a triprotic three, and the plateaux between them are the buffer regions — the same $\mathrm{p}K_a$ ladder read as a picture instead of a table. Away from a crossing the minor band falls with a slope of $\pm 1$ (or $\pm 2$ for a species two protons away), which is why the diagram is drawn on a *logarithmic* concentration axis: on a linear one, four of the five curves are indistinguishable from zero.

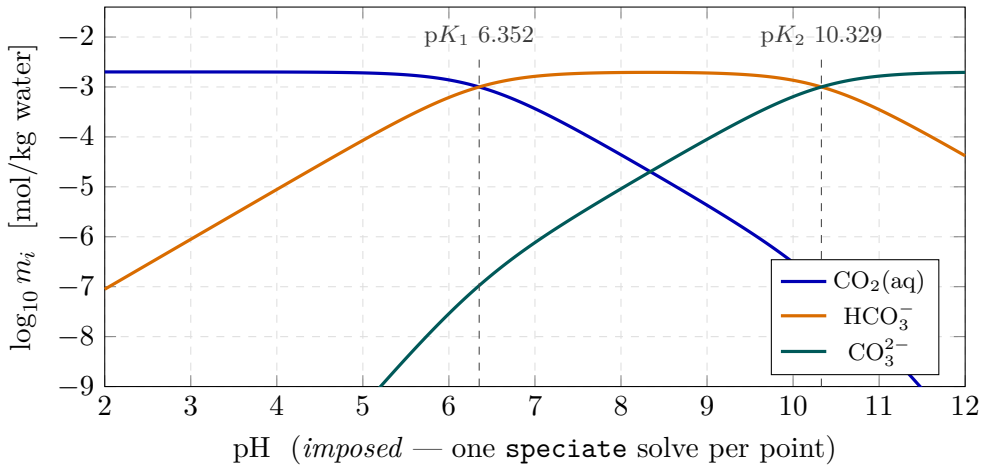



**Figure 17:** A distribution diagram of the carbonate system, 2 mmol/kg total carbonate: the three bands hand the material over at the two $\mathrm{p}K_a$'s and separate by one decade per pH unit, (135), while a band two protons from the dominant one falls at two decades per unit. These curves are the *closed form* at $\gamma_i = 1$, drawn from the two constants the curated records carry (`chemistry/CO2aq-formation.dat`, `chemistry/CO3-formation.dat`; USGS PHREEQC, public domain). A Choupo run does not draw this formula: its answer moves off these curves by the single-ion activity coefficients, by $K_s(T)$ away from 25 °C, and by any ion pair the feed makes reachable (a sodium bicarbonate water forms $\mathrm{NaHCO_3(aq)}$, which this expression has no term for). That gap is the reason the Explore lens plots the engine's own molalities instead — but the *shape*, and where the crossings sit, is this.

**The two axes are not the same kind of thing.** Read (135) once more and notice what it does *not* say: it fixes the *ratio* of the bands, not their position on the axis. What sets that is the total, and what sets the pH is — in this diagram — nothing. The pH is an *input*. That is the whole difference between the two closures of Figure 16, and it is easy to lose sight of because the axis of a distribution diagram looks exactly like the axis of a measured titration.

**Common pitfall. An imposed pH is a strong acid or base that no total declares.** Under the given-pH closure, $H^+$ is known and its mole balance is *removed* from the system (131). Nothing then forces $\sum_i z_i m_i$ to zero — the electroneutrality row (132) is precisely the equation that was dropped. The converged answer therefore carries a *net charge*,

$$q \equiv \frac{\sum_i z_i\, m_i}{\sum_i |z_i|\, m_i}, \tag{136}$$

summed over the species of the answer ($H^+$ and $OH^-$ included), and $q \neq 0$ is not an error: it is the charge that a strong acid or base would have had to bring in for the solution to sit at that pH. Drive a 2 mmol/kg bicarbonate water to pH 4 and the model reports $q \approx +0.99$ — almost *all* the ionic charge in the answer is unbalanced, because reaching pH 4 meant adding hydrochloric acid and the chloride was never declared. The sign says which reagent: $q > 0$ means an anion is missing (an acid was implied), $q < 0$ means a cation is (a base). The classical Bjerrum plot is drawn exactly this way and is not wrong for it; *drawing it without saying so* is the error, because the reader then takes an input for a measurement.

Do not confuse $q$ with the percent error $\varepsilon_{\mathrm{CB}}$ of the previous subsection. They answer different questions and are computed over different sets. $\varepsilon_{\mathrm{CB}}$ is taken *before* the solve, over the *master totals* of the analysis, with $H^+/OH^-$ deliberately excluded, and it measures whether a *lab sheet* balances; $q$ is taken *after* the solve, over *every species* including the protons, and under an imposed pH it measures the titrant the model silently supplied. A perfectly balanced analysis ($\varepsilon_{\mathrm{CB}} = 0$) still gives a large $q$ the moment the pH is imposed away from its natural value — which is the normal condition of every point of a distribution diagram except one. Under the *solved*-pH closure the same quantity means the third thing: electroneutrality is then an equation of the system, so $q$ is its residual and belongs at round-off. Choupo prints $q$ beside every speciation answer and states, in the same line, which of the two meanings it is carrying.

**Key idea.** A distribution diagram is the given-pH closure *swept*. Its crossings are the p$K_a$'s and its slopes are the stoichiometries, so the picture contains the whole mass-action network. Its horizontal axis, however, is an imposed input, not an observation: every point off the natural pH of the system is held there by an implicit strong acid or base, and the converged net charge (136) is how much of it.

**Where the two closures meet.** There is one pH on the axis at which no titrant is needed, and it is worth finding because it joins the two halves of this section. Take a real feed — 2 mmol/kg of bicarbonate *with* its 2 mmol/kg of sodium, so the analysis is electroneutral to begin with — and sweep it with the pH imposed. The net charge (136) falls from +0.991 at pH 4 through +0.000687 at pH 8.25 to −0.0059 at pH 8.50, vanishing at pH 8.276. Hand the *same* feed to the solved-pH closure and it reports pH = 8.277 from electroneutrality. They agree because they are one statement read two ways: the point where the implicit titrant goes to zero is the point where $\sum_i z_i m_i = 0$ is satisfied without being imposed, which is the definition of the solved pH. Everywhere else on the diagram, the vertical distance of the titrant curve from zero is how much strong acid or base the picture is quietly assuming. A feed declared as a *bare* ion — bicarbonate with no cation at all — has no such point: its net charge sits at one whole equivalent at every pH, correctly, because such a solution does not exist.

**In Choupo.** There is no pH-sweep operation and there does not need to be one: a distribution diagram is a series of ordinary `speciate` operations that differ only in their `pH` entry — a number, never the word `solve` — so a sweep is written as an operation list and each point is a full speciation with its own announcements. Opening the system, `atmosphere { CO2 4.2e-4 atm; }`, changes what the diagram is about: the pinned family's total stops being an input and becomes an outcome of Henry's law, so the bands move with the gas rather than conserving a declared inventory (the run says which mole balance the pin replaced). Temperature moves everything at once — $K_s(T)$ for each equilibrium and $A_{\mathrm{DH}}(T)$ in the activity model — and across 0–100 °C the two do not travel on the same footing: some $K_s$ carry a fitted analytic $\log_{10} K(T)$, some only a constant-$\Delta H$ van't Hoff slope, and some are held at 25 °C for want of either. The run classifies and *announces* each of them, and raises an advisory when an analytic form is used outside its fitted range, so the temperature knob of a distribution diagram is a demonstration of what the catalogue knows rather than a hazard. The Explore workspace's species-distribution lens drives exactly this: one `speciate` run per pH point, the engine's own molalities on the log axis, and the net charge (136) drawn beside them.

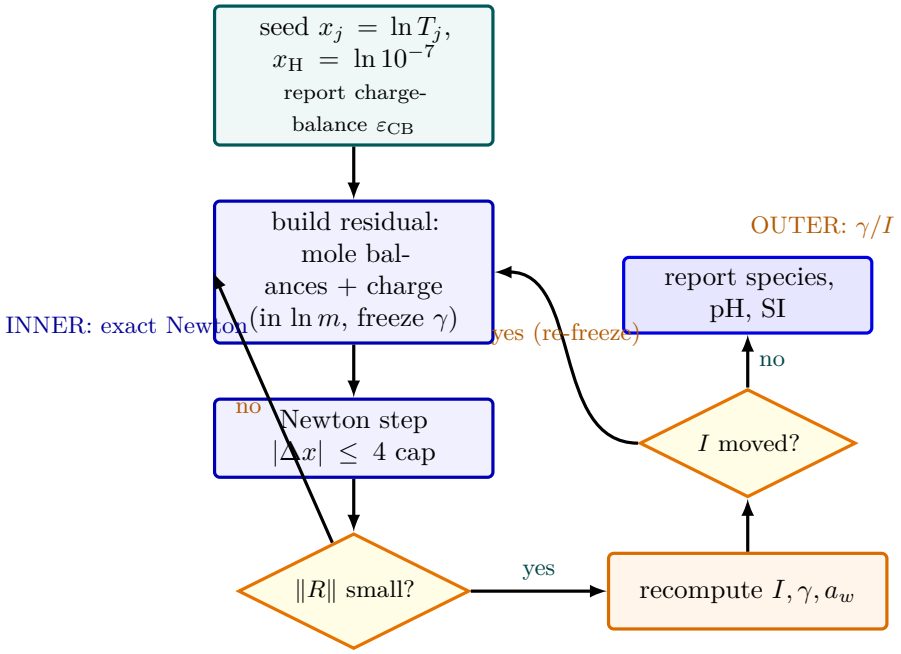



**Figure 18:** The speciation solver as a two-loop machine—the same freeze-the-slow-coupling pattern as a recycle tear. The *inner* loop (blue) is an exact Newton on the mole balances (plus electroneutrality when the pH is solved), written in $\ln m$ so molalities stay positive and the mass-action laws turn linear; each step is capped at $|\Delta x| \le 4$ so the iteration cannot leap out of its basin. The *outer* loop (orange) freezes the activity coefficients, solves the inner system exactly, then re-evaluates $\gamma$ and $a_w$ at the new ionic strength, repeating until $I$ stops moving. Before any of it runs, the charge-balance error $\varepsilon_{CB}$ is reported (teal)—a solved pH is no better than the analysis it balances.

## 8.6 The solution algorithm: Newton in log-molality

The mole balances (130) are stiff: molalities span many orders of magnitude ($Na^+$ at $10^{-2}$, $CO_3^{2-}$ at $10^{-5}$, $H^+$ at $10^{-8}$) and must stay strictly positive. Both problems vanish under the substitution

$$x_j \equiv \ln m_j \qquad \Longrightarrow \qquad m_j = e^{x_j} > 0 \text{ automatically,} \tag{137}$$

and the mass-action law (129) becomes *linear* in the log unknowns:

$$\ln m_s = \ln 10 \, \log_{10} K_s - \ln \gamma_s + \nu_{s,H_2O} \ln a_w + \sum_j \nu_{sj} \big( \ln \gamma_j + x_j \big). \tag{138}$$

The residual vector is the set of mole balances (plus, when $H^+$ is solved, the electroneutrality row (132) with $x_H$ as its unknown). Because each $m_s$ depends on $x_j$ through (138), the Jacobian is *analytic*: a mole-balance row has $\partial R_j / \partial x_k = -\delta_{jk} m_j - \sum_s \nu_{sj} \nu_{sk} m_s$, and the dense $n \times n$ (or $(n{+}1) \times (n{+}1)$) system is solved by Gaussian elimination (Chapter 3). Newton steps in $\ln m$ are capped at $|\Delta x| \le 4$ ($m$ may change by at most $e^4 \approx 55\times$ per step) so the iteration cannot leap out of the basin.

The activity coefficients are the wrinkle: $\gamma_j$ depends on the ionic strength $I = \frac{1}{2} \sum_i z_i^2 m_i$, which depends on the very molalities being solved. Choupo breaks this with a *frozen-$\gamma$ fixed point*: hold all $\gamma$ (and the water activity $a_w$) constant, run Newton to convergence on the molalities, recompute $I$, $\gamma$ and $a_w$, and repeat the outer pass until $I$ stops moving. The two-loop structure — an outer $\gamma/I$ self-consistency wrapping an inner exact Newton — is the same pattern used for the recycle tear (Chapter 6): freeze the slow coupling, solve the fast system exactly, update, repeat.

> **Key idea.** Speciation is solved as a **two-loop** problem. The *inner* loop is an exact Newton on mole balances (and electroneutrality, if pH is solved) written in $\ln m$ so molalities stay positive and the mass-action laws become linear. The *outer* loop freezes the activity coefficients and water activity, then re-evaluates them at the new ionic strength — iterating $I \to \gamma \to m \to I$ to self-consistency. The default $\gamma$ model is Davies (the teaching rung below Pitzer); Pitzer HMW is selectable for high ionic strength.

## 8.7 The Davies equation: a parameter-free single-ion $\gamma_i$

The speciation network of the previous subsections needs, for every charged species, a *single-ion* activity coefficient $\gamma_i$ — not the mean $\gamma_\pm$ that the eNRTL (§4) and the osmotic Pitzer (§3) deliver. The cheapest rung that still

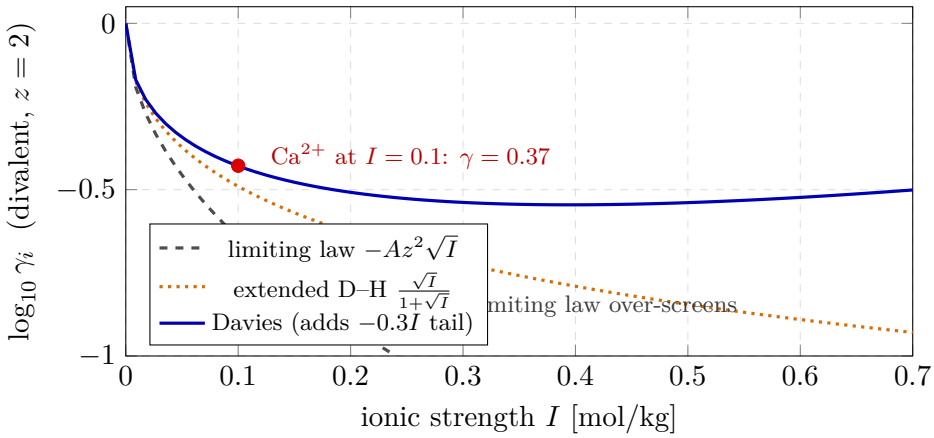



**Figure 19:** The three rungs of single-ion non-ideality for a divalent ion, showing why Davies is the right teaching default. The bare Debye–Hückel limiting law (dashed grey) over-screens at finite $I$, driving $\gamma$ unphysically low. The extended law's closest-approach denominator $1+\sqrt{I}$ (dotted orange) tames it. Davies (blue) adds one universal, salt-*in*dependent empirical tail $-0.3I$ that turns the decline back upward—mimicking the short-range repulsion a per-salt $\beta^{(0)}$ would otherwise carry, with *no* fitted parameter. At $I=0.1$ the worked $Ca^{2+}$ value (red, $\gamma=0.37$) sits 40 % above the limiting-law answer—exactly the gap that decides whether $CaCO_3$ reads as supersaturated at a membrane wall.

gets the electrostatics right is the *Davies equation* [22], the model Choupo's `davies` kernel (`AqueousActivity`) implements and the speciation solver freezes once per outer iteration. This subsection derives it, states exactly what the code evaluates, and is honest about where it runs out.

**From the limiting law to Davies.** The Debye–Hückel theory (1923) treats each ion as a point charge surrounded by a diffuse *ionic atmosphere* of opposite net charge. Solving the linearised Poisson–Boltzmann equation for the work of charging that atmosphere gives the celebrated *limiting law*, valid only as $I \to 0$,

$$\log_{10}\gamma_i \;=\; -A\, z_i^2 \sqrt{I}, \tag{139}$$

where the square-root in ionic strength $I = \tfrac{1}{2}\sum_i m_i z_i^2$ is the electrostatic signature no mole-fraction polynomial can reproduce. The slope $A$ is a property of the *solvent only* (no salt-specific input). The limiting law over-screens at finite concentration; the *extended* Debye–Hückel law tames it with a closest-approach denominator $1+\mathring{a}B\sqrt{I}$, and the **Davies** fix takes the universal choice $\mathring{a}B = 1$ and adds a single empirical linear tail $-0.3\,I$ that turns the otherwise monotone decline back upward, mimicking the short-range repulsion that a per-salt $\beta^{(0)}$ would otherwise carry:

$$\boxed{\log_{10}\gamma_i \;=\; -A\, z_i^2 \left( \frac{\sqrt{I}}{1+\sqrt{I}} \;-\; 0.3\, I \right).} \tag{140}$$

The remarkable thing about (140) is that it carries *no adjustable, salt-specific parameter at all* — only $A$ (the solvent) and the ion charge $z_i$. That is its entire pedagogical and practical appeal: a defensible $\gamma_i$ for *any* ion in water from charge alone.

**The slope $A$ and its temperature scaling.** $A$ is the $\log_{10}$-base Debye–Hückel slope; Choupo anchors it at the textbook

$$A \;=\; 0.51\ \mathrm{kg}^{1/2}\,\mathrm{mol}^{-1/2} \quad (25\,\text{řC}),$$

which is exactly $3A_\phi/\ln 10$ with the Pitzer $A_\phi = 0.3915$, so the Davies and Pitzer rungs share one consistent Debye–Hückel constant. Off 25 °C, $A$ inherits its temperature dependence from the *solvent* through

$$A(T) \;=\; 0.51 \times f_{\mathrm{DH}}(T), \qquad f_{\mathrm{DH}}(T) = \left(\frac{\rho_w(T)}{\rho_w^{25}}\right)^{1/2} \left(\frac{\varepsilon_w^{25}\, T_{25}}{\varepsilon_w(T)\, T}\right)^{3/2}, \tag{141}$$

the same dimensionless factor (`SolventProperties::debyeHuckelFactor`) that scales the Pitzer $A_\phi$ and the eNRTL $A_{\mathrm{DH}}$ — so the three electrolyte kernels move together with temperature, never independently. The factor is built from the static permittivity $\varepsilon_w(T)$ (Malmberg–Maryott [24]) and the density $\rho_w(T)$ (Kell [25]), each written as a *ratio* anchored so that $f_{\mathrm{DH}}(298.15\,\mathrm{K}) \equiv 1$: at 25 °C the value is byte-identically $A = 0.51$, and only off 25 °C does the physics of a colder, denser, more polar solvent (a *larger* $A$, hence stronger non-ideality) appear. The $T^{-3/2}\varepsilon^{-3/2}$ form is the textbook Debye–Hückel scaling of the charging integral.

**What the code actually evaluates.** The `davies` kernel is deliberately tiny and glass-box (`AqueousActivity::evaluate`):

- it computes $A = 0.51\, f_{\mathrm{DH}}(T)$ once, then returns a closure $\gamma(z) = 10^{-Az^2(\sqrt{I}/(1+\sqrt{I})-0.3I)}$ evaluated with `std::pow(10.0,...)` (not $e^{x \ln 10}$, to keep the golden masters byte-stable);
- **neutral species** ($z = 0$) and the $I \le 0$ guard return $\gamma = 1$ exactly — there is *no* Setchenow salting-out term ($\log_{10} \gamma_{\mathrm{neutral}} = b\, I$) in this rung, stated honestly rather than faked;
- it is a pure function of $(|z|, I, A)$, so the solver freezes the whole $\gamma$ table at the iteration's $I$ (the Newton in log-molality stays linear in the activities — see the solution algorithm above).

**Worked example 8.1.** Davies vs. the limiting law for $Ca^{2+}$Take a brackish water at $I = 0.10$ mol/kg, 25 °C, so $A = 0.51$ and $\sqrt{I} = 0.316$. For a divalent ion ($z = 2$, $z^2 = 4$):

$$\frac{\sqrt{I}}{1+\sqrt{I}} - 0.3I = \frac{0.316}{1.316} - 0.030 = 0.240 - 0.030 = 0.210,$$

$$\log_{10} \gamma_{\mathrm{Ca}} = -0.51 \times 4 \times 0.210 = -0.428, \qquad \gamma_{\mathrm{Ca}} = 10^{-0.428} = 0.373.$$

The bare limiting law (139) would give $\log_{10} \gamma = -0.51 \times 4 \times 0.316 = -0.645$, i.e. $\gamma = 0.226$ — a *40 % lower* activity coefficient. The denominator and the $-0.3I$ tail together pull the divalent ion back from the unphysically strong screening the limiting law predicts; this is precisely the difference that decides whether $CaCO_3$ shows up as supersaturated ((144)) at the membrane wall. A monovalent ion at the same $I$ has only a quarter of the exponent, $\gamma_{\mathrm{Na}} = 10^{-0.107} = 0.78$ — the $z^2$ weighting in action.

**Power and limit.** *Power:* a single, parameter-free equation gives every ion a defensible $\gamma_i$ from its charge alone, with the correct $\sqrt{I}$ electrostatic limit and a sensible turn-up at moderate $I$ — the ideal teaching rung between "ideal solution" ($\gamma_i = 1$) and a fully fitted Pitzer. *Limit:* the $-0.3I$ tail is empirical, so Davies is trustworthy only to $I \approx 0.5$ mol/kg; the speciation solver therefore raises an explicit advisory once $I$ exceeds $\approx 0.7$ mol/kg and points the student at the multi-ion Pitzer HMW kernel (§6, [11]), which carries the salt-specific virial coefficients Davies lacks. Neutral species are ideal ($\gamma = 1$), and the calorimetric ($T$-derivative) accuracy of the slope is only as good as the solvent correlations (141). The model is honest about all three.

## 8.8 Activity model and water activity

The default single-ion $\gamma$ is the Davies equation (140) of §8.7, with $A \approx 0.51$ at 25 °C and the $\varepsilon_w(T), \rho_w(T)$ scaling (141) shared with the Pitzer kernel; it is trustworthy to $I \approx 0.5$ mol/kg, the solver raises an advisory above $I \approx 0.7$, and Pitzer HMW [11] is the documented replacement. Neutral species take $\gamma = 1$. The water activity is

$$\ln a_w \;=\; -\,M_w\, \phi \sum_i m_i, \tag{142}$$

with $M_w$ the molar mass of water in kg/mol and $\phi$ the osmotic coefficient — the definition (87). Under **Davies** the solver keeps the dilute approximation $\phi = 1$ (declared in the run header), consistent with Davies being indicative beyond $I \approx 0.5$ mol/kg. Under **Pitzer** $\phi$ is the *real* multi-ion osmotic coefficient of (117), built from the same virials as the $\gamma_i$, so the brine's water activity — the gypsum $a_w^2$ leg, the boiling-point elevation, the osmotic pressure — is Pitzer-consistent and not a $\phi = 1$ guess (§6.4).

## 8.9 Temperature dependence of $K$

Formation constants are anchored at 25 °C and corrected to the run temperature by a three-tier priority, *all* pinned so that $\log_{10} K(298.15\,\mathrm{K}) \equiv \log_{10} K_{25}$ exactly (the 25 °C run is byte-stable by construction):

1. **Analytic** (PHREEQC `-analytical_expression`): the fitted shape $\mathrm{ana}(T) = A_1 + A_2 T + A_3/T + A_4 \log_{10} T + A_5/T^2 + A_6 T^2$ is applied as an *anchored* correction, $\log_{10} K(T) = \log_{10} K_{25} + [\mathrm{ana}(T) - \mathrm{ana}(298.15)]$, so the published fit's absolute offset cancels and only its $T$-shape is borrowed; outside the fitted range the run announces "$K(T)$ extrapolated" (no silent extrapolation).
2. **van't Hoff** (constant $\Delta H_r$): $\log_{10} K(T) = \log_{10} K_{25} + \dfrac{\Delta H_r}{R_g \ln 10}\left(\dfrac{1}{298.15} - \dfrac{1}{T}\right)$.
3. **Flat**: a genuinely bare entry holds $K$ at its 25 °C value.

Each form is listed at verbosity $\ge 2$, so the student sees exactly which constants were temperature-corrected and how.

## 8.10 Open systems: the Henry gas pin

Everything above conserves every master total — a *closed* vessel. A water in contact with a large gas reservoir (the atmosphere, a blanketing gas) obeys a different constraint: the dissolved gas is set by gas–liquid equilibrium, not by

an inventory. Henry's law fixes the *activity* of the dissolved species,

$$a_{\text{dissolved}} \;=\; K_H(T)\, p_{\text{gas}}, \tag{143}$$

with $K_H$ from the gas–liquid catalogue (`chemistry/ (recordType gasLiquidEquilibrium)` , PHREEQC `PHASES` form, $\log_{10} K_H$ anchored and $T$-corrected exactly like every other constant of the previous subsection). The case declares the reservoir per gas:

```
atmosphere { pCO2 4.2e-4 atm;  O2 0.2095 atm;  N2 0.7808 atm; }
```

**The swap.** For each listed gas the solver drops ONE mole balance and puts equation (143) in its place — in log space, $\ln a_{\text{dissolved}} = \ln K_H + \ln p$, which is *linear* in the unknowns $x_j = \ln m_j$, so Newton absorbs it without ceremony. Which balance is dropped follows from what the gas dissolves *into*:

- a **computed species** — $CO_2$(g) dissolves to $CO_2$(aq), a mass-action species formed from the $HCO_3^-$ master: the *family's* balance ($HCO_3^-$) is replaced, and the whole carbonate ladder equilibrates with the reservoir. Total dissolved carbon becomes a solved *outcome*; the `totals` entry is only the initial guess (announced, and excluded from the charge-imbalance check — a guess is not an analysis);
- an **inert neutral master** — $O_2$, $N_2$, $CH_4$, $H_2$ dissolve to themselves (identity records `species/O2.dat`, `species/N2.dat`, … — `modelSpecies`, charge 0, so `formula` not `ion`): the pin replaces the master's *own* balance and the dissolved concentration is set by the gas phase alone.

Several gases pin simultaneously (one row each; two gases claiming the same master are refused). A gas whose dissolution spans several aqueous legs at once — $H_2S(g) = H^+ + HS^-$, because the neutral $H_2S$(aq) species is not yet curated — is refused loudly as a named gap rather than pinned approximately.

Two textbook numbers fall out of pure water open to air with no fitting anywhere (`rainwater_air`): carbonic acid from 420 ppm $CO_2$ alone gives rain its natural pH $\approx 5.6$, and the dissolved-oxygen saturation runs retrograde with temperature — ∼8.5 mg/L at 25 °C rising to ∼11 mg/L at 10 °C — which is why cold streams hold more oxygen than warm ones.

## 8.11 From speciation to saturation: the scaling link

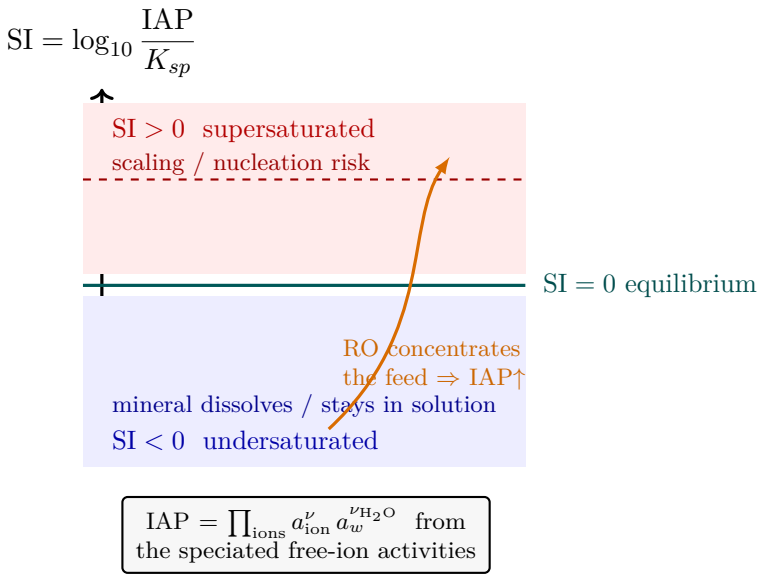



**Figure 20:** The saturation index turns speciated free-ion activities into a single scaling verdict. Once the solver knows every free-ion activity, the ion-activity product IAP is compared to the solubility product $K_{sp}(T)$: $\text{SI} = \log_{10}(\text{IAP}/K_{sp})$. Below the teal equilibrium line the mineral is undersaturated (blue, it dissolves or stays dissolved); above it the water is supersaturated (red, scaling and nucleation risk). As a reverse-osmosis stage concentrates the feed the activities climb and the operating point walks *up* the ladder (orange)—the `scalingScan` op traces exactly this trajectory, and the `equilibrate` option drops the precipitate once a mineral crosses SI = 0.

Once the free-ion activities are known, the *saturation index* of any mineral the case admits follows immediately (the $K_{sp}$ lives in that component's own `solidPhases{}` block — a phase belongs to the substance whose phase it is):

$$\text{SI} \;=\; \log_{10}\frac{\text{IAP}}{K_{sp}(T)}, \qquad \text{IAP} = \prod_{\text{ions}} a_{\text{ion}}^{\nu}\, a_w^{\nu_{H_2O}}, \tag{144}$$

$\mathrm{SI} < 0$ undersaturated, $= 0$ at equilibrium, $> 0$ supersaturated (scaling risk). The `scalingScan` op walks this index along an RO concentration trajectory; the `equilibrate` option goes further and finds the precipitated amounts by an *active-set* complementarity solve (admit the most-supersaturated phase, drive its SI to zero, evict any phase whose precipitated amount goes negative) layered on top of exactly the Newton above. Saturation indices are the bridge from the chemistry of this section to the membrane and crystalliser chapters.

**Power and limit.** The power: from seven total concentrations and a temperature the model recovers the entire distribution of dozens of species, a defensible pH (closed by physics, not by a stale measurement), and the saturation state of every candidate scale mineral — the same calculation a regulatory water-chemistry report rests on, with every $\log K$, every $\gamma$, and every Newton iteration visible. The limits are deliberate and announced: the Davies default is trustworthy only to $I \approx 0.5\,\mathrm{mol/kg}$ (use Pitzer beyond); the water activity uses the $\phi = 1$ dilute approximation; neutral species are ideal ($\gamma = 1$); a solved pH is no better than the charge balance of its input, which is precisely why $\varepsilon_{\mathrm{CB}}$ is reported and gated before the answer is offered.

**Worked example 8.2.** A brackish RO feed, pH solvedA brackish feed lists (mg/L) Ca 84, Mg 27, Na 363, K 12, Cl 510, $SO_4$ 250, $HCO_3$ 183 at 25 °C, with no reliable pH. Run with `pH solve;`. The solver first reports the charge-balance error $\varepsilon_{\mathrm{CB}} = 200\,|S_+ - S_-|/(S_+ + S_-)$ over the master equivalents; here it is a couple of percent (a clean analysis), well under the 5 % advisory. It then seeds $x_j = \ln T_j$, $x_{\mathrm{H}} = \ln 10^{-7}$, freezes Davies $\gamma$ at the initial $I$, and Newton-solves the seven mole balances plus electroneutrality in $\ln m$; recomputing $I$ and $\gamma$ and re-solving a few times lands $I \approx 0.03\,\mathrm{mol/kg}$ and a solved pH $\approx 7.72$ — close to the lab's reported 7.8, the charge balance vindicating the analysis. The carbonate now distributes into $HCO_3^-$, $CO_3^{2-}$ and $CaHCO_3^+$, and the calcite SI from (144) reads the scaling risk that drives the downstream RO audit (`tutorials/props/electrolyte/scaling_ro_brackish`).

**Runnable case.** `tutorials/props/electrolyte/scaling_ro_brackish` — the worked feed above, with analytic "pins" (pure-water $K_w$, gypsum-saturated $CaSO_4$, neutral pure water) that nail the solver against hand-checkable answers, then the feed speciated with `pH solve;` and a closed-vs-open-$CO_2$ scaling scan. `tutorials/props/electrolyte/softener_brackish` adds Gaines–Thomas cation exchange on the same Newton.

# 9 Fugacity and the vapour phase

*What you'll learn.* The activity coefficient $\gamma_i$ of the previous chapter corrects the liquid-side chemical potential for non-ideality. On the vapour side, the analogous correction is the *fugacity coefficient* $\varphi_i$. Together $(\gamma_i, \varphi_i)$ are the two multiplicative gauges that take the ideal-mixture reference of Chapter 1 all the way to real-fluid phase equilibrium. fixes $\varphi_i = 1$ (ideal-gas vapour), which is adequate at the operating pressures of the shipped tutorials; the cubic equations of state that produce $\varphi_i \neq 1$ at higher pressures are on the v1.x roadmap.

## 9.1 Why we need fugacity at all

The chemical potential of a real gas is not $\mu = \mu^0(T) + RT\ln(P/P_0)$. That expression is the ideal-gas case where the molecules do not interact. Real molecules feel attractive and repulsive forces; their chemical potential depends on more than just pressure in a logarithmic-of-ratio way.

G. N. Lewis (1901) defined the *fugacity* $f$ as the auxiliary variable that restores the simple logarithmic form for any fluid:

$$\boxed{\mu(T,P) \;=\; \mu^{0}(T) \;+\; RT\ln\frac{f(T,P)}{f^{0}}.} \tag{145}$$

Reference $f^0 = 1$ bar by convention. The defining condition is that $f \to P$ as $P \to 0$: in the dilute-gas limit, the fugacity becomes the ordinary pressure. The *fugacity coefficient* is the ratio

$$\varphi(T,P) \;\equiv\; \frac{f(T,P)}{P}, \tag{146}$$

with $\varphi \to 1$ as $P \to 0$ and $\varphi = 1$ identically for an ideal gas at any $P$. Deviations of $\varphi$ from unity measure how far the real gas departs from the ideal-gas equation of state.

*Intuition.* Fugacity is the engineering trick that lets every formula derived under the ideal-gas assumption survive in a real-gas world: replace $P$ with $f$ everywhere and the algebra continues to apply. All of the complexity of non-ideal molecular interactions is bundled into the single multiplicative correction $\varphi$, which is then computed from an equation of state. $\varphi < 1$ for fluids dominated by attraction (compression denser than ideal); $\varphi > 1$ for fluids dominated by repulsion (e.g. helium near $T_c$).

## 9.2 Fugacity of a pure liquid

For a pure substance at saturation $(T, P^{\mathrm{sat}}(T))$, liquid and vapour coexist and their chemical potentials are equal, so their fugacities are equal too:

$$f^{\mathrm{L}}(T,P^{\mathrm{sat}}) \;=\; f^{\mathrm{V}}(T,P^{\mathrm{sat}}) \;=\; \varphi^{\mathrm{sat}}(T)\,P^{\mathrm{sat}}(T). \tag{147}$$

For a pure liquid at a different $P$, the small pressure dependence is the *Poynting correction* $\exp\!\big[v^{\mathrm{L}}(P-P^{\mathrm{sat}})/(RT)\big]$, which is $\sim 1$ until $P$ exceeds several bar above $P^{\mathrm{sat}}$. At operating conditions the simulator drops the Poynting factor entirely:

$$f_i^{\mathrm{L},*}(T) \;\approx\; \varphi_i^{\mathrm{sat}}(T)\,P_i^{\mathrm{sat}}(T). \tag{148}$$

With the ideal-gas $\varphi_i^{\mathrm{sat}} = 1$ assumption, this collapses further to $f_i^{\mathrm{L},*} \approx P_i^{\mathrm{sat}}$.

## 9.3 Partial fugacity in mixtures

For component $i$ in a mixture, the partial fugacity $\hat{f}_i$ plays the role of an "effective partial pressure". In a gas mixture

$$\hat{f}_i^{\mathrm{V}}(T,P,\mathbf{y}) \;=\; y_i\,\hat{\varphi}_i(T,P,\mathbf{y})\,P, \tag{149}$$

where $\hat{\varphi}_i$ is the partial fugacity coefficient of $i$ in the mixture (note the hat: it depends on composition, unlike the pure-component $\varphi_i$). In an ideal-gas mixture $\hat{\varphi}_i = 1$ for every $i$, recovering Dalton's law $\hat{f}_i = y_i P$ that we already derived in Chapter 1.

In a liquid mixture the partial fugacity is built from the activity coefficient and the pure-component liquid fugacity:

$$\hat{f}_i^{\mathrm{L}}(T,P,\mathbf{x}) \;=\; x_i\,\gamma_i(T,\mathbf{x})\,f_i^{\mathrm{L},*}(T) \;\approx\; x_i\,\gamma_i\,\varphi_i^{\mathrm{sat}}\,P_i^{\mathrm{sat}}. \tag{150}$$

The two corrections separate cleanly: $\gamma_i$ absorbs *composition-driven* non-ideality (next-neighbour interactions in the liquid); $\varphi_i^{\mathrm{sat}}$ absorbs the *vapour-phase* non-ideality at the saturation pressure of the pure component. In both are taken at the simplest end ($\gamma_i$ from NRTL/Wilson but $\varphi_i^{\mathrm{sat}} = 1$).

## 9.4 Phase-equilibrium criterion in fugacity form

Chapter 2 laid out the criterion: $\mu_i^{\mathrm{L}} = \mu_i^{\mathrm{V}}$ for every component across the phase boundary. Substituting (145) into both sides and cancelling $\mu^0$ and $f^0$,

$$\boxed{\hat{f}_i^{\mathrm{L}}(T,P,\mathbf{x}) \;=\; \hat{f}_i^{\mathrm{V}}(T,P,\mathbf{y}) \quad \text{for every component } i.} \tag{151}$$

Substituting (149) and (150) produces the $\gamma$-$\varphi$ **formulation**:

$$x_i\,\gamma_i(\mathbf{x},T)\,\varphi_i^{\mathrm{sat}}(T)\,P_i^{\mathrm{sat}}(T) \;=\; y_i\,\hat{\varphi}_i(\mathbf{y},T,P)\,P. \tag{152}$$

This is the equation the manual's current Part I Chapter 2 presents out of context. Now you can read it: liquid-side fugacity equals vapour-side fugacity, both built up from the ideal-mixture baseline ($x_i$ or $y_i$ times a reference) times two real-fluid corrections (the $\gamma$'s and the $\varphi$'s).

Setting $\hat{\varphi}_i = \varphi_i^{\mathrm{sat}} = 1$ (the ideal-gas-vapour assumption) reduces (152) to

$$\boxed{x_i\,\gamma_i(\mathbf{x},T)\,P_i^{\mathrm{sat}}(T) \;=\; y_i\,P.} \tag{153}$$

Often called *modified Raoult's law*. It is the closed-form phase-equilibrium relation every flash, bubble-T, dew-T and distillation stage ultimately uses. Setting $\gamma_i = 1$ on top gives plain Raoult.

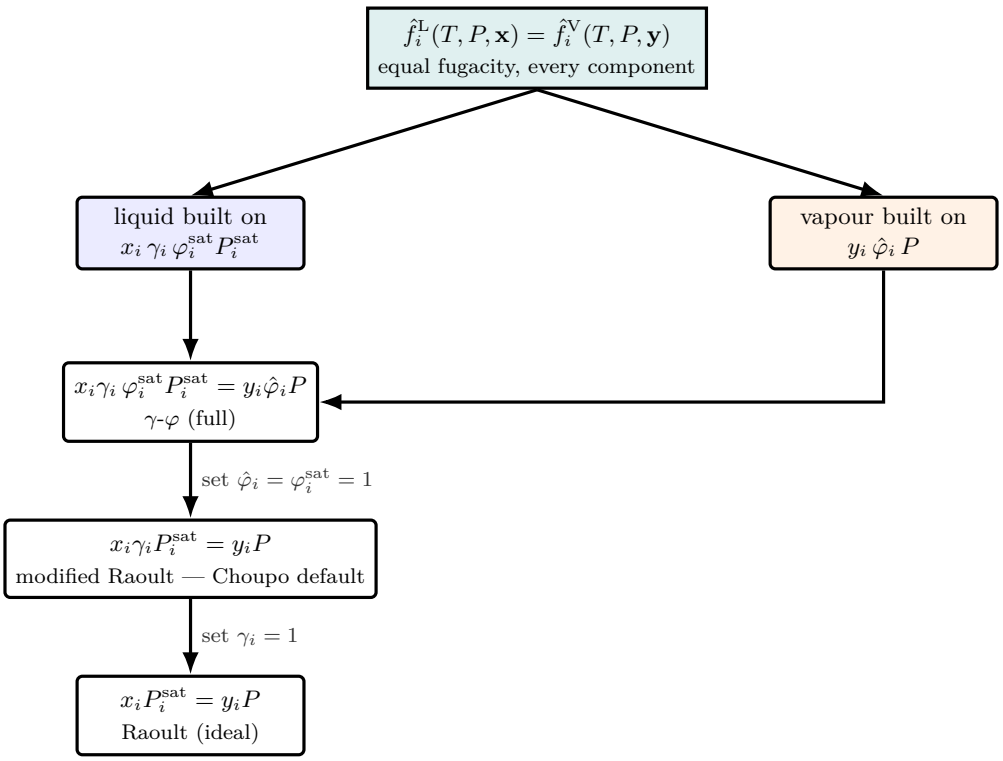



**Figure 21:** The fugacity route from the exact equilibrium criterion down to the closed form every flash actually solves. Both phases start from the same statement — liquid fugacity equals vapour fugacity — each built from an ideal-mixture baseline ($x_i$ or $y_i$) times real-fluid corrections (the $\gamma$'s on the liquid, the $\varphi$'s on the vapour). Dropping the two vapour corrections ($\hat{\varphi}_i = \varphi_i^{\mathrm{sat}} = 1$) collapses the $\gamma$-$\varphi$ formula to *modified Raoult's law* — the relation Choupo's `ThermoPackage::Kvec` evaluates — and dropping $\gamma_i$ too gives plain Raoult. Each downward step is an assumption made visible, not a hidden default.

## 9.5 When $\varphi_i = 1$ fails

Three regimes where the ideal-gas simplification breaks down, and the simulator quietly mispredicts:

- **High pressure** ($P \gtrsim 10$ bar for most species, $\gtrsim 1$ bar for water near $T_c$). Attractive interactions condense the gas below ideal density; $\varphi < 1$ by 5–20%, pushing the predicted bubble pressure of a flash low by a similar margin.
- **Dimerising vapours** (acetic acid, formic acid, nitrogen tetroxide). The vapour contains an equilibrium fraction of dimers, halving the apparent number of moles. No EoS short of an explicit dimerisation model gets this right. $\omega = 0.47$ for acetic acid is the indirect symptom; the apparent $\varphi$ for the monomer can sit around 0.5–0.7.
- **Supercritical and near-critical**. The very concept of a vapour breaks down, $\varphi$ varies dramatically with small changes in $T$ and $P$, and cubic EoS take over from the simpler approximations.

## 9.6 Where $\varphi_i$ comes from when not 1

When the ideal-gas assumption breaks down, Choupo replaces the `idealGas` block with a cubic equation of state for the vapour:

$$P \;=\; \frac{RT}{v-b} \;-\; \frac{a(T)}{v^2 + u\,b\,v + w\,b^2},$$

parametrised by $T_c$, $P_c$ and $\omega$ for each species (the data files already carry these; see Chapter 1). Soave-Redlich-Kwong (SRK)[26]: $u = 1, w = 0$. Peng-Robinson (PR)[27]: $u = 2, w = -1$. Both produce $\hat{\varphi}_i$ in closed form from the mixture parameters $a_{\mathrm{mix}}, b_{\mathrm{mix}}$ via the standard derivation $\ln \hat{\varphi}_i = (\partial(n\,\bar{A}^{\mathrm{R}}/RT)/\partial n_i)_{T,V,n_{j\neq i}}$, where $\bar{A}^{\mathrm{R}}$ is the residual Helmholtz energy. The next chapter (Chapter 10) derives the SRK form Choupo implements end to end — the parameters, the cubic in $Z$, the fugacity coefficient, and the enthalpy/entropy departure functions — and contrasts it with Peng-Robinson.

The point of carrying $T_c$, $P_c$ and $\omega$ in every data file: those three numbers *become* the cubic EoS the moment a case selects `model SRK;` (or `PR;`). The thermodynamic stack is forward-compatible by design.

## 9.7 The $\gamma$-$\varphi$ implementation

The simulator's `ThermoPackage::Kvec` computes

$$K_i(\mathbf{x}, \mathbf{y}, T, P) \;=\; \frac{\gamma_i(\mathbf{x}, T)\, P_i^{\mathrm{sat}}(T)}{\varphi_i(\mathbf{y}, T, P)\, P} \tag{154}$$

exactly as Eq. (153), with $\gamma_i$ from NRTL/Wilson/UNIQUAC (or 1 for the ideal-solution model) and $\varphi_i$ from the declared `equationOfState` block — the `idealGas` model simply returns $\varphi_i = 1$ at run time, and selecting `SRK` puts the real-vapour correction into the same formula with no other change. For the systems where the liquid too is EoS-priced (light gases, no meaningful $P^{\mathrm{sat}}$ anchor), a second VLE world is declared as `vleWorld phiPhi;`: $K_i = \hat{\varphi}_i^{L}(\mathbf{x})/\hat{\varphi}_i^{V}(\mathbf{y})$, the same cubic evaluated in both phases (Chapter 10).

## 9.8 Henry's law: the dilute-solute K-value

Modified Raoult's law, Eq. (153), anchors the liquid-phase fugacity of component $i$ on the *pure liquid* $i$: the reference is $x_i \to 1$, where $\gamma_i \to 1$ and the liquid escapes at its own saturation pressure $P_i^{\mathrm{sat}}(T)$. That reference is useless for a gas that is only sparingly soluble. A bubble of oxygen, $CO_2$ or $NH_3$ never *is* a liquid at the temperature of the water it dissolves in — $P_{\mathrm{O_2}}^{\mathrm{sat}}(298\ \mathrm{K})$ does not even exist below the critical point — so the pure-liquid anchor is a fiction. The physically natural reference for a solute is the *opposite* limit, $x_i \to 0$.

**The constant.** In a sufficiently dilute solution the partial pressure of the solute is linear in its liquid mole fraction. This is **Henry's law** (Henry, *Phil. Trans. R. Soc.* 1803):

$$\boxed{\;p_i \;=\; H_i(T)\, x_i, \qquad x_i \to 0.\;} \tag{155}$$

$H_i$ is the **Henry's-law constant** (here in pressure per mole-fraction units, Pa — Choupo's $H_{xp}$ convention). It is the slope of the $p_i$–$x_i$ curve at the origin, exactly as $P_i^{\mathrm{sat}}$ is the slope at $x_i = 1$ under Raoult. The two pictures are the two tangents of the same real $p_i(x_i)$ curve, taken at its two ends; for a species that obeys $\gamma_i^{\infty} \neq 1$ they relate through the infinite-dilution activity coefficient,

$$H_i(T) \;=\; \gamma_i^{\infty}(T)\, P_i^{\mathrm{sat}}(T), \tag{156}$$

which is why a large positive deviation ($\gamma_i^{\infty} \gg 1$, the solute "dislikes" the solvent) gives a large $H_i$ and a low solubility. Eq. (156) is also the bridge a curator uses to cross-check a tabulated $H_i$ against an activity model.

**The K-value.** Combining $p_i = y_i P$ (Dalton, ideal vapour) with Eq. (155) gives the equilibrium ratio in the same $K_i = y_i/x_i$ form every other model produces:

$$\boxed{\;K_i \;=\; \frac{y_i}{x_i} \;=\; \frac{H_i(T)}{P}.\;} \tag{157}$$

Set side by side with the $\gamma$-$\varphi$ result $K_i = \gamma_i\, P_i^{\mathrm{sat}}/(\varphi_i P)$, the only change is the *reference fugacity*: $P_i^{\mathrm{sat}}$ (pure-liquid anchor) is replaced by $H_i$ (infinite-dilution anchor), and $\gamma_i$ drops out because it has been folded into $H_i$ at $x_i \to 0$ via Eq. (156). Both leave $K_i \propto 1/P$ — a solute is always squeezed into the liquid by raising the pressure.

**Temperature dependence: van't Hoff.** Dissolution is a phase change with a molar enthalpy of dissolution $\Delta H_{\mathrm{diss}}$ (the heat released when one mole of solute leaves the gas and enters the dilute liquid). Applying the van't Hoff relation to the dissolution equilibrium constant $K_{\mathrm{sol}} = 1/H$,

$$\frac{\mathrm{d}\ln K_{\mathrm{sol}}}{\mathrm{d}T} = \frac{\Delta H_{\mathrm{diss}}}{RT^2} \quad\Longrightarrow\quad \frac{\mathrm{d}\ln H}{\mathrm{d}T} = -\,\frac{\Delta H_{\mathrm{diss}}}{RT^2},$$

and integrating from a reference temperature $T_{\mathrm{ref}}$ with $\Delta H_{\mathrm{diss}}$ taken constant over the (narrow) validity window gives the form Choupo implements:

$$\boxed{H(T) \;=\; H_{\mathrm{ref}}\,\exp\!\left[\frac{\Delta H_{\mathrm{diss}}}{R}\left(\frac{1}{T}-\frac{1}{T_{\mathrm{ref}}}\right)\right].} \tag{158}$$

The sign deserves care, and the source comment in `HenrysLaw::H` flags a historical bug here. For an *exothermic* dissolution $\Delta H_{\mathrm{diss}} < 0$ (the common case — $CO_2$, $NH_3$, $H_2S$, $O_2$ all release heat on dissolving). For $T > T_{\mathrm{ref}}$ both factors in the exponent are negative, so $H$ *rises*: the gas becomes *less* soluble when the liquid is warmed. This is the everyday fact that warm soda goes flat and that ammonia boils out of warm water — and it is why a *stripper* runs hot and an *absorber* runs cold (Chapter 16).

**Worked example: $CO_2$ in water.** The standard catalogue ships `parameters/Henry/CO2-water.dat` with $H_{\mathrm{ref}} = 1.64\times10^{8}$ Pa (= 1640 bar) at $T_{\mathrm{ref}} = 298.15$ K and $\Delta H_{\mathrm{diss}} = -19\,400$ J mol$^{-1}$ (Sander, 2015). At 25 °C and $P = 1$ atm $= 1.013\times10^{5}$ Pa,

$$K_{\mathrm{CO_2}} = \frac{H_{\mathrm{ref}}}{P} = \frac{1.64\times10^{8}}{1.013\times10^{5}} \approx 1.6\times10^{3} \quad (\gg 1:\ \text{strongly favours the vapour}).$$

Equivalently, the liquid mole fraction in equilibrium with a pure $CO_2$ atmosphere is $x = p/H = 1.013\times10^{5}/1.64\times10^{8} \approx 6\times10^{-4}$ — about 0.034 mol L$^{-1}$, the familiar carbonation figure. Warming to 40 °C (313.15 K),

$$H(313.15) = 1.64\times10^{8}\,\exp\!\left[\tfrac{-19400}{8.314}\left(\tfrac{1}{313.15}-\tfrac{1}{298.15}\right)\right] \approx 2.39\times10^{8}\ \text{Pa},$$

a 46% rise in $H$ over a 15 K warming — the soda goes flat.

**In the simulator.** `ThermoPackage::Kvec` branches to the Henry path for a component whose data file declares `role solute;` *and* for which a `(solute, solvent)` pair file exists for the case's named solvent; it then equates the FULL unsymmetric-convention fugacities (Prausnitz, *Molecular Thermodynamics*, Ch. 10),

$$y_i\,\varphi_i^{V}\,P \;=\; x_i\,\gamma_i^{*}\,H_i(T)\,\exp\!\left[\frac{\bar v_i^{\infty}\,(P-P_{\mathrm{solv}}^{\mathrm{sat}})}{RT}\right] \quad\Longrightarrow\quad K_i \;=\; \frac{\gamma_i^{*}\,H_i(T)\,\mathcal{P}}{\varphi_i^{V}\,P}, \tag{159}$$

with three physically distinct corrections on top of the textbook $H_i/P$ (each optional in the pair file, each defaulting to the dilute low-pressure limit):

- $\varphi_i^{V}$ — the *vapour* fugacity coefficient from the package EoS: the gas phase is real at 200 bar.
- $\mathcal{P} = \exp[\bar v_i^{\infty}(P - P^{\mathrm{sat}})/RT]$ — the **Poynting** term of **Krichevsky–Kasarnovsky** (1935) [28]: compressing the *liquid* raises the dissolved gas's fugacity through its partial molar volume $\bar v_i^{\infty}$ (`v_inf` in the pair file).
- $\gamma_i^{*}$ — the **unsymmetric activity coefficient** of **Krichevsky–Ilinskaya** (1945), normalised so $\gamma^{*}\to1$ as $x_i\to0$; a two-suffix Margules, $\ln\gamma_i^{*} = (A/RT)(x_{\mathrm{solv}}^{2}-1)$ (`margulesA`). This is where the liquid's activity model lives in the Henry world — not NRTL on the solvent, but the solute's departure from its own dilute limit.

**Worked fit: hydrogen in ammonia to 1000 atm.** Plot Wiebe & Tremearne's isotherms as $\ln(f_2/x_2)$ against $P$ (the KK coordinates) and the data collapse onto straight lines: slope $\bar v_{\mathrm{H_2}}^{\infty} = 24$ cm$^3$ mol$^{-1}$ (25 and 50 °C agree), intercept $H(298) = 1.18\times10^{9}$ Pa — the apparent "drift of $H$ with pressure" IS the Poynting term, and the two-constant line reproduces the full 1000-atm isotherm to ∼2% ($x_{\mathrm{H_2}}$ at 400 atm: computed 0.0287 vs. measured 0.0282). The same plot on Larson & Black's nitrogen data gives $\bar v_{\mathrm{N_2}}^{\infty} = 23$ cm$^3$ mol$^{-1}$. Both fits keep $\gamma^{*} = 1$: hydrogen stays on the KK line up to $x\sim9\%$, so the Margules slot is available but the data do not demand it. Everything downstream (flash, the absorber/stripper tridiagonal of Chapter 16) consumes that $K_i$ identically — the Henry model is invisible to the stage solver, which only ever sees $K_i$. The same data file also feeds the heat-of-absorption energy balance: the absorber reads $\Delta H_{\mathrm{abs}} = -\Delta H_{\mathrm{diss}}$ straight from the pair file (`Absorber.cpp`), so the constant that sets the solubility *and* the constant that warms the column are one number, never two.

**Supercritical solutes: the ammonia loop.** Henry's law is not merely convenient at high pressure — above a solute's critical temperature it is the *only* honest anchor. In the ammonia synthesis loop the separator condenses $NH_3$ at 250 K and 200 bar with $H_2$ ($T_c = 33$ K) and $N_2$ ($T_c = 126$ K) far above their critical points: neither HAS a vapour pressure there, so the Raoult anchor $P_i^{\mathrm{sat}}$ does not exist, and extrapolating an Antoine fit above $T_c$ manufactures a number with no physical meaning (it once predicted 13% dissolved $H_2$ — forty times the measured value). The shipped pairs are fitted to the classical syntheses-loop measurements: `H2-NH3` to Wiebe & Tremearne (1934), and `N2-NH3` to Larson & Black (1925) as critically evaluated in the IUPAC Solubility Data Series (Vol. 5/6), whose evaluation confirms the two gases dissolve *independently*, each linear in its partial pressure — exactly the independent-Henry-pairs model. Both carry $\Delta H_{\mathrm{diss}} > 0$: unusually, hydrogen and nitrogen are *more* soluble in hot ammonia. With these constants the separator liquid carries ∼0.5% $H_2$ + ∼0.2% $N_2$ — the physical numbers (`tutorials/plant/ammonia02_full_plant`).

**Power and limit.** The power: with two numbers ($H_{\mathrm{ref}}$, $\Delta H_{\mathrm{diss}}$) per pair, Henry's law gives a quantitative, temperature-correct K-value for any sparingly soluble gas — precisely the regime where the pure-liquid Raoult anchor does not exist and a cubic EoS would be a heavy, poorly constrained hammer. The limit: Eq. (155) is the *tangent at* $x_i \to 0$. It is accurate only in the dilute limit; for a water-miscible species (an alcohol, ketone, amine, or acid that spans the full composition range) the linear law is just the $x_i \to 0$ end of a curved $p_i(x_i)$, and an activity-coefficient model (NRTL/UNIQUAC, §2.3–§2.5) is the right tool. For strong electrolytes or high loadings, switch to a Pitzer / eNRTL model (Chapter 2). The shipped constants are infinite-dilution values from Sander's compilation (Sander, R., *Atmos. Chem. Phys.* **15** (2015) 4399–4981, CC-BY), valid roughly 273–373 K near 1 atm; $\Delta H_{\mathrm{diss}}$ is taken constant across that window, so do not extrapolate far beyond it.

## 9.9 Why the simulator uses two solver layers, not one

In code, Eq. (153) is a strong coupling between $\mathbf{x}$, $\mathbf{y}$, $T$, and $P$. Solving it in one shot with full Newton on the joint variable would work in principle, but the simulator splits the work into two nested layers, exploiting the fact that $\gamma_i$ depends on the liquid composition only:

1. **Outer loop on $\gamma_i$.** Hold $\gamma_i$ fixed at the current estimate. Then the $K_i = \gamma_i P_i^{\mathrm{sat}}/P$ are also fixed and the phase-fraction-plus-composition problem becomes a single scalar equation in the vapour fraction $V$ (the Rachford-Rice equation; see Chapter 13). Solve it by Newton-1D.
2. **Recompute $\gamma_i$.** With the updated $\mathbf{x}$ from step 1, evaluate $\gamma_i(\mathbf{x}, T)$ from NRTL or Wilson. Go back to step 1. Wegstein (Chapter 5) accelerates this outer loop.

The reason this beats a joint Newton: $\gamma_i$ is smooth in $\mathbf{x}$ but mildly non-monotone — a finite-difference Jacobian on a joint $(V, \mathbf{x}, \mathbf{y})$ system is expensive to build per iteration and easy to mis-step on. The inner Rachford-Rice problem, in contrast, is one-dimensional in $V$ with a strictly monotone residual; Newton-1D nails it in $\sim 5$ iterations. Splitting buys cost-per-iteration plus robustness, at the price of an outer convergence that needs Wegstein to be fast.

> **Runnable case.** In any tutorial's `constant/thermoPhysPropDict` you will find, inside the `equilibrium` block:
>
> ```
> vapour { fugacityModel idealGas; }
> ```
>
> That single line is the assumption $\varphi_i = 1$. Switching to a real-gas vapour is a one-word edit — the line becomes `fugacityModel SRK;` or `fugacityModel PR;` — and nothing else in the case file changes. Forward compatibility by design.

# 10 The SRK cubic equation of state

*What you'll learn.* How Choupo turns the three numbers $(T_c, P_c, \omega)$ into a real-fluid vapour model: the Soave-Redlich-Kwong $a(T,\omega)$ and $b$, the Cardano solution of the cubic in $Z$, the fugacity coefficient $\hat{\varphi}_i$ that replaces the ideal-gas $\varphi_i = 1$, and the enthalpy/entropy *departure functions* $H^{\mathrm{R}}, S^{\mathrm{R}}$ that feed the energy balance at elevated pressure — all in closed form, exactly as `src/thermo/equationOfState/SRK.cpp` computes them. Then where Peng-Robinson differs by one pair of constants.

## 10.1 From van der Waals to Soave

Every cubic EoS descends from van der Waals (1873): a repulsive term $R_{\mathrm{g}}T/(v-b)$ that keeps molecules from overlapping (the co-volume $b$), and an attractive term $-a/v^2$ that pulls them together. Redlich and Kwong (1949) sharpened the attractive denominator to $v(v+b)$ and made $a$ scale as $T^{-1/2}$. Soave's 1972 contribution [26] was to replace that fixed temperature scaling with a fitted function $\alpha(T;\omega)$ tied to the acentric factor, so the EoS reproduces pure-component vapour pressures along the whole saturation curve — the single change that made cubic EoS accurate enough for VLE. The result is the Soave-Redlich-Kwong (SRK) equation Choupo implements:

$$P \;=\; \frac{R_{\mathrm{g}}T}{v-b} \;-\; \frac{a(T)}{v\,(v+b)}. \tag{160}$$

This is the $u=1,\ w=0$ member of the generalised cubic $P = R_{\mathrm{g}}T/(v-b) - a(T)/(v^2 + u\,b\,v + w\,b^2)$ introduced in Chapter 9.

## 10.2 Pure-component parameters

The co-volume $b$ and the critical attraction $a_c$ come from forcing Eq. (160) to satisfy the critical conditions $(\partial P/\partial v)_{T_c} = (\partial^2 P/\partial v^2)_{T_c} = 0$. For SRK this fixes the two universal constants $\Omega_a = 0.42747$ and $\Omega_b = 0.08664$:

$$a_{c,i} = \Omega_a \,\frac{R_{\mathrm{g}}^2 T_{c,i}^2}{P_{c,i}} = 0.42747\,\frac{R_{\mathrm{g}}^2 T_{c,i}^2}{P_{c,i}}, \tag{161}$$

$$b_i = \Omega_b \,\frac{R_{\mathrm{g}} T_{c,i}}{P_{c,i}} = 0.08664\,\frac{R_{\mathrm{g}} T_{c,i}}{P_{c,i}}. \tag{162}$$

The temperature dependence lives entirely in Soave's $\alpha$:

$$a_i(T) \;=\; a_{c,i}\,\alpha_i(T), \qquad \alpha_i(T) = \left[\,1 + m_i\,(1 - \sqrt{T/T_{c,i}})\,\right]^2, \tag{163}$$

with the Soave slope a quadratic in the acentric factor [26]:

$$m_i \;=\; 0.48508 + 1.55171\,\omega_i - 0.15613\,\omega_i^2. \tag{164}$$

Note $\alpha_i(T_{c,i}) = 1$ exactly (the curve is anchored at the critical point), and $\alpha$ rises as $T$ falls below $T_c$, strengthening attraction in the subcritical region where the vapour pressure must be matched.[3]

The temperature derivative $\mathrm{d}a_i/\mathrm{d}T$ — needed for the departure functions below — follows by the chain rule on Eq. (163):

$$\frac{\mathrm{d}a_i}{\mathrm{d}T} = -\,a_{c,i}\,\frac{m_i\left[1 + m_i(1-\sqrt{T/T_{c,i}})\right]}{\sqrt{T\,T_{c,i}}}. \tag{165}$$

## 10.3 Mixing rules

A mixture needs single effective $a_{\mathrm{mix}}, b_{\mathrm{mix}}$. Choupo uses the classical van der Waals one-fluid rules with a binary interaction parameter $k_{ij}$:

$$a_{\mathrm{mix}}(T) = \sum_i \sum_j y_i\,y_j\,(1-k_{ij})\,\sqrt{a_i(T)\,a_j(T)}, \tag{166}$$

$$b_{\mathrm{mix}} = \sum_i y_i\,b_i. \tag{167}$$

The $k_{ij}$ are small empirical corrections (zero by default, symmetric, $k_{ii}=0$) read from the optional `binaryInteractions` sub-list. They are the only fitted mixture quantity; everything else is pure-component. The mixture derivative required later is

$$\frac{\mathrm{d}a_{\mathrm{mix}}}{\mathrm{d}T} = \sum_i \sum_j y_i y_j (1-k_{ij})\,\tfrac{1}{2}\left(\sqrt{\tfrac{a_j}{a_i}}\,\frac{\mathrm{d}a_i}{\mathrm{d}T} + \sqrt{\tfrac{a_i}{a_j}}\,\frac{\mathrm{d}a_j}{\mathrm{d}T}\right). \tag{168}$$

[3] Choupo uses the original Soave slope of Eq. (164); some texts quote the rounded $0.480 + 1.574\,\omega - 0.176\,\omega^2$. The difference is within the model's own accuracy.

## 10.4 The cubic in $Z$ and its Cardano root

Writing $Z = Pv/R_{\mathrm{g}}T$ and the dimensionless groups

$$A = \frac{a_{\mathrm{mix}}\,P}{(R_{\mathrm{g}}T)^2}, \qquad B = \frac{b_{\mathrm{mix}}\,P}{R_{\mathrm{g}}T}, \tag{169}$$

Eq. (160) rearranges to the cubic

$$\boxed{Z^3 - Z^2 + (A - B - B^2)\,Z - A\,B = 0} \tag{170}$$

Choupo solves Eq. (170) *analytically* by Cardano's method (`cardano_root`), not by iteration — a cubic has a closed-form solution, so there is no Newton loop and no initial guess to get wrong. The discriminant $\Delta = R_d^2/4 + Q_d^3/27$ of the depressed cubic decides the branch: $\Delta > 0$ gives one real root (single fluid phase); $\Delta \le 0$ gives three real roots, evaluated through the trigonometric form $w_k = 2\sqrt[3]{r}\cos\!\big((\phi + 2\pi k)/3\big)$. When three roots exist the *largest above $B$* is the vapour root (largest molar volume); the smallest is the liquid root. The $\gamma$-$\varphi$ path keeps the **vapour root** (`SRK::phi`) as the $\hat{\varphi}_i$ supplier for the gas phase; the phi-phi world (`vleWorld phiPhi;`, Chapter 9) also takes the **liquid root** (`SRK::phiLiquid`) for its $K_i = \hat{\varphi}_i^L/\hat{\varphi}_i^V$ ratio.

## 10.5 The fugacity coefficient

Differentiating the residual Helmholtz energy with respect to mole number (the recipe of Chapter 9) gives the SRK partial fugacity coefficient in closed form:

$$\boxed{\ln\hat{\varphi}_i = \frac{b_i}{b_{\mathrm{mix}}}(Z-1) - \ln(Z-B) - \frac{A}{B}\left(\frac{2\sum_j y_j(1-k_{ij})\sqrt{a_i a_j}}{a_{\mathrm{mix}}} - \frac{b_i}{b_{\mathrm{mix}}}\right)\ln\!\left(1+\frac{B}{Z}\right)} \tag{171}$$

This is exactly the expression in `SRK::phi`; the inner sum $\sum_j y_j(1-k_{ij})\sqrt{a_i a_j}$ is the $i$-th row of the same matrix that built $a_{\mathrm{mix}}$, so it costs nothing extra. In the ideal-gas limit ($A, B \to 0$, $Z \to 1$) every term vanishes and $\hat{\varphi}_i \to 1$, recovering the assumption the tutorials ship with.

## 10.6 Departure (residual) functions for energy balances

The flash gives compositions; the *energy* balance needs how far real-gas enthalpy and entropy sit from the ideal-gas reference at the same $T, P$. These are the *departure* (residual) functions. For the SRK form ($\sigma = 1, \varepsilon = 0$ in the generalised cubic), Choupo evaluates [29]

$$H^{\mathrm{R}}(T,P,\mathbf{y}) = R_{\mathrm{g}}T\,(Z-1) + \frac{T\,\dfrac{\mathrm{d}a_{\mathrm{mix}}}{\mathrm{d}T} - a_{\mathrm{mix}}}{b_{\mathrm{mix}}}\,\ln\frac{Z+B}{Z}, \tag{172}$$

$$S^{\mathrm{R}}(T,P,\mathbf{y}) = R_{\mathrm{g}}\,\ln(Z-B) + \frac{1}{b_{\mathrm{mix}}}\,\frac{\mathrm{d}a_{\mathrm{mix}}}{\mathrm{d}T}\,\ln\frac{Z+B}{Z}. \tag{173}$$

These are `SRK::H_residual` and `SRK::S_residual` line for line. The real-fluid enthalpy used in the energy ledger is then the ideal-gas enthalpy (from the heat-capacity integral of Chapter 5) *plus* $H^{\mathrm{R}}$; likewise for entropy. Both departures vanish as $P \to 0$ ($Z \to 1$, $B \to 0$), so the high-pressure correction switches itself off where it is not needed — the same self-consistency as $\hat{\varphi}_i \to 1$.

## 10.7 Peng-Robinson: the same machine, two constants

Peng-Robinson (1976) [27] keeps the entire structure above and changes only the attractive denominator to $v^2 + 2bv - b^2$, i.e. the generalised-cubic constants $\varepsilon = 1 - \sqrt{2}$, $\sigma = 1 + \sqrt{2}$ (so $u = 2, w = -1$). Three substitutions convert SRK into PR:

| | **SRK** | **Peng-Robinson** |
|---|---|---|
| $\Omega_a$ | $0.42747$ | $0.45724$ |
| $\Omega_b$ | $0.08664$ | $0.07780$ |
| $m(\omega)$ / $\kappa(\omega)$ | $0.48508 + 1.55171\omega - 0.15613\omega^2$ | $0.37464 + 1.54226\omega - 0.26992\omega^2$ $\quad(\omega \le 0.491)$<br>$0.379642 + 1.48503\omega - 0.164423\omega^2 + 0.016666\omega^3$ $\quad(\omega > 0.491)$ |
| attractive term | $a/[v(v+b)]$ | $a/[v^2 + 2bv - b^2]$ |
| ln factor | $\ln\dfrac{Z+B}{Z}$ | $\dfrac{1}{2\sqrt{2}}\ln\dfrac{Z+(1+\sqrt{2})B}{Z+(1-\sqrt{2})B}$ |

In the code this is literally the difference between `SRK.cpp` and `PR.cpp`: the constants `OMEGA_A`, `OMEGA_B`, the slope function, and the single logarithm `ln_genB`. PR's third critical condition forces $Z_c = 0.307$ versus SRK's $0.333$ — both above the true $Z_c \approx 0.27$ of most fluids, but PR's is closer, which is why PR gives better *liquid* densities.

SRK is often marginally better for light-gas *vapour* fugacities. Choupo ships both; the case picks one with a single keyword.[4]

**Worked example 10.1.** SRK $a, b$ for $CO_2$ at 320 KWith $T_c = 304.13$ K, $P_c = 73.77$ bar $= 7.377 \times 10^6$ Pa, $\omega = 0.225$ and $R_g = 8.314$ J mol$^{-1}$K$^{-1}$:

$$b = 0.08664 \frac{8.314 \times 304.13}{7.377 \times 10^6} = 2.97 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1},$$

$$a_c = 0.42747 \frac{8.314^2 \times 304.13^2}{7.377 \times 10^6} = 0.370 \text{ Pa m}^6 \text{ mol}^{-2}.$$

The Soave slope $m = 0.48508 + 1.55171(0.225) - 0.15613(0.225)^2 = 0.826$. At $T = 320$ K, $T_r = 1.052$, so $\sqrt{T_r} = 1.026$ and $\alpha = [1 + 0.826(1 - 1.026)]^2 = 0.958$, giving $a(320) = 0.355$ Pa m$^6$ mol$^{-2}$. At $P = 50$ bar these yield $A = aP/(R_gT)^2 = 0.251$, $B = bP/(R_gT) = 0.0558$, and the Cardano vapour root of Eq. (170) is $Z \approx 0.78$ — a 22% departure from the ideal gas, exactly the regime where keeping `model idealGas` would corrupt both the energy balance and the fugacity coefficient.

**Power and limit.** Three numbers per species $(T_c, P_c, \omega)$ and one optional $k_{ij}$ per pair buy a closed-form real-gas model: $\hat{\varphi}_i \neq 1$, real $Z$, and the $H^R, S^R$ departures that make the energy balance correct at tens of bar. No iteration (Cardano is analytic), no per-species curve fitting beyond the acentric factor, and the exact same algebra serves SRK and PR. The **limit**: cubic EoS are vapour/non-polar workhorses. They are poor for strongly hydrogen-bonded or associating fluids (water, alcohols, acids — where the activity-coefficient $\gamma$-route of Chapter 2 is the right tool for the liquid), and the van der Waals one-fluid mixing rule degrades for very asymmetric or polar mixtures. Choupo's cubics serve the *fugacity* routes (the vapour root under $\gamma$-$\varphi$, both roots under the phi-phi world); liquid *density* still comes from Rackett, it is not (yet) a full-VLLE EoS engine, and a single $k_{ij}$ cannot rescue a system the underlying $\alpha$-function gets wrong. As always: all models are wrong, some are useful — SRK/PR earn their keep precisely where the ideal gas stops being so and association has not yet taken over.

**Runnable case.** Select the cubic in any case's `constant/thermoPhysPropDict`: a real-gas vapour inside `gammaPhi` is `vapour { fugacityModel SRK; }` (or `PR`), and the full $\varphi$–$\varphi$ world declares the ONE equation of state both phase slots route through:

```
equilibrium
{
    formulation phiPhi;

    equationOfState
    {
        model      SRK;                  // or  PR;
        mixingRule vanDerWaalsOneFluid;
        binaryInteractions               // optional; k_ij = 0 announced
        {
            N2-CH4 { source "data/standards/parameters/SRK/N2-CH4.dat"; }
        }
    }

    liquid { fugacityRoute equationOfState; root liquid; }
    vapour { fugacityRoute equationOfState; root vapour; }
}
```

The components must carry `Tc`, `Pc` and `omega` in their `.dat` files (the loader rejects the EoS otherwise — see `requiredComponentProperties`).

[4] Choupo's PR is the 1978 form (PR78): the slope $\kappa(\omega)$ has the two branches in the table, the 1976 polynomial up to $\omega = 0.491$ and the cubic of Robinson & Peng (1978, GPA RR-28) above it, as the PPR78 literature restates them (Jaubert, Privat & Mutelet, *AIChE J.* 56 (2010) 3225, Eq. 2). Below the break the two coincide, so a light mixture is unchanged; a component above it (methanol, $\omega = 0.565$; the $C_{11+}$ fractions of a crude) is announced on the run with the slope it received (`PR.cpp`). The PPR78 group-contribution $k_{ij}$ are defined on this slope, which is why the model is not an option.

# 11 COSMO-SAC: an activity coefficient from a charge surface

Every activity model in the preceding sections describes a molecule by numbers fitted to how it *behaves*: an interaction energy, a group count, a pair parameter. COSMO-SAC (Lin & Sandler, *Ind. Eng. Chem. Res.* 41 (2002) 899)[30] starts one level down. A quantum-chemistry calculation places the molecule in a **perfect conductor**, which screens its charge exactly, and the screening charge density $\sigma$ is read off the cavity surface. Choupo does *not* perform that calculation: it consumes the result.

**The $\sigma$-profile.** The geometry is discarded and only the statistics kept. $p_i(\sigma)$ is the fraction of molecule $i$'s cavity area carrying screening charge density $\sigma$, on a fixed grid of 51 points from $-0.025$ to $0.025\,\mathrm{e/\AA^2}$. With the cavity area $A_i$ and volume $V_i$ that profile is the *entire* input: no binary data, no group table. The price of discarding the geometry is that two molecules whose profiles coincide cannot be told apart.

**Restoring reality.** In the conductor every segment was screened perfectly; in a liquid a segment of density $\sigma_m$ must sit against one of density $\sigma_n$, and unless they cancel that costs energy,

$$\Delta W(\sigma_m,\sigma_n) = \tfrac{1}{2}\alpha'(\sigma_m+\sigma_n)^2 + c_{\mathrm{HB}}\max(0,\sigma_{\mathrm{acc}}-\sigma_{\mathrm{HB}})\min(0,\sigma_{\mathrm{don}}+\sigma_{\mathrm{HB}}), \tag{174}$$

with $\sigma_{\mathrm{acc}} = \max(\sigma_m,\sigma_n)$ and $\sigma_{\mathrm{don}} = \min(\sigma_m,\sigma_n)$. The first term is the **misfit** and is symmetric in sign. The second is the **hydrogen bond**: non-zero only when one segment lies above $+\sigma_{\mathrm{HB}}$ and the other below $-\sigma_{\mathrm{HB}}$, and then negative. Nobody declares that a hydrogen bond exists; a cutoff of $0.0084\,\mathrm{e/\AA^2}$ decides it.

**The segment activity coefficient is implicit.** How disfavoured a segment of density $\sigma_m$ is depends on what it meets, which depends on how disfavoured *those* are:

$$\Gamma(\sigma_m) = \Big[\sum\nolimits_n p(\sigma_n)\,\Gamma(\sigma_n)\,\exp\!\big(-\Delta W(\sigma_m,\sigma_n)/RT\big)\Big]^{-1}. \tag{175}$$

This is a fixed point, solved from $\Gamma \equiv 1$ with one-half damping to $10^{-8}$. Note that the *only* place temperature enters the model is the $RT$ in that exponential: neither the profiles nor $\Delta W$ depend on $T$.

**Residual and combinatorial.** Solving (175) for the mixture's area-weighted profile $p_S$ and for each pure $p_i$ gives the cost of moving molecule $i$'s segments from its own environment into the mixture,

$$\ln\gamma_i^{\mathrm{res}} = \frac{A_i}{a_{\mathrm{eff}}}\sum_k p_i(\sigma_k)\big[\ln\Gamma_S(\sigma_k) - \ln\Gamma_i(\sigma_k)\big], \tag{176}$$

where $A_i/a_{\mathrm{eff}}$ is the segment count. For a pure component every bracket vanishes identically, so $\gamma = 1$ exactly rather than approximately. Size and shape are carried by the Staverman–Guggenheim combinatorial term of the UNIQUAC family, with $r_i = V_i/r_0$ and $q_i = A_i/q_0$ taken from the same cavity, and $\ln\gamma_i = \ln\gamma_i^{\mathrm{comb}} + \ln\gamma_i^{\mathrm{res}}$. That the two parts simply *add* is itself an assumption, made by the plus sign rather than by any term in it.

**Constants, and a unit trap.** Choupo compiles in the 2002 set: $a_{\mathrm{eff}} = 7.5\,\mathrm{\AA^2}$, $\alpha' = 16466.72$ and $c_{\mathrm{HB}} = 85580\,\mathrm{kcal\,\AA^4/(mol\,e^2)}$, $\sigma_{\mathrm{HB}} = 0.0084\,\mathrm{e/\AA^2}$, $z = 10$, $r_0 = 66.69\,\mathrm{\AA^3}$, $q_0 = 79.53\,\mathrm{\AA^2}$, and $R = 0.001987\,\mathrm{kcal/(mol\,K)}$. The gas constant is in kcal because the interaction constants were fitted there; mixing unit systems inside (175) is the commonest way to obtain a plausible wrong answer from this model.

**What ships, and what does not.** Only the 2002 variant is implemented, and a profile declaring another variant is refused rather than run against these constants — a profile and its constants are one fitted object. **The public tree ships synthetic teaching surrogates for the profiles**: the VT-2005 set cannot be redistributed here, so the shipped witness demonstrates the mechanism and its numbers are not validated against measurement. A curator's own copy installs via `bin/choupo-import-cosmo`; without it the model refuses by name.

# 12 PC-SAFT: a molecular equation of state

The cubic chapter closed on its own limit: hydrogen-bonded fluids, where much of the physics *is* the directional bond. PC-SAFT (Gross & Sadowski, *Ind. Eng. Chem. Res.* 40 (2001) 1244)[31] approaches the problem from statistical mechanics instead of from a patched van der Waals loop: a molecule is a **chain of $m$ tangent spherical segments** of diameter $\sigma$ and dispersion energy $\varepsilon/k$, and the residual Helmholtz energy is built as a sum of physically named contributions,

$$\frac{a^{\mathrm{res}}}{RT} \;=\; a^{\mathrm{hc}} \;+\; a^{\mathrm{disp}} \;\big(+\, a^{\mathrm{assoc}}\big), \tag{177}$$

a hard-*chain* reference (hard spheres plus the bonding of $m$ segments into a chain, priced through the hard-sphere contact value $g_{ij}^{\mathrm{hs}}$), a two-term dispersion perturbation with universal constants, and — for fluids whose molecules carry donor and acceptor sites — the Wertheim association term[32] of the next subsection. Everything downstream is a *derivative* of this one function: $Z = 1 + \rho\,(\partial\tilde{a}/\partial\rho)$, the pressure closes $P = Z\rho kT$, and $\ln\hat{\varphi}_i$ is a mole-number derivative at fixed $(T, V)$. In Choupo the derivative is taken *numerically from the analytic* $a^{\mathrm{res}}$ — a declared choice: the physics lives once, and a derivative cannot disagree with it by a transcription error. The price (a central-difference step) is measured, not assumed: every construction runs `verify()`, which checks the ideal-gas limit, the Euler identity $\sum x_i \ln\hat{\varphi}_i = g^{\mathrm{res}}/RT$, and the step-convergence plateau of the numeric derivative, and *refuses* a parameter set that fails any of the three.

Unlike the cubics, PC-SAFT's parameters cannot be derived from $\{T_c, P_c, \omega\}$: each component carries its fitted $\{m, \sigma, \varepsilon/k\}$ in its own record's `pcsaft{}` block, primary-cited. With no binary correction ($k_{ij} = 0$, announced) mixture behaviour is *predicted* from pure-component parameters alone — the $CH_4$/propane flash of `tutorials/steady/flash/flash10_ch4propane_pcsaft` splits within a few percent of the cubic route with nothing fitted to the pair.

## 12.1 Association: Wertheim's theory, visible

A hydrogen bond is short-ranged, directional and saturable — one donor bonds one acceptor, and a site, once bonded, is spent. Wertheim's first-order perturbation theory turns exactly those three facts into a closed contribution. Let $X_i^A$ be the fraction of molecules $i$ *not* bonded at site $A$. A mass-action closure determines it,

$$X_i^A \;=\; \frac{1}{1 + \rho \sum_j x_j \sum_{B \in \mathrm{sites}(j)} X_j^B\, \Delta^{A_i B_j}}, \qquad \Delta^{A_i B_j} = \sigma_{ij}^3\, g_{ij}^{\mathrm{hs}}\, \kappa^{A_i B_j} \left[ e^{\varepsilon^{A_i B_j}/kT} - 1 \right], \tag{178}$$

and the Helmholtz contribution follows,

$$\frac{a^{\mathrm{assoc}}}{RT} = \sum_i x_i \Big[ \sum_{A \in \mathrm{sites}(i)} \big( \ln X_i^A - \tfrac{1}{2} X_i^A \big) + \tfrac{1}{2} M_i \Big]. \tag{179}$$

Two new fitted constants per associating fluid — the association energy $\varepsilon^{AB}/k$ and volume $\kappa^{AB}$ — and a **site scheme** naming how many donors and acceptors the molecule carries (Huang–Radosz vocabulary: `2B` = one donor + one acceptor, the alcohols; `4C` = two + two). Cross-association in mixtures needs *no* new fitted constant: the Wolbach–Sandler combining rules build $\Delta$ between unlike sites from the pure constants. The closure (178) is an inner fixed point solved at every $(T, \rho, x)$; for a pure fluid with equal site counts it collapses to a quadratic solved in closed form, and `verify()` checks the iteration against that closed form — machine zero, printed — beside the site fractions themselves: $X^A(\mathrm{water}) \approx 0.04$ at ambient conditions says 96% of water's sites are hydrogen-bonded, which *is* the physics the term adds.

**The scheme is part of the fit.** A published $(\varepsilon^{AB}, \kappa^{AB})$ pair is meaningful only under the site scheme it was regressed with, and the curator reads the paper's own site count rather than assuming one from the molecule's chemistry. Choupo paid for this lesson once: the Gross & Sadowski (2002) water set — which the paper fits with *two* sites — was first curated as `4C`. The pure liquid density then read +0.6%, a coincidence that masked the error, while the ethanol/water mixture flash collapsed and pure-water $P^{\mathrm{sat}}$ left the physical range. Re-curated as `2B`, the set shows its true published face: $P^{\mathrm{sat}}$ accurate to a fraction of a percent near the normal boiling point, ambient liquid density 7.5% light — the trade-off the original regression chose, now stated on the record instead of discovered downstream. The general moral is the same one this guide repeats about every model: a fitted parameter set is one object — values, scheme, and validity range together — and splitting it silently produces a model nobody ever validated.

**Predictive vs. fitted, measured.** The mixture witness `tutorials/steady/flash/flash20_ethanol_water_pcsaft` flashes one 30/70 ethanol/water feed at 358 K, 1 atm through two thermodynamic worlds side by side: predictive PC-SAFT (association on, $k_{ij} = 0$, nothing fitted to the pair) and the data-fitted NRTL $\gamma$–$\varphi$ route. The predictive route lands $K_{\mathrm{ethanol}}$ within ∼2% of the fit and $K_{\mathrm{water}}$ within ∼11% — from pure parameters alone, a genuinely strong showing for a hydrogen-bonded pair — and that residual water error still moves $V/F$ from 0.51 to 0.65.

Predictive means *no mixture data used*; it does not mean *as accurate as a fit*. The fitted route remains the design tool where data exist; the molecular EoS is the screen for systems with nothing to fit.

**Runnable case.** The case declares only `equationOfState { model PCSAFT; }` inside `formulation phiPhi`; whether association is active is decided by the FACTS on the component records — a record carrying the trio associates, one without it does not, and the engine announces the roster it found (`[pcsaft] associating: water (2B), ethanol (2B); cross: Wolbach-Sandler`). The component block, one home:

```
pcsaft
{
    m           1.0656;     sigma  3.0007;    epsilonK 366.51;
    assocScheme 2B;         // 2B | 4C  (absent = non-associating)
    epsAB_K     2500.7;     // K
    kappaAB     0.034868;
    source      "Gross & Sadowski, IECR 41 (2002) 5510, Table 1";
}
```

A partial trio refuses at load; an unknown scheme refuses listing the known ones. Pure-fluid witnesses: `tutorials/props/molecular/pcsaft01_pure_nhexane` (non-associating core), `tutorials/props/molecular/pcsaft03_association_pure` (2B water + 2B ethanol, site fractions and the closed-form oracle echoed at verbosity 3).

# 13 $K$-values and Rachford-Rice

*What you'll learn.* The single equation at the heart of every flash, bubble-point, dew-point, and column stage in the simulator — and why it has exactly one root when a two-phase region exists.

## 13.1 Defining the $K$-value

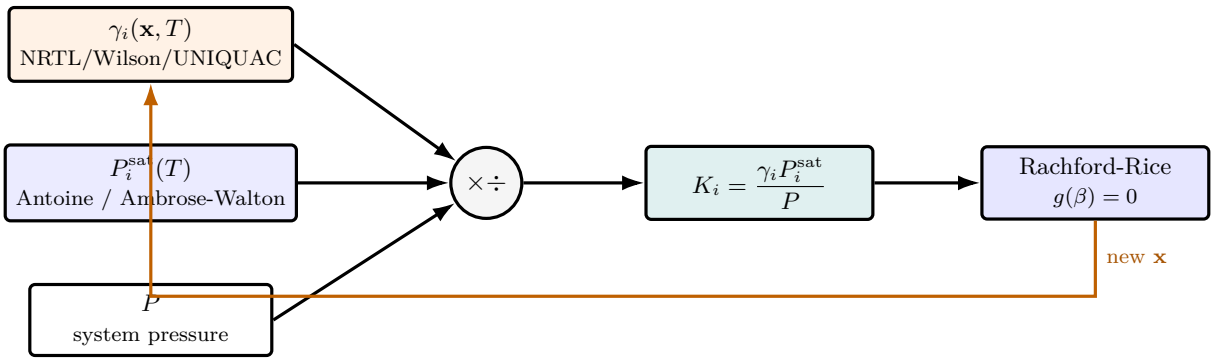



**Figure 23:** What the $K$-value is, and where it sits in the flash loop. Modified Raoult's law assembles each $K_i = \gamma_i P_i^{\text{sat}}/P$ from three glass-box ingredients: the activity coefficient (from the chosen $G^E$ model), the pure-component vapour pressure (Antoine), and the system pressure. Once $\mathbf{K}$ is fixed, the flash is a *pure mass-balance* problem — Rachford-Rice for $\beta$. But $\gamma_i$ depends on the liquid composition $\mathbf{x}$, which Rachford-Rice updates, so the orange arrow closes an *outer* loop: re-evaluate $\mathbf{K}$ at the new $\mathbf{x}$ and repeat. This composition–$K$ coupling is exactly why the flash needs two solver layers, not one.

The $K$-value of component $i$ is the partition ratio between vapour and liquid:

$$K_i \equiv \frac{y_i}{x_i}. \tag{180}$$

Combining with (153):

$$\boxed{K_i = \frac{\gamma_i(\mathbf{x}, T)\, P_i^{\text{sat}}(T)}{P}} \tag{181}$$

At each outer iteration, $K_i$ is a function of $T, P, \mathbf{x}$ alone — once $\mathbf{K}$ is fixed, the flash becomes a pure mass-balance problem.

## 13.2 Material balance and Rachford-Rice

A single-stage flash takes feed $\mathbf{z}$ at $F$ and splits it into vapour $V$ at composition $\mathbf{y}$ and liquid $L$ at composition $\mathbf{x}$. The component balance is:

$$F z_i = V y_i + L x_i = F[\beta y_i + (1-\beta) x_i]. \tag{182}$$

Cancelling $F$ and using $y_i = K_i x_i$:

$$z_i = \beta K_i x_i + (1-\beta) x_i = x_i\,[1 + \beta(K_i - 1)],$$

so

$$x_i = \frac{z_i}{1 + \beta(K_i - 1)}, \qquad y_i = \frac{K_i z_i}{1 + \beta(K_i - 1)}. \tag{183}$$

Imposing $\sum_i y_i - \sum_i x_i = 0$ (the closure) and collecting terms yields the *Rachford-Rice equation*:

$$\boxed{g(\beta) \equiv \sum_{i=1}^{N_c} \frac{z_i\,(K_i - 1)}{1 + \beta(K_i - 1)} = 0} \tag{184}$$

## 13.3 Why exactly one root, in $(0, 1)$, when there is two-phase

Two properties make Rachford-Rice numerically pleasant:

**Monotonicity.** Differentiating (184):

$$g'(\beta) = -\sum_i \frac{z_i\,(K_i - 1)^2}{[1 + \beta(K_i - 1)]^2} < 0.$$

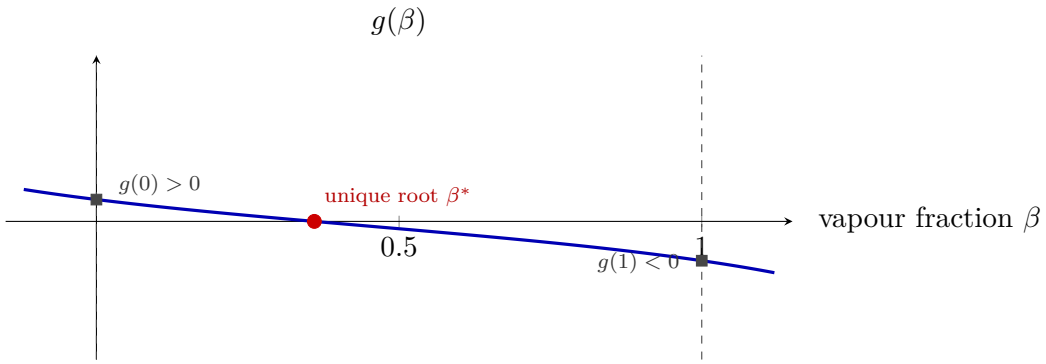



**Figure 24:** Why Rachford-Rice is numerically tame. When a two-phase region exists, some $K_i > 1$ (volatiles) and some $K_i < 1$ (heavies), so $g(0) > 0$ and $g(1) < 0$ — the function changes sign across the physical window $[0, 1]$. And $g'(\beta) < 0$ everywhere it is defined (each term in the derivative is a negative square), so $g$ is *strictly decreasing*. A monotone function with one sign change has exactly one root, and Newton-Raphson from any interior $\beta_0$ converges to it quadratically. If the flash will not converge, the bug is upstream — a bad $K_i$ (Antoine out of range, $\gamma_i$ blowing up), never Rachford-Rice itself.

Since each term in the sum is non-negative, $g'(\beta) < 0$ wherever the sum is defined — $g$ is strictly decreasing.

**Sign change.** If a two-phase region exists, some $K_i > 1$ (the volatile components) and some $K_i < 1$ (the heavy ones). Then $g(0) = \sum_i z_i(K_i - 1)$ has the sign of "are most components volatile?", and $g(1) = \sum_i z_i(1 - 1/K_i)$ has the opposite sign. Both endpoints differ in sign, so a root exists in $(0, 1)$.

The combination of monotonicity and a sign change in $(0, 1)$ means Newton-Raphson starting from *any* interior $\beta_0$ will converge, quadratically, to the unique root.

> **Key idea.** Rachford-Rice is convex enough to make Newton-Raphson tame. If your flash refuses to converge, the bug is upstream — almost always a bad $K_i$ (Antoine outside its range; $\gamma_i$ blowing up; $T$ below the lower phase boundary).

## 13.4 Initialisation: the Wilson $K$-value

A robust initial guess for $\mathbf{K}$ before any activity-coefficient information is available comes from Wilson's correlation:

$$K_i^{(0)}(T, P) \;=\; \frac{P_{c,i}}{P}\,\exp\!\Big[\,5.373\,(1+\omega_i)\,\big(1 - T_{c,i}/T\big)\Big]. \tag{185}$$

This uses only the critical point and the acentric factor — both in the component database — and works astonishingly well as a starting point.

## 13.5 Worked example

> **Worked example 13.1.** Flash of benzene/toluene at 370 K, 1 barA feed at $\mathbf{z} = (0.40, 0.60)$ (benzene then toluene), $T = 370\,K$,
> $P = 1$ bar. At this temperature, Antoine (the coefficients in the component `.dat` files) gives $P_1^{\text{sat}} \approx 1.651$ bar and $P_2^{\text{sat}} \approx 0.674$ bar. Ideal solution: $\gamma_i = 1$, so $K_1 = 1.651$ and $K_2 = 0.674$.
> Rachford-Rice:
>
> $$g(\beta) \;=\; \frac{0.40 \cdot 0.651}{1 + 0.651\beta} + \frac{0.60 \cdot (-0.326)}{1 - 0.326\beta} \;=\; 0.$$
>
> At $\beta = 0$: $g = 0.260 - 0.196 = 0.065$. At $\beta = 1$: $g = 0.158 - 0.290 = -0.132$. The endpoints straddle zero — the feed flashes. Newton on $g$ from any interior start lands on $\beta \approx 0.304$ in three iterations, and then (183) gives $x_1 = 0.40/(1 + 0.651 \cdot 0.304) = 0.334$ and $y_1 = K_1 x_1 = 0.551$. This is exactly the `flash01_benzene_toluene` setup; the tutorial's golden master pins $V/F = 0.30398$.

> **Runnable case.** `tutorials/steady/flash/flash01_benzene_toluene` — run with `verbosity 3` and watch the Newton trace converge in 3–4 iterations.

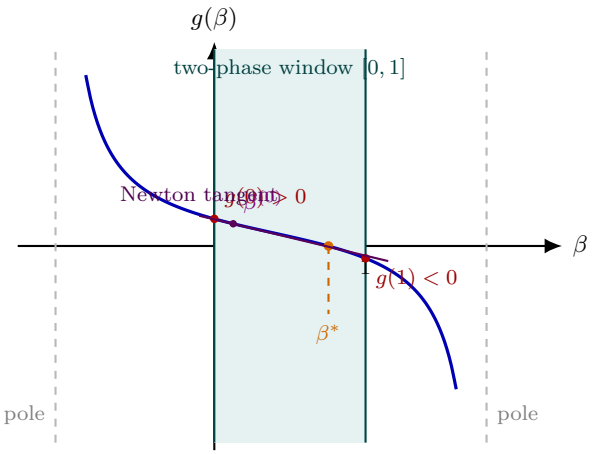



**Figure 25:** Why Rachford–Rice is tame. Each term has a pole where $1 + \beta(K_i - 1) = 0$; for a genuine two-phase mixture all the poles lie *outside* the physical window $[0, 1]$ (shaded), so on that window $g$ is a single, smooth, strictly *decreasing* branch ($g'(\beta) < 0$ always). When a split exists the endpoints straddle zero — $g(0) > 0$, $g(1) < 0$ — guaranteeing exactly one root $\beta^*$ in $(0, 1)$. Monotone plus a sign change is the textbook recipe for a trouble-free Newton: the tangent from *any* interior $\beta^{(0)}$ slides to $\beta^*$ and converges quadratically. If the endpoints share a sign, the root is pushed outside $[0, 1]$ — the subcooled / superheated case the flash reports honestly.

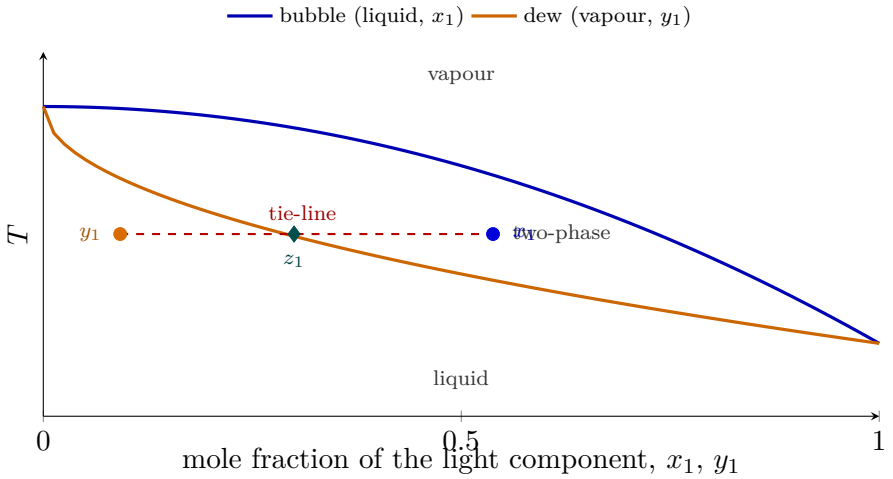



**Figure 26:** A temperature–composition ($Txy$) diagram, the phase-diagram face of the flash. The lower (blue) curve is the bubble locus, plotted against the *liquid* composition $x_1$; the upper (orange) curve is the dew locus, plotted against the *vapour* composition $y_1$. Between them lies the two-phase region. At a flash temperature, the horizontal *tie-line* (red) connects the coexisting liquid $x_1$ and vapour $y_1$ in equilibrium ($y_1 = K_1 x_1$). A feed of composition $z_1$ (diamond) lands inside the lens, and the lever rule along the tie-line fixes the vapour fraction $\beta$ that Rachford-Rice computes. Outside the lens there is a single phase — below it sub-cooled liquid, above it superheated vapour.

# Part IV — Numerical methods

*The purpose of computing is insight, not numbers.*
Richard W. Hamming, *Numerical Methods for Scientists and Engineers* (1962)

# 1 Numerical methods, gently

*What you'll learn.* A first look at root-finding, with no Jacobians and no Taylor expansions yet. We will pose one concrete chemical-engineering problem — the bubble temperature of a benzene/toluene mixture — and solve it four times: by hand, by bisection, by secant, and finally by Newton-Raphson. By the end you will have a feel for *why* the simulator uses Newton everywhere instead of just guessing, and the heavier chapters that follow (Newton-1D, Newton in $n$ dimensions, Levenberg-Marquardt) will read as natural generalisations of what you just did by hand.

## 1.1 One problem, four strategies

Mix 50 mol% benzene and 50 mol% toluene in a sealed vessel at $P = 1$ **bar** and slowly heat the liquid. At what temperature does the first vapour bubble form?

For an ideal Raoult mixture the bubble condition is

$$x_b\, P_b^{\mathrm{sat}}(T) \;+\; x_t\, P_t^{\mathrm{sat}}(T) \;=\; P, \tag{186}$$

which with $x_b = x_t = 0.5$ and $P = 1$ bar reduces to $P_b^{\mathrm{sat}}(T) + P_t^{\mathrm{sat}}(T) = 2$ bar. Both saturation pressures come from Antoine,

$$\log_{10} P^{\mathrm{sat}}(T) \;=\; A \;-\; \frac{B}{T + C},$$

with the constants shipped in `data/standards/components/`: $A_b = 4.0181$, $B_b = 1203.84$, $C_b = -53.23$ K for benzene; $A_t = 4.0783$, $B_t = 1343.94$, $C_t = -53.77$ K for toluene.

Define the **residual**

$$f(T) \;\equiv\; \tfrac{1}{2}\big[P_b^{\mathrm{sat}}(T) + P_t^{\mathrm{sat}}(T)\big] - 1, \tag{187}$$

which is zero exactly at the bubble temperature. Knowing $T_b \approx 353$ K and $T_t \approx 384$ K, the answer must lie somewhere between those, but where?

*Intuition.* Every numerical method in this book solves the same shape of problem: **given a function $f$, find a $T$ for which $f(T) = 0$.** The strategies differ only in how cleverly they use the available information about $f$. The dumbest method needs only the ability to evaluate $f$; the smartest also needs $f'$. More information you exploit, fewer evaluations you need.

## 1.2 Strategy 1 — guess and adjust

Pick a temperature, compute $f$, see whether it is positive or negative, adjust. Honest, slow, but absolutely transparent:

| $T$ [K] | $f(T)$ [bar] |
|---|---|
| 340 | $-0.55$ |
| 360 | $-0.13$ |
| 380 | $+0.53$ |
| 365 | $+0.0053$ |
| 364 | $-0.024$ |

After five tries we know the root is between 364 and 365 K and closer to the upper bound. Two more refinements get us to within 0.05 K of the answer. The procedure works but you do all the thinking.

## 1.3 Strategy 2 — bisection

Trial-and-error implicitly used a fundamental fact: $f$ is continuous, $f(340) < 0$, $f(390) > 0$, so by the intermediate value theorem the root sits somewhere in $[340, 390]$. Bisection mechanises that intuition: take the midpoint, evaluate $f$, keep the half that still brackets the root.

**Worked example 1.1.** Bisection on Eq. (186)Starting bracket $[340, 390]$. Midpoint $T_1 = 365$, $f(T_1) = +0.0053$. Since $f$ changed sign on the lower half, replace $b$ with $T_1$ and repeat:

| iter | $T$ [K] | $f$ [bar] |
|---|---|---|
| 1 | 365.000 | $+5.34 \times 10^{-3}$ |
| 2 | 352.500 | $-3.15 \times 10^{-1}$ |
| 3 | 358.750 | $-1.67 \times 10^{-1}$ |
| 5 | 363.438 | $-4.02 \times 10^{-2}$ |
| 8 | 364.805 | $-4.43 \times 10^{-4}$ |
| 12 | 364.817 | $-8.24 \times 10^{-5}$ |
| 20 | 364.820 | $\pm 7 \times 10^{-7}$ |

After 20 iterations $T \approx 364.820$ K.

Bisection is a tortoise: every iteration halves the bracket, so the error decays exactly by a factor 2. To get from a 50 K wide bracket down to a $10^{-7}$ K tolerance takes roughly $\log_2(50/10^{-7}) \approx 29$ steps. Never blows up, never overshoots — but never fast either.

## 1.4 Strategy 3 — secant

Bisection only uses the *sign* of $f$ at each midpoint. We can do better by exploiting the *value*: if $f$ has changed by a known amount between two trial $T$'s, a linear extrapolation predicts where it would hit zero. Geometrically, draw the secant line through the last two points and read off the $T$-intercept:

$$T_{k+1} \;=\; T_k \;-\; f(T_k)\,\frac{T_k - T_{k-1}}{f(T_k) - f(T_{k-1})}. \tag{188}$$

**Worked example 1.2.** Secant on Eq. (186)Starting from the same two points $(T_0, T_1) = (340, 390)$ K:

| iter | $T$ [K] | $f$ [bar] |
|---|---|---|
| 1 | 357.817 | $-1.91 \times 10^{-1}$ |
| 2 | 363.009 | $-5.24 \times 10^{-2}$ |
| 3 | 364.975 | $+4.60 \times 10^{-3}$ |
| 4 | 364.816 | $-9.74 \times 10^{-5}$ |
| 5 | 364.81968 | $-1.76 \times 10^{-7}$ |
| 6 | 364.81969 | $+6.7 \times 10^{-12}$ |

Six iterations to machine precision.

Each step uses two function evaluations of history, no derivatives. The order of convergence is $\varphi = (1+\sqrt{5})/2 \approx 1.618$ — the golden ratio. In practice this means the number of correct digits multiplies by 1.6 each iteration: 1 digit, 2, 3, 5, 8, 12 — so it doubles roughly every two iterations.

**Common pitfall.** The secant formula breaks if $f(T_k) = f(T_{k-1})$ (division by zero). Bisection cannot break that way. In practice the simulator falls back to bisection whenever the secant step would carry the iterate outside the current bracket.

## 1.5 Strategy 4 — Newton-Raphson, in one breath

Secant approximated the slope of $f$ from two adjacent evaluations. What if we knew the slope exactly — the derivative $f'(T)$? Then we could shoot a line off the current point and intersect zero in *one* step:

$$\boxed{T_{k+1} \;=\; T_k \;-\; \frac{f(T_k)}{f'(T_k)}.} \tag{189}$$

That is Newton-Raphson. The next chapter derives it rigorously from a Taylor expansion; here we just apply it.

**Worked example 1.3.** Newton on Eq. (186)Start from $T_0 = 340$ K with $f'(T)$ computed numerically (or, more typically in this simulator, analytically from $\mathrm{d}P^{\mathrm{sat}}/\mathrm{d}T$):

| iter | $T$ [K] | $f$ [bar] |
|---|---|---|
| 1 | 374.942 | $+3.36 \times 10^{-1}$ |
| 2 | 365.872 | $+3.15 \times 10^{-2}$ |
| 3 | 364.832 | $+3.74 \times 10^{-4}$ |
| 4 | 364.8197 | $+5.5 \times 10^{-8}$ |
| 5 | 364.8197 | $+2.4 \times 10^{-15}$ |

Four iterations to $10^{-8}$, five to machine precision.

Look at the residual column above. At iteration 3 it is $3.7 \times 10^{-4}$; at iteration 4 it is $5.5 \times 10^{-8}$. The exponent jumped from $-4$ to $-8$ — the error **squared**. That is the signature of **quadratic convergence**, and it is the reason every iterative solver in this simulator is Newton-based when a derivative is available.

> *Intuition.* "Quadratic" here means the number of correct decimal digits *doubles* every iteration: 2 digits, 4, 8, 16, machine precision. Bisection adds one bit per step; secant multiplies digits by 1.6; Newton multiplies digits by 2. This is why even though Newton needs the extra information $f'$, it pays back so dramatically that it is essentially never not worth it.

## 1.6 The four strategies side by side

Figure 27 plots $|f(T_k)|$ versus iteration number for the four runs on a semi-log axis. The slope of each line is the convergence rate.

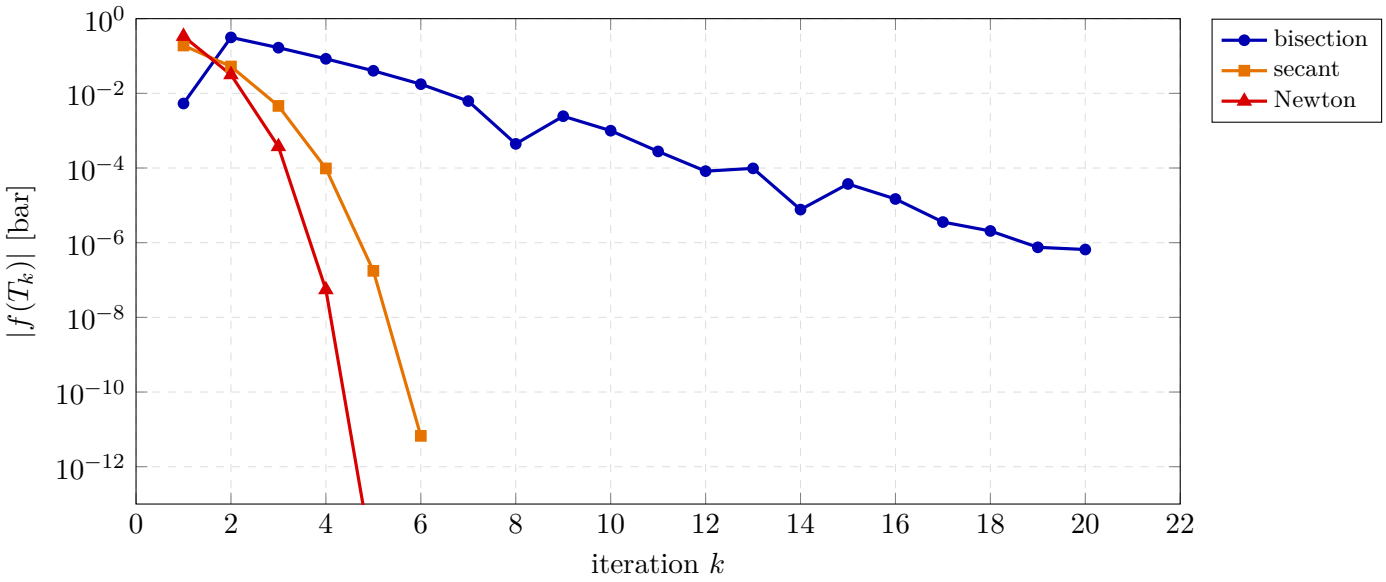



**Figure 27:** Residual decay for the four strategies on the same benzene/toluene bubble-temperature problem (initial bracket $[340, 390]$ K, target $T_b \approx 364.82$ K). Bisection is straight on the semi-log plot — a constant factor of $\sim 2$ per step. Secant accelerates with the slope steepening over time. Newton dives off the chart in five steps.

| method | uses | per-iter cost | order $p$ | iters to $10^{-8}$ |
|---|---|---|---|---|
| trial & error | $f$ values + intuition | 1 eval | — | many |
| bisection | $f$ signs | 1 eval | 1 (linear) | ∼30 |
| secant | last 2 $f$ values | 1 eval | 1.618 | ∼6 |
| Newton | $f$ *and* $f'$ | 1 eval + 1 deriv | 2 (quadratic) | ∼4 |

## 1.7 Why the rest of Part III matters

Every solver in the simulator is a generalisation of these four

moves:

- Chapter 2 re-derives Newton from Taylor's theorem, formalises "convergence," and shows how the simulator damps an overshoot or falls back to bisection when the local linear model misleads.

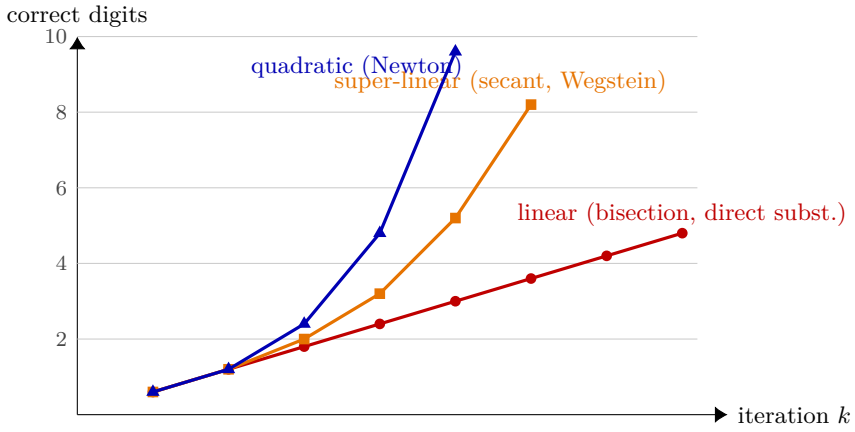



**Figure 28:** The three convergence *orders* that organise every solver in this manual, read as "how many correct digits each step buys." *Linear* convergence (red — bisection, direct substitution) adds a roughly constant number of digits per step, so the digit count climbs along a straight line. *Super-linear* (orange — the secant and Wegstein) speeds up as it goes: the per-step gain itself grows. *Quadratic* (blue — Newton and its $n$-D and tear-loop descendants) *doubles* the correct digits every step once it is near the root, which is why Newton clears twelve digits in a handful of iterations. Damping and finite-difference Jacobians are the price paid to keep a method on its fast curve without falling off into divergence far from the root.

- The next chapter generalises Newton to vector equations — the kind that pop up the moment you have multiple unknowns (multi-component flash, adiabatic flash, rigorous distillation). Replace "$f'(T_k)$" by a *Jacobian matrix*; replace "divide" by "solve a linear system." Nothing else changes.
- Levenberg-Marquardt is Newton plus a damping knob that lets it gracefully degrade to gradient descent when the linear model is bad. We use it for parameter estimation from experimental data.
- Wegstein and direct substitution attack a different shape of problem ($x = f(x)$ rather than $f(x) = 0$) and accelerate fixed-point iterations. We use them on activity-coefficient outer loops where computing a derivative is awkward.

Keep the four strategies in mind as you read on. Every more elaborate algorithm in this manual is, at heart, one of them dressed up.

# 2 Newton-Raphson, one dimension

*What you'll learn.* The single algorithm that solves every scalar fixed-point in the simulator. Why it converges quadratically when it converges, and what damping is for.

## 2.1 Derivation

Given a continuously differentiable $g\colon \mathbb{R} \to \mathbb{R}$ and a root $x^*$ with $g(x^*) = 0$, Taylor-expand around an estimate $x_k$:

$$0 \;=\; g(x^*) \;=\; g(x_k) + g'(x_k)\,(x^* - x_k) + O((x^* - x_k)^2).$$

Solving for $x^*$ and dropping the higher-order term defines the *Newton iteration*:

$$\boxed{\,x_{k+1} \;=\; x_k \;-\; \frac{g(x_k)}{g'(x_k)}\,} \tag{190}$$

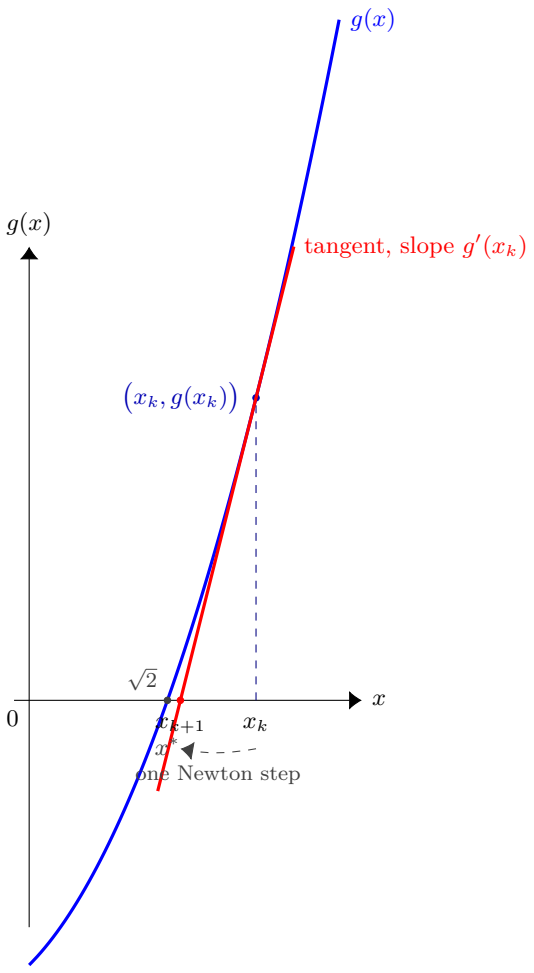



**Figure 29:** One Newton iteration on $g(x) = x^2 - 2$ (root $x^* = \sqrt{2} \approx 1.4142$). At the current estimate $x_k = 2$, draw the tangent to $g$ at $(x_k, g(x_k))$ with slope $g'(x_k) = 4$; the tangent's $x$-intercept is the next estimate $x_{k+1} = 1.5$. Two more iterations from there converge to $x^*$ to twelve digits — the quadratic doubling of correct digits per step that gives Newton its reputation.

## 2.2 Convergence

Define the error $e_k \equiv x_k - x^*$. A second-order Taylor expansion of $g$ around $x^*$ gives

$$g(x_k) \;=\; g'(x^*)\,e_k \;+\; \tfrac{1}{2}g''(x^*)\,e_k^2 \;+\; O(e_k^3),$$

and similarly $g'(x_k) = g'(x^*) + g''(x^*)e_k + O(e_k^2)$. Substituting into (190) and keeping leading order in $e_k$:

$$e_{k+1} \;=\; \frac{g''(x^*)}{2\,g'(x^*)}\,e_k^2 \;+\; O(e_k^3). \tag{191}$$

This is *quadratic convergence*: the error squares at each step, provided $g'(x^*) \neq 0$ and we start close enough to $x^*$.

## 2.3 Damping

Far from the root, the linearisation can over-shoot — pushing $x_{k+1}$ into a region where $g'$ has the wrong sign, or even outside the domain of $g$. A damping factor $\lambda \in (0, 1]$ tames this:

$$x_{k+1} \;=\; x_k - \lambda \, \frac{g(x_k)}{g'(x_k)}, \tag{192}$$

with $\lambda$ reduced (typically halved) until $|g(x_{k+1})| < |g(x_k)|$. This loses quadratic convergence *near the root* (where $\lambda$ returns to 1 anyway), but buys global robustness. The simulator's 1-D `NewtonRaphson` guards differently — an optional step cap (`maxStep`) plus a bracket whose bisection fallback catches any step that leaves it; the $n$-D `NewtonND` (§3) is the one that backtracks with a halving schedule.

## 2.4 Worked example

**Worked example 2.1.** Bubble pressure of benzene/tolueneSolve $g(P) = \sum_i x_i \gamma_i P_i^{\mathrm{sat}}(T)/P - 1 = 0$ for $P$ at $T = 365\,\mathrm{K}$, $x_1 = 0.4$ benzene, ideal solution.
At $T = 365\,\mathrm{K}$ Antoine gives $P_1^{\mathrm{sat}} \approx 1.435$ bar, $P_2^{\mathrm{sat}} \approx 0.575$ bar. So $g(P) = (0.4 \cdot 1.435 + 0.6 \cdot 0.575)/P - 1 = 0.919/P - 1$.
This is linear in $1/P$ — the root is $P^* = 0.919$ bar in one "Newton step". Real cases with $\gamma_i = \gamma_i(\mathbf{x}, T)$ are non-linear: typical convergence is 3–5 steps to $|g| < 10^{-8}$.

# 3 Newton-Raphson, $n$ dimensions

*What you'll learn.* Most non-trivial equations in a process simulator have more than one unknown. This chapter generalises the Newton step of §2 to a system of $n$ coupled equations in $n$ unknowns. The key conceptual leap is that a derivative becomes a *Jacobian matrix*, and the division $g/g'$ becomes a *linear-system solve* $\mathbf{J}\,\Delta\mathbf{x} = -\mathbf{F}$. Everything else — damping, finite-difference approximation, convergence behaviour — carries over from the scalar case once that leap is made.

## 3.1 Why we need a multi-dimensional Newton

The 1-D Newton iteration solves one equation in one unknown. Lots of chemistry is not like that. A few representative examples from this simulator:

- **Multi-component flash.** At fixed $(T, P, \mathbf{z})$, the Rachford-Rice equation determines $V/F$ *given* the $K_i$'s; but the $K_i$'s themselves depend on the unknown liquid composition $\mathbf{x}$, which in turn depends on $V/F$. If we close this loop simultaneously instead of with an outer Wegstein loop, we get $n_c$ coupled equations in $n_c$ unknowns.
- **Adiabatic flash.** Both $T$ and $V/F$ are unknown; the system is two equations (component-balance closure + energy balance) in two unknowns.
- **Rigorous distillation.** Each stage contributes its own MESH equations (Mass, Equilibrium, Summations, Heat balance); a column with $N_s$ stages and $n_c$ components builds a system of $O(N_s n_c)$ coupled equations. Naphtali–Sandholm uses Newton-ND on the whole block.
- **Reactive distillation.** A reaction on the catalytic stages adds, per reactive stage $j$, the source $\nu_i\,\xi_j$ to that stage's component balance, with the molar extent $\xi_j$ set either by chemical *equilibrium*, $\sum_i \nu_i \ln\!\big(\gamma_i\, x_{j,i}\big) = \ln K_a(T_j)$, (one extra unknown $\xi_j$ and one extra equation per stage), or by a *rate* law $\xi_j = r_j\, m_{\mathrm{cat},j}$ (no extra unknown — $\xi_j$ is a function of $x_j, T_j$). Because the reaction makes the residual far more nonlinear, it is converged by *homotopy*: solve the column non-reactive first, then switch the reaction on, seeded from that profile. For a mole-conserving ($\sum_i \nu_i = 0$) reaction the constant-molal-overflow flow profile is undisturbed; the worked methyl-acetate case is in the User Guide, under *Multiple feeds, side draws, reactive stages.* (Named rather than numbered: `ref` cannot reach a label in another document, and this one rendered as `??` in every build of this guide until 2026-08-04.)

The pattern is universal: residuals $F_i$ that all depend on all the unknowns $x_j$, and zero of them can be solved independently.

## 3.2 Generalising Taylor: the Jacobian matrix

We want to solve

$$\mathbf{F}(\mathbf{x}) \;=\; \mathbf{0}, \qquad \mathbf{F}\colon \mathbb{R}^n \to \mathbb{R}^n. \tag{193}$$

Each $F_i$ is a function of all $n$ variables $x_1, \dots, x_n$. Around an estimate $\mathbf{x}_k$, the analog of the scalar Taylor expansion $g(x^*) = g(x_k) + g'(x_k)(x^* - x_k) + \cdots$ is the *matrix-vector* Taylor expansion

$$\mathbf{0} \;=\; \mathbf{F}(\mathbf{x}^*) \;=\; \mathbf{F}(\mathbf{x}_k) + \mathbf{J}(\mathbf{x}_k)\,(\mathbf{x}^* - \mathbf{x}_k) + O(|\mathbf{x}^* - \mathbf{x}_k|^2), \tag{194}$$

where the role of $g'$ is now played by the *Jacobian matrix* $\mathbf{J} \in \mathbb{R}^{n\times n}$ whose entries are all the partial derivatives,

$$\mathbf{J}_{ij}(\mathbf{x}_k) \;\equiv\; \left.\frac{\partial F_i}{\partial x_j}\right|_{\mathbf{x}_k}. \tag{195}$$

Row $i$ of $\mathbf{J}$ is the gradient of $F_i$; column $j$ describes how all $n$ residuals change when we move $x_j$ alone.

**Geometric reading.** The map $\mathbf{x} \mapsto \mathbf{F}(\mathbf{x})$ sends $\mathbb{R}^n \to \mathbb{R}^n$; the Jacobian $\mathbf{J}(\mathbf{x}_k)$ is the *best linear approximation* of that map at $\mathbf{x}_k$. Equation (194) says that $\mathbf{F}$ is, to first order, $\mathbf{J}\cdot\Delta\mathbf{x}$ plus a constant shift.

## 3.3 The Newton step in $n$ dimensions

Dropping the higher-order term in (194) and solving for $\mathbf{x}^*$ gives, formally, $\mathbf{x}^* \approx \mathbf{x}_k - \mathbf{J}^{-1}\mathbf{F}(\mathbf{x}_k)$. In practice we do *not* compute $\mathbf{J}^{-1}$; we solve the linear system

$$\boxed{\mathbf{J}(\mathbf{x}_k)\,\Delta\mathbf{x} \;=\; -\mathbf{F}(\mathbf{x}_k), \qquad \mathbf{x}_{k+1} \;=\; \mathbf{x}_k + \lambda\,\Delta\mathbf{x},} \tag{196}$$

where $\lambda \in (0, 1]$ is the damping factor (next subsection). Solving the system is both *cheaper* (one LU factorisation, $O(n^3/3)$ flops) and *numerically more stable* than forming the explicit inverse ($O(n^3)$ flops, more accumulated round-off, hard to keep accurate when $\mathbf{J}$ is poorly conditioned).

*Intuition.* The scalar Newton step $x_{k+1} = x_k - g(x_k)/g'(x_k)$ asks *"how far do I have to slide $x$ to make the linear model of $g$ hit zero?"*. The $n$-D version asks the same question, but the answer is a *direction vector* in $\mathbb{R}^n$ rather than a single scalar, and we find it by solving a linear system instead of doing a division. Setting $n = 1$ in (196) recovers (190): $\mathbf{J}$ becomes the scalar $g'$, and the "linear system" degenerates to one equation $g'(x)\Delta x = -g(x)$.

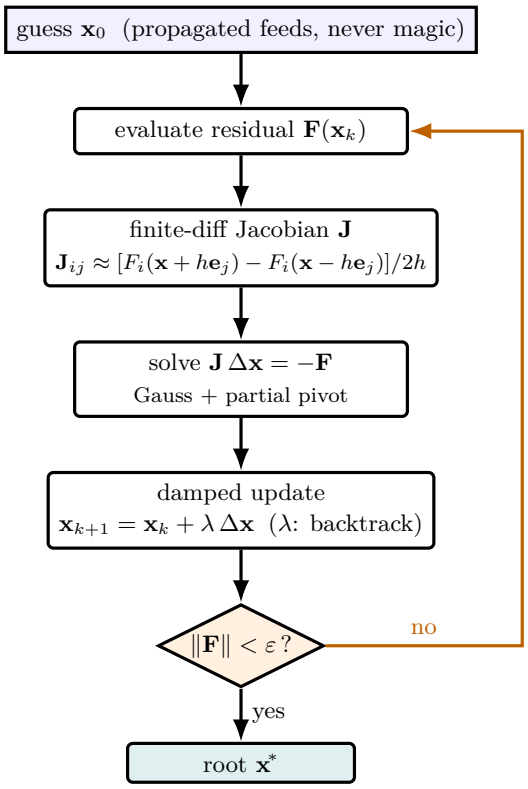



**Figure 30:** The $n$-dimensional Newton loop, exactly as `src/solver/NewtonND.cpp` runs it. Read it top to bottom: start from an honest guess (feeds propagated through the topology, never a universal constant), measure how wrong you are ($\mathbf{F}$), build the local linear model ($\mathbf{J}$, by finite differences so the engineer only codes $\mathbf{F}$), solve *one* linear system for the step direction $\Delta\mathbf{x}$, walk a damped fraction $\lambda$ of it, and test convergence. If $\|\mathbf{F}\|$ is still too big, the orange arrow loops back and the whole pass repeats with the improved $\mathbf{x}$. The damping $\lambda$ is the only knob that distinguishes "which direction" (the Jacobian's job) from "how far" (safe progress far from the root).

## 3.4 Finite-difference Jacobian

Analytic Jacobians for chemical models (NRTL $\gamma$ derivatives, say, or distillation MESH equations) are tedious to derive and a maintenance burden once derived: every time the activity model is extended, the Jacobian code has to change too. The simulator opts for the universal approach of *finite differences* on the residuals. For each column $j$, perturb $x_j$ by $\pm h$ keeping all other components fixed and compute

$$\mathbf{J}_{ij}(\mathbf{x}) \;\approx\; \frac{F_i(\mathbf{x} + h\,\mathbf{e}_j) - F_i(\mathbf{x} - h\,\mathbf{e}_j)}{2h}, \tag{197}$$

where $\mathbf{e}_j$ is the $j$-th standard basis vector. The step size is taken relative to the magnitude of $x_j$,

$$h \;=\; \max\!\big(\epsilon_{\text{rel}}\,|x_j|,\, \epsilon_{\text{abs}}\big), \qquad \epsilon_{\text{rel}} \approx 10^{-6}.$$

Cost: two extra residual evaluations per Jacobian column, so $2n$ per full Jacobian. Benefit: the engineer only has to implement $\mathbf{F}$, never $\mathbf{J}$ — decisive for pedagogical visibility and for adding new unit ops.

**Common pitfall.** Choosing $h$ is a delicate compromise. Truncation error of the central-difference formula grows as $h^2$; round-off error grows as $\epsilon_{\text{machine}}/h$. For double precision ($\epsilon_{\text{machine}} \approx 10^{-16}$) the optimum sits near $h \sim \epsilon_{\text{machine}}^{1/3} \approx 10^{-5}$. Pick $h$ too small (e.g. $10^{-12}$) and the subtraction in the numerator of (197) loses most of its digits to cancellation.

## 3.5 Solving the linear system

The Jacobian is dense, square, and small ($n \lesssim 50$ in nearly every simulator unit op). Iterative methods (CG, GMRES) shine on large sparse systems; for our size they are slower than direct factorisation. The simulator uses Gauss elimination with partial pivoting on $\mathbf{J}\,\Delta\mathbf{x} = -\mathbf{F}$ followed by back-substitution[33] — the same algorithm taught in undergraduate linear algebra, implemented in $\sim 60$ LOC in `src/solver/NewtonND.cpp`.

**Partial pivoting.** At each elimination step, swap the

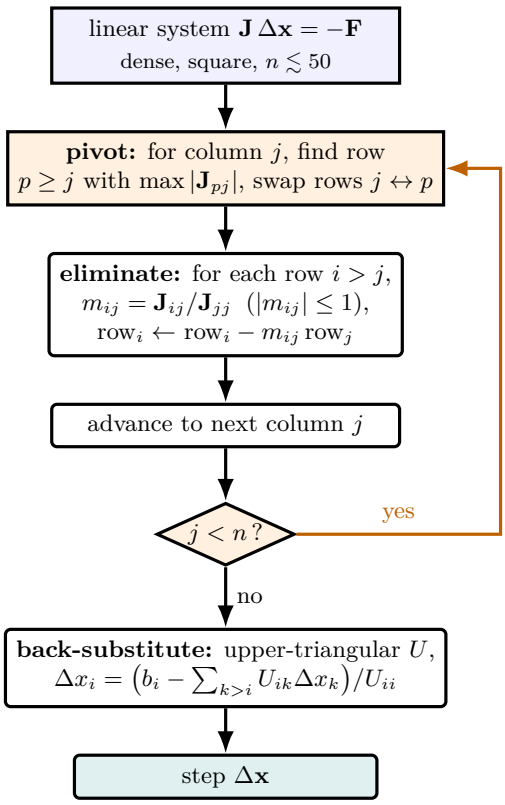



**Figure 31:** Gauss elimination with partial pivoting — the ∼60-line direct solver that every Newton step in the simulator calls to turn $\mathbf{J}\,\Delta\mathbf{x} = -\mathbf{F}$ into a step. Column by column it first *pivots* (swaps in the row with the largest entry in the active column, the orange box) so the multipliers $m_{ij}$ never exceed one in magnitude — the guard against round-off blow-up — then *eliminates* everything below the diagonal, leaving an upper-triangular system. When the last column is reached, one *back-substitution* sweep reads off $\Delta\mathbf{x}$ from the bottom up. No matrix inverse is ever formed: the factorisation is both cheaper ($O(n^3/3)$) and steadier than $\mathbf{J}^{-1}$.

current pivot row with whichever row below has the largest absolute entry in the pivot column. This keeps the multipliers $\leq 1$ in magnitude and prevents catastrophic growth of round-off in the elimination. Without pivoting, a Jacobian with widely varying entries can produce numerically meaningless results.

## 3.6 Damping by backtracking line search

The full Newton step ($\lambda = 1$) is correct to second order in $\Delta\mathbf{x}$. Far from the root, second-order error can be huge and $\mathbf{x}_k + \Delta\mathbf{x}$ can land somewhere worse than $\mathbf{x}_k$ — in chemistry, "worse" often means a non-physical state (negative concentration, pressure below the saturation curve, temperature outside the Antoine range). The fix is the same as in 1-D (§2, Damping): try $\lambda = 1$; if $\|\mathbf{F}(\mathbf{x}_{k+1})\|_2^2$ does not decrease, halve $\lambda$ and try again, down to a minimum $\lambda_{\min} \sim 2^{-10}$.

> *Intuition.* The Jacobian decides *which direction* to walk; damping decides *how far* to walk in that direction. When the linear model is trustworthy — close to the root, where higher-order terms are small — $\lambda \to 1$ and Newton recovers its quadratic-doubling speed. When the linear model is not trustworthy — far from the root, or in a region where $\mathbf{J}$ is nearly singular — $\lambda$ shrinks until the step is short enough to stay in a safe basin.

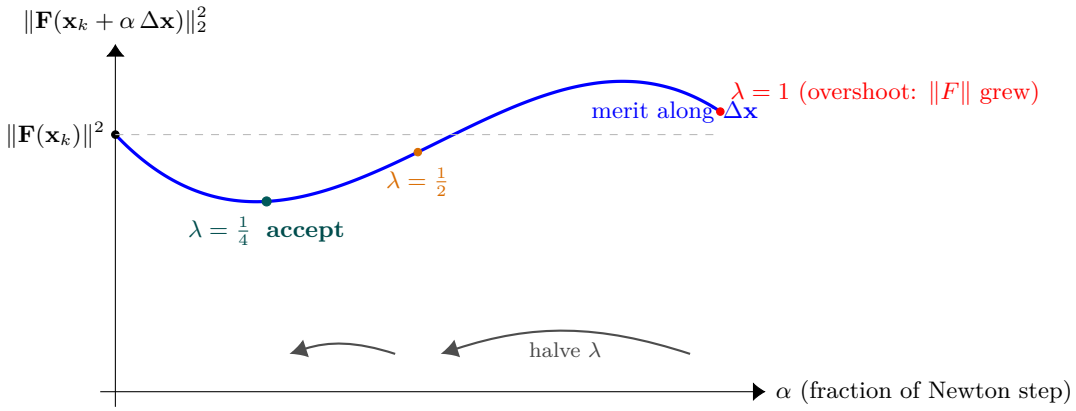



**Figure 32:** Backtracking damping seen as a one-dimensional slice of the residual norm *along* the fixed Newton direction $\Delta\mathbf{x}$. The Jacobian has already chosen the direction; damping only chooses how far. The full step ($\lambda = 1$, red) overshoots into a region where $\|\mathbf{F}\|^2$ is *larger* than at $\mathbf{x}_k$ — in chemistry often a non-physical state. The simulator halves $\lambda$ ($1 \to \frac{1}{2} \to \frac{1}{4}$) until the trial point drops below the dashed current-residual line (the sufficient-decrease test), and accepts the first $\lambda$ that does. Near the root the curve is monotone down from $\alpha = 0$, the first try $\lambda = 1$ passes, and quadratic convergence is recovered untouched.

## 3.7 Worked example: a $2 \times 2$ system

**Worked example 3.1.** Two coupled algebraic equationsSolve the system

$$F_1(\mathbf{x}) \;=\; x_1^2 + x_2^2 - 5 \;=\; 0, \qquad F_2(\mathbf{x}) \;=\; x_1 x_2 - 2 \;=\; 0.$$

The pair has solutions $(2, 1)$ and $(1, 2)$ (and the two reflections through the origin). We seek $(2, 1)$ from the initial guess $\mathbf{x}_0 = (2.5, 0.5)$.

**Step 1.** Evaluate $\mathbf{F}$ and $\mathbf{J}$:

$$\mathbf{F}(\mathbf{x}_0) = \begin{pmatrix} 2.5^2 + 0.5^2 - 5 \\ 2.5 \cdot 0.5 - 2 \end{pmatrix} = \begin{pmatrix} 1.50 \\ -0.75 \end{pmatrix}, \quad \mathbf{J}(\mathbf{x}_0) = \begin{pmatrix} 2x_1 & 2x_2 \\ x_2 & x_1 \end{pmatrix}\Bigg|_{\mathbf{x}_0} = \begin{pmatrix} 5.0 & 1.0 \\ 0.5 & 2.5 \end{pmatrix}.$$

Solve $\mathbf{J}\,\Delta\mathbf{x} = -\mathbf{F}$:

$$\begin{pmatrix} 5.0 & 1.0 \\ 0.5 & 2.5 \end{pmatrix} \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \end{pmatrix} \;=\; \begin{pmatrix} -1.50 \\ 0.75 \end{pmatrix}.$$

Gauss elimination: pivot row 1, multiplier $0.5/5.0 = 0.1$. Subtract 0.1 times row 1 from row 2: row 2 becomes $(0, 2.4 \mid 0.90)$. Back-substitute: $\Delta x_2 = 0.375$, $\Delta x_1 = (-1.50 - 1.0 \cdot 0.375)/5.0 = -0.375$. So $\mathbf{x}_1 = (2.5, 0.5) + (-0.375, +0.375) = (2.125, 0.875)$.

**Step 2.** $\mathbf{F}(\mathbf{x}_1) = (0.281, -0.141)$, with $\|\mathbf{F}\|_2 = 0.314$ (down from 1.677); take another Newton step. After it, $\mathbf{x}_2 \approx (2.012, 0.988)$, $\|\mathbf{F}\|_2 \approx 0.025$. Two more steps recover machine precision. The error is squaring roughly each iteration — the expected quadratic convergence of Newton near the root.

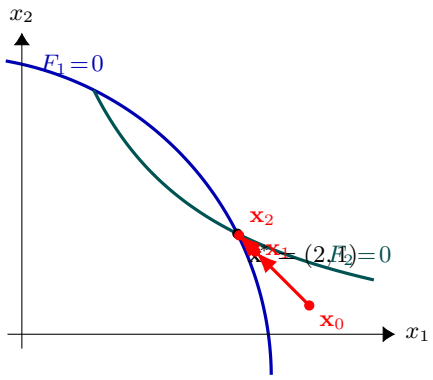



**Figure 33:** The two Newton steps of the worked $2 \times 2$ example, drawn in the $(x_1, x_2)$ plane. The two residual equations each define a zero-curve — a circle ($F_1 = x_1^2 + x_2^2 - 5 = 0$, blue) and a hyperbola ($F_2 = x_1 x_2 - 2 = 0$, teal) — and the root is where they cross. Each Newton step solves $\mathbf{J}\,\Delta\mathbf{x} = -\mathbf{F}$ for a step *direction*, and the iterates $\mathbf{x}_0 \to \mathbf{x}_1 \to \mathbf{x}_2$ march straight at the intersection, the gap to $\mathbf{x}^*$ roughly squaring every step (quadratic convergence). The same picture in $n$ dimensions is a path on the intersection of $n$ residual hypersurfaces — only the drawing gets harder, never the algorithm.

# 4 Parameter estimation: Levenberg-Marquardt

*What you'll learn.* How the simulator turns a table of experimental bubble-temperature data into a fitted set of NRTL (or Wilson) interaction parameters. We will derive each piece from scratch: why *least squares*; why the gradient of $\chi^2$ is exactly $\mathbf{J}^\top\mathbf{r}$; why its Hessian splits into two terms and why one of them can be dropped; how the resulting Gauss-Newton step can overshoot; Levenberg's damping fix; Marquardt's invariance fix; and the simple trust-region rule the simulator uses to update the damping. The runnable case is `tutorials/steady/optimisation/fitNRTL01_ethanol_water/`.

## 4.1 A different question

Every solver in Part III so far has had the same shape: *given a model, find the inputs that satisfy a residual.* Flash, bubble-$T$, CSTR, distillation — all instances of "solve $\mathbf{F}(\mathbf{x}) = 0$".

Parameter estimation flips the question. We have:

- a *model with adjustable knobs* — here, NRTL with its four binary-interaction parameters $a_{ij}, b_{ij}, a_{ji}, b_{ji}$ (the $\alpha_{ij}$ non-randomness is held fixed by convention);
- a *table of experimental data* that the model should reproduce — here, $N$ pairs $(x_i^{\text{exp}}, T_i^{\text{exp}})$ at fixed pressure $P$;
- and we want to find the knob setting that makes the model agree with the data as closely as possible.

This is an *over-determined* problem: $N \gg m$, so we cannot satisfy every data point exactly, only minimise the total mismatch.

## 4.2 Problem statement

Let $\mathbf{p} \in \mathbb{R}^m$ be the parameter vector (for an NRTL binary, $m = 4$: $a_{ij}, b_{ij}, a_{ji}, b_{ji}$). The simulator provides a *forward model*: for each experimental composition $x_i^{\text{exp}}$, run a bubble-$T$ calculation with the current $\mathbf{p}$ and return the predicted temperature,

$$T_i^{\text{pred}}(\mathbf{p}) \;=\; \text{bubble-}T \text{ at } (x_i^{\text{exp}},\, 1 - x_i^{\text{exp}},\, P) \text{ with NRTL}(\mathbf{p}). \tag{198}$$

The **residual** at point $i$ is the prediction error,

$$r_i(\mathbf{p}) \;\equiv\; T_i^{\text{pred}}(\mathbf{p}) - T_i^{\text{exp}}, \tag{199}$$

and we minimise the sum of squared residuals,

$$\boxed{\chi^2(\mathbf{p}) \;\equiv\; \tfrac{1}{2}\sum_{i=1}^{N} r_i(\mathbf{p})^2 \;=\; \tfrac{1}{2}\,\mathbf{r}(\mathbf{p})^\top\mathbf{r}(\mathbf{p}).} \tag{200}$$

*Intuition.* Why squared, not absolute value or fourth power? Squaring is the unique choice that is everywhere differentiable, penalises large mismatches disproportionately (so a single outlier still pulls the fit), and is the maximum-likelihood estimator when the experimental errors are independent Gaussians of equal variance. The factor $\frac{1}{2}$ is bookkeeping — it cancels the 2 that drops out of $\frac{d}{dp_j}(r_i^2) = 2\,r_i\,\partial r_i/\partial p_j$ in the gradient.

**Common pitfall. The engine prints $\chi^2$ WITHOUT the $\frac{1}{2}$.** `FitParameters.cpp` states `chi2 =` sum `residual_i^2` and accumulates exactly that, so the number in `fit_history.csv` and on the console is *twice* the boxed definition above. The minimiser is unaffected — a constant factor moves no minimum, and the $\frac{1}{2}$ is chosen here only so that the gradient below carries no stray 2 — but a student checking a printed $\chi^2$ against Eq. (200) by hand will find a factor of two, and it is this convention, not an error in the fit. (Stated 2026-09-25, when the two were found to disagree; the derivation was left on its own convention rather than rewritten.)

## 4.3 The gradient of $\chi^2$: deriving $\mathbf{J}^\top\mathbf{r}$

At a minimum, every partial derivative of $\chi^2$ vanishes. Let us compute $\partial\chi^2/\partial p_j$ explicitly. Using the chain rule on Eq. (200),

$$\frac{\partial\chi^2}{\partial p_j} \;=\; \frac{\partial}{\partial p_j}\Big[\tfrac{1}{2}\sum_{i=1}^{N} r_i^2\Big] \;=\; \tfrac{1}{2}\sum_{i=1}^{N} 2\,r_i\,\frac{\partial r_i}{\partial p_j} \;=\; \sum_{i=1}^{N} r_i\,\frac{\partial r_i}{\partial p_j}. \tag{201}$$

Now collect the partial derivatives of the residual into the **residual Jacobian**

$$\mathbf{J}_{ij}(\mathbf{p}) \;\equiv\; \frac{\partial r_i}{\partial p_j}, \qquad \mathbf{J} \in \mathbb{R}^{N\times m}, \tag{202}$$

so that the $j$-th component of the gradient is exactly $\sum_i r_i \mathbf{J}_{ij} = (\mathbf{J}^\top \mathbf{r})_j$. Stacking all $m$ partials:

$$\boxed{\nabla_{\mathbf{p}} \chi^2(\mathbf{p}) \;=\; \mathbf{J}(\mathbf{p})^\top \, \mathbf{r}(\mathbf{p}).} \tag{203}$$

A minimum therefore satisfies $\mathbf{J}^\top \mathbf{r} = 0$, the *normal-equation condition*. This is the same shape of system we have already met: $m$ equations, $m$ unknowns. The natural thing to do is run a Newton iteration on it.

## 4.4 The Hessian splits in two: keep one piece, drop the other

To apply Newton-Raphson to $\nabla\chi^2 = 0$ we need the Hessian $\nabla^2\chi^2 \in \mathbb{R}^{m\times m}$. Differentiate $(\mathbf{J}^\top \mathbf{r})_j = \sum_i r_i \, \partial r_i/\partial p_j$ once more, with respect to $p_k$:

$$\frac{\partial^2 \chi^2}{\partial p_j \partial p_k} \;=\; \sum_{i=1}^{N} \frac{\partial r_i}{\partial p_k}\,\frac{\partial r_i}{\partial p_j} \;+\; \sum_{i=1}^{N} r_i \,\frac{\partial^2 r_i}{\partial p_j \partial p_k}. \tag{204}$$

In matrix form,

$$\nabla^2\chi^2 \;=\; \underbrace{\mathbf{J}^\top \mathbf{J}}_{\text{first-derivative information}} \;+\; \underbrace{\sum_{i=1}^{N} r_i \, \nabla^2 r_i}_{\text{second-derivative information}} \qquad . \tag{205}$$

The first term costs us nothing extra — we already have $\mathbf{J}$. The second term requires the $m \times m$ Hessian of every single residual: $N$ matrices of $m^2$ second partials, each obtained from forward model calls. Forming it by finite differences would multiply the cost of one LM iteration by roughly a factor of $m$.

The *Gauss-Newton approximation* is to drop the second term:

$$\nabla^2\chi^2 \;\approx\; \mathbf{J}^\top \mathbf{J}. \tag{206}$$

> *Intuition.* Why is dropping $\sum_i r_i \nabla^2 r_i$ defensible? Look at the sum: it is weighted by the residuals themselves. At the optimum $\mathbf{p}^*$ the residuals are small (the whole point of the fit), so the dropped term is small *compared to* $\mathbf{J}^\top\mathbf{J}$ (which is a sum of squares and grows with $N$). Near $\mathbf{p}^*$, $r_i$ errors of $\sim 0.4$ K with a $\mathbf{J}$ whose columns have norm $\sim 1$ make the dropped term roughly $100\times$ smaller than the kept one. Far from $\mathbf{p}^*$ the approximation can mislead — and that is exactly where Levenberg's damping (next subsection) saves us.

With the Gauss-Newton Hessian, Newton-Raphson on $\nabla\chi^2 = 0$ becomes

$$\left(\mathbf{J}^\top \mathbf{J}\right) \Delta\mathbf{p} \;=\; -\mathbf{J}^\top \mathbf{r}, \qquad \mathbf{p}_{k+1} \;=\; \mathbf{p}_k + \Delta\mathbf{p}. \tag{207}$$

This $m \times m$ linear system is the **normal equations** of the linear least-squares problem $\min_{\Delta\mathbf{p}} \|\mathbf{r} + \mathbf{J}\,\Delta\mathbf{p}\|^2$. You can read it either way: as a Newton step on $\nabla\chi^2 = 0$ (the derivation above), or as the closed-form minimiser of the linearised residual $\mathbf{r}(\mathbf{p} + \Delta\mathbf{p}) \approx \mathbf{r} + \mathbf{J}\,\Delta\mathbf{p}$.

> **Common pitfall.** Gauss-Newton is locally a descent direction for $\chi^2$, but only *locally*. If the linearisation $\mathbf{r}(\mathbf{p} + \Delta\mathbf{p}) \approx \mathbf{r} + \mathbf{J}\,\Delta\mathbf{p}$ is poor — because we are far from $\mathbf{p}^*$, or some $r_i$ are large, or the residual surface curves sharply — the full Gauss-Newton step can overshoot wildly and *increase* $\chi^2$. You will see this in the simulator's LM trace as iterations where the trial $\chi^2$ jumps by orders of magnitude before being rolled back.

## 4.5 Levenberg's idea: a knob between two extremes

We have two algorithms in hand to minimise $\chi^2$, with opposite failure modes:

- **Gauss-Newton**, Eq. (207), takes *long* steps in the direction the linearised model prefers. Near $\mathbf{p}^*$ it converges quadratically. Far from $\mathbf{p}^*$ it overshoots.
- **Steepest descent**, $\Delta\mathbf{p} = -\alpha\,\nabla\chi^2 = -\alpha\,\mathbf{J}^\top\mathbf{r}$, takes *short* steps in the direction $\chi^2$ decreases fastest. It provably reduces $\chi^2$ for a small enough $\alpha$. But near $\mathbf{p}^*$ it zigzags through narrow valleys and crawls.

Levenberg [34] proposed to interpolate between the two by adding a tunable damping term to the normal-equation matrix:

$$\left(\mathbf{J}^\top \mathbf{J} + \lambda\,\mathbf{I}\right) \Delta\mathbf{p} \;=\; -\mathbf{J}^\top \mathbf{r}. \tag{208}$$

The two limits of the $\lambda$-knob are illuminating:

- $\lambda \to 0$: the damping vanishes and the system reduces to Eq. (207) — pure Gauss-Newton, with local quadratic convergence.
- $\lambda \to \infty$: the damping term dominates and $\Delta\mathbf{p} \approx -\lambda^{-1}\,\mathbf{J}^\top\mathbf{r} = -\lambda^{-1}\,\nabla\chi^2$ — an arbitrarily short step along the steepest-descent direction. Linear convergence, but *guaranteed* to decrease $\chi^2$ if the step is short enough.

Notice the dual role $\lambda$ plays. Increasing it does not only *rotate* the step toward the gradient: it also *shrinks* the step, because the diagonal of $\mathbf{J}^\top\mathbf{J} + \lambda\mathbf{I}$ grows and divides $\Delta\mathbf{p}$ down. That is why $\lambda$ doubles as an implicit trust-region radius: big $\lambda$ means "I do not trust the linear model out to a long step, take a short careful one."

## 4.6 Marquardt's invariance fix

Levenberg's $\lambda \mathbf{I}$ has a subtle defect: it is not invariant under a diagonal rescaling of the parameters. Suppose we re-expressed every $p_j$ in units that doubled it: $p_j \to 2p_j$. The columns of $\mathbf{J}$ would halve, $\mathbf{J}^\top\mathbf{J}$ would scale by $1/4$ along its diagonal, but $\lambda\mathbf{I}$ would not change. A $\lambda$ that was small compared to one diagonal entry of $\mathbf{J}^\top\mathbf{J}$ might now dominate it.

For NRTL parameters this matters in practice: $a_{ij}$ is a dimensionless $\mathcal{O}(1)$ number, but $b_{ij}$ carries units of K and runs $\mathcal{O}(100\text{–}1000)$. A single $\lambda$ that damps the $b$ direction reasonably will completely lock the $a$ direction (or vice versa).

Marquardt [35] fixed this by scaling the damping with the diagonal of $\mathbf{J}^\top\mathbf{J}$ itself:

$$\boxed{\left(\mathbf{J}^\top\mathbf{J} + \lambda\ \mathrm{diag}(\mathbf{J}^\top\mathbf{J})\right)\Delta\mathbf{p} \;=\; -\mathbf{J}^\top\mathbf{r}.} \tag{209}$$

This is invariant under diagonal reparametrisation: doubling all $b_{ij}$ shrinks both terms in the matrix by the same factor, so their relative weighting is preserved.

The simulator implements Eq. (209) in the *multiplicative* form that is common in practice: each diagonal entry of $\mathbf{J}^\top\mathbf{J}$ is multiplied by $1+\lambda$. That is the same equation with the $\lambda$ knob shifted by one — $\lambda = 0$ in the multiplicative form is pure Gauss-Newton; $\lambda \to \infty$ is pure scaled gradient descent.

> *Intuition.* The diagonal entry $(\mathbf{J}^\top\mathbf{J})_{jj} = \sum_i (\partial r_i/\partial p_j)^2$ is the local curvature of $\chi^2$ along the $p_j$ axis — the second derivative of $\chi^2$ in direction $j$, in the Gauss-Newton approximation. Damping in proportion to this curvature means: take small steps in well-determined (stiff) parameter directions, take longer steps in poorly-determined (slack) ones. Without the scaling, one fixed $\lambda$ ends up calibrated for the stiffest direction and is needlessly conservative everywhere else — which is what makes plain Levenberg's convergence so unsteady on real problems.

## 4.7 The trust-region update: 3 down, 5 up

The damping $\lambda$ is not fixed — it is adjusted every iteration based on whether the proposed step *actually* decreased $\chi^2$. The logic is simple:

- If $\chi^2(\mathbf{p}_k + \Delta\mathbf{p}) < \chi^2(\mathbf{p}_k)$, the linearisation was good enough. *Accept* the step ($\mathbf{p}_{k+1} = \mathbf{p}_k + \Delta\mathbf{p}$) and *shrink* $\lambda$ to push toward Gauss-Newton next time. The simulator uses $\lambda \leftarrow \lambda/3$.
- Otherwise the linearisation overshot. *Reject* the step ($\mathbf{p}_{k+1} = \mathbf{p}_k$) and *grow* $\lambda$ to take a shorter, more gradient-aligned step next time. The simulator uses $\lambda \leftarrow 5\,\lambda$.

> *Intuition.* Read $\lambda$ as a trust-region radius: small $\lambda$ = "I trust the linear model out to a long step"; large $\lambda$ = "I do not, keep close to the current $\mathbf{p}$." Each LM iteration is one experiment that tests how far the linear model can be trusted; the trust radius shrinks or grows based on the experimental outcome. This is the single algorithmic ingredient that distinguishes LM from plain Gauss-Newton — and it is the reason LM converges on problems where Gauss-Newton diverges.

The 3-down/5-up asymmetry is conservative on purpose: a rejected step has already cost a full Jacobian rebuild ($2m$ extra forward solves), so we want to avoid the *next* rejection more aggressively than we want to chase Gauss-Newton speed. The exact factors are tunable; 3 and 5 follow Press *et al.* [36] with one click more caution.

## 4.8 Finite-difference Jacobian

The model $T_i^{\mathrm{pred}}(\mathbf{p})$ is a black-box solver call, not a closed-form expression. We compute $\mathbf{J}$ by central differences in $\mathbf{p}$:

$$\mathbf{J}_{ij}(\mathbf{p}) \;\approx\; \frac{r_i(\mathbf{p} + h_j\,\mathbf{e}_j) - r_i(\mathbf{p} - h_j\,\mathbf{e}_j)}{2\,h_j}, \tag{210}$$

with a relative step

$$h_j \;=\; \epsilon_{\mathrm{rel}}\ \max(1, |p_j|), \qquad \epsilon_{\mathrm{rel}} \;=\; 10^{-4}. \tag{211}$$

Each column of $\mathbf{J}$ costs 2 forward solves of the simulator. For the canonical ethanol-water case ($m = 4$ parameters, $N = 11$ data points), one LM iteration is therefore $1 + 2\,m = 9$ vector forward evaluations, each of which loops over $N$ bubble-$T$ calls.

> **Common pitfall.** The step size $h_j$ has two error sources working against each other: **truncation error** (the second derivative we ignore in the FD formula) grows as $h^2$; **round-off error** grows as $\epsilon_{\mathrm{machine}}/h$ for an arithmetic resembling double precision. The optimum is near $h \sim \epsilon_{\mathrm{machine}}^{1/3} \approx 10^{-5}$ for central differences. The simulator's $10^{-4}$ is a shade larger — chosen because the inner bubble-$T$ solve is itself accurate to about $10^{-8}$ K, so a smaller $h$ would not buy more accuracy and would amplify the inner solver's noise.

## 4.9 Stopping criteria

The LM loop exits successfully when the relative improvement in $\chi^2$ falls below `tolerance` (default $10^{-6}$):

$$\frac{\chi^2_{k-1} - \chi^2_k}{\chi^2_{k-1}} < \texttt{tolerance}. \tag{212}$$

It exits unsuccessfully on either `maxIter` iterations (default 30) or a damping blow-up $\lambda > 10^{10}$, which signals that the trust region has collapsed — the Hessian approximation is indefinite or the residual surface is too non-smooth for the FD Jacobian. In practice the second failure mode shows up when the dataset contains near-duplicate $x_i$ values or when the initial guess is so far from $\mathbf{p}^*$ that the inner bubble-$T$ Newton itself fails to converge.

## 4.10 The simulator's implementation

The operation is registered as `type fitParameters;` in `system/propsDict` (its choupoSolve-era predecessor `fitBinaryPair` is retired — the factory throws, naming the replacement). Its main loop lives in `src/propertyOps/FitParameters.cpp`, which calls into:

- `Dictionary::setScalarAtPath()` to overwrite the four NRTL parameters in `thermoDict_` for each forward evaluation;
- the existing `BubblePoint` unit-op (which emits `T_bubble` as a KPI from onwards) for the inner bubble-$T$;
- `solver::gaussSolve` (the same Gauss elimination used by `NewtonND`, just on the $4 \times 4$ normal-equations system).

No external library, no auto-differentiation, no hand-coded analytical Jacobian: a deliberate constraint for pedagogical visibility.

**Runnable case.** `runCase tutorials/steady/optimisation/fitNRTL01_ethanol_water` produces the full LM trace, the per-point residual breakdown, and writes the proposal file `constant/parameters/NRTL/ethanol-water.fit-<date>.dat`. See the case's `README.md` for the promote-by-`mv` step.

**Worked example 4.1.** Reading the LM trace for ethanol-water at 1 atmThe simulator prints one row per iteration:

```
  iter     chi^2          RMS(K)     lambda    step
  ----  ------------  ----------  --------  --------
  init  6.0195e+01    2.3393e+00  1.0e-03               (DECHEMA literature)
  1     54.0505       2.2167      1.0e-03   523.9       (accepted, lambda /= 3)
  2     17990.5239    40.4413     3.3e-04   0.0009      (REJECTED, lambda *= 5)
  3     6020.4407     23.3947     1.7e-03   0.0009      (REJECTED)
...
  18    1.8006        0.4046      4.5e+00   6.8971      (accepted)
  19    6109.9721     23.5680     1.5e+00   0.0000      (REJECTED)
  20    1.8006        0.4046      7.4e+00   0.0000      (converged: rel<tol)
```

Read row by row:

- **Initial $\chi^2 = 60.2$, RMS $= 2.34$ K.** With the literature DECHEMA parameters, the model overpredicts $T_b$ by about 2 K on average across this dataset.
- **Iter 1 ($\lambda = 10^{-3}$):** small damping, pure Gauss-Newton step. $\chi^2$ drops marginally; step norm is large ($\sim 524$) — the gradient was steep.
- **Iter 2 ($\lambda = 3.3 \times 10^{-4}$):** the previous step shrank $\lambda$, so the next is even more Gauss-Newton-ish. But the linearisation no longer holds at this displaced point; $\chi^2$ rockets to $1.8 \times 10^4$. *Rejected*; $\lambda$ jumps to $1.7 \times 10^{-3}$.
- **Iters 3–17:** the loop oscillates between accept (with $\lambda$ decreasing 3-fold) and reject (with $\lambda$ increasing 5-fold). $\chi^2$ drifts down on the accept rows. The `step` column shrinks as the active subspace of $\mathbf{p}$ narrows.
- **Iter 18:** an accepted step that drops $\chi^2$ to 1.80, RMS = 0.40 K. Five-fold improvement over literature for THIS dataset.
- **Iter 19:** a final overreach, rejected.
- **Iter 20:** same $\chi^2$ as iter 18, but with the relative improvement now $< 10^{-6}$. Converged.

The lesson visible in the trace: the trust region tightens the more non-linear the residual surface is around the current $\mathbf{p}$, and relaxes when the linearisation holds. That dance is exactly what distinguishes LM from plain Gauss-Newton — the latter would have diverged at iteration 2.

*Intuition.* Levenberg-Marquardt is best understood not as a single algorithm but as a continuously-tuned *mixture* of two limits: Gauss-Newton (fast but fragile) and gradient descent (slow but bullet-proof). The $\lambda$ knob slides between them automatically based on whether the last step worked. The Marquardt diagonal scaling makes the knob behave the same regardless of how you re-express the parameters.

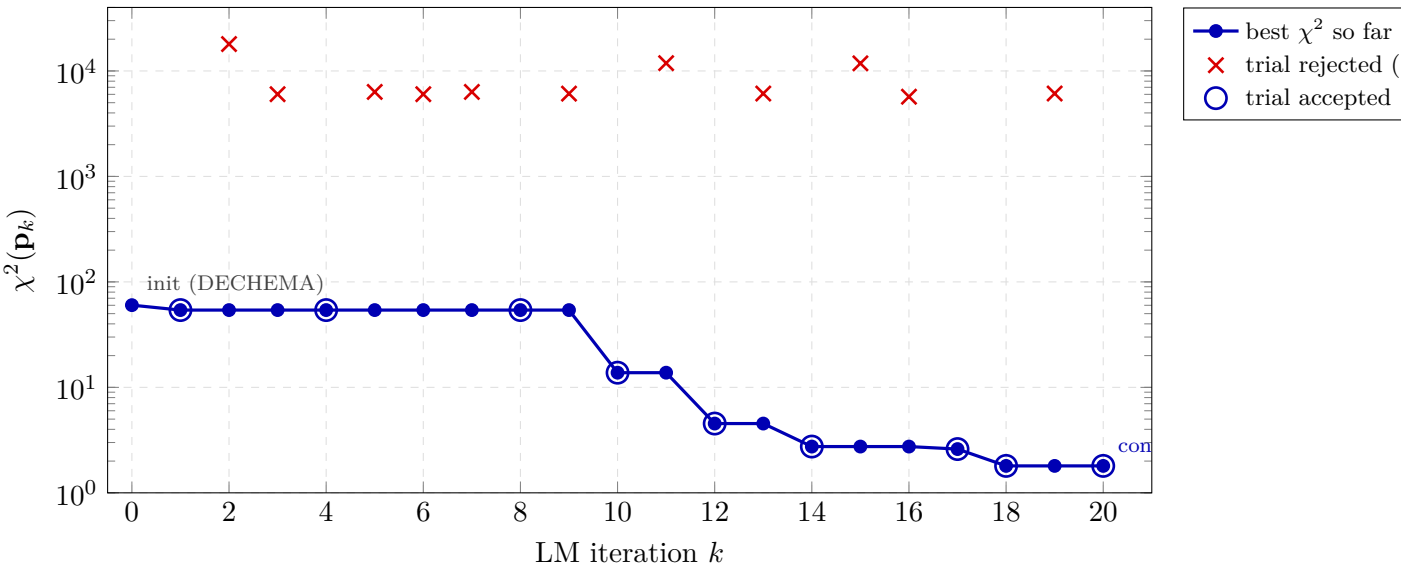



**Figure 34:** Levenberg-Marquardt $\chi^2$ history for the ethanol/water NRTL fit in `tutorials/steady/optimisation/fitNRTL01_ethanol_water` (real run, all 20 iterations). The solid blue line tracks the best $\chi^2$ achieved so far — the parameters that are carried forward. The red crosses are trial $\chi^2$ values the LM proposed but *rejected* because they were larger than the best, each triggering $\lambda \leftarrow 5\lambda$. The blue circles are the accepted trials, each triggering $\lambda \leftarrow \lambda/3$. The 3-up/5-down trust-region update keeps the accepted sequence monotone even though per-iteration trials oscillate over three orders of magnitude.

# 5 Wegstein acceleration

*What you'll learn.* *Fixed-point* problems — equations of the form $x = f(x)$ — are at the heart of every simulator's outer loops. The naive solution method, called *direct substitution*, can be glacially slow. This chapter derives Wegstein's acceleration from first principles, step by step: from the slowness of direct substitution, to the idea of approximating $f$ as linear, to the closed-form formula that the simulator uses for its activity-coefficient loop.

## 5.1 The setting: fixed-point problems

A *fixed-point problem* is the search for a value $x^*$ that satisfies

$$x^* \;=\; f(x^*), \tag{213}$$

where $f\colon \mathbb{R} \to \mathbb{R}$ is some function we can evaluate. Geometrically, $x^*$ is the value where the curve $y = f(x)$ crosses the diagonal $y = x$. In a process simulator a typical $f$ takes a current estimate of the activity coefficients and returns an updated estimate (we'll get to the details in §13 and §2); what matters here is the abstract problem.

## 5.2 Direct substitution and why it can be slow

The most obvious way to solve $x = f(x)$ is to start somewhere, plug in, and repeat:

$$x_{k+1} \;=\; f(x_k), \qquad k = 0, 1, 2, \ldots \tag{214}$$

This is *direct substitution*. When it works, it works because each step shrinks the error. Let $e_k \equiv x_k - x^*$. Linearising $f$ around $x^*$,

$$f(x_k) \;=\; f(x^*) + f'(x^*)\, e_k + O(e_k^2) \;=\; x^* + f'(x^*)\, e_k + O(e_k^2),$$

so

$$e_{k+1} \;=\; x_{k+1} - x^* \;=\; f'(x^*)\, e_k + O(e_k^2). \tag{215}$$

The error shrinks by the factor $|f'(x^*)|$ each step. If this *contraction factor* is 0.1, ten iterations give six digits; if it is 0.9, ten iterations give roughly half a digit.

**Worked example 5.1.** Direct substitution on a slow problemConsider $f(x) = 0.9\,x + 0.5$, with fixed point $x^* = 5$ (you can verify by substitution). Here $f'(x) = 0.9$ everywhere, so the contraction factor is exactly 0.9. Starting at $x_0 = 0$:

$$x_0 = 0,\;\; x_1 = 0.5,\;\; x_2 = 0.95,\;\; x_3 = 1.36,\;\; x_4 = 1.72,\;\; \ldots,\;\; x_{20} \approx 4.39,\;\; x_{30} \approx 4.79.$$

After thirty iterations we are still 0.21 away from the answer. This kind of slow contraction is exactly what happens to the activity-coefficient loop near an azeotrope.

## 5.3 The Wegstein idea: model $f$ as a straight line

Wegstein's observation [3] is this: *if we already have two iterates $x_{k-1}$ and $x_k$, we know not just one point on the curve $y = f(x)$ but* two. *Two points define a line. We can use that line as a local linear approximation to $f$, and solve the linear problem $x =$ (line) exactly in one step.*

**The model.** Approximate the curve $y = f(x)$ by the secant line through $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$. Its slope is

$$s_k \;\equiv\; \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}. \tag{216}$$

The line itself can be written, point-slope form anchored at $(x_k, f(x_k))$, as

$$y \;=\; f(x_k) + s_k\,(x - x_k).$$

**The next iterate.** Treat this line as if it *were* the true $f$ and solve the linear fixed-point equation $y = x$:

$$x \;=\; f(x_k) + s_k\,(x - x_k).$$

Collect $x$-terms on the left,

$$x\,(1 - s_k) \;=\; f(x_k) - s_k\,x_k,$$

and divide by $(1 - s_k)$:

$$x_{k+1} \;=\; \frac{f(x_k) - s_k\,x_k}{1 - s_k}. \tag{217}$$

**Rearranging into Wegstein's form.** Equation (217) is the answer, but it is rarely written this way. Define a single bookkeeping coefficient

$$q_k \equiv \frac{s_k}{s_k - 1} \tag{218}$$

(so $1 - q_k = -1/(s_k - 1) = 1/(1 - s_k)$, and $1 - q_k = 1 + 1/(1 - s_k)$ when written symmetrically — the algebra is in the next paragraph). Multiplying numerator and denominator of (217) by $-1$ and using $q_k$:

$$x_{k+1} = \frac{s_k\, x_k - f(x_k)}{s_k - 1} = \frac{s_k}{s_k - 1}\, x_k - \frac{1}{s_k - 1}\, f(x_k) = q_k\, x_k + (1 - q_k)\, f(x_k),$$

because $-1/(s_k - 1) = 1 - s_k/(s_k - 1) = 1 - q_k$. This is the Wegstein step:

$$\boxed{x_{k+1} = q_k\, x_k + (1 - q_k)\, f(x_k), \quad \text{with} \quad q_k = \frac{s_k}{s_k - 1}, \quad s_k = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}.} \tag{219}$$

**Reading the formula.** The Wegstein step is a *linear combination* of the current iterate $x_k$ and the direct-substitution next iterate $f(x_k)$. The coefficient $q_k$ controls the mix. Three limiting cases give the intuition:

- $s_k = 0$ (the function is locally flat): $q_k = 0$. The step becomes $x_{k+1} = f(x_k)$ — ordinary direct substitution. Makes sense: a flat $f$ has nothing to accelerate.
- $s_k \to 1^-$ (the contraction is becoming slow: $f'(x^*)$ near 1): $q_k$ becomes a large negative number, $1 - q_k$ a large positive one. The step throws the iterate far past $f(x_k)$ in the same direction. This is the whole point: direct substitution would take many steps to cover that distance; Wegstein takes one.
- $s_k > 1$: the secant slope is steeper than the diagonal, so the linear model is not a contraction at all. The formula gives a positive $q_k$ that can be arbitrarily large, throwing the iterate in the *wrong* direction. We will fix this in §5.5.

> *Intuition.* Direct substitution says "my next guess is wherever $f$ took me". Wegstein says "$f$ is approximately linear locally; let me solve the linear problem directly and jump straight to the answer". When $f$ really is linear, Wegstein hits $x^*$ in one step (the secant is the curve itself). When $f$ is non-linear, the secant is only a chord; we land near $x^*$, take a new secant from the latest two iterates, and try again. Each round the secant is closer to the local tangent at $x^*$, so convergence accelerates as we close in. This is *super-linear* convergence: the per-step error reduction itself improves over time, unlike direct substitution's fixed contraction factor.

## 5.4 A worked iteration

Take the slow problem from the earlier example: $f(x) = 0.9\,x + 0.5$, $x^* = 5$. Start from two direct-substitution iterates $x_0 = 0$ and $x_1 = f(x_0) = 0.5$. Then:

$$\begin{aligned} s_1 &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{0.95 - 0.5}{0.5 - 0} = 0.9, \\ q_1 &= \frac{s_1}{s_1 - 1} = \frac{0.9}{-0.1} = -9, \\ x_2 &= q_1\, x_1 + (1 - q_1)\, f(x_1) = -9 \cdot 0.5 + 10 \cdot 0.95 = -4.5 + 9.5 = 5.0. \end{aligned}$$

*One* Wegstein step reaches the fixed point exactly — whereas direct substitution needed thirty iterations to get within 0.21. The miracle is that $f$ is linear, so Wegstein's linear model is *exact*, and the linear problem is solved in closed form.

## 5.5 When the linear model misleads: the $q$-clip

The miracle case ($f$ linear) is rare; the typical case is $f$ non-linear and the secant slope $s_k$ a noisy approximation to $f'(x^*)$. Two pathologies show up.

**Pathology 1: the wrong contraction direction ($s_k > 1$).** If the secant slope is greater than 1, the linear model is not a contraction. Geometrically, the secant line is steeper than the diagonal $y = x$ and intersects it on the *far side* of both data points — the extrapolation lands somewhere unrelated to the actual fixed point. Plug $s_k > 1$ into (218): $q_k = s_k/(s_k - 1)$ is positive and possibly large, and the formula sends the iterate in the wrong direction.

**Pathology 2: over-extrapolation.** Even with $s_k < 1$ (genuine contraction), if $s_k$ is very close to 1 the denominator $s_k - 1$ is tiny and $q_k$ becomes a huge negative number. The Wegstein step then throws the iterate *far* past the secant intercept — correct in direction but wildly overshooting. This is fine when $f$ really is approximately linear over a wide range; it is destabilising when it is not.

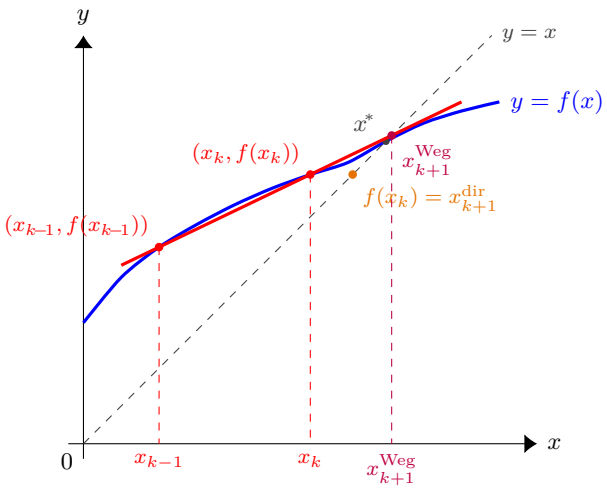



**Figure 35:** Geometric construction of one Wegstein step. Two prior iterates lie on the curve $y = f(x)$ as $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$. Their *secant line* (red) approximates $f$ locally as an implicit linear model; the next iterate $x_{k+1}^{\text{Weg}}$ is where this secant crosses the fixed-point diagonal $y = x$. Direct substitution would jump only as far as $f(x_k)$ along the diagonal (orange) — visibly short of $x^*$ — while the secant extrapolation lands almost exactly on the fixed point. For a *linear* $f$ the secant coincides with the curve and Wegstein hits $x^*$ in one step; for the non-linear $f$ shown, the secant is a chord and one step gets within $\sim 2\%$ of the root.

**The fix: clip $q_k$ to a safe interval.** The simulator forces

$$q_k \in [\, q_{\min},\, q_{\max} \,], \qquad \text{typical defaults: } q_{\min} = -5, \;\; q_{\max} = 0. \tag{220}$$

These bounds carry concrete meaning:

- $q_{\max} = 0$ disables wrong-direction steps. With $q = 0$ Wegstein degrades to direct substitution ($x_{k+1} = f(x_k)$) — slow but never wrong.
- $q_{\min} = -5$ caps the maximum acceleration to a factor of 6: the step travels at most six times further than direct substitution would. When the secant flattens ($s_k \to 1^-$), this is what keeps the iterate from escaping the basin of attraction.

> *Intuition.* The clip turns Wegstein into a kind of adaptive method that uses the secant when it is informative (modest $s_k$, $-5 \le q_k < 0$) and falls back to safe direct substitution when it is not ($s_k > 1 \Rightarrow q_k \to 0$). The student should picture $q_k$ as a knob between two regimes, with $q_{\min}$ and $q_{\max}$ as safety limits chosen empirically across decades of process simulation.

## 5.6 Where this matters in the simulator: the $\gamma$-loop

Wegstein is the workhorse of the simulator's activity-coefficient outer loop. After a Rachford-Rice solve, the calculated compositions $\mathbf{x}^*$ depend on the assumed $\gamma$-values; we recompute $\gamma_i(\mathbf{x}^*, T)$, get new $K$-values, redo Rachford-Rice, and so on. Symbolically the structure is

$$\gamma^{(k+1)} \;=\; \text{Wegstein}\big(\gamma^{(k)},\, f(\gamma^{(k)})\big), \qquad f(\gamma) \;\equiv\; \gamma_i\big(\mathbf{x}^*(\gamma),\, T\big),$$

i.e. $f$ is the composition of the Rachford-Rice solver with a re-evaluation of the activity model. Wegstein is applied *component-wise*: $N_c$ independent scalar accelerations, one per chemical species, not a single vector norm and not the residual. This component-wise design is important because different components in a non-ideal mixture can have very different local contraction factors — a single global $q_k$ would slow the fast ones to match the slow ones.

In simulation runs near the ethanol/water azeotrope, where the ideal-mixing contraction factor is close to 1, Wegstein routinely converges in $\sim 7$ outer iterations against the $\sim 29$ that direct substitution would need on the same case. That four-fold speed-up across thousands of flash calculations per simulator pass is what makes a non-ideal flowsheet practical to run on a laptop.

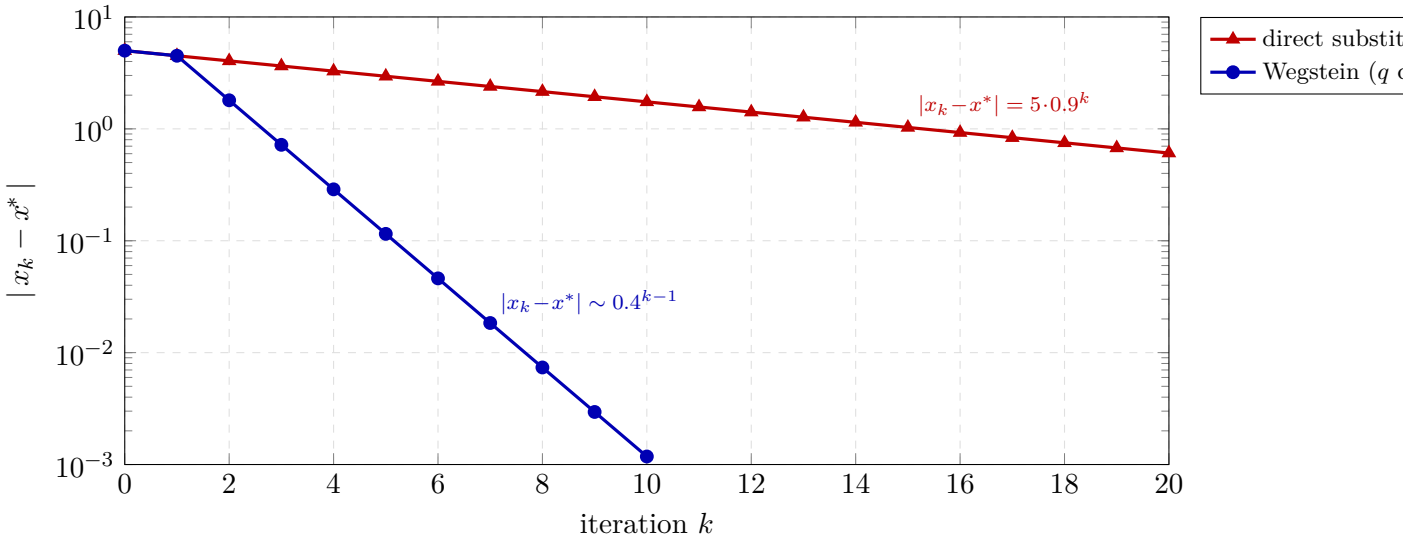



**Figure 36:** Wegstein vs. direct substitution on the test problem $g(x) = 0.9\,x + 0.5$ with fixed point $x^* = 5$ (chosen so that $g'(x^*) = 0.9$ is close to one — direct substitution converges slowly). Direct substitution contracts at rate 0.9 (geometric, dotted red); Wegstein with $q$ clipped to $[-5, 0]$ contracts at rate $\sim 0.4$, two-and-a-half times faster on a logarithmic scale. After ten iterations Wegstein has reduced the error by three orders of magnitude where direct substitution has barely halved it; after twenty iterations the gap is six orders.

# 6 Closing a recycle: Newton on the tear stream

*What you'll learn.* A flowsheet with a recycle has a directed cycle: a downstream unit feeds an upstream one, so no unit can be solved until the stream entering the loop is already known — a chicken-and-egg the simulator breaks by *tearing* the loop, guessing the torn stream, sweeping once, and iterating until the guess reproduces itself. The previous chapter accelerated that iteration scalar-by-scalar with Wegstein. This chapter takes the alternative the simulator now uses by default: treat the whole torn stream as one vector and drive its residual to zero with the $n$-dimensional Newton of Chapter 3. We derive the tear residual, justify the choice of tear variables, and — via a real bug the change exposed — explain why the residual must be *relative*.

## 6.1 The recycle as a fixed-point in the tear stream

Recall the tearing picture from Chapter 1. Cut one stream in the cycle — the *tear stream* — and let $\mathbf{x}$ be its state (flows, temperature). Guessing $\mathbf{x}$ makes the loop acyclic: every unit now has a known inlet, so one pass of the ordinary sequential-modular executive can be run, unit by unit, all the way around the loop until it recomputes the torn stream. Call the result of that one full sweep

$$\mathbf{x}^{\text{new}} \;=\; G(\mathbf{x}), \tag{221}$$

where $G$ is "run the whole flowsheet once, starting from this tear guess". The recycle is converged when the guess reproduces itself,

$$G(\mathbf{x}^*) \;=\; \mathbf{x}^* \qquad\Longleftrightarrow\qquad \mathbf{r}(\mathbf{x}^*) \;\equiv\; G(\mathbf{x}^*) - \mathbf{x}^* \;=\; \mathbf{0}. \tag{222}$$

Wegstein[3] attacks the left form (a fixed point $\mathbf{x} = G(\mathbf{x})$, accelerated component-wise). Newton attacks the right form (a root $\mathbf{r}(\mathbf{x}) = \mathbf{0}$), all components at once. Same $G$, same sweep, same per-unit pedagogy — only the outer iteration differs.

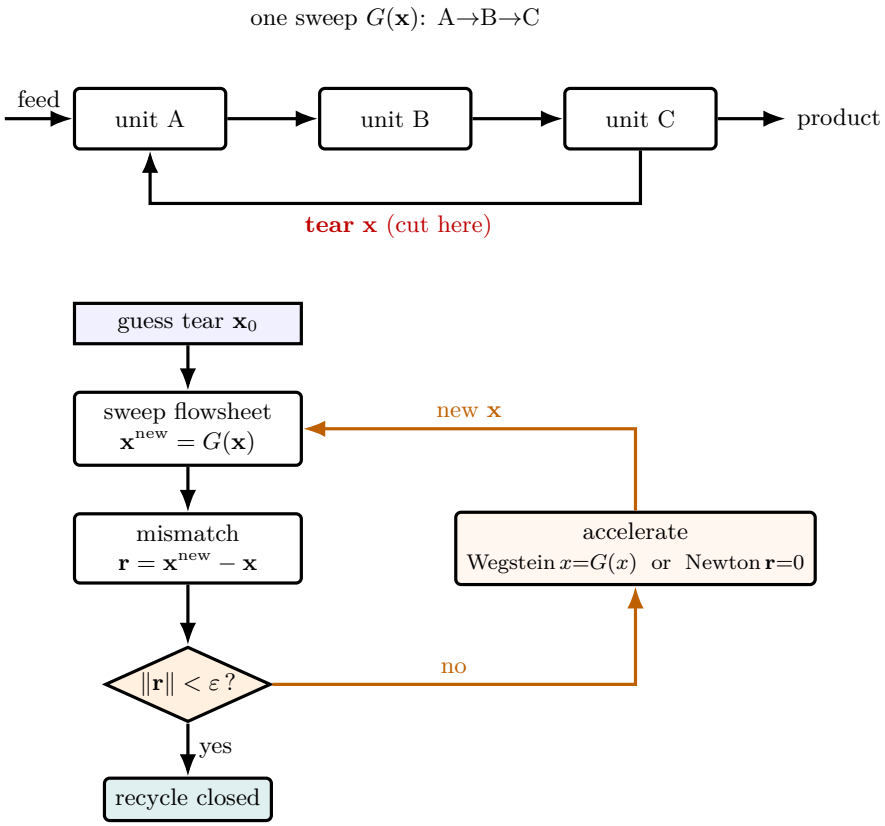



**Figure 37:** Closing a recycle. *Left:* the physical flowsheet has a cycle (C feeds back to A); cutting the recycle at the dashed **tear** turns it into an open chain whose one sweep $G(\mathbf{x})$ = "run A→B→C from this tear guess" is just the ordinary sequential pass. *Right:* the outer loop guesses the tear, sweeps once, measures the mismatch $\mathbf{r} = G(\mathbf{x}) - \mathbf{x}$, and — if it is still too large — hands $\mathbf{r}$ to an accelerator (Wegstein on the fixed point, or Newton on the root) to propose a better tear, then sweeps again. The convergence is reached when the guess reproduces itself. Crucially, the per-unit physics inside $G$ never changes; only the box that proposes the *next* tear differs between Wegstein and Newton.

## 6.2 What goes in the tear vector

A process stream carries a total flow, a composition, a temperature, a pressure and a phase split — but these are not all independent, and a good tear vector should use only independent coordinates. The tempting choice "total flow $F$ plus all mole fractions $z_i$" is *redundant*: the fractions obey $\sum_i z_i = 1$, so $(F, z_1, \dots, z_{N_c})$ carries $N_c + 1$ numbers for a stream whose physical content is only its $N_c$ component flows. The redundancy is not catastrophic — it does *not*, on its own, make the Jacobian singular: the surplus direction (the one that holds every component flow $Fz_i$ fixed while trading $F$ off against the $z_i$) leaves $G$ unchanged, so $\partial G/\partial \mathbf{x}$ annihilates it and the Jacobian $\mathbf{J} = \partial G/\partial \mathbf{x} - \mathbf{I}$ maps it to the harmless eigenvalue $-1$, not 0. It is simply wasteful and awkward: one extra variable (hence an extra pair of Jacobian sweeps), and the iterate's $z_i$ must be re-normalised onto the simplex every step or they drift and $F$ loses its meaning.

Choupo sidesteps this by tearing on *component molar flows* plus temperature,

$$\mathbf{x} \;=\; (F_1,\, F_2,\, \dots,\, F_{N_c},\, T), \qquad F_i \equiv F z_i, \tag{223}$$

which are mutually independent — there is no normalisation tying them together (the total $F = \sum_i F_i$ and the fractions $z_i = F_i/F$ are recovered afterwards). Pressure and phase split are not torn: pressure is set by the units, and the phase fractions follow from a flash on the converged $(F_i, T)$. For $N_c$ components the tear vector has $N_c + 1$ entries.

## 6.3 Newton on the tear residual

With the residual (222) in hand, the rest is the machinery of Chapter 3 applied verbatim. The Jacobian

$$\mathbf{J} \;=\; \frac{\partial \mathbf{r}}{\partial \mathbf{x}} \;=\; \frac{\partial G}{\partial \mathbf{x}} - \mathbf{I} \tag{224}$$

is built by *central* finite differences: perturb tear variable $j$ by $\pm\delta$, run a full sweep $G$ at each, and the symmetric difference $[\,G(\mathbf{x} + \delta\mathbf{e}_j) - G(\mathbf{x} - \delta\mathbf{e}_j)\,]/2\delta$ gives column $j$. That is *two* sweeps per variable, $2(N_c + 1)$ sweeps per Jacobian — the price Newton pays that Wegstein (one sweep per step) does not. Central differences cost twice what forward ones would, but are $O(\delta^2)$ accurate rather than $O(\delta)$, which keeps the Jacobian — and the convergence rate it drives — clean. Each Newton step then solves the linear system

$$\mathbf{J}\,\Delta\mathbf{x} \;=\; -\,\mathbf{r}(\mathbf{x}), \tag{225}$$

by the same Gauss elimination with partial pivoting, takes $\mathbf{x} \leftarrow \mathbf{x} + \alpha\,\Delta\mathbf{x}$ under a backtracking line search (start at $\alpha = 1$ and halve it until the residual norm $\|\mathbf{r}\|$ actually drops, so the method stays robust far from the solution), and repeats. In code this is literally a call to `solver::newtonND` with $\mathbf{r}$ as the residual functor — the recycle reuses the project's one general-purpose $n$-D Newton, no special-case algebra. Near the solution it converges essentially *quadratically* — the number of correct digits roughly doubles each step, the $O(\delta^2)$ central-difference Jacobian holding that rate down to a very low floor — where Wegstein only improves super-linearly.

## 6.4 Why the residual must be relative — a bug story

There is a subtlety that bites hard, and the simulator learned it the painful way. Write the convergence test naively as an *absolute* norm, $\|\mathbf{r}\| < \varepsilon$ with, say, $\varepsilon = 10^{-5}$. Now look at what is inside $\mathbf{r}$ for a typical tear: a temperature near 365 and component flows that, for a small recycle, can be of order $2 \times 10^{-4}$ kmol/s. The norm is *dominated by the temperature residual*. A solver can satisfy $\|\mathbf{r}\| < 10^{-5}$ with the temperature pinned to six digits while the tiny tear flow is still off by 5 % — and report success.

That is exactly what happened. The `process03_recycle` tutorial ran for versions with its recycle flow converged to only a few percent; the recorded golden value was itself under-converged, and nobody noticed because the exit code was zero and the temperature looked perfect. The fix is to make every component of the residual *dimensionless and comparable* by scaling each by the magnitude of its own variable:

$$\boxed{\;\tilde{r}_i \;=\; \frac{G_i(\mathbf{x}) - x_i}{s_i}, \qquad s_i = \begin{cases} F_{\text{tot}} & \text{for a component-flow entry,} \\ T & \text{for the temperature entry,} \end{cases}\;} \tag{226}$$

so that "converged" means *every* variable — the dominant temperature and the faint trace flow alike — has stopped moving to the same relative precision. With (226) the process03 recycle converges to its true fixed point ($F \approx 2.008 \times 10^{-4}$ kmol/s, $\|\tilde{\mathbf{r}}\| \approx 5 \times 10^{-9}$ in five Newton steps), and the golden master was re-recorded against that genuinely-converged answer.

> *Intuition.* A residual norm is only as trustworthy as its worst-scaled component. Mixing a temperature ($\sim 10^2$) and a trace flow ($\sim 10^{-4}$) in one absolute norm is like checking that a bridge is level by averaging the height of the towers and the rivets — the towers swamp the average and a crooked rivet passes. Scale first, then compare.

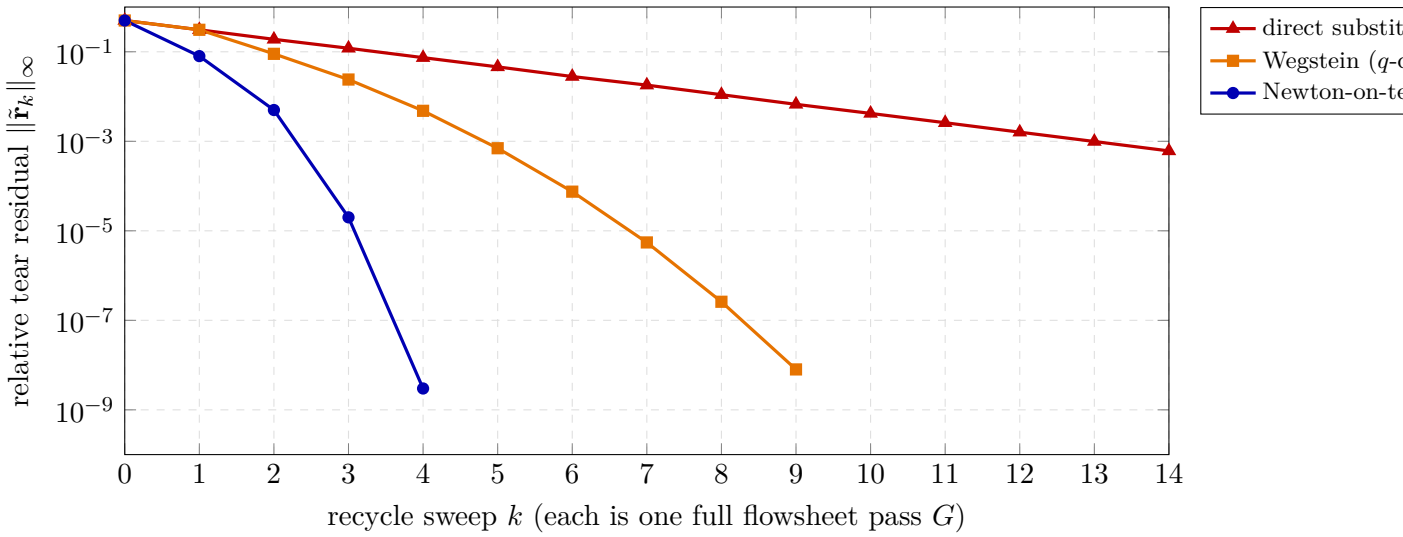



**Figure 38:** The three ways to close the same coupled recycle, plotted as relative tear residual versus outer sweep (`process03_recycle`). Direct substitution contracts at a fixed geometric rate — a straight line on the log scale, slow whenever the loop gain is near one. Wegstein bends downward (super-linear: each step's reduction improves as the secant sharpens), and Newton-on-tears dives quadratically, clearing machine precision in $\sim$5 outer steps. The catch hidden by the $x$-axis: a Newton step costs $2(N_c{+}1)$ sweeps to build its central-difference Jacobian, while Wegstein and direct substitution cost one sweep each — so Newton wins on *coupled* tears where its Jacobian pays for itself, and Wegstein wins when sweeps are dear and the tear is nearly diagonal.

## 6.5 Wegstein or Newton? Choosing the recycle solver

Both are available; `recycleSolver Newton;` is the default and

`recycleSolver Wegstein;` keeps the accelerator. They trade off cleanly:

| | **Wegstein** | **Newton-on-tears** |
|---|---|---|
| cost per outer step | one sweep | $2(N_c + 1)$ sweeps (central-difference Jacobian) |
| convergence | super-linear | quadratic |
| coupling between tear variables | ignored (component-wise scalars) | captured (full Jacobian) |
| robustness on counter-current / strongly-coupled tears | needs damping ($q$-clip) | strong |
| best when | loosely-coupled single tear | coupled and/or multiple tears |

Newton wins when the tear variables are coupled — a change in one component flow swings the others through the units — because its Jacobian sees that coupling, which Wegstein's independent scalars cannot. Wegstein can still be the better tool when sweeps are expensive and the tear is nearly diagonal, or as a robust damped fallback: the counter-current double-effect evaporator `evaporator05` runs damped Wegstein ($q_{\min} = -0.7$) to tame the over-shoot endemic to backward-feed vapour recycles, where an undamped step would oscillate.

## 6.6 This is not equation-oriented — and that is the point

It is tempting to call Newton-on-tears "equation-oriented", but it is not. A true equation-oriented (EO) solver collapses *every* unit's internal equations and *every* stream connection into one giant sparse system and Newtons on all of it at once; the units lose their identity as solvable modules. Choupo deliberately does not do that (see the v1.0 note in `CLAUDE.md`): full EO needs hand-rolled sparse linear algebra, and it dissolves the per-unit transparency that is the whole pedagogical point — a student can no longer watch a single flash or reactor converge in isolation.

Newton-on-tears keeps the sequential-modular skeleton intact — each unit still solves itself, $G$ is still one honest sweep — and wraps *only the recycle* in a Newton. It is the Pareto step: most of the convergence robustness of EO, at none of its architectural cost, on the handful of variables (the tears) where the coupling actually lives. If the project ever adds a true EO mode it will be an opt-in alternative (`solver eo;`) offered *alongside* the modular path, precisely so the manual can teach the contrast — not a replacement for it.

**Runnable case.** `tutorials/steady/flowsheets/process03_recycle` — reactor–separator with a recycle, `recycleSolver Newton;` explicit; the genuinely-tight fixed point of the bug story above. `tutorials/steady/flowsheets/process05_isomerization_recycle` — a second Newton recycle. `tutorials/steady/evaporation/evaporator05_counter_current` — two tear streams closed with damped `recycleSolver Wegstein;`, the counter-current case where damping earns its keep.

# 7 Direct search: Nelder-Mead

*What you'll learn.* Newton and Levenberg-Marquardt need derivatives. Wegstein needs a fixed-point iteration. When the objective is the output of a long, discontinuous, or noisy simulator pass — as it is whenever we ask "what reflux ratio minimises the reboiler duty?" — derivatives may not exist as smooth analytical objects, and finite differences may amplify noise. Nelder-Mead is the canonical *derivative-free* optimiser: it maintains a geometric simplex of $n+1$ points in $\mathbb{R}^n$ and crawls downhill by reflecting and contracting that shape. It is the algorithm behind `outerDict type optimization`.

## The problem

We want to minimise a scalar

$$f : \mathbb{R}^n \longrightarrow \mathbb{R},$$

where $f$ is provided as a black box — evaluating $f(\mathbf{x})$ runs one full simulator pass on a flowsheet with the design variables $\mathbf{x}$ written into the dictionary, and (optionally) a post-processing chain (sizing + costing) on top of the result. We have:

- no symbolic derivative;
- finite differences are expensive (each one is another full simulator pass) *and* noisy when the inner solvers carry their own tolerance;
- bounds on each $x_i$ (refluxes cannot be negative, reactor volumes cannot exceed your budget).

The Nelder-Mead simplex algorithm [37] sidesteps all three obstacles.

## The simplex

A simplex in $\mathbb{R}^n$ is a set of $n+1$ vertices,

$$\mathcal{S} = \{\mathbf{x}^{(0)}, \mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\},$$

in general position — they span $\mathbb{R}^n$ as an affine basis. In two dimensions a simplex is a triangle; in three, a tetrahedron. At each iteration we order the vertices by their objective values:

$$f(\mathbf{x}_{\text{best}}) \;\le\; f(\mathbf{x}_{\text{second}}) \;\le\; \cdots \;\le\; f(\mathbf{x}_{\text{worst}}).$$

We then try to replace the worst vertex by something better, leaving the other $n$ alone.

## The four moves

Let $\mathbf{x}_c$ be the centroid of all vertices *except* the worst:

$$\mathbf{x}_c \;=\; \frac{1}{n} \sum_{k \neq \text{worst}} \mathbf{x}^{(k)}.$$

The candidates are formed along the line from the worst point through the centroid. In order of cost (each move is one extra evaluation):

### 1. Reflection.

$$\mathbf{x}_r \;=\; \mathbf{x}_c + \alpha\,(\mathbf{x}_c - \mathbf{x}_{\text{worst}}), \qquad \alpha = 1.$$

The worst vertex is reflected through the centroid. If $f(\mathbf{x}_r)$ falls between $f(\mathbf{x}_{\text{best}})$ and $f(\mathbf{x}_{\text{second}})$, the reflection is accepted and the iteration ends.

### 2. Expansion.

If $f(\mathbf{x}_r) < f(\mathbf{x}_{\text{best}})$, the reflection direction looks *very* promising; we extend further:

$$\mathbf{x}_e \;=\; \mathbf{x}_c + \gamma\,(\mathbf{x}_r - \mathbf{x}_c), \qquad \gamma = 2.$$

Whichever of $\mathbf{x}_e, \mathbf{x}_r$ has the lower $f$ replaces the worst vertex.

### 3. Contraction.

If $f(\mathbf{x}_r) \ge f(\mathbf{x}_{\text{second}})$, the reflection landed on a hill; we shrink toward the centroid instead:

$$\mathbf{x}_o \;=\; \mathbf{x}_c + \rho\,(\mathbf{x}_{\text{worst}} - \mathbf{x}_c), \qquad \rho = 0.5.$$

(In Choupo we use *outside contraction* — $\mathbf{x}_c + \rho\,(\mathbf{x}_r - \mathbf{x}_c)$ — when $f(\mathbf{x}_r) < f(\mathbf{x}_{\text{worst}})$, otherwise the inside form above. Either way the new point is on the line through $\mathbf{x}_c$, closer to it.)

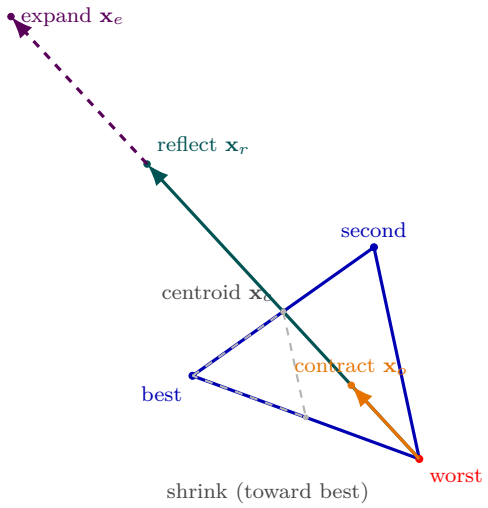



**Figure 39:** The four Nelder–Mead moves as geometry, on a triangular simplex ($n = 2$). All candidates lie on the line through the worst vertex and the centroid $\mathbf{x}_c$ of the kept vertices. *Reflection* (teal) flips the worst point through $\mathbf{x}_c$; if that is wildly successful, *expansion* (violet, dashed) pushes further along the same ray; if reflection lands on a hill, *contraction* (orange) pulls back toward $\mathbf{x}_c$; and if even that fails, *shrink* (grey, dashed) drags every vertex but the best halfway in, making the whole simplex smaller and more local. Only shrink re-evaluates more than one vertex — it is the algorithm conceding the reflection ray has stalled. The decision tree of Fig. 40 is exactly the rule for picking which of these four points to keep.

**4. Shrink.** If contraction also fails, every vertex except the best is pulled halfway toward the best:

$$\mathbf{x}^{(k)} \leftarrow \mathbf{x}_{\text{best}} + \sigma\,(\mathbf{x}^{(k)} - \mathbf{x}_{\text{best}}), \qquad \sigma = 0.5,$$

for $k \neq$ best. This is the only move that re-evaluates multiple vertices. It is the algorithm's confession that progress along the reflection line has stalled and the simplex needs to become smaller and more local.

## Box bounds without changing the algorithm

Nelder-Mead is, by construction, unconstrained. When the user supplies bounds $\mathbf{x} \in [\mathbf{x}_{\text{lo}}, \mathbf{x}_{\text{hi}}]$ for each variable, Choupo *clips* every candidate vertex back into the box element-wise *before* calling $f$:

$$x_i \leftarrow \min(\max(x_i, \text{lo}_i), \text{hi}_i).$$

The simplex learns very quickly that probing the violation region pays the same as the wall, so it contracts away from the wall. This is the simplest defensible handling; if you want a smooth penalty term or a true active-set strategy, that is the job of an SQP or interior-point wrapper, not Nelder-Mead.

## Stopping criteria

Two natural tests:

- **tolX** — the simplex has shrunk. We compute the maximum coordinate-wise extent of the simplex, normalised *per axis* by that axis's bound width:

  $$\text{tolX trip} \;:\; \max_i \frac{\max_k |x_i^{(k)} - x_i^{\text{best}}|}{\text{hi}_i - \text{lo}_i} \;<\; \text{tolX}.$$

  The per-axis normalisation is crucial: if one variable is a reactor volume in $\text{m}^3$ and the other a temperature in K, a global $\|\Delta\mathbf{x}\|$ would be dominated by the larger scale and would stop the algorithm before the small axis has converged. Choupo uses this scaled criterion.
- **tolF** — the range of $f$ over the simplex is small relative to $|f_{\text{best}}|$. Useful when $f$ is flat near the optimum.

Plus the usual safety net of `maxIter`.

## When Nelder-Mead is the right hammer

- Modest number of design variables (say $n \leq 10$). The simplex needs $n + 1$ vertices and each iteration costs at least one new evaluation, so the wall-clock cost grows gracefully in $n$.
- Black-box, possibly noisy, possibly discontinuous objective. The simulator pass is exactly that.
- Local search. Nelder-Mead converges to a local minimum near the initial simplex; for a global search you would restart with several random simplexes (*multi-start*) and keep the best.

## When to reach for something else

- High dimension ($n \gg 10$). Simplex moves become inefficient because each move only affects one vertex; on a 50-D problem, a quasi-Newton method with finite differences typically wins despite the gradient cost.
- Smooth, differentiable objective with cheap derivatives. Use a Newton-family method directly (Choupo's NewtonND, or LM for least-squares, both already in §3 and §4).
- Hard constraints, especially non-linear ones. Use SQP — Choupo's constrained optimiser, §8 — or an interior-point wrapper (Choupo offers no interior-point method).

## What you see in Choupo

The `OptimizationDriver` prints one line per simplex iteration, showing the move type (`reflect` / `expand` / `contract` / `shrink`), the current best $f$ value, the current best `x`, and the simplex size. Identical rows go into `optimization_history.csv` so the student can plot the convergence trajectory. On termination the simulator is replayed once at the optimum with full verbosity, so the case directory ends with the converged final state on standard output — exactly as a direct (non-outer) run would leave it.

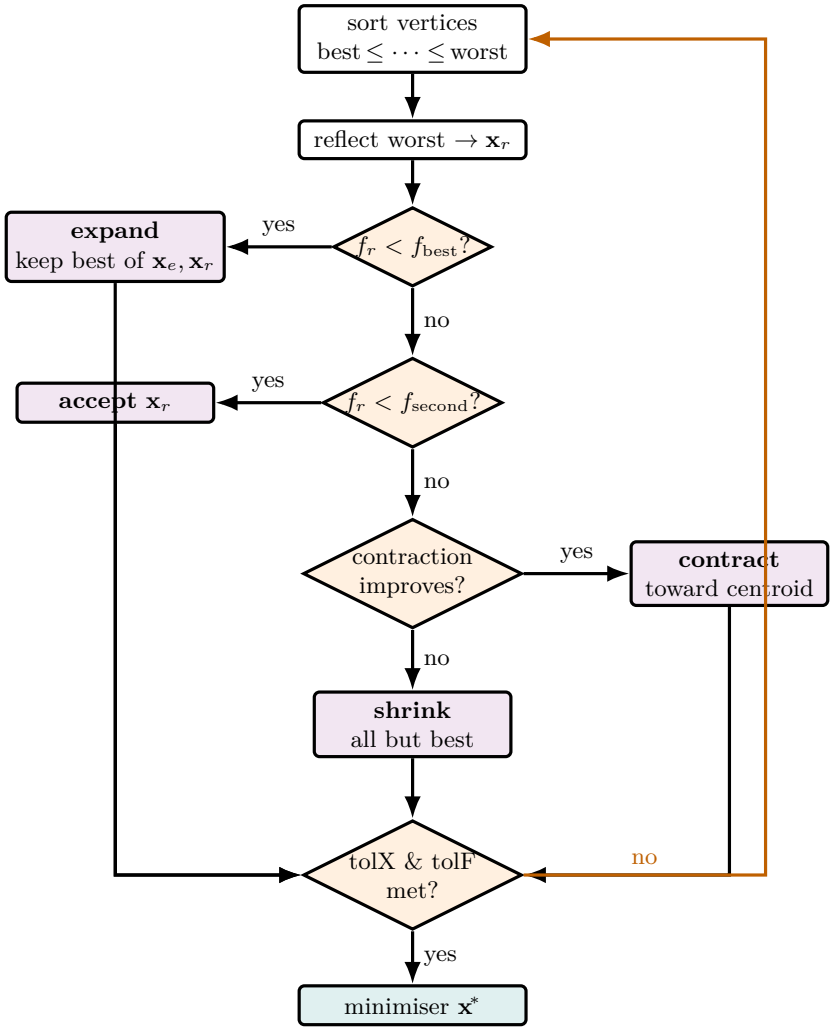



**Figure 40:** The Nelder–Mead move decision tree — a derivative-free descent that needs only $f$ values, used in Choupo wherever a Jacobian is treacherous: the *direct* Gibbs-energy minimisation of an LL or VLL flash (§11), and parameter-fitting in `OptimizationDriver`. Each pass orders the $n{+}1$ vertices, reflects the worst through the centroid, and then — by a short cascade of cheap tests on the reflected value — chooses the *expand* / *accept* / *contract* / *shrink* move, each just a point on the line through the centroid. Only *shrink* re-evaluates several vertices; it is the algorithm conceding the reflection line has stalled. The simplex crawls downhill until it is both small (tolX) and flat (tolF). For the symmetric $\gamma$-models where a Newton flash would collapse onto the trivial $K = 1$ saddle, this "crawl the Gibbs surface" search — multi-started — is what finds the true two-phase split.

# 8 Constrained optimisation: line-search SQP

*What you'll learn.* Nelder-Mead minimises a black box but cannot honour a hard constraint. Sequential Quadratic Programming (SQP) does: at every iterate it linearises the constraints, builds a convex *quadratic* model of the Lagrangian, solves that quadratic program (QP) for a step, then shortens the step with an $\ell_1$ merit function until it makes honest progress. We derive the QP from the KKT conditions, the damped-BFGS curvature update that keeps the QP convex, the merit function that arbitrates objective against feasibility, the Armijo backtracking line search, and the five-residual KKT certificate Choupo prints at the optimum. The inner convex QP is solved by an active-set method, derived in §8.7.

The flowsheet optimiser often comes with real constraints: a purity that must be *at least* 0.99, a duty that must *equal* a target, a recovery bounded above. Nelder-Mead (§7) cannot see any of these — it only ranks objective values, so a constraint can only enter as a hand-tuned penalty, which distorts the objective and is brittle. SQP [38, 39] is the classical answer: it treats the constraints as first-class citizens and converges to a point that provably satisfies the Karush–Kuhn–Tucker (KKT) optimality conditions. Choupo's `solver::sqp` is a hand-rolled, finite-difference, line-search SQP — no external library, every quantity inspectable.

## 8.1 The problem and the KKT conditions

The general nonlinear program (NLP) Choupo's SQP solves is

$$\min_{\mathbf{x}\in\mathbb{R}^n} \; f(\mathbf{x}) \quad \text{s.t.} \quad \begin{cases} h_j(\mathbf{x}) = 0, & j = 1,\dots,m_{\mathrm{e}}, \\ g_k(\mathbf{x}) \le 0, & k = 1,\dots,m_{\mathrm{i}}, \\ \mathrm{lo} \le \mathbf{x} \le \mathrm{hi}, \end{cases} \tag{227}$$

with equalities $h_j$ (e.g. a DesignSpec residual), inequalities $g_k$ (written in the $\le 0$ convention — a purity floor $x_{\mathrm{purity}} \ge 0.99$ becomes $g = 0.99 - x_{\mathrm{purity}} \le 0$), and box bounds. The Lagrangian collects every constraint with a multiplier:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \;=\; f(\mathbf{x}) \;+\; \sum_j \lambda_j \, h_j(\mathbf{x}) \;+\; \sum_k \mu_k \, g_k(\mathbf{x}). \tag{228}$$

A point $\mathbf{x}^\star$ is a (first-order) constrained minimiser when the **KKT conditions** hold:

$$\nabla_{\mathbf{x}}\mathcal{L} = \nabla f + \textstyle\sum_j \lambda_j \nabla h_j + \sum_k \mu_k \nabla g_k = \mathbf{0} \qquad \text{(stationarity)}, \tag{229}$$

$$h_j(\mathbf{x}^\star) = 0, \quad g_k(\mathbf{x}^\star) \le 0 \qquad \text{(primal feasibility)}, \tag{230}$$

$$\mu_k \ge 0 \qquad \text{(dual feasibility)}, \tag{231}$$

$$\mu_k \, g_k(\mathbf{x}^\star) = 0 \qquad \text{(complementary slackness)}. \tag{232}$$

Complementarity (232) says: a constraint that is not

binding ($g_k < 0$) carries no multiplier ($\mu_k = 0$); only *active* constraints ($g_k = 0$) can push back with $\mu_k > 0$.

## 8.2 Shadow prices: what the multiplier is worth

The multipliers are not bookkeeping — they answer the question a process engineer actually asks: *"what is this constraint costing me, and what would I pay to loosen it?"* Perturb the right-hand side of an active inequality, $g_k(\mathbf{x}) \le b_k$, by a small $\Delta b_k$. The feasible set opens up, the optimum slides along the relaxed boundary, and — because at the optimum $\nabla_{\mathbf{x}}\mathcal{L} = \mathbf{0}$, so only the *explicit* dependence on $b_k$ survives (the envelope / sensitivity theorem) — the optimal objective moves by

$$\frac{\partial f^\star}{\partial b_k} \;=\; -\,\mu_k \qquad \Longleftrightarrow \qquad \Delta f^\star \;\approx\; -\,\mu_k \, \Delta b_k. \tag{233}$$

That is the whole reason the multiplier is called a **shadow price**. Read it economically — with the sign flipped for a *maximisation*, since Choupo maximises NPV by minimising −NPV — and $\mu_k$ is the **marginal € value of one more unit of the capped resource**: one more m$^3$ of reactor, one more € of capital budget, one more kg/h of market demand. A binding capital budget with $\mu = 0.8$ says the next euro of CAPEX would buy €0.80 of extra NPV (build it); a binding cap with $\mu = 0.05$ is barely worth relaxing; a *slack* constraint has $\mu = 0$ and can be ignored entirely. This is why `solver::sqp` prints the active set and its multipliers every iteration and again in the closing KKT certificate — the shadow prices *are* the design answer, not a diagnostic. The tutorial `tutorials/steady/optimisation/optim05_reactor_npv` maximises a small plant's NPV against a capital budget and a capacity cap and reads the binding cap's shadow price straight off the certificate.

The multipliers are *shadow prices* — $\lambda_j$ (or $\mu_k$ for an inequality) is the marginal change in the optimal $f$ per unit relaxation of its constraint — which is exactly the sensitivity a process engineer wants, so Choupo prints them every iteration.

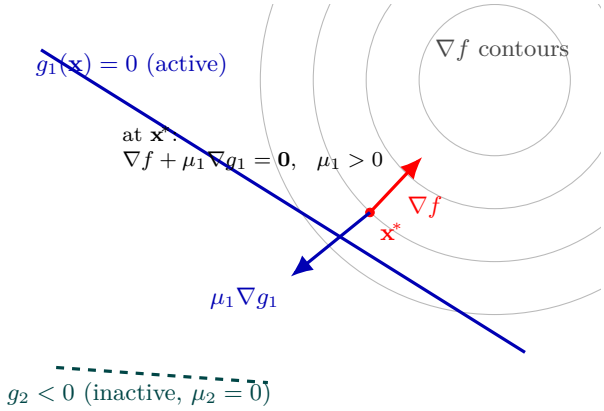



**Figure 41:** The KKT stationarity condition as a force balance at a constrained minimiser. The grey arcs are level sets of the objective $f$; absent constraints, $f$ would keep decreasing toward its unconstrained minimum (upper right). The active inequality $g_1 = 0$ (blue line) blocks the way, so the constrained optimum $\mathbf{x}^*$ sits *on* that boundary, where an objective contour is tangent to it. There the objective gradient $\nabla f$ (red, pointing uphill) is exactly cancelled by the constraint's push $\mu_1 \nabla g_1$ (blue) — stationarity $\nabla f + \mu_1 \nabla g_1 = \mathbf{0}$ with multiplier $\mu_1 > 0$. The second constraint $g_2$ (dashed teal) is slack: it does not touch $\mathbf{x}^*$, so by complementary slackness its multiplier is zero and it exerts no force. The active multiplier $\mu_1$ is the *shadow price* — how much $f$ would improve per unit of relaxing $g_1$ — which is why Choupo prints it.

## 8.3 From KKT to a quadratic program

Newton's method on the nonlinear KKT system (229)–(230) is the heart of SQP. Linearise the constraints about the current $\mathbf{x}$ and replace the Lagrangian by a quadratic model in the step $\mathbf{d} = \mathbf{x}_+ - \mathbf{x}$, with curvature matrix $\mathbf{B} \approx \nabla^2_{\mathbf{xx}} \mathcal{L}$ (note: the Hessian of the *Lagrangian*, not of $f$ — this is the single most common SQP mistake; see §8.4). The step solves the convex QP

$$\min_{\mathbf{d} \in \mathbb{R}^n} \ \tfrac{1}{2}\, \mathbf{d}^\top \mathbf{B}\, \mathbf{d} + \nabla f^\top \mathbf{d} \quad \text{s.t.} \quad \begin{cases} \nabla h_j^\top \mathbf{d} + h_j = 0, \\ \nabla g_k^\top \mathbf{d} + g_k \le 0, \\ \mathrm{lo} - \mathbf{x} \le \mathbf{d} \le \mathrm{hi} - \mathbf{x}. \end{cases} \tag{234}$$

The constant $f(\mathbf{x})$ is dropped (it does not move the minimiser). The linear-in-$\mathbf{d}$ constraint rows are precisely the first-order Taylor predictions $h_j(\mathbf{x} + \mathbf{d}) \approx h_j + \nabla h_j^\top \mathbf{d}$ driven to feasibility. The QP multipliers returned for (234) *are* the SQP's estimate of $(\boldsymbol{\lambda}, \boldsymbol{\mu})$ for the next iterate — the QP solve does double duty, producing both the primal step $\mathbf{d}$ and the dual estimate.

**Finite-difference gradients.** Choupo carries no analytic sensitivities — the objective and every constraint come out of one simulator pass. So $\nabla f$ and the full constraint Jacobian are built by finite differences from a *single* perturbation of each variable: forward differences

$$\frac{\partial f}{\partial x_j} \approx \frac{f(\mathbf{x} + \delta_j \mathbf{e}_j) - f(\mathbf{x})}{\delta_j}, \qquad \delta_j = \varepsilon_{\mathrm{FD}} \max(|x_j|,\, x_j^{\mathrm{typ}}), \tag{235}$$

cost $n + 1$ response evaluations per iteration ($n$ perturbations + the base point); the opt-in *central* scheme costs $2n$ but is $O(\delta^2)$ accurate. Crucially, one perturbation of $x_j$ yields column $j$ of $\nabla f$ *and* of the whole Jacobian at once — the responses $f$, $\mathbf{h}$, $\mathbf{g}$ are read off the same pass.

## 8.4 Keeping the QP convex: damped-BFGS on the Lagrangian

The QP (234) is convex — and the active-set solver of §8.7 only works — if $\mathbf{B}$ is symmetric positive definite (SPD). Choupo never forms the true Lagrangian Hessian (it would need second derivatives by finite difference: $O(n^2)$ extra passes). Instead it carries a quasi-Newton approximation $\mathbf{B}$, started at $\mathbf{B}_0 = \mathbf{I}$ and updated each step from the curvature pair

$$\mathbf{s} = \mathbf{x}_+ - \mathbf{x}, \qquad \mathbf{y} = \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}_+, \boldsymbol{\lambda}_+) - \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}_+). \tag{236}$$

The subtlety that students miss: $\mathbf{y}$ is the difference of the *Lagrangian* gradient (229) (objective *plus* constraint terms) evaluated with the *same* new multipliers $\boldsymbol{\lambda}_+$ at *both* points. Using $\nabla f$ alone silently builds the wrong curvature, solves the wrong QP, and stalls on active constraints.

The plain BFGS update preserves SPD only when the curvature condition $\mathbf{s}^\top \mathbf{y} > 0$ holds, which a nonconvex

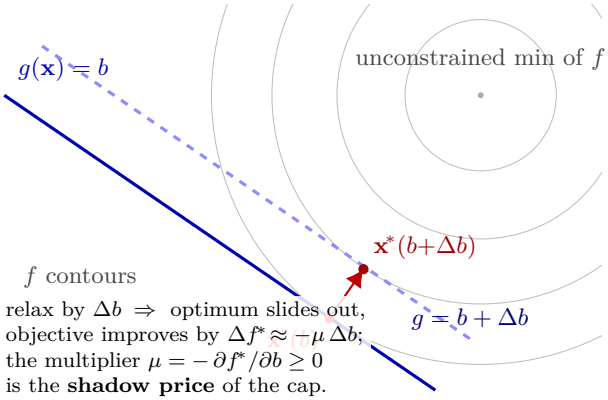



**Figure 42:** The multiplier as a price. Contours of the objective $f$ (grey) tighten toward its unconstrained minimum (upper right), which the active cap $g = b$ (solid blue) keeps out of reach, so the constrained optimum $\mathbf{x}^*(b)$ sits on that boundary, tangent to the outer contour. Loosen the cap to $g = b + \Delta b$ (dashed) and the boundary shifts toward the unconstrained minimum; the optimum slides to $\mathbf{x}^*(b+\Delta b)$ on a *smaller* contour — a better objective. The rate of that improvement per unit of relaxation is exactly the Lagrange multiplier, $\Delta f^* \approx -\mu\,\Delta b$ (Eq. 233): $\mu$ is the shadow price. A constraint that does not touch the optimum is slack, its multiplier is zero, and moving it changes nothing — complementary slackness, seen geometrically.

Lagrangian can violate. Powell's *damped* BFGS [39] repairs this by replacing $\mathbf{y}$ with a blend toward $\mathbf{Bs}$:

$$\theta = \begin{cases} 1, & \mathbf{s}^\top\mathbf{y} \geq 0.2\,\mathbf{s}^\top\mathbf{Bs}, \\ \dfrac{0.8\,\mathbf{s}^\top\mathbf{Bs}}{\mathbf{s}^\top\mathbf{Bs} - \mathbf{s}^\top\mathbf{y}}, & \text{otherwise}, \end{cases} \qquad \mathbf{r} = \theta\,\mathbf{y} + (1-\theta)\,\mathbf{Bs}, \tag{237}$$

which guarantees $\mathbf{s}^\top\mathbf{r} \geq 0.2\,\mathbf{s}^\top\mathbf{Bs} > 0$, then applies the BFGS rank-two update with $\mathbf{r}$ in place of $\mathbf{y}$:

$$\mathbf{B} \;\leftarrow\; \mathbf{B} - \frac{\mathbf{Bs}\,\mathbf{s}^\top\mathbf{B}}{\mathbf{s}^\top\mathbf{Bs}} + \frac{\mathbf{r}\,\mathbf{r}^\top}{\mathbf{s}^\top\mathbf{r}}. \tag{238}$$

$\theta = 1$ recovers ordinary BFGS; $\theta < 1$ damps toward the positive-definite $\mathbf{Bs}$ direction. Choupo prints $\theta$ each step — a run of $\theta \ll 1$ warns that the Lagrangian curvature is fighting the update (often finite-difference noise), and the update is *skipped* entirely if even the damped $\mathbf{s}^\top\mathbf{r}$ collapses.

## 8.5 The $\ell_1$ merit function and the Armijo line search

The QP step $\mathbf{d}$ is a direction, not a commitment. A full step $\alpha = 1$ may overshoot, especially far from the optimum where the linearised constraints lie. SQP needs a scalar that decides whether a trial step is *good* — but "good" must balance two competing goals: lower the objective *and* reduce infeasibility. Choupo uses the exact $\ell_1$ *merit function* [38]

$$\phi_\mu(\mathbf{x}) \;=\; f(\mathbf{x}) \;+\; \mu\left(\sum_j |h_j(\mathbf{x})| + \sum_k \max(0,\, g_k(\mathbf{x}))\right), \tag{239}$$

where the penalty parameter $\mu$ weights the total constraint violation. "Exact" means: for $\mu$ large enough, an unconstrained minimiser of $\phi_\mu$ coincides with the constrained minimiser of (227) — no $\mu \to \infty$ limit is needed. Choupo raises $\mu$ monotonically to stay above the multiplier estimates,

$$\mu \;\geq\; \mu_{\text{safety}}\,\max(\|\boldsymbol{\lambda}\|_\infty, \|\boldsymbol{\mu}\|_\infty), \qquad \mu_{\text{safety}} = 1.1, \tag{240}$$

the standard rule that makes $\mathbf{d}$ a descent direction for $\phi_\mu$.

A backtracking line search then picks $\alpha$. The QP step is a descent direction with directional derivative

$$D\phi_\mu(\mathbf{x};\mathbf{d}) \;=\; \nabla f^\top\mathbf{d} \;-\; \mu\left(\sum_j |h_j| + \sum_k \max(0, g_k)\right), \tag{241}$$

(the second term is the predicted reduction of the violation the QP constraints drive to zero). Choupo halves $\alpha$ from 1 until the **Armijo** sufficient-decrease condition holds:

$$\phi_\mu(\mathbf{x} + \alpha\mathbf{d}) \;\leq\; \phi_\mu(\mathbf{x}) + c_1\,\alpha\,D\phi_\mu(\mathbf{x};\mathbf{d}), \qquad c_1 = 10^{-4}. \tag{242}$$

If $\alpha$ falls below $\alpha_{\min} = 10^{-8}$ without acceptance the

solver stops *honestly* ("line search failed — no merit decrease"), rather than pretend convergence — usually a sign the finite-difference gradients have hit their noise floor.

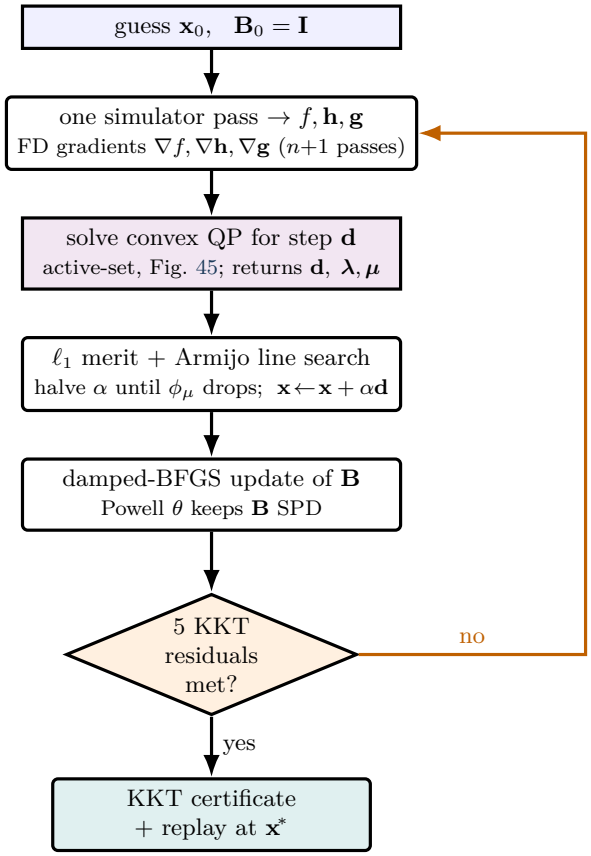



**Figure 43:** One pass of Choupo's line-search SQP, top to bottom. Each iterate runs *one* simulator pass to read off the objective and all constraints, builds finite-difference gradients from it, then solves a convex quadratic program (the inner active-set solver of Fig. 45) for both a step $\mathbf{d}$ and fresh multiplier estimates. An $\ell_1$ merit function with an Armijo backtracking line search decides how far along $\mathbf{d}$ to actually move — balancing lower objective against less constraint violation — and a damped-BFGS update keeps the curvature matrix $\mathbf{B}$ symmetric positive definite so the next QP stays convex. The loop exits only when all five KKT residuals clear tolerance, and the printed certificate lets the student verify optimality by hand.

**The Maratos effect and the watchdog.** A known pathology: near the optimum the $\ell_1$ merit can reject the *full* Newton step even though it improves both objective and feasibility — the *Maratos effect* [38] — which destroys the fast local rate. Choupo's v1 mitigation is deliberately minimal: a *watchdog* that forces the full $\alpha = 1$ step every $K = 5$ iterations regardless of the merit, breaking the merit's veto. It also *reports* the symptom (full step improved feasibility yet the merit rejected it) so the student sees the effect, not just its cure.

## 8.6 Termination: the five-residual KKT certificate

SQP stops when the KKT conditions (229)–(232) are met within tolerance. Choupo measures, at the current iterate with the latest QP multipliers, five residuals:

$$\begin{aligned} r_{\text{stat}} &= \|\nabla_{\mathbf{x}} \mathcal{L}\|_\infty && (\le 10^{-5}), \qquad r^{\text{eq}}_{\text{feas}} = \max_j |h_j| && (\le 10^{-6}), \\ r^{\text{ineq}}_{\text{feas}} &= \max_k \max(0, g_k) && (\le 10^{-6}), \quad r_{\text{compl}} = \max_k |\mu_k\, g_k| && (\le 10^{-6}), \\ r_{\text{dual}} &= \max_k \max(0, -\mu_k) && (\ge 0). \end{aligned} \tag{243}$$

for stationarity, equality and inequality feasibility, complementary slackness, and dual feasibility. When the first four clear their tolerances the solver returns `converged`, and prints all five as a *certificate* the student can verify by hand. Active box bounds are folded in as ordinary inequality rows in the QP, so a binding bound contributes its multiplier into $r_{\text{stat}}$ — a bound that holds is a genuine active constraint, not a free direction.

## 8.7 The inner active-set QP

Every SQP iteration must solve the strictly-convex QP (234). Choupo does this with a hand-rolled **active-set** method (`solver::activeSetQP`), the classical algorithm for convex QPs [38, Alg. 16.3]. The idea: maintain a *working set* $W$ of constraints treated as equalities (all true equality rows, plus the inequality rows currently guessed

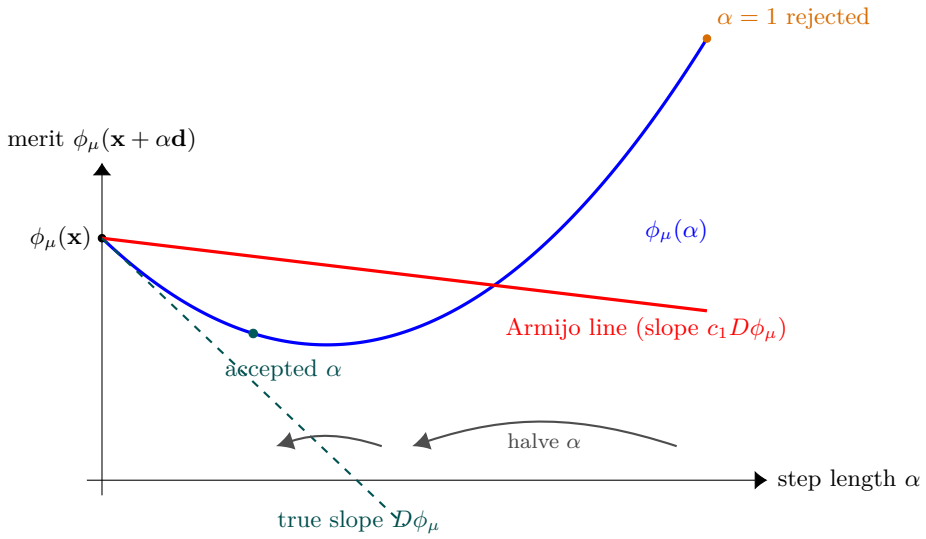



**Figure 44:** The Armijo backtracking line search on the $\ell_1$ merit function, the SQP's "how far" decision. The QP gives a descent direction $\mathbf{d}$; we walk along it and watch the merit $\phi_\mu(\mathbf{x} + \alpha\mathbf{d})$, which trades objective against constraint violation. The full step $\alpha = 1$ overshoots the dip and sits *above* the gentle Armijo sufficient-decrease line (red, slope $c_1 D\phi_\mu$ with $c_1 = 10^{-4}$), so it is rejected; the simulator halves $\alpha$ until a trial point falls *below* that line — a small but genuine merit decrease, enough to guarantee progress. If $\alpha$ shrinks past $10^{-8}$ with no acceptance, SQP stops honestly rather than fake convergence — usually the finite-difference gradients have hit their noise floor.

active), and at each step solve the equality-constrained QP over $W$ exactly via its KKT linear system

$$\begin{bmatrix} \mathbf{B} & A_W^\top \\ A_W & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{p} \\ \boldsymbol{\lambda}_W \end{bmatrix} = \begin{bmatrix} -\mathbf{g}_W \\ -\mathbf{c}_W \end{bmatrix}, \tag{244}$$

solved by the same Gauss elimination with partial pivoting as the rest

of the simulator (§3). Then:

- If the resulting sub-step $\mathbf{p}$ keeps every *inactive* inequality feasible, take it and inspect the multipliers of the active inequality rows. If all $\lambda_W \geq 0$ the KKT conditions hold over the full set — **optimal**, return. Otherwise *drop* the inequality with the most-negative multiplier from $W$ (it wants to move off its bound).
- If $\mathbf{p}$ would violate an inactive inequality, take the maximal feasible fraction $\alpha \in [0, 1)$ toward the nearest *blocking* constraint, stp, and *add* that constraint to $W$.

Strict convexity of $\mathbf{B}$ guarantees a strict decrease at every working-set change, hence *finite* termination [38, Sec. 16.5]. Box bounds $\text{lo} \leq \mathbf{d} \leq \text{hi}$ are folded in as ordinary inequality rows, so one machinery handles everything.

**Honest failure modes.** A singular KKT matrix (244) means the active constraint gradients are linearly dependent — a *LICQ failure*. Choupo recovers when the deficiency is among active *inequalities* (drop the most-recently-added, and forbid it from re-entering this solve, breaking the drop/re-add cycle two nearly-parallel constraints would form); if the deficiency is in the equality rows it is genuine and unrecoverable, and the solver throws a clear diagnostic rather than returning garbage. A working-set-change cap aborts loudly on suspected cycling — never a silent spin.

**An infeasible start, and a failed computation.** Algorithm 16.3 assumes a *feasible* starting point, and the QP an SQP iteration hands over starts at $\mathbf{d} = 0$, which violates every constraint the iterate itself violates. Until 2026-10-27 such a row was neither in $W$ nor a blocking row, so a zero step was certified optimal at a point that broke it (min $\frac{1}{2}d^2$ s.t. $d \geq 1$ returned $d = 0$). Now the most violated inactive row joins $W$ at each pass — the KKT solve then drives its residual to zero exactly as it drives an equality's — a linearly dependent working set is found before the solve by Gram–Schmidt instead of by hoping the elimination throws, and **optimal** is declared only once *every* row, active or not, is checked feasible; a QP whose rows cannot all hold is returned *not converged, infeasible*, naming the row. The SQP answers an infeasible linearisation (an iterate far from the feasible region, a constraint with a small slope) with Powell's relaxation [39]: the residuals the step must remove are scaled by the largest $\theta \in [0, 1]$ for which the QP is consistent, found by bisection and announced in the trace; $\theta = 0$ with a vanishing

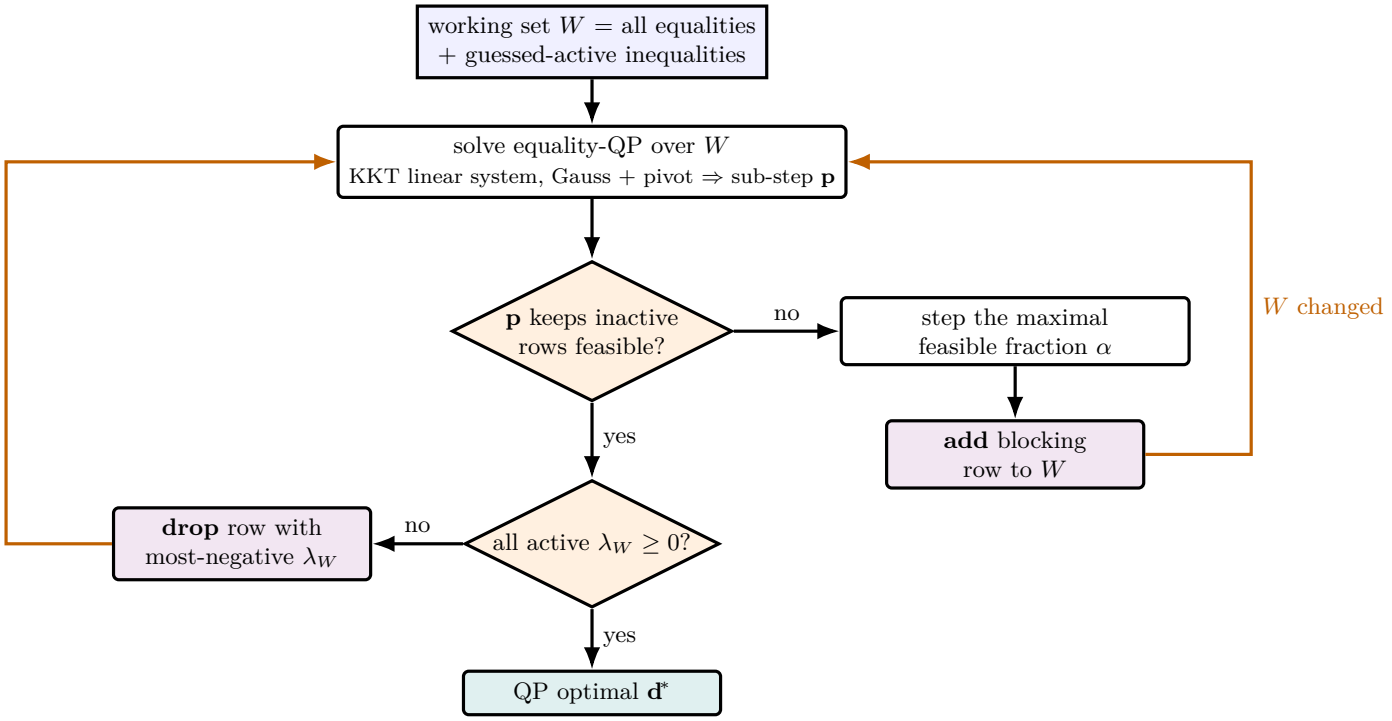



**Figure 45:** The inner active-set method that solves each SQP quadratic program. It maintains a *working set* $W$ — the constraints it currently treats as equalities — and at every pass solves the equality-constrained QP over $W$ exactly through its KKT linear system (the same Gauss-with-pivoting solver, Fig. 31). Then two questions steer it: if the sub-step would cross an inactive inequality, take the largest feasible fraction toward it and *add* that blocking row to $W$; otherwise check the active multipliers — if any is negative, that constraint wants to leave, so *drop* the most-negative one; if all are non-negative, the KKT conditions hold and the QP is optimal. Because $\mathbf{B}$ is strictly convex, every working-set change strictly lowers the QP objective, so the loop terminates in finitely many add/drop steps.

step is reported as a *local infeasibility*. Before, the solver stepped past the box instead — one integer-enumeration tutorial reported reflux ratios of 73, 14 and 6.8 as optima of a problem declared on $R \leq 6$. Finally, a finite difference is never built from a failed simulator evaluation: the other side is taken, announced, and a variable whose two sides both fail stops the SQP, unconverged, naming it.

## 8.8 A worked self-check: Hock–Schittkowski

Because the SQP is finite-difference and the simulator pass is noisy, Choupo *re-proves the algorithm* before every constrained run, on three analytic problems from the Hock–Schittkowski test set [40] (`verifySQP`, a merge-blocking gate). Take **HS35** (Beale):

$$\begin{aligned} &\min_{\mathbf{x}\geq 0} \; 9 - 8x_1 - 6x_2 - 4x_3 + 2x_1^2 + 2x_2^2 + x_3^2 + 2x_1x_2 + 2x_1x_3 \\ &\text{s.t.} \;\; x_1 + x_2 + 2x_3 \leq 3, \end{aligned} \tag{245}$$

whose published optimum is $f^\star = \frac{1}{9} = 0.1111\ldots$ at $\mathbf{x}^\star = (\frac{4}{3}, \frac{7}{9}, \frac{4}{9})$. From $\mathbf{x}_0 = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ Choupo's SQP linearises (one constraint row plus three lower-bound rows), solves the convex QP, damped-BFGS updates $\mathbf{B}$, and converges to $|f - f^\star| < 10^{-4}$. The companion `verifyActiveSetQP` solves a hand-worked three-inequality QP whose active set $\{d_1 + d_2 \leq 1\}$, primal solution $\mathbf{d}^\star = (0, 1)$ and multipliers $\boldsymbol{\mu} = (1, 0, 0)$ are known by hand, and asserts the inner solver reproduces them to $10^{-9}$. If either self-check fails the run aborts — the tolerance is never loosened to mask a bug.

**Power and limit.** **Power:** SQP is the method of choice for smooth, equality- and inequality-constrained NLPs — it honours hard constraints exactly at the solution, returns shadow-price multipliers for free, and converges superlinearly near the optimum because $\mathbf{B}$ approximates the Lagrangian curvature. A flowsheet DesignSpec, a purity floor, or a duty-equals-target all map directly onto (227). **Limit:** it is a *local* method (converges to the nearest KKT point — multi-start for global search), the gradients are finite-difference so it inherits the simulator's noise floor (a noisy pass poisons $\mathbf{B}$ and the merit), and it assumes the responses are smooth in $\mathbf{x}$ — a discrete decision or a phase-appearance discontinuity breaks the linearisation. For a black-box *unconstrained* objective with no derivatives, reach back for Nelder-Mead (§7); SQP earns its keep precisely when the constraints are real.

## What you see in Choupo

Selecting `method sqp;` in an `outerDict` optimisation block makes `OptimizationDriver` print, per iteration, the damped-BFGS $\theta$, the merit penalty $\mu$, the active set, the Lagrange multipliers (shadow prices), the Armijo step $\alpha$, and the four KKT residuals — the whole inner workings, not just a final answer. The self-checks `verifySQP` / `verifyActiveSetQP` run first and their pass is announced. On convergence the five-line KKT certificate is printed and the simulator is replayed once at the optimum with full verbosity, exactly as the Nelder-Mead path does.

# 9 Runge-Kutta 4 for the PFR

*What you'll learn.* The PFR model integrates concentrations along the reactor volume by the canonical fourth-order Runge-Kutta scheme. The independent variable is $V$, not $t$ — a small but important reinterpretation.

## 9.1 The reactor as an ODE

In a steady-state PFR with molar flow rate $F$ and reaction extent per unit volume $\dot{r}$:

$$\frac{d\mathbf{x}}{dV} = \frac{\boldsymbol{\nu} \cdot r(\mathbf{x}, T)}{F}, \tag{246}$$

with $\boldsymbol{\nu}$ the stoichiometric vector and $\mathbf{x}$ the mole-fraction vector along the reactor. The state $\mathbf{x}$ at $V = V_R$ is the outlet.

## 9.2 RK4 step

For a step of size $h = V_R/N$:

$$\mathbf{k}_1 = f(\mathbf{x}_n) \tag{247}$$

$$\mathbf{k}_2 = f(\mathbf{x}_n + \tfrac{h}{2}\,\mathbf{k}_1) \tag{248}$$

$$\mathbf{k}_3 = f(\mathbf{x}_n + \tfrac{h}{2}\,\mathbf{k}_2) \tag{249}$$

$$\mathbf{k}_4 = f(\mathbf{x}_n + h\,\mathbf{k}_3) \tag{250}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \tfrac{h}{6}\,(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \tag{251}$$

Local truncation error is $O(h^5)$ per step; global error is $O(h^4)$. At $N = 100$ (the simulator's default) this is well below the significance of any kinetics one can realistically know.

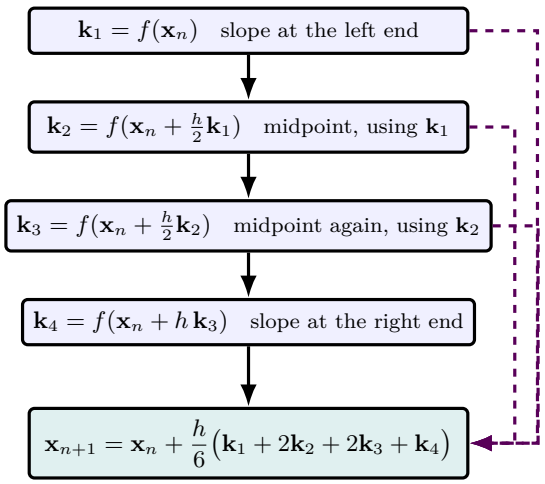



**Figure 46:** The four-stage structure of classical RK4 for one step $\mathbf{x}_n \to \mathbf{x}_{n+1}$ of size $h$ (here $h = V_R/N$ marching down the PFR). Each stage is one evaluation of the right-hand side $f$: a slope at the left end ($\mathbf{k}_1$), two probes of the midpoint slope that reuse the previous stage ($\mathbf{k}_2, \mathbf{k}_3$), and a slope at the right end ($\mathbf{k}_4$). The solid arrows are the data dependency — each $\mathbf{k}$ feeds the next — and the dashed violet arrows show all four slopes returning to be combined with Simpson-like weights 1:2:2:1. That weighting is exactly what buys the $O(h^5)$ local accuracy from only four function calls. The same packed-state machinery drives the batch integrators of Chapter 1.

# Part V — Transport properties and mixing rules

*Far better an approximate answer to the* right question*, which is often vague, than an* exact *answer to the* wrong question*, which can always be made precise.*

John W. Tukey, "The Future of Data Analysis", *Ann. Math. Statist.* 33 (1962) 1–67

# 1 Why transport: where pressure drop, heat transfer, and HETP come from

*What you'll learn.* A flash needs no transport coefficients — the answer depends only on phase equilibrium. A packed column, a tube-side heat exchanger or a fixed-bed reactor lives or dies on viscosity, thermal conductivity and diffusivity. This Part introduces those three property families in the Reid-Prausnitz-Poling tradition (classic, non-cubic models, two or three parameters per pure component, plus a mixture rule per property), documents the planned extension of the `*.dat` schema, and points forward to the unit-operation correlations they feed.

## 1.1 Three coefficients, three modes of irreversibility

Thermodynamics tells you where a system wants to go. Transport tells you how fast it gets there and what it costs. Three Newtonian constitutive laws — one per mode of irreversibility — show up in every unit operation that involves flow, heat or mass transfer:

| mode | flux | gradient driver | coefficient |
|---|---|---|---|
| momentum | $\tau$ (Pa) | $\partial u/\partial y$ | viscosity $\mu$ |
| heat | $q$ (W/m²) | $\partial T/\partial y$ | thermal conductivity $\kappa$ |
| mass (i) | $J_i$ (mol/m²s) | $\partial c_i/\partial y$ | diffusivity $D_i$ |

Newton's law of viscosity ($\tau = -\mu\,\mathrm{d}u/\mathrm{d}y$), Fourier's law of heat conduction ($q = -\kappa\,\mathrm{d}T/\mathrm{d}y$) and Fick's first law of diffusion ($J_i = -D_i\,\mathrm{d}c_i/\mathrm{d}y$) share the same linear-response form. Each transport coefficient is a *thermophysical* property of the fluid: it depends on $T$, mildly on $P$ (strongly for gases above ∼10 bar), and on composition. Unlike thermodynamic properties they describe irreversible processes; they do not derive from an energy.

*Intuition.* A common student trap is to expect that, just like enthalpy follows from $c_p$, viscosity should follow from something equally fundamental. It does not. Transport coefficients ultimately come from the underlying kinetic theory — collision integrals for gases, free-volume theory for liquids — but for engineering work we use *correlations*: 2-3-parameter fits to empirical data, identical in spirit to Antoine for vapour pressure.

## 1.2 Where each property is consumed

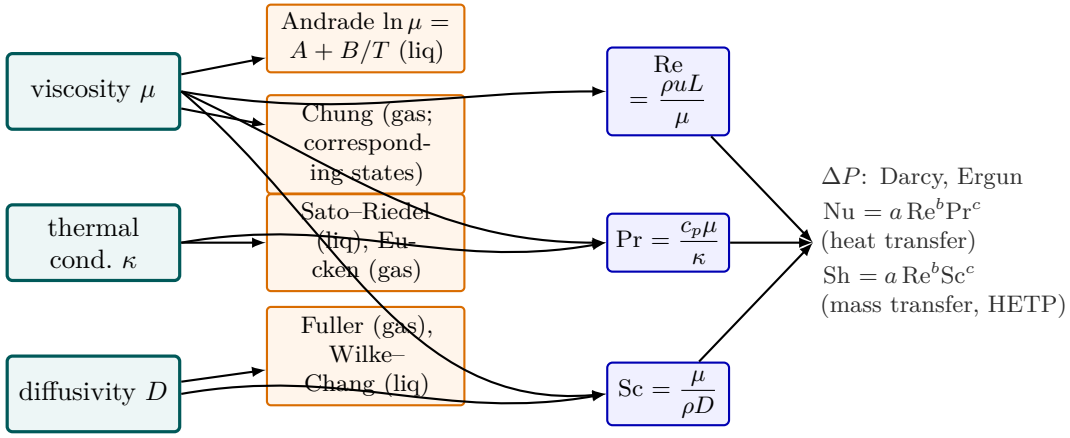



**Figure 47:** The transport map: three coefficients, their correlations, and where they are consumed. Each Newtonian coefficient (teal) is supplied by a two- or three-parameter correlation (orange)—Andrade/Vogel [41, 42] or Chung/Chapman–Enskog for viscosity, Sato–Riedel/Eucken/modified Eucken for conductivity, Fuller and Wilke–Chang for diffusivity (the models the engine registers; Sutherland is taught below as theory and is NOT implemented). The coefficients then assemble into the three dimensionless groups Re, Pr, Sc (blue), and nearly every downstream design equation is a power law in them ($\mathrm{Nu} = a\,\mathrm{Re}^b\mathrm{Pr}^c$, $\mathrm{Sh} = a\,\mathrm{Re}^b\mathrm{Sc}^c$). Note that viscosity feeds all three groups—one good $\mu$ correlation unlocks pressure drop, heat *and* mass transfer.

| property | feeds | through correlation |
|---|---|---|
| viscosity $\mu$ | pressure drop in tubes, packings | Darcy-Weisbach, Ergun |
| | shear-controlled mixing | power number correlations |
| | convective heat transfer | Dittus-Boelter (turbulent), Sieder-Tate |
| | column flooding limits | Souders-Brown coefficient |
| thermal cond. $\kappa$ | conductive heat transfer | Fourier across pipe walls / fins |
| | convective heat transfer | via Nusselt = $f$(Re, Pr) |
| diffusivity $D$ | mass-transfer coefficient | Sherwood = $f$(Re, Sc) |
| | HETP / NTU in absorption, dist. | Onda, Eduljee |
| | gas-side concentration polarisation | Sherwood again |

*Intuition.* The three dimensionless groups Re ($= \rho u L/\mu$), Pr ($= c_p\mu/\kappa$) and Sc ($= \mu/(\rho D)$) capture each property's role in a heat- or mass-transfer correlation, normalised against the flow. Most empirical Nusselt or Sherwood relations are functions of the form $\mathrm{Nu} = a\ \mathrm{Re}^b\ \mathrm{Pr}^c$ or $\mathrm{Sh} = a\ \mathrm{Re}^b\ \mathrm{Sc}^c$, so *one good correlation per coefficient* unlocks the entire family of design equations downstream.

## 1.3 Status note

None of this is implemented yet. Pressure drops in the simulator are zero; heat-exchanger duties bypass the $UA$ product (a Tout or $Q$ is given, not derived from area); column efficiency is taken as 100% per stage. Part IV documents the theory and the planned data-file extension; the C++ wiring is on the v1.x roadmap. Every chapter that follows closes with a candid status note.

# 2 Viscosity

*What you'll learn.* The viscosity of a pure liquid follows the Andrade equation $\ln \mu = A + B/T$ — a two-parameter fit that captures the strong Arrhenius-like decay of liquid viscosity with temperature. Gases take the opposite trend ($\mu$ grows with $T$); the classical model is Sutherland. The kinetic-theory alternative is the Chapman-Enskog formula in terms of Lennard-Jones $\sigma$ and $\epsilon/k$. Choupo SHIPS two gas models: `ChapmanEnskog`, which reads a record's `lennardJones` block (Svehla's 1962 fits to measured viscosities, curated by page), and `Chung`, the corresponding-states realisation of the same kinetic theory from $T_c$, $P_c$, $\omega$ alone. Sutherland stays theory: no record carries $S$. Reid-Prausnitz-Poling Chapter 9.

## 2.1 Pure liquid: Andrade

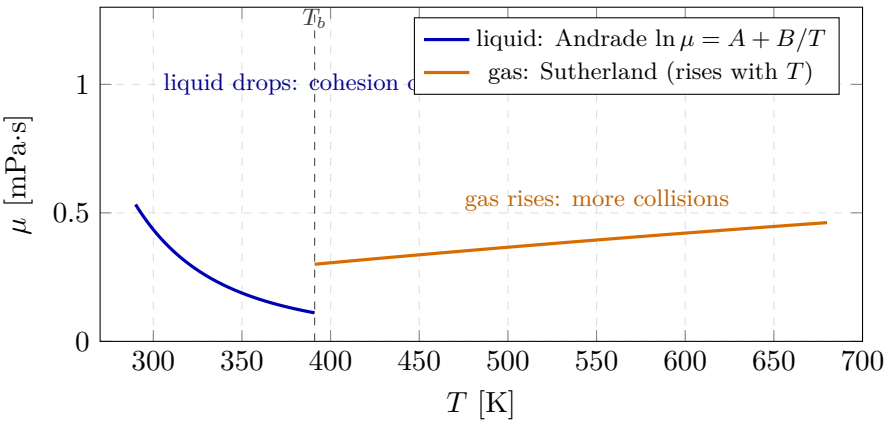



**Figure 48:** Liquid and gas viscosity move in *opposite* directions with temperature—the most counter-intuitive fact in transport. Below the boiling point the liquid follows Andrade ($\ln \mu = A + B/T$, blue): viscosity falls steeply as thermal energy lets molecules slip past one another. Above it the gas follows Sutherland (orange): viscosity *rises* because hotter molecules carry more momentum across the shear gradient. Curves are illustrative (acetic-acid liquid constants $A = -13.61$, $B = 1761$ K; the gas branch is a representative Sutherland shape, not to the same vertical scale).

Liquid viscosity drops sharply with temperature because molecules move past each other more easily once thermal energy starts to overcome the cohesive potential. The empirical fit is Andrade's[41],

$$\boxed{\ln \mu^{\text{liq}}(T) \;=\; A \;+\; \frac{B}{T},} \qquad \mu \text{ in Pa·s, } T \text{ in K.} \tag{252}$$

Two parameters, fitted to two or more $(\mu, T)$ data points. For the practically interesting range $0.4\,T_c < T < 0.95\,T_c$ this two-parameter form is within ∼5% of experimental viscosity. For wider ranges (cryogenic to near-critical) Reid recommends the extended four-parameter Andrade

$$\ln \mu^{\text{liq}}(T) \;=\; A \;+\; \frac{B}{T} \;+\; C\,\ln T \;+\; D\,T^{E}, \tag{253}$$

and Choupo ships *both*: `Andrade` for a pair fitted to two points, and `chemsepEq101`[43] for the four-parameter form read straight from a published DIPPR-style fit ($\mu = \exp(A + B/T + C\ln T + D\,T^{E})$), which the great majority of the imported components carry. Note what is *not* done: the five published parameters are never refitted down to Andrade's two. That would replace a published fit with an approximation to it, and the residual would be invisible — the number would still look like a viscosity.

**Worked example 2.1.** Acetic acid at 300 K and 380 KFitting Andrade to two literature data points ($\mu(300) = 1.07$ mPa · s, $\mu(380) = 0.38$ mPa · s) gives $A = -13.61$, $B = 1761$ K. At an intermediate $T = 340$ K, Eq. (252) predicts $\mu(340) \approx 0.63$ mPa · s, against a measured $\approx 0.60$ mPa · s — a 5% prediction without any 340 K measurement.

## 2.2 Pure gas: Sutherland

Gas viscosity *increases* with temperature: more thermal energy means more momentum-carrying collisions across a flow gradient. The classical fit is Sutherland (originally derived from a square-well potential),

$$\boxed{\mu^{\text{gas}}(T) \;=\; \mu_0 \left(\frac{T}{T_0}\right)^{3/2} \frac{T_0 + S}{T + S},} \tag{254}$$

where $\mu_0$ is the viscosity at the reference temperature $T_0$ (commonly 273.15 K) and $S$ is the Sutherland constant (a positive material parameter that controls the deviation from $T^{1/2}$ behaviour of hard-sphere theory). Three parameters per component. Air: $\mu_0 = 1.716 \times 10^{-5}$ Pa · s, $T_0 = 273.15$ K, $S = 110.4$ K.

**Status: NOT implemented.** No component record carries $S$, and a Sutherland constant entered without a primary source per gas would be a number nobody can check. It ships the day a cited table of $S$ is curated, and not before.

## 2.3 Either phase: Chapman-Enskog with Lennard-Jones

A corresponding-states alternative falls out of kinetic theory. For a dilute gas of molecules interacting through a Lennard-Jones (12–6) pair potential characterised by $\sigma$ (collision diameter, in ångström) and $\epsilon/k$ (well depth normalised to Boltzmann's constant, in K),

$$\mu^{\text{gas}}(T) \;=\; 2.6693 \times 10^{-6}\,\frac{\sqrt{M\,T}}{\sigma^2\,\Omega_\mu(T^*)} \qquad \text{[Pa·s; } M \text{ in kg/kmol; } T \text{ in K; } \sigma \text{ in Å]}, \tag{255}$$

where $T^* = kT/\epsilon$ and the dimensionless collision integral $\Omega_\mu$ is tabulated (Reid Table 9-2[4]) or approximated by the Neufeld correlation[44]

$$\Omega_\mu(T^*) \;\approx\; \frac{1.16145}{(T^*)^{0.14874}} \;+\; \frac{0.52487}{e^{0.7732\,T^*}} \;+\; \frac{2.16178}{e^{2.43787\,T^*}}. \tag{256}$$

The appeal of the LJ approach is that *two* parameters ($\sigma$, $\epsilon/k$) determine viscosity, thermal conductivity and diffusivity — not separately fitted per property. Tabulations exist for most common species.

**Status: IMPLEMENTED (2026-09-06)** as `model ChapmanEnskog;`, in the form Svehla (1962, NASA TR R-132, eq. (1), report page 2)[45] states it: $\eta\,[\mu\text{P}] = 26.693\sqrt{MT}/(\sigma^2\Omega^{(2,2)*})$, the same expression as eq. (255) in other units. The Neufeld fit above has ONE home in the engine (`src/thermo/transport/NeufeldOmega.H`), called by Chung at its reduced temperature $T^* = 1.2593\,T/T_c$ and by Chapman-Enskog at $T^* = kT/\epsilon$. The model reads the component record's block,

```
lennardJones                                  // in the component's .dat
{
    sigma      [0 1 0 0 0] 3.798e-10;         // m  (printed: 3.798 A)
    epsOverK   71.4 K;
    provenance
    {
        sigma    { origin measuredFit; method "Svehla 1962 method 1: ..."; }
        epsOverK { origin measuredFit; method "Svehla 1962 method 1: ..."; }
        source   "Svehla, R. A. (1962). ... NASA TR R-132, Table I(a), report page 23 (PDF page
    ↪ 25), row 'N2'";
        licence  publicDomain;
    }
}
```

and is selected in the case with `transport { vapour { viscosity { model ChapmanEnskog; } } }`. $\sigma$ is written in metres in the bracket form because ångström is not a named unit in `src/core/Units.cpp`; $\epsilon/k$ in kelvin. The provenance is PER VALUE, because Svehla codes $\sigma$ and $\epsilon/k$ by separate methods: codes 1–4 are fits to measured data (`measuredFit`), code 20 a radiometer measurement (`measured`), everything else an estimate (`estimated`) — and the run ANNOUNCES the origin of both constants for every component it prices. A component WITHOUT the block REFUSES by name; there is no default $\sigma$. The blocks are drafted from the committed transcription of Svehla's Table I(a) (`bin/curate/svehla1962/`) by `bin/curate/propose_lennard_jones.py` into `data/local/lennardJones/`, matched to records by parsed elemental formula with an MW cross-check, never by name; promotion into the public catalogue is a review. The model's own anchor is Svehla's printed viscosity for $N_2$ at 300 K (report page 90) from his own constants, reproduced within a declared 1 % (0.40 % observed: the Neufeld fit against the Hirschfelder 1954[46] $\Omega$ tables he interpolated) — an arithmetic check of the constant and the unit chain, not a comparison with a measurement Choupo holds. Tutorial `transport02_chapman_enskog`. **Binary diffusivity by Chapman-Enskog is NOT implemented**: the $\Omega^{(1,1)*}$ fit it needs is not citable from any source in the tree, and a constant typed from memory is a number nobody can check.

## 2.4 The Chung correlation — what Choupo actually uses

The Lennard-Jones form needs $\sigma$ and $\epsilon/k$, which are tabulated for only some species. Choupo instead ships the **Chung et al. (1988)**[47] corresponding-states realisation of the SAME kinetic theory, which ESTIMATES the collision parameters from the critical properties every component already carries — so a gas viscosity follows from $T_c, P_c, \omega, M$ alone, with no separate viscosity datum:

$$\mu^{\text{gas}}(T) \;=\; 40.785 \times 10^{-7}\,\frac{F_c\,\sqrt{M\,T}}{V_c^{2/3}\,\Omega_\mu(T^*)} \qquad \text{[Pa·s; } M \text{ in kg/kmol; } V_c \text{ in cm}^3\text{/mol]}, \tag{257}$$

with the critical volume from the acentric-corrected compressibility $V_c = Z_c RT_c/P_c$, $Z_c = 0.2918 - 0.0928\,\omega$; the reduced temperature $T^* = 1.2593\,T/T_c$ feeding the SAME Neufeld collision integral $\Omega_\mu$; and the polarity factor $F_c = 1 - 0.2756\,\omega$ (non-polar here — the factor does not capture strong dipoles, so Chung degrades for polar species). For a simple non-polar gas the accuracy is excellent: at 298 K it returns $\mu_{N_2} = 1.78 \times 10^{-5}$ Pa·s, within ∼0.3% of the measured value. The model is DECLARED (the `transport` block of `thermoPhysPropDict`) and the engine REFUSES to guess a viscosity without it — no silent default. (Water and steam take the higher-accuracy IAPWS transport formulation instead.)

## 2.5 Declaring the models — the grammar the engine reads

The Chung route above needs no per-component viscosity datum. The case declares WHICH correlation serves each phase in the `transport` block of `thermoPhysPropDict`; the coefficients a FITTED liquid model needs live in the component's own record, under the model's name:

```
transport                                // constant/thermoPhysPropDict
{
    vapour
    {
        viscosity           { model Chung; }    // gas mu   (Tc, Pc, omega, M)
        thermalConductivity { model Eucken; }   // gas k    (Cp + mu)
        diffusivity         { model Fuller; }   // binary D (diffusionVolume)
    }
    liquid
    {
        viscosity           { model Andrade; }  // or Vogel, chemsepEq101
    }
}

liquidViscosity                          // in the component's .dat
{
    andrade { A -13.03;   B 1796.1; }              // ln(mu/Pa.s) = A + B/T
    vogel   { A -10.5366; B 541.4;  C 144.0; }     // ln(mu/Pa.s) = A + B/(T-C)
}
```

An earlier printing of this section showed `viscosityGas`, `Sutherland` and `lennardJones` keys. No engine reader existed for any of them at the time; the listing was corrected on 2026-09-05, and on 2026-09-06 a `lennardJones` reader arrived (the Chapman-Enskog model above, whose listing shows the grammar the loader reads). The gate (`check_transport_correlations`) refuses the guide a `Sutherland` or `viscosityGas` example while the engine has no reader for one, and holds any `lennardJones` example here to the record grammar the engine parses. Every registered transport model states its validity window, its PRIMARY citation and a self-check anchor, and `choupoProps transportBench` runs them all (tutorial `transport01_gas_bench`); record: `docs/design/transport-correlations-as-objects.md`.

> **Runnable case.** **Status note.** Viscosity is wired in: the call surface is `ThermoPackage::viscosityGas(T, y)` / `viscosityLiquid(T, x)`, fed by the declared `transport { viscosity { model Chung; } }` and `liquidViscosity { model Andrade; }` blocks of `thermoPhysPropDict`. Consumers include the pipe (Reynolds number and friction factor, with an announced `operation.viscosity` override), the heat-exchanger HTC correlations, the spray dryer, and the packed-bed Ergun pressure drop. A unit that needs a viscosity no declared model provides refuses loudly with the remedy — no silent default.

# 3 Thermal conductivity

*What you'll learn.* Liquid thermal conductivity is well represented by the Sato-Riedel correlation — a corresponding-states formula that needs only $M$, $T_c$ and $T_b$ (data the simulator already has). Gas conductivity follows from the viscosity through Eucken's relation $\kappa = (\mu/M)(c_p + 5R/4)$ or the modified Eucken with a correction for polyatomics. Reid Chapter 10.

## 3.1 Pure liquid: Sato-Riedel

$$\kappa^{\mathrm{liq}}(T) \;=\; \frac{1.11}{\sqrt{M}}\,\frac{3+20\,(1-T_r)^{2/3}}{3+20\,(1-T_{br})^{2/3}} \qquad [\mathrm{W/(m{\cdot}K)}], \tag{258}$$

where $T_r = T/T_c$, $T_{br} = T_b/T_c$ and $M$ is in g/mol (equivalently kg/kmol). The formula contains zero fitted constants per component — it is pure corresponding states, calibrated once across the Reid-Prausnitz-Poling reference set[4]. Accuracy: ∼8% on average across non-polar liquids, ∼15% for polar ones (water is an outlier and gets its own tabulation).

**Worked example 3.1.** Acetic acid at 350 KWith $M = 60.052$, $T_c = 591.95$, $T_b = 391.05$: $T_r = 0.591$, $T_{br} = 0.661$. Eq. (258) gives $\kappa \approx 0.162$ W/(m · K) against a literature value of ∼0.158 W/(m · K).

## 3.2 Per-component fits, and how far corresponding states gets you

Sato–Riedel asks for nothing but $M$, $T_b$ and $T_c$, which is its whole appeal: it answers for a substance nobody has measured. The price is that it cannot know anything a published fit knows. Choupo therefore also carries `chemsepEq16`[43], a five-parameter per-component form

$$\kappa^{\mathrm{liq}}(T) \;=\; A + \exp\!\Big(\frac{B}{T} + C + D\,T + E\,T^2\Big) \qquad [\mathrm{W/(m{\cdot}K)}], \tag{259}$$

declared per component rather than per package, so a substance with a fit uses it and a substance without one still gets an answer.

*How far apart are they?* Comparing the two across every catalogue substance that has both, at 298.15 K and inside each fit's own declared window — 351 of them — the median disagreement is 9.3% and the mean 12.1%, with 250 inside 15%. That is the number the ∼8–15% claim above deserves to be read as: it holds, and it holds across hundreds of substances rather than a handful.

*Where it does not hold is the interesting part.* The eight largest disagreements are, without exception, hydrogen-bonding or strongly polar liquids: ammonia (+123%), water (+106%), hydrogen chloride (+105%), sulfur trioxide (+89%), glycerol (+79%), nitric acid (+73%), sulfur dioxide (+60%). Corresponding states works by assuming that all substances behave alike once scaled by their critical constants, and an associating liquid is precisely the case where that assumption fails — the hydrogen-bond network conducts heat by a route the scaling never sees. Water at 25 °C is about 0.61 W/(m·K); Eq. (258) says 0.30.

Two lessons worth carrying past this page. First, a correlation's stated accuracy is an average over the set it was calibrated on, and your substance may not be in the middle of that set. Second — and this is why the comparison is presented as a comparison and not as a validation — *neither of these is a measurement.* Two correlations agreeing tells you the arithmetic and the units are right. It does not tell you either one is true. To learn that you need measured data, and the honest statement is that Choupo's catalogue does not currently carry measured liquid conductivities to check against.

## 3.3 Pure gas: Eucken

Eucken[48] combined heat conduction and viscosity through the kinetic energy budget of a colliding-molecule model:

$$\kappa^{\mathrm{gas}}(T) \;=\; \frac{\mu(T)}{M}\,\Big[\,c_p(T) + \tfrac{5}{4}R\,\Big] \qquad [\mathrm{W/(m{\cdot}K)}]. \tag{260}$$

The factor $\frac{5}{4}R$ corrects an Euler-type derivation that otherwise overpredicts $\kappa$ for monatomics. Polyatomic molecules carry extra internal degrees of freedom; the *modified Eucken* separates the translational and internal contributions and weights the internal one by 1.32. In the form Svehla (1962, NASA TR R-132, eq. (2), report page 3)[45] used for every polyatomic gas in his tables,

$$\kappa \;=\; \frac{R}{M}\left[\frac{15}{4} + 1.32\left(\frac{c_p}{R} - \frac{5}{2}\right)\right]\mu \qquad [\mathrm{W/(m{\cdot}K)}], \tag{261}$$

which on $c_v = c_p - R$ is the textbook $(\mu/M)(1.32\,c_v + 1.77\,R)$. Either way, gas thermal conductivity is a derived quantity in this scheme — no new fitted parameter beyond those already in $\mu$ and $c_p$. **Status:** Choupo registers BOTH, `model Eucken;` and `model modifiedEucken;` (since 2026-09-06). They coincide EXACTLY for a monatomic gas (with $c_p = 5R/2$ both give the Chapman-Enskog $\kappa = \frac{15}{4}\,R\mu/M$[49], which is what each one's own `verify()` anchor checks) and differ on every polyatomic gas, which the `transportBench` shows side by side — the conductivity spread in tutorial `transport02_chapman_enskog` is that difference. Plain Eucken is flagged per component as an approximation from the record's own $c_p$; the modified form's window is Svehla's own sentence, that the correction "may be rather poor at low temperatures" and improves as $T$ rises.

## 3.4 Declaring the models

Sato-Riedel needs no coefficients; Eucken needs none either once $\mu$ and $c_p$ are in place. `chemsepEq16` needs five, and takes them from the component's own record rather than from the case. The case's `thermoPhysPropDict` just declares which model to use, inside the `transport` block:

```
transport
{
    thermalConductivity  { model Eucken; }      // gas (needs a viscosity model)
 // thermalConductivity  { model modifiedEucken; } // gas, Svehla eq. (2), polyatomics
    liquidConductivity   { model SatoRiedel; }  // liquid, predictive
 // liquidConductivity   { model chemsepEq16; } // liquid, per-component fit
}
```

> **Runnable case. Status note.** $\kappa$ is wired in: the call surface is `ThermoPackage::thermalConductivityGas(T, y)` / `thermalConductivityLiquid(T, x)`, consumed by the heat-transfer-coefficient correlations of the heat exchanger and by the spray dryer. Eucken needs the gas viscosity, so a viscosity model must be declared alongside; a unit that needs $\kappa$ with no declared model refuses loudly.

# 4 Diffusivity

*What you'll learn.* Diffusivity is fundamentally a *pair* property: it describes the motion of species $i$ relative to species $j$, and an isolated component cannot carry a single number. For binary diffusion in gases the workhorse is Fuller-Schettler-Giddings[50]; for binary diffusion in dilute liquids it is Wilke-Chang[4]. Multicomponent extensions (Vignes, Maxwell-Stefan) build on the binary values. Reid Chapters 11–12.

## 4.1 Gas binary: Fuller-Schettler-Giddings

$$D_{AB}(T,P) \;=\; \frac{1.013\times10^{-7}\,T^{1.75}\,[(M_A+M_B)/(M_AM_B)]^{1/2}}{P\left[(\Sigma v)_A^{1/3}+(\Sigma v)_B^{1/3}\right]^2} \qquad [\mathrm{m^2/s};\ T \text{ in K},\ P \text{ in bar},\ M \text{ in g/mol}]. \tag{262}$$

The $(\Sigma v)_i$ are *atomic diffusion volumes*: a small table of element-additive contributions (C: 16.5, H: 1.98, O: 5.48, N: 5.69; complete tables in Reid 11-1). No experimental binary diffusivity is needed — the prediction is geometric, from atomic group contributions only. Accuracy: ∼10% for gases at ∼1 bar; degrades sharply above 10 bar where high-pressure corrections become necessary.

## 4.2 Liquid binary, dilute: Wilke-Chang

$$D_{AB}^{\infty} \;=\; 7.4\times10^{-12}\,\frac{(\phi_B\,M_B)^{1/2}\,T}{\mu_B\,V_A^{0.6}} \qquad [\mathrm{m^2/s};\ T \text{ in K},\ M \text{ in g/mol},\ \mu_B \text{ in Pa}\cdot\text{s},\ V_A \text{ in cm}^3\text{/mol}], \tag{263}$$

where $V_A$ is the molar volume of solute $A$ at its boiling point (LeBas group-contribution table, or just $V_A^{\mathrm{liq}}$ from the data file as a first approximation) and $\phi_B$ is the association factor of solvent $B$ (water: 2.6; methanol: 1.9; ethanol: 1.5; benzene/other unassociated: 1.0). The result is the *infinite-dilution* diffusivity — valid as $x_A \to 0$.

*Intuition.* Liquid diffusivities are six to seven orders of magnitude lower than gas diffusivities at 1 bar ($\sim 10^{-9}$ vs $\sim 10^{-5}$ m²/s). This single fact is why almost every mass-transfer-limited unit op is dominated by the liquid film resistance.

## 4.3 Composition dependence: Vignes

Wilke-Chang gives only the limiting values $D_{AB}^{\infty}$ (solute $A$ in solvent $B$) and $D_{BA}^{\infty}$ (the reverse). Across the full composition range Vignes interpolates logarithmically:

$$D_{AB}(x_A) \;=\; \left(D_{AB}^{\infty}\right)^{1-x_A}\left(D_{BA}^{\infty}\right)^{x_A}. \tag{264}$$

For multicomponent systems the Maxwell-Stefan framework promotes each binary $D_{ij}$ to a matrix that combines them rigorously. Reid Chapter 12 treats this in depth; we will not need it before v1.x of the simulator.

## 4.4 Where the ingredient data live

The pair property itself is computed, never stored; what each component's `.dat` carries are the *per-component* ingredients — the Fuller atomic-volume sum and, for a solvent, the Wilke-Chang association factor:

```
diffusionVolume    50.36;   // Fuller Sigma-v, in the component's .dat
associationFactor  2.6;     // Wilke-Chang phi (water; solvents only)
```

The case then declares the models in the `transport` block of `thermoPhysPropDict`: `diffusivity { model Fuller; }` for the gas and `liquidDiffusivity { model WilkeChang; }` for the dilute liquid (the latter needs a `liquidViscosity` model alongside, since Wilke-Chang divides by the solvent viscosity).

**Runnable case. Status note.** Diffusivity is wired in: the call surface is `ThermoPackage::diffusivityGas(T, P, i, j)` / `diffusivityLiquid(T, i, j)`, consumed by the spiral-wound membrane's mass-transfer correlations, the fixed-bed adsorber, the spray dryer, and the electrodialysis stack. A missing `diffusionVolume` refuses by name with the remedy.

# 5 Mixture rules

*What you'll learn.* Pure-component transport properties are useless on their own — every real stream is a mixture. Reid-Prausnitz-Poling Chapter 9[4] recommends Wilke's rule for gas viscosity, a log-mole-fraction average for liquid viscosity, a mass-fraction average for liquid thermal conductivity, and the Vignes interpolation for binary diffusivity. Each rule has a domain of validity and a ten-line implementation.

## 5.1 Gas viscosity: Wilke

$$\mu_{\text{mix}}^{\text{gas}} \;=\; \sum_i \frac{y_i\,\mu_i}{\sum_j y_j\,\phi_{ij}}, \qquad \phi_{ij} \;=\; \frac{\left[1+(\mu_i/\mu_j)^{1/2}(M_j/M_i)^{1/4}\right]^2}{\left[8(1+M_i/M_j)\right]^{1/2}}. \tag{265}$$

The $\phi_{ij}$ are dimensionless interaction terms[51]. Reid reports $\sim$2% accuracy for binary gas mixtures across the full composition range, with the rule reducing exactly to $\mu_i$ in the pure-component limit by construction.

## 5.2 Liquid viscosity: log-mole-fraction average

$$\ln \mu_{\text{mix}}^{\text{liq}} \;=\; \sum_i x_i\,\ln \mu_i. \tag{266}$$

The motivation is empirical (Arrhenius rule for ideal liquid mixtures) but the form is well behaved: it is exact when the mixing is athermal and the simplest non-trivial answer otherwise. For strongly hydrogen-bonded mixtures the Grunberg-Nissan correction adds a binary interaction term $\sum_i \sum_j x_i x_j G_{ij}$ but we will not need it.

## 5.3 Liquid thermal conductivity: mass-fraction average

$$\kappa_{\text{mix}}^{\text{liq}} \;=\; \sum_i w_i\,\kappa_i, \qquad w_i \;=\; \frac{x_i M_i}{\sum_j x_j M_j}. \tag{267}$$

The mass weighting is empirical; Filippov's binary correlation is more accurate but needs an interaction term per pair. For engineering accuracy mass-fraction mixing is enough.

## 5.4 Gas thermal conductivity: Wassiljewa with Mason-Saxena

Same shape as Wilke for viscosity[52, 53], with $\phi_{ij}$ rebuilt from the ratio of thermal conductivities:

$$\kappa_{\text{mix}}^{\text{gas}} \;=\; \sum_i \frac{y_i\,\kappa_i}{\sum_j y_j\,A_{ij}}. \tag{268}$$

The $A_{ij}$ are usually approximated as $\phi_{ij}$ of Wilke — the kinetic-theory connection between $\mu$ and $\kappa$ ensures the two rules share the same interaction structure.

## 5.5 Diffusivity in multicomponent: Vignes plus Maxwell-Stefan

The binary Vignes formula (264) handles ternary and beyond if generalised to a full Maxwell-Stefan diffusivity matrix $\left[D\right]$ whose $i,j$ entries come from the binaries $D_{ij}^{\infty}$. Practical implementation requires inverting that matrix to recover Fick's-law-equivalent diffusivities. This is non-trivial; Reid Chapter 12 covers it in depth and we will adopt his Eq. (12-3.5) when v1.x of the simulator needs it.

*Intuition.* Notice the pattern: every mixture rule reduces to the pure-component value when $x_i = 1$, and most are weighted means (linear, logarithmic, or harmonic) of the pure values with an interaction correction. The engine ships exactly the rules of this chapter — Wilke, the log-mole-fraction (Grunberg-Nissan) average, mass-fraction $\kappa$, Wassiljewa/Mason-Saxena — and a `mixingRule` key is deliberately *not* selectable yet: the builder refuses it by name rather than letting a refinement switch on by accident.

# 6 Where transport feeds the downstream chapters

*What you'll learn.* A brief look forward: once the transport properties of the previous chapters are wired in, they unlock pressure-drop, heat-transfer and mass-transfer correlations in the unit operations. This chapter sketches each consumer without rigorously deriving it — those correlations live in Part V (unit operations) and the post-processing chapters — so you know where the cables go before v1.x starts plugging them.

## 6.1 Pressure drop

- **Pipes (single phase):** Darcy-Weisbach, $\Delta P = f\,\rho u^2 L/(2D)$, with $f = f(\mathrm{Re}, \epsilon/D)$ via Colebrook or Churchill. $\mathrm{Re} = \rho u D/\mu$ ties $\Delta P$ directly to viscosity.
- **Packed beds (catalyst pellets, structured packing):** Ergun, $\Delta P/L = 150(1-\varepsilon)^2\mu u/(\varepsilon^3 d_p^2) + 1.75(1-\varepsilon)\rho u^2/(\varepsilon^3 d_p)$. Viscous and inertial terms, both viscosity-dependent in the laminar regime.
- **Two-phase flow (column trays, slug flow):** Lockhart-Martinelli phenomenological correlation in $X^2 = (\Delta P_L/\Delta P_G)$, both legs computed with their own viscosity.

## 6.2 Heat-transfer coefficients

Convective $h$ on either side of a tube wall comes from a Nusselt correlation $\mathrm{Nu} = f(\mathrm{Re}, \mathrm{Pr})$. Two workhorses:

- **Dittus-Boelter (turbulent, smooth tubes)**[54, 55]**:** $\mathrm{Nu} = 0.023\,\mathrm{Re}^{0.8}\,\mathrm{Pr}^n$, with $n = 0.4$ for heating, 0.3 for cooling.
- **Sieder-Tate** (same with a wall-viscosity correction $(\mu/\mu_w)^{0.14}$ for liquids with steep $\mu(T)$).

With Nu in hand, $h = \mathrm{Nu}\,\kappa/D$ and the overall coefficient $U$ falls out of the standard series resistance $1/(UA) = 1/(h_i A_i) + R_{\mathrm{wall}} + 1/(h_o A_o)$. The heat-exchanger sizing chapter (Part VI Chapter 1's sibling) currently uses a hardcoded $U$; with transport in place, $U$ will be a result, not a constant.

## 6.3 Mass-transfer coefficients, NTU and HETP

For absorption or distillation, the mass-transfer film coefficient $k_c$ comes from a Sherwood correlation $\mathrm{Sh} = f(\mathrm{Re}, \mathrm{Sc})$ of the same shape as Nu(Re, Pr). HETP for a packed column then follows from Onda's or Eduljee's correlations. For the distillation column each stage is assumed perfectly efficient (Murphree $E_{\mathrm{MV}} = 1$); the v1.x extension will introduce $E_{\mathrm{MV}}$ as a Sherwood-derived quantity, varying with composition and flow regime.

**Runnable case. Status note (consolidated).** No transport coefficient is read or computed by any unit op. Pressure drops are zero, heat-exchanger U is constant (or absent — the user supplies $T_{\mathrm{out}}$ or $Q$), column efficiency is 100% per stage. This Part is the theoretical scaffolding for the v1.x rollout; the data-file extensions listed in Chapters 2–4 are the contract the implementation will honour.

# Part VI — Selected unit operations

*A complex system that works is invariably found to have evolved from a simple system that worked.*

John Gall, *Systemantics* (1975)

# 1 A unit operation is a physical equipment

> *What you'll learn.* The framing Choupo adopts for every unit operation in this Part: the **unit op object models a real piece of hardware** (a vessel with a heat-exchange surface, a column with $N$ trays, a tube of length $L$), with inputs the engineer can point to in the plant ("this feed line", "this heating-steam header") and outputs the engineer can sample ("this concentrated stream", "this distillate drum"). Anything that is *not* a property of the equipment itself — a target outlet concentration, an equal-area design rule, a desired conversion — is handled by an outer **DesignSpec controller** that wraps the simulation, not by the unit-op spec.

## 1.1 The friction we want to remove

A student writing her first single-effect evaporator dict tends to freeze on the question "*which* variable do I specify — duty, vapour fraction, area, both?" The same paralysis returns for every unit op in a typical undergraduate flowsheeting course: distillation columns asking the student to choose between "$R, N$ and $F$" or "$D/F, R$ and $N$" or "$D, R$ and $N$"; heat exchangers asking "T-out or duty"; reactors asking "volume or $X_A$". These degree-of-freedom puzzles are mathematically equivalent and every chem-eng curriculum eventually explains why — but in the first *week* of a course on simulators, the puzzle is friction. The student does not yet own the mental model that justifies the choice; she fills out some boxes, gets an unreasonable answer, and concludes (often correctly) that "the simulator is being weird again".

Choupo takes a different stance, learned from twenty-plus years of Choupo's convention: the dict that defines a unit op describes the **hardware**, not a math problem. An evaporator is a heat-exchange surface of area $A$ with an overall transfer coefficient $U$ operating at a process-side pressure $P$ and heated by a stream coming in on the chest. A heat exchanger is a tube bank of area $A$ with a coefficient $U$. A distillation column has $N$ trays at a feed stage $f$ with a reflux ratio $R$. A reactor is a vessel of volume $V$ at temperature $T$ with a named reaction. None of these dicts contains the word "target" or "conversion" or "$V/F$". Those are *outcomes* of the simulation, not inputs to it.

Concretely, the evaporator (and the case files in `tutorials/steady/evaporation/evaporator0*`) keeps the operation block to four scalars:

```
operation
{
    P     1.0  bar;
    area  20.0 m2;
    U   2000.0 W/m2/K;
    Tref 298.15 K;          // optional
}
```

and the student reads off "V/F", "Q", "T_boil", "steam demand" and "economy" from the run log. The mode-switch between `duty` and `vaporFraction` that carried is *gone* — a single physical specification in, all the operating quantities out.

## 1.2 Where this idea comes from

The "unit op = physical equipment" stance is older than sequential-modular simulation itself. Rudd & Powers' *Process Synthesis* [56] opens with the observation that a process flowsheet is a directed graph of *real-world equipment connected by real-world streams* — the design problem is then framed as choosing the equipment shape and the stream topology, not as solving an algebraic system. Westerberg *et al.*'s *Process Flowsheeting* [1] formalises this view: every node in the graph is an object that owns its connections, owns its sizing parameters, and exposes a small, fixed set of physical inputs to the user. The book makes the further claim — still true — that "the user should not be required to know which variables the solver will choose to iterate on".

This is the standard layered model, where each unit op block has a small fixed set of physical parameters and external *Design Specs* sit on top. "Calculator"- and "Design-Specification"-style blocks let the engineer write "manipulate the column reflux ratio so that the distillate methanol mole fraction equals 0.95" *without* altering the column block itself — the block still owns $R, N, f$; the Design Spec owns the targeting. Stephanopoulos' textbook treatment of the sequential-modular paradigm [57] draws this distinction explicitly: *operating parameters* belong on the block (the engineer sets the hardware), *design constraints* belong on a controller (the engineer sets the target).

Biegler, Grossmann, and Westerberg's *Systematic Methods* [58] go on to formalise the **decomposition** that this layering enables: the inner problem ("simulate this fixed hardware with these fixed inputs") is well-posed and fast, the outer problem ("find the hardware that satisfies these constraints") is the *design* problem, and the two solvers do not need to know about each other's variables. Choupo is built on the same decomposition: the unit ops in this Part are inner-problem solvers; the `DesignSpec` driver scheduled for is the outer problem, and it manipulates the same dict-level scalars that the user could have edited by hand.

## 1.3 What the student sees

In each unit-op section below, the dict block carries only physical quantities:

- **Flash**: $T, P$ and a phase set (the vessel sees the same $T, P$ on inlet and outlet by definition; nothing computed enters the dict).
- **Evaporator**: $P$ on the process side, area $A$, overall HTC $U$, and (via the heating-steam input) the chest temperature. V/F, $Q$, $T_{\text{boil}}$ and the steam demand are RESULTS.
- **Heat exchanger**: hot-side $T_{\text{in}}$, cold-side $T_{\text{in}}$, $UA$ (and a flow arrangement); duty $Q$ and both outlet temperatures come out.
- **CSTR / PFR**: vessel volume $V$ (or tube length $L$ + cross-section $A$), $T$ (isothermal) or thermal mode, plus a named reaction. Conversion and outlet composition are results.
- **Distillation column**: $N, f, R, D$ (or some physical analogue). Tray temperatures and the full $\mathbf{x}, \mathbf{y}$ profile come out.

and the student's homework consists of **describing the hardware**, not solving the math. Where the assignment is "find the area such that the outlet concentration is 25%", that statement maps onto a `DesignSpec` on top of the unit op — the same way a real plant engineer thinks: I have an evaporator with area $A$; the question is, "*what $A$* delivers the spec the customer wrote into the PO?".

> *Intuition.* The discriminator between an operating parameter (lives on the unit op) and a design target (lives on an outer DesignSpec) is the question: *would you walk out to the plant and measure this with a wrench or a thermometer?* If yes, it is hardware — write it on the block. If you would only see it on the DCS trend afterwards, it is a result — let the simulator compute it and a DesignSpec target it.

# 2 A stream is a thermodynamic state, not five numbers

*What you'll learn.* The companion framing for the unit-op pattern of §1: a process **stream** is the thermodynamic state of a fluid at a point in the plant. Every state has a fixed number of degrees of freedom, and the engineer specifies just enough of them — not all at once — to fix the state uniquely. The simulator computes the rest, including a sanity-check pass for the variables left unspecified. This "flash on import" is what lets the case file carry only the *given data* of a problem instead of pre-computed answers.

## 2.1 Degrees of freedom for a single-phase pure-component stream

For one mole of fluid at thermodynamic equilibrium, the Gibbs phase rule fixes the variance. For a pure ($C = 1$) single-phase ($\pi = 1$) system,

$$F = C - \pi + 2 = 2,$$

so two intensive variables (any two of $T$, $P$, density, molar enthalpy, ...) fix the state completely. Multiply the stream by the extensive total molar flow $F_{\text{total}}$ and you have the full description: $\{F_{\text{total}}, T, P\}$ (3 numbers; one extensive + two intensive).

If the stream is two-phase (still $C = 1$, $\pi = 2$, $F = 1$), one intensive plus the phase fraction $\beta$ is enough — the other intensive collapses to the saturation curve $P_{\text{sat}}(T)$. So $\{F_{\text{total}}, T \text{ or } P, \beta\}$.

## 2.2 The Choupo stream record carries six fields

The C++ struct (`src/streams/ProcessStream.H`) is:

```
struct ProcessStream {
    std::string name;
    scalar      F;          // total molar flow      [kmol/s]
    scalar      T;          // temperature            [K]
    scalar      P;          // pressure               [Pa]
    sVector     z;          // mole fractions       (Sigma = 1)
    scalar      vf;         // vapour fraction        [-]
    bool        demandDriven;   // utility-tap marker
};
```

For a multi-component stream the degrees of freedom are $C - \pi + 2 + (C - 1) = 2C - \pi + 1$ if you count the composition (the $C - 1$ independent mole fractions). Choupo's record exposes $T$, $P$, $\mathbf{z}$, $\beta$ explicitly — one intensive variable too many for a single-phase stream, exactly right for a two-phase stream, and one too few only if the user also wants enthalpy and entropy stored explicitly (Choupo recomputes those from the equation-of-state layer on demand).

## 2.3 Specification + consistency check on import

A stream's state is written in its own file under a state directory (`0/<stream>`); `system/flowsheetDict` carries the topology and no numbers. The implemented closure is $T + P + \mathbf{z}$: the package checks whether the state is one- or two-phase and fills $\beta$.

| What you write | What the simulator does | Status |
|---|---|---|
| $T$ + $P$ + $\mathbf{z}$ | flash → $\beta$, phase | implemented |
| $P$ + `vaporFraction` | solve $T$ so the flash gives $\beta$ | recognised, resolution deferred |
| $T$ + `vaporFraction` | solve $P$ so the flash gives $\beta$ | recognised, resolution deferred |
| $P + H$ / $P + S$ | isenthalpic / isentropic closure | recognised, resolution deferred |

**Common pitfall. There is no `state` keyword on a stream, and this table used to say there was.** Earlier printings documented `state saturatedVapour`, `saturatedLiquid`, `subcooled` and `superheated` as stream specifications, with a parse-time Antoine inversion behind them. Those were readers of the `flowsheetDict streams {}` block, which was retired in the 2026-07-10 stream-state migration; the block is now refused loudly and its two solvers (`invertPsat`, `solveStateForVf`) were deleted with it, having lost their only caller. Write the saturation temperature you mean, or let a `phaseChanger` produce it — the four saturated-state words survive there, as its `outletState`, where a *unit operation* rather than a parser shortcut does the work. The consequence for the author is real and not merely lexical: the engine no longer inverts $P_{\text{sat}}$ for you at import, so a hand-entered $T$ that drifts from $T_{\text{sat}}(P)$ silently makes a "saturated" feed superheated or sub-cooled, and no check catches it.

## 2.4 `vaporFraction`: the general two-spec rule (why $\beta$ is often essential)

The mixture-aware statement of the degrees of freedom is: **a stream is fixed by z + $F$ + exactly TWO of** {$T$, $P$, `vaporFraction`} (Duhem's theorem; `vaporFraction` is the vapour fraction $\beta$; it is spelled in full — `vf` is the name of the field inside `ProcessStream`, not a dict key). That is the *theory*, and since 2026-10-03 the engine enforces the "exactly TWO" half of it on the one combination the grammar can spell: a third value is refused. What the engine implements of the OTHER pairs is narrower, and the difference is worth stating precisely because it is the kind of gap a reader would otherwise discover by getting a wrong answer:

| What you write (mixture) | What the engine does today | |
|---|---|---|
| $T$ + $P$ | Rachford–Rice flash → $\beta$; the table shows it | the implemented closure |
| $T$ + $P$ + `phase gas\|liquid` | prices the DECLARED phase (R-E2) | the single-phase pin |
| $T$ + $P$ + `vaporFraction` | — | **REFUSED**: over-specified (three values c |
| $P$ + `vaporFraction` | — | recognised; resolution deferred |
| $T$ + `vaporFraction` | — | recognised; resolution deferred |

**Common pitfall.** Earlier printings of this table said that all three variables together were *carried, not refused*, and the printing before that said they were refused while no refusal existed. Since 2026-10-03 the refusal IS implemented, for a mixture: a `0/<stream>` that declares $T$, $P$ and `vaporFraction` on more than one present component is refused by name, and the message quotes the two-variable rule, the three values it found and the two remedies — delete the line and let $(T, P)$ resolve the split, or, if the quality is the specification, declare the $T$ at which the $(T, P)$ flash gives it (the engine does not solve $T$ from $\beta$ today). What caused the change is worth knowing: `flash01_benzene_toluene`, the landing case, shipped all three, and with the line edited to `vaporFraction 0.9` the run still exited 0, the stream table printed 0.900 and the first law priced the feed at its resolved 0.304 — the number shown was not the number used. The case now declares $T$ and $P$ only, and the table shows the resolved 0.304. Gate: `check_overspecified_stream`.

*Intuition.* Why the exception is a PURE component, and only that: on the phase boundary, $T$ and $P$ are **not independent**. For a pure component a two-phase state has $P = P_{\text{sat}}(T)$, so $(T, P)$ is degenerate — you *cannot* pin a saturated state with $T$ and $P$; you must give $\beta$ with one of them, and `vaporFraction q` stays the pin there (wet steam, a partly condensed pure vapour). For a mixture $(T, P)$ between bubble and dew fix $\beta$ uniquely, so writing $\beta$ beside them is either repeating what the flash will say or contradicting it. A saturated liquid or vapour feed is spelled `phase liquid;` / `phase gas;` — a declaration the engine honours and prices as declared, and a `vaporFraction 0` or `1` on a mixture is refused naming that spelling. This is the most common point of confusion in stream specification, and the refusal exists so the engine never shows one number while using another.

A resolver that brackets and bisects on the (monotone) flash $\beta$ — $\beta$=0 the bubble point, $\beta$=1 the dew point, in between a flash-at-$\beta$ — is what the two deferred closures need, and is the shape the retired `Flowsheet.cpp::solveStateForVf` had before it lost its only caller with the `streams {}` block. When those closures return they return through the `0/` grammar, and the announcement obligation returns with them. **A feed's thermal state lives in the STREAM, never duplicated on the consuming unit.** A distillation column reads its feed quality directly from the inlet stream (§34): $q = 1 - \beta$ for a two-phase feed, the enthalpic $q$ of the constant-molar-overflow section for a single-phase one; a redundant `operation.feeds.quality` that *disagrees* with the stream is refused at import, so the two can never drift apart (the silent "feed declared liquid but flashing to vapour" bug that broke an early multi-feed energy balance).

## 2.5 The flow $F$: extensive, source-driven vs. demand-driven

For feed streams the case author always knows $F$ — it is the basis of the design problem. For utility streams (heating steam, cooling water) the situation is reversed: the engineer knows the *conditions* (pressure header, temperature) but the *quantity* is whatever the consuming unit operation demands. Choupo has the machinery for that reversal: a stream carries a `ProcessStream::demandDriven` flag, and when it is set the Flowsheet writes back the $F$ the consuming unit publishes through `consumedInputs()` after its `solve()` returns, so the Streams panel and the structured JSON show the actual consumption rather than a placeholder zero.

**Common pitfall. Nothing currently sets that flag.** The only way to raise it was to declare a stream with `state saturatedVapour` / `saturatedLiquid` and no $F$, and that keyword went with the `streams {}` block — so the write-back at `Flowsheet.cpp` reads a flag that is never true, and a utility stream must be given its $F$ like any other. This is recorded here rather than quietly dropped because a reader who finds the flag in the source is entitled to know it is unreachable; restoring demand-driven flow means giving the `0/` grammar its own way to declare the intent, which is a design act and not a rename.

*Intuition.* The mental model the student should leave with: a stream declaration is a *question* asked of the thermodynamic package ("what state is this fluid in?"), not a five-number data dump that has to be filled out completely. This is the standard contract, and Choupo adopts it too. The pedagogical payoff is that the case file shows the engineer's intent — *this stream is the live steam at 170 kPa from the plant header* — and not the result of a pre-computed arithmetic exercise (Antoine inversion to find $T = 388.36\,\mathrm{K}$).

## 2.6 Where in the code this happens

The single entry point for stream import is `readSourceStream()` in `src/unitOperations/flowsheet/Flowsheet.cpp`, called by `Flowsheet::initialiseFromDict()` once per top-level stream in `flowsheetDict.streams {... }`. Its work breaks into three numbered phases visible in the source:

1. **Composition.** Read one of the five composition flavours (`molarComposition`, `massComposition`, `composition`, `molarFlows`, `massFlows`). Convert to canonical mole-fraction vector $\mathbf{z}$; record the sum of per-species flows when applicable.
2. **$T$, $P$, $\beta$ via `state`.** If the `state` key is present, find the dominant pure component (the one with $x_i \geq 0.999$; mixtures fall through to the explicit-$T$-and-$P$ branch). Invert Antoine if needed via `invertPsat()` — a Newton-1D on $f(T) = P_{\mathrm{sat}}(T) - P$ that converges in $\sim 5{-}10$ iterations because $P_{\mathrm{sat}}$ is monotone increasing.
3. **$F$.** Pick the source: $F_{\text{from per-species}}$, the explicit `F` entry, or *omit* (`demandDriven = true`) when the stream is utility-shaped.

A failed Antoine inversion (component below its triple point, above its critical point, or with an Antoine block whose Trange does not cover the given $P$) throws a runtime_error with a message that names the stream, the dominant component, and the out-of-range value.

# 3 Balances: mass, elements, energy

*What you'll learn.* Three conservation statements judge every solved flowsheet, and they are not the same statement. Total *mass* is conserved always; *moles* are not (a reaction changes them legally); the invariant of a reacting system is the *atoms of each element*. This chapter derives the plant-boundary element balance the engine writes for every converged steady run, explains why the closure seal must test every element separately, and states the honesty rules (`FULL` / `PARTIAL` / `UNAVAILABLE`) that keep a diagnostic from ever lying. The implementation is `src/reporting/BalanceMath.H` + `src/reporting/ElementBalanceReport.cpp`; the shared formula parser is `src/thermo/ElementComposition.cpp`.

## 3.1 Why moles are not conserved, and atoms are

Let stream $s$ carry component molar flows $\dot{n}_{i,s}$ [kmol s$^{-1}$]. A reaction with stoichiometry $\boldsymbol{\nu}$ changes the total molar flow by $\dot{\xi}\sum_i \nu_i$ — for A $\rightarrow$ 2B the moles *double* at full conversion, and nothing is wrong. What a closed reaction cannot change is the count of each element's atoms: if $a_{ie}$ is the number of atoms of element $e$ in one molecule of species $i$ (the *atom matrix*, parsed once from each component's formula), then stoichiometric consistency means $\sum_i \nu_i\, a_{ie} = 0$ for every $e$ — the same condition the Gibbs reactor imposes as its element-abundance constraint (§14).

The plant-boundary element balance follows by summing over the boundary streams (feeds $\mathcal{F}$, products $\mathcal{P}$ — roles inferred from the topology, never declared):

$$\dot{N}_e^{\text{in}} \;=\; \sum_{s\in\mathcal{F}}\sum_i \dot{n}_{i,s}\, a_{ie}, \qquad \dot{N}_e^{\text{out}} \;=\; \sum_{s\in\mathcal{P}}\sum_i \dot{n}_{i,s}\, a_{ie}, \tag{269}$$

in kmol-atom per unit time, with the particulate (solid-phase) flows included alongside the fluid flows. At a converged steady state $\dot{N}_e^{\text{in}} = \dot{N}_e^{\text{out}}$ for *every* element — reacting or not. The report prints one row per element: in, out, residual, and the closure percentage $100\,\dot{N}_e^{\text{out}}/\dot{N}_e^{\text{in}}$. Total mass conservation is the molar-mass-weighted consequence ($M_i = \sum_e a_{ie}\,A_e$ with $A_e$ the standard atomic weights), so a passing element balance implies a passing mass balance — but not conversely: a stoichiometry error that conserves mass by accident (isomer bookkeeping, lumped species) still shows up in the atoms.

## 3.2 The seal tests every element, not the total

It is tempting to compress the table into one number — total atoms in vs total atoms out. That number is shown (it is the headline bar chart in the GUI), but it can *lie*: a +10 kmol-atom/h error in carbon cancels a −10 kmol-atom/h error in oxygen and the signed total closes exactly. The conservation claim ("all elements conserved") is therefore granted only when *every* element's closure is within tolerance; the compact indicator is the *worst* per-element deviation. This is the same anti-cancellation discipline as the per-component mass balance: aggregates are displays, never proofs.

## 3.3 Honesty states: FULL, PARTIAL, UNAVAILABLE

An element balance is only as good as the atom matrix. Three states, written into the report's metadata sidecar, keep the claim honest:

- **FULL** — every species on the boundary resolved its composition (formula, or a declared composition; see below), and every declared boundary stream is present in the result.
- **PARTIAL** — every species resolved, but at least one only through a *declared* composition with an unaccounted mass fraction (a petroleum cut whose assay covers, say, 99.5% of its mass). The declared elements are still usable; a *complete* closure is not stamped, and the sidecar names each partial species with its unaccounted fraction.
- **UNAVAILABLE** — some species could not resolve at all (a pseudo-component with no parseable formula and no declaration), or a boundary stream is missing. The claim is *withheld*, naming the species or stream and the reason — never a silent zero, never a fabricated row.

The composition resolves formula-first: the shared parser reads the component's `formula` (nested groups, hydrates, isotopes, charges). A substance with no meaningful formula — a petroleum cut, a lumped biomass, an ash — may instead carry a declared `elementalComposition{}` block in its `.dat` (mass-fraction basis with a mandatory `unaccountedMassFraction`, or a formula-unit basis whose atom counts must reproduce the molar mass); the declaration is validated at component read, and its provenance is required — an invented composition cannot wear a trustworthy badge.

## 3.4 The three regimes

- **Steady** (`choupoSolve`): the element balance is a *default* diagnostic of every converged run — the report writes `reports/balances/elementBalance.csv` + `.meta` without being asked (opt out per report block with `enabled false;`).
- **Batch** (`choupoBatch`): the campaign ledgers carry the same laws as exact state differences — every transfer record prices its elements, and the campaign verdict is granted only when every piece is accounted.
- **Dynamic** (`choupoCtrl`): the ledger integrates *accepted* integrator states — mass inventory and per-element conservation residuals over time. The *physical* energy closure for the dynamic CSTR honestly refuses: its energy equation is a $c_p$/convective model, not the exact derivative of a stored $U(n, T)$, and claiming a first-law closure over it would be a lie (§4).

The GUI draws these artefacts and never recomputes them: the engine owns the physics; the browser owns the pixels.

# 4 Isothermal flash, vapour–liquid

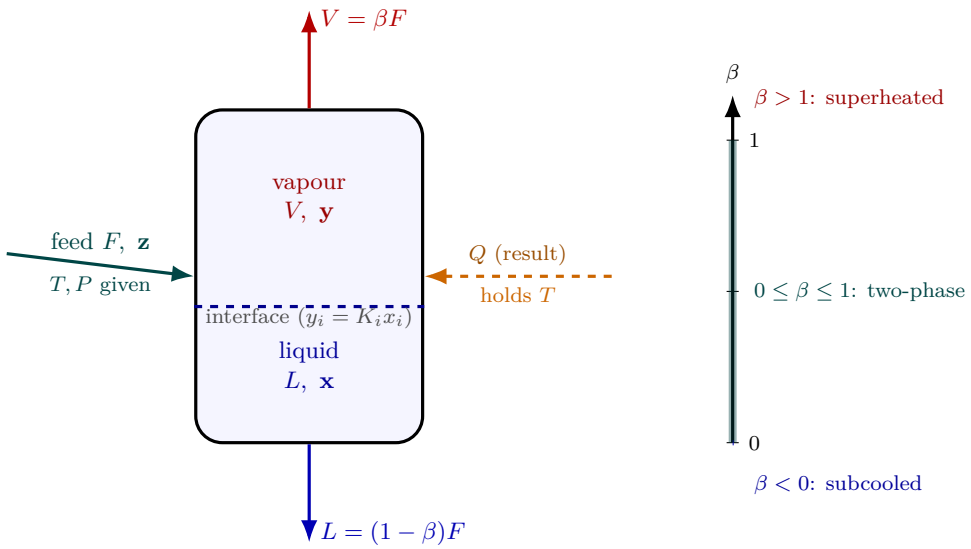



**Figure 49:** The flash drum as hardware. One feed $(F, \mathbf{z})$ at the specified $(T, P)$ enters and disengages into a vapour $V = \beta F$ off the top and a liquid $L = (1-\beta)F$ off the bottom, the two phases meeting at the interface where $y_i = K_i x_i$. The vapour fraction $\beta = V/F$ is *not* an input — it is the Rachford–Rice root forced by the mass balance. The duty $Q$ (dashed, orange) is whatever heat must cross the wall to *hold* $T$ while part of the feed changes phase: a *result* of the converged energy balance, shown as a stream, never a hidden number. The ruler on the right reads $\beta$ back as a regime tag: a $\beta$ outside $[0, 1]$ is the honest signal that the feed is single-phase (subcooled or superheated) at the chosen $T, P$.

> *What you'll learn.* The end-to-end algorithm the simulator runs for a V-L flash: outer Wegstein loop on $\gamma$, inner Newton on $\beta$, exit conditions and diagnostics.

## 4.1 Problem statement

**Given**: $\mathbf{z}$, $T$, $P$, a thermo package (Antoine, activity model).
**Find**: $\beta$, $\mathbf{x}$, $\mathbf{y}$ such that (181) and (184) hold simultaneously.

## 4.2 Degrees of freedom: what you specify, what the flash computes

A flash vessel takes *one* feed and splits it, at equilibrium, into a vapour and a liquid product. Counting variables against equations tells you exactly how many numbers the engineer must fix — and which ones the simulator is then free to move. This is the single most common point of confusion, so we spell it out.

**The feed is given, not a degree of freedom.** Its total flow $F$, composition $\mathbf{z}$ (with $\sum_i z_i = 1$, hence $C-1$ independent mole fractions), and thermal state all arrive on the inlet stream. Nothing about the feed is yours to choose at the flash — it is whatever the upstream unit delivers.

**With the feed fixed, exactly two degrees of freedom remain.** The unknowns of the split are $\{\beta,\ \mathbf{x},\ \mathbf{y},\ T,\ P,\ Q\}$. Against them stand the equilibrium relations $y_i = K_i x_i$, the $C$ component balances $z_i = \beta\, y_i + (1-\beta)\, x_i$, the summation condition $\sum_i x_i = \sum_i y_i$ (the Rachford-Rice constraint), and the energy balance $Q = H_{\text{out}} - H_{\text{in}}$. The bookkeeping leaves *two* free. This is *Duhem's theorem*: the equilibrium state of a closed system of known overall composition is fixed by exactly two independent variables, regardless of how many components or phases are present. (Equivalently via Gibbs' phase rule: a two-phase $C$-component system has $\mathcal{F} = C - 2 + 2 = C$ intensive degrees of freedom; the feed composition consumes $C-2$ of them through the lever rule, leaving two.) You close those two by specifying *any two* of:

| You specify | Flash type |
|---|---|
| $T$ and $P$ | *isothermal* flash — Choupo's `isothermalFlash` |
| $Q$ and $P$ | *adiabatic* ($Q = 0$) or fixed-duty flash; $T$ is found |
| $\beta$ and $P$ | fixed vapour fraction ($V/F$); $T$ is found |

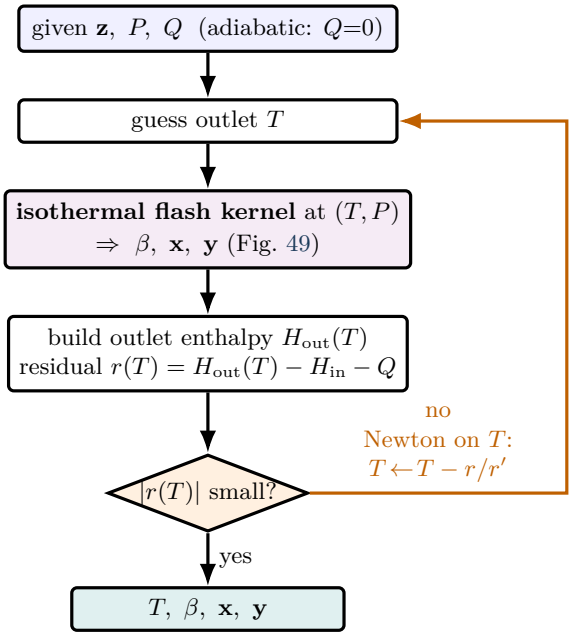



**Figure 50:** The adiabatic (or fixed-duty) flash is the isothermal flash wearing one more loop. Here $T$ is *not* given — the spec is the duty $Q$ (zero for a true adiabatic throttle) and the pressure $P$. The outer Newton (orange) searches for the outlet temperature: at each trial $T$ it runs the *whole* isothermal kernel of Fig. 49 (itself two loops) to get the split, builds the outlet enthalpy $H_{\text{out}}(T)$, and forms the energy residual $r(T) = H_{\text{out}}(T) - H_{\text{in}} - Q$. Newton on $T$ drives $r$ to zero — the temperature at which the phase split absorbs exactly the specified duty. The same kernel, reused: an isothermal flash fixes $T$ and *reports* $Q$; the adiabatic flash fixes $Q$ and *finds* $T$. This is Duhem's two degrees of freedom, closed from the other side.

**Specified vs. manipulated, for the isothermal flash.** In the `operation` block you **specify** $T$ and $P$. The simulator then **manipulates** the remaining unknowns until every equation holds:

- the vapour fraction $\beta = V/F$ (the root of Rachford-Rice),
- the phase compositions $\mathbf{x}$ and $\mathbf{y}$,
- the heat duty $Q$ that must cross the boundary to *hold* $T$ constant while part of the feed changes phase — read off the converged energy balance, $Q = H_{\text{out}} - H_{\text{in}}$. On the flowsheet this $Q$ appears as the heat *stream* feeding the unit (a docked utility stub + dashed wire), never as a hidden number.

**Where the duty lives — a flash *gives* a $Q$, a heater *takes* one.** Because $T, P$ already spend both degrees of freedom, $Q$ is a *result* of the isothermal flash, never an input — the `operation` block has no $Q$ key (it was removed on purpose). The unit whose hardware knob *is* the duty is the `heater`: there you **specify** $Q$ and the outlet $T$ is the result. So "inject a known 500 kW, then separate" is a `heater` → `flash` chain (the heat enters on a visible wire), and "hold the outlet at a target $T$" is a *design spec* that manipulates the heater's $Q$ in a visible outer loop — exactly as outlet pressure is handled on a compressor (specify $W_{\text{shaft}}$, target $P_{\text{out}}$). This is the "heat is a stream, not a hidden number" rule applied to the flash.

> *Intuition.* The trap is to count $T$, $P$ *and* $\beta$ as three inputs. Once $T$ and $P$ are set, $\beta$ is *not* free — it is whatever the two-phase mass balance forces, and it may even land outside $[0, 1]$ (subcooled liquid or superheated vapour, see the regime tags below). Symmetrically, the duty $Q$ is *never* specified for an isothermal flash; it is the answer, not the question. Pick the two specifications that match the equipment: a flash drum held by a thermostat is $T, P$; an adiabatic throttle is $Q$=0, $P$; a partial condenser sized for a target recovery is $\beta, P$.

## 4.3 Algorithm

1. Initialise $\mathbf{K}$ from Wilson (185).
2. Solve Rachford-Rice (184) for $\beta$ by Newton-1D.
3. Compute $\mathbf{x}, \mathbf{y}$ from (183).
4. Compute $\boldsymbol{\gamma}^{(k+1)} = \gamma(\mathbf{x}, T)$.
5. Update $K_i = \gamma_i \, P_i^{\text{sat}}(T)/P$.
6. Wegstein-accelerate $\mathbf{K}$ (component-wise; (219)).
7. If $\|\mathbf{K}^{(k+1)} - \mathbf{K}^{(k)}\|_\infty < \varepsilon$, exit; else goto 2.

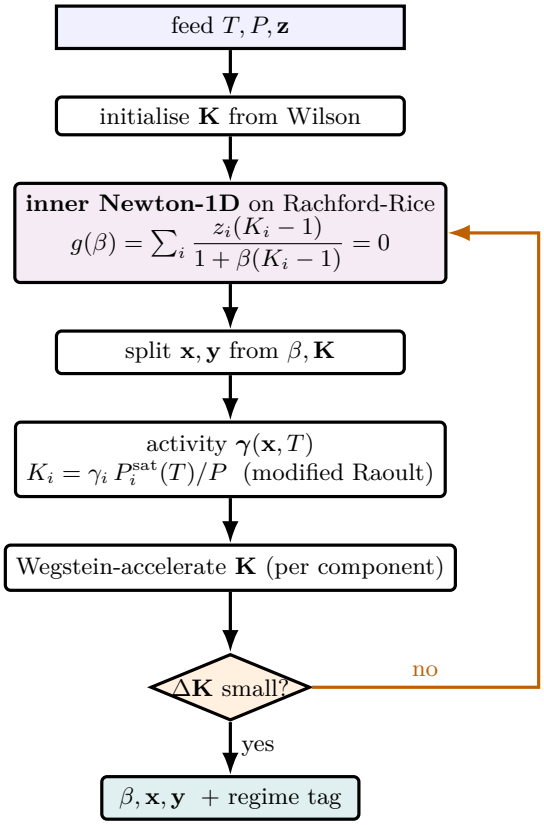



**Figure 51:** The isothermal V–L flash is *two nested loops*. The **inner** loop (violet) is a Newton-1D that solves the Rachford–Rice equation for the vapour fraction $\beta$ at the current $K$-values — a clean monotone root. The **outer** loop (orange) updates the $K$-values: from the trial split it re-evaluates the activity coefficients $\gamma_i(\mathbf{x}, T)$, rebuilds $K_i = \gamma_i P_i^{\mathrm{sat}}/P$ through modified Raoult, and Wegstein-accelerates them. When the $K$-values stop moving the two loops are mutually consistent and the flash is converged; the final $\beta$ also classifies the regime (subcooled $\beta < 0$, two-phase $0 \le \beta \le 1$, superheated $\beta > 1$). For an *ideal* system the activity step is a no-op and only the inner loop runs once — the same code, fewer passes.

## 4.4 Diagnostics in the log

At `verbosity 3`, the simulator prints:

- Inner Newton trace on Rachford-Rice: $(\beta^{(j)}, g(\beta^{(j)}), g'(\beta^{(j)}))$ per iteration.
- Outer Wegstein trace: $\gamma_i^{(k)}$ and $q_i^{(k)}$ per component.
- Final regime tag: `V-L (subcooled)` if $\beta < 0$, `V-L (superheated)` if $\beta > 1$, `V-L (two-phase)` otherwise.

> *Intuition.* The most informative number to watch is $q_i^{(k)}$. If $q$ sits near $q_{\min}$, the system is strongly non-linear (azeotrope nearby) and the simulator is choosing the most aggressive accepted extrapolation. If $q \approx 0$, the system is mildly non-linear and direct substitution would have been fine.

# 5 Phase changer: the dome-crossing boiler / condenser

*What you'll learn.* How Choupo's `phaseChanger` (and its aliases `boiler` and `condenser`) takes one stream across the saturation dome — subcooled liquid, anywhere inside the two-phase region, or superheated vapour — by specifying *one* outlet coordinate and letting the heat duty $Q$ fall out as the result. How the saturation temperature is read off $P^{\text{sat}}$ (or an IF97 $T^{\text{sat}}(P)$ kernel for a pure fluid), how the latent/sensible split is recovered, and why "boiler" and "condenser" are not separate models but emergent labels.

## 5.1 Where it sits between a heater and a flash

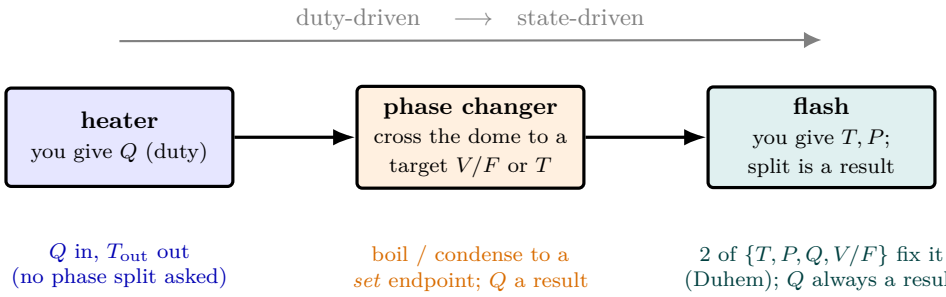



**Figure 52:** Three single-vessel thermal units, ordered by what the student specifies. The **heater** is duty-driven: you furnish $Q$ and the outlet temperature follows — no phase split is even asked. The **flash** is state-driven: you fix two of $\{T, P, Q, V/F\}$ (Duhem's rule), and the vapour–liquid split is the result, with $Q$ *always* an output, never an input key. The **phase changer** sits between them — a boiler/condenser that crosses the saturation dome to a *set* endpoint (a target $V/F$ or $T$), reporting the duty it took. Reading the units this way — duty-driven on the left, state-driven on the right — keeps the degrees of freedom straight and avoids the classic "is $Q$ an input or an output?" confusion.

A `heater` (Sec. 39) is *sensible-only*: you give it a duty $Q$, it returns the outlet $T$, and it must *not* cross the saturation line — it refuses loudly if the energy you inject would start vaporising the stream. An `isothermalFlash` (Sec. 4) fixes $T$ and $P$ and *reports* the duty $Q$ that holds $T$ while the feed splits. The `phaseChanger` is the unit built for the case in between: a piece of hardware whose explicit *job* is to land the outlet somewhere on the $(T, \text{vf})$ surface that the heater forbids and the isothermal flash cannot target directly. One stream in, one (two-phase-aware) stream out, $Q$ a result.

## 5.2 Degrees of freedom: one outlet coordinate

By Duhem's theorem (Sec. 4) the outlet of a single feed at known **z** is fixed by exactly two numbers. The `phaseChanger` spends the first on pressure — $P_{\text{out}}$ is *held* from the inlet unless you set `P` explicitly — and asks you to spend the second by picking **exactly one** outlet coordinate:

| You specify | Meaning | Result |
|---|---|---|
| `Q` | duty added/removed (kW) | $T_{\text{out}}$, $\text{vf}_{\text{out}}$ |
| `outletT` | outlet temperature | $Q$ |
| `outletQuality` | target vapour fraction $\text{vf} \in [0, 1]$ | $Q$ |
| `outletState` | `saturatedLiquid` / `saturatedVapour` | $Q$ |
| `superheat` | $T = T^{\text{sat}}(P) + \Delta K$ (vapour side) | $Q$ |
| `subcool` | $T = T^{\text{sat}}(P) - \Delta K$ (liquid side) | $Q$ |

Supplying zero or more than one coordinate is a hard error — the degree-of-freedom count is enforced, never silently patched. Every mode but the first is an inverse problem: the unit *self-targets*, solving for the $Q$ that lands the outlet on the coordinate you declared, and *announces* that inverse solve in its report (the no-silent-crutch rule — there is no hidden `outerDict` or design-spec loop, the solve is built in but visible). The single-phase states `subcooledLiquid` / `superheatedVapour` are refused as bare `outletState` words precisely because they need that extra distance from saturation; you must give `subcool`/`superheat` instead.

## 5.3 The saturation temperature

Every spec mode needs the position of the dome at $P_{\text{out}}$, i.e. the saturation temperature $T^{\text{sat}}(P)$. Choupo locates it from the *dominant* component (the one with the largest mole fraction) by two routes:

- **Pure-fluid route.** If the dominant species carries an IF97-class pure-fluid kernel *and* the feed is effectively pure, the kernel's closed-form $T^{\mathrm{sat}}(P)$ is used directly.
- **Generic route.** Otherwise the dominant component's vapour-pressure curve is inverted by bisection,

$$P^{\mathrm{sat}}_{\mathrm{dom}}\big(T^{\mathrm{sat}}\big) = P_{\mathrm{out}}, \tag{270}$$

over $T \in [150, 1200]\,\mathrm{K}$ (the Antoine root, Sec. 2).

For a *pure* (or dominant-component) dome the saturation plateau is a single $T = T^{\mathrm{sat}}$ and the vapour fraction vf is the free coordinate along it: $H$ spans $[h_f, h_g]$ at *constant* $T$. For a genuine multicomponent mixture the dome has *width* in $T$ (the bubble-to-dew interval), so $\mathrm{vf}(T)$ rises smoothly from 0 at the bubble point to 1 at the dew point, and the engine brackets the sign-changing interval on a flash scan before driving Newton inside it. This split — step-at-$T^{\mathrm{sat}}$ for a pure dome, smooth-vf$(T)$ for a mixture — is the one subtlety that distinguishes the two cases throughout the solve.

## 5.4 Dome-aware enthalpy and the vf closure

The whole unit is built on the existing flash kernel: `IsothermalFlash::solveCore` returns the split $(\beta, \mathbf{x}, \mathbf{y})$ at any $(T, P_{\mathrm{out}})$, and the stream enthalpy is the lever-rule blend of the equilibrium phases,

$$H(T,P) \;=\; (1-\beta)\, h_f(\mathbf{x},T,P) \;+\; \beta\, h_g(\mathbf{y},T,P), \tag{271}$$

the very enthalpies the produced liquid and vapour streams carry. The molar latent heat at saturation is $h_g - h_f$.

**The vf=1 trap, and the $\varepsilon$-step that avoids it.** Evaluating the enthalpy at *exactly* $\mathrm{vf} = 1$ on the saturation line is dangerous: at $P = P^{\mathrm{sat}}$ the property routing falls onto the *liquid* branch (region 1 of an IF97 kernel; the single-phase side of a cubic EoS) and silently *loses the latent heat*. Choupo therefore probes a tiny step $\varepsilon = 10^{-4}$ *into* the two-phase region and extrapolates the endpoints linearly back out:

$$h_{\mathrm{lo}} = H(T^{\mathrm{sat}}, P,\ \varepsilon), \qquad h_{\mathrm{hi}} = H(T^{\mathrm{sat}}, P,\ 1-\varepsilon),$$
$$h_f = \frac{h_{\mathrm{lo}} - \varepsilon\, h_{\mathrm{hi}}}{1-2\varepsilon}, \qquad h_g = \frac{h_{\mathrm{hi}} - \varepsilon\, h_{\mathrm{lo}}}{1-2\varepsilon}. \tag{272}$$

This recovers the true $h_f, h_g$ (hence the latent heat) without ever asking the thermo for an ambiguous on-line point.

**Duty mode — closing on vf.** In duty mode you give $Q$; the target outlet enthalpy is

$$H_{\mathrm{target}} \;=\; H_{\mathrm{in}} \;+\; \frac{Q}{F}, \tag{273}$$

($F$ in $\mathrm{mol\,s^{-1}}$, $Q$ in W). On a pure dome the engine first tests whether $H_{\mathrm{target}}$ falls on the plateau, $h_f \le H_{\mathrm{target}} \le h_g$; if so the outlet sits at $T^{\mathrm{sat}}$ and the vapour fraction is read straight off the lever rule,

$$\mathrm{vf}_{\mathrm{out}} \;=\; \frac{H_{\mathrm{target}} - h_f}{h_g - h_f}, \tag{274}$$

clipped to $[0,1]$. Only *outside* the plateau — where $H(T)$ is smooth and monotone — does a single Newton-on-$T$ run, with the pure-fluid kernel's triple-point floor ($T \ge 274\,\mathrm{K}$) protecting the search from the kernel's forbidden region. A multicomponent dome has no plateau, so duty mode Newtons on $T$ throughout.

## 5.5 The duty is always a result; latent / sensible split

Whatever coordinate you picked, the duty is computed last, from the converged endpoints:

$$Q \;=\; F\,\big(H_{\mathrm{out}} - H_{\mathrm{in}}\big), \qquad Q>0 \text{ added}, \quad Q<0 \text{ removed}. \tag{275}$$

It is reported as `DUTY (result)` and, on the flowsheet, appears as a heat *stream* (a utility stub + dashed wire), never as a number painted on the node. Choupo further attributes the duty to a *latent* share — the part tied to the change in vapour fraction at saturation — and a *sensible* remainder:

$$Q_{\mathrm{latent}} \;=\; F\,\big(\mathrm{vf}_{\mathrm{out}} - \mathrm{vf}_{\mathrm{in}}\big)\,(h_g - h_f), \qquad Q_{\mathrm{sensible}} \;=\; Q - Q_{\mathrm{latent}}, \tag{276}$$

with $h_g - h_f$ the latent heat from (272) at $T^{\mathrm{sat}}$.

## 5.6 Boiler and condenser are emergent labels

There is *one* class. After convergence Choupo reads the sign of $Q$ and the outlet vapour fraction and prints a *role*: a positive duty landing on saturated vapour is a `boiler`, partway up the dome a `partial boiler`; a negative duty landing on saturated liquid is a `condenser`, partway down a `partial condenser`; a duty that stays single-phase is just a heater/cooler. The aliases `boiler` and `condenser` register the *same* class — the direction is emergent from the physics, not a model switch, so a case that mis-labels a condensing duty as a `boiler` still computes the correct (negative) $Q$ and says so.

**Worked example 5.1.** Saturated-vapour boiler on a pure domeFeed: saturated-or-subcooled water at $P = 1\,\text{bar}$, $F = 1\,\text{kmol}\,\text{s}^{-1}$, with `outletState saturatedVapour;`. The engine finds $T^{\text{sat}}(1\,\text{bar}) \approx 372.8\,\text{K}$ from the pure-fluid kernel, sets $T_{\text{out}} = T^{\text{sat}}$ and forces $\text{vf}_{\text{out}} = 1$ (vf is the free coordinate on the plateau). With $H_{\text{out}} = h_g$ and $H_{\text{in}} = h_f$ the latent heat $h_g - h_f \approx 40.7\,\text{kJ}\,\text{mol}^{-1}$ gives $Q = F(h_g - h_f) \approx 1000 \cdot 40.7 = 4.07 \times 10^4\,\text{kW}$, reported as a positive duty, role `boiler`, $Q_{\text{latent}} \approx Q$ and $Q_{\text{sensible}} \approx 0$. Re-specify `outletQuality 0.5;` and the same plateau gives half the latent duty.

**Power and limit.** **Power:** one class spans every dome-crossing service with a single, honest degree of freedom; the duty is always a visible result; the latent/sensible split and the boiler/condenser role come for free from the converged state; and the same code path serves both the generic cubic-EoS + Antoine route and the high-accuracy pure-fluid (IF97) route, the dispatch hidden inside the thermo. **Limit:** the saturation dome is located from the *dominant* component, so the single $T^{\text{sat}}$ used for the latent split is a pure/dominant-component approximation — for a wide multicomponent dome it is only the bracket centre, and a strongly non-ideal mixture's true bubble-to-dew width is captured by the flash but not by that one plateau temperature. The unit is a thermodynamic phase-change *specifier*, not a rating model: it does not size area or compute a real heat-transfer coefficient unless you switch to `model geometry;` (the Nusselt film-condensation / Rohsenow nucleate-boiling branch with a Zuber CHF ceiling), documented in the heat-transfer rating material.

*Primary references:* the saturation curve and Clausius–Clapeyron latent heat (Sec. 2); Smith, Van Ness & Abbott, *Introduction to Chemical Engineering Thermodynamics*, on the lever rule and dome enthalpies; IAPWS R7-97 (IF97)[59] for the pure-fluid $T^{\text{sat}}(P)$ kernel; for the geometry branch, Nusselt (1916)[60] film condensation and Rohsenow (1952)[61] / Zuber (1959)[62] nucleate boiling and critical heat flux.

# 6 Continuous stirred-tank reactor (CSTR)

*What you'll learn.* A single-reaction steady-state CSTR collapses to a one-dimensional Newton solve in the reaction extent $\xi$. The simulator does exactly that.

## 6.1 Material balance with reaction extent

For one independent reaction with stoichiometry $\boldsymbol{\nu}$ and rate $r(\mathbf{x}, T)$ (units mol/(m$^3$ s)), the steady-state component balance over the reactor is

$$F_{\text{in}}\,\mathbf{z}_{\text{in}} \;=\; F_{\text{out}}\,\mathbf{z}_{\text{out}} \;-\; V_R\,\boldsymbol{\nu}\,r(\mathbf{z}_{\text{out}}, T). \tag{277}$$

Using the extent of reaction $\xi$ (mol/h) we write $\mathbf{z}_{\text{out}} F_{\text{out}} = \mathbf{z}_{\text{in}} F_{\text{in}} + \boldsymbol{\nu}\,\xi$. Eliminating $\mathbf{z}_{\text{out}}$:

$$g(\xi) \;\equiv\; \xi - V_R\,r\big(\mathbf{z}_{\text{out}}(\xi), T\big) \;=\; 0. \tag{278}$$

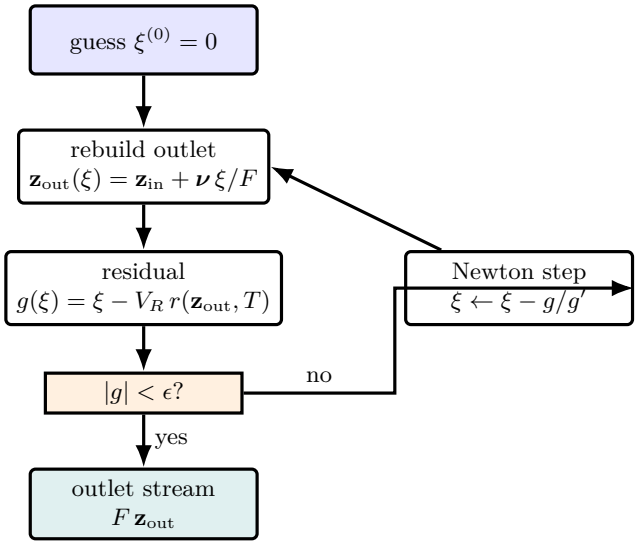



**Figure 53:** The steady CSTR is *not* a big system — it collapses to one scalar root-find. The single unknown is the reaction extent $\xi$; every outlet mole fraction is rebuilt from it through stoichiometry $\mathbf{z}_{\text{out}} = \mathbf{z}_{\text{in}} + \boldsymbol{\nu}\,\xi/F$, the rate $r(\mathbf{z}_{\text{out}}, T)$ is evaluated there, and the residual $g(\xi) = \xi - V_R r$ measures how far the guessed extent is from production-equals-consumption. A damped **Newton-1D** (right loop) drives $g \to 0$ in 3–5 iterations. The whole reactor is one equation in one number.

A single non-linear equation in a single unknown $\xi$: **Newton-1D in one variable**, with damping. Convergence is typically 3–5 iterations.

## 6.2 First-order kinetics — analytic check

For an Arrhenius first-order reaction $r = k\,c_A$ with $k = k_0\,e^{-E_a/R_g T}$, the steady-state CSTR has the analytic solution

$$X_A \;\equiv\; 1 - \frac{c_A}{c_{A,0}} \;=\; \frac{\text{Da}}{1 + \text{Da}}, \qquad \text{Da} = k\,\tau, \tag{279}$$

where $\tau = V_R/Q$ is the residence time. The simulator reproduces this to machine precision in `tutorials/steady/reactors/cstr01_fi`

## 6.3 Worked example

**Worked example 6.1.** First-order $A \to B$ in a 0.5 m$^3$ CSTRInlet: $c_{A,0} = 100$ mol/m$^3$, volumetric flow $Q = 0.01$ m$^3$/s, kinetic constant $k = 0.02$ s$^{-1}$. Then $\tau = V_R/Q = 50$ s, Da $= 1.0$, $X_A = 0.5$. Newton starting from $\xi^{(0)} = 0$ converges in 4 iterations to $\xi = 0.5\,c_{A,0}\,Q = 0.5$ mol/s.

**Runnable case.** `tutorials/steady/reactors/cstr01_first_order` — includes the analytic check in the case output.

# 7 The rate law: from mass action to Langmuir–Hinshelwood

*What you'll learn.* Why a power law cannot describe a catalytic reaction at any order, where the denominator of an LHHW rate law comes from, and why a rate constant means nothing without the rate law and the activity model it was regressed with.

Every reactor so far has been handed a rate $r(\mathbf{c}, T)$ and asked what to do with it: a CSTR sets $\xi = rV_R$, a PFR integrates it along the volume, a batch vessel integrates it in time. None of them cares what $r$ is. This section is about $r$ itself, and in Choupo it is one object — the shared `RateLaw` — so a batch vessel and a continuous one cannot disagree about it.

## 7.1 Where mass action stops

For a homogeneous elementary step the rate is a product of concentrations raised to the stoichiometric orders: mass action, the law of a well-mixed molecular collision. A *catalytic* reaction is not that. The molecules do not meet in the fluid; they meet on a surface, and the surface has a finite number of sites.

Follow Langmuir. Let $\theta_i$ be the fraction of sites carrying species $i$ and $\theta_v$ the fraction left vacant. Adsorption and desorption of $i$ balance at equilibrium,

$$k_{a,i}\, c_i\, \theta_v \;=\; k_{d,i}\, \theta_i \qquad \Longrightarrow \qquad \theta_i \;=\; K_i\, c_i\, \theta_v,$$

with $K_i = k_{a,i}/k_{d,i}$ the *adsorption equilibrium constant* of $i$. Every site is either vacant or occupied, so $\theta_v + \sum_i \theta_i = 1$ and therefore

$$\theta_v \;=\; \frac{1}{1 + \sum_i K_i c_i}, \qquad \theta_i \;=\; \frac{K_i c_i}{1 + \sum_i K_i c_i}. \tag{280}$$

Now let the rate-determining step be a surface reaction between adsorbed species. Its rate is proportional to the coverages of the reactants — not to their bulk concentrations — so for a bimolecular step $A_{\mathrm{ads}} + B_{\mathrm{ads}}$ occupying two sites,

$$r \;=\; k\, \theta_A\, \theta_B \;=\; \frac{k\, K_A c_A\, K_B c_B}{\left(1 + \sum_i K_i c_i\right)^2}. \tag{281}$$

That denominator is the whole point, and no power law contains it. Look at what it does. A *product* that adsorbs strongly appears in $\sum_i K_i c_i$ but not in the numerator: as the reaction makes it, it covers the surface, crowds the reactants off, and **throttles the reaction that produced it**. This is product inhibition. You cannot reproduce it by tuning the order of a power law, because a power law's dependence on a product's concentration is monotone in one direction only. Equation (281) is the Langmuir–Hinshelwood–Hougen–Watson (LHHW) form [63, 64, 65].

## 7.2 The one formula Choupo evaluates

Choupo does not carry a taxonomy of LHHW shapes. It carries one expression, and the shapes are what you get by choosing its parts. Write $\theta_i$ for the intensive variable of species $i$ — a concentration, or an *activity* $a_i = \gamma_i x_i$ when `basis activity` — and weight it by its adsorption constant to get the species *as the surface sees it*,

$$\bar{a}_i \;=\; K_i(T)\, \theta_i, \qquad K_i(T) = K_i^0\, \mathrm{e}^{B_i/T}$$

($B_i$ the van 't Hoff slope; $K_i = 1$ for a species that declares no adsorption). Then

$$\boxed{\; r \;=\; \frac{k_f \prod_i \bar{a}_i^{\,\alpha_i} \;-\; k_r \prod_i \bar{a}_i^{\,\beta_i}}{\left(v + \sum_{i \in \mathrm{ads}} \bar{a}_i\right)^p} \;} \tag{282}$$

Three dials, and every rate law in this guide is a setting of them.

- $p = 0$ (no `adsorption` block): the denominator is 1 and what remains is the plain power law, `type Arrhenius`. Mass action.
- $p > 0$: LHHW. The exponent is the number of sites the rate-determining step occupies — $p = 2$ in Eq. (281).
- $v$ is the vacant-site term. It is 1 in the classical Hougen–Watson forms, and 0 where the surface is taken as saturated — an ion-exchange resin swollen with liquid, say, on which every site holds *something* and only the competition matters.

Note that $\bar{a}_i$ sits in the numerator too. That is not a convention: the rate-determining surface step sees the *adsorbed* species, as Eq. (281) shows. Any constant $K_i$ can of course be absorbed into $k_f$, which is exactly why textbook LHHW forms look different from one another — and it is the first hint of the pitfall below.

The reverse leg is either **detailed balance** (`reversible true`: $k_r = k_f/K_c$, so equilibrium sits precisely where the thermodynamics puts it, and cannot drift with a fitted number), or an **explicitly regressed** Arrhenius pair $(k_r^0, E_{a,r})$ — which is what a kinetics paper reports. Choupo refuses both at once, and refuses detailed balance on an activity basis, since $K_c$ is a concentration-basis constant.

## 7.3 Per gram of catalyst

A heterogeneous rate constant is reported per gram of dry catalyst, and a reactor is measured in cubic metres. Declare the bulk loading $\rho_{\mathrm{cat}}$ and the conversion is exact:

$$r\left[\tfrac{\mathrm{kmol}}{\mathrm{m^3\,s}}\right] \;=\; \rho_{\mathrm{cat}}\left[\tfrac{\mathrm{kg}}{\mathrm{m^3}}\right]\cdot r\left[\tfrac{\mathrm{mol}}{\mathrm{g\,s}}\right].$$

It is a declared, printed number — change the packing and the reactor changes with it.

## 7.4 A worked, cited example, and a trap

Methyl-acetate synthesis over the ion-exchange resin Amberlyst 15, $\mathrm{CH_3OH + CH_3COOH \rightleftharpoons CH_3COOCH_3 + H_2O}$, follows Eq. 16 of Pöpken, Götze and Gmehling [66] — the adsorption-based model the authors recommend (mean relative error 5.4 %, against 13.9 % for the pseudo-homogeneous power law on the same data). It is Eq. (282) with $v = 0$, $p = 2$, activities from UNIQUAC, and $\bar{a}_i = K_i a_i / M_i$.

Water and methanol adsorb hardest ($K_i/M_i = 0.29$ and 0.18, against 0.052 for the acid). So the water the reaction *makes* crowds the acid off the resin. That is why reactive distillation, which pulls the water out as it forms, pays for its capital — and here it is a line in a dictionary rather than a sentence in a textbook.

> **Common pitfall.** Do not try to see the inhibition by deleting the denominator and keeping $k_f$. It shows nothing. In this parameterisation the reaction gets *faster*, because the adsorbed activities are small numbers and dividing by their square is a multiplication. That is an arithmetic accident, not physics: the constants were regressed *for* that rate law, on *those* activities. **A rate constant without its rate law and its activity model is a number without a meaning.** Swap UNIQUAC for UNIFAC in `cstr07_lhhw_methylAcetate` and $k_1 = 8.497 \times 10^6$ stops being the authors' $k_1$.
> To *see* the inhibition, hold the reactants fixed and add the product. In `batch07_lhhw_methylAcetate`, charging water at unchanged acid and alcohol fractions halves the initial rate, from 58.3 to 29.0 mmol/(L s). The water is not a reactant. It is sitting on the sites.

Three reactors evaluate the same Eq. (282). The steady CSTR (`cstr07`) runs kinetically limited at $X = 49\,\%$; grow its volume and it climbs to 76 %. The batch vessel (`batch07`) integrates to $X = 76.6\,\%$ and stops. That plateau is asserted nowhere: it is where the numerator of Eq. (282) vanishes, i.e. the activity-based $K_a = 25.50$ that the authors' *independent* Eq. 17 recovers from the same eight constants. Two reactors, two integration schemes, one equilibrium — and a cheap, honest check that the parameters were entered right.

> **Runnable case.** `cstr07_lhhw_methylAcetate` (steady, activity basis) · `batch07_lhhw_methylAcetate` (the same law integrated in time, plus the water-spike experiment) · `ctrl07_lhhw_inhibition` (a PID loop against a reaction whose exotherm fades as its own product covers the catalyst).

# 8 Plug-flow reactor (PFR)

*What you'll learn.* The CSTR's continuous-flow counterpart with *no* backmixing. A single differential balance integrated along the reactor volume by the fourth-order Runge-Kutta scheme of Chapter 9. For first-order kinetics it admits the textbook closed-form $X = 1 - e^{-k\tau}$ that the simulator's tutorial reproduces.

## 8.1 The differential material balance

A PFR is a tube in which the molar flow of each component changes continuously with the cumulative volume $V$ ($0 \le V \le V_R$). For a single liquid-phase reaction $\sum_i \nu_i A_i = 0$ proceeding at rate $r(\mathbf{x}, T)$ (mol per unit volume per unit time), a shell balance on a differential slab of volume $\mathrm{d}V$ yields

$$\boxed{\frac{\mathrm{d}F_i}{\mathrm{d}V} \;=\; \nu_i\, r(\mathbf{x}, T), \qquad i = 1, \ldots, N_c,} \tag{283}$$

where $F_i = F_i(V)$ is the molar flow of $i$, and $x_i = F_i / \sum_k F_k$ rebuilds the mole fractions inside $r$ at each volume. Equation (283) is an autonomous, first-order ODE system in $V$; the inlet condition $F_i(0) = F_{i,0}$ closes the problem.

*Intuition.* The CSTR is a single zero-dimensional point where everything is at the outlet conditions. The PFR is a one-dimensional continuum where conditions evolve from inlet to outlet — mathematically a *plug* of fluid sliding down the tube without mixing axially. The CSTR's algebraic equation becomes the PFR's ODE because the time the plug spends between $V$ and $V + \mathrm{d}V$ is $\mathrm{d}V/Q$.

## 8.2 Algorithm: RK4 integration

The simulator integrates (283) from $V = 0$ to $V_R$ with the classical fourth-order Runge-Kutta scheme of §9. With `nSteps` sub-intervals (default 100) the step size is $h = V_R/\texttt{nSteps}$ and the local truncation error is $O(h^5)$ — amply sufficient for any pedagogical kinetic. Per-step cost: four evaluations of $r(\mathbf{x}, T)$, which dominates because each evaluation calls the kinetics object's `rate()` method.

The reactor outlet $\mathbf{F}(V_R)$ feeds whatever is downstream in `flowsheetDict`, exactly like a CSTR outlet — the unit-op contract does not distinguish.

## 8.3 First-order kinetics: the analytic check

For an irreversible first-order reaction $A \to B$ at constant $T$, $r = k\, c_A = k\, F_A/Q$, and (283) reduces (for the limiting reactant $A$) to

$$\frac{\mathrm{d}F_A}{\mathrm{d}V} \;=\; -\,k\,\frac{F_A}{Q}. \tag{284}$$

Integrating,

$$\boxed{X_A \;\equiv\; 1 - \frac{F_A(V_R)}{F_{A,0}} \;=\; 1 - e^{-k\,\tau},} \tag{285}$$

with the residence time $\tau = V_R/Q$. The PFR achieves higher conversion than a CSTR with the same $\tau$: at $k\tau = 1$, $X^{\mathrm{PFR}} = 1 - e^{-1} = 0.632$ versus $X^{\mathrm{CSTR}} = 1/(1 + k\tau) = 0.500$ — the canonical "no backmixing wastes less feed at the front of the reactor"

result.

## 8.4 Worked example

**Worked example 8.1.** Same kinetics as `cstr01_first_order`, in a PFR$k\tau = 1.10$. The PFR conversion from (285) is $X = 1 - e^{-1.10} = 0.667$. The CSTR of the same volume gives $X = 1/(1 + 1.10) = 0.476$. The simulator's RK4 integration (100 steps) returns $X = 0.667$ to six digits — the difference between CSTR and PFR for this kinetic is exactly the backmixing effect.

**Runnable case.** `tutorials/steady/reactors/pfr01_first_order` — prints conversion every `writeInterval` steps along the axial coordinate; the final $X = 0.667$ matches (285) to RK4 truncation error.

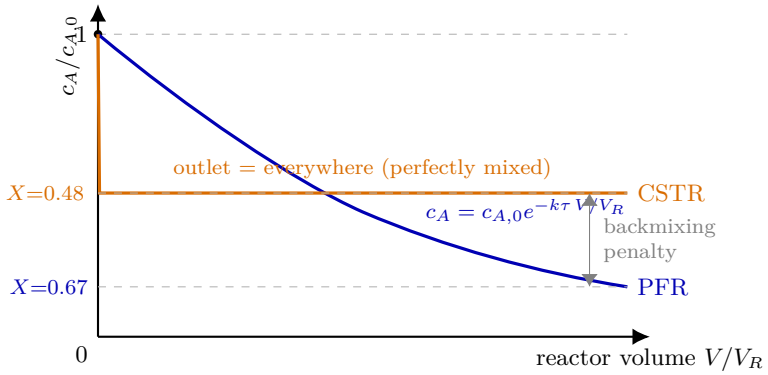



**Figure 54:** Same first-order kinetics, same residence time $\tau$ ($k\tau = 1.1$), two reactor idealisations. The **PFR** (blue) is a plug sliding down the tube: $c_A$ falls continuously from inlet to outlet, so the reaction runs at high concentration near the front — $X = 1 - e^{-k\tau} = 0.67$. The **CSTR** (orange) mixes the feed *instantly* to the outlet state, so the *entire* vessel reacts at the low outlet concentration: the residual horizontal line is the whole reactor at once, $X = k\tau/(1 + k\tau) = 0.48$. The shaded gap is the price of backmixing — the same volume converts less feed when it is stirred to uniformity.

# 9 The catalyst pellet: the Thiele modulus and the effectiveness factor

*What you'll learn.* Every reactor chapter so far has evaluated the rate at the conditions of the fluid *outside* the catalyst. Here we open the pellet. Diffusion and reaction compete inside it, the reactant never reaches the centre at full strength, and the observed rate is the intrinsic rate multiplied by a number between 0 and 1: the effectiveness factor $\eta$. You will learn where the Thiele modulus comes from (it is a ratio of two rates, not a fitted group), why the three pellet shapes nearly collapse onto one curve when the modulus is built on the right length, what the two asymptotes mean physically, and how the Weisz–Prater criterion lets you decide from *measurable* quantities alone whether any of this matters in your reactor.

## 9.1 The balance inside one pellet

Take one porous pellet, isothermal, sitting in a fluid whose concentration of the reactant is $c_s$ at the pellet's external surface. Inside the pellet the reactant must diffuse through the pore network while it is being consumed. A shell balance on a differential element gives, for a first-order reaction and an effective diffusivity $D_e$ that does not vary with position,

$$D_e \left( \frac{\mathrm{d}^2 c}{\mathrm{d}r^2} + \frac{s}{r}\frac{\mathrm{d}c}{\mathrm{d}r} \right) \;=\; k\,c, \tag{286}$$

where $r$ is measured from the centre and the *shape factor* $s$ is 0 for a slab sealed on its edges, 1 for a long cylinder and 2 for a sphere. It is worth noticing that the shape enters as one integer and nothing else.

Two boundary conditions close it, and both are physical rather than mathematical: the pellet is symmetric about its centre, so no net flux crosses it,

$$\left.\frac{\mathrm{d}c}{\mathrm{d}r}\right|_{r=0} = 0,$$

and the surface is in contact with the fluid, so

$$c(R) = c_s.$$

The second one carries an assumption that is easy to forget: it says the *external* film offers no resistance. If it does, $c_s$ is not the bulk concentration and a second resistance sits in series with everything below.

## 9.2 The modulus is a ratio of rates

Normalise: let $u = c/c_s$ and $\xi = r/R$, with $R$ the characteristic dimension — the half-thickness $L$ of a slab, the radius of a cylinder or a sphere. Equation (286) becomes

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\xi^2} + \frac{s}{\xi}\frac{\mathrm{d}u}{\mathrm{d}\xi} \;=\; \phi^2\, u, \qquad \left.\frac{\mathrm{d}u}{\mathrm{d}\xi}\right|_{0} = 0, \quad u(1) = 1, \tag{287}$$

and *one* group survives:

$$\phi \;=\; R\sqrt{\frac{k}{D_e}}. \tag{288}$$

This is the Thiele modulus [67], and it is not an empirical correlation group. Square it and read it as

$$\phi^2 \;=\; \frac{k\,c_s\,R^2}{D_e\,c_s} \;=\; \frac{\text{rate of reaction at surface conditions}}{\text{rate of diffusion into the pellet}}.$$

Small $\phi$: diffusion wins the race and the pellet is uniform. Large $\phi$: the reaction consumes the reactant before it has travelled far, and the interior sits idle.

**Key idea.** The Thiele modulus compares two *rates*, and both of them are rates the pellet would exhibit if the other did not exist. That is why it is dimensionless without anyone making it so, and why a pellet twice as large has four times the squared modulus.

## 9.3 The three closed forms

Equation (287) is linear, so it can be solved in closed form for each shape. The effectiveness factor is defined as

$$\eta \;=\; \frac{\text{observed rate over the whole pellet}}{\text{rate the whole pellet would show at } c_s} \;=\; \frac{1}{V_p}\int_{V_p} u\,\mathrm{d}V \;=\; (s+1)\int_0^1 \xi^{\,s} u(\xi)\,\mathrm{d}\xi, \tag{289}$$

and the three results are

$$\text{slab:}\quad u = \frac{\cosh(\phi\,\xi)}{\cosh\phi}, \qquad \eta = \frac{\tanh\phi}{\phi}, \tag{290}$$

$$\text{cylinder:}\quad u = \frac{I_0(\phi\,\xi)}{I_0(\phi)}, \qquad \eta = \frac{2\,I_1(\phi)}{\phi\,I_0(\phi)}, \tag{291}$$

$$\text{sphere:}\quad u = \frac{\sinh(\phi\,\xi)}{\xi\,\sinh\phi}, \qquad \eta = \frac{3}{\phi}\left(\frac{1}{\tanh\phi} - \frac{1}{\phi}\right), \tag{292}$$

with $I_0$ and $I_1$ the modified Bessel functions of the first kind. The sphere's field has a removable singularity at the centre, where $u(0) = \phi/\sinh\phi$; it is a finite number, and it is the concentration a measurement at the pellet's core would find.

There is a second, equivalent route to $\eta$ that is worth carrying because it is an independent check: everything that reacts inside must have entered through the surface, so

$$\eta \;=\; \frac{(s+1)}{\phi^2}\,\left.\frac{\mathrm{d}u}{\mathrm{d}\xi}\right|_{\xi=1}. \tag{293}$$

The volume integral and the surface gradient are different functionals of the same field. If they disagree, the field is not a solution of the equation it was meant to solve — which is exactly how Choupo checks its own numerical answer.

## 9.4 One length that almost erases the shape

Comparing Eqs. (290)–(292) at the same $\phi$ is misleading, because $R$ means a different thing in each. Aris [68] showed that the right length is the pellet's volume-to-surface ratio

$$\Lambda \;=\; \frac{V_p}{S_p} \;=\; \begin{cases} L & \text{slab (half-thickness } L\text{)},\\ R/2 & \text{cylinder},\\ R/3 & \text{sphere}, \end{cases} \qquad \phi_\Lambda \;=\; \Lambda\sqrt{\frac{k}{D_e}}. \tag{294}$$

On $\phi_\Lambda$ the three curves lie almost on top of one another over the whole range — within a few per cent everywhere. This is the modulus most textbooks and every Choupo message mean when they write $\phi$ without qualification; the sphere's result reads

$$\eta \;=\; \frac{1}{\phi_\Lambda}\left(\frac{1}{\tanh 3\phi_\Lambda} - \frac{1}{3\phi_\Lambda}\right),$$

which is Eq. (292) with $\phi = 3\phi_\Lambda$.

> **Common pitfall. Two moduli, one symbol.** A value of $\phi$ quoted without saying which length it was built on is a factor of three away from the other convention for a sphere. Choupo publishes both and names them: `phi_char` is Eq. (288) on the characteristic dimension, `phi` is Eq. (294) on $\Lambda$. When you compare a Choupo number with a textbook one, check which is which before concluding anything.

## 9.5 The two asymptotes, and what they say about design

Expand Eq. (292) at small and large argument:

$$\eta \;\longrightarrow\; 1 - \frac{3\,\phi_\Lambda^2}{5} + \dots \quad (\phi_\Lambda \ll 1), \qquad \eta \;\longrightarrow\; \frac{1}{\phi_\Lambda} \quad (\phi_\Lambda \gg 1). \tag{295}$$

The first says a small enough pellet is fully used, and that the penalty for growing it is second order at first — which is why a modest increase in pellet size to cut pressure drop is often free. The second is more consequential. In the strong-diffusion limit $\eta \propto 1/\Lambda$, so the *observed* rate per unit volume of bed becomes

$$\eta\,k \;\propto\; \frac{1}{\Lambda}\sqrt{kD_e}\,,$$

i.e. it depends on $\sqrt{k}$ and not on $k$. Two things follow, and both are laboratory experience: the apparent activation energy is *half* the true one, because $k$ enters under a square root; and doubling the catalyst's intrinsic activity buys only a factor of $\sqrt{2}$ until the pellet is made smaller.

> *Intuition.* In the strong-diffusion regime the reaction lives in a shell of thickness about $\Lambda/\phi_\Lambda$ under the surface, and the rest of the pellet is inert ballast. Everything the asymptote says follows from that picture: the rate is proportional to the surface area of the shell, so it scales with $1/\Lambda$; and the shell thickness itself is set by how far a molecule diffuses before reacting, $\sqrt{D_e/k}$, which is where the square root comes from.

## 9.6 $\eta$ multiplies the rate — and the size of the error

Wherever a reactor model writes a heterogeneous rate, the honest form is

$$r_{\text{observed}} \;=\; \eta\, r(c_{\text{bulk}}, T). \tag{296}$$

Taking $\eta = 1$ is a modelling assumption, not a neutral default. At $\phi_\Lambda = 5$ the sphere gives $\eta = 0.1867$: a bed sized as though the reactant reached every pellet's centre would be undersized by more than a factor of five. Choupo's four catalytic reactors do currently evaluate the rate at bulk conditions, and each of them says so out loud rather than leaving it to be discovered (Section 7); the pellet calculation of this chapter is what will eventually supply the multiplier.

## 9.7 The Weisz–Prater criterion: deciding without knowing $k$

There is a catch in everything above. The Thiele modulus contains the *intrinsic* rate constant $k$ — and if the pellet is diffusion-limited, $k$ is precisely the quantity the experiment cannot measure. Weisz and Prater [69] removed the circularity by forming a group out of observable quantities only:

$$\Phi_{\text{WP}} \;=\; \eta\,\phi_\Lambda^2 \;=\; \frac{r_{\text{observed}}\,\Lambda^2}{D_e\, c_s}. \tag{297}$$

The middle expression contains $k$; the right-hand one does not. Every quantity in it — the measured rate per unit pellet volume, the pellet's volume-to-surface length, the effective diffusivity, the surface concentration — is available without ever knowing the intrinsic kinetics. That is what makes it a *criterion* rather than a result: it is computed after the experiment, from the experiment.

The conventional readings are $\Phi_{\text{WP}} \ll 1$ for negligible internal resistance and $\Phi_{\text{WP}} \gg 1$ for severe limitation, with the usual first-order thresholds quoted as 0.3 and 6. Choupo publishes $\Phi_{\text{WP}}$ and does not apply a threshold to it, for the same reason it announces an extrapolated vapour pressure rather than rejecting it: what counts as negligible depends on what the number is going to be used for, and that is the reader's judgement, not the simulator's.

## 9.8 What is deliberately not modelled

- **The non-isothermal pellet.** Release heat inside a particle with a finite thermal conductivity and the interior runs hotter than the surface; the Arrhenius factor then feeds back into Eq. (286) through the Prater number, $\eta$ can exceed 1, and for some parameter ranges *three* steady states satisfy the same boundary-value problem at one $\phi$. A Newton iteration returns whichever of the three the initial guess was nearest, which is a plausible number arrived at silently — so this case is named rather than approximated. The multiplicity lesson is carried, one scale up, by the Van Heerden construction for the CSTR.
- **Orders other than first.** Equation (287) becomes nonlinear, $u$ can reach zero at a finite depth, and no closed form exists to check a numerical answer against.
- **Knudsen diffusion.** $D_e$ is obtained here as $(\varepsilon_p/\tau)\, D_{\text{molecular}}$. In pores comparable with the mean free path a molecule collides with the wall more often than with another molecule, and the bulk value overstates the transport.
- **The external film, and a distribution of pellet sizes.** One pellet, one size, $c_s$ equal to the bulk value.

**Common pitfall.** **$D_e$ is not a property of the catalyst.** It belongs to a *species* diffusing in a *particular* pellet at a *particular* temperature. Choupo therefore refuses to store it on the catalyst record, where it would silently stand for every species in the mixture; the pellet carries $\varepsilon_p$ and $\tau$, the fluid model supplies $D_{\text{molecular}}$, and the product is derived and announced at the point of use. There is no default tortuosity: values between 2 and 7 are all conventional, and that spread alone moves $\eta$ by a factor of three.

**Runnable case.** `tutorials/props/reactor/thiele01_sphere_first_order` — solves Eq. (287) numerically by central differences on the same Newton and LU routines used everywhere else, prints the field beside Eqs. (290)–(292) node by node, reports $\eta$ both ways (Eqs. (289) and (293)), sweeps the $\eta(\phi_\Lambda)$ family over all three shapes, and measures its own order of convergence. The pellet in it is a declared teaching surrogate: what the case verifies is the numerics against the closed form, not the catalyst.

# 10 Single-effect evaporator: the credo case study

*What you'll learn.* The evaporator is the unit operation where Choupo's credo earned its scars. Three iterations of the unit op design (hardware- spec F-spec Mode-2) ended in a credo-pure interface that is now the template every other unit op is being aligned to. The credo: *the operation block holds only hardware parameters; the operating state of the vessel (pressure, boiling temperature, vapour fraction) is computed by the simulator from the heat balance and the saturation curve.* Any design question that the textbook poses — "find the area", "find the steam flow", "find the intermediate pressures" — is handled by an outer `DesignSpec`, never by switching modes inside the unit op.

## 10.1 What the evaporator solves

Inputs (ports):

- One **feed** liquid stream (process side): $F_{\text{feed}}, T_{\text{feed}}, P_{\text{feed}}, \mathbf{z}_{\text{feed}}$.
- One **heating steam** stream (chest side): $F_{\text{steam}}, T_{\text{steam}}, P_{\text{steam}}, \mathbf{z}_{\text{steam}}$. The chest-side $T$ is the saturation temperature at $P_{\text{steam}}$ and is *written* into the stream file by the author: the engine no longer inverts the saturation curve at import (§2), so the number is yours to compute and to keep consistent with $P$.

Parameters (operation block):

- **area** — heat-exchange surface $A$ [$\text{m}^2$]
- **U** — overall heat-transfer coefficient [$\text{W}/(\text{m}^2\,\text{K})$]
- **Tref** (optional) — reference temperature for the sensible-enthalpy integrals.

Conspicuously *absent* from the operation block: the operating pressure $P_{\text{op}}$, the boiling temperature $T_{\text{boil}}$, the vapour fraction $V/F$, the duty $Q$, the steam mass demand — everything else. Those are all **computed**.

## 10.2 The Mode-2 equations

The unit op solves four equations in this order, each carrying

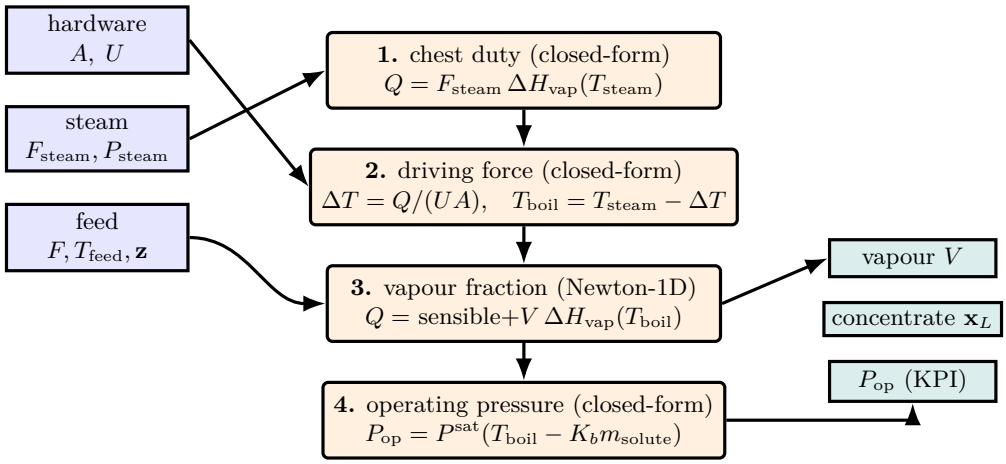



**Figure 55:** The credo evaporator solves four equations *in order*, and the operating pressure $P_{\text{op}}$ — which the student never writes — falls out at the end. Only hardware $(A, U)$ and the streams enter (blue). **1.** The condensing steam fixes the duty $Q$. **2.** The hardware turns $Q$ into a driving force $\Delta T$ and hence the boiling temperature $T_{\text{boil}}$, set by chest and geometry *alone*. **3.** A single Newton-1D closes the process-side energy balance for the vapour flow $V$. **4.** The Antoine curve, corrected for boiling-point elevation $K_b m$, inverts $T_{\text{boil}}$ back to the pressure the vessel must sit at. No mode switch, no $P_{\text{op}}$ in the dict — the crown jewel of the credo is a computed output, not an input.

its own physical meaning, none of them touching the user dict:

**1. Chest duty (closed-form).** The heating steam supplies its full latent heat as it condenses on the chest tubes:

$$Q \;=\; F_{\text{steam}} \cdot \Delta H_{\text{vap, solv}}(T_{\text{steam}}). \tag{298}$$

**2. Driving force from hardware (closed-form).** The geometry of the heat exchanger gives the LMTD-equivalent driving force at the achievable boiling point:

$$\Delta T \;=\; \frac{Q}{U \cdot A}, \qquad T_{\text{boil}} \;=\; T_{\text{steam}} \;-\; \Delta T. \tag{299}$$

Note that $T_{\text{boil}}$ is independent of the process-side composition in Mode 2 — it is fixed by chest + hardware alone.

**3. Vapour fraction from energy balance (Newton-1D).** The duty must close the process-side energy balance: heating the feed sensibly from $T_{\text{feed}}$ to $T_{\text{boil}}$, plus vaporising the produced solvent at $T_{\text{boil}}$:

$$Q \;=\; F_{\text{feed}} \sum_i z_i \left[ H_i^{l}(T_{\text{boil}}) - H_i^{l}(T_{\text{feed}}) \right] \;+\; V \cdot \Delta H_{\text{vap, solv}}(T_{\text{boil}}). \tag{300}$$

Solving for $V$ (the vapour molar flow) is one Newton-1D iteration: the residual is monotone in $V/F$ at fixed $T_{\text{boil}}$ and $Q$, so the bracket $[0, 0.999]$ is safe.

**4. Operating pressure from saturation (closed-form).** $T_{\text{boil}}$ is the boiling point of the liquid at the vessel's pressure, corrected for the boiling-point elevation caused by the non-volatile solute:

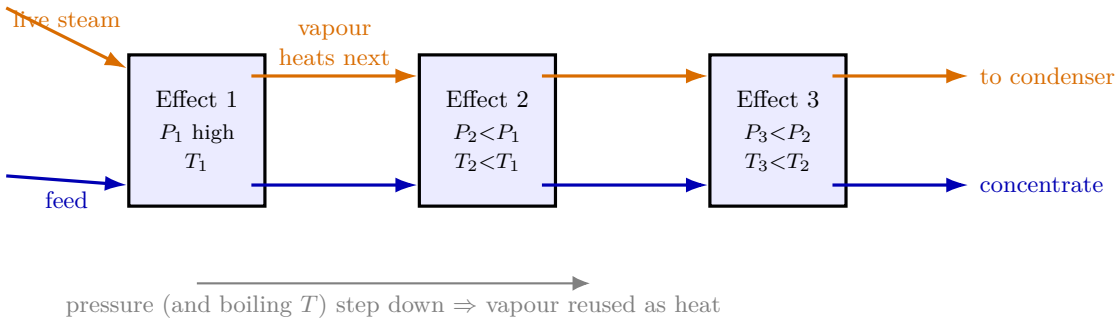



**Figure 56:** Why evaporators are run in a train. Each effect's **vapour** (orange) becomes the heating steam in the *next* effect's chest, so one unit of live steam boils off water roughly $N$ times across $N$ effects — the steam economy. This works only because the pressure (and therefore the boiling temperature) *steps down* from effect to effect: the cooler next effect can be heated by the vapour the warmer one produced. The **liquor** (blue) cascades forward, growing more concentrated at each stage until it leaves as product. Each box is exactly one single-effect credo evaporator of Figure 55; Choupo builds the train by piping the vapour port of one into the steam port of the next in `flowsheetDict`.

$$T_{\text{boil}} \;=\; T_{\text{sat, pure}}(P_{\text{op}}) \;+\; K_b \cdot m_{\text{solute}}(\mathbf{x}_L), \tag{301}$$

where $m_{\text{solute}}$ is the molality of the solute in the concentrated outlet $\mathbf{x}_L$ and $K_b$ is the ebulioscopic constant. Inverting (301) gives

$$P_{\text{op}} \;=\; P_{\text{sat, pure}}\big(T_{\text{boil}} - K_b\, m_{\text{solute}}\big),$$

a single call to the solvent's Antoine equation. This is the *credo's crown jewel*: the user never wrote $P_{\text{op}}$; the simulator computes it from physics and emits it as a KPI (`evap.P`) and as the pressure stamped on both liquid and vapour output streams.

## 10.3 What *you* write vs what the simulator gives back

| Variable | In Mode 2, this is… | Where it shows up |
|---|---|---|
| $F_{\text{feed}}, T_{\text{feed}}, \mathbf{z}_{\text{feed}}$ | USER INPUT | `0/feed` |
| $P_{\text{steam}}$ | USER INPUT | `P` in `0/liveSteam` |
| `phase gas` | USER PIN | `phase` in `0/liveSteam` |
| $T_{\text{steam}}$ | USER INPUT ($= T_{\text{sat}}(P)$, computed by the author) | `T` in `0/liveSteam` |
| $F_{\text{steam}}$ | USER INPUT (often via $variable) | `F` in `0/liveSteam` |
| $A$, $U$ | USER INPUT (hardware) | `operation.{area,U}` |
| $Q$ | COMPUTED, eq. (298) | KPI `evap.duty` |
| $\Delta T$ | COMPUTED, eq. (299) | KPI `evap.dT` |
| $T_{\text{boil}}$ | COMPUTED, eq. (299) | KPI `evap.T_boil` |
| $V/F$ | COMPUTED, eq. (300) | KPI `evap.V_over_F` |
| $P_{\text{op}}$ | COMPUTED, eq. (301) | KPI `evap.P` |
| $\mathbf{x}_L$ (concentrate composition) | COMPUTED, species mass balance | `converged/productConc` |
| BPE | COMPUTED, $K_b\, m_{\text{solute}}$ | KPI `evap.BPE` |

The student writes a small handful of physically tangible quantities; everything else falls out. The same code path works for every evaporator case — there is no mode switch.

## 10.4 Why mode-switching inside the unit op was wrong

The path to this form went through two redesigns whose common mistake was offering *alternative spec modes* inside the unit op:

- hardware-spec: $A$, $U$, $P_{\text{op}}$ given, $Q$ and $F_{\text{steam}}$ computed.
- F-spec: $F_{\text{steam}}$ given, $Q$ computed from chest condensation; $A$ became a consistency report (`A_needed` KPI).
- Mode-2: $A$, $U$ given, $P_{\text{op}}$ computed from the heat balance.

The first two flavours each had a defensible physics narrative. But they violated the credo because the *operation* block contained, at different times, a vessel pressure ($P_{\text{op}}$) or a chest-flow placeholder, neither of which is hardware: $P_{\text{op}}$ is a stream property of the outputs; $F_{\text{steam}}$ is a stream property of the chest. Different design questions ("find $A$ for a given $P_{\text{op}}$", "find $P_{\text{op}}$ for a given $A$") were being answered by silently switching which variable the `solve()` was iterating on. Pedagogically, the student could not tell from the dict alone what was input and what was output.

The redesign removes the choice: *always* hardware in, *always* state computed. Every design variation is hoisted to an outer `DesignSpec` that manipulates user-level $variables — including $A$ itself when the textbook makes it the unknown.

## 10.5 The three canonical design questions, one unit op

The textbook problems map directly onto outer-driver formulations. All three use the same evaporator underneath; only the outerDict changes.

**Q1 — "Given hardware, what comes out?"** No outerDict. The cascade runs once, sequentially. Tutorial `evaporator02_triple_effect_sugar` demonstrates this for a fixed $A = 100\,\text{m}^2$ cascade.

**Q2 — "Given product spec, find the missing hardware."** 1-D or 2-D `DesignSpec`: manipulate $A$ (and possibly $F_{\text{steam}}$); target the outlet concentration (and possibly the textbook operating pressure). Tutorial `evaporator04_orange_juice_simple` is the canonical example — the dict expresses the textbook textbook Example 10.2 problem in 50 lines.

**Q3 — "Given product spec and terminal vacuum, find the multi-effect design."** $N$-D `DesignSpec` on $A$ and $F_{\text{steam}}$ with two targets (product mass + terminal $P_N$). Co-current is a DAG, no recycle; counter-current is a recycle problem with two tear streams (one per inter-effect pair, see §5). Tutorials `evaporator03_co_current` and `evaporator05_counter_current` cover both wirings.

> *Intuition.* The mental model the student should leave with: the evaporator is a black box with two stream inputs, three stream outputs, and two parameters ($A$ and $U$). It always computes the same set of unknowns from the same set of inputs. Every "what if" becomes a question for the outer DesignSpec, not a modification of the operation block. The credo enforces this single contract — and the rest of the unit ops in Choupo are being incrementally aligned to the same shape.

# 11 Liquid-liquid equilibrium by Gibbs minimisation

*What you'll learn.* Two-phase liquid-liquid equilibrium is, classically, solved by writing the equal-activity constraint $\gamma_{i,\alpha}x_{i,\alpha} = \gamma_{i,\beta}x_{i,\beta}$ for every component and feeding the resulting $2n+1$ algebraic system to Newton. For LLE landscapes with the same activity model on both phases — and water / butanol is the textbook example — this method fails: the trivial $K = 1$ solution at $\mathbf{x}_\alpha = \mathbf{x}_\beta = \mathbf{x}_z$ is a *saddle* of the Gibbs landscape but a stable fixed point of the residual map, and damped Newton from any nearby start collapses to it. Direct minimisation of the per-mole Gibbs energy on the composition simplex bypasses the saddle entirely. This is what `IsothermalFlash` does for the `phaseSet LL` branch. But before paying for that multi-start minimisation, the flash first asks a cheaper, sharper question — *is the feed even unstable?* — with Michelsen's tangent-plane-distance test, the subject of the next subsection.

## 11.1 The phase-stability pre-check: Michelsen's tangent plane

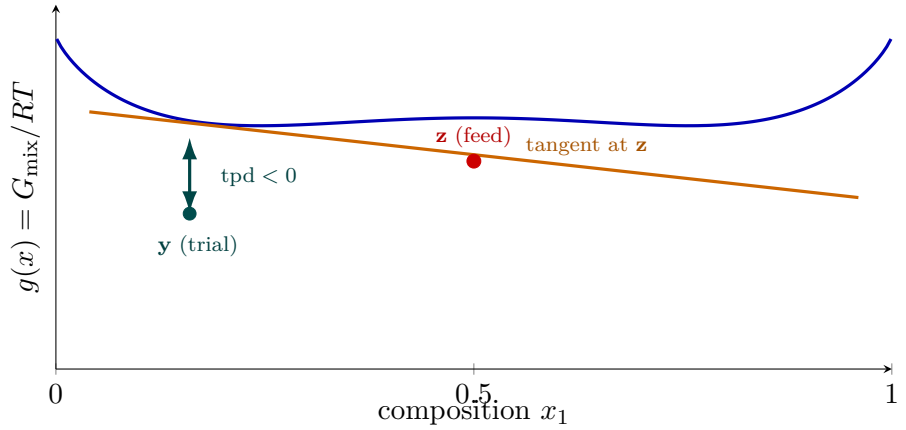



**Figure 57:** Michelsen's tangent-plane stability test, read off the molar-Gibbs-of-mixing surface $g(x)$. Draw the hyperplane tangent to $g$ at the feed $\mathbf{z}$ (orange). The tangent-plane distance $\mathrm{tpd}(\mathbf{y})$ is the vertical gap between the surface and that tangent, measured at a trial composition $\mathbf{y}$. Here the surface dips *below* the tangent near $\mathbf{y}$: $\mathrm{tpd}(\mathbf{y}) < 0$, which *proves* the single feed is unstable and a liquid–liquid split exists — and that $\mathbf{y}$ is a ready-made guess for the incipient phase. A feed is stable (one phase) iff $\mathrm{tpd} \geq 0$ everywhere. At a converged stationary point this whole picture collapses to one number, $\mathrm{tm} = 1 - \sum_i Y_i$, the quantity `michelsenTPD` actually tests.

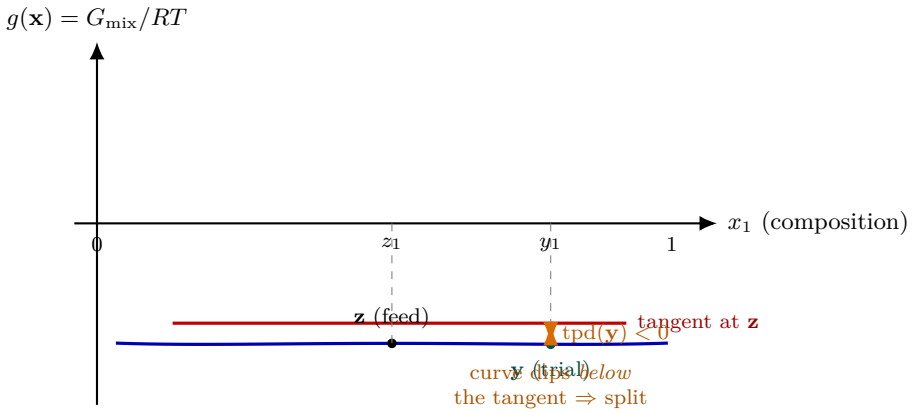



**Figure 58:** Michelsen's tangent-plane-distance test, drawn for a binary. $g(\mathbf{x}) = G_{\mathrm{mix}}/RT$ is the molar Gibbs energy of mixing; the feed sits at $\mathbf{z}$. Draw the line tangent to $g$ at $\mathbf{z}$ (red). The tangent-plane distance $\mathrm{tpd}(\mathbf{y})$ is the *vertical gap* from that tangent down to the surface at a trial composition $\mathbf{y}$. If the surface stays *above* the tangent everywhere, $\mathrm{tpd} \geq 0$ and the single feed is stable. But here the curve has a hump and dips *below* the tangent near $\mathbf{y}$: $\mathrm{tpd}(\mathbf{y}) < 0$ (equivalently $\mathrm{tm} < 0$, or $\sum_i Y_i > 1$) — one negative value proves a phase split exists, and that $\mathbf{y}$ is a ready-made guess for the incipient phase that the Gibbs-min flash then refines.

*What you'll learn.* Before any split is computed, the flash must decide *whether a split exists at all*. A single liquid is stable when no trial phase can lower the total Gibbs energy; it is unstable — a split exists — when some trial composition $\mathbf{y}$ lies *below* the tangent plane drawn to the molar-Gibbs surface at the feed. Michelsen (1982)[70] turned this geometric statement into a one-number criterion, the tangent-plane distance tm, and a cheap successive-substitution iteration that finds the deepest-cutting trial phase. `michelsenTPD` in `src/solver/StabilityTest.cpp` implements it as the gate of the `phaseSet LL` branch.

## The tangent-plane distance

Fix $T$ and $P$ and let $g(\mathbf{x}) = G_{\text{mix}}(\mathbf{x})/RT$ be the molar Gibbs energy of mixing, in units of $RT$, of a homogeneous liquid of composition $\mathbf{x}$. The feed sits at $\mathbf{x} = \mathbf{z}$. Now imagine pinching off an infinitesimal amount of a *trial* phase of composition $\mathbf{y}$ from the feed. The change in total Gibbs energy, per mole of trial phase, relative to the value the same material had *at the feed composition*, is the **tangent-plane distance**

$$\operatorname{tpd}(\mathbf{y}) \;=\; \sum_{i=1}^{n} y_i \Big[\, \mu_i(\mathbf{y}) - \mu_i(\mathbf{z}) \,\Big] / RT \;=\; \sum_{i=1}^{n} y_i \big[\, \ln\!\big(y_i\gamma_i(\mathbf{y})\big) - \ln\!\big(z_i\gamma_i(\mathbf{z})\big) \,\big], \tag{302}$$

where the second form uses the symmetric activity-coefficient expression $\mu_i/RT = \mu_i^{\circ}/RT + \ln(x_i\gamma_i)$, so the reference chemical potentials $\mu_i^{\circ}$ cancel between the two phases. Geometrically, $\operatorname{tpd}(\mathbf{y})$ is the vertical gap between the surface $g(\mathbf{y})$ and the hyperplane tangent to $g$ at $\mathbf{z}$, measured at $\mathbf{y}$. The stability statement of Baker–Pierce–Luks and Michelsen is then a single line:

> **Key idea.** The feed $\mathbf{z}$ is **stable** (one phase) if and only if $\operatorname{tpd}(\mathbf{y}) \geq 0$ for *every* trial composition $\mathbf{y}$ on the simplex. A single $\mathbf{y}$ with $\operatorname{tpd}(\mathbf{y}) < 0$ proves a phase split exists — and that $\mathbf{y}$ is a good guess for the incipient phase.

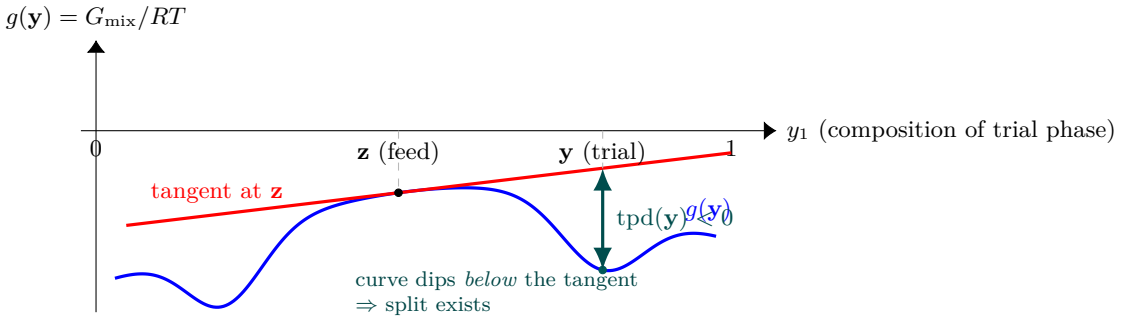



**Figure 59:** Michelsen's tangent-plane distance, drawn for a binary feed whose Gibbs-of-mixing curve $g(\mathbf{y})$ has two wells (an LLE system). Draw the plane tangent to $g$ at the feed composition $\mathbf{z}$ (red). The tangent-plane distance $\operatorname{tpd}(\mathbf{y})$ is the *vertical gap* from the curve down to (or up to) that plane, measured at a trial composition $\mathbf{y}$. Here the curve dips *below* the tangent at the right-hand well, so $\operatorname{tpd}(\mathbf{y}) < 0$: a trial phase there lowers the total Gibbs energy, which *proves* the single feed is unstable and a split exists. If the curve stayed on or above the tangent everywhere ($\operatorname{tpd} \geq 0$), the feed would be one stable phase and the flash would short-circuit. Michelsen finds the deepest such dip by a cheap successive-substitution walk to the curve's stationary points, never a brute simplex sweep.

## From a global search to stationary points

Searching the whole simplex for a negative value is impractical, so Michelsen searches only the *stationary points* of tpd. Setting $\partial \operatorname{tpd}/\partial y_i = 0$ subject to $\sum_i y_i = 1$ gives the condition

$$\ln y_i + \ln \gamma_i(\mathbf{y}) - h_i \;=\; k \qquad \text{for all } i, \qquad h_i \;\equiv\; \ln\!\big(z_i\,\gamma_i(\mathbf{z})\big), \tag{303}$$

with the same Lagrange constant $k$ for every component. Following Michelsen, introduce *unnormalised* mole numbers $Y_i = y_i\, e^{-k}$, so that $\sum_i Y_i = e^{-k}$ and $y_i = Y_i / \sum_j Y_j$. In these variables the stationarity condition (303) becomes the fixed point

$$\ln Y_i \;=\; h_i - \ln \gamma_i(\mathbf{y}) \qquad\Longleftrightarrow\qquad Y_i \;=\; \exp\!\big[\, h_i - \ln \gamma_i(\mathbf{y}) \,\big], \tag{304}$$

which suggests the **successive-substitution** iteration that `runProbe` performs:

$$Y_i^{(m+1)} \;=\; \exp\!\Big[\, h_i - \ln \gamma_i\big(\mathbf{y}^{(m)}\big) \,\Big], \qquad y_i^{(m+1)} \;=\; \frac{Y_i^{(m+1)}}{\sum_j Y_j^{(m+1)}}. \tag{305}$$

The $\gamma_i(\mathbf{z})$ in $h_i$ is computed *once*; each iteration costs one activity-coefficient evaluation at the trial composition. At a converged stationary point Michelsen shows the tangent-plane distance reduces to a function of the unnormalised $Y$ alone,

$$\mathrm{tm} \;=\; 1 \;+\; \sum_{i=1}^{n} Y_i \left(\ln Y_i - 1\right) \;-\; \sum_{i=1}^{n} Y_i\, h_i, \tag{306}$$

which is exactly the `evalTm` helper in the source. The useful identity is $\mathrm{tm} = 1 - \sum_i Y_i$ at convergence (because $\ln Y_i = h_i - \ln \gamma_i$ there), so the classic shortcut "$\sum_i Y_i > 1 \Rightarrow$ unstable" is the same test as $\mathrm{tm} < 0$. Choupo reports and decides on tm directly, declaring instability when

$$\mathrm{tm}_{\min} \;<\; \mathrm{tm}_{\mathrm{tol}} \qquad (\text{default } \mathrm{tm}_{\mathrm{tol}} = -10^{-3}), \tag{307}$$

the small negative threshold absorbing round-off so that a marginally negative tm at the trivial point is not mistaken for a true split.

## Multi-component probing: which starting points

Equation (305) only finds *a* stationary point —

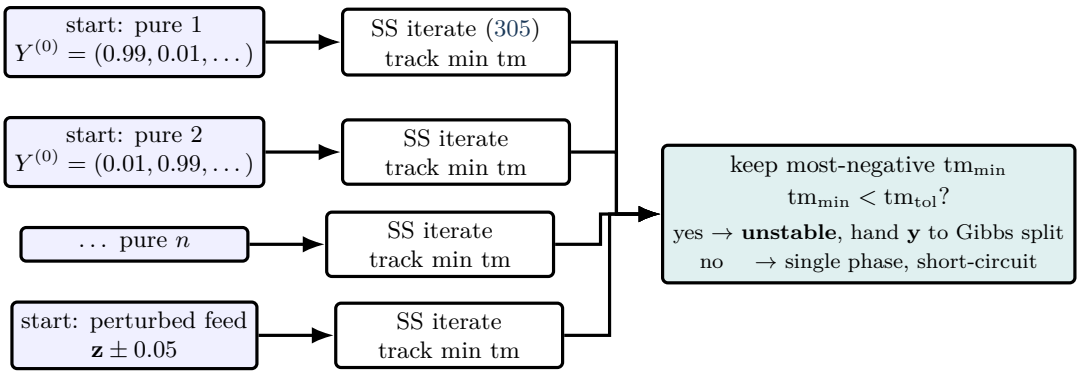



**Figure 60:** Why Michelsen's test launches *several* probes. The tangent-plane-distance surface usually has more than one stationary point — one per liquid well, plus the trivial feed point that is always stationary — and successive substitution (305) only finds the one in whose basin it starts. So `michelsenTPD` fires $n$+1 starts: one heavily biased toward each pure component (to land in that species' well) and one mildly perturbed feed (to catch shallow off-feed instabilities). Each probe records the *lowest* tm visited *along its whole trajectory*, not just the endpoint — the cheap guard against drifting back to the trivial saddle on a symmetric system. The most-negative tm across all probes decides: below the tolerance it declares a split and hands its $\mathbf{y}$ to the Gibbs minimisation; otherwise the feed is one stable phase and the expensive search is skipped.

the one in whose basin the start $\mathbf{y}^{(0)}$ lives. The TPD surface generally has several stationary points (one per liquid "well", plus the trivial point $\mathbf{y} = \mathbf{z}$ which is always stationary), and *any* of them with $\mathrm{tm} < 0$ proves instability. To hit every well at least once, `michelsenTPD` launches $n + 1$ probes for an $n$-component feed:

- $n$ **pure-component starts.** Probe $k$ begins with a heavy bias toward pure $k$: $Y_k^{(0)} = 0.99$ and the remaining 0.01 split evenly. These find the wells dominated by each species — the water-rich and butanol-rich liquids in an LLE gap.
- **one mildly perturbed-feed start.** $Y^{(0)} = \mathbf{z}$ nudged by $\pm 0.05$ in the first component, then renormalised. This catches "weak" instabilities where a pure-component start overshoots the shallow off-feed well.

The probe with the lowest (most negative) tm wins; its converged $\mathbf{y}$ is returned as `trialY`, the incipient-phase seed handed to the Gibbs-minimisation split of §11, and `probeIndex` records which start found it (pedagogically: *which* pure component the second liquid is rich in).

**Common pitfall.** For a *symmetric* LLE — the same NRTL model on both phases, as in water/$n$-butanol — the feed itself, $\mathbf{y} = \mathbf{z}$, is a stationary point of (305) where the gradient of tpd vanishes. Plain successive substitution can drift *back* to this trivial point even from a strongly biased start, reporting $\mathrm{tm} \approx 0$ and missing the real split. Choupo's `runProbe` guards against this without paying for Michelsen's full second-order Newton: it records the *lowest* tm *visited along the whole trajectory*, not just the endpoint. The path from a pure-component start *passes through* the genuine off-diagonal well on its way back to the trivial saddle, so the trajectory minimum captures the true incipient composition. This is a deliberate, cheap substitute —

adequate for the binary and ternary LLE/VLLE cases Choupo targets; a high-component industrial flash would want the real Newton-on-tpd of Michelsen Part I.

**Worked example 11.1.** TPD verdict on water/1-butanol at 25 °CTake the `extraction01_waterButanol` feed, $\mathbf{z} = (0.70, 0.30)$ (water, butanol), with the gap-tuned NRTL. The reference vector is $h_i = \ln(z_i\gamma_i(\mathbf{z}))$, evaluated once. The pure-water probe ($Y^{(0)} = (0.99, 0.01)$) iterates Eq. (305) and its trajectory dips to a markedly negative tm as $\mathbf{y}$ passes through the aqueous well ($x_{\text{water}} \approx 0.98$); the pure-butanol probe finds the organic well. The reported $\text{tm}_{\min} \ll 0$ clears the threshold (307), so the flash prints

```
Michelsen TPD stability test on phase 'liquid':
     tm_min = -X.XXXXe-02   unstable = YES
```

and proceeds to the two-liquid Gibbs minimisation, seeded with the winning `trialY`. Had the same routine been run on an ideal or fully-miscible feed, every probe would converge to $\text{tm} \geq 0$, the test would print `unstable = no`, and the flash would short-circuit to "single liquid (no LL split detected by TPD)" without ever paying for the simplex search — the whole point of a *pre*-check.

*Intuition.* The TPD test and the Gibbs minimisation are the two halves of one honest answer. The minimisation *would* return the feed itself ($\beta \to 0$) when no split exists, so in principle it could decide stability on its own — but only after paying for a full multi-start search. The TPD test answers the cheap yes/no question first, and *when* the answer is yes it hands over a vetted starting phase, turning the expensive search into a short polish. Glass-box pedagogy: the student sees the tm number that justified attempting a split, not a silent branch.

**Power and limit.** The power of the tangent-plane test is that it converts a global, geometric question — "does any trial phase undercut the tangent plane?" — into a handful of cheap fixed-point iterations whose endpoints are the only places a negative value can hide, and it does so using *only* the activity model already in hand (no second derivatives, no EoS). Its limit is inherited from successive substitution: it locates stationary points but does not *guarantee* it has found the global minimum of tpd unless every basin is seeded. Choupo's $n + 1$ canonical starts plus the trajectory-minimum guard cover the binary and ternary landscapes it targets; a feed with many near-degenerate liquid wells could still slip a split past a too-sparse probe set — which is why Michelsen Part I's second-order Newton on tpd remains the rigorous tool on the roadmap.

### The Gibbs surface

For a two-phase split of a feed of composition $\mathbf{x}_z$ at temperature $T$ and pressure $P$, with $\beta \in (0, 1)$ moles of phase $\beta$ per mole of feed, the Gibbs energy of mixing per mole of feed (in units of $RT$) is

$$\frac{G_{\text{mix}}}{RT} \;=\; (1-\beta)\sum_i x_{\alpha,i}\left[\ln x_{\alpha,i} + \ln\gamma_{\alpha,i}(\mathbf{x}_\alpha)\right] \;+\; \beta\sum_i x_{\beta,i}\left[\ln x_{\beta,i} + \ln\gamma_{\beta,i}(\mathbf{x}_\beta)\right].$$

The first term is the ideal-mixing contribution, $x\ln x$, summed over each phase; the second is the excess part $x\ln\gamma$ that encodes the activity model. Overall mass balance ties the two phase compositions:

$$z_i \;=\; (1-\beta)\,x_{\alpha,i} + \beta\,x_{\beta,i} \quad\Longrightarrow\quad x_{\beta,i} \;=\; \frac{z_i - (1-\beta)\,x_{\alpha,i}}{\beta}.$$

We minimise $G_{\text{mix}}$ over the free variables $(\beta, x_{\alpha,1}, \ldots, x_{\alpha,n-1})$; the last alpha component follows from $\sum_i x_{\alpha,i} = 1$, and the entire $\mathbf{x}_\beta$ from the mass balance above. An infeasible probe (one where some $x_{\beta,i}$ would come out negative) gets a large penalty so the optimiser steers back into the physical simplex.

## 11.2 The common tangent: where the two liquids come from

For a binary at fixed $T$ and $P$ the whole of liquid–liquid equilibrium can be read off one curve. Write the molar Gibbs energy of mixing as a function of the mole fraction $x \equiv x_1$,

$$g(x) \;=\; RT\left[\,x\ln(\gamma_1 x) + (1-x)\ln(\gamma_2(1-x))\,\right],$$

the ideal part $RT[x\ln x + (1-x)\ln(1-x)]$ plus the excess part $G^E = RT[x\ln\gamma_1 + (1-x)\ln\gamma_2]$ that the activity model supplies.

**The tangent line *is* the pair of chemical potentials.** The two partial molar quantities follow from $g$ and its slope alone,

$$\mu_1 - \mu_1^\circ \;=\; g + (1-x)\,\frac{\mathrm{d}g}{\mathrm{d}x}, \qquad \mu_2 - \mu_2^\circ \;=\; g - x\,\frac{\mathrm{d}g}{\mathrm{d}x}.$$

Both are values of the same straight line: the tangent to $g$ at $x$, extended, meets the axis $x = 1$ at $\mu_1 - \mu_1^\circ$ and the axis $x = 0$ at $\mu_2 - \mu_2^\circ$. The tangent is therefore not an illustration of the chemical potentials — it is a plot of them.

**Equilibrium therefore forces one shared tangent.** Two liquid phases $\alpha$ and $\beta$ coexist when each component has the same chemical potential in both, $\mu_{i,\alpha} = \mu_{i,\beta}$ for $i = 1, 2$. Read through the identities above, those two equations say that the tangent drawn at $x_\alpha$ and the tangent drawn at $x_\beta$ have the same intercept at $x = 1$ *and* the same intercept at $x = 0$ — two lines agreeing at two points are one line. Hence the *double-tangent* (or common-tangent) construction: the coexisting compositions are the two points at which a single straight line touches $g(x)$. Equality of $\mu_1$ and $\mu_2$ is exactly equality of slope and of intercept, so the geometric statement and the thermodynamic one carry the same information; neither is an approximation of the other.

**Why the split lowers $G$.** Take a feed of composition $z$ lying between $x_\alpha$ and $x_\beta$. If it stays homogeneous its molar Gibbs energy is $g(z)$, a point on the curve. If it splits, the lever rule fixes the phase fraction,

$$\beta \;=\; \frac{z - x_\alpha}{x_\beta - x_\alpha}, \qquad g_{\text{split}}(z) \;=\; (1-\beta)\, g(x_\alpha) + \beta\, g(x_\beta),$$

which is the height of the *chord* through the two tangent points, evaluated at $z$. Wherever that chord runs below the curve, $g_{\text{split}}(z) < g(z)$ and two liquids beat one. The chord is the common tangent, so the construction simultaneously gives the two mutual solubilities (its touching points) and the amount of each phase (the lever rule along it).

**Convexity, and what the curve must do.** A homogeneous mixture is locally stable against small composition fluctuations while

$$\frac{\mathrm{d}^2 g}{\mathrm{d}x^2} \;>\; 0.$$

The ideal contribution alone gives $\mathrm{d}^2 g/\mathrm{d}x^2 = RT/[x(1-x)] > 0$ everywhere, and diverges at both pure ends: an ideal solution can never demix, and no real one demixes right up against a pure component. A split therefore requires the excess part to be sufficiently positive — repulsive unlike-pair interactions — *and* requires the model form to be capable of the resulting double well at all. That is a property of the functional form, not of the fitted numbers: Wilson's single logarithm per term cannot produce it for any parameters, while NRTL and UNIQUAC can (Figure 11).

**Binodal, spinodal, and the band between them.** Two different conditions mark two different loci, and they must not be confused. The *binodal* is the common-tangent locus above: global, non-local, a statement about two compositions at once. The *spinodal* is where $\mathrm{d}^2 g/\mathrm{d}x^2 = 0$: local, a statement about one composition. The spinodal always lies inside the binodal. Between them sits the metastable band — a mixture there is stable to infinitesimal fluctuations (the curvature is still positive) yet has a lower-Gibbs two-phase state available to it, so it separates only by nucleation, and can be held homogeneous indefinitely if none occurs. Inside the spinodal the curvature is negative and even an infinitesimal fluctuation grows: separation is spontaneous everywhere at once (spinodal decomposition). The two loci meet only at the critical solution point, where $\mathrm{d}^2 g/\mathrm{d}x^2 = \mathrm{d}^3 g/\mathrm{d}x^3 = 0$ and the two coexisting compositions merge. Neither locus can be recovered from the other: knowing the two tangent points says nothing about where the curvature changes sign, and knowing the spinodal does not locate the tangent.

**Beyond two components.** For $n$ components the tangent line becomes a tangent *hyperplane* to the $g$ surface over the composition simplex, and coexistence is the statement that one hyperplane touches the surface at two (or more) points. This is precisely the object Michelsen's tangent-plane-distance test of Section 11.1 probes: $\mathrm{tpd}(\mathbf{x}) < 0$ says some trial composition dips below the tangent plane at the feed, so the feed cannot be the whole story. The common-tangent picture and the stability test are the same geometry seen from two sides — one constructs the answer, the other detects that an answer is missing.

## Why direct minimisation succeeds where Newton fails

At $\mathbf{x}_\alpha = \mathbf{x}_\beta = \mathbf{x}_z$ (the K = 1 trivial), the equal-activity constraint $\gamma_i(\mathbf{x}_z)\, z_i = \gamma_i(\mathbf{x}_z)\, z_i$ is satisfied for every $i$, the mass balance is trivially closed for any $\beta$, and the normalisation constraints are met. All residuals are zero. A Newton method tracking $\mathbf{F}(\mathbf{v}) = \mathbf{0}$ has no way to know that this point is a saddle of $G$ rather than a minimum — residual zero is residual zero — so it accepts the point. Even worse, damped Newton from $(\mathbf{x}_\alpha, \mathbf{x}_\beta)$ biased toward pure components overshoots and lands on the trivial because the basin of attraction extends out into the entire simplex along the equal-composition diagonal.

Direct minimisation never makes this mistake. The simplex algorithm samples $G$ values; a saddle point of $G$ has neighbouring points with *lower* $G$, so any simplex vertex placed there will be displaced on the next move. Multi-start gives global search — each start follows a different downhill path — and the lowest $G$ across all converged points is the global minimum (or the feed-single phase, in which case $G(\mathbf{x}_z) < G_{\text{split}}$ and we report "no LL split exists for these parameters").

## What Choupo reports

For `extraction01_waterButanol` (`phaseSet LL`, NRTL parameters tuned to the 25 °C water / 1-butanol gap), the routine converges in 4 multi-starts × ∼ 30 simplex iterations to:

| Phase | $x_{\text{water}}$ | $x_{\text{butanol}}$ | flow [kmol/h] |
|---|---|---|---|
| feed ($\mathbf{x}_z$) | 0.700 | 0.300 | 100.0 |
| aqueous ($\alpha$) | 0.984 | 0.016 | 50.6 |
| organic ($\beta$) | 0.408 | 0.592 | 49.4 |

The aqueous phase carries essentially all the water; the organic phase has substantial water dissolved in butanol, consistent with the literature tie-line at 25 °C. The 2n+1 Newton that the same case ran returned $\beta = -0.5$ (clamped to zero on output) and the regime "two-phase liquid" with both phases indistinguishable — the classic K = 1 trap.

## Beyond two liquids: VLLE (delivered)

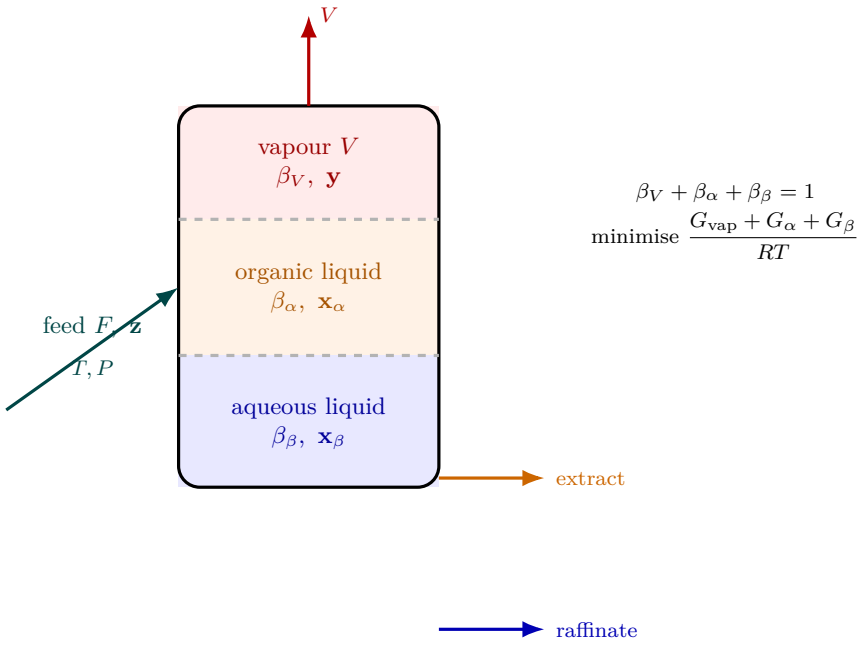



**Figure 61:** The three-phase (VLLE) flash: one feed disengages into a vapour and *two* immiscible liquids that the Gibbs-min flash finds by adding a vapour term (ideal-gas reference relative to pure liquid) to the two-liquid surface. The free variables grow to $2n$ — the phase fractions $\beta_V, \beta_\alpha$ (with $\beta_\beta = 1 - \beta_V - \beta_\alpha$) plus the vapour and $\alpha$-liquid compositions — and direct Nelder–Mead minimisation of the *total* $G/RT$ over the simplex locates the global split. Six multi-starts deliberately bias the aqueous liquid to water-rich corners ($x_{\alpha,\text{water}} \approx 0.97$–$0.99$); without that seeding the simplex misses the narrow heteroazeotrope basin and collapses to a two-phase answer. For a binary this coexistence is a single curve in $(T, P)$ (phase rule $F = 2 + 2 - 3 = 1$).

The same direct-minimisation approach extends to three coexisting phases (vapour plus two liquids) by adding a vapour Gibbs term with the ideal-gas reference state relative to pure liquid:

$$\frac{G_{\text{vap}}}{RT} = \beta_V \sum_i y_i \left[ \ln y_i + \ln\left( \frac{P}{P_i^{\text{sat}}(T)} \right) \right].$$

Free variables grow to $2n$: $(\beta_V, \beta_\alpha, y_0, \ldots, y_{n-2}, x_{\alpha,0}, \ldots, x_{\alpha,n-2})$, with $\beta_\beta$ from the phase fractions summing to 1 and the entire $\mathbf{x}_\beta$ from overall mass balance per component. The `IsothermalFlash::PhaseSet::VLLE` branch implements exactly this, with six multi-starts that explicitly bias the aqueous liquid simplex to extreme water-rich compositions ($x_{\alpha,\text{water}} \approx 0.97$–$0.99$) — without this seeding, Nelder-Mead misses the narrow heteroazeotrope basin and the optimum collapses to two-phase V+L or LL. Binary three-phase coexistence is a one-dimensional curve in $(T, P)$ by Gibbs phase rule ($F = 2 + 2 - 3 = 1$); ternaries give a two-dimensional region ($F = 2$) and are pedagogically more robust.

## Beyond Gibbs-minimisation Nelder-Mead

Nelder-Mead works well up to about ten dimensions. For very high-component LLE/VLLE, or for cases that need quadratic convergence, the Naphtali-Sandholm formulation of Newton on log-$K$-values (a piece) reaches a smaller

cost per iteration once the basin is correctly identified by Gibbs minimisation. The minimisation thus plays the role of a globally robust pre-conditioner; Newton finishes the polish.

## 11.3 Ternary phase diagrams: sweeping the simplex

A ternary diagram is not a new piece of physics — it is the flash of this chapter (and the bubble point of §4) evaluated over a grid of feed compositions on the composition triangle $x_1 + x_2 + x_3 = 1$. At each interior node the engine calls the *same* solver a single `isothermalFlash` would, so the map and a one-off flash can never disagree:

- **Solubility (LLE) map.** The liquid–liquid flash classifies each node as one phase or two liquids; the boundary between them traces the *binodal* (miscibility) curve, and the equilibrium pair of liquid compositions at a split node is a *tie-line*. Using the liquid–liquid branch (no vapour) gives the clean extraction/decanter triangle that students recognise — water / ethanol / benzene, say, with a large two-liquid region and tie-lines sloping toward the water–benzene edge.
- **Boiling surface (VLE).** Bubble-point at each node gives a temperature *surface* over the triangle; it needs only the three vapour pressures, so even ideal binaries produce a meaningful map.

**Predicting the gap without fitted pairs.** The miscibility gap is governed by the activity model (§2). NRTL/Wilson need a *regressed* binary pair for every edge; absent one, that edge defaults to ideal and the gap vanishes. **UNIFAC** sidesteps this: it predicts $\gamma$ from molecular *groups* ($CH_2$, OH, ACH, $H_2O$, ...) with no system-specific fit, so a real ternary shows a real liquid–liquid gap with zero curated pair data — the practical key that makes the solubility map usable for arbitrary mixtures whose groups are tabulated. A component without a group decomposition simply falls back to $\gamma = 1$ (ideal) for itself; the model never invents data.

## 11.4 The tie-triangle and the Hunter–Nash construction

The ternary map of §11.3 is a picture of *phase behaviour*. Laid over it, the same triangle becomes a picture of a *process*: the whole material balance of a counter-current extraction column can be drawn on it with a straight-edge, and the number of equilibrium stages read off. That construction is due to Hunter and Nash, and this section derives it — as thermodynamics and as method — independently of any particular implementation.

**Two kinds of line, and they must not be confused.** A *tie-line* joins two compositions that are *in equilibrium* with each other: it is a statement about chemical potentials, and it comes from the activity model. An *operating line* joins two compositions related by a *material balance* around some cut of the apparatus: it is a statement about bookkeeping, and it would be the same line if the thermodynamics were different. The entire construction is the alternation of the two, exactly as McCabe–Thiele's staircase alternates an equilibrium curve with an operating line; the ternary case merely replaces "curve" by "family of tie-lines" and "line" by "pencil of lines through one point".

**The lever rule on a triangle.** Let a stream be described by its total flow $F$ and its composition vector $\mathbf{x}$ with $\sum_i x_i = 1$. Because barycentric coordinates map affinely onto the plane, mixing is *linear* on the diagram: combining $\mathbf{A}$ and $\mathbf{B}$ gives

$$\mathbf{M} = \mathbf{A} + \mathbf{B}, \qquad \mathbf{x}_M = \frac{F_A\,\mathbf{x}_A + F_B\,\mathbf{x}_B}{F_A + F_B}, \tag{308}$$

so the mixing point $M$ lies *on the segment* $AB$, and divides it in the inverse ratio of the flows: $\overline{AM}/\overline{MB} = F_B/F_A$. That is the lever rule, and it is the only quantitative tool the construction needs.

**The overall balance fixes $M$ twice.** A column contacts a feed $\mathbf{F}$ with a fresh solvent $\mathbf{S}$ and delivers an extract $\mathbf{E}_1$ and a raffinate $\mathbf{R}_N$ (Fig. 62 for the stage numbering). The overall balance is

$$\mathbf{F} + \mathbf{S} \;=\; \mathbf{E}_1 + \mathbf{R}_N \;\equiv\; \mathbf{M}, \tag{309}$$

so the *same* point $M$ is the mix of the two feeds and the mix of the two products. Geometrically: $M$ lies on the line $FS$ *and* on the line $E_1R_N$, hence at their intersection. This is the first and most useful thing the triangle does — given three of the four terminal compositions, the fourth is a ruler construction, and given all four it is a *check* on the balance rather than a determination.

**The difference point $\Delta$: a stream that is not there.** Rearranging (309) and, more generally, the balance around the envelope containing stages 1 through $j$,

$$\mathbf{F} + \mathbf{E}_{j+1} \;=\; \mathbf{R}_j + \mathbf{E}_1 \quad\Longrightarrow\quad \boxed{\Delta \;\equiv\; \mathbf{F} - \mathbf{E}_1 \;=\; \mathbf{R}_j - \mathbf{E}_{j+1} \;=\; \mathbf{R}_N - \mathbf{S}} \tag{310}$$

for *every* $j = 1, \ldots, N$. The net flow $\Delta$ — what passes any cut of the column in the raffinate direction, minus what passes it in the extract direction — is the same at every cut. It is not a physical stream: its flow $F_\Delta = F_F - F_{E_1}$

may be negative and its "composition" $\mathbf{x}_\Delta = (\mathbf{F} - \mathbf{E}_1)/F_\Delta$ generally has components outside $[0, 1]$, which is why $\Delta$ almost always plots *outside* the triangle. A difference obeys the lever rule with the same algebra as a sum, so the geometric content of (310) is:

> $\Delta$, $\mathbf{R}_j$ and $\mathbf{E}_{j+1}$ are *collinear*, for every $j$.

Every operating line of the column is therefore a line through one fixed point. $\Delta$ is located as the intersection of the line $FE_1$ with the line $SR_N$ — two instances of (310) — and the remaining $N - 1$ instances are then predictions that can be checked.

**The stage staircase.** With $\Delta$ in hand the stages are stepped off by alternating the two kinds of line:

1. start at $\mathbf{E}_1$, the extract leaving stage 1;
2. follow the *tie-line* through $\mathbf{E}_1$ to its conjugate $\mathbf{R}_1$ — one equilibrium stage, and the only step in the whole construction that uses thermodynamics;
3. draw the *operating line* $\Delta\mathbf{R}_1$ and follow it to where it re-enters the binodal on the extract side: that intersection is $\mathbf{E}_2$;
4. repeat until the raffinate composition reaches the specified $\mathbf{R}_N$. The number of tie-lines used is the number of theoretical stages.

Two limits are visible on the same drawing. As the solvent rate falls, $M$ slides along $FS$ toward $F$, $\Delta$ moves, and at some rate an operating line coincides with a tie-line: the driving force vanishes there and the stage count diverges. That is the *minimum solvent* condition, and it is a *pinch* in exactly the sense the minimum-reflux pinch is one. At the other end, a duty whose $M$ falls outside the two-phase region cannot be run at all: the mixture does not split, and no number of stages helps.

**What the construction assumes, and where it stops being exact.** Step 2 above requires a tie-line *through an arbitrary composition*. A chart carries finitely many, so the classical hand method *interpolates* — conventionally by a conjugate (Hand or Othmer–Tobias) curve fitted through the plotted tie-line ends. That interpolation is the construction's real error source, and it is worst exactly where the tie-lines are shortest, near the plait point, because there the conjugate curve is steep and the tie-lines are crowding into nothing. A computed cascade (Chapter 12) has no such step: it flashes each stage at the composition the stage actually has. The honest relationship between the two is therefore not "the diagram approximates the column"; it is that the diagram and the column share (310) exactly, and differ only in how a conjugate composition is obtained — read off a fitted curve, or solved.

**Why this is worth drawing when a rigorous cascade is available.** Because (310) is a *structural* fact about counter-current contacting and the diagram makes it visible: one point controls every operating line, its position is set by the terminal specifications alone, and moving a specification moves it in a way the eye can follow. A converged stage table shows that the answer came out; the triangle shows why it had to.

# 12 Counter-current liquid–liquid extraction

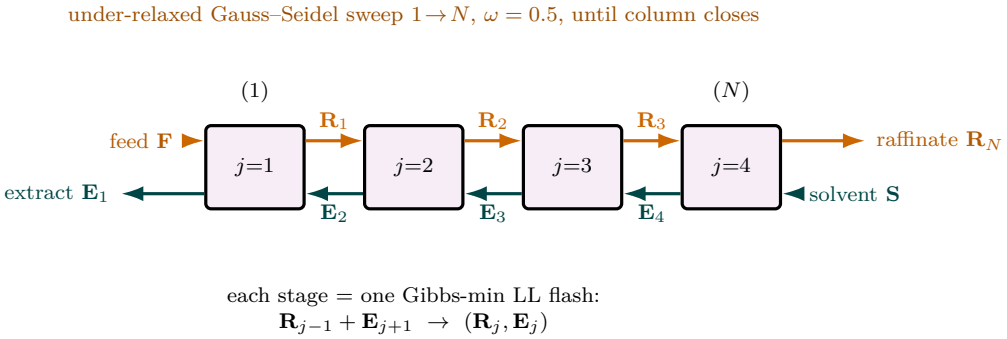



**Figure 62:** The counter-current extraction cascade. Feed **F** enters stage 1 and fresh solvent **S** enters stage $N$; the feed-derived *raffinate* (orange) flows $1 \to N$ while the solvent-rich *extract* (teal) flows $N \to 1$, so the leaving extract has met the richest feed and the leaving raffinate the cleanest solvent. Each stage is *exactly* one liquid–liquid Gibbs-min flash (Fig. 61's two-liquid cousin): it takes the combined inlet $\mathbf{R}_{j-1} + \mathbf{E}_{j+1}$ and splits it into the two equilibrium liquids. Because the two directions are coupled, the cascade is a fixed-point problem; Choupo tears it by under-relaxed Gauss–Seidel sweeps ($\omega = 0.5$, the raw update oscillates) and stops not on a noisy per-stage delta but on the honest test that what leaves the *column* equals what entered it.

> *What you'll learn.* A single liquid–liquid flash (Chapter 11) splits one mixed feed into two equilibrium liquids — a decanter. A *column* does better: a feed and a fresh solvent enter at *opposite* ends and pass each other counter-currently over $N$ theoretical stages, so the leaving extract has seen the richest feed and the leaving raffinate has seen the cleanest solvent. The chemistry of each stage is exactly the Gibbs-minimisation LL flash you already know; the new content is the *cascade* — how the stages are linked and torn. Choupo solves it by under-relaxed stage-by-stage successive substitution (Gauss–Seidel sweeps), the liquid–liquid twin of the Wang–Henke bubble-point loop used for distillation (Chapter 15) and of the coupled absorber/stripper sweep (Chapter 16). The implementation is in `src/unitOperations/separation/Extractor.cpp`; the type is `extractor` (with `extract` as an alias). It calls the shared flash kernel `IsothermalFlash::solveCore` read-only — it never re-implements the equilibrium split.

## 12.1 The cascade and its variables

Number the stages $j = 1, \dots, N$. The **feed** enters at stage 1 and the **fresh solvent** at stage $N$. Two liquid streams flow in opposite directions:

- the *raffinate* (feed-derived, carrier-rich) cascades $1 \to N$ and leaves the column at stage $N$;
- the *extract* (solvent-rich) cascades $N \to 1$ and leaves at stage 1.

Write $\mathbf{R}_j$ and $\mathbf{E}_j$ for the component-molar flow vectors ($n$ species each) of the raffinate and extract *leaving* stage $j$. The two streams arriving at stage $j$ are the raffinate descending from $j-1$ and the extract rising from $j+1$, with the column ends supplied by the external feeds:

$$\text{into stage } j \;=\; \underbrace{\mathbf{R}_{j-1}}_{(\text{feed } \mathbf{F} \text{ if } j=1)} \;+\; \underbrace{\mathbf{E}_{j+1}}_{(\text{solvent } \mathbf{S} \text{ if } j=N)} . \tag{311}$$

The two leaving streams $(\mathbf{R}_j, \mathbf{E}_j)$ are the two liquid phases that this combined inlet splits into at the column temperature $T$ and pressure $P$. Counting unknowns: $2nN$ component flows, tied by $N$ two-phase equilibria. This is precisely the structure of the MESH equations restricted to two liquid phases with no vapour and (by the isothermal assumption below) no per-stage energy balance.

## 12.2 One stage is one Gibbs-min flash

Stage $j$ takes its combined inlet (311), of total moles $F_j = \sum_i (\text{into stage } j)_i$ and overall composition $z_{j,i} = (\text{into stage } j)_i / F_j$, and hands it verbatim to the liquid–liquid flash of Chapter 11:

$$z_{j,i} \;=\; (1-\beta_j)\, x_{\alpha,i} + \beta_j\, x_{\beta,i}, \qquad \beta_j = \arg\min_{\beta,\,\mathbf{x}_\alpha} \frac{G_{\text{mix}}(\beta, \mathbf{x}_\alpha, \mathbf{x}_\beta)}{RT}, \tag{312}$$

solved by multi-start Nelder–Mead on the Gibbs surface, *not* by Newton on the equal-activity residual — this is the whole point of Chapter 11, and the Extractor inherits it for free. Of the two phases the flash returns, the one

richer in the designated **solvent species** $s$ (the largest component of the solvent feed, or the user's `solvent` key) is labelled the extract; the other is the raffinate:

$$\mathbf{E}_j \;=\; \text{phase with the larger } x_s, \qquad \mathbf{R}_j \;=\; \text{the other phase.} \tag{313}$$

The same species choice seeds the flash's $\alpha/\beta$ basin bias ($s$ for the extract-rich phase, the feed's dominant carrier for the raffinate-rich phase), so each stage's Gibbs minimisation starts in the correct well. **Robust degeneracy handling:** if a stage's flash finds no genuine split ($\beta_j \le 10^{-6}$ or $\ge 1-10^{-6}$, or it fails to converge), the whole inlet leaves as raffinate ($\mathbf{R}_j = $ inlet, $\mathbf{E}_j = \mathbf{0}$) so the cascade mass balance stays exact; the count of two-phase stages is reported (`splitStages`) so a column operating below its solubility window is *visible* rather than silently wrong.

## 12.3 Tearing the cascade: under-relaxed Gauss–Seidel

The inter-stage flows in (311) are coupled in both directions, so the cascade is a fixed-point problem. Choupo tears it by successive substitution exactly as a distillation column tears its bubble-point loop:

1. **Initialise** honestly by topology propagation — every stage's extract $\mathbf{E}_j \leftarrow \mathbf{S}$ (the fresh solvent) and every raffinate $\mathbf{R}_j \leftarrow \mathbf{F}$ (the feed). No magic constant; the feeds themselves are the seed (cf. the no-silent-crutch credo).
2. **Sweep** $j = 1 \to N$ (Gauss–Seidel: each stage uses the *freshest* available neighbours). Build the inlet (311), flash it (312), and *under-relax* the update with factor $\omega = 0.5$:

$$\mathbf{R}_j \leftarrow (1-\omega)\,\mathbf{R}_j + \omega\,\mathbf{R}_j^{\text{flash}}, \qquad \mathbf{E}_j \leftarrow (1-\omega)\,\mathbf{E}_j + \omega\,\mathbf{E}_j^{\text{flash}}. \tag{314}$$

   Under-relaxation is essential: a raw Gauss–Seidel update of counter-current cascades oscillates, because changing $\mathbf{E}_j$ changes the inlet of stage $j-1$, which changes *its* extract, and so on back up the column.
3. **Iterate** until the *column mass balance closes*.

**Why the convergence test is the overall closure, not $|\Delta F|$.** Each per-stage flash splits its own inlet to machine tolerance, but the Nelder–Mead simplex stops at a finite `tolX` $\sim 10^{-5}$ on compositions, so the inter-stage flows carry a small flash-noise floor that the under-relaxed sweeps grind down. The per-step residual $\max_j |\Delta\mathbf{R}_j|, |\Delta\mathbf{E}_j|$ limit-cycles on that floor and is a poor convergence proxy. The physically meaningful metric is whether what leaves the *column* equals what entered it:

$$\varepsilon_{\text{close}} \;=\; \frac{\big|\,(F_{\mathbf{F}} + F_{\mathbf{S}}) - (F_{\mathbf{E}_1} + F_{\mathbf{R}_N})\,\big|}{F_{\mathbf{F}} + F_{\mathbf{S}}}, \tag{315}$$

the relative gap between the two feeds in and the extract-plus-raffinate out. The sweep stops when $\varepsilon_{\text{close}} < 10^{-4}$, or when it *stalls* (60 sweeps with no closure improvement). On a stall the column reports the best state reached but only flags `converged` if the closure it achieved is still acceptable ($< 10\,\varepsilon_{\text{close}}^{\text{tol}}$); otherwise the `converged` KPI stays 0 and the run warns aloud — numerical honesty over a green light. The closure $\varepsilon_{\text{close}}$ is pushed to `recordResidual` each sweep so the convergence plot shows the cascade settling.

## 12.4 What the column reports

The two products and the per-species accounting follow directly from the torn cascade:

$$\text{extract product} = \mathbf{E}_1 \qquad \text{(solvent-rich, leaves stage 1)}, \tag{316}$$

$$\text{raffinate product} = \mathbf{R}_N \qquad \text{(feed-derived, leaves stage } N\text{)}, \tag{317}$$

$$\text{recovery}_i = \frac{E_{1,i}}{F_{\mathbf{F},i} + F_{\mathbf{S},i}} \qquad \text{(fraction of species } i \text{ leaving in the extract)}, \tag{318}$$

$$\text{distribution}_i = \frac{z_{\mathbf{E}_1,i}}{z_{\mathbf{R}_N,i}} \qquad \text{(the partition ratio } K_i = x_{i,E}/x_{i,R}\text{)}. \tag{319}$$

The **solute** is identified automatically as the species — other than the solvent $s$ and the feed carrier — most enriched into the extract, and its recovery and distribution ratio are headlined; a full per-species breakdown follows. Note the credo-pure interface: the `operation` block carries only *hardware* — `stages` $N$ and the optional shared `temperature` — while the solute recovery, the distribution ratios and the product flows are all *results* of turning the crank, never inputs.

## Power and limit

**Power.** The column reuses the saddle-proof Gibbs-min flash, so it is robust on the same symmetric-$\gamma$ systems that defeat equal-activity Newton; combined with UNIFAC (§11.3) it can screen extraction of a solute by an arbitrary solvent with *zero* fitted pair data. The counter-current cascade captures the central design trade-off — more stages or more solvent buy more recovery — and the stage profile (`xE_`/`xR_` versus stage) lets a student watch the solute concentrate toward stage 1 exactly as in a McCabe–Thiele construction, but read off a rigorous activity model rather than a single constant $K$.

**Limit.** The model is *isothermal*: one temperature for the whole column (the feed's, unless overridden), because LLE is weakly enthalpic and a per-stage energy balance is deliberately out of scope — the same fallback the absorber and stripper take when $c_p$ data are absent. Stages are *theoretical* (Murphree-style stage efficiency and the HETP/mass-transfer rate that fix a real packed-column height are not modelled). The two products are taken to be the two declared `liquid` phases, so the declared system must carry $\geq 2$ liquid phases or the unit refuses to run. There is no entrainer/decanter solvent-recovery loop — the extract leaves with its solvent, and recovering it (e.g. by distillation) is a downstream unit in the flowsheet. Finally, the cascade is solved by successive substitution, not the quadratically convergent Naphtali–Sandholm Newton; for routine ternary extraction it converges in tens of sweeps, but very stiff, many-component columns would benefit from the Newton polish noted at the end of Chapter 11.

## Worked sketch: a 3-stage solvent wash

Take a feed of 100 kmol/h with mole fractions $(\text{carrier}, \text{solute}) = (0.80, 0.20)$ contacted counter-currently with 150 kmol/h of pure solvent over $N = 3$ stages at fixed $T$. Initialise $\mathbf{E}_j = \mathbf{S}$ and $\mathbf{R}_j = \mathbf{F}$ on all three stages. Sweep 1 flashes stage 1 (feed + extract-from-2): the solute distributes between the carrier-rich raffinate and the solvent-rich extract per (312), and under-relaxation moves the flows halfway. Stages 2 and 3 follow, stage 3 seeing the fresh solvent directly. Each subsequent sweep tightens the inter-stage flows; the closure (315) falls (modulo the flash-noise floor) and the run stops when it drops below $10^{-4}$. The extract leaving stage 1 carries the solute that three equilibrium contacts could strip from the carrier; pushing $N$ or the solvent-to-feed ratio up raises the headline `soluteRecovery` — the lever the student turns. The natural runnable analogue is the water/1-butanol gap of `extraction01_waterButanol` promoted from a single decanter to a multistage column.

# 13 Stoichiometric conversion reactor

*What you'll learn.* The CSTR and PFR ask *how far* a reaction goes given a rate law and a residence time. Sometimes you do not have — and do not want — a rate law: you simply *know* the per-pass conversion of a reactant (measured in a pilot plant, set by the catalyst, quoted in a textbook problem). The `conversionReactor` takes that single number and turns the crank of stoichiometry: one specified conversion fixes the reaction extent $\xi$, and every outlet flow follows from $n_{i,\mathrm{out}} = n_{i,\mathrm{in}} + \nu_i\,\xi$. No kinetics, no iteration, no liquid volume — an *algebraic* reactor. This chapter derives that one line, shows why a *finite* conversion is exactly what makes a recycle loop meaningful, and contrasts it with the Gibbs reactor of the next chapter. The implementation is in `src/unitOperations/reactor/ConversionReactor.cpp`; the runnable case is `tutorials/steady/reactors/conversion01_hda/`.

## 13.1 The extent of reaction, made primary

In every reactor chapter so far the extent $\xi$ was an *unknown* solved for — a Newton root (CSTR), the endpoint of an integration (PFR), or the minimiser of a free energy (Gibbs). The conversion reactor inverts the question: it makes $\xi$ a *direct* consequence of one user-supplied number, the fractional conversion $X$ of a chosen **limiting reactant** $\ell$.

Recall the definition of conversion for reactant $\ell$ over a single reaction with stoichiometry $\boldsymbol{\nu}$ (negative for reactants, positive for products, exactly as in Section 6.3):

$$X \;\equiv\; \frac{n_{\ell,\mathrm{in}} - n_{\ell,\mathrm{out}}}{n_{\ell,\mathrm{in}}}, \qquad 0 \le X \le 1. \tag{320}$$

The extent that delivers this conversion follows by combining (320) with the stoichiometric outlet relation for the limiting species, $n_{\ell,\mathrm{out}} = n_{\ell,\mathrm{in}} + \nu_\ell\,\xi$:

$$\boxed{\;\xi \;=\; \frac{X\,n_{\ell,\mathrm{in}}}{-\nu_\ell}\;} \qquad (\nu_\ell < 0). \tag{321}$$

The denominator $-\nu_\ell > 0$ converts "moles of $\ell$ consumed" into "moles of reaction-as-written". With the feed molar flow $F_{\mathrm{in}}$ and feed mole fractions $\mathbf{z}$, the inlet flow of the limiting reactant is $n_{\ell,\mathrm{in}} = z_\ell\,F_{\mathrm{in}}$, exactly the line `xi = X * molesLim_in / (-nu[iLim])` in the source.

Once $\xi$ is known, *every* species — reactants, products, inerts — is propagated by the one stoichiometric balance:

$$\boxed{\;n_{i,\mathrm{out}} \;=\; n_{i,\mathrm{in}} + \nu_i\,\xi \;=\; z_i\,F_{\mathrm{in}} + \nu_i\,\xi, \qquad i = 1,\dots,N_c\;} \tag{322}$$

with $\nu_i = 0$ for any species not in the reaction (inerts pass through untouched). The outlet flow and composition are then $F_{\mathrm{out}} = \sum_i n_{i,\mathrm{out}}$ and $z_{i,\mathrm{out}} = n_{i,\mathrm{out}}/F_{\mathrm{out}}$. That is the *entire* mass balance: no equation to solve, no Jacobian, no tolerance — the reactor is one pass of arithmetic.

*Intuition.* The extent $\xi$ is the single "turn of the crank" of the balanced equation. Conversion $X$ measures how far one specific reactant turned; $\xi$ measures how far the reaction-as-written turned. Equation (321) is just the gearing between the two — $-\nu_\ell$ is the gear ratio. Choose the gear ($\ell$), read off how far it turned ($X$), and (322) reports where every other gear ended up. Because all species share the *same* $\xi$, stoichiometry — hence atom balance — is satisfied exactly and for free.

## 13.2 Honesty guards in the algorithm

The glass-box implementation refuses to silently produce nonsense:

- The named `limitingReactant` must have $\nu_\ell < 0$ (it must actually be consumed); a non-negative $\nu_\ell$ is a hard error, because (321) would otherwise divide by a product or give a negative extent.
- $X$ is range-checked to $[0, 1]$.
- Every outlet flow is checked for negativity after (322). If $n_{i,\mathrm{out}} < 0$, the conversion was placed on the *wrong* (non-limiting) reactant — some other reactant ran out first — and the run aborts with a message saying exactly that. A tiny negative round-off (within $10^{-12}$) is tidied to zero.

This last guard is the pedagogical payoff of naming a limiting reactant: the code will not let you ask for 90 % conversion of a species that is only present in 50 % of the stoichiometric amount.

## 13.3 The isothermal duty

The conversion reactor runs *isothermal*: the outlet temperature is the operating temperature $T$ (defaulting to the feed temperature if none is given). The duty needed to hold $T$ is the heat of reaction times the extent. Using the elements (formation) datum of Section 6.3 — the only datum that survives a reactor — the heat of reaction per mole of extent is

$$\Delta h_{\mathrm{rxn}}(T) \;=\; \sum_i \nu_i \, h_i^{\mathrm{ig}}(T), \tag{323}$$

evaluated at the reactor temperature from each species' absolute ideal-gas enthalpy `Component::h_pure_ig(T)` (which carries $\Delta h_f^\circ$ as its floor). The isothermal duty is then

$$Q \;=\; \xi \, \Delta h_{\mathrm{rxn}}(T), \tag{324}$$

negative (heat removed) for an exothermic reaction, positive (heat added) for an endothermic one. If any participating species lacks ideal-gas $c_p$ data, the duty is simply not reported rather than faked — the mass balance still stands on its own.

> *Intuition.* Notice what the conversion reactor does *not* do: it does not let the heat of reaction feed back on the conversion. $X$ is yours to set; $Q$ is the consequence. This is the deliberate opposite of the *adiabatic* Gibbs reactor, where $T$ floats until the released heat and the equilibrium are mutually consistent. Here the chemistry is decoupled from the energy balance by construction — which is exactly right when the conversion is fixed by a catalyst bed cooled to a set temperature, and wrong if you meant "let it run adiabatically".

## 13.4 Why a *finite* conversion is the whole point of a recycle

A Gibbs reactor drives to equilibrium; a conversion reactor stops at the number you give it. The second is what real recycle processes need. Take hydrodealkylation of toluene (HDA), the canonical recycle flowsheet:

$$\mathrm{C_7H_8} + \mathrm{H_2} \;\longrightarrow\; \mathrm{C_6H_6} + \mathrm{CH_4},$$

run with hydrogen in excess so toluene is limiting. Per pass only ∼75 % of the toluene reacts; the unconverted 25 % is separated downstream and *recycled*. The per-pass conversion is set by the reactor (residence time, temperature) — it is an input, not an equilibrium output. If you modelled this reactor as a Gibbs reactor it would convert essentially *all* the toluene (the equilibrium lies far to the right), the recycle stream would shrink to nothing, and the whole reason for the loop would vanish. The conversion reactor's *finite, specified* $X$ is precisely what leaves material in the recycle and makes the tear-stream solve of Chapter 6 non-trivial.

**Worked example 13.1.** One pass of the HDA reactorFeed: $F_{\mathrm{in}} = 400$ kmol/h at 900 K, 35 bar, mole fractions $z_{\mathrm{tol}} = 0.25$, $z_{\mathrm{H_2}} = 0.75$ (benzene and $\mathrm{CH_4}$ zero). Toluene is the limiting reactant with $\nu_{\mathrm{tol}} = -1$; specified conversion $X = 0.75$.
*Extent.* Inlet toluene $n_{\mathrm{tol,in}} = 0.25 \times 400 = 100$ kmol/h, so by (321)

$$\xi = \frac{0.75 \times 100}{-(-1)} = 75 \text{ kmol/h}.$$

*Outlets* by (322) (kmol/h):

| species | $n_{\mathrm{in}}$ | $\nu_i\,\xi$ | $n_{\mathrm{out}}$ |
|---|---|---|---|
| toluene | 100 | −75 | 25 |
| $\mathrm{H_2}$ | 300 | −75 | 225 |
| benzene | 0 | +75 | 75 |
| $\mathrm{CH_4}$ | 0 | +75 | 75 |
| total | 400 | 0 | 400 |

Total moles are conserved here only because $\sum_i \nu_i = -1 - 1 + 1 + 1 = 0$ for this reaction; in general the outlet total shifts by $\xi \sum_i \nu_i$. The 25 kmol/h of unconverted toluene is what a downstream separator returns to the feed — the recycle. No Newton iteration was performed; the answer is exact stoichiometry.

> **Runnable case.** `tutorials/steady/reactors/conversion01_hda` — the single-pass reactor above. The full recycle loop (with separation and tear-stream Newton) is `tutorials/plant/hda`.

## 13.5 Conversion vs. Gibbs: two opposite idealisations

The conversion reactor and the Gibbs reactor of the next chapter are

the two extreme idealisations of a reactor, and choosing between them is a modelling decision the student should make consciously:

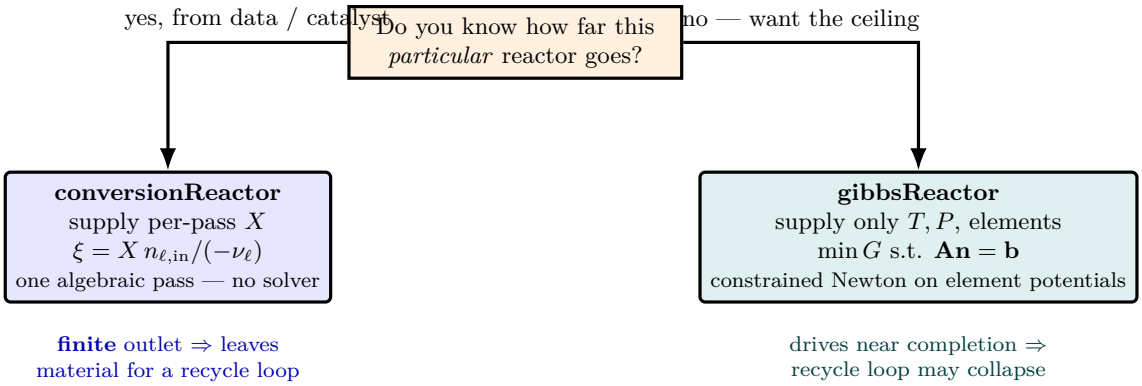



**Figure 63:** The two extreme reactor idealisations, and the conscious modelling choice between them. The **conversion reactor** answers "what does *this* reactor actually do?" — you supply the per-pass conversion $X$ (known from a catalyst or plant data) and the outlet falls out of stoichiometry in a single algebraic pass, no solver. The **Gibbs reactor** answers "what is the *best* this chemistry could do?" — you supply nothing about extent, only $T$, $P$ and the element inventory, and the outlet is the composition that minimises $G$ subject to atom conservation. The recycle consequence is the deciding factor: a finite conversion leaves material for the loop, while equilibrium often drives to near-completion and the loop collapses.

| | **conversionReactor** | **gibbsReactor** |
|---|---|---|
| what you supply | a per-pass conversion $X$ | nothing about extent — only $T$, $P$, elements, species |
| what fixes the outlet | stoichiometry, $\xi = X n_{\ell,\mathrm{in}}/(-\nu_\ell)$ | minimisation of $G$ over the atom-conserving set |
| computation | one algebraic pass | constrained Newton on element potentials |
| reactions | one specified reaction | all combinations the species list allows |
| recycle behaviour | **finite outlet** — leaves material for the loop | drives (often) to near-completion — loop may collapse |
| when it is right | conversion known from data / catalyst; recycle processes | fast kinetics, conversion unknown, want the equilibrium ceiling |

*Intuition.* The Gibbs reactor answers "what is the *best* this chemistry could do?"; the conversion reactor answers "what does this *particular* reactor actually do?". A recycle flowsheet lives or dies on the second question — which is why the conversion reactor, despite being the simplest unit in the whole simulator (no solver at all), is the one that makes the most interesting flowsheets work.

# 14 Gibbs reactor: equilibrium by free-energy minimisation

*What you'll learn.* The CSTR and PFR of the previous chapters are *kinetic* reactors: they tell you what composition you get from a given residence time and rate law, irrespective of whether the result is at equilibrium. The Gibbs reactor is the opposite limit — it assumes kinetics are infinitely fast and computes the composition that minimises the total Gibbs energy at the specified $T$ and $P$, subject to the atomic elements being conserved. Useful whenever the timescale for reaction is much shorter than the residence time (reforming, high-temperature combustion, ammonia synthesis at scale), or as an upper bound on conversion when kinetic data are unavailable. This chapter derives the algorithm Choupo uses — Lagrangian multipliers on the element-balance constraints, one $\pi_j$ per element — and walks through the water-gas-shift tutorial end-to-end. The implementation is in `src/unitOperations/reactor/GibbsReactor.cpp`, and the runnable case is `tutorials/steady/gibbs/gibbs01_water_gas_shift/`.

## 14.1 The problem

Given a feed of mole numbers $\mathbf{n}^{\text{in}}$ over $N$ species, a reactor temperature $T$, a reactor pressure $P$ and the element-composition matrix $\mathbf{A} \in \mathbb{R}^{M\times N}$ (so $A_{ji}$ is the number of atoms of element $j$ in species $i$), find the equilibrium mole numbers $\mathbf{n}$ that minimise the total molar Gibbs energy of the (single-phase, ideal-gas) mixture:

$$\min_{\mathbf{n}} \; G(\mathbf{n}) \;=\; \sum_{i=1}^{N} n_i\,\mu_i(T,P,\mathbf{n}), \qquad \text{subject to} \quad \mathbf{A}\,\mathbf{n} \;=\; \mathbf{b}, \quad n_i \ge 0, \tag{325}$$

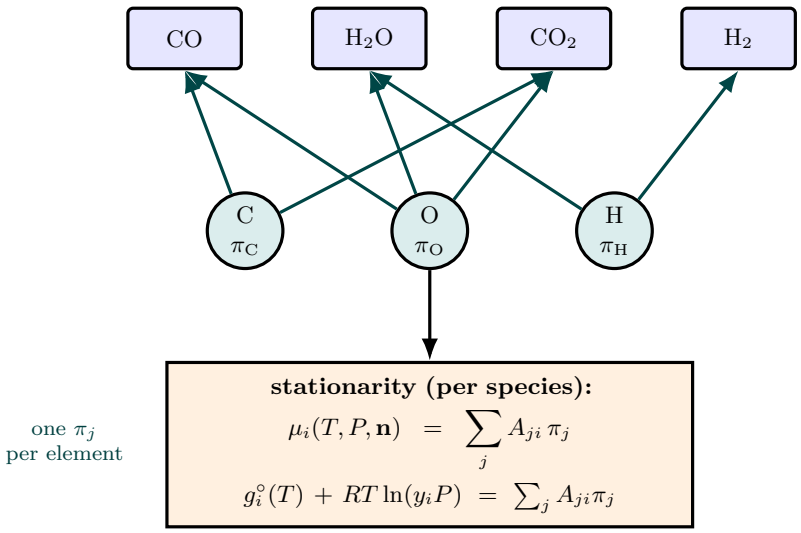



**Figure 64:** The Gibbs reactor never enumerates reactions. Instead it attaches one Lagrange multiplier $\pi_j$ — an *element chemical potential* — to each conserved element (C, O, H …, teal circles), and the atom-balance constraints $\mathbf{A}\mathbf{n} = \mathbf{b}$ wire every species (blue) to the elements it contains. At the constrained minimum of $G$ each species' chemical potential equals the sum of its element potentials, $\mu_i = \sum_j A_{ji}\pi_j$ (orange box). That is $N$ stationarity equations plus $M$ element balances — a small Newton system in $(\mathbf{n}, \boldsymbol{\pi})$ — replacing the combinatorial mess of "which reactions occur". The water-gas-shift species shown here resolve themselves automatically.

where $\mathbf{b} = \mathbf{A}\,\mathbf{n}^{\text{in}}$ records the total atoms of each element present in the feed (conserved across the reaction). Note that the feed itself trivially satisfies the constraint, so the optimisation lives in a non-empty affine set.

For an ideal-gas mixture at the reactor pressure $P$, the chemical potential is the one we derived in Chapter 9:

$$\mu_i(T,P,\mathbf{n}) \;=\; g_i^{\circ}(T) \;+\; RT\,\ln\!\big(y_i\,P\big), \qquad y_i = \frac{n_i}{N_{\text{tot}}}, \quad N_{\text{tot}} = \sum_i n_i, \tag{326}$$

with $P$ taken in bar and the standard reference $P^{\circ} = 1$ bar suppressed. The standard-state $g_i^{\circ}(T)$ is the absolute ideal-gas Gibbs energy computed by the chain $h_f^{\circ} + s^{\circ} + \int c_p^{\text{ig}}$ of Chapter 6, available in `Component::g_pure_ig(T)`.

## 14.2 The Lagrangian

Introduce a multiplier $\lambda_j$ for each of the $M$ element-balance constraints:

$$L(\mathbf{n}, \boldsymbol{\lambda}) \;=\; G(\mathbf{n}) \;-\; \sum_{j=1}^{M} \lambda_j \left( \sum_{i=1}^{N} A_{ji}\, n_i - b_j \right). \tag{327}$$

Stationarity with respect to $n_i$ gives, at an interior solution ($n_i > 0$ for every $i$),

$$\frac{\partial L}{\partial n_i} \;=\; \mu_i \;-\; \sum_{j=1}^{M} \lambda_j\, A_{ji} \;=\; 0 \qquad \forall i. \tag{328}$$

Substituting Eq. (326) and dividing through by $RT$:

$$\frac{g_i^{\circ}(T)}{RT} + \ln y_i + \ln P \;=\; \sum_{j=1}^{M} \pi_j\, A_{ji}, \qquad \pi_j \equiv \frac{\lambda_j}{RT}. \tag{329}$$

Solving for $n_i = N_{\mathrm{tot}} y_i$ gives the closed form the simulator uses to reconstruct mole numbers from the iteration variables:

$$\boxed{\; n_i \;=\; \exp\!\left( \ln N_{\mathrm{tot}} - \ln P - \frac{g_i^{\circ}(T)}{RT} + \sum_{j=1}^{M} \pi_j\, A_{ji} \right). \;} \tag{330}$$

> *Intuition.* Each element multiplier $\pi_j$ has units of inverse $RT$ and ought to be thought of as the "thermodynamic activity" of an atom of element $j$ in the equilibrium mixture. Species $i$ that contains more of an element with a high $\pi_j$ is correspondingly more populated at equilibrium. At $T = 0$ K only the species with the lowest $g_i^{\circ}$ survive; at $T \to \infty$ the entropy term $\ln y_i$ wins and all species become equally populated.

## 14.3 The residual system

Define the iteration variables

$$\mathbf{x} \;=\; (\, \pi_1,\, \ldots,\, \pi_M,\, \ln N_{\mathrm{tot}} \,) \;\in\; \mathbb{R}^{M+1}. \tag{331}$$

There are $M$ element-balance residuals and one mass-conservation residual:

$$f_j(\mathbf{x}) \;=\; \sum_{i=1}^{N} A_{ji}\, n_i(\mathbf{x}) \;-\; b_j, \qquad j = 1, \ldots, M, \tag{332}$$

$$f_{M+1}(\mathbf{x}) \;=\; \sum_{i=1}^{N} n_i(\mathbf{x}) \;-\; \exp(\ln N_{\mathrm{tot}}), \tag{333}$$

with $n_i(\mathbf{x})$ from Eq. (330). These $M+1$ non-linear equations in $M+1$ unknowns are solved by the `solver::newtonND` routine of Chapter 3 with a finite-difference Jacobian; one Gauss-eliminated linear system per iteration, backtracking line search.

## 14.4 The initial guess that makes this work

Equation (330) contains an exponential of a quantity that, at typical reactor temperatures, has $g_i^{\circ}/RT$ of order $-100$ for products of formation. A naive initial guess of $\pi_j = 0$ then asks the simulator to evaluate $\exp(+100) \approx 10^{43}$ on the very first residual call, drowning all gradient information and refusing to converge.

The simulator's trick is to seed $\boldsymbol{\pi}$ from a **least-squares fit** of the stationarity condition (328) to the *feed* composition. Treat the feed mole fractions $y_i^{\mathrm{feed}}$ (perturbed up from zero to a floor of $10^{-6}$ for inert species so the log stays finite) as if they were already the equilibrium answer, and solve the over-determined $N \times M$ linear system

$$\sum_{j=1}^{M} \pi_j\, A_{ji} \;=\; \frac{g_i^{\circ}(T)}{RT} \;+\; \ln y_i^{\mathrm{feed}} + \ln P \qquad i = 1, \ldots, N \tag{334}$$

in the least-squares sense by forming the $M \times M$ normal equations

$$(\mathbf{A}\,\mathbf{A}^{\top})\,\boldsymbol{\pi} \;=\; \mathbf{A}\,\mathbf{b}_{\mathrm{rhs}}, \tag{335}$$

and solving by `solver::gaussSolve`. After this seed the Newton iteration converges with visibly quadratic decay in five to ten steps.

**When the feed is far from its equilibrium.** The fit treats the feed as if it were the answer, so it inherits the whole affinity of the reactions that separate the two. Ten per cent ammonia in air below 800 K is such a feed: water, absent from it, is seeded at $e^{17}$ to $e^{45}$ mol/s against about 4 mol/s at the answer, two species then carry every atom, the $(M+1)$ Jacobian is singular in double precision, and the Newton stops at its first iteration. The simulator then re-seeds the *same* Newton from a primal iteration on $\ln n_i$ started from equal amounts (the reduced iteration equations of Gordon and McBride, NASA Reference Publication 1311, 1994), whose linear system is weighted by the current amounts rather than by $\exp(\boldsymbol{\pi}\cdot\mathbf{A}_i)$, and says so in the run's caveat block. The answer is still the Newton's, on the same residual and to the same tolerance.

## 14.5 Worked example: water-gas shift at 800 K

The water-gas shift reaction

$$\mathrm{CO} + \mathrm{H_2O} \;\rightleftharpoons\; \mathrm{CO_2} + \mathrm{H_2}$$

runs over a $\mathrm{Cu/ZnO/Al_2O_3}$ low-temperature catalyst and an $\mathrm{Fe_2O_3/Cr_2O_3}$ high-temperature catalyst; the equilibrium composition is the practical upper bound on hydrogen production. Take $T = 800$ K, $P = 1$ bar, equimolar $\mathrm{CO/H_2O}$ feed.

The atom matrix over elements (C, H, O) for species ($\mathrm{CO}$, $\mathrm{H_2O}$, $\mathrm{CO_2}$, $\mathrm{H_2}$) is

$$\mathbf{A} \;=\; \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 0 \end{pmatrix},$$

so $M = 3$, $N = 4$. With $b_\mathrm{C} = b_\mathrm{H} = 2\,n_0$, $b_\mathrm{O} = 2\,n_0$ where $n_0 = 0.5\,F$ is the half-feed.

The Gibbs energies at 800 K the engine uses, read from the run's own `newtonLog` (its `standardState` rows; the reactor `shift` of `tutorials/steady/gibbs/gibbs12_element_potential_newton` poses this same problem and logs them), are

| species | $g^\circ(800\,\mathrm{K})$ [kJ/mol] | $g^\circ/RT$ |
|---|---|---|
| $\mathrm{CO}$ | $-277.2$ | $-41.67$ |
| $\mathrm{H_2O}$ | $-402.9$ | $-60.58$ |
| $\mathrm{CO_2}$ | $-576.7$ | $-86.70$ |
| $\mathrm{H_2}$ | $-112.9$ | $-16.98$ |

(This table carried $-282.0$, $-389.3$, $-577.5$ and $-117.8$ kJ/mol until 2026-09-27: numbers no run of the engine produced, and ones that do not give the $K_\mathrm{eq}$ quoted below, which the engine's own values do: $\ln K = -\sum_i \nu_i g_i^\circ/RT = 1.433$.)

Running `./choupoSolve tutorials/steady/gibbs/gibbs01_water_gas_shift` gives, after six Newton steps with residual decay [$0.499 \to 0.235 \to 0.142 \to 0.015 \to 2.1{\times}10^{-4} \to 4.3{\times}10^{-8}$]:

| species | $y_i^\mathrm{feed}$ | $y_i^\mathrm{eq}$ |
|---|---|---|
| $\mathrm{CO}$ | 0.500 | 0.164 |
| $\mathrm{H_2O}$ | 0.500 | 0.164 |
| $\mathrm{CO_2}$ | 0.000 | 0.336 |
| $\mathrm{H_2}$ | 0.000 | 0.336 |

CO conversion is 67.2%. The equilibrium constant

$$K_\mathrm{eq}(800\,\mathrm{K}) \;=\; \frac{y_{\mathrm{CO_2}}\,y_{\mathrm{H_2}}}{y_{\mathrm{CO}}\,y_{\mathrm{H_2O}}} \;=\; \frac{0.336\times 0.336}{0.164\times 0.164} \;\approx\; 4.19, \tag{336}$$

against the tabulated value 4.04 from Smith-Van Ness-Abbott Table 13.2 — agreement at the 3.7% level, which is the residual error in the $\Delta h_f^\circ$ and $s^\circ$ tabulations themselves.

> *Intuition.* The independent reaction count for a Gibbs problem is $R = N - M$ (species minus elements). Water-gas shift has $R = 4 - 3 = 1$: one independent reaction. Steam reforming of methane ($\mathrm{CH_4}$ + $\mathrm{H_2O}$, five species: $\mathrm{CH_4}$, $\mathrm{H_2O}$, $\mathrm{CO}$, $\mathrm{CO_2}$, $\mathrm{H_2}$, three elements C, H, O) has $R = 5 - 3 = 2$ — two independent reactions, typically written as reforming itself and water-gas shift. Combustion of methane in air with possible incomplete burning ($\mathrm{CH_4}$, $\mathrm{O_2}$, $\mathrm{CO}$, $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{N_2}$, six species, four elements C, H, O, N) has $R = 6 - 4 = 2$. The simulator does not care about $R$ — it solves the $M + 1$ unknowns regardless of how many independent reactions that corresponds to.

## 14.6 When the ideal-gas Gibbs reactor stops being right

- **Kinetic limitation.** If the residence time is shorter than the slowest elementary step (e.g. NO formation in a furnace, ammonia synthesis below 700 K), the equilibrium composition is an upper bound but the actual exit composition is dictated by kinetics. Run a CSTR or PFR instead.
- **Non-ideal gas.** Above $\sim$ 10 bar the ideal-gas $\varphi_i = 1$ approximation breaks down. Under a declared real equation of state the element-potential method folds $\ln\varphi_i$ into $g_i^\circ/RT$ and repeats the Newton until $\varphi$ settles (`GibbsMethod.cpp`, `gibbsGasSolve`); the ideal-gas package is one pass. (This bullet said until 2026-09-27 that the reactor ignored $\varphi$; it had not for some time.)
- **Condensation, and a solid.** Single-phase Gibbs is correct only if all species stay in one phase. The element-potential method TESTS the gas answer ($y_iP > P_i^{\mathrm{sat}}$ for a condensable, $\sum_k \pi_k A_{ks} > g_s^\circ/RT$ for a declared pure solid) and, when one phase appears, finds how much of it forms by a one-dimensional bisection on its amount, each step a gas-only Newton on the atoms that remain — not by enlarging the Newton. One condensable at a time (`reactiveFlash` and `directMin` carry a multi-component liquid); two pure solids at once are refused. (This bullet said until 2026-09-27 that no multi-phase Gibbs was done.) The EduTool *How a Gibbs reactor solves* draws both searches from the engine's own log.
- **Element rank deficiency.** If **A** does not have full row rank (e.g. a redundant element constraint), the normal-equation matrix in the initial-guess step becomes singular. The implementation falls back to $\pi = 0$ in that case and hopes the iteration recovers — in practice the rank deficiency means the user has set the problem up wrong.

> **Runnable case.** `runCase tutorials/steady/gibbs/gibbs01_water_gas_shift` executes the worked example above. Try editing `system/flowsheetDict` to change the reactor $T$ to 1000 K and watch $K_{\mathrm{eq}}$ drop (the WGS is mildly exothermic, so the forward reaction is favoured at lower temperature). At 1000 K the simulator returns $K_{\mathrm{eq}} \approx 1.4$, against the literature 1.40 — the same accuracy as the 800 K case.

## 14.7 Equilibrium maps and the temperature approach

A single Gibbs solve answers "what is the equilibrium composition at *this* $(T, P)$?". Sweep the solve over a $T \times P$ grid and you get an *equilibrium map* — iso-lines of a chosen quantity (a product mole fraction, an element yield) over the operating plane. This is the picture that explains, at a glance, why an industrial reactor fixes its temperature and pressure in a narrow window. Take ammonia synthesis, $N_2 + 3H_2 \rightleftharpoons 2NH_3$:

- the reaction is exothermic, so by Le Chatelier the equilibrium $NH_3$ fraction *rises as $T$ falls* — the contours climb toward the cold side;
- it consumes moles ($\Delta n = -2$), so the fraction *rises with pressure* — the contours climb with $P$;
- yet the plant runs *hot* (400–500 °C) because the iron catalyst is inert below ∼400 °C: without a rate there is no approach to that beautiful cold equilibrium at all.

The map makes this tension a single image: the equilibrium optimum sits in one corner, the industrial window (a labelled box) sits where the catalyst can work, and the arrow between them *is* the compromise every ammonia plant lives with. Choupo draws the map with the `gibbsMap` property operation (the Explorer's "Equilibrium map" view is the same engine, interactive); unconverged grid cells are marked, never interpolated over.

**The temperature approach to equilibrium.** A real reactor does not reach equilibrium at its outlet temperature; a rate-limited, catalyst-limited bed falls short. The classical engineering device for this — the one a commercial equilibrium reactor exposes without explaining — is the *temperature approach* $\Delta T$: evaluate the *reaction* equilibrium at $T + \Delta T$ while the physical state (enthalpy, any condensation, the energy balance) stays at the real $T$. For an exothermic reaction the equilibrium conversion falls with temperature, so the conversion a real bed reaches equals the equilibrium conversion of a *higher* temperature: $\Delta T$ is that distance, in kelvin, and industry quotes it routinely (ammonia beds ∼10–20 °C, methanol ∼10–30, shift converters ∼5–25).

Choupo implements it exactly (Song's convention, ratified by the design forum, `docs/design/gibbs-map-forum-2026-07-02.md`): the outlet is a legitimate *non-equilibrium* state — composition $n(T + \Delta T)$, physical state $(T, P)$ — so its enthalpy is $H = \sum_i n_i(T + \Delta T)\, h_i(T, P)$, evaluated at the physical $T$, and its affinity at $T$ is nonzero (that nonzero affinity *is* the distance from equilibrium, not a bug). Two honesty rules the forum demanded:

> *Intuition.* **$\Delta T$ is calibrated, never predicted.** It lumps kinetics, catalyst activity and transport into one empirical number you fit against a plant measurement — *all models are wrong, some are useful.* A single global $\Delta T$ is meaningful for one dominant reaction (ammonia, methanol); applied to a multi-reaction pool it shifts reactions in physically inconsistent directions (in a reformer, methane reforming and the water-gas shift have *opposite* thermicity and different real approaches), so Choupo announces the global-only limitation aloud. And at high pressure an ideal-gas Gibbs kernel will let $\Delta T$ silently absorb the missing fugacity corrections — announced too, so a student never calibrates non-ideality and calls it kinetics. The value is echoed at every point of consumption — the streams table, the KPI epilogue, a column in the map CSV, the anchor-KPI names — because a $\Delta T$ that survives unnoticed in a copied case file is the classic silent-fudge accident.

# 15 Distillation column, Wang-Henke bubble-point

*What you'll learn.* The simulator's column model: stage-by-stage equilibrium, CMO (constant molar overflow), bubble-point inner loop, $K$-value outer loop. Robust on ideal systems; struggles near azeotropes.

## 15.1 Stage model

For a column with $N_s$ equilibrium stages numbered 1 (condenser) to

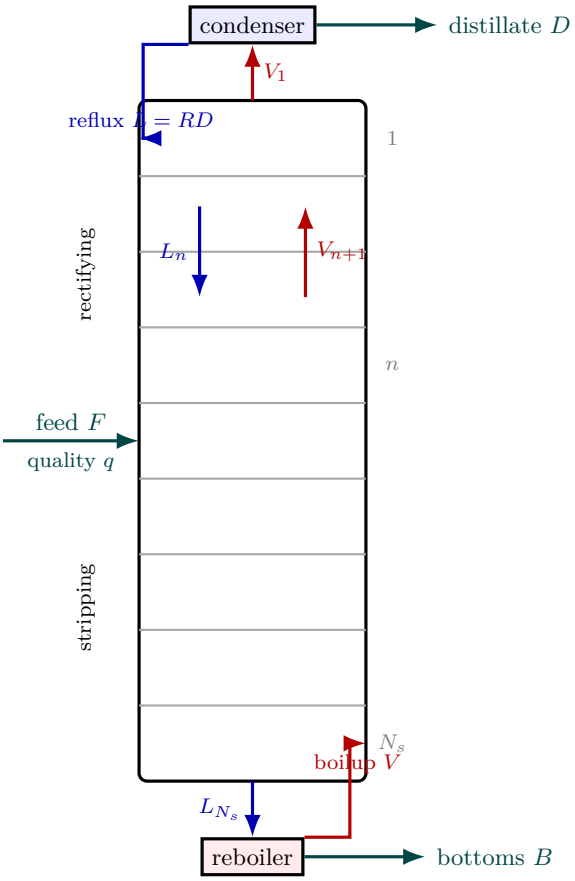



**Figure 65:** Anatomy of the distillation column the stage model describes. Stages are numbered 1 (condenser) at the top to $N_s$ (reboiler) at the bottom. Liquid (blue) cascades *down* stage to stage, vapour (red) rises *up*; on every tray the leaving streams are in equilibrium, $y_{n,i} = K_{n,i} x_{n,i}$. The condenser returns reflux $L = RD$ and draws distillate $D$; the reboiler returns boilup and draws bottoms $B$. The feed enters at its own stage and divides the column into a *rectifying* section above and a *stripping* section below — the feed quality $q$ sets how the internal flows step across it. *Constant molar overflow* is the simplifying assumption that the internal $L_n, V_n$ are constant *within* each section ($L = RD$, $V = (R{+}1)D$ in the rectifier), replacing a per-stage energy balance — exact for equal molar latent heats, forgiving otherwise.

$N_s$ (reboiler), each interior stage $n$ satisfies, at steady state:

- Component balance: $L_{n-1}\, x_{n-1,i} + V_{n+1}\, y_{n+1,i} = L_n\, x_{n,i} + V_n\, y_{n,i}$.
- Equilibrium: $y_{n,i} = K_{n,i}(T_n, P_n, \mathbf{x}_n)\, x_{n,i}$.
- Stoichiometry: $\sum_i x_{n,i} = 1$, $\sum_i y_{n,i} = 1$.
- Energy balance (here simplified: CMO).

## 15.2 Constant molar overflow

CMO replaces the per-stage energy balance with the assumption that liquid and vapour internal flows are constant within each section:

$$L_n = L = R\,D, \quad V_n = V = (R+1)\,D \quad \text{(rectifying section)}, \tag{337}$$

where $R$ is the reflux ratio and $D$ the distillate rate. For the stripping section, $L' = L + qF$, $V' = V - (1-q)F$ with $q$ the feed quality. This is exact only for ideal-enthalpy mixtures, but remarkably forgiving even when it is wrong.

The feed quality is the fraction of a mole of feed that joins the liquid on the feed stage,

$$q = \frac{H_{V,\mathrm{sat}} - H_F}{H_{V,\mathrm{sat}} - H_{L,\mathrm{sat}}}, \tag{338}$$

with $H_F$ the feed's enthalpy and $H_{L,\mathrm{sat}}$, $H_{V,\mathrm{sat}}$ its composition as a saturated liquid (at its bubble point) and a saturated vapour (at its dew point), all on the enthalpy surface the column's streams are priced on: $q > 1$ for a subcooled liquid, which condenses vapour on the feed stage, $q < 0$ for a superheated vapour. The engine takes it so for every single-phase stream feed and announces it whenever it differs from $1 - \beta$; a two-phase feed keeps $q = 1 - \beta$, its resolved liquid fraction (and so does a feed whose package resolves reactive equilibrium, announced, because its saturation points are the reactive flash's). The equilibrium $K_{n,i}$ of every stage is the package's at the vapour that stage sends up, $K_{n,i}(T_n, P_n, \mathbf{x}_n, \mathbf{y}_n)$: under a non-ideal vapour the fugacity coefficients belong to $\mathbf{y}_n$, not to $\mathbf{x}_n$.

## 15.3 Wang-Henke bubble-point method

The algorithm (Wang & Henke 1966, [71]) breaks the coupled system into two nested loops:

1. Guess a temperature profile $T_n$, $n = 1, \ldots, N_s$.
2. For each stage, given $T_n$ and the current $\mathbf{x}_n$, compute $K_{n,i}$ and a new $\mathbf{x}_n$ from the tri-diagonal

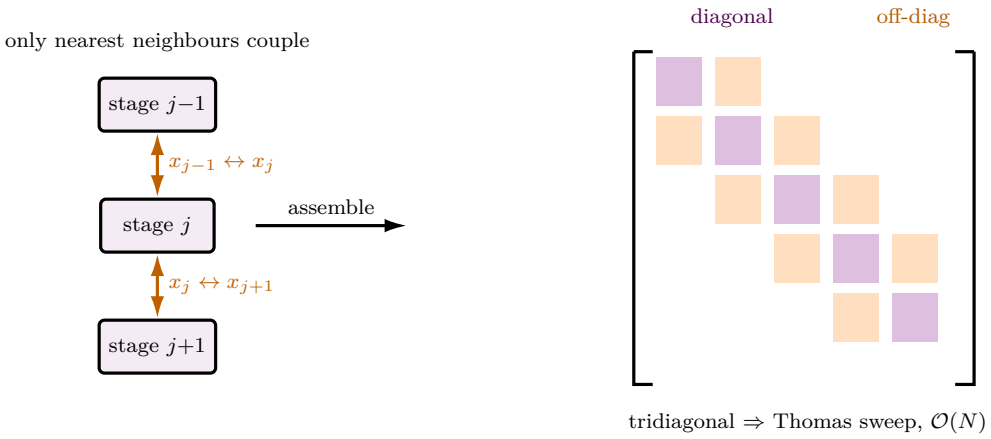



**Figure 66:** Why a column's component balance is *tridiagonal*. Around stage $j$ the steady balance ties only three liquid compositions — $x_{j-1}$ descending in, $x_j$ leaving, $x_{j+1}$ rising in (one species shown). Because each stage couples to *just its two neighbours*, stacking the $N$ balances gives a matrix with non-zeros only on the main diagonal and the two adjacent ones: a banded, tridiagonal system, one per component. Such a system is solved exactly by the Thomas algorithm in a single forward-then-backward sweep at $\mathcal{O}(N)$ cost — no general matrix inversion — which is what Wang–Henke's inner step and the absorber cascade both exploit. The simultaneous MESH Newton keeps the same band structure in its Jacobian, which is why even the rigorous column stays cheap per iteration.

   component-balance system.
3. For each stage, update $T_n$ by solving the bubble-point equation $\sum_i K_{n,i}(T_n) x_{n,i} = 1$ (Newton-1D in $T_n$).
4. Iterate steps 2–3 until the state the next pass would start from satisfies the *balances*, not merely a small step: every stage's component balance and the column's balance over its ports closed to $10^{-9}$ of the feed, every bubble point $|\sum_i K_{n,i} x_{n,i} - 1| \le 10^{-9}$, and the step $\max |x^{(k+1)} - x^{(k)}|$ under the declared `compositionTol`.

This is successive substitution, and it converges *linearly*: the error shrinks by a roughly constant factor $\rho$ per pass (about 0.65 on `column01`, close to 1 on a close-boiling splitter). The distance to the answer is then about $\Delta x/(1-\rho)$, so a stop taken on the step alone leaves the balances open by far more than the step whenever $\rho$ is near 1 — which is why the stop is taken on the balances, normalised by the feed they telescope to.

The simulator's implementation is in `src/unitOperations/distillation/DistillationColumn.cpp`.

## 15.4 When this method struggles

- **Azeotropes.** If $K_i \to 1$ for the key components, the bubble-T solve becomes ill-conditioned and the outer loop can pass through the azeotrope into a non-physical region. The ethanol/water case at $x_1 \approx 0.94$ is the canonical trouble spot.
- **High reflux.** The CMO assumption is exact only if molar latent heats are equal; at $R \gg 1$ small enthalpy differences accumulate, and stage temperatures drift.

Both remedies are shipped. For the azeotrope, `model simultaneous;` — the equation-oriented CMO Newton — stays physical where the sweep does not (`column03_azeotrope_mesh`). For the CMO drift, `model fullMESH;` (alias

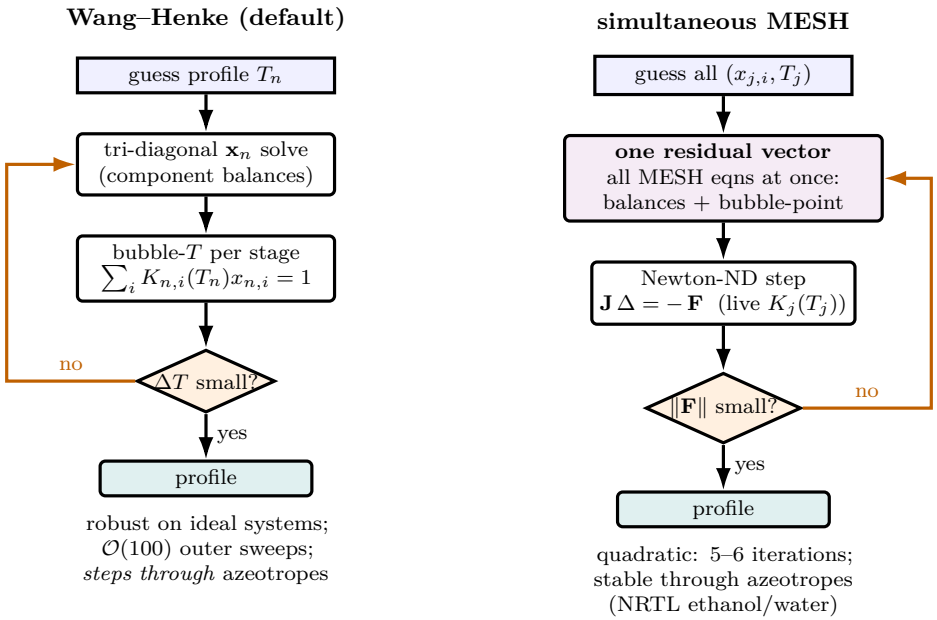



**Figure 67:** Two ways to solve the same column. *Left* — **Wang–Henke** (the default) decouples the problem into nested loops: with a guessed temperature profile it solves a tri-diagonal component-balance system for the stage compositions, then updates each stage temperature from its own bubble-point equation, and sweeps until the profile stops moving. Simple and robust on ideal mixtures, but slow (∼100 sweeps) and prone to stepping *through* an azeotrope into a non-physical region. *Right* — `model simultaneous` (the rigorous MESH Newton) packs *every* stage equation into one residual vector and drives the whole thing to zero with the project's one $n$-D Newton (live $K_j(T_j)$). It converges quadratically in 5–6 iterations and stays physical on azeotropic systems — refusing to invent a "past the azeotrope" answer that does not exist. Same column, same thermodynamics; only the *shape* of the iteration differs.

`NaphtaliSandholm`) adds the per-stage energy balances of Naphtali-Sandholm [72], seeded from the converged CMO profile and validated in `column07_naphtaliSandholm` against the original 1971 paper's Problem 2.

> **Runnable case.** `tutorials/steady/distillation/column01_benzene_toluene` — the ideal case, converges in 30–50 outer iterations.
> `tutorials/steady/optimisation/sensitivity01_column_reflux` — the same column swept over $R \in [1.5, 5]$ by the outer driver.

# 16 Gas absorption and stripping: the coupled stage cascade

*What you'll learn.* An absorber washes a solute out of a gas with a liquid solvent; a stripper does the reverse, blowing a solute out of a liquid with a gas. Both are counter-current multistage cascades, and both are governed by one dimensionless group — the absorption factor $A = L/(KV)$[73]. This chapter sets up the per-stage balance as a tridiagonal system (solved by the Thomas algorithm, not the closed-form Kremser), then adds the *energy* balance that the textbook Kremser ignores — and which turns out to dominate: the heat of absorption warms the column, raises $K$, and self-limits the very absorption that releases it.

## 16.1 The counter-current cascade and the absorption factor

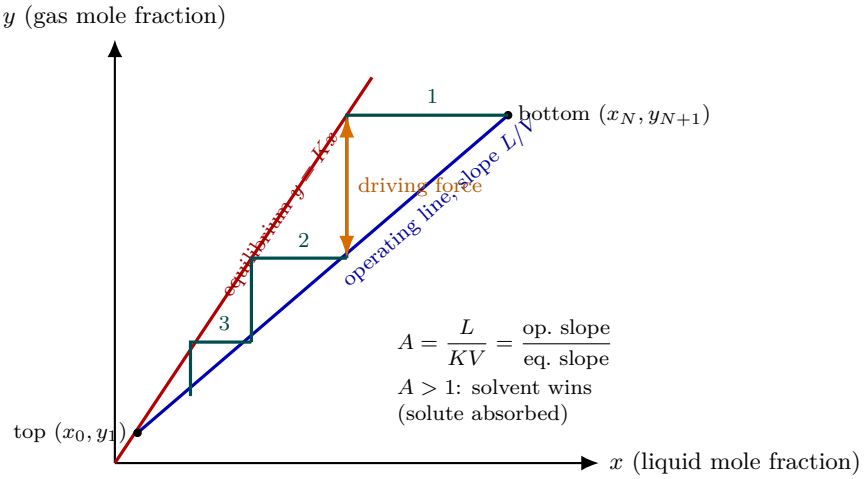



**Figure 68:** The McCabe–Thiele staircase for a counter-current absorber. The *equilibrium line* $y = Kx$ (red) says what gas and liquid *leaving a stage* must satisfy; the *operating line* (blue), slope $L/V$, is the mass balance tying the streams *passing between* stages. For an absorber the operating line sits *above* the equilibrium line, and the vertical gap between them is the driving force that pulls solute from gas into liquid. Each theoretical stage is one step of the staircase — horizontal to the operating line (a passing mass balance), vertical to the equilibrium line (a stage equilibrium) — so the number of steps from bottom to top is the stage count. The single group that governs it all is the absorption factor $A = L/(KV)$, exactly the ratio of the two slopes: $A > 1$ widens the staircase and the solvent wins; $A < 1$ pinches it shut and the solute escapes. Choupo solves the same balance as a tridiagonal (Thomas), then *couples in the self-heating energy balance* the dry staircase ignores.

Number the equilibrium stages 1 (top) to $N$ (bottom). Lean solvent ($L$) enters the top, rich gas ($V$) enters the bottom; they pass counter-currently. On each stage the leaving streams are in equilibrium, $y_{i,j} = K_i(T_j)\, x_{i,j}$. A component balance around stage $j$ (constant molar $L$ and $V$) relates three neighbouring liquid compositions — $x_{j-1}, x_j, x_{j+1}$ — so the whole column is a *tridiagonal* system in $\mathbf{x}_i$, one per component. The controlling group is the *absorption factor*

$$A_i \;=\; \frac{L}{K_i V}, \tag{339}$$

the ratio of the solvent's capacity to carry the solute ($L$) to the gas's tendency to keep it ($K_i V$). $A_i > 1$ means the solvent wins and the solute is absorbed; $A_i < 1$ means it escapes. The classic Kremser equation writes the recovery in closed form as a function of $A_i$ and $N$; Choupo instead solves the tridiagonal directly (the Thomas algorithm), which costs nothing extra and generalises cleanly to the non-isothermal case below.

## 16.2 Why the energy balance matters

Dissolving a gas releases its heat of absorption (the negative of the

heat of solution — here recovered from the Henry's-law van't Hoff slope, Chapter 3's sibling). Choupo solves a *second* tridiagonal, in the stage temperatures $T_j$, with that heat as a source, and iterates the two balances to convergence (falling back to isothermal cleanly when no heat data exist). The coupling is a vicious circle:

$$\text{absorb} \;\Rightarrow\; \text{heat released} \;\Rightarrow\; T\!\uparrow \Rightarrow\; K(T)\!\uparrow \Rightarrow\; A = \tfrac{L}{KV}\!\downarrow \Rightarrow\; \text{less absorption.} \tag{340}$$

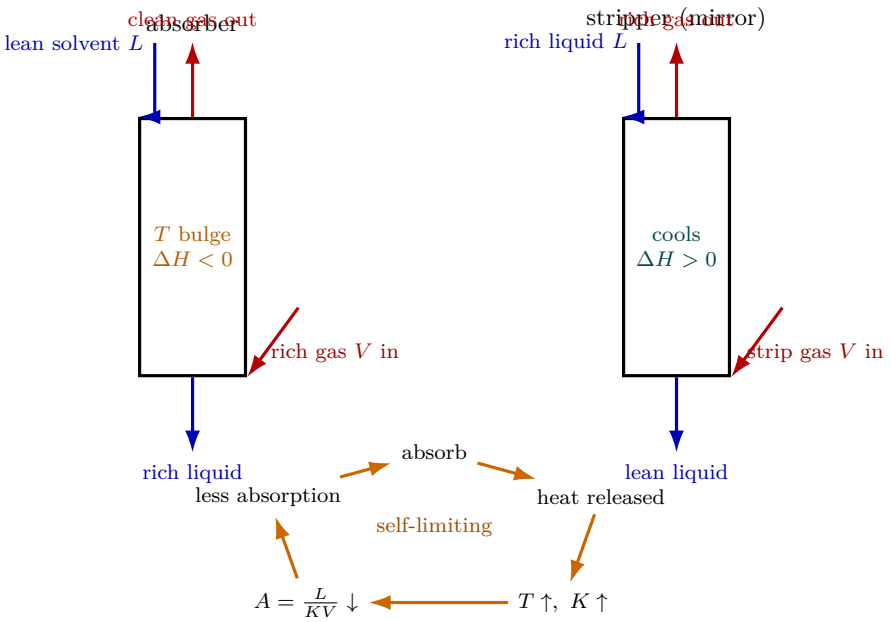



**Figure 69:** The absorber and its mirror, the stripper, plus the feedback loop that makes the energy balance non-optional. In the absorber lean solvent enters the top, rich gas the bottom; the dissolving gas releases its heat of absorption ($\Delta H < 0$), so the liquid *warms* — a "temperature bulge". The stripper is the same cascade inverted (rich liquid down, strip gas up) and its desorption is *endothermic*, so it *cools*. The loop at the bottom is why the isothermal Kremser is dangerously optimistic: absorbing releases heat, which raises $T$, which raises $K$, which *lowers* the absorption factor $A = L/(KV)$, which suppresses the very absorption that released the heat. The column throttles itself exactly where it works hardest — the $NH_3$ scrubber heats ~17 K and recovers 67.5%, not the ~97% an isothermal balance would promise.

> *Intuition.* The isothermal Kremser is dangerously optimistic for a soluble, exothermic gas. The column heats itself up exactly where it is working hardest (a "temperature bulge"), and the warmer it gets the less it can hold — a built-in throttle. This is why real ammonia and amine absorbers are cooled, and why an energy balance is not optional for them. The same machinery, run in reverse, gives the stripper its *endothermic* cooling.

**Worked example 16.1.** Ammonia scrubbing, `absorber01_NH3_water`Six stages, $NH_3$ from air into water. The exotherm ($\Delta H \approx -34$ kJ/mol) heats the liquid by ~17 K (top 309, peak ~318, bottom 315 K), pushing $K$ up and $A$ down, so the recovery is **67.5 %** — far below the ~97 % an isothermal Kremser would promise. The exotherm self-limits the absorption; the energy balance is the whole story.

## 16.3 The stripper: the same cascade, mirrored

A stripper (`stripper01_NH3_water`) is the absorber inverted: liquid feed enters the top, stripping gas the bottom, and the rich stream is now the *gas*. The tridiagonal is identical; only the reported quantities differ. Its controlling group is the *stripping factor*

$$S_i = \frac{K_i V}{L} = \frac{1}{A_i}, \tag{341}$$

and its energy balance is the absorber's with the sign flipped: desorption is endothermic, so the liquid *cools* as it is stripped (the $NH_3$ case drops ~16.5 K, the mirror of the absorber's +17 K). $S_i > 1$ strips, $S_i < 1$ retains.

> **Runnable case.** `tutorials/steady/absorption/absorber01_NH3_water` — 6-stage $NH_3$ scrubber, the exothermic temperature bulge.
> `tutorials/steady/absorption/stripper01_NH3_water` — the mirror, $NH_3$ stripped from water by air, endothermic cooling.

# 17 Crystallisation: the population balance and the method of moments

*What you'll learn.* Every unit operation so far balances how much of each species is present. A crystalliser must also track *how big the crystals are* — the particle-size distribution (PSD) is the product specification, not a by-product. That demands a balance over a new, *internal* coordinate, crystal size $L$: the *population balance*. This chapter builds it from the supersaturation driving force, solves the steady continuous (MSMPR) case in closed form $n(L) = n^0 e^{-L/G\tau}$, and shows how the *method of moments* collapses that distribution to a handful of ordinary numbers — mean size, coefficient of variation, magma density — with no size grid at all. It is the first unit in Choupo whose state lives in a coordinate other than composition.

## 17.1 The driving force: solubility and supersaturation

A solute crystallises only when its dissolved concentration $c$ exceeds the saturation value $c_{\text{sat}}$ — read either from a measured solubility curve on a molecular component (`solubility { coefficients (...) }`, a polynomial in $T$) or, for a dissociating salt, from the ion-activity product derived in §17.2. The ratio

$$S \equiv \frac{c}{c_{\text{sat}}(T)} \tag{342}$$

is the *supersaturation*: $S > 1$ drives both the birth of new crystals (nucleation) and the growth of existing ones. It is the crystalliser's analogue of the distillation column's relative volatility — the single number the whole unit turns on.

The simplest model (`model equilibrium`, the cooling crystalliser) assumes the kinetics are *infinitely fast*: the mother liquor leaves exactly saturated, $S = 1$, and the yield is pure thermodynamics,

$$\text{crystals} \;=\; \text{solute}_{\text{in}} \;-\; c_{\text{sat}}(T_{\text{op}}) \cdot \text{solvent}. \tag{343}$$

This answers "how much" but says nothing about "how big". For that we need the kinetics, and the kinetics need the population balance.

## 17.2 Where the saturation comes from for a salt: the solubility product

Equation (343) — and every model below — needs $c_{\text{sat}}$. For a molecular solute (sucrose, a pharmaceutical) it is read directly from a measured `solubility { coefficients (...) }` curve, a polynomial $c_{\text{sat}}(T)$. For a dissociating *salt* there is no such single curve, and — this is the crux — the electrolyte activity model of §3 *cannot supply one.*

**An activity model is blind to the solid.** The eNRTL or Pitzer model computes the mean ionic activity coefficient $\gamma_\pm(m, T)$ — how non-ideal the *liquid* solution of ions is. It knows nothing about the crystal: the lattice, the chemical potential of the solid. But solubility is a *solid–liquid* equilibrium,

$$\mathrm{NaCl(s)} \rightleftharpoons \mathrm{Na^+} + \mathrm{Cl^-}, \qquad \mu_{\mathrm{NaCl(s)}} = \mu_{\mathrm{Na^+}} + \mu_{\mathrm{Cl^-}},$$

and the position of that equilibrium — the *solubility product*

$$K_{sp} \;=\; a_{\mathrm{Na^+}}\, a_{\mathrm{Cl^-}} \;=\; \big(\gamma_\pm\, m_{\text{sat}}\big)^{\nu} \;=\; \exp\!\Big(-\frac{\Delta G^\circ_{\text{dissol}}}{R_{\text{g}} T}\Big) \tag{344}$$

($\nu = \nu_+ + \nu_-$, $= 2$ for NaCl) depends on $\Delta G^\circ_{\text{dissol}}$, the free energy of the hydrated ions *relative to the crystal.* That difference is a property of the *solid*, to which a liquid-phase $\gamma$-model has no access.

*Intuition.* The salt's solubility is to the crystalliser what a volatile's vapour pressure is to the flash: the pure-phase anchor the activity model does *not* provide. NRTL gives you $\gamma$ for the bubble point but you still supply Antoine's $P^{\text{sat}}$; the eNRTL gives you $\gamma_\pm$ for the saturation but you still supply one $K_{sp}$ (equivalently, one measured solubility). An activity model predicts *non-ideality*, never the reference state of a phase it cannot see.

**One anchor, a whole surface.** Choupo takes the practical route: anchor $K_{sp}$ to a single *measured* solubility $m^\circ_{\text{sat}}$ at 25°C (`solubility 6.144;` in the salt's `electrolyte{}` block), so that, with $\gamma_\pm$ from the model,

$$K_{sp} \;=\; \big(\gamma_\pm(m^\circ_{\text{sat}})\, m^\circ_{\text{sat}}\big)^{\nu}. \tag{345}$$

This looks circular — a solubility in to get a solubility out. The payoff is that the *same* $K_{sp}$ now fixes saturation at *every other* condition: solving

$$\big(\gamma_\pm(m, T, \mathbf{x}_{\text{solv}})\, m\big)^{\nu} \;=\; K_{sp} \tag{346}$$

for $m$ returns $m_{\text{sat}}$ as a function of temperature *and solvent composition.* The model turns one number into the full solubility surface — whereas the molecular route would need a freshly measured curve for every solvent.

**Why store the solubility and not $K_{sp}$ itself.** It is tempting to tabulate the "fundamental" constant $K_{sp}$ directly. Resist it: the $K_{sp}$ back-computed from a solubility is *model-contaminated*. The *same* measured $m^{\circ}_{\mathrm{sat}} = 6.144$ gives $K_{sp} = 34.8$ through the eNRTL ($\gamma_{\pm} = 0.96$) but 38.2 through Pitzer ($\gamma_{\pm} = 1.01$) — a 10% split for one salt, because the two models disagree about non-ideality at 6 molal. A solubility product and its activity-coefficient model are an *inseparable pair*: serious brine parameterisations (Harvie–Møller–Weare; the $\log K$ values in PHREEQC) regress the mineral $K_{sp}$ *jointly* with the ion-interaction terms on the same data, so the $K_{sp}$ is never portable to another convention. The *solubility* is the model-independent observable — you read it off a balance — so it is what the curated `.dat` stores; the engine derives the model-consistent $K_{sp}$ at load time and reports it, so a student who swaps Pitzer↔ eNRTL *sees* $K_{sp}$ shift while the solubility holds. (The standard $\Delta G^{\circ}_{\mathrm{dissol}}$ of (344) is the one genuinely model-free anchor, but routing it through an imperfect $\gamma_{\pm}$ would inject the model's $\sim$2% error into the predicted solubility at the very reference point. We keep the observable exact there and make the model earn its keep on the *extrapolation* away from 25°C and pure water.)

**Temperature: van't Hoff on $K_{sp}$.** Two distinct things move $m_{\mathrm{sat}}$ with $T$: the activity coefficient $\gamma_{\pm}(T)$ (the liquid; §3) and the solubility product $K_{sp}(T)$ itself (the solid–liquid equilibrium constant). The latter obeys van't Hoff,

$$\frac{d\ln K_{sp}}{dT} = \frac{\Delta H_{\mathrm{sol}}}{R_{\mathrm{g}}T^2} \implies K_{sp}(T) = K_{sp}(T_0)\,\exp\!\Big[-\frac{\Delta H_{\mathrm{sol}}}{R_{\mathrm{g}}}\Big(\tfrac{1}{T}-\tfrac{1}{T_0}\Big)\Big], \tag{347}$$

with $\Delta H_{\mathrm{sol}}$ the enthalpy of solution (`dissolutionEnthalpy`; optional — omit it and $K_{sp}$ is held flat in $T$). For NaCl $\Delta H_{\mathrm{sol}} \approx +3.9$ kJ/mol is small, which is precisely why NaCl's solubility is famously almost flat with temperature — and why NaCl is purified by evaporation or by antisolvent, never by cooling.

## 17.3 The two-layer $K_{sp}(T)$: which heat of solution drives van't Hoff?

Equation (347) hides a subtlety that the caustic-soda case forces into the open. The constant `dissolutionEnthalpy` is the *standard* (infinite-dilution) enthalpy of solution $\Delta H^{\infty}_{\mathrm{sol}}$ — the heat absorbed when one mole of the crystal dissolves into an *infinite* amount of water. But van't Hoff on a *saturated* solution is not about diluting to infinity; it is about the equilibrium between the crystal and a liquid that is already at $m_{\mathrm{sat}}$. The thermodynamically correct driver of $d\ln K_{sp}/dT$ at saturation is the *differential* (partial-molar) heat of solution *at that molality*,

$$\Delta H_{\mathrm{diff}}(m_{\mathrm{sat}}) \;=\; \Delta H^{\infty}_{\mathrm{sol}} \;+\; \overline{L}_2(m_{\mathrm{sat}}), \qquad \overline{L}_2 \;=\; L_{\phi} + m\,\frac{\partial L_{\phi}}{\partial m}, \tag{348}$$

where $L_{\phi}(m,T)$ is the salt's *relative apparent molar enthalpy* (the heat-of-dilution curve) and $\overline{L}_2$ its partial-molar sibling — the heat associated with adding one more mole of salt to a solution *already at* $m$, rather than to pure water. At infinite dilution $L_{\phi} \to 0$ and (348) collapses back to $\Delta H^{\infty}_{\mathrm{sol}}$; at a real saturation $\overline{L}_2$ can be large and of *either* sign.

> *Intuition.* Why this matters so much for NaOH. The infinite-dilution figure is $\Delta H^{\infty}_{\mathrm{sol}} \approx -44.5$ kJ/mol — strongly *exothermic*, which through (347) alone would predict solubility *falling* with temperature. Caustic soda does the opposite: its solubility *rises* steeply with $T$ (you concentrate it by evaporation, and it crash-crystallises on *cooling*). The resolution is $\overline{L}_2(m_{\mathrm{sat}})$: at $\sim$29 molal the partial-molar relative enthalpy is large and *positive*, enough to overturn the sign, so $\Delta H_{\mathrm{diff}} > 0$ and the solubility climbs. Use the wrong (infinite-dilution) enthalpy and the model gets the *direction* of the temperature effect backwards.

**Where $L_{\phi}$ comes from: $-RT^2\,\partial g^{ex}/\partial T$.** The heat-of-dilution curve is not a fourth dataset — it is the *temperature derivative of the very activity surface* already in hand. By the Gibbs–Helmholtz identity, writing the dimensionless excess Gibbs energy per mole of salt as $g^{ex} = \nu\,[(1-\phi) + \ln\gamma_{\pm}]$ (§3),

$$L_{\phi}(m,T) \;=\; -\,R_{\mathrm{g}}\,T^2\,\frac{\partial g^{ex}}{\partial T}, \tag{349}$$

which Choupo evaluates by a central finite difference ($\Delta T = 0.05$ K) *on the same Pitzer kernel* that returns $\phi$ and $\gamma_{\pm}$ — one implementation, used identically by the props bench, the activity adapter and the curation fit (`PitzerSingleSalt::Lphi`). The partial-molar $\overline{L}_2$ of (348) is then a second central difference of $L_{\phi}$ in $m$ (`partialMolarRelativeEnthalpy`). Glass-box and sign-safe: no hand-coded enthalpy expansion can drift out of step with the activity coefficient it must be consistent with.

**The Silvester–Pitzer $T$-slots.** Equation (349) needs the *temperature slope* of the virial parameters, not just their 25°C values. The single-salt kernel carries three optional derivatives applied *linearly* (Silvester & Pitzer 1977 [74]),

$$\beta^{(0)}(T) = \beta^{(0)} + \beta^{(0)}_T\,(T-T_0), \quad \beta^{(1)}(T) = \beta^{(1)} + \beta^{(1)}_T\,(T-T_0), \quad C^{\phi}(T) = C^{\phi} + C^{\phi}_T\,(T-T_0), \tag{350}$$

with $T_0 = 298.15$ K, so the 25°C $\gamma_{\pm}/\phi$ anchor is *never moved* (every slope vanishes at $T_0$) and the slots carry *only* the heat-of-dilution information. The dominant long-range slope is already in $A_{\phi}(T)$ (the Born–Bjerrum factor, §3); the three slots add the short-range correction. With all slots zero, $L_{\phi}$ is still computed — but from $A_{\phi}(T)$ alone, the *calorimetrically uncalibrated* guess.

**The `calorimetricFit` gate.** Here is the discipline that keeps the model honest. A $\beta(T)$ slope of zero gives a non-zero $L_\phi$ that is *Gibbs–Helmholtz-consistent but calorimetrically wrong* — consistency is not correctness. So $L_\phi$ is admitted into the energy balance (and into $\Delta H_{\text{diff}}$ above) *only* when the salt's pair row carries `calorimetricFit true;`: the flag is set *by* the curation tool `bin/curate/fit_lphi_pitzer.py`, which regresses the three slots of (350) by exact linear least squares (they enter $L_\phi$ linearly) against the measured relative apparent molar enthalpy $\Phi_L(m, 25°\text{C})$ of Parker's NSRDS–NBS 2 compilation [75], and sets the flag only if the fit passes the gate (relative AAD $< 5\%$ over $m = 0.1$–$6$ mol/kg, skipping the $|\Phi_L| < 50$ cal/mol band where the curve crosses zero twice and a ratio is noise). The row also records `lphiValidityMax`, the upper molality of the fit window; a saturation *beyond* it (NaOH's ∼29 molal is far past a 6-molal fit) still gets the sign-correct $\overline{L}_2$, but the run *announces* the extrapolation in the advisory log rather than pretending to accuracy it does not have. Without the flag, $K_{sp}(T)$ reverts byte-for-byte to the infinite-dilution form (347) — every legacy golden master is untouched. The eNRTL package deliberately does *not* inherit the flag (§4): its $\tau(T) \propto 1/T$ is the uncalibrated form, and the $T$-slots were regressed for the Pitzer surface, not the eNRTL one.

> **Common pitfall.** The gate is per-*kernel* and per-*salt*, not global. Turning on `calorimetricFit` for a salt whose pair row has zero $T$-slots does *not* make the enthalpy correct — it makes the uncalibrated $A_\phi(T)$-only $L_\phi$ believed. The flag is an output of the fitting tool's gate, never a knob the case author flips by hand to "enable enthalpy". If `fit_lphi_pitzer.py` fails its 5% gate, the flag stays `false` and the salt keeps the infinite-dilution van't Hoff form: an honest reduced model, not a silent wrong one.

**Power and limit.** Power: a *single* measured solubility plus a *single* calorimetric refit turn one number into the full $m_{\text{sat}}(T, \mathbf{x}_{\text{solv}})$ surface *with the correct temperature direction* — and the same $L_\phi$ surface feeds the crystalliser's heat of crystallisation $\Delta H_{\text{cryst}}(m_{\text{sat}}) = -\Delta H_{\text{diff}}(m_{\text{sat}})$ and any evaporator's heat of concentration, all from one Gibbs–Helmholtz-consistent kernel. Limit: the $T$-slots are *linear* (Silvester–Pitzer's first correction), fitted to 25°C $\Phi_L$ data on the 0–6 molal window; a saturation far outside it is honestly flagged, and the heat-*capacity* of solution ($\partial^2 g^{ex}/\partial T^2$, the $J_\phi$ channel) is a later slice. The model gets the *sign and order of magnitude* of the $T$-dependence right where it matters — it is not a substitute for a full $\beta(T)$ surface regressed over a wide temperature range.

**The antisolvent: drowning-out.** The most striking use of (346) is *antisolvent* crystallisation. Adding ethanol to a brine lowers the mixture dielectric constant $\varepsilon$ (from ∼78 to ∼24); the ions, less screened, attract more strongly, so $\gamma_\pm$ *rises* steeply and (346) forces $m_{\text{sat}}$ *down*. The mixed-solvent eNRTL captures this through two terms: the Debye–Hückel coefficient $A \propto (\varepsilon T)^{-3/2}$, which grows as $\varepsilon$ falls, and a *Born* term for the energy of moving an ion from water into the poorer solvent. No cooling and no boil-off: the supersaturation $S = c/c_{\text{sat}}$ is created purely by displacing the saturation. In Choupo this is the difference between $m_{\text{sat}} = 6.14$ mol/kg in water and 0.88 mol/kg at 40 % ethanol — the *same* $K_{sp}$, a different $\gamma_\pm$.

With $c_{\text{sat}}$ in hand — by either route — the supersaturation $S = c/c_{\text{sat}}$ is defined and everything below proceeds identically. The kinetics see only $S$; the crystalliser does not care whether it came from a sugar solubility polynomial or from an eNRTL ion-activity product.

## 17.4 The population balance: a balance over size

Let $n(L)\,dL$ be the number of crystals per unit suspension volume with size in $[L, L + dL]$; $n(L)$ is the *population density* [$\#\,\text{m}^{-4}$]. Crystals enter and leave a size interval not only by flowing in and out of the vessel but by *growing* across its boundaries, at the linear growth rate $G = dL/dt$. Writing the balance on $[L, L + dL]$ for a well-mixed vessel of volume $V$ with volumetric throughput $Q$ gives the *population balance equation* (Randolph & Larson [76]):

$$\frac{\partial n}{\partial t} + \frac{\partial (G\,n)}{\partial L} = \underbrace{B^0\,\delta(L)}_{\text{nucleation}} + \frac{Q_{\text{in}} n_{\text{in}} - Q\,n}{V}. \tag{351}$$

The $\partial(Gn)/\partial L$ term is the novelty: a *convection in size space*, crystals marching up the $L$-axis as they grow. Nucleation appears as a source of zero-size particles at $L = 0$ (a boundary condition $n(0) = n^0 = B^0/G$, the nuclei density), where $B^0$ is the nucleation rate [$\#\,\text{m}^{-3}\text{s}^{-1}$].

> *Intuition.* A composition balance is bookkeeping over a *finite* list of species. A population balance is bookkeeping over a *continuum* of sizes — the same conservation logic, but the "species" is now "a crystal of size $L$", and growth lets one continuously turn into the next. This is why crystallisation (and aerosols, polymers, bubbles, cells) needs its own balance: the property that defines the product varies continuously and the process moves material along that continuum.

## 17.5 The MSMPR steady state: the famous straight line

The reference idealisation is the *mixed-suspension, mixed-product-removal* (MSMPR) crystalliser: perfectly mixed, steady, clear feed ($n_{\text{in}} = 0$), product withdrawn at the suspension composition ($Q_{\text{in}} = Q$), and *size-independent*

*growth* (McCabe's $\Delta L$ law, $G \neq f(L)$) with no breakage or agglomeration. With $\tau = V/Q$ the residence time, (351) collapses to

$$G\,\frac{dn}{dL} + \frac{n}{\tau} = 0 \qquad \Longrightarrow \qquad \boxed{n(L) = n^0\, e^{-L/(G\tau)}.} \tag{352}$$

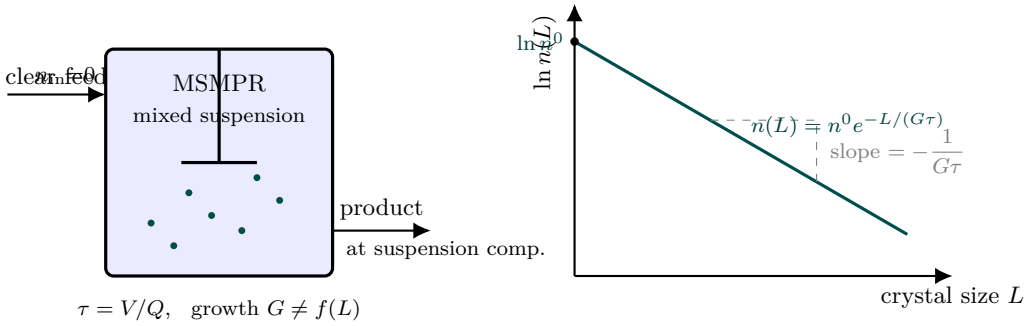



**Figure 70:** The MSMPR crystalliser and its signature diagnostic. Left: a perfectly mixed, steady vessel fed clear liquor ($n_{\text{in}} = 0$) and withdrawn at the suspension composition, with size-independent growth ($G \neq f(L)$). The population balance then collapses to a single exponential $n(L) = n^0 e^{-L/(G\tau)}$. Right: plot $\ln n$ against crystal size $L$ and it is a *straight line* — the single most recognisable plot in crystallisation. The **intercept** hands you $\ln n^0 = \ln(B^0/G)$ (the nucleation-to-growth ratio) and the **slope** hands you $-1/(G\tau)$. Measure a product PSD, plot it this way, and the two kinetic numbers fall straight out.

Plotting $\ln n$ against $L$ gives a *straight line* of slope $-1/(G\tau)$ and intercept $\ln n^0$ — the single most recognisable diagnostic in crystallisation practice. Measure a product PSD, plot it this way, and the slope hands you $G\tau$ while the intercept hands you the nucleation-to-growth ratio $n^0 = B^0/G$.

## 17.6 The method of moments

We rarely need the whole curve; we need numbers — a mean size, a spread, a mass. Define the moments

$$\mu_j \;=\; \int_0^{\infty} L^j\, n(L)\, dL. \tag{353}$$

For the MSMPR exponential (352) they evaluate in closed form (Hulburt & Katz [77]):

$$\mu_j \;=\; n^0\,(G\tau)^{j+1}\, j!. \tag{354}$$

Crucially, when $G$ is size-independent the moment equations *close*: each $\mu_j$ depends only on lower moments and the kinetics, so we never discretise the $L$-axis. This is the method of moments, and it is exactly the kind of hand-rolled, no-grid solution Choupo favours. The moments carry direct physical meaning:

| moment | meaning | MSMPR value |
|---|---|---|
| $\mu_0$ | total number / volume | $n^0 G\tau = B^0\tau$ |
| $\mu_2$ | $\propto$ total surface area | $2n^0(G\tau)^3$ |
| $\mu_3$ | $\propto$ total volume | $6n^0(G\tau)^4$ |

From them the engineering numbers fall out:

$$\underbrace{L_{10} = \frac{\mu_1}{\mu_0} = G\tau}_{\text{number mean}}, \quad \underbrace{L_{43} = \frac{\mu_4}{\mu_3} = 4G\tau}_{\text{mass mean}}, \quad \underbrace{L_{\text{d}} = 3G\tau}_{\text{dominant (mass mode)}}, \tag{355}$$

the magma (suspension) density

$$M_T \;=\; k_v\, \rho_c\, \mu_3, \tag{356}$$

($k_v$ the volume shape factor, $\rho_c$ the crystal density — the *number↔mass bridge* on the component's `solid { rho_p; k_v; }` block), and a coefficient of variation that is a pure number:

$$\text{CV} = \sqrt{\frac{\mu_0\mu_2}{\mu_1^2} - 1} = \sqrt{2-1} = 1 \quad (= 100\%). \tag{357}$$

**Key idea.** An MSMPR crystalliser always produces a PSD with a coefficient of variation of exactly 100 % — independent of kinetics, residence time, everything. It is a structural property of the perfectly-mixed,

size-independent-growth exponential, and a sharp sanity check: if your moments do not give CV = 1, the arithmetic is wrong. Choupo's `crystalliser02_msmpr` reports CV = 1.00.

## 17.7 Kinetics and closing for the supersaturation

Growth and nucleation are empirical power laws in the supersaturation (the FitParameters calibration targets), the standard data-anchored forms:

$$G = k_g\,(S-1)^{g}, \qquad B^0 = k_b\,(S-1)^{b}\,M_T^{j}, \tag{358}$$

the magma exponent $j$ capturing secondary (crystal-assisted) nucleation. Substituting $\mu_3 = 6B^0G^3\tau^4$ and $M_T = k_v\rho_c\mu_3$ closes $\mu_3$ explicitly in $S$:

$$\mu_3^{1-j} = 6\,G^3\tau^4\,k_b\,(S-1)^{b}\,(k_v\rho_c)^{j} \qquad (j<1). \tag{359}$$

The one remaining unknown is the steady supersaturation $S$ itself, fixed by the *solute mass balance*: at steady state the crystal mass produced must equal the solute removed from the liquor,

$$\underbrace{Q\,M_T(S)}_{\text{crystallised}} \;=\; \underbrace{\text{solute}_{\text{in}} - S\,c_{\text{sat}}\,\text{solvent}}_{\text{removed from liquor}}. \tag{360}$$

This is one nonlinear equation in $S$, solved by a bracketed Newton-1D (Chapter 2) on $[1, S_{\text{feed}}]$ — the left side rises with $S$, the right side falls, so the root is unique. The residence time itself is a *result* of the hardware, $\tau = V/Q$, with $Q = \sum_i F_i\bar{V}_{\text{liq},i}$ the volumetric throughput.

> *Intuition.* The finite-kinetics MSMPR refines the equilibrium model in a physically honest way. Because crystallisation takes time and the vessel offers only $\tau$ seconds, the liquor cannot reach saturation — it leaves *supersaturated* ($S>1$). So the MSMPR yield is always *less* than the equilibrium prediction (343); the gap is the price of running at a finite rate. Slower kinetics or a smaller vessel ⇒ higher residual $S$ ⇒ less yield, but also (through $G\tau$) a different crystal size. The same hardware knob, $V$, trades yield against size — the central crystalliser design tension.

**Worked example 17.1.** Sugar MSMPR, `crystalliser02_msmpr`A near-saturated sucrose syrup ($S_{\text{feed}} = 1.66$) is fed to a $V = 5\,\text{m}^3$ cooled vessel at 20 °C; $Q = 1.31\times10^{-3}\,\text{m}^3/\text{s}$ so $\tau \approx 3820\,\text{s}$. The Newton-1D closes (360) at $S = 1.195$, giving $G = 1.95\times10^{-8}\,\text{m/s}$, hence $G\tau = 74.6\,\mu\text{m}$ and

$$L_{\text{d}} = 3G\tau = 224\,\mu\text{m}, \quad L_{43} = 4G\tau = 298.6\,\mu\text{m}, \quad \text{CV} = 1.00,$$

a magma density $M_T = 303\,\text{kg/m}^3$ and a yield of 27.8 % — below the equilibrium figure, because the liquor leaves supersaturated. The ratio $L_{43}/L_{\text{d}} = 4/3$ and CV = 1 are the analytic MSMPR signatures, reproduced to printer precision.

## 17.8 The batch crystalliser: integrating the moments in time

The MSMPR is the *steady* population balance. Its dynamic sibling is the *batch* crystalliser (`batchCrystalliser`, under choupoBatch): a closed vessel charged supersaturated, where the moments evolve in time. With no flow terms, the population balance (351) reduces to the *moment ODEs*

$$\frac{d\mu_0}{dt} = B^0, \qquad \frac{d\mu_j}{dt} = j\,G\,\mu_{j-1} \;\; (j\ge1), \tag{361}$$

nucleation feeding the count $\mu_0$ at $L=0$, growth convecting each moment up from the one below. They close (size-independent $G$) and are integrated by RK4 — the same time-stepping as the batch reactor (Chapter 1). The solute mass balance couples in: the dissolved solute lost is exactly the crystal mass grown,

$$\frac{dn_{\text{solute}}}{dt} = -\,\frac{V\,\rho_c\,k_v}{M_w}\,\frac{d\mu_3}{dt}, \tag{362}$$

so the supersaturation $S = c/c_{\text{sat}}$ falls as crystals form. The packed RK4 state is $(\mu_0,\mu_1,\mu_2,\mu_3,n_{\text{solute}})$. Unseeded, primary nucleation ($j=0$) bootstraps the population from the initial $S>1$; the supersaturation then relaxes toward equilibrium ($S\to1$) — the classic *desupersaturation curve*.

**Worked example 17.2.** Batch sugar, `batch05_crystalliser`A charge at $S_0 = 1.23$ (20 °C) relaxes over 3 h: $S$ falls $1.23\to1.01$, the Sauter mean grows $0\to105\,\mu$m, and 229 kg of the 1249 kg dissolved sucrose crystallises. The dissolved-plus-crystal sucrose mass is conserved to the gram at every step — the moment↔solute coupling closes the balance.

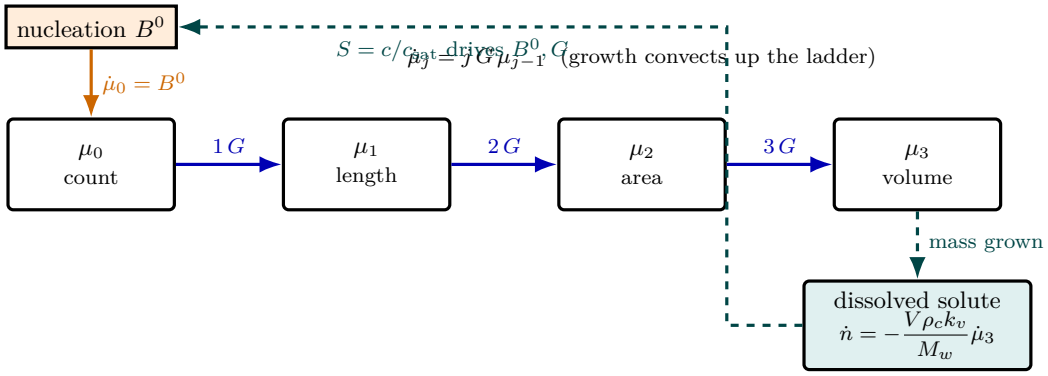



**Figure 71:** The method of moments turns the population-balance PDE into a small ladder of ODEs — no size grid. The state is just four moments: $\mu_0$ (total crystal count), $\mu_1$ (total length), $\mu_2$ (area), $\mu_3$ (volume $\propto$ mass). **Nucleation** $B^0$ injects new crystals at $L$=0, feeding only $\mu_0$ (orange). **Growth** $G$ convects every moment up from the one below, $\dot{\mu}_j = jG\mu_{j-1}$ (blue) — this is why the system *closes* when $G$ is size-independent: each rate needs only lower moments. The crystal volume $\mu_3$ sets the mass that leaves solution, so the dissolved solute falls (teal), the supersaturation $S = c/c_{\text{sat}}$ relaxes toward 1, and as $S$ drops it feeds back to throttle $B^0$ and $G$ — the desupersaturation loop. Choupo packs $(\mu_0, \mu_1, \mu_2, \mu_3, n_{\text{solute}})$ into one RK4 state and marches it in time.

## 17.9 When the moments do not close: the discretised PBE

The method of moments is exact, but only when the closure works. Let the growth law be *size-dependent* —

$$G(L) \;=\; G_0\,(1+\alpha L), \qquad \alpha = \text{sizeFactor}\;\;[1/m] \tag{363}$$

— and the cascade $d\mu_j/dt = j\,G\,\mu_{j-1}$ no longer closes: multiplying the population balance (351) by $L^j$ and integrating, the $\alpha L$ term pulls in a moment one step *above* the row we are integrating,

$$\frac{d\mu_j}{dt} \;=\; j\,G_0\,\big(\mu_{j-1} + \alpha\,\mu_j\big), \tag{364}$$

which is closed only by coincidence ($\alpha = 0$). The same trouble shows up in agglomeration, breakage, and any law where $G$ depends on $L$ explicitly. We need a representation of the PDF $n(L)$ that is *not* a finite list of moments.

The *finite-volume method* (FVM) discretises $n(L)$ on $N$ evenly-spaced size bins,

$$L_k = (k+\tfrac{1}{2})\,\Delta L, \quad k = 0,\dots,N{-}1, \quad \Delta L = L_{\max}/N, \tag{365}$$

with $n_k \equiv n(L_k)$ the cell-centred number density. The steady MSMPR PBE (351) for an arbitrary $G(L)$ reads

$$\frac{d}{dL}\big[G(L)\,n(L)\big] + \frac{n(L)}{\tau} \;=\; 0. \tag{366}$$

First-order upwind on the bin boundary gives the cell-to-cell recurrence

$$n_k \;=\; n_{k-1}\,\frac{G_{k-1}}{G_k + \Delta L/\tau}, \qquad k \ge 1, \tag{367}$$

seeded by the boundary condition $n_0 = B^0/G(0)$. Moments are then *post-computed* by quadrature,

$$\mu_j \;\approx\; \sum_k L_k^j\,n_k\,\Delta L, \tag{368}$$

and the outer closure on the supersaturation $S$ is the same Newton-1D on the solute mass balance ($\rho_c k_v \mu_3$ = solute consumed) that closes the MSMPR.

**Sanity vs MSMPR.** With $\alpha = 0$ the FVM scheme above reduces to a geometric sequence whose continuous limit is $n(L) = n^0\,e^{-L/(G\tau)}$ — the MSMPR exponential. The discretisation error is $\mathcal{O}(\Delta L/(G\tau))$, so 64 bins already give $\sim 5\,\%$ tail error and 256 bins close the gap to $\sim 1\,\%$. Tutorial `crystalliser03_pbe_size_independent` runs the FVM with $\alpha = 0$ against the same kinetics as `crystalliser02_msmpr` and reproduces every observable ($L_{43}$, CV, $M_T$, yield) to better than $1\,\%$ — the numerical sanity check before the model is trusted in the regime where the analytic answer does not exist.

**The payoff.** Tutorial `crystalliser04_size_dependent_growth` repeats the same case with $\alpha = 500\,\mathrm{m}^{-1}$ (+50 % growth enhancement at $L = 1\,\mathrm{mm}$). The big crystals out-grow the small ones: the $n(L)$ profile shifts visibly toward larger sizes, $L_{43}$ climbs $298.6 \to 329.4\,\mu\mathrm{m}$ (+10 %), CV widens $1.00 \to 1.04$, and the magma density rises $306 \to 311\,\mathrm{kg/m}^3$. The closed-form MoM cannot deliver any of these numbers; the FVM does, on the same scaffold of kinetics + vessel + supersaturation closure.

**What does NOT do.** The current FVM is restricted to primary nucleation ($j = 0$ in $B^0 = k_b(S-1)^b M_T^j$); a secondary $j > 0$ would couple $B^0$ to $\mu_3$ through $M_T$, demanding an inner iteration on the boundary condition. Agglomeration and breakage (integral source terms in the PBE, the third reason the moments do not close) are also out of scope — the FVM scaffold is ready for them (per-cell birth and death terms), but the kernel infrastructure is not.

> **Runnable case.** `tutorials/steady/crystallisation/crystalliser01_sugar` — the equilibrium (infinite-kinetics) yield model, `model equilibrium`.
> `tutorials/steady/crystallisation/crystalliser02_msmpr` — the continuous MSMPR population balance by the method of moments, `model MSMPR`; the profile is the $n(L)$ line and the product PSD.
> `tutorials/steady/crystallisation/crystalliser03_pbe_size_independent` — the discretised PBE on a size grid, `model FVM` with $\alpha = 0$: sanity validation against MSMPR, same observables within $\sim 1\,\%$.
> `tutorials/steady/crystallisation/crystalliser04_size_dependent_growth` — the FVM with $\alpha > 0$: the cut that justifies the discretised solver, where the MoM closure fails.
> `tutorials/batch/reactor/batch05_crystalliser` — the dynamic batch PBE (choupoBatch): the moments + solute integrated in time, the desupersaturation curve.

# 18 Spiral-wound membrane: solution-diffusion and concentration polarisation

*What you'll learn.* Pressure-driven membranes (reverse osmosis, nanofiltration) separate without a phase change: push a feed against a film and water crosses while salt is held back. Two coupled physics set the performance — *solution-diffusion* (how water and solute permeate the membrane) and *concentration polarisation* (how rejected solute piles up at the wall and fights back through osmotic pressure). This chapter derives the local flux, shows why it needs an inner Newton, and marches it down the feed channel. It is the research bridge of the simulator — Vítor's NF/RO domain — and the consumer of the Pitzer osmotic model of Chapter 3.

## 18.1 Solution-diffusion: two independent fluxes

In the solution-diffusion picture [78] the membrane is a dense film; water and solute dissolve into it and diffuse down their own potential gradients, so their fluxes are *independent*. The water (volume) flux is driven by the net pressure — the applied $\Delta P$ minus the osmotic back-pressure $\Delta\pi$ it must overcome:

$$J_w \;=\; A_w\,(\Delta P - \Delta\pi), \tag{369}$$

with $A_w$ the membrane's water permeability. The solute flux is driven by the concentration difference across the membrane, independent of pressure:

$$J_s \;=\; B_s\,(c_m - c_p), \tag{370}$$

with $B_s$ the solute permeability, $c_m$ the concentration at the membrane wall and $c_p$ in the permeate. These two coefficients $(A_w, B_s)$ — carried per membrane in `data/standards/assets/` — are the whole material model; a tight RO membrane has a tiny $B_s$, a loose NF membrane a larger one (hence NF's partial rejection).

The permeate concentration is not independent: the solute that crosses is swept along by the water, so $c_p = J_s/J_w$. Substituting (370),

$$c_p = \frac{B_s(c_m - c_p)}{J_w} \quad\Longrightarrow\quad \boxed{\,c_p = \frac{B_s}{J_w + B_s}\,c_m.\,} \tag{371}$$

Already a key result: as $J_w$ grows (high pressure) $c_p \to 0$ — more water dilutes the fixed solute leak, so *rejection improves with pressure*.

## 18.2 Where the coefficients come from — and where the model ends

Equations (369)–(370) look phenomenological, but they follow from ONE structural premise that Wijmans & Baker [78] place at the centre of the model: **inside a dense membrane the pressure is uniform** (equal to the feed-side pressure), so the chemical-potential gradient that drives permeation is expressed entirely as a *concentration* (activity) gradient. A pore-flow membrane is the exact mirror — uniform concentration, pressure gradient — and the two pictures give different, testable flux laws. From that premise, water transport integrates to $J_w \propto \left(1 - e^{-\bar{V}_w(\Delta P - \Delta\pi)/RT}\right)$, whose linearisation (the exponent is $\ll 1$ at process conditions) is (369) with

$$A_w \;=\; \frac{D_w\,K_w\,c_w\,\bar{V}_w}{R\,T\,\ell}, \tag{372}$$

diffusivity × solubility × molar volume over $RT$ and film thickness $\ell$: the "permeability" is not a fudge factor but a composite of measurable membrane properties — which is why $A_w$ is a *material* constant in `data/standards/assets/` and not a per-case knob. The same integration for the solute (whose $\bar{V}_s\Delta P/RT$ term is negligible) yields (370): solute flux blind to pressure, the signature the classic RO experiments confirm.

*Intuition.* **Know where the model ends.** Wijmans & Baker close their review exactly where nanofiltration lives: between dense RO films (solution-diffusion) and open UF pores (pore-flow) sits a transition at virtual pore sizes of roughly 5–10 Å. A loose NF membrane (the catalogue's NF270) is *in* that transition: solution-diffusion with a fitted $(A_w, B_s)$ still describes its water flux and partial rejection, but charge effects and convective coupling grow underneath — which is why the catalogue also carries `NF270_dspmde` (the Donnan-steric, charge-aware sibling, §19) and why the two give different answers for multivalent ions; and why a feed declared ion by ion has a third law, `transport SDEM;` (§20), which keeps the per-ion permeances and adds only the electric field that makes the permeate electroneutral. Choosing between them IS a modelling decision the case author owns.

## 18.3 Concentration polarisation: the film model

Water leaves through the membrane; the solute it rejects does not, so solute accumulates in a thin boundary layer at the wall. At steady

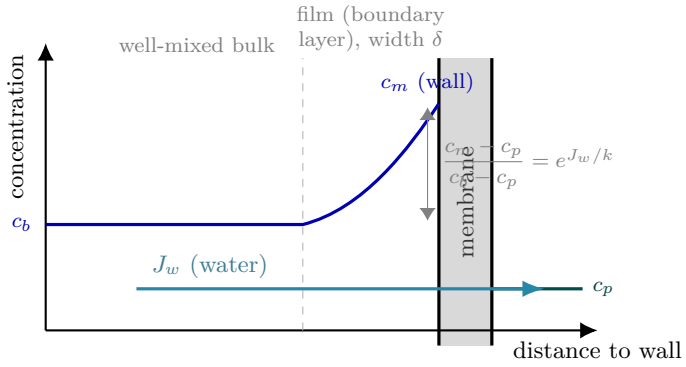



**Figure 72:** Concentration polarisation, the wall fighting back. Water crosses the membrane ($J_w$, cyan), but the rejected solute does not — it piles up in a thin film of width $\delta$ at the wall, so the wall concentration $c_m$ swells *above* the well-mixed bulk $c_b$ (blue). The film model balances convection toward the wall against back-diffusion (coefficient $k$) and gives the polarisation factor $E = e^{J_w/k}$: poor mixing (small $k$) or high flux makes $c_m$ tower over $c_b$. Because the osmotic back-pressure is set by $c_m$, not $c_b$, polarisation is self-limiting — pushing harder raises $J_w$, which raises $c_m$, which raises $\Delta\pi$ and pushes back. The permeate $c_p$ (teal) is the small leak that the inner Newton closes against.

state the convective solute carried toward the membrane balances back-diffusion plus the solute that permeates. Integrating across the film of mass-transfer coefficient $k$ gives the *film model*:

$$\frac{c_m - c_p}{c_b - c_p} \;=\; \exp\!\left(\frac{J_w}{k}\right), \tag{373}$$

where $c_b$ is the bulk (channel) concentration. The exponential $E = e^{J_w/k}$ is the *polarisation factor*: poor mixing (small $k$) or high flux makes $c_m$ swell far above $c_b$.

Combining the permeate relation (371) with the film model (373) eliminates $c_p$ and gives the wall concentration in closed form,

$$c_m = \frac{E\, c_b}{1 + \dfrac{B_s}{J_w + B_s}\,(E - 1)}, \tag{374}$$

exactly the expression `localFluxes` evaluates (with $\varphi = J_w + B_s$).

## 18.4 Polarisation of a multi-ionic solution: one field at the interface

The film model above has one coefficient $k$ and therefore one diffusivity. A solution declared ion by ion — $Na^+$, $Cl^-$, a trivalent dye — has several, and ions of different mobility cannot pile up at the wall independently: the fast one would run ahead, charge would separate, and the resulting field would pull the slow one along. Two answers exist. One integrates the extended Nernst–Planck equations across a film of assumed thickness $\delta$, and the thickness is the trouble: it is not measured, and Bowen and Mohammad had to assign it from the slowest ion. The other, due to Geraldes and Afonso [79], keeps the conventional mass-transfer correlation of the module and adds *one* unknown, the electrical potential gradient $\xi$ at the feed-solution/membrane interface. It is the model Choupo computes the wall from, in one place (`membrane/massTransfer/Polarisation`), for every transport law.

At the interface the total flux of ion $i$, which the permeate carries away as $J_v C_{i,p}$ (Eq. 1 of the paper), is the sum of back-diffusion, convection and migration:

$$J_v\, C_{i,p} \;=\; -\,k^*_{c,i}\,(C_{i,m} - C_{i,b}) \;+\; J_v\, C_{i,m} \;-\; z_i\, C_{i,m}\, D_i\, \frac{F}{R_g T}\,\xi, \tag{375}$$

with $k^*_{c,i} = k_{c,i}\,\Xi_i$ the mass-transfer coefficient of ion $i$ corrected for the suction of the permeation (Eqs. 2–4, 7), and electroneutrality of the wall closing the system (Eq. 8):

$$\sum_i z_i\, C_{i,m} = 0. \tag{376}$$

Each $k_{c,i}$ comes from the module's own correlation, $Sh_i = a\, Re^{b}\, Sc_i^{c}$, evaluated on the ion's *own* diffusivity ($Sc_i = \nu/D_i$) — the main assumption of the model is that the coefficient of one ion does not depend on the others. Two correction factors are available:

$$\Xi_i = \varphi_i + \left(1 + 0.26\,\varphi_i^{1.4}\right)^{-1.7}, \qquad \varphi_i \equiv \frac{J_v}{k_{c,i}}, \quad \varphi_i < 20, \tag{377}$$

$$\Xi_i = \varphi_i + \frac{\varphi_i}{e^{\varphi_i} - 1}, \tag{378}$$

the first the semi-empirical correlation of Geraldes and Afonso [80] (independent of flow regime and geometry), the second the film theory's, which the paper reports usable at low fluxes ($\varphi_i \leq 1$). In the case dictionary they are `suctionCorrection GeraldesAfonso2006` and `filmTheory` (`none` keeps the uncorrected impermeable-wall coefficient), and the coupling is `ionCoupling electroneutral` or `none`, declared in a `polarisation {}` block of the unit's operation.

**Solving it.** For a given $\xi' \equiv F\xi/(R_gT)$ every wall concentration is explicit,

$$C_{i,m} \;=\; \frac{k^*_{c,i}\,C_{i,b} - J_v\,C_{i,p}}{k^*_{c,i} - J_v + z_i D_i\,\xi'}, \tag{379}$$

and $k^*_{c,i} - J_v = k_{c,i}(\Xi_i - \varphi_i) > 0$ for both corrections. Electroneutrality (376) is then one scalar equation $g(\xi') = 0$; on the interval where every denominator keeps its sign, bounded below by the cations and above by the anions, $g$ falls monotonically from $+\infty$ to $-\infty$, so the root is bracketed, bisected and polished by Newton. If no such interval leaves every $C_{i,m}$ positive the wall is refused by name — a wall concentration *below* the bulk is legitimate (the paper's own Table 1 has $Cl^-$ with negative rejection and negative $\Gamma$), a negative one is not. What is published per ion is the polarisation index $\Gamma_i \equiv (C_{i,m} - C_{i,b})/C_{i,b}$ and, when coupled, $\xi$ in V/m.

**The reduction that keeps the film model.** With the film-theory correction (378) and $z_i = 0$ (or $\xi = 0$), $k^* - J_v = J_v/(e^{\varphi} - 1)$ and $k^*C_b - J_vC_p = J_v\big(e^{\varphi}C_b/(e^{\varphi} - 1) - C_p\big)$, so (379) collapses to $c_m = c_p + (c_b - c_p)\,e^{J_v/k}$ — exactly (373). The defaults (`filmTheory`, `none`) are therefore the film model of the previous subsection to round-off, which is how every membrane case recorded before the model existed keeps its answer.

**The single-salt identity.** For one salt $A^{z_1}B^{z_2}$ the two wall concentrations are tied by electroneutrality and the coupled solution has the closed form (the paper's Eq. 20)

$$\Gamma = \frac{R}{\eta\,\Xi_1/\varphi_1 + (1-\eta)\,\Xi_2/\varphi_2 - R}, \qquad \eta = \frac{1}{1 + (z_1/|z_2|)(D_1/D_2)}, \tag{380}$$

with $R$ the intrinsic rejection and $\varphi_1, \varphi_2$ related to the salt's $\varphi = J_v/k_c$ (on the salt diffusivity $D_S = D_1D_2(z_1 + |z_2|)/(z_1D_1 + |z_2|D_2)$) by $\varphi_1 = [\eta(1 + z_1/|z_2|)]^{1-c}\varphi$ and $\varphi_2 = [(1-\eta)(1 + |z_2|/z_1)]^{1-c}\varphi$ (Eqs. 19, 23, 24). Against the exact single-salt solution the paper finds the approximation within 10 % for $0.16 < D_1/D_2 < 5.5$ and $\varphi < 3$, the deviation growing at high fluxes. The gate `check_polarisation_coupling` holds the engine to (380) to $10^{-10}$ and to an independent solve of Eqs. (375)–(376) to $10^{-8}$.

**The case study.** The paper applies the model to the nanofiltration of $Na^+/Cl^-/dye^{3-}$ solutions in a stirred cell (data of Bowen and Mohammad; $r = 2.4$ cm, $\omega = 41.9$ rad/s, $Re = 26\,907$, $Sh_i = 0.23\,Re^{0.567}Sc_i^{0.33}$): four algebraic equations per operating point, no film thickness. The `polarisationIndex` bench of tutorial `props/membrane/polarisation01_geraldes_afonso_table1` runs that table and prints, per row, $k_{c,i}$, $\varphi_i$, $\Xi_i$, $C_{i,m}$, $\Gamma_i$ and $\xi$: the dye, slowest and most rejected, polarises most; $Na^+$ is dragged after it by the field; $Cl^-$, less rejected than the dye and pulled by the same field, sits *below* its bulk concentration. In a module the same object prices one $k_{c,i}$ per ion from the spacer correlation (tutorial `membrane14_polarisation_multiionic`, which publishes one `k_film_<ion>` column per ion, `Gamma_<ion>` and `xi_V_per_m` beside the membrane's own $\psi$). What the model does not contain: any film thickness, any diffusion potential *inside* a film (the interface is a plane between the electrical double layer and the bulk, and the solution is assumed dilute), and concentration-dependent diffusivities.

## 18.5 The coupling, and why it needs an inner Newton

Equations (369), (371), (374) are circular: $J_w$ depends on $\Delta\pi$, which depends on $c_m$ and $c_p$, which depend on $J_w$. The osmotic back-pressure is supplied by the selected osmotic model (Chapter 3) evaluated at each face,

$$\Delta\pi = \pi(c_m) - \pi(c_p) \quad \big[\text{van't Hoff } \nu R_gTc, \text{ or Pitzer } \phi(I)\,\nu R_gTc\big]. \tag{381}$$

Choupo closes the loop at each channel station with a one-dimensional Newton in $J_w$ on the residual

$$F(J_w) = J_w - A_w\big(\Delta P - \Delta\pi(J_w)\big) = 0, \tag{382}$$

the $\Delta\pi$ derivative taken by central difference (the $A_w\Delta P$ part is constant), damped so $J_w$ stays positive and bounded. It converges in a handful of iterations from the no-osmotic guess $J_w = A_w\Delta P$.

> *Intuition.* The osmotic term is what makes a membrane non-trivial. Without it $J_w = A_w\Delta P$ — a one-line answer. With it, pushing harder raises $J_w$, which polarises the wall (higher $c_m$), which raises $\Delta\pi$, which pushes back on $J_w$: a self-limiting feedback. The inner Newton finds the balance point. This is exactly why a real RO plant cannot simply crank the pressure — the wall fights back.

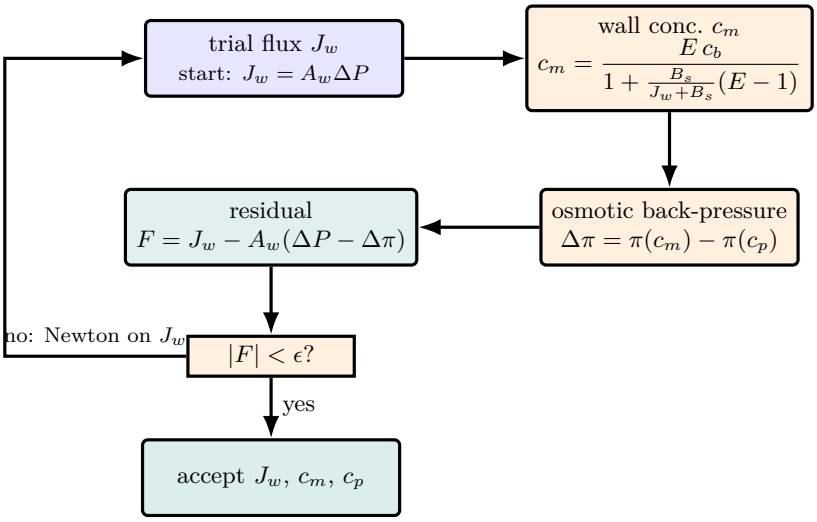



**Figure 73:** The local membrane flux is circular, so Choupo closes it with a one-dimensional Newton at every channel station. A trial water flux $J_w$ sets the polarisation $E$ and hence the wall concentration $c_m$ (orange); $c_m$ sets the osmotic back-pressure $\Delta\pi$; and $\Delta\pi$ feeds back into the solution-diffusion law through the residual $F = J_w - A_w(\Delta P - \Delta\pi)$ (teal). Starting from the no-osmotic guess $J_w = A_w \Delta P$, the Newton converges in a handful of iterations. This loop is exactly the physical feedback that stops a real RO plant from simply cranking the pressure: more flux → more polarisation → more back-pressure → the wall fights back.

## 18.6 Marching down the channel

A spiral-wound element is not a point: as feed flows along it, water is removed, so the bulk concentration $c_b$ rises and the pressure $P_b$ falls. Choupo discretises the feed channel into stations and *marches*: at each station it solves the local flux above, removes the permeated water and solute from the bulk, updates $c_b$ and $P_b$, and steps on — a 1-D plug-flow integration along the leaf. The profile (`J_w(z)`, $c_b(z)$, $P_b(z)$) is the output a membrane engineer reads. Two observed performance numbers summarise it:

$$\text{recovery} = \frac{\text{permeate flow}}{\text{feed flow}}, \qquad R_{\text{obs}} = 1 - \frac{c_p}{c_b} \ \text{(vs the } \textit{true}\ 1 - c_p/c_m). \tag{383}$$

The gap between observed and true rejection *is* the polarisation penalty.

Three pieces are selectable sub-models (the same factory pattern as the EoS or the cyclone), all living in the optional membrane module: $k$ (constant, or a Schock–Miquel Sherwood correlation from the spacer hydraulics), $\Delta P$ along the channel (constant, or Schock–Miquel friction), and the osmotic model (van't Hoff or Pitzer). An `elements N` keyword chains $N$ elements in series (the retentate of one is the feed of the next), with an inter-element $\Delta P$ loss — a full train.

**Worked example 18.1.** Seawater RO, `membrane01_RO_NaCl_seawater`35 g/L NaCl at 55 bar through an SW30HR-style element ($A_w$, $B_s$ for NaCl, $k = 50\,\mu$m/s): the inner Newton balances $J_w$ against the ∼28 bar osmotic back-pressure, giving $J_{w,\text{avg}} \approx 13\,\mu$m/s, ∼15.6 % water recovery and **99.78 % observed NaCl rejection**; the NaCl mass balance closes to $< 0.04$ %. Switching the osmotic block to Pitzer (`membrane06_pitzer`) lowers $\Delta\pi$ by ∼8 % and lifts recovery to 17.2 %.

**Runnable case.** `tutorials/steady/membranes/membrane01_RO_NaCl_seawater` — canonical seawater RO.
`tutorials/steady/membranes/membrane02_NF_sugar` — loose NF, the defining *partial* rejection.
`tutorials/steady/membranes/membrane03_concentration_polarization` — sweeps $k$ to draw the polarisation curve.
`tutorials/steady/membranes/membrane05_train` — a 3-element train, the diminishing-returns recovery down the series.

## 18.7 Modules as records: a spiral element and a laboratory cell

Everything above takes the channel's geometry — the active area $A$, the leaf length $L$, the spacer height $h$ and its porosity $\varepsilon$ — from the operation dictionary, typed by the author. Since 2026-09-15 a case may instead name the hardware: `module NF270-4040;` reads a record of kind `membraneModule` from the assets catalogue (`data/standards/assets/`), and the record supplies what the author used to type, plus two things an author never

had a place for: the manufacturer's operating *limits* and the standard *rated test* the data sheet quotes. Three ship: the DuPont FilmTec NF270-4040 and SW30HR-380 elements (`format spiralWound;`, each wound with its membrane record) and the Sterlitech SEPA CF cell (`format flatSheetCell;`), a laboratory flat-sheet crossflow cell that takes any coupon — there the *case* declares `membrane` beside `module`.

**One face or two.** A spiral leaf is an envelope: membrane on both sides of the feed spacer, the permeate carrier inside. So the channel the feed sees has *two* membrane faces and, for a leaf of length $L$ and total area $A$, the channel width is

$$W_{\text{spiral}} = \frac{A}{2L}, \qquad W_{\text{cell}} = \frac{A}{L}, \qquad L_{\text{cell}} = \frac{A}{w_{\text{slot}}}, \tag{384}$$

while in a flat cell the coupon lies on its permeate carrier under a steel lid: *one* face. The crossflow section $Wh\varepsilon$ and with it the velocity, the Reynolds number, $k$ and the friction all follow from which of the two the engine takes, so the module reads the record's `format` and prints which it used. The cell's channel length is not stored: it is the active area over the slot width, derived where it is used (the tree never stores a derivative).

**What the sheet does not say is an estimate, and the run says so.** A data sheet publishes the area and the limits. It does not publish the leaf length, the spacer porosity, or — for most elements — the spacer thickness; the SEPA CF's channel height is whatever spacer the user installs. Each such value in a record is an *educated* one, marked `origin estimated; reviewStatus unverified;` with a note saying how it is to be verified (weigh a spacer coupon for its porosity; unroll a leaf for its length; measure the spacer), and **every run that reads the record announces every one of them**, in the log and in the end-of-run caveat block. A data-sheet value is never marked so; a record with no estimate produces no line. Declaring `area`, `length` or a `channelHeight` beside `module` is refused by name: one home.

**Limits announce; they do not refuse.** The record's `limits {}` (maximum pressure, temperature, pressure drop per element, feed flow, pH band) are checked against the run and a violation is printed as `[limit]` — a student may run an element outside its rating on purpose, and the run then says so. The `ratedTest {}` (for the NF270-4040: 2,000 ppm $MgSO_4$, 4.8 bar, 25 °C, 15 % recovery, 9.5 $m^3$/d, > 97 % rejection) is not asserted by the engine but *measured* by the gate, which runs the element at those conditions and prints the engine's permeate flow and rejection beside the sheet's: the membrane records' $A_w$ are fits to teaching cases, so the ratio is a finding, never tuned away.

> **Runnable case.** `tutorials/steady/membranes/membrane15_module_nf270_4040` — the NF270-4040 element as a record, its three estimates announced.
> `tutorials/steady/membranes/membrane16_module_sw30hr_8040` — the 8-inch seawater element beside membrane01.
> `tutorials/steady/membranes/membrane17_sepa_cf_flat_cell` — a laboratory cell in five lines: one membrane face, 0.2 m/s crossflow, the coupled multi-ionic film.

## 18.8 Stages, arrays and the feed spacer as a record

**One unit is one stage.** A plant is an *array*: pressure vessels, each holding several elements in series, the vessels of a stage in parallel, the concentrate of one stage boosted and fed to the next on fewer vessels. A `spiralWoundModule` is one stage — `vessels` $n$`;` in parallel, `elements` $m$`;` in series in each. Identical vessels see the same feed state, so the engine solves *one* vessel on $F/n$ and scales its retentate and permeate by $n$: exact, not an approximation, and announced. The array is built in the flowsheet, one unit per stage, the interstage booster a `pump` of its own. Each element publishes its own flux, friction drop, feed, inlet Reynolds number and polarisation modulus per solute, and a manufacturer's pressure-drop limit is checked element by element.

**The spacer sets the channel.** Schock and Miquel (Desalination 64 (1987) 339–352) characterise a feed spacer by its thickness $h$, its filament diameter $d_F$ and its mesh length $l_M$, and take the channel height equal to the spacer thickness. The velocity is the flow over the open section, $v = V/(b\,h\,\varepsilon)$, and the hydraulic diameter of a spacer-filled channel is four times the open volume over the wetted surface:

$$d_h = \frac{4\varepsilon}{2/h + (1-\varepsilon)\,S_{V,sp}}, \qquad S_{V,sp} = \frac{4}{d_F}, \qquad \varepsilon = 1 - \frac{\frac{1}{2}\pi d_F^2 l_M}{l_M^2\,h}, \tag{385}$$

their Eqs. (17), (23) and (20)–(22). With $\varepsilon \to 1$ it is the empty flat channel, $2h$; with $h \to \infty$ it is a packed bed. A feed spacer is a record (`kind feedSpacer`) that stores what was measured — the thickness, the filament, and *either* a weighed porosity *or* the mesh it follows from — and the engine derives $S_{V,sp}$ and $d_h$. For the FilmTec FT30 spacer of their Tab. 1 ($h = 0.77$ mm, $d_F = 0.385$ mm, $\varepsilon = 0.89$) Eq. (385) gives $d_h = 0.952$ mm; an inline channel's $2h\varepsilon = 1.37$ mm is not that diameter, and a run that reads it says so. The record carries the two correlations the paper fits,

$$Sh = 0.065\,Re^{0.875}\,Sc^{0.25}, \qquad \lambda = 6.23\,Re^{-0.3}, \tag{386}$$

their Eqs. (30) and (24), *with* the window they were measured over, $100 < Re < 1000$: a run outside it is announced as an extrapolation and continues. For a mixture of ions the Sherwood law is evaluated per ion on that ion's own $Sc_i = \nu/D_i$, so $k_i \propto D_i^{0.75}$, before the Geraldes–Afonso coupling of the film. The paper applies no wall viscosity correction (its numbers use bulk properties), and neither does the engine.

**Standard sizes are records too.** The four 40-inch sizes (2.5, 4, 8 and 16 in) are `Generic-2540` … `Generic-16040`: an area, a leaf count and a leaf length, every value an announced estimate, wound with the FT30 spacer, the membrane the case's. The leaf *width* is derived, $W = A/(2\,n\,L)$, never stored.

> **Runnable case.** `tutorials/steady/membranes/membrane21_two_stage_nf_array` — a 4:2 array of six `Generic-8040` elements per vessel with an interstage booster, multi-ion (SDEM and the multi-ionic film), per element.

## 18.9 Batch diafiltration: the same flux law, once per instant

Everything above serves a steady module marching down a channel. At laboratory and pilot scale a membrane is run the other way round: a stirred *retentate* vessel behind the membrane, the permeate leaving, and either nothing replacing it (**concentration** — the vessel shrinks and the solutes concentrate) or solvent made up as fast as it permeates (**constant-volume diafiltration** — the volume holds and the solutes wash out). `batchDiafilter` in `choupoBatch` adds no physics to what this chapter already has: it asks the same local flux law (solution-diffusion or DSPM-DE), with the same membrane record, the same film coefficient and the same osmotic model, once per instant instead of once per channel node. The vessel is well mixed, so there is one bulk composition; with $V$ the retentate volume, $A$ the area and $Q_d$ the make-up rate,

$$\frac{dn_i}{dt} = -J_{s,i}\,A + Q_d\,c_{d,i} \quad \text{(solutes)}, \qquad Q_d = 0 \text{ (concentration)} \;\text{ or }\; Q_d = Q_p = J_w A \text{ (constant volume)},$$

and the solvent line closes the permeate's *mass* on the same declared solution density the concentrations were built from, which is why the vessel conserves mass to machine precision rather than leaking the percent that a pure-water permeate concentration would.

**The washout law, and where it fails.** The hand derivation every student meets for constant-volume diafiltration is

$$\frac{c}{c_0} = \exp\!\big[-(1-R)\,N\big], \qquad N = \frac{1}{V_0}\int Q_d\,dt \;\; \text{(diavolumes)}, \tag{387}$$

and it assumes the rejection $R$ is a *constant*. It is not: the solute leaves, the osmotic pressure falls, the flux climbs, the wall concentration changes and the observed rejection rises with it. So the unit publishes the observed rejection `R_obs` at every instant and, in constant-volume mode, `washoutActual` beside `washoutIdeal` — the ideal evaluated from *this run's* own initial $R$ and its own diavolumes, computed and never declared, so the gap between the two cannot have been arranged. Where they agree the hand derivation was safe; where they part the $R$ trajectory says which assumption failed and when.

> **Worked example 18.2.** NF desalting, `diafilter01_nf_desalting`Ten litres of a dilute NaCl/$MgSO_4$ brine behind 5 m$^2$ of NF270 at 10 bar, constant volume for 160 s: 5.44 diavolumes; NaCl washed down to 50.2 % of its charge while $MgSO_4$ is held at 99.2 %; the flux *rises* from 233 to 256 LMH as the osmotic back-pressure falls, and the observed NaCl rejection climbs from 0.8714 to 0.8748. The washout pair reads 0.5015 actual against 0.4970 ideal — the constant-$R$ derivation is safe to about one percent here, and the run says so.

**Fouling: resistance in series, declared as a mechanism.** A rising flux is what the model has; a real rig's flux *falls*, because the membrane fouls, and every schedule a student writes is sized on that decline. The unit carries it where the transport context is assembled, so both flux laws see it and neither is modified:

$$\frac{1}{A_{\text{eff}}} = \frac{1}{A_w} + r_f(v), \qquad v = \frac{V_{\text{permeated}}}{A},$$

with $r_f$ growing by one of Hermia's blocking laws (Hermia, *Trans. IChemE* 60 (1982) 183): `cake` ($r_f = k\,v$, a deposited layer) or `intermediate` ($1/A_{\text{eff}} = e^{kv}/A_w$, pores progressively covered). A blocking law is a claim about a *mechanism*, not a fitting exponent, so the block requires a written `reason`; the `standard` and `complete` laws describe pore-internal geometry a lumped permeance cannot represent and refuse by name. The constant $k$ is equipment-and-feed data: it lives in the case and never on a membrane record. Absent the block there is no fouling and the run is byte-identical. In `diafilter02_fouling_decline` — the same run with a *hypothetical* cake law, declared as such in its own header, because a fully dissolved brine has nothing to deposit — the flux turns round (233 → 196 LMH), the permeance ends at 72.5 % of clean, and the same 160 s buys 4.74 diavolumes instead of 5.44, leaving 53.4 % of the NaCl behind against 50.2 %.

**What is not modelled, said plainly.** Any correlation for the film coefficient (the case declares $k_{\text{film}}$ and the unit refuses without it), temperature transients, critical flux and gel layers, cleaning or backwash cycles, any prediction of $k$ itself — the tree carries no fouling data at all — and the pump work, which the campaign's energy ledger names as a gap rather than closing with a term quietly zero.

> **Runnable case.** `tutorials/batch/membrane/diafilter01_nf_desalting` — constant-volume NF desalting; the washout pair.
> `tutorials/batch/membrane/diafilter02_fouling_decline` — the same run under a declared cake law; the difference is the fouling.

# 19 Nanofiltration with charge: the Donnan-Steric Pore Model with Dielectric Exclusion (DSPM-DE)

*What you'll learn.* Solution-diffusion (§18) treats every solute the same: one permeability $B_s$ per salt, fitted, not predicted. It cannot say *why* a nanofiltration membrane rejects $MgSO_4$ at ∼98 % while letting half the NaCl through at the same pore size. The answer is *charge*. An NF membrane is a bundle of nm-scale pores carrying a fixed wall charge; ions partition into those pores against three barriers — steric, electrostatic (Donnan), and dielectric (Born) — and then move under the coupled gradient of concentration *and* electric potential (extended Nernst–Planck). This chapter derives the Donnan-Steric Pore Model with Dielectric Exclusion [81, 82], the charge-aware sibling of solution-diffusion in the same transport factory (`transport DSPM-DE;`), and shows how the divalent-co-ion rejection ordering falls out of the partition with no per-salt fit.

**How to read this chapter.** §19.1 and §19.2 give the physics of each term; §19.3 writes the *whole* boundary-value problem one channel station solves — every equation, every unknown, both faces and the wall — and §19.4 puts it in dimensionless form. The numerical method that solves it is §19.5 (the discretisation, the unknown vector and its block structure, the Newton, the seed and the continuation, with a pseudo-algorithm); §19.6 is the grid and its observed order, §19.7 the evidence that the equations are solved right, and §19.8 what a user needs to decide how many nodes to ask for. Read in that order, the chapter is meant to be enough to re-implement the pore solver without opening its source.

## 19.1 The pore picture and the three exclusions

DSPM-DE replaces the dense film with a bundle of cylindrical pores of effective radius $r_p$, fixed volumetric charge density $X_d$ (signed; negative for a polyamide NF carrying carboxylate groups), porosity $A_k$ and active-layer thickness $\Delta x$ — the geometry is carried in the membrane `.dat poreModel{}` block as $r_p$, $A_k/\Delta x$, $X_d$ and the pore dielectric $\varepsilon_p$. Only the ratio $A_k/\Delta x$ enters the equations; its inverse $\Delta x_e = \Delta x/A_k$ is the *equivalent thickness*. Each *ion* $i$ (the salt is split into its cation and anion through the component's declared `dissociatesTo { ...}` stoichiometry) carries a radius $r_i$, a free-solution diffusivity $D_i$ and a signed charge $z_i$. $D_i$ is the species record's $D_{i,0}$ at 25 °C, corrected to the run's temperature by Stokes–Einstein on the package's declared liquid viscosity, $D_i = D_{i,0}\,(T/298.15)\,\mu(298.15)/\mu(T)$; with no viscosity model declared the factor is 1, and the run says so.

Throughout the chapter the electric potential is **dimensionless**: $\psi = F\phi/(R_\mathrm{g}T)$, with $\phi$ the potential in volts, measured from the feed solution at the wall ($\psi = 0$ there). Whatever is published in volts (the potential across the membrane, `dPsi_m_V`) is $(R_\mathrm{g}T/F)\,\psi$.

An ion entering a pore pays three independent partition penalties. Their product with the activity ratio of the two sides of the face is the *partition coefficient* of the face (Bowen & Welfoot, Eq. 7); it is written out, with the concentration it multiplies, in (400) below.

**Steric.** A finite ion cannot reach within $r_i$ of the pore wall. The geometric partition for a sphere in a cylinder is

$$\Phi_{S,i} = (1-\lambda_i)^2, \qquad \lambda_i = \frac{r_i}{r_p}, \tag{388}$$

the same $\lambda$ that sets the *hindrance* factors of §19.2. An ion as large as the pore ($\lambda_i \geq 1$) does not enter it at all; how it is carried is said in §19.3. A purely steric membrane ($X_d = 0$, $\varepsilon_p = \varepsilon_b$) is a sieve — the uncharged starting point of the staged build.

**Donnan.** The fixed pore charge $X_d$ must be balanced: inside the pore, at *every* position,

$$\sum_i z_i\, c_i + X_d = 0. \tag{389}$$

At each pore mouth the potential jumps by a Donnan potential that makes the pore side of the face satisfy (389): at the feed face the jump is $\psi(0) - 0 = \psi(0)$, at the permeate face $\psi(1) - \psi_p$, with $\psi_p$ the permeate's potential; the Donnan factor is $e^{-z_i\psi(0)}$ at one face and $e^{-z_i(\psi(1)-\psi_p)}$ at the other. With $X_d < 0$ the pore expels the *co-ion* (the anion) and draws in the counter-ion, and a divalent co-ion ($|z| = 2$) feels the jump *twice* as hard as a monovalent one — the origin of the NF fingerprint.

**Born (dielectric exclusion).** Water confined in a nm pore is more ordered, so its dielectric constant $\varepsilon_p$ is lower than the bulk $\varepsilon_b$ ($\varepsilon_b = \varepsilon_b(T)$ from the solvent-properties table, or the case's `bulkDielectric`; $\varepsilon_p$ from the membrane `.dat`). Moving a charged ion into the lower-$\varepsilon$ medium costs Born solvation energy

$$\Delta W_i = \frac{z_i^2 e^2}{8\pi\varepsilon_0 r_i}\left(\frac{1}{\varepsilon_p} - \frac{1}{\varepsilon_b}\right), \tag{390}$$

(Bowen & Welfoot, Eq. 16; $e$ the elementary charge, $\varepsilon_0$ the vacuum permittivity) — the very same Born form already met in the eNRTL solvent-shift of §3. It scales with $z_i^2$, where the Donnan exponent is LINEAR in $z_i$: Donnan excludes a co-ion and attracts a counter-ion by the sign of its charge, while Born excludes both signs alike and divalent ions four times as strongly as monovalent ones. So it reinforces divalent exclusion — and, unlike Donnan, it rejects *both* ions of a divalent salt, which is why $MgSO_4$ can be held above what charge alone would give. Set $\varepsilon_p < \varepsilon_b$ in the `.dat` to turn it on. Steric and Born do not depend on the solution, so they combine into one constant per ion,

$$S_i \;=\; \Phi_{S,i}\,\exp\!\big(-\Delta W_i/k_BT\big), \tag{391}$$

the same at both faces.

## 19.2 Hindered transport inside the pore

A confined ion diffuses and is convected more slowly than in free solution — the theory of hindered transport in liquid-filled pores [83]. Two factors carry it, $K_{d,i}$ on diffusion and $K_{c,i}$ on convection. Choupo takes, in $\lambda = r_i/r_p$, the cross-section-averaged pair of Dechadilok & Deen [84]:

$$\begin{aligned} H(\lambda) &= 1 + \tfrac{9}{8}\lambda\ln\lambda - 1.56034\,\lambda + 0.528155\,\lambda^2 + 1.91521\,\lambda^3 - 2.81903\,\lambda^4 \\ &\quad + 0.270788\,\lambda^5 + 1.10115\,\lambda^6 - 0.435933\,\lambda^7, \end{aligned} \tag{392}$$

$$K_{d,i} = H(\lambda)/\Phi_{S,i} \quad (\lambda \le 0.95), \qquad K_{d,i} = 0.984\Big(\frac{1-\lambda}{\lambda}\Big)^{5/2} \quad (0.95 < \lambda < 1), \tag{393}$$

$$K_{c,i} = \frac{1 + 3.867\lambda - 1.907\lambda^2 - 0.834\lambda^3}{1 + 1.867\lambda - 0.741\lambda^2}, \tag{394}$$

the pair NanoFiltran [85] evaluates (`hindrance DechadilokDeen2006;`, the only accepted word and the default). The polynomial (392) is *not* a $K_d$: it is the product $H = \Phi\,K_d$ of the diffusive hindrance and the steric partition — which one can check without the paper, because only $H/\Phi$ meets the lubrication form of (393) at $\lambda = 0.95$. Since $\Phi$ is already in the partition, the diffusive factor is $H/\Phi$. Note that $K_c > 1$ for every ion of the corpus: a small ion is carried by the fast centre-line flow faster than the cross-section-average velocity.

The two factors and the layer combine into one *diffusive conductance* per ion, the quantity the discrete equations are written in:

$$P_i \;=\; \frac{K_{d,i}\,D_i\,A_k}{\Delta x} \quad [\mathrm{m/s}]. \tag{395}$$

## 19.3 The model: the whole boundary-value problem

This subsection is the problem one channel station solves, complete. The exclusions behind each partition factor are §19.1; the method that solves it is §19.5. Table 2 collects the symbols.

**Table 2:** The symbols of the DSPM-DE problem. $i$ indexes the ions that enter the pore, $k$ the grid nodes.

| symbol | meaning | unit |
|---|---|---|
| $x$, $\zeta = x/\Delta x$ | position across the active layer from the feed face, and its dimensionless form, $\zeta \in [0,1]$ | m, – |
| $c_i(\zeta)$ | concentration in the pore | mol/m$^3$ |
| $\psi(\zeta)$ | potential in the pore, $F\phi/(R_gT)$, 0 in the feed solution at the wall | – |
| $C_{b,i}$, $C_{m,i}$ | bulk (feed channel) and wall concentration | mol/m$^3$ |
| $c_{p,i}$, $\psi_p$ | permeate concentration and potential | mol/m$^3$, – |
| $V$, $J_v = A_kV$ | solvent velocity in the pore; volume flux per *membrane* area | m/s |
| $j_i$, $N_i = A_k\,j_i$ | ion flux per *pore* area (in $x$); per *membrane* area (in $\zeta$) | mol/(m$^2$ s) |
| $P_i$ | diffusive conductance $K_{d,i}D_iA_k/\Delta x$, (395) | m/s |
| $S_i$ | steric × Born partition, (391) | – |
| $g_i^{(0)}$, $g_i^{(1)}$ | activity ratio (solution side over pore side) at the feed and permeate faces | – |
| $k_{c,i}$, $\Xi_i$, $k^*_{c,i}$ | film coefficient, suction correction, $k_{c,i}\Xi_i$ | m/s, –, m/s |
| $\xi'$ | wall field $F\xi/(R_gT)$, $\xi$ the potential gradient at the interface (§18.4) | 1/m |
| $A_w$, $\Delta P$ | water permeance, applied pressure difference | m/(s Pa), Pa |
| $\mathrm{Pe}_i$ | pore Péclet number $K_{c,i}J_v/P_i$, (404) | – |

**Inside the pore: extended Nernst–Planck.** Each ion moves under the gradient of its own concentration, the electric field and convection with the water (Bowen & Welfoot, Eq. 5). Per unit *pore* cross-section, in $x$,

$$j_i = -K_{d,i}\,D_i\Big(\frac{dc_i}{dx} + z_i\,c_i\,\frac{d\psi}{dx}\Big) + K_{c,i}\,c_i\,V. \tag{396}$$

The module and the permeate count fluxes per unit *membrane* area, and the pores are a fraction $A_k$ of it: $N_i = A_k\, j_i$ and $J_v = A_k V$. Multiplying (396) by $A_k$ and writing $x = \Delta x\, \zeta$ gives the form the solver uses,

$$N_i \;=\; -P_i\Big(\frac{dc_i}{d\zeta} + z_i\, c_i\, \frac{d\psi}{d\zeta}\Big) + K_{c,i}\, J_v\, c_i. \tag{397}$$

At steady state nothing accumulates and nothing reacts, so $N_i$ is the same at every $\zeta$, and the whole of it leaves with the permeate water:

$$N_i \;=\; J_v\, c_{p,i} \qquad \text{for every } \zeta \in [0,1]. \tag{398}$$

The pore is electroneutral at every $\zeta$ (389), and no current flows through an NF element (there is no external circuit):

$$\sum_i z_i\, N_i = 0 \quad\Longleftrightarrow\quad \sum_i z_i\, c_{p,i} = 0 \quad (J_v > 0), \tag{399}$$

so the permeate is electroneutral too.

**The two faces.** The pore concentration at each mouth is the partition coefficient times the concentration on the solution side of that face — the *wall* concentration $C_{m,i}$ at the feed face, not the bulk, and the permeate concentration at the other:

$$\begin{aligned} c_i(0) &= S_i\, g_i^{(0)}\, C_{m,i}\, e^{-z_i\,\psi(0)} && \text{(feed face, } \zeta = 0\text{)},\\ c_i(1) &= S_i\, g_i^{(1)}\, c_{p,i}\, e^{-z_i\,(\psi(1)-\psi_p)} && \text{(permeate face, } \zeta = 1\text{)}. \end{aligned} \tag{400}$$

The activity ratios $g$ are a declared choice (`transportParameters { activity ...; }`):

- `wallDavies` (the default): $g_i^{(0)} = \gamma_i(I_b)$, Davies (§3) with the speciation solver's $A(T)$ at the *bulk* ionic strength, evaluated once per channel station and held fixed in its Newton; $g_i^{(1)} = 1$ (an ideal permeate, an ideal pore);
- `ideal`: $g_i^{(0)} = g_i^{(1)} = 1$;
- `bulkOverPoreDavies`: the ratio on both faces, $g_i^{(0)} = \gamma_i(I_m;\varepsilon_b)/\gamma_i(I(0);\varepsilon_p)$ and $g_i^{(1)} = \gamma_i(I_p;\varepsilon_b)/\gamma_i(I(1);\varepsilon_p)$, each on the ionic strength of its own side (the wall, the first and last pore nodes, the permeate), with $\log_{10}\gamma_i = -Az_i^2\big(\sqrt{I}/(1+\sqrt{I}) - 0.3\,I\big)$, $A(\varepsilon,T) = 1.82483\times10^6\,(\varepsilon T)^{-3/2}$ and $I$ in mol/L; these depend on the answer, so they are evaluated *inside* the station's Newton, with their derivatives.

The ions are ideal between the faces, as in NanoFiltran.

**The wall.** The wall concentration is an unknown of the same station: it depends on the permeate the pore makes and on the water flux. Choupo takes it from the multi-ionic interface balance of Geraldes and Afonso [79] (§18.4), written here because it is solved in the same Newton:

$$J_v\, c_{p,i} \;=\; -\,k^*_{c,i}\,(C_{m,i} - C_{b,i}) + J_v\, C_{m,i} - z_i\, D_i\, \xi'\, C_{m,i} \quad\Longrightarrow\quad C_{m,i} = \frac{k^*_{c,i}\, C_{b,i} - J_v\, c_{p,i}}{k^*_{c,i} - J_v + z_i D_i\, \xi'}, \tag{401}$$

$k^*_{c,i} = k_{c,i}\,\Xi_i(\varphi_i)$, $\varphi_i = J_v/k_{c,i}$, with the suction correction of Geraldes and Afonso [80], $\Xi_i = \varphi_i + (1+0.26\,\varphi_i^{1.4})^{-1.7}$ (`suctionCorrection GeraldesAfonso2006`, valid for $\varphi_i < 20$ and announced beyond), or the film theory's $\Xi_i = \varphi_i + \varphi_i/(e^{\varphi_i}-1)$ (`filmTheory`, the default), or $\Xi_i = 1$ (`none`). $k_{c,i}$ is the declared `k_film`, or the module's mass-transfer correlation on the ion's own $D_i$. The field $\xi'$ is fixed by electroneutrality of the wall,

$$\sum_i z_i\, C_{m,i} = 0 \qquad (\texttt{ionCoupling electroneutral};\ \text{with the default } \texttt{none},\ \xi' = 0). \tag{402}$$

This is the default `wallClosure porePermeate`: the wall is closed with the permeate of *this* pore. The alternative `wallClosure solutionDiffusion` closes it with the salts' solution-diffusion $B_s$ instead, $C_{m,i} = k^*_{c,i}C_{b,i}/(k^*_{c,i} - J_v + z_iD_i\xi' + J_vB_i/(J_v+B_i))$, a permeate the pore does not make.

**The water flux.** Either imposed (the `poreTransport` bench is handed $J_v$), or closed by the solution-diffusion water law with a van't Hoff osmotic difference over every ion on both sides,

$$J_v \;=\; A_w\Big[\Delta P - R_{\mathrm{g}}T\Big(\sum_i C_{m,i} - \sum_i c_{p,i}\Big)\Big]. \tag{403}$$

When the osmotic difference at vanishing flux — the pore in Donnan equilibrium with both baths, $c_{p,i} = g_i^{(0)}C_{b,i}$, so $\Delta\pi_0 = R_{\mathrm{g}}T\sum_i C_{b,i}(1-g_i^{(0)})$ — is not below $\Delta P$, (403) has no positive root; the station has no water flux, and says so. (With `bulkOverPoreDavies`, whose $g$ depends on the answer, a converged $J_v \le 0$ is caught after the solve instead.)

**An ion the pore excludes.** An ion with $\lambda_i \geq 1$ (Bowen and Mohammad's trivalent dye in a CA30 pore) cannot enter a pore no larger than itself — (388) does not apply there, and its partition is taken as zero: it carries no flux and its permeate is zero. It is *not* a pore unknown — carried as one, with NanoFiltran's floor partition of $10^{-15}$, it would sit fifteen orders below the other ions and the Newton could not step. It stays in the film: its wall concentration is (401) with $c_{p,i} = 0$, and it enters the wall's electroneutrality (402), the wall's ionic strength and the osmotic difference (403). Everything below indexes only the ions that enter the pore.

**The unknowns, counted.** With $n$ ions in the pore the continuous problem has the profiles $c_i(\zeta)$ and $\psi(\zeta)$, the $n$ permeate concentrations $c_{p,i}$, the permeate potential $\psi_p$, the field $\xi'$ and the water flux $J_v$ (the wall concentrations are explicit in them by (401), and $N_i$ is eliminated by (398)). They are fixed by:

- each (397)–(398) is a first-order ODE in $c_i$ with one unknown constant $c_{p,i}$; its *two* face conditions (400) fix both the profile and $c_{p,i}$;
- $\psi(\zeta)$ has no equation of its own: it is the algebraic unknown that (389) fixes at every $\zeta$ (the problem is a differential-algebraic one); at $\zeta = 0$ the same condition with the feed partitions gives the feed Donnan potential $\psi(0)$;
- $\psi_p$ is fixed by zero current (399), $\xi'$ by the wall's electroneutrality (402), $J_v$ by (403) (or imposed).

## 19.4 Dimensionless form and the Péclet numbers

The coordinate $\zeta$ and the potential $\psi$ are already dimensionless. Dividing (397) by $P_i$ and using (398),

$$\frac{dc_i}{d\zeta} \;=\; \mathrm{Pe}_i\Big(c_i - \frac{c_{p,i}}{K_{c,i}}\Big) - z_i\, c_i\, \frac{d\psi}{d\zeta}, \qquad \mathrm{Pe}_i \;\equiv\; \frac{K_{c,i}\, J_v}{P_i} = \frac{K_{c,i}\, J_v\, \Delta x}{K_{d,i}\, D_i\, A_k}, \tag{404}$$

so each ion is governed by ONE number, its pore Péclet number — convection against hindered diffusion across the layer — plus its charge and the field. The field itself follows by differentiating (389) ($\sum_i z_i\, dc_i/d\zeta = 0$) and substituting (404):

$$\frac{d\psi}{d\zeta} \;=\; \frac{\sum_i z_i\, \mathrm{Pe}_i\, \big(c_i - c_{p,i}/K_{c,i}\big)}{\sum_i z_i^2\, c_i}. \tag{405}$$

Two consequences decide how hard the numerics are. An ion with $\mathrm{Pe}_i \ll 1$ diffuses across faster than it is carried: its profile is nearly linear between its two face values. An ion with $\mathrm{Pe}_i \gg 1$ is carried: its profile is flat where convection rules and steep where the field turns it round. And when the $\mathrm{Pe}_i$ of two counter-ions are far apart — the ratio of their *hindered* diffusivities $K_{d,i}D_i$ is what sets it — (405) couples a fast and a slow ion through one field, and the electromigration flux of an excluded ion, $P_i|z_i|c_i\,|d\psi/d\zeta|$, becomes far larger than the net flux $N_i$ it nets with diffusion. The run prints that ratio as the *stiffness* of the answer. §19.8 measures both numbers on the reference case.

## 19.5 The numerical method: every node at once

Choupo — `poreSolver { method finiteDifference; nodes 101; }` when written out, each default announced — solves the problem of §19.3 as NanoFiltran [85] does in outline: the extended Nernst–Planck equations discretised by finite differences inside the membrane, linearised, and every node solved simultaneously, because the problem is stiff (§19.4) and a method that marches across the pore from one face has to hit the other face through that cancellation. The four ingredients follow: the discrete flux, the unknown vector and its block structure, the Newton, and how it is started.

### 19.5.1 The discrete flux: Scharfetter–Gummel, derived

Take $M$ intervals with nodes $0 = \zeta_0 < \zeta_1 < \cdots < \zeta_M = 1$ (the case declares the node count $n_{\mathrm{nodes}} = M + 1$), $h_k = \zeta_{k+1} - \zeta_k$. On one interval freeze the coefficients and take the potential linear, $d\psi/d\zeta = (\psi_{k+1} - \psi_k)/h_k$. Then (397) with $N_i$ constant is a linear first-order ODE,

$$\frac{dc_i}{d\zeta} = \alpha_i\, c_i - \frac{N_i}{P_i}, \qquad \alpha_i = \frac{K_{c,i} J_v}{P_i} - z_i\, \frac{\psi_{k+1} - \psi_k}{h_k},$$

whose exact solution from $c_i(\zeta_k) = c_{i,k}$ is $c_i(\zeta) = N_i/(P_i\alpha_i) + \big(c_{i,k} - N_i/(P_i\alpha_i)\big)e^{\alpha_i(\zeta-\zeta_k)}$. Setting $c_i(\zeta_{k+1}) = c_{i,k+1}$ and solving for $N_i$, with the *cell Péclet number* $a_{i,k} = \alpha_i h_k$,

$$N_i \;=\; \frac{P_i}{h_k}\Big[\,B(-a_{i,k})\, c_{i,k} - B(a_{i,k})\, c_{i,k+1}\Big], \qquad a_{i,k} = \mathrm{Pe}_i\, h_k - z_i\,(\psi_{k+1} - \psi_k), \qquad B(x) = \frac{x}{e^x - 1}, \tag{406}$$

using $B(-x) = xe^x/(e^x - 1)$. This is the Scharfetter–Gummel (exponentially fitted) flux of semiconductor device simulation, where the same drift-diffusion equations are solved. Its two limits show why it is used: as $a \to 0$, $B \to 1$ and (406) is the central difference $P_i(c_{i,k} - c_{i,k+1})/h_k$; as $a \to +\infty$, $B(-a) \to a$ and $B(a) \to 0$, and it is the upwind convective flux $K_{c,i}J_v c_{i,k}$ (minus the field term). A central difference of a drift-diffusion flux is free of oscillation only while $|a| < 2$; (406) is exact for a uniform drift at *any* $|a|$, on a uniform or a non-uniform grid. What it leaves

out is the *curvature* of $\psi$ on the cell, so its local error is $O(h_k^2|\psi''|)$ and the scheme is second order on a smooth profile (measured, §19.6).

$B$ must be evaluated without cancellation near 0 and without overflow at large $|x|$: $B(x) = 1 - x/2 + x^2/12$ for $|x| < 10^{-4}$; $x\,e^{-x}/(1-e^{-x})$ for $x > 0$; $x/(e^x - 1)$ (with $e^x - 1$ from `expm1`) for $x < 0$. Its derivative, needed by the Jacobian, follows from $B(-x) = B(x) + x$:

$$B'(x) = \frac{B(x)\big(1 - x - B(x)\big)}{x}, \qquad B'(x) = -\tfrac{1}{2} + \tfrac{x}{6} - \tfrac{x^3}{180} + \tfrac{x^5}{5040} \quad (|x| < 10^{-2}). \tag{407}$$

### 19.5.2 The unknowns, the equations and the block structure

The concentrations are carried as logarithms, $u_{i,k} = \ln c_{i,k}$, so they are positive by construction. The difficulty is that the problem has *global* unknowns — $c_{p,i}$, $\psi_p$, $J_v$, $\xi'$ — that rows at every node read (every flux row (398) reads $c_{p,i}$ and $J_v$; the permeate face reads $\psi_p$; the wall at node 0 reads $c_{p,i}$, $J_v$ and $\xi'$). A global unknown read by every node fills a whole column of the Jacobian. The cure is to give every node its own *copy* of each global unknown, held equal to its neighbour's by a row: each copy then couples only to the adjacent node, and the Jacobian is *block-tridiagonal* with blocks of size $2n + 4$. The unknown vector of node $k$ is

$$\mathbf{x}_k = \big(\underbrace{u_{1,k},\dots,u_{n,k}}_{\ln c_{i,k}},\ \psi_k,\ \underbrace{v_{1,k},\dots,v_{n,k}}_{\text{copies of } \ln c_{p,i}},\ \underbrace{\pi_k}_{\text{copy of } \psi_p},\ \underbrace{\jmath_k}_{\text{copy of } J_v/J_{\text{ref}}},\ \underbrace{\chi_k}_{\text{copy of } \xi'/\xi'_{\text{ref}}}\big) \in \mathbb{R}^{2n+4}, \tag{408}$$

$k = 0,\dots,M$, with the scales $J_{\text{ref}} = A_w\Delta P$ under the closure (the imposed $J_v$ otherwise) and $\xi'_{\text{ref}} = \sum_i k_{c,i}/\sum_i D_i$, so that every unknown is $O(1)$. The $2n + 4$ equations of block $k$ are, in this order (rows and unknowns share slot numbers):

| slot | $k = 0$ | $0 < k < M$ | $k = M$ |
|---|---|---|---|
| $i = 1..n$ | flux (406) on $[\zeta_0, \zeta_1] = J_v c_{p,i}$ | flux on $[\zeta_k, \zeta_{k+1}] = J_v c_{p,i}$ | permeate partition (400), in logs |
| $n + 1$ | EN (389) at node 0 | EN at node $k$ | EN at node $M$ |
| $n + 1 + i$ | feed partition (400) with (401), in logs | $v_{i,k} - v_{i,k-1} = 0$ | $v_{i,M} - v_{i,M-1} = 0$ |
| $2n + 2$ | $\pi_0 - \pi_1 = 0$ | $\pi_k - \pi_{k+1} = 0$ | zero current (399) |
| $2n + 3$ | water flux (403) (or $\jmath_0 = J_v/J_{\text{ref}}$) | $\jmath_k - \jmath_{k-1} = 0$ | $\jmath_M - \jmath_{M-1} = 0$ |
| $2n + 4$ | wall EN (402) (or $\chi_0 = 0$) | $\chi_k = 0$ (idle) | $\chi_M = 0$ (idle) |

The flux row of block $k$ reads $u_{i,k}$, $\psi_k$, $v_{i,k}$, $\jmath_k$ of its own node and $u_{i,k+1}$, $\psi_{k+1}$ of the next; a copy row reads its own node and one neighbour; everything else reads its own node only. So block row $k$ touches block columns $k - 1$, $k$ and $k + 1$ and nothing else (Figure 74). The field copy is read at node 0 only; its slot is idle elsewhere ($\chi_k = 0$) because all blocks share one size. In the log rows the feed partition reads $u_{i,0} - \ln\big(S_i g_i^{(0)} C_{m,i}\big) + z_i\psi_0 = 0$ and the permeate partition $u_{i,M} - \ln\big(S_i g_i^{(1)}\big) - v_{i,M} + z_i(\psi_M - \pi_M) = 0$.

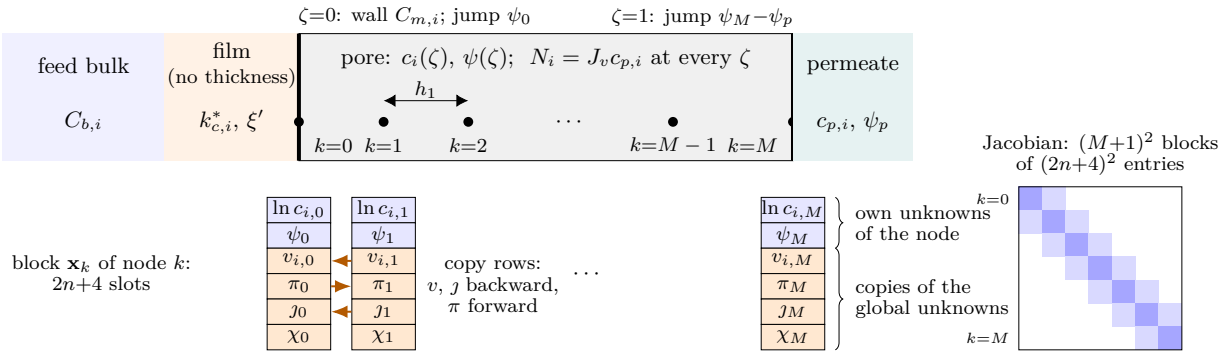



**Figure 74:** The DSPM-DE station. *Top*: the feed bulk, the film (an interface balance with no thickness), the wall, the pore nodes $k = 0,\dots,M$ and the permeate; the Donnan jumps sit at the two faces. *Bottom*: the unknowns of a node (408) — its own (blue) and its copies of the global unknowns (orange), each copy held equal to a neighbour's by a row; the arrows show the direction of each chain ($v$ and $\jmath$ backward, $\pi$ forward; $\chi$ is read at node 0 only, so it is not chained). *Right*: the resulting Jacobian, block-tridiagonal — the diagonal blocks dark, the off-diagonal blocks light, nothing else.

**Why the order of the copy rows matters.** The linear solve is the block-Thomas factorisation (block LU of a block-tridiagonal matrix): with $\mathbf{A}_k$, $\mathbf{D}_k$, $\mathbf{C}_k$ the sub-, main and super-diagonal blocks,

$$\mathbf{D}'_0 = \mathbf{D}_0, \qquad \mathbf{L}_k = \mathbf{A}_k\,\mathbf{D}'^{-1}_{k-1}, \qquad \mathbf{D}'_k = \mathbf{D}_k - \mathbf{L}_k\,\mathbf{C}_{k-1},$$

then forward and back substitution; each $\mathbf{D}'_k$ is factorised by LU with partial pivoting *inside* the block, but nothing pivots *between* blocks. So every pivot block must be non-singular on its own — and $\mathbf{D}'_0 = \mathbf{D}_0$ is the first block's own rows and columns, untouched by anything else. The first layout of the rows held the $\psi_p$ copy equal *backward* ($\pi_k - \pi_{k-1}$) and put zero current at node 0. Then no row of block 0 read $\pi_0$ (zero current reads only the permeate copies $v$): the $\pi_0$ column of $\mathbf{D}_0$ was empty, $\mathbf{D}_0$ singular, and every first solve failed — measured, not supposed. The rule that fixes it: each global unknown's own equation sits at the node where it is *read* by a physical row, and its copy chain runs *away* from that node, so that every copy row has its $+1$ on its own block's diagonal. $\psi_p$ is read at the permeate face, so zero current sits at node $M$ and the chain $\pi_k - \pi_{k+1}$ runs forward; $J_v$ and $\xi'$ are fixed at the wall, so their equations sit at node 0 and their chains run backward ($\jmath_k - \jmath_{k-1}$; the field is not chained, its slot idle). The permeate copies $v$ chain backward too: their defining rows are the permeate partitions at node $M$, but every flux row reads its own node's $v_{i,k}$, so each diagonal block holds its $v$ columns regardless.

### 19.5.3 The Newton, its Jacobian and its stopping rule

**The residual, normalised.** Every row is divided by the terms it balances (the project's one convergence rule, `solver/Convergence.H`), so each is $O(1)$ and a single norm compares them:

- a flux row, $F = \big(t_1 - t_2 - J_v c_{p,i}\big)/\big(t_1 + t_2 + |J_v c_{p,i}|\big)$, with $t_1 = (P_i/h_k)B(-a)c_{i,k}$ and $t_2 = (P_i/h_k)B(a)c_{i,k+1}$ the two terms of (406);
- electroneutrality, $(X_d + \sum_i z_i c_{i,k})/(|X_d| + \sum_i |z_i| c_{i,k})$;
- zero current, $\sum_i z_i c_{p,i} / \sum_i |z_i| c_{p,i}$; wall electroneutrality, $\sum z_i C_{m,i} / \sum |z_i| C_{m,i}$ (the excluded ions included);
- the water flux, $\big(J_v - A_w \Delta P + A_w R_{\mathrm{g}} T(\sum C_{m,i} - \sum c_{p,i})\big)/(A_w \Delta P)$;
- the partitions (in logs) and the copy rows are already $O(1)$.

**The analytic Jacobian.** Every row has closed-form derivatives, assembled straight into the three bands: the flux terms through $\partial t_1/\partial a = -(P_i/h_k)B'(-a)c_{i,k}$ and $\partial t_2/\partial a = (P_i/h_k)B'(a)c_{i,k+1}$ with (407), $\partial a/\partial\psi_{k+1} = -z_i$, $\partial a/\partial\psi_k = z_i$, $\partial a/\partial\jmath = K_{c,i}h_k J_{\mathrm{ref}}/P_i$, and $\partial c/\partial u = c$; the wall through the derivatives of (401) with respect to $c_{p,j}$, $J_v$ (including $d\Xi/d\varphi$) and $\xi'$; the Davies coefficients through $d\ln\gamma/dI$. The *normalisation is differentiated too*, $d(r/s) = \big(dr - (r/s)\,ds\big)/s$: the Jacobian of a differently normalised function would make the line search compare the norms of two different functions. The claim is audited, not assumed: at the first station of a unit, on an 11-node copy of the problem, the first iteration also builds the dense finite-difference Jacobian and reports the largest entry outside the bands (exactly 0 on every corpus case; a declared structure that is false by more than $10^{-4}$ of the in-band entries is refused by name) and the departure of the analytic entries from it ($10^{-9}$ to $10^{-6}$ of each row's largest entry, the difference quotient's own error).

**The step.** $\delta = -\mathbf{J}^{-1}\mathbf{F}$ by the block-Thomas solve above, $O\big(M(2n+4)^3\big)$ operations.

**The line search.** From $\alpha = 1$, the trial $\mathbf{x} + \alpha\delta$ is accepted as soon as $\|\mathbf{F}\|_2$ decreases; otherwise $\alpha$ is halved. A trial that cannot be evaluated — a logarithm beyond $\pm 700$, a wall concentration (401) that is not positive — is *infeasible* and halves $\alpha$ the same way; it is never clamped. The halving stops below $\alpha_{\min} = 10^{-4}$: if no trial down to $\alpha = 2^{-13}$ decreased the norm, the Newton takes the smallest step tried and goes on; if no trial at all could be evaluated it stops and says so (and the station turns to the continuation below).

**The stopping rule.** Converged when $\|\mathbf{F}\|_2 < 10^{-12}$ on the normalised rows, within 60 iterations. On the answer the run also measures what the norm does not show: the largest departure of any interval's flux from $J_v c_{p,i}$, relative to the two terms it nets (round-off, $10^{-15}$) and relative to the net flux itself (larger by the stiffness: a co-ion carrying terms many orders larger than the flux they net loses $10^{-16}$ of the large terms to round-off, the representable limit of the problem); the zero-current and electroneutrality residuals; and the stiffness ratio.

### 19.5.4 The start: a local Donnan seed, and continuation in $J_v$

A Newton is only as good as its start, and here the naive one fails: log-linear concentrations with a *flat* potential leave electroneutrality to the Newton, and on a strongly charged pore with a divalent co-ion the first step overflowed (measured). The seed Choupo builds satisfies both partitions and electroneutrality at every node *exactly*, and leaves the Newton only the flux rows and zero current. Each ion gets a *reservoir* value that runs log-linear, in the node

index, from the feed side to the permeate side, and at every node the potential is the one that makes the pore electroneutral against those reservoirs:

$$\ell_{i,k} = (1-t_k)\ln\!\big(S_i g_i^{(0)} C^s_{m,i}\big) + t_k \ln\!\big(S_i\, c^s_{p,i}\big), \quad t_k = k/M; \qquad \sum_i z_i\, e^{\ell_{i,k} - z_i\psi_k} + X_d = 0, \tag{409}$$

then $u_{i,k} = \ell_{i,k} - z_i\psi_k$. The left side of the second equation falls monotonically in $\psi_k$ (its derivative is $-\sum_i z_i^2 e^{\ell_{i,k}-z_i\psi_k} < 0$) from $+\infty$ to $-\infty$ when both signs of charge are present, so a bisection on $[-150, 150]$ (to a width of $10^{-13}$, the exponent capped at 700) cannot miss the root — and, unlike a Newton step, cannot fall onto the trivial $\psi = 0$ branch when a non-trivial one exists. The copies start at $v_{i,k} = \ln c^s_{p,i}$, $\pi_k = 0$ (so the last node's jump is its own local Donnan potential, consistent with the permeate partition by construction), $\jmath_k = J^s/J_{\text{ref}}$, $\chi_k = 0$. The seed permeate and flux are the unrejected $c^s_{p,i} = g_i^{(0)} C_{b,i}$ and half the pure-water flux, $J^s = A_w \Delta P/2$ (the imposed $J_v$ on the bench); $C^s_{m,i}$ is (401) at that permeate and flux with $\xi' = 0$, or $C_{b,i}$ if that gives no positive wall. This seed is used once per unit; every later station (or bench point) starts from the previous converged answer.

**Continuation in the drive.** When the Newton fails from its start, the station is solved again from the one state that is exact without any Newton: at $J_v \to 0$ nothing crosses, the wall is the bulk and the pore is in Donnan equilibrium with both baths, so the permeate is $c_{p,i} = g_i^{(0)} C_{b,i}$ — which is what the seed (409) builds when handed that permeate. From a drive small enough that every ion has $\mathrm{Pe}_i \le 10^{-4}$, the drive (the imposed $J_v$, or $\Delta P$ under the closure) is raised geometrically to its target, each solve warm-started from the last; a failed step shortens the ratio. It is a solver aid, so it is announced and counted.

```
STATION(problem, previous):
  if closed and ΔP ≤ Δπ₀: return "no water flux"   // eq. (403)
  if a previous converged state exists:              // next station / next point
      x ← previous state, J_v copies rescaled to this J_ref
  else:                                              // the unit's first station
      AUDIT on an 11-node copy (structure, analytic vs FD Jacobian)
      x ← SEED(c^s_{p,i} = g_i^(0) C_{b,i},  J^s = A_w ΔP/2)
  (x, ok) ← NEWTON(x)
  if not ok: (x, ok) ← CONTINUATION()
  if not ok: REFUSE the run (naming J_v, X_d, the node count, why)
  return x

SEED(c^s_p, J^s):                                    // eq. (409)
  C^s_m ← wall (401) at (c^s_p, J^s, ξ' = 0);  if not positive: C^s_m ← C_b
  for k = 0..M:    t ← k/M
      ℓ_i ← (1 − t) ln(S_i g_i^(0) C^s_{m,i}) + t ln(S_i c^s_{p,i})
      ψ_k ← bisection on [−150, 150] of Σ_i z_i e^{ℓ_i − z_i ψ} + X_d = 0
      u_{i,k} ← ℓ_i − z_i ψ_k;   v_{i,k} ← ln c^s_{p,i};   π_k ← 0
      ȷ_k ← J^s/J_ref;   χ_k ← 0

NEWTON(x):                                  // at most 60 iterations
  repeat:
      if ‖F(x)‖₂ < 10^−12: return (x, ok)
      assemble the analytic bands of J(x);   δ ← −J^−1 F (block Thomas)
      α ← 1
      while α ≥ 10^−4:
          if F(x + αδ) is infeasible: α ← α/2;  continue
          if ‖F(x + αδ)‖₂ < ‖F(x)‖₂: break
          α ← α/2
      if no trial was evaluable: return (x, failed)
      x ← x + αδ          // the accepted α, else the smallest tried

CONTINUATION():                                  // drive D = imposed J_v, or ΔP
  t ← min(1,  min_i(10^−4 P_i/K_{c,i})/D_J)    // D_J = J_v, or A_w ΔP
  x ← NEWTON( SEED(c_{p,i} = g_i^(0) C_{b,i},  J^s = t D_J) ) at drive tD
  r ← 4
  while converged and t < 1 and at most 400 solves:
      t' ← min(1, rt);  try NEWTON from x at drive t'D
```

```
    if converged: t ← t′;   r ← min(2r, 100)
    else:         r ← √r;   stop if r < 1.001
return (x, converged and t = 1), with the number of solves
```

**Listing 1:** One channel station of DSPM-DE (the procedure the module and the bench run).

On the corpus the continuation is rare and cheap: in `membrane10_dspmde_divalent` it is needed once, at the first station of the $MgSO_4$ unit (5 solves, 44 Newton iterations in all; the line search had found no evaluable step from the seed), and nowhere else.

## 19.6 The mesh and its observed order

The nodes are uniform by default, 101 of them (100 intervals), each default announced. Two other grids are declared choices. `grading` (0–6) clusters the nodes toward *both* faces by $\zeta_k = \frac{1}{2}\big[1 + \tanh\big(\beta(2k/M - 1)\big)/\tanh\beta\big]$; it keeps the second order but was *measured* not to help on any case in the corpus or in the NanoFiltran matrix (from $\beta = 1.5$ up, 1.1 to 8 times the uniform grid's error at the same node count), because the steep part of a mixed-counter-ion pore is an *interior* front — where a trace monovalent counter-ion hands the fixed charge over to a divalent one — not a face layer: the Donnan and Born jumps are partitions, not layers. `grid adaptive` moves the nodes to the solved profile. Since the fitted flux is exact on a cell where $\psi$ is linear, its truncation error on cell $k$ is $O(h_k^2|\psi''|)$, and the node count is spent best with $h_k \propto |\psi''|^{-1/3}$: the nodes equidistribute the monitor $M_k = |\psi''|_k^{1/3} + \overline{|\psi''|^{1/3}}$ (the mean is a floor that keeps at least half of them uniformly spread), smoothed by two passes of a $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ filter, and the pore is solved again on the moved grid — the state carried by linear interpolation of $u$ and $\psi$, the copies kept — until no node moves by more than a tenth of its cell (`gridTolerance`, at most `gridPasses` = 8 re-solves). On the interior front it leaves 0.76 of the uniform grid's error at the same node count, on the corpus cases the same error: a margin and not an order, because the curvature of $\psi$ there is spread over a third of the pore. So it is a declared option, not the default.

**The order.** On every problem measured — the ambipolar limit of §19.7 at Péclet 0.44 and 4.4, a stiff pore ($X_d = -200\,\mathrm{mol/m^3}$, migration $2.9 \times 10^5$ times the net flux) and the reference case of §19.8 — the observed order of the uniform grid is 2.00 (1.97–2.01). At Péclet 440, a cell Péclet number of 17.6 on 25 intervals, the fitted flux is exact to round-off on every grid, which a central difference is not. Because the order is clean, two grids give a Richardson extrapolation (error $\propto h^2$), which is how the reference values of §19.8 are obtained.

## 19.7 Verification: is the equation solved right?

**An exact limit.** For one 1:1 salt in an uncharged pore, (389) makes $c_+ = c_- = c$ and (399) makes $c_{p,+} = c_{p,-} = c_p$; eliminating $d\psi/d\zeta$ between the two flux equations leaves one *ambipolar* equation, $J_v c_p = -P_s\, dc/d\zeta + K_{c,s} J_v c$, with

$$P_s = \frac{2P_+P_-}{P_+ + P_-}, \qquad K_{c,s} = \frac{K_{c,+}P_- + K_{c,-}P_+}{P_+ + P_-}, \qquad \frac{c_p}{\gamma\, C_m} = \frac{SK_{c,s}}{1 - (1 - SK_{c,s})\, e^{-\mathrm{Pe}_s}}, \tag{410}$$

$S = \sqrt{\Phi_+\Phi_-}$, $\mathrm{Pe}_s = K_{c,s}J_v/P_s$ — the neutral solute's classical result with the salt's ambipolar coefficients. The finite-difference solve converges to (410) at second order (1.97–2.00 from 25 to 400 intervals), within $10^{-8}$ at 1600 intervals; and end to end, on an uncharged, Born-free copy of the witness module, the engine reproduces it within $4 \times 10^{-8}$ at every channel station with 101 nodes (the hindrance of (393)–(394), the wall closed with the pore's own permeate). The gate `check_pore_solver` holds all of this, and the analytic Jacobian against a finite-difference one on random states ($4.5 \times 10^{-8}$ of a row at worst, exactly 0 off the bands).

**The same model in another program.** NanoFiltran [85] solves the same model with central differences and an under-relaxed Newton on a dense matrix. With three declared choices — `activity bulkOverPoreDavies;`, `bulkDielectric 80;` and the default `wallClosure porePermeate;` — Choupo's DSPM-DE is NanoFiltran's model, and `tutorials/props/membrane/nanofiltran_geraldesAlves2008` (part A) runs it at imposed $J_v$ (the `poreTransport` bench) beside NanoFiltran on its shipped example, a 1:1 salt, a 2:1 salt with a divalent co-ion, Born and Davies on and off, $J_v$ from $10^{-6}$ to $5 \times 10^{-5}$ m/s and a polarised mixture: the two programs' grid-limit permeates agree within $8 \times 10^{-7}$ on every one of 35 rows, against NanoFiltran's own discretisation error of up to $4 \times 10^{-4}$ at its 205-node limit. That is *verification* — two implementations of one model — not validation. Reaching it needed three of NanoFiltran's *constants* matched through its inputs (its Vogel correction of $D$ at 298.15 K, its rounded Born constant, and single-precision literals in the hindrance polynomial that move $K_d$ of $Mg^{2+}$ by $1.5 \times 10^{-4}$) and showed that NanoFiltran evaluates Davies with exp on the $\log_{10}$ constant, so its $\ln\gamma$ is $\ln 10$ too small; Choupo has deliberately no option that reproduces it. Every difference and how it was handled is in `docs/design/a-stiff-pore-solved-at-once.md` §9.

**What was abandoned, and why.** Before the pore was solved, a closed form carried each ion across alone,

$$\frac{c_{\mathrm{out},i}}{c_{\mathrm{in},i}} = \frac{K_{c,i}}{K_{c,i} + \big(1 - K_{c,i}\, e^{-\mathrm{Pe}_i}\big)R_i}, \tag{411}$$

$R_i$ the ratio of the two faces' partition coefficients, with electroneutrality imposed only at the faces and no current condition between them. It is not an integral of (397) under (389)–(399), not even in its own uncharged limit (a point solute is rejected by up to 50 % where the ENP rejects nothing), and with $K_c > 1$ it tends to a permeate richer than the pore inlet as $\mathrm{Pe} \to 0$; on the corpus record it rejected $MgSO_4$ at 58 % where the solved pore gives 99.98 %. `poreSolver { method closedForm; }` now refuses by name, and so does `hindrance legacy;` (which took $H$ itself as $K_d$, counting the steric partition twice); the record of both, and of the nested wall and water-flux loops the single Newton replaced, is `docs/design/a-stiff-pore-solved-at-once.md`.

## 19.8 Practical guidance: when the pore is stiff, and how many nodes

**When the pore is stiff.** Compute $\mathrm{Pe}_i$ (404) for each ion and the ratio of the hindered diffusivities $K_{d,i}D_i$ of the counter-ions. The three studies of the NanoFiltran case are the examples, each $\mathrm{Pe}_i$ computed from the case's own records with (392)–(394):

| study | pore | $J_v$ [m/s] | $\mathrm{Pe}_i$ | what the profile does |
|---|---|---|---|---|
| C (Table 2) | $r_p = 0.40\,\mathrm{nm}$, $\Delta x_e = 2.5\,\mu\mathrm{m}$, $X_d = -30$ | $5 \times 10^{-5}$ | $Mg^{2+}$ 33, $Na^+$ 0.56, $Cl^-$ 0.20 | interior front; stiffness $1.65 \times 10^3$ |
| A (matrix) | the same | $10^{-6}$–$5 \times 10^{-5}$ | $Mg^{2+}$ 0.67–33 | linear at low $J_v$, front at high |
| B (Fig. 3) | $r_p = 0.63\,\mathrm{nm}$, $\Delta x_e = 6.2\,\mu\mathrm{m}$ | $(4.3\text{–}9.9) \times 10^{-6}$ | $Na^+$ 0.06–0.14, $Cl^-$ 0.03–0.07 | nearly linear; stiffness 5–40 |

In study C the two counter-ions' hindered diffusivities differ 70 times ($K_dD = 3.0 \times 10^{-10}$ for $Na^+$ against $4.3 \times 10^{-12}\,\mathrm{m^2/s}$ for $Mg^{2+}$, whose $\lambda = 0.875$): $Mg^{2+}$ is carried, the trace $Na^+$ diffuses, and the pore holds an interior front where the trace $Na^+$ hands the fixed charge over to $Mg^{2+}$: $Na^+$ falls from 12 to $0.16\,\mathrm{mol/m^3}$ between $\zeta = 0.3$ and the permeate face while $Mg^{2+}$ rises from 9 to $15\,\mathrm{mol/m^3}$ to keep the pore neutral. That is the stiff case. In study B every $\mathrm{Pe}_i$ is about 0.1 and the profiles are nearly straight: the physics of that study is not in the pore but at the wall, where the excluded trivalent dye ($\lambda = 1$) accumulates and sets the wall's electroneutrality, which drives $Cl^-$ to $R_{\mathrm{obs}} \approx -2$. The run prints the stiffness (migration over net flux) of every first station.

**Reading a solved pore on screen.** The `poreTransport` bench publishes, for every operation, the numbers of this chapter beside each other, so none has to be recomputed by hand: per ion, $\lambda_i$, $K_{d,i}$, $K_{c,i}$, $\Phi_{S,i}$ and the Born factor $e^{-\Delta W_i/k_BT}$; per ion, face and $J_v$ point, the activity ratio $g_i$, the Donnan factor and the partition coefficient (400) as read off the solved pore (so that the product of the factors equalling it is a check, not a construction), the pore's face concentrations and $\mathrm{Pe}_i$ (404); per point, the potential across the membrane split into the feed Donnan jump, the drop inside the pore and the permeate Donnan jump, the stiffness, and the Newton's final $\|\mathbf{F}\|_2$ against its tolerance with the closure checks of §19.5.3 (electroneutrality, zero current, flux constancy). An excluded ion is published as excluded. The console prints the same as a per-ion "why" table.

**How many nodes — it depends on what is reported.** Measured on study C (the case's six node counts, plus 801 and 1601 nodes, each operation writing its pore profile; the reference is the case's finest grid, $n_{\mathrm{nodes}} = 350$, itself within $10^{-4}$ relative of the Richardson limit of 801/1601 nodes on every row):

| quantity (its value at 350 nodes) | error at $n_{\mathrm{nodes}} =$ 10 | 20 | 50 | 100 | 200 |
|---|---|---|---|---|---|
| $R_{\mathrm{obs}}$ of $Na^+$ (99.4457 %), in points | −0.073 | −0.017 | −0.0025 | −0.0006 | −0.0001 |
| permeate $Na^+$ ($0.05543\,\mathrm{mol/m^3}$) | +13.2 % | +3.03 % | +0.45 % | +0.10 % | +0.019 % |
| permeate $Mg^{2+}$ ($8.6317\,\mathrm{mol/m^3}$) | −0.64 % | −0.15 % | −0.023 % | −0.005 % | −0.001 % |
| permeate $Cl^-$ ($17.3188\,\mathrm{mol/m^3}$) | −0.59 % | −0.14 % | −0.021 % | −0.005 % | −0.001 % |
| $Mg^{2+}$ at the pore's permeate end ($15.0235\,\mathrm{mol/m^3}$) | −0.077 % | −0.018 % | −0.0026 % | −0.0006 % | −0.0001 % |
| $\psi$ at the pore's permeate end (0.07217 V) | −4.52 % | −1.08 % | −0.16 % | −0.038 % | −0.007 % |
| potential across the membrane, $\Delta\Psi_m$ (0.14496 V) | −2.21 % | −0.53 % | −0.080 % | −0.018 % | −0.003 % |
| Newton iterations (12 at 350) | 7 | 9 | 9 | 10 | 11 |

The lesson: *the rejection forgives a coarse grid; the permeate of a trace ion — and therefore any selectivity built on it — does not.* Ten nodes put the $Na^+$ rejection within 0.1 point and its permeate 13 % high. The error falls as $h^2$ (each doubling of the node count divides it by about four), so a target accuracy $\varepsilon$ on the reported quantity needs $n_{\mathrm{nodes}} \approx n_0\sqrt{e_0/\varepsilon}$ from any measured pair $(n_0, e_0)$. On the corpus modules the 101-node default is within

$1.4 \times 10^{-7}$ of the converged grid; an interior front like study C's at its hardest needs some 400 nodes for $10^{-4}$ on the trace ion, and the code-to-code case runs 801. Measure it for your case: declare `nodes` at $n$ and $2n$ and compare the quantity you report.

**What it costs, against NanoFiltran.** On NanoFiltran's 13-problem matrix (each problem solved cold and alone, wall clock including process start, one machine; `docs/design/a-stiff-pore-solved-at-once.md` §11.5), NanoFiltran took 0.16–0.61 s per problem at its 102-node grid on the eleven it converged there, and 2.7–32 s at its 204-node grid; Choupo took 0.008–0.019 s at 101 nodes — 18 to 39 times faster on those eleven, about 10 ms of it the process start — and 0.025–0.21 s at 801 nodes, four times NanoFiltran's finer grid (up to some 580 times faster than NanoFiltran at 204 nodes, on NaCl at $10^{-5}$ m/s). NanoFiltran stopped at its iteration limit, short of its tolerance, in 6 of its 26 runs (the 1:1 salt at 204 nodes, the 2:1 salt at both grids), taking 236–4001 iterations of its under-relaxed Newton; Choupo converged on all 13 problems in 4 to 24 Newton iterations to $\|\mathbf{F}\|_2 \le 10^{-12}$, with no continuation at 801 nodes (one problem needed it at 101 nodes, 5 solves). A whole module — `membrane10_dspmde_divalent`, three units of 51 stations at 101 nodes — runs in about a third of a second, a station taking about three Newton iterations from the previous station's answer.

## 19.9 Closing the water flux and marching the channel

In a spiral-wound module the station of §19.3 is solved at every node of the channel march of §18, each with its own local bulk $C_{b,i}$ and pressure difference $\Delta P$, the water flux (403) one row of its Newton. Each station starts from the previous station's converged state (the channel moves a little between stations), so only a unit's first station uses the seed. The per-ion permeate concentrations are then collapsed back to the per-salt $c_p$ the march carries; in a single-salt feed both legs agree by electroneutrality, in a *mixed* feed the pore conserves charge and not salts, so a salt's two legs can differ — the module publishes the cation leg and the run names every salt whose legs differ.

**Key idea. Staged exclusions — avoiding the $K = 1$ trap.** A symmetric charged solve can collapse onto the trivial Donnan $= 0$ saddle (the same pathology as symmetric LLE, §11). DSPM-DE is therefore built as *visible switches*: **(a)** steric only ($X_d = 0$, $\varepsilon_p = \varepsilon_b$); **(b)** +Donnan ($X_d \neq 0$); **(c)** +Born ($\varepsilon_p < \varepsilon_b$). Which stage is active is decided by the fields present in the membrane `.dat` and *announced in the run header* — never a silent fallback. No stage is seeded from another at run time: the trap is avoided by how the solve starts (§19.5.4) — local Donnan equilibrium at every node, each node's potential found by a monotone bisection, which *cannot* land on the $\psi = 0$ branch when a non-trivial one exists.

**Worked example 19.1.** The divalent rejection ordering, `membrane10_dspmde_divalent`Three parallel NF270-archetype elements ($r_p = 0.60$ nm, $X_d = -5$ mol/m$^3$, $\varepsilon_p = 35$ vs $\varepsilon_b \approx 78$), each fed 5 mM of a single salt at 10 bar. With *no* per-salt permeability fit, the DSPM-DE partition reproduces the canonical NF fingerprint

$$R_{\mathrm{obs}}(\mathrm{MgSO_4}) \;>\; R_{\mathrm{obs}}(\mathrm{CaCl_2}) \;>\; R_{\mathrm{obs}}(\mathrm{NaCl}),$$

Choupo returning $R_{\mathrm{obs}}(\mathrm{NaCl}) \approx 50\,\%$ and $R_{\mathrm{obs}}(\mathrm{CaCl_2}) \approx 70\,\%$ against near-total rejection of the divalent-co-ion salt $\mathrm{MgSO_4}$ (99.97 %), with the pore solved and the wall closed with its own permeate (the defaults). The charge is the whole story — linear in $z$ in the Donnan factor of (400), squared in the Born term (390): the divalent co-ion $\mathrm{SO_4^{2-}}$ is excluded far harder than the monovalent $\mathrm{Cl^-}$ at the same pore. Plain solution-diffusion, with one $B_s$ per salt, can be *fitted* to this ordering but cannot *predict* it. Toggling $\varepsilon_p$ off (`membrane11_dspmde_born_toggle`) isolates the Born contribution.

**Power and limit.** The power is mechanism: DSPM-DE predicts the salt-by-salt and divalent rejection ordering from a handful of *physical* membrane parameters ($r_p$, $X_d$, $\varepsilon_p$, $A_k/\Delta x$) plus tabulated ion radii and diffusivities — no per-salt $B_s$ to fit. It is the mechanistic NF model and the research bridge of the simulator's membrane domain. The limit is that every parameter is an *effective* fitted quantity: the pore radius depends on the ion-radius convention (Choupo's catalogue uses hydrated/Nightingale radii, so $r_p = 0.6$ nm is consistent with *those*, not with the Stokes-radius fits of the original papers); the charge density $X_d$ is in reality pH- and concentration-dependent (it grows on dilution and with rising pH) but is a constant scalar here; $\varepsilon_p$ is a single fitted dielectric; the pore is solved on a grid (101 nodes unless the case declares otherwise) at every channel station, a few hundred milliseconds per module; and the model is for an aqueous electrolyte, not a general mixed-charge speciation solver. Treat the numbers as a mechanistic ordering tool, calibrated per operating point — *all models are wrong, some are useful.*

**Runnable case.** `tutorials/steady/membranes/membrane10_dspmde_divalent` — the divalent > mixed > monovalent rejection signature, three salts on one membrane.
`tutorials/steady/membranes/membrane11_dspmde_born_toggle` — the same case with the Born (dielectric) term toggled on/off to isolate its contribution.
`tutorials/steady/membranes/membrane20_dspmde_pore_finite_difference` — the same membrane and

feeds as membrane10, the solved pore DECLARED on NaCl and $MgSO_4$, its wall closed with the pore's own permeate and, beside it, with the solution-diffusion $B_s$ (NaCl $R_{obs}$ 49.7 against 30.8 %; $MgSO_4$ equal). The code-to-code comparison with NanoFiltran is `tutorials/props/membrane/nanofiltran_geraldesAlves2008`, the NanoFiltran paper's one case, whose propsDict `views` generate the paper's tables and figures.

# 20 Nanofiltration without a fixed charge: solution-diffusion with one electric field (SDEM)

*What you'll learn.* Solution-diffusion (§18) gives every solute its own permeability and couples nothing. Applied to a feed declared *ion by ion* — $Na^+$, $Cl^-$, $Mg^{2+}$, $SO_4^{2-}$ — it lets each ion cross at its own rate, and the permeate comes out carrying a net electric charge, which no permeate does. DSPM-DE (§19) answers with a pore, a fixed wall charge and three partition penalties. This chapter derives the other answer, the one that adds *no parameter*: the ions keep their own permeances but share *one* electric field, fixed by the condition that no current flows. The single-salt case collapses exactly to the classical rejection law with an *ambipolar* permeance, which is why per-ion permeances can be calibrated from the single-salt tests that manufacturers actually publish; a trace ion in a dominant salt can be pushed *through* the membrane by that field, and rejected negatively. Selected by `transport SDEM;`.

## 20.1 The silence of the uncoupled law

Write a brackish feed as the water analysis gives it — seven ions and water — and run it through `transport solutionDiffusion;`. Each ion obeys (371) with its own $B_s$, so a fast anion crosses faster than a slow cation and the permeate carries a net charge. Measured on `membrane07` the imbalance was $-0.82\,\%$ of the cation equivalents against the $-0.35\,\%$ the analysis itself carries: small, physically impossible, and until 2026-09-15 said only inside the scaling block. The engine now announces *per-ion solution-diffusion without charge coupling* on every such feed and names the remedy.

The remedy is not a new coefficient. A membrane that lets $Na^+$ through faster than $Cl^-$ builds up, within nanoseconds, an electric potential across its active layer that holds the cation back and pulls the anion forward until the two cross at one common rate — the diffusion potential of every electrochemistry course. Adding that potential as one state variable per station, and zero net current as the one equation that fixes it, is the whole model.

## 20.2 One field, zero current: the equations

The active layer is a slab of dimensionless thickness $x \in [0,1]$, the wall at $x=0$ and the permeate face at $x=1$. Each ion $i$ carries a signed charge $z_i$, a membrane permeance $P_i$ (m/s) and a *virtual* concentration $c_i(x)$: the concentration of the bulk solution that would be in equilibrium with the membrane at $x$ [86]. The flux is Nernst–Planck on the virtual solution, with $\psi = F\varphi/R_gT$ the dimensionless potential:

$$j_i \;=\; -P_i\Big(\frac{dc_i}{dx} + z_i\,c_i\,\frac{d\psi}{dx}\Big). \tag{412}$$

Two constraints close it. The virtual solution is electroneutral at every $x$, and no current flows:

$$\sum_i z_i\,c_i(x) = 0, \qquad \sum_i z_i\,j_i = 0. \tag{413}$$

Differentiate the first along $x$, substitute (412), and the field is an algebraic consequence of the two with nothing to fit:

$$\boxed{\;\frac{d\psi}{dx} \;=\; -\,\frac{\sum_i z_i\,j_i/P_i}{\sum_i z_i^2\,c_i}\;} \tag{414}$$

The boundary conditions are the wall concentrations $c_i(0) = c_{m,i}$ (the polarisation object of §18: the film model applied per ion by default, the coupled interface model of §18.4 when the case declares it) and the convective closure at the permeate, $j_i = J_v\,c_{p,i}$ with $c_i(1) = c_{p,i}$, where $J_v$ is the volume flux. A neutral solute ($z_i = 0$) drops out of (414) and follows the plain law (371) unchanged.

**Where the partition coefficient went.** Nowhere new. The permeance is the solution-diffusion composite $P_i = D_iK_i/\delta$ — diffusivity, partition coefficient, thickness — exactly as $B_s$ is in (370) and $A_w$ in §18; because (412) is written on *virtual* concentrations the partition is already inside $P_i$ and nothing is declared for it. A separate $K$ is needed only when something makes the two faces different: a fixed charge (the Donnan jump of DSPM-DE) or electroneutrality imposed *inside* the membrane rather than on the virtual solution. SDEM assumes neither, and the formulation is not the constant-field (Goldman) integration it can be mistaken for — the two differ, and the difference is checkable, because only this one gives the exact single-salt limit that follows.

## 20.3 The single-salt limit: an ambipolar permeance, derived

Take one salt $M_{\nu_+}X_{\nu_-}$ at virtual concentration $c(x)$, so $c_+ = \nu_+c$, $c_- = \nu_-c$ and $z_+\nu_+ = |z_-|\nu_- \equiv e$ (equivalents per formula unit). The salt flux $j_s$ carries $j_+ = \nu_+j_s$ and $j_- = \nu_-j_s$, which satisfies zero current identically. The field (414) becomes

$$\frac{d\psi}{dx} = -\,\frac{j_s\,e\,(1/P_+ - 1/P_-)}{c\,e\,(z_+ + |z_-|)} = -\,\frac{j_s}{c}\,\frac{1/P_+ - 1/P_-}{z_+ + |z_-|},$$

and substituting it into the cation's own equation $j_s = -P_+(dc/dx + z_+ c\, d\psi/dx)$ gives, after one line of algebra,

$$j_s = -P_s \frac{dc}{dx}, \qquad \boxed{P_s \;=\; \frac{(z_+ + |z_-|)\, P_+ P_-}{z_+ P_+ + |z_-|\, P_-}} \tag{415}$$

The salt diffuses with a single *ambipolar* permeance, the harmonic-type mean of the two ion permeances weighted by charge, and the profile is linear. With $j_s = P_s(c_m - c_p)$ and $j_s = J_v c_p$,

$$c_p = \frac{P_s}{J_v + P_s}\, c_m, \qquad R = \frac{J_v}{J_v + P_s}, \tag{416}$$

which is (371) with $B_s \to P_s$: the classical law, recovered exactly. The potential across the layer also has a closed form. Integrating the field over the linear profile, $\int_0^1 dx/c = \ln(c_m/c_p)/(c_m - c_p)$, so

$$\Delta\psi \;=\; \psi(1) - \psi(0) \;=\; \frac{P_+ - P_-}{z_+ P_+ + |z_-|\, P_-}\, \ln\frac{c_m}{c_p}, \qquad \frac{c_m}{c_p} = 1 + \frac{J_v}{P_s}. \tag{417}$$

Its sign is the physics: when the cation is the faster ion the permeate side goes *positive* relative to the wall, which is the field that holds the cation back and pulls the anion along.

**Key idea. Calibratable from what is actually measured.** (415) is the reason this law costs no data. A manufacturer's datasheet and most of the literature report *single-salt* rejections, never per-ion permeate analyses. Each single-salt test gives one $P_s$, i.e. one equation in two ion permeances; $n$ ions and $n$ salts close the system and an extra salt over-determines it and checks. The NF270 permeances the witness cases carry [87] were fitted that way, and their Table 1 salt permeances reproduce from their own ion columns through (415) to the table's rounding — an identity the gate `check_sdem` recomputes in Python, independently of the engine, on all four rows.

## 20.4 A trace ion in a dominant salt: negative rejection

Add a trace ion at a concentration far below the dominant salt. Its own term in the denominator of (414) vanishes, so it does not shape the field: it *rides* the field the dominant pair sets, and (412) for the trace is linear in $c_i$ with a known coefficient. (That linear problem integrates in closed form; the 2013 paper gives the solution and the authors of [87] use it as the lines through their data.) The consequence is asymmetric by sign. Under NaCl on NF270 the sodium is far more permeable than the chloride, so the field is positive toward the permeate: a trace *anion* is pulled through and rejected less than the chloride even at equal permeance — and nitrate, three and a half times more permeable than chloride, is lightly *negatively* rejected at low flux — while a trace *cation* is held back and rejected more than its own permeance would say. Under $MgSO_4$ both dominant ions are slow, the magnesium seven times faster than the sulfate, and the same field acting on a nitrate that is three hundred times more permeable than the sulfate pushes more nitrate into the permeate than the water carries: the permeate is *richer* in nitrate than the feed, $R < 0$, with no parameter for the effect.

**Common pitfall. A trace is not a trace when the dominant anion is a hundred times slower.** The closed forms above hold in the *limit* of vanishing trace concentration. At the concentrations the validation source actually used — dominant salt 0.01 mol/L, traces $2 \times 10^{-4}$ mol/L — a trace anion at 42 $\mu$m/s carries more current than 0.01 mol/L of sulfate at 0.13 $\mu$m/s, so the traces relax the field the sulfate would otherwise impose. The engine integrates the full system and needs no such limit; the analytical lines do. The example below measures the gap on the witness cases, and the Tutorials Guide's validation section measures it against the published figures.

## 20.5 Numerics: shoot from the face where the unknown is

Per channel station the module solves an outer Newton in $J_w$ (the osmotic back-pressure, exactly as in §18) around an inner Newton in the permeate composition $c_p$. Each inner residual evaluation integrates (412) with (414) by a fixed-step RK4 across the layer at the fluxes $j_i = J_v c_{p,i}$ the trial $c_p$ implies, and requires the profile to arrive at the wall concentrations the polarisation object gives for that same $c_p$ (the per-ion film by default, the coupled interface model of §18.4 when the case declares it). The unknowns are the $n$ permeate concentrations; the equations are $n - 1$ arrival residuals plus permeate electroneutrality — the arrival residual of the ion carrying the most charge is dropped, because the shot conserves charge exactly and the wall is electroneutral, so it is implied by the others. The Jacobian is finite differences (one shot per ion; the largest ionic feed in the corpus has seven), the step is halved until every concentration stays positive and then backtracked on the residual norm.

The direction of the shot decides whether this converges. The first version integrated from the wall *toward* the permeate and converged at the two witnesses' operating points and almost nowhere else across the validation source's 2–14 bar range: a well-rejected salt arrives at a permeate value a hundred times smaller than it started from, and the field (414) divides by an ionic strength that is smallest exactly there. From the *permeate* face the

state at the start of the shot *is* the unknown, the ionic strength there is known exactly, and the profile grows toward the wall — an error at the start is compared against the large wall values, which is the stable direction. Where both converge they agree to the last digit (it is one ODE); after the change every (feed, flux) pair the source publishes converges in three to seven iterations. The step count is a measurement too: 40 steps put the sharpest profile in the corpus (nitrate under $MgSO_4$) $2 \times 10^{-3}$ off its converged rejection, above the goldens' $10^{-4}$; the engine takes 320, and says so in the source beside the number.

**Refused by name.** A solute with no declared aqueous bridge (the charge is read through `aqueousMapping`, never off a name); an ion the membrane record gives no permeance (an impermeable ion cannot sit in a zero-current balance); a feed that is not electroneutral (the law's own premise — reconcile the analysis first, `aqueousAnalysis {}` does it); fewer than two ions. **Announced:** the model and its assumptions once per run, and the polarisation film, which is computed per ion *without* a field in the film unless `ionCoupling electroneutral;` is declared, which applies the interface model of §18.4 (the diffusion potential exists in the film too; the coupled film carries the interface's, never one inside the film). **Published:** the potential drop per station (`psi_mV` in the profile) and its average (`membranePotential_mV_avg`), because the pedagogical point of the law is that the student *sees* the field that makes chloride wait for sodium.

**Worked example 20.1.** Two witnesses and the closed form`membrane12_sdem_nf270_nacl_traces` is the NaCl witness: 0.01 mol/L of NaCl with $2 \times 10^{-4}$ mol/L each of $NO_3^-$ and $NH_4^+$, 20 °C, the NF270 permeances of [87]. At $J_v = 25.9$ $\mu$m/s the run gives $R(Cl^-) = 0.775$, $R(Na^+) = 0.766$, $R(NO_3^-) = 0.31$, $R(NH_4^+) = 0.78$ and a potential of 34.2 mV. The single-salt law (416) with the ambipolar $P_s = 7.54$ $\mu$m/s predicts $R = 0.774$, and (417) predicts 35.1 mV.
`membrane13_sdem_nf270_mgso4_traces` is the $MgSO_4$ witness: 0.01 mol/L of $MgSO_4$ with four traces. At 24.8 $\mu$m/s: $R(SO_4^{2-}) = 0.992$, $R(Mg^{2+}) = 0.981$, $R(Cl^-) = 0.40$, $R(NO_3^-) = \mathbf{-0.15}$, 24.2 mV; the closed form predicts $R = 0.991$ and 44.4 mV.
The two potentials disagree by under 3 % on NaCl and by a factor of two on $MgSO_4$, and the difference is the pitfall above, measured. Divide the four trace flows in `0/feed` by a thousand and run again: the engine reports 35.2 mV against 35.2 from (417) on NaCl, and 44.4 against 44.4 on $MgSO_4$ — the closed form recovered to 0.1 mV on both — while nitrate under $MgSO_4$ drops to $R = -0.64$. At the paper's own trace level the fast traces carry a share of the current and halve the field the slow sulfate would impose; the rejections published in that paper's figures are compared against both readings, trace limit and actual feed, in the Tutorials Guide.

**Power and limit.** The power is that the electroneutral permeate, the single-salt law, the calibration from single-salt tests and the negative rejection of a fast trace anion all follow from one constraint and no new datum. The limits are named in the run banner and worth repeating. The permeances are constants: no dependence on concentration, so the concentration polarisation of a fitted permeance is whatever the fit absorbed. There is no fixed charge, no Donnan partition and no dielectric exclusion — whatever of those NF270 has is *inside* the fitted permeances, which is also why the source fits a different permeance set for each dominant salt and each witness case carries the record of its own salt. The film is per ion and field-free unless `ionCoupling electroneutral;` is declared, which applies the model of §18.4. The virtual solutions are ideal and dilute. And two of the source's permeances are *bounds* ("> 60", "> 0.8"), so the runs that use them are conditional on a value the source did not fix. Between the two charge-aware laws the choice is a modelling decision the case author owns: DSPM-DE predicts the rejection *ordering* from pore geometry and charge with no per-salt fit; SDEM reproduces measured single-salt permeances exactly and predicts what a *mixture* of those salts does.

**Runnable case.** `tutorials/steady/membranes/membrane12_sdem_nf270_nacl_traces` — NaCl with nitrate and ammonium traces; the single-salt identity and the 34 mV field.
`tutorials/steady/membranes/membrane13_sdem_nf270_mgso4_traces` — $MgSO_4$ with four traces; nitrate negatively rejected.
The Tutorials Guide's *Validation study: ion rejection in nanofiltration* compares the law against the digitised figures of [87], table by table, regenerated from the engine.

# 21 Electrodialysis: Faraday transfer, the Nernst stack, and the limiting current

*What you'll learn.* Electrodialysis (ED) separates ions *electrically*, not by pressure or phase change. A stack of $N$ alternating cation- (CEM) and anion- exchange (AEM) membranes is sandwiched between two electrodes; an applied DC current pulls cations one way and anions the other, draining a *diluate* channel and enriching a *concentrate* channel. This chapter shows the four equations the `electrodialysisStack` unit solves: **(i)** Faraday's law for the ion transfer, **(ii)** the Nernst membrane potential built from *real* Davies activities per channel, **(iii)** the ohmic drop across the membrane and solution resistances, and **(iv)** the limiting current density — the hard ceiling above which the boundary layer runs out of ions, which you will meet twice: as the classical single-salt form on a *declared* transport number, and then as a *prediction* from the ionic diffusivities alone, with no transport number read at all. You will see why ED reuses the electrolyte stack (Chapter 3) wholesale rather than re-deriving an activity model, and why the stack *refuses to clamp* an over-limiting current.

## 21.1 The stack: where charge meets selectivity

A *cell pair* is the repeating unit: one CEM, one diluate channel, one AEM, one concentrate channel. Stack $N$ of them between two electrodes and the same current $I$ passes through every membrane in series. Because a CEM is nearly impermeable to anions and an AEM nearly impermeable to cations, the cations migrating toward the cathode are trapped by the next AEM and the anions migrating toward the anode are trapped by the next CEM: salt accumulates in alternate channels (the concentrate) and is stripped from the others (the diluate). The electrodes themselves run the redox reactions that source and sink the electrons, but ED gathers *many* cell pairs per electrode pair, so the fixed electrode overpotential is amortised — the reason ED is an $N$-times-cheaper-per-membrane process than a single electrochemical cell.

Choupo's unit takes *two* feeds (`inputs ( diluateFeed concentrateFeed )`) and returns two products (`outputs ( diluate concentrate )`), plus an electrical-work stream published on an energy wire ($P = UI$), mirroring the shaft-work convention of the rotating gear in Chapter 24. The components must be ionic species with a row in `ions.dat` (`Na`, `Cl`, ...) plus `water` as the carrier — exactly the analysis the `ionExchanger` expects.

## 21.2 Faraday's law: charge in, moles out

The current is a flux of charge; each mole of a $z$-valent counter-ion carried across a membrane consumes $zF$ coulombs, where $F = 96\,485$ C/mol is Faraday's constant[88]. Inverting, the molar transfer rate of a counter-ion per cell pair is

$$\dot{n}_i \;=\; \xi\,\frac{I}{z_i F}, \tag{418}$$

and the whole stack moves $N\dot{n}_i$ moles per second out of the diluate and into the concentrate. Here $\xi \in (0, 1]$ is the *current efficiency* (default 0.9, announced in the log): not all of the charge does useful work. The shortfall $(1 - \xi)$ is co-ion leakage (a real membrane is not perfectly selective), electro-osmotic water drag, and any parasitic electrode reaction. Choupo treats $\xi$ as a lumped, explicit operating number the student owns — there is no hidden permselectivity sub-model in v1.

**Several counter-ions on one membrane.** Equation (418) gives a membrane's whole current to its one counter-ion. When a membrane has several (a cation-exchange membrane facing $Na^+$ and $K^+$, an anion-exchange membrane facing $Cl^-$ and $SO_4^{2-}$), the current is *shared*: since 2026-10-05 (Vítor's ruling, no new parameter) counter-ion $i$ carries the fraction

$$t_i \;=\; \frac{|z_i|\,D_i\,c_i}{\sum_j |z_j|\,D_j\,c_j}, \qquad \dot{n}_i \;=\; t_i\,\xi\,\frac{I}{|z_i|F}, \tag{419}$$

the sum running over that membrane's counter-ions, with $D_i$ the ion's diffusivity at infinite dilution. These are the limiting transport numbers of Geraldes and Afonso[89] (their Eqs. 12 and 13, §21.7), applied at every current. Because $\sum_i |z_i|\dot{n}_i = \xi I/F$ on each membrane, the cation and anion membranes move the same equivalents and the diluate stays electroneutral; with one counter-ion (419) is (418). Dividing by the ion's own amount, $\mathrm{d}\ln n_i/\mathrm{d}q = -D_i/\sum_j |z_j| D_j n_j$ with $q$ the equivalents passed: the *fractional* removal of a counter-ion depends on its diffusivity alone, while its *share* of the current is $zDc$. Along one pass of a flowing stack this integrates in closed form, $n_i = n_{i,\mathrm{in}}\,e^{-D_i\tau}$ with the single $\tau$ fixed by $\sum_i |z_i|(n_{i,\mathrm{in}} - n_i) = \xi I N/F$; a recirculating batch rig integrates the instantaneous rates in time. Any preference of a real membrane between counter-ions (a Donnan preference for divalent ions, a monovalent-selective layer) is *not* modelled.

Equation (418) is a hard upper bound on what current can remove: the diluate cannot give up more of an ion than it carries. The solver therefore *caps* the transfer at the diluate inflow and emits a LOUD warning when the cap binds (the diluate is fully demineralised for that ion) — the "no silent clamp" credo of Chapter 1 applied to charge.

**Key idea.** Three operating modes, exactly one of which you give. `current <I> A;` fixes the current and the demineralisation emerges; `targetDemin <fraction>;` fixes the demineralisation of the reference cation (the first cation in component order) and Choupo solves (418) for the current $I$ with a glass-box 1-D Newton (the residual $\dot{n}_{\mathrm{removed}}/\dot{n}_{\mathrm{in}} -$ target, printed); `targetConductivityRemoval <fraction>;` fixes the fraction of the diluate's *conductivity* removed, $1 - \kappa_{\mathrm{out}}/\kappa_{\mathrm{in}}$, with both $\kappa$ priced by Nernst–Einstein on every ion's $D^0$ at the two ends of the channel — conductivity is the quantity a cellar or a plant actually measures (the tartaric-stability practice on wine is stated as a per-cent drop in it), and every ion counts with its own $z_i^2 D_i^0 c_i$, so this target does not depend on the order the components are listed in. Giving two of the three, or none, is a degrees-of-freedom error and the unit says so.

## 21.3 The membrane potential: Nernst on real activities

Even at zero current a cell pair sustains a potential, because each membrane separates two solutions of different ionic activity. A perfectly selective CEM is reversible to its counter-cation, so its equilibrium (Donnan + diffusion) potential is the Nernst expression[90]

$$E \;=\; \frac{R_{\mathrm{g}} T}{z F}\,\ln\!\frac{a_{\mathrm{conc}}}{a_{\mathrm{dil}}}, \tag{420}$$

evaluated on the *activity ratio* of the counter-ion across the membrane, not the bare concentration ratio. The activity is $a_i = \gamma_i\, m_i$, and the activity coefficient $\gamma_i$ is the very same **Davies** model the electrolyte stack uses (Chapter 3):

$$\log_{10}\gamma_i \;=\; -A\, z_i^2\left(\frac{\sqrt{I}}{1+\sqrt{I}} - 0.3\, I\right), \qquad I = \tfrac{1}{2}\sum_k m_k z_k^2, \tag{421}$$

with $A = 0.51$ at 25°C (scaled in $T$ by the Debye–Hückel factor) [22]. The unit speciates each channel *once* per pass, freezes the $\gamma_i$, and reuses them everywhere — there is **no parallel $\gamma$ implementation** for ED. The two membranes of a cell pair add their potentials (the CEM on the mean cation ratio, the AEM on the mean anion ratio, $z < 0$ flipping the ratio):

$$E_{\mathrm{mem}} \;=\; E_{\mathrm{CEM}} + E_{\mathrm{AEM}} \quad \text{(V per cell pair)}. \tag{422}$$

This $E_{\mathrm{mem}}$ is the *back-EMF*: it opposes the applied current, growing as the concentrate/diluate ratio widens — the thermodynamic "price" of separating against an activity gradient.

## 21.4 The ohmic drop: membranes and solutions in series

Driving the current $I$ also pays an ohmic toll across the four resistances of a cell pair: the two membranes and the two solution channels. The membrane areal resistances $R^A_{\mathrm{CEM}}, R^A_{\mathrm{AEM}}$ [$\Omega\,\mathrm{m}^2$] come straight from the IEM `.dat` (measured in 0.5 M NaCl); divided by the active area they give resistances in ohms. The solution resistance of a channel of gap $h$ and conductivity $\kappa$ is

$$R_{\mathrm{sol}} \;=\; \frac{h}{\kappa\, A_{\mathrm{mem}}}, \qquad \kappa \;=\; \sum_i c_i\,\lambda_i, \qquad \lambda_i \;=\; \frac{z_i^2 F^2 D_i^0}{R_{\mathrm{g}} T}, \tag{423}$$

where the equivalent ionic conductance $\lambda_i$ is built from the infinite-dilution diffusivity $D_i^0$ (the `transport` block of each species record: the 25 °C tracer diffusion coefficient of the public-domain USGS PHREEQC database `phreeqc.dat`, its `-dw` line [23], each record citing the line it was read from; the hydrated radius beside it is Nightingale's [91], a separate column that this equation does not read) via the **Nernst–Einstein** relation, and $c_i \approx m_i \rho_w$ in the dilute-carrier approximation ($\rho_w \approx 1000$ kg/m$^3$, announced). The per-cell-pair resistance is the series sum[90]

$$R_{\mathrm{pair}} = \frac{R^A_{\mathrm{CEM}}}{A_{\mathrm{mem}}} + \frac{R^A_{\mathrm{AEM}}}{A_{\mathrm{mem}}} + R_{\mathrm{dil}} + R_{\mathrm{conc}}. \tag{424}$$

## 21.5 Assembling the stack voltage and power

The terminal voltage of the whole stack is the sum, over $N$ cell pairs, of the back-EMF plus the ohmic drop, with the lumped electrode overpotential added once:

$$U \;=\; N\big(E_{\mathrm{mem}} + I\, R_{\mathrm{pair}}\big) + E_{\mathrm{electrodes}}, \qquad P = U I = W_{\mathrm{electric}}. \tag{425}$$

$E_{\mathrm{electrodes}}$ is a single ANNOUNCED number, not a resolved Butler–Volmer kinetics: ED v1 does not model the electrode reactions (the shared `electrochem` kernel ships a fully-implemented `butlerVolmer` stub for a future electrolysis unit, so the kernel need not fork). The electrical power $P$ is published on the energy wire; divided by the diluate volumetric product it gives the headline *specific energy* in kWh/m$^3$.

## 21.6 The limiting current: what Cowan and Brown actually gave us

The defining constraint of ED is the **limiting current density**. As current rises, the diluate-side boundary layer at the membrane face is depleted of ions faster than diffusion can replenish it; at the limiting current $i_{\mathrm{lim}}$ the interfacial concentration hits zero and the ohmic resistance there diverges. Past it, the only charge carriers left are $\mathrm{H}^+$ and $\mathrm{OH}^-$ from water splitting — current efficiency collapses, the pH swings, and the membranes are stressed.

**Key idea.** Cowan and Brown [92] gave an *experimental method*, not a predictive model. Plot the stack's electrical resistance $\Delta V/I$ against the reciprocal current $1/I$ and read $i_{\lim}$ off the *minimum*: below the limiting current the stack is ohmic, above it the depleted film adds resistance, and the turning point is the transition. Geraldes and Afonso [89] measure their limiting currents exactly that way, refining the minimum with the fourth-order polynomial fit of Rapp and Pfromm [93]. The single-salt *formula* usually attached to the name is what you get when you integrate Nernst–Planck across a stagnant film for one salt; it is worth knowing separately from the measurement.

That classical single-salt form is

$$i_{\lim} \;=\; \frac{zF\,k\,c_{\mathrm{dil}}}{t_{\mathrm{cu}} - t_{\mathrm{co}}}, \qquad t_{\mathrm{co}} = 1 - t_{\mathrm{cu}}, \tag{426}$$

where $c_{\mathrm{dil}}$ is the bulk equivalent concentration of the *depleting* (diluate) channel and $t_{\mathrm{cu}}$ the counter-ion transport number *in the membrane*. Two things about it should make you uneasy. It is written for *one salt*, and real feeds are not; and $t_{\mathrm{cu}}$ is a **declared parameter** — a number the student supplies, typically measured in a different solution at a different temperature. Choupo keeps (426) as the `CowanBrown` route for a case that names no stack record, and the run says so.

## 21.7 The multi-ionic limiting current, predicted

*What you'll learn.* The rest of this section removes that parameter. It follows Geraldes and Afonso, *J. Membr. Sci.* **360** (2010) 499–508 [89], whose result is that the limiting transport numbers of a multi-ionic solution are not data at all: they follow explicitly from the ionic diffusivities and the bulk composition.

**The film, and what is transported in it.** The ions cross a stagnant film of thickness

$$\delta_{\mathrm{eff}} \;=\; \frac{D_{\mathrm{eff}}}{k_{c,\mathrm{eff}}}, \tag{427}$$

where $D_{\mathrm{eff}}$ is the *effective diffusivity of the multi-ionic solution* (below) and $k_{c,\mathrm{eff}}$ comes from a Sherwood correlation built on that same $D_{\mathrm{eff}}$. Inside the film each ion obeys Nernst–Planck,

$$N_j \;=\; -D_j \frac{\mathrm{d}C_j}{\mathrm{d}x} - z_j C_j D_j \frac{F}{R_{\mathrm{g}} T}\frac{\mathrm{d}\psi}{\mathrm{d}x}, \qquad \sum_{j=1}^{n} z_j C_j = 0, \tag{428}$$

and inside a strongly charged membrane the counter-ion moves by *electromigration only*, so its flux is fixed by the current, $N_j^m = t_j^m i/(z_j F)$.

**The linearisation, and what it buys.** Solving (428) exactly needs an iteration over both the current and the membrane transport numbers. Linearising on the assumption that the concentration profiles through the film are *nearly linear* — and they are, close to the limiting current — collapses that into closed form. Taking the co-ion transport numbers as null, the **limiting transport numbers** of the counter-ions come out as ratios of the ions' own $|z|DC$ products:

$$t_{j,\lim}^m = \frac{z_j D_j C_{j,b}}{\sum_{j=1}^{n_c} z_j D_j C_{j,b}} \qquad \text{(cation-exchange membrane)}, \tag{429}$$

$$t_{j,\lim}^m = \frac{|z_j| D_j C_{j,b}}{\sum_{j=n_c+1}^{n_c+n_a} |z_j| D_j C_{j,b}} \qquad \text{(anion-exchange membrane)}, \tag{430}$$

and the limiting current density itself is

$$i_{\lim} \;=\; F\,\frac{k_{c,\mathrm{eff}}}{D_{\mathrm{eff}}}\,\frac{\sum_{j=1}^{n} C_{j,b}}{\sum_{j=1}^{n} t_{j,\lim}^m \big/ (z_j D_j)}. \tag{431}$$

Equation (431) gives a *different* number for each membrane — positive for the CEM and negative for the AEM, in the paper's sign convention — and **the applicable one is the lowest absolute value**. Choupo computes both and prints which membrane set the answer.

**Key idea.** **Nothing in (429)–(431) is fitted.** The inputs are the ionic diffusivities at infinite dilution — catalogue data, the same `data/standards/species/` tier the conductivity of (423) reads — and the bulk composition of the stream. The only equipment datum is the Sherwood correlation, and that lives on the stack record where it belongs. *A model that removes a parameter is worth more than one that fits better.*

**The effective diffusivity.** $D_{\text{eff}}$ is the multi-ionic generalisation of the Nernst–Hartley salt diffusivity. The paper derives it over $n$ salts (Appendix A), salt $k$ made of cation $k$ and anion $n+k$ with equivalent fraction $x_k$. Written out, the $x$-weights collapse onto the ions themselves:

$$D_{\text{eff}} \;=\; \frac{\displaystyle\sum_{i\in\text{cat}} X_i^{+} D_i \sum_{j\in\text{an}} X_j^{-}\,(z_i + |z_j|)\,D_j}{\displaystyle\sum_{i\in\text{cat}} X_i^{+} z_i D_i + \sum_{j\in\text{an}} X_j^{-}|z_j| D_j}, \qquad X^{\pm} = \frac{|z|C}{C_S}, \tag{432}$$

with the equivalent fractions taken on the total of their own sign. This matters practically: a salt decomposition is *not unique* from a list of ions (Na, Mg, Cl, $SO_4$ is NaCl + $MgSO_4$, or $Na_2SO_4$ + $MgCl_2$, or any blend), and (432) needs none. For one salt it collapses to $D_S = (z_1 + |z_2|)D_1D_2/(z_1D_1 + |z_2|D_2)$, the Nernst–Hartley result.

**One salt: the two forms must agree.** For a single salt (431) reduces *exactly* to the classical

$$i_{\text{lim}} \;=\; \frac{F k_c C_b}{t^{m}_{1,\text{lim}} - z_1 D_1/(z_1 D_1 + |z_2| D_2)}, \tag{433}$$

and Choupo prints both on a single-salt feed rather than asserting the reduction:

```
single salt: Eq. 16 gives 643.0407 A/m2 against Eq. 15's 643.0407 A/m2
             -- the reduction, seen
```

**The assumptions, which are what you must weigh.** Negligible steric exclusion, so the limiting current is reached when *every* counter-ion's interfacial concentration vanishes; null co-ion transport numbers; strongly charged membranes, so diffusion inside them is negligible against electromigration; nearly linear film profiles; dilute solutions, so only ion–solvent friction enters (individual ionic diffusivities, no Onsager cross terms); and film theory itself, which means *no electroconvection* — exactly the regime that ends at $i_{\text{lim}}$. The paper compares the linearised result against a numerical solution of the full non-linear Nernst–Planck system and finds the limiting current within about 15 % even in an extreme case where the highest and lowest ionic diffusivities differ tenfold.

## 21.8 Where the mass-transfer coefficient comes from — and the velocity

$k_{c,\text{eff}}$ is the one number in (431) that is neither thermodynamics nor catalogue data. It comes from a Sherwood correlation

$$\text{Sh} = a\,\text{Re}^{b}\,\text{Sc}^{c}, \qquad \text{Sh} = \frac{k_{c,\text{eff}} h}{D_{\text{eff}}}, \quad \text{Re} = \frac{\rho u h}{\mu}, \quad \text{Sc} = \frac{\mu}{\rho D_{\text{eff}}}, \tag{434}$$

on the diluate *channel height* $h$ — and that correlation is **equipment-and-spacer data, never a universal law**. The one Choupo ships, $\text{Sh} = 0.29\,\text{Re}^{0.5}\text{Sc}^{0.33}$, was fitted by [89] to one stack with one mesh spacer from its own NaCl limiting-current measurements, with an average error of 7 %, and its Schmidt exponent was *postulated* rather than fitted. So it lives on the stack record, with its citation and with that postulate marked as an estimate, and the run announces it.

**Key idea. The crossflow velocity is not an operating knob.** Until 2026-09-16 a case declared `linearVelocity` beside a diluate flow the unit already had on its inlet stream — two homes for one fact, nothing checking them against each other, and the velocity setting $k$ and therefore $i_{\text{lim}}$. With a stack record it is *derived*, and the run prints the arithmetic:

$$u \;=\; \frac{Q_{\text{diluate}}}{(N_{\text{cell pairs}}/n_{\text{passes}})\,W\,h}, \qquad L_{\text{path}} = L_{\text{channel}} \times n_{\text{passes}}.$$

$u$ here is the *superficial* velocity, in the empty channel cross-section; the interstitial value $u/\varepsilon$ is printed beside it, and the record declares which of the two its correlation was fitted on. Declaring a velocity beside `stack` refuses by name.

## 21.9 The stack as a record

```
stack EUR2C-7P18;
```

names a `kind edStack` record in `data/standards/assets/` which carries the membrane pair, the cell pairs, the active area, the channel and spacer, the hydraulic passes, the Sherwood correlation and the operating limits. Two conventions in it are worth stating explicitly.

**`activeArea` is the cell-pair area of the whole stack.** A 50 $m^2$ stack has 50 $m^2$ of anion-exchange membrane *and* 50 $m^2$ of cation-exchange membrane. The area per cell pair ($A/N$), the area per membrane kind ($A$) and the total membrane area ($2A$) are all derived and printed; none is stored, and a record that stores one is refused.

**The hydraulic pass belongs to the equipment**, not to the operation: it is a fact about how the stack is plumbed, so `hydraulicPasses` is on the record and a case that declares it is refused.

The limits are *announced* when a run exceeds them and never refuse — a student may drive a stack past its rating on purpose — and every value the source does not state is marked `origin estimated; reviewStatus unverified;` with a note saying how to verify it, announced on every run that reads it.

Two things a stack record may leave *out*, and in both cases the absence is a fact. A record whose source does not name the membrane pair does not name one, and the *case* then declares `membrane <name>;` beside `stack`; declaring one beside a record that does name a pair refuses instead, so the record decides which way the refusal points and neither side guesses. And where nobody states the channel width or its length, **exactly one of the two is stored** — the width, as the estimate — and the length is $A/(N\,W)$, derived and printed as derived. Storing a back-calculated length *beside* a back-calculated width would be two homes for one guess, and the second would read like a declaration.

A record that borrows another stack's correlation carries the Reynolds band that correlation was fitted over, `massTransfer { validity { Re ( 50 86 ); } }` — equipment data, like every other validity window in this tree — and a run outside the band is **announced as an extrapolation**, never refused.

> **Common pitfall.** Choupo **never clamps** the current at $i_{\mathrm{lim}}$. If you specify a current with $i > i_{\mathrm{lim}}$, the unit returns the unclamped Faraday transfer of (418) *and* prints a loud over-limiting warning with the ratio $i/i_{\mathrm{lim}}$. Silently capping would hide the single most important operating fault of an ED stack — glass-box honesty means showing you the physically-meaningless answer *labelled as such*, not a quietly "fixed" one.

## 21.10 The limiting current in TIME: the batch rig, and in SPACE: the train

Everything above is written for a stack that sees ONE diluate composition. That is a steady stack, and under it the whole apparatus of §21.7 delivers a single number: a better estimate of $i_{\mathrm{lim}}$ than a declared transport number would give. The apparatus earns its keep somewhere else.

$i_{\mathrm{lim}}$ in (431) is a function of the *diluate composition*. Strip the diluate and it falls. There are exactly two ways to do that, and Choupo has a case for each.

**In time: a recirculating batch rig.** Put the diluate in a tank, pump it round through the stack, and leave the rectifier on. The tank depletes, so $i_{\mathrm{lim}}$ falls; hold the current constant and $i$ does not move; therefore they *must* cross. For a single salt the mass-transfer coefficient is fixed by the declared recirculation flow and $D_{\mathrm{eff}}$ of (432) does not depend on composition at all, so (431) reduces to

$$\frac{i_{\mathrm{lim}}(t)}{i_{\mathrm{lim}}(0)} \;=\; \frac{c(t)}{c(0)}, \tag{435}$$

which is the cleanest statement of the whole chapter: the limiting current of a batch desalination is a straight line down to the origin while the current a student holds is a horizontal one. The `batchElectrodialysis` unit of `choupoBatch` integrates this, holding both tanks so the closed rig conserves matter exactly and so the Nernst term of (420) sees the concentrate strengthening as the diluate empties.

What crosses matters. Below $i_{\mathrm{lim}}$, Faraday's law with a constant current efficiency integrates to a straight line,

$$n(t) \;=\; n(0) \;-\; \frac{\xi\, I\, N}{z F}\, t, \tag{436}$$

and the tank is predicted empty at a finite $t$. Above $i_{\mathrm{lim}}$ the counter-ion concentration vanishes at the diluate/membrane interface and the film can deliver no more of it; a larger current must be carried by something else, and what is available is $\mathrm{H^+}$ and $\mathrm{OH^-}$ from water dissociation there. Freezing the counter-ion transfer at its limiting value is the first-order statement of that,

$$\xi_{\mathrm{eff}} \;=\; \xi \min\!\left(1, \frac{i_{\mathrm{lim}}}{i}\right), \tag{437}$$

and it introduces *no new parameter*. Combine it with (435) and the depletion stops being linear and becomes exponential: past the crossing $\mathrm{d}n/\mathrm{d}t \propto -n$, so the straight line reaches zero and the tank never does. Choupo treats (437) as a *model of a regime* and makes it opt-in; without it the current efficiency stays at $\xi$ and the run announces that it is extrapolating past the crossing. Either way it does not clamp the current — the pitfall above holds in time as it does at steady state.

Hold the *voltage* constant instead and the crossing never happens. As the diluate loses conductivity $R_{\mathrm{pair}}$ of (424) rises, and as the two tanks separate the Nernst back-EMF of (420) rises with it, so (425) solved for $I$ gives a current that decays. $i$ and $i_{\mathrm{lim}}$ then fall together and the ratio barely moves — which is why a constant-voltage rig is self-limiting, and why the price of that safety is time. The straight line (436) fails here too, and for a reason that has nothing to do with the limiting current: $I$ is simply not what it was.

**In space: stages in series.** The same physics read along the other axis. A plant that must strip a diluate deeply puts several stacks in series; each sees a weaker diluate than the one before, so each has a lower $i_{\mathrm{lim}}$. This needs no electrical topology in the flowsheet: an *electrical stage* is a set of cell pairs served by one electrode pair on its own rectifier, so stages in series are units in series, each with its own `electricLoad`. Ask each stage for the same fractional demineralisation and its current falls by the same factor its limiting current does, so the whole train runs at ONE margin $i/i_{\mathrm{lim}}$ — and a train held at a single current density would drive its last stage deep into the over-limiting regime.

**And in a loop: feed-and-bleed.** A recycle does in time what the train does in space, and brings its own trap. With a fraction of the product returned to the stack inlet, the stack never sees the FEED composition: it sees the mixed loop, which sits much nearer the product. Its limiting current is correspondingly lower than the feed analysis suggests, its per-pass demineralisation is far below the plant's overall figure, and a specific energy quoted per cubic metre *through the stack* is not the plant's specific energy at all.

**Worked example 21.1.** NaCl desalination, `steady/electrodialysis/ed01_nacl_desalination`A 100-cell-pair Neosepta CMX/AMX stack ($R^A_{\mathrm{CEM}} = 3.0\ \Omega\,\mathrm{cm}^2$, $t_{\mathrm{cu}} = 0.98$; $R^A_{\mathrm{AEM}} = 2.4\ \Omega\,\mathrm{cm}^2$) desalinates a 0.1 mol/kg NaCl diluate against a 0.5 mol/kg concentrate, 0.2 $\mathrm{m}^2$ per membrane, 0.5 mm channel gap, $v = 0.05$ m/s. With $I = 8$ A and $\xi = 0.9$, Eq. (418) removes $\xi IN/(zF) = 0.9 \cdot 8 \cdot 100/(1 \cdot 96485) \approx 7.5 \times 10^{-3}$ mol/s of $Na^+$, against a diluate $Na^+$ inflow of $5.0 \cdot 0.001795 \approx 9.0 \times 10^{-3}$ mol/s — about 83 % demineralisation. This case names *no* stack record, so it takes the `CowanBrown` route on the membrane record's declared $t_{\mathrm{cu}}$ and the run says so in two `[legacy]` lines: the velocity it could not check against the flow, and the fallback with its reason.

**Worked example 21.2.** The limiting current against a measurement, `steady/electrodialysis/ed04_limiting_current_multiionic`Six units, one per $MgCl_2$ + $MgSO_4$ row of Table 4 of [89], on the paper's own bench stack at its own flow rates and 20 °C. $Mg^{2+}$ is the only cation, so its limiting transport number at the CEM is exactly 1 by (429), and it has the lowest $|z|D$ of the three ions — which is why the *cation*-exchange membrane sets the limiting current here. The engine's predictions sit +1 to +12 % from the measurements, 6.9 % on average, against the 9 % the paper reports for this system. The golden pins them as `anchor` rows with the paper's own 13 % band.
Two findings are recorded rather than tuned away. The paper's stated Reynolds number at its highest flow (86) does not follow from its own stated flow rate — 230 L/h through seven channels of $0.114 \times 7.7 \times 10^{-4}$ m gives 0.104 m/s and Re ≈ 80, not 0.11 m/s and 86 — and Choupo derives the velocity from the flow, which is the measurable. And the predictions are biased *high* by about 7 % on this system where the paper's own are biased *low* by 13 % on the single-salt $MgCl_2$ rows of the same table; with a mass-transfer correlation of 7 % average error behind it, neither is surprising.

**Runnable case.** `tutorials/steady/electrodialysis/ed01_nacl_desalination` — canonical sub-limiting NaCl desalination; `current` mode.
`tutorials/steady/electrodialysis/ed02_over_limiting_current` — the same stack driven past $i_{\mathrm{lim}}$; the LOUD over-limiting warning, no silent clamp.
`tutorials/steady/electrodialysis/ed03_stack_record` — the same chemistry through a `kind edStack` record: the velocity derived from the flow, the area rule printed as arithmetic, the four estimates announced, and Eq. (431) seen reducing to Eq. (433).
`tutorials/steady/electrodialysis/ed04_limiting_current_multiionic` — the multi-ionic limiting current against Table 4 of [89], six operating points, as `anchor` rows.
`tutorials/steady/electrodialysis/ed05_industrial_stack` — a 100-cell-pair, 50 $\mathrm{m}^2$ industrial unit: two hydraulic passes, a record that names no membrane pair (so the case does) and stores no channel length (so it is derived), and a borrowed correlation used inside its own Reynolds band.
`tutorials/steady/electrodialysis/ed06_stages_in_series` — four stages, four rectifiers, the current solved per stage: $i_{\mathrm{lim}}$ falls 15.6-fold along the train while the margin $i/i_{\mathrm{lim}}$ stays at 0.158 (§21.10).
`tutorials/steady/electrodialysis/ed07_feed_and_bleed` — a recycle: the stack sees 0.0187 mol/kg where the feed is 0.0500, so its limiting current is 2.68 times below what the feed analysis suggests.
`tutorials/batch/electrodialysis/edbatch01_constant_current` — the crossing, in time: $i_{\mathrm{lim}}$ from 643.04 to 88.26 A/$\mathrm{m}^2$, $i$ flat at 175, and the hand calculation (436) parting from the run at $t = 1593$ s.
`tutorials/batch/electrodialysis/edbatch02_constant_voltage` — the same rig at fixed voltage: the current decays 4.365 to 2.018 A, $i/i_{\mathrm{lim}}$ never reaches 1, and (436) fails for the other reason.

# 22 Ion-exchange softening: Gaines–Thomas mass action

*What you'll learn.* A water softener trades the *hardness* cations ($Ca^{2+}$, $Mg^{2+}$) for the sodium that a strong-acid cation (SAC) resin carries. The exchange is not a black-box "percent removal" — it is a *competitive chemical equilibrium* between the dissolved ions and a fixed-capacity solid phase, governed by the same mass-action thermodynamics as an aqueous complex. This chapter shows how Choupo writes that equilibrium as ONE extra unknown and ONE extra balance bolted onto the speciation Newton of Chapter 3, why the bound-species activity is an *equivalent fraction* rather than a concentration, where the $CaCO_3$-equivalent hardness and the unavoidable "salt penalty" come from, and — the honest limit — why this predicts the *limiting* effluent of a fully-equilibrated contactor, not the rising-leakage breakthrough curve of a bed in service.

The `ionExchanger` unit op (and its beaker-scale twin, the `exchange` property op) sits downstream of the aqueous-speciation machinery. Its purpose is concrete: take a brackish or hard water analysis, equilibrate it with a softening resin, and report the softened effluent — the feed a downstream NF/RO element (§18) sees once the scale-formers are stripped out. The case `membrane08_softened_scaling` chains exactly this: soften, then watch the calcite/gypsum saturation indices of Chapter 3 collapse.

## 22.1 The exchange half-reaction and the selectivity constant

A SAC resin is a cross-linked polymer skeleton hung with fixed, negatively charged sulfonate groups; we write the free site as $X^-$. Electroneutrality forces every site to carry a counter-cation, so the resin is best read as a separate, fixed-capacity *pseudo-phase*. A divalent cation $Me^{z+}$ binds by displacing $z$ singly-charged sites:

$$Me^{z+} + z\,X^- \rightleftharpoons MeX_z, \qquad K = \frac{a_{MeX_z}}{a_{Me}^{1}\, a_X^{z}}. \tag{438}$$

For a softener the players are $Na^+, K^+, Ca^{2+}, Mg^{2+}$ (plus $Sr^{2+}, Ba^{2+}$ when scaling matters), with bound species $NaX, KX, CaX_2, MgX_2, \ldots$. Choupo reads this network from `exchange.dat` (curated from the USGS PHREEQC `EXCHANGE_SPECIES` block, public domain [23]); the $Na^+$ exchange is the reference reaction with $\log K = 0$, so each $\log K_{25}$ is a *selectivity relative to sodium*. The representative values in the catalogue are

$$\log K_{CaX_2} = 0.80, \quad \log K_{MgX_2} = 0.60, \quad \log K_{KX} = 0.70, \quad \log K_{NaX} = 0.00, \tag{439}$$

which encode the textbook affinity ordering $Ca^{2+} > Mg^{2+} > K^+ > Na^+$ for a sulfonated SAC resin. An optional van't Hoff $\Delta H$ slot lets $\log K(T)$ drift with temperature exactly as in the $K$-value chapter (§13).

## 22.2 The Gaines–Thomas convention: activity = equivalent fraction

The novelty of (438) is the activity of a *bound* species. We cannot use its molality — a sulfonate group is not free to roam — so we need a reference state for the solid pseudo-phase. The Gaines–Thomas convention [94] takes the activity of a bound species to be its *equivalent fraction* on the resin:

$$a_{MeX_z} = \beta_{MeX_z} = \frac{z\, m_{MeX_z}}{\mathrm{CEC}}, \qquad a_X = \beta_X = \frac{m_X}{\mathrm{CEC}}, \tag{440}$$

where $m_{MeX_z}$ is the bound molality [mol/kg water], the factor $z$ converts moles of bound cation to *equivalents* (charges occupied), and the cation-exchange capacity

$$\mathrm{CEC} = \sum_s z_s\, m_{sX} \qquad \text{[eq/kg water]} \tag{441}$$

is the total number of charged sites. The equivalent fractions sum to one, $\sum_s \beta_s + \beta_X = 1$, with the convention that the *free* site $X^-$ carries the residual. The crucial point: in Choupo's kernel the resin is treated as always electroneutral — there is no thermodynamically "empty" free site that competes for cations. $\beta_X$ is therefore not a populated species but the activity variable that the mass actions (438) solve *from*, while the conserved capacity (441) is the closure that pins it. The aqueous-ion activity $a_{Me} = \gamma_{Me}\, m_{Me}$ keeps its Davies $\gamma$ from Chapter 3; only the resin phase uses $\gamma = 1$ with the equivalent-fraction reference.

*Intuition.* A bound cation is not dissolved — it is parked on a site. "How much is parked" is naturally a *fraction of the sites*, not a concentration in a volume that has no meaning for a solid. The equivalent fraction is the resin's mole fraction, weighted by charge because a divalent ion eats two sites. That single change of reference state is the whole difference between an aqueous-complexation equilibrium and an ion-exchange equilibrium; everything else — the mass action, the Newton, the Jacobian — is reused unchanged.

## 22.3 One extra unknown, one extra balance — folded into the speciation Newton

The implementation (`src/thermo/electrolyte/SpeciationSolver.cpp`) is deliberately frugal. The free site $X^-$ joins the log-molality unknown vector as one extra coordinate $x_X = \ln m_X$, and the capacity (441) joins the residual system as one extra *equality*:

$$R_{\mathrm{CEC}} \;=\; \mathrm{CEC} \;-\; \sum_s z_s\, m_{sX} \;=\; 0. \tag{442}$$

Each bound species reuses the same mass-action machinery as an aqueous complex. Taking logs of (438) with (440) and solving for the bound molality,

$$\ln m_{sX} \;=\; \ln 10\,\log K \;+\; \ln\gamma_{\mathrm{Me}} + \ln m_{\mathrm{Me}} \;+\; z\,\ln m_X \;+\; \underbrace{\left[\ln\tfrac{\mathrm{CEC}}{z} - z\ln \mathrm{CEC}\right]}_{\text{CEC constant, folded into } \log K}, \tag{443}$$

so the constant CEC term is pre-folded into an *effective* $\log K$ at build time, and the species is presented to the solver with stoichiometry $\{(\mathrm{Me},1),(X,z)\}$ and $\gamma \equiv 1$ — byte-identically to how an aqueous complex with the same legs is handled. The free site is *excluded* from the ionic strength, the water activity, and the aqueous electroneutrality balance, because it lives on the resin, not in solution. When the exchange spec is empty the whole branch is skipped and the result is identical to the no-exchange speciation, by construction — the glass-box guarantee that adding a feature cannot silently move an unrelated answer.

The same gamma-fixed-point / Newton solver of Chapter 3 then converges molalities, the free site, and (when `pH solve;`) the proton together. A resin shipped in the sodium form starts holding CEC equivalents of Na; that initial load is added to the conserved Na total, so the Na not retained on the sites is *released* to the water eq-for-eq — the model's account of where the displaced sodium goes.

## 22.4 Hardness, leakage, and the salt penalty

The headline output is *hardness*, reported in the practitioner's universal currency, mg/L as $CaCO_3$:

$$H \;=\; \big(m_{\mathrm{Ca}} + m_{\mathrm{Mg}}\big) \times M_{\mathrm{CaCO_3}} \times 1000, \qquad M_{\mathrm{CaCO_3}} = 100.09\ \mathrm{g/mol}, \tag{444}$$

with molalities [mol/kg] read as mg/L at the dilute closure $\rho \approx 1\,\mathrm{kg/L}$. Expressing both $Ca^{2+}$ and $Mg^{2+}$ as if they were calcium carbonate is the convention that lets a single number stand for "total hardness" regardless of which divalent cation supplies it. The removal is $100\,(H_{\mathrm{in}} - H_{\mathrm{out}})/H_{\mathrm{in}}$; the residual $Ca^{2+}, Mg^{2+}$ in the effluent is the *leakage*. Every equivalent of hardness removed is balanced by an equivalent of sodium released, the *salt penalty*

$$\Delta m_{\mathrm{Na}} \;=\; m_{\mathrm{Na}}^{\mathrm{out}} - m_{\mathrm{Na}}^{\mathrm{in}} \;\approx\; 2\big(\Delta m_{\mathrm{Ca}} + \Delta m_{\mathrm{Mg}}\big) \quad \text{[eq-for-eq]}, \tag{445}$$

which is why a softener lowers hardness but *raises* the sodium load — a real cost for sodium-restricted feed waters and for any downstream RO that must then reject the added Na. Choupo also reports the resin loadings $\beta_{\mathrm{CaX_2}}, \beta_{\mathrm{MgX_2}}, \beta_{\mathrm{NaX}}$ and the CEC utilised ($\approx 100\,\%$, since the sites are always full).

**Power and limit.** **Power.** The competitive equilibrium falls straight out of the selectivity constants (439) — no fitted "removal efficiency". Because it reuses the speciation Newton, the softened water is a fully consistent analysis: its activities, ionic strength, and saturation indices are immediately available to the scaling and membrane chapters. The salt penalty (445) emerges automatically from charge conservation. **Limit.** The result is the *limiting effluent* — water *fully* equilibrated with the resin at its stated CEC, i.e. the best a fresh or fully-loaded contactor can do per pass. A real fixed bed leaks rising hardness as the exchange front migrates toward breakthrough, then must be regenerated; cycle length, bed-volumes-to-breakthrough, and regeneration chemistry are *transient* and are not modelled. The safe reading is therefore a bound: real leakage $\geq$ this equilibrium leakage — if the equilibrium effluent is already hard, no bed will soften it. The bed volume, if given, is announced but does not enter the per-pass equilibrium (it sets the cycle length, not the limiting composition). The Davies $\gamma$ is trusted only to $I \lesssim 0.5\,\mathrm{mol/kg}$; beyond that the equilibrium is flagged *indicative*. Choupo prints all of this as a non-suppressible honesty banner — the limiting-case caveat is part of the answer, not a footnote.

**Worked example 22.1.** SAC softener on a hard water, `softener_brackish`A hard water carrying $Ca^{2+}$, $Mg^{2+}$, $Na^+$ and the matching anions is equilibrated with the `SAC_Na` resin — a sulfonated polystyrene-DVB strong-acid cation exchanger in the sodium form, nameplate CEC = 2.0 eq/L of settled bed (taken as eq/kg water at the dilute closure) [95, 96]. The kernel adds one unknown ($\ln m_X$) and one balance (the CEC capacity (442)) to the existing speciation Newton and converges in a handful of steps. At equilibrium the high-selectivity divalents (439) drive almost entirely onto the resin ($\beta_{\mathrm{CaX_2}}, \beta_{\mathrm{MgX_2}}$ dominate; the residual $\beta_{\mathrm{NaX}}$ is small), the hardness (444) collapses by well over 90 %, and the sodium rises by the eq-for-eq salt penalty (445). Feeding the softened analysis to the calcite/gypsum saturation check of Chapter 3 (case `membrane08_softened_scaling`) then shows the scaling indices dropping below zero — the point of softening ahead of an RO element.

**Runnable case.** `tutorials/props/electrolyte/softener_brackish` — the beaker-scale `exchange` property op: Gaines–Thomas softening of a hard water, with the resin-loading isotherm $\beta_s$ vs. solution equivalent fraction written for the GUI to plot.
`tutorials/steady/membranes/membrane08_softened_scaling` — the `ionExchanger` unit op in a flowsheet, soften-then-watch-the-scaling coupled to the saturation indices of Chapter 3.

# 23 Adsorption: isotherms, LDF uptake, and the fixed bed

*What you'll learn.* A gas molecule sticking to a porous solid is an *equilibrium* (the isotherm, a curated pair datum: adsorbent × species), a *rate* (the linear driving force, an equipment datum: this particle, in this bed), and a *contactor* (the closed vessel or the 1-D fixed bed). This chapter gives the Henry and Langmuir isotherms with their van't Hoff temperature dependence, the extended (competitive) Langmuir together with its *declared* thermodynamic inconsistency, the LDF rate and why its $k$ can never live in a standards catalogue, the conservative finite-volume bed balance with Danckwerts boundaries, the stoichiometric time every breakthrough curve must obey before a single time step is taken — and, closing the loop, how a "steady-state" pressure- or temperature-swing unit reads as the *time average of a twin-bed pair*.

## 23.1 The equilibrium loading: Henry and Langmuir

The state of the solid is the *loading* $q_i$ — moles of adsorbate $i$ per kilogram of regenerated, dry adsorbent (mol/kg) — and the driving force the isotherm sees is the partial pressure $p_i$ (Pa). The extensive solid inventory is always $m_{\mathrm{ads}}\, q_i$, with $m_{\mathrm{ads}}$ the adsorbent mass of the *equipment*, never of the catalogue.

At vanishing coverage every isotherm is linear (Henry's law of the adsorbed phase):

$$q = H\, p, \qquad H = \lim_{p \to 0} \frac{q}{p}. \tag{446}$$

The Langmuir model[63] adds the finite site count: a balance over identical, independent sites with fractional coverage $\theta = q/q_{\max}$, adsorption rate $\propto p\,(1-\theta)$ and desorption rate $\propto \theta$, gives at equilibrium

$$q(T,p) \;=\; q_{\max}\, \frac{b(T)\, p}{1 + b(T)\, p}, \tag{447}$$

with the two limits that make it teachable: $p \to 0$ recovers (446) with $H = q_{\max}\, b$, and $p \to \infty$ saturates at the monolayer $q_{\max}$. The affinity $b$ carries the temperature dependence through van't Hoff on the adsorption equilibrium:

$$b(T) \;=\; b_{\mathrm{ref}}\, \exp\!\left[ -\frac{\Delta H_{\mathrm{ads}}}{R} \left( \frac{1}{T} - \frac{1}{T_{\mathrm{ref}}} \right) \right], \tag{448}$$

with $\Delta H_{\mathrm{ads}} < 0$ for exothermic adsorption (the usual case), so $b$ *falls* as $T$ rises — beds regenerate hot. $T_{\mathrm{ref}}$ is a mandatory part of the curated record, never an implicit 298 K. The equilibrium record (its $q_{\max}$, $b_{\mathrm{ref}}$, $\Delta H_{\mathrm{ads}}$, bases, validity window and primary citation) is a *pair* datum — adsorbent × species — and lives in the standards catalogue; the evaluation bench (`tutorials/props/adsorption/isotherm01_eval_13x_co2`) runs the Henry-limit, saturation, pin and unit-invariance gates aloud on every record it touches, and the regression bench (`isotherm02_fit_jaschik`) shows how such a record is *earned* from data — including the identifiability refusals (single-temperature data cannot yield $\Delta H_{\mathrm{ads}}$; Henry-regime data yields only the product $q_{\max} b$).

## 23.2 Competition: the extended Langmuir, honestly

For a mixture the workhorse teaching model is the *extended* (competitive) Langmuir — all species compete for the same sites, each term in the denominator crowding everyone else out:

$$q_i \;=\; q_{\max,i}\, \frac{b_i(T)\, p_i}{1 + \sum_j b_j(T)\, p_j}. \tag{449}$$

Now the honesty clause, stated here exactly as it is stated in the run headers: (449) is thermodynamically consistent *only* if all the $q_{\max,i}$ are equal. With distinct $q_{\max,i}$ the model violates the Gibbs–Duhem relation of the adsorbed phase — there is no single free-energy surface (no consistent spreading pressure) from which all the $q_i$ derive, so the predicted loadings depend on a path that thermodynamics says must not matter. It remains a *useful, declared* teaching model, and Choupo keeps it as the default mixing rule precisely because its error is announced rather than hidden. The honest extension is the Ideal Adsorbed Solution Theory (IAST), which builds the mixture from the *pure-component* isotherms by equating spreading pressures — an explicit extension point of the mixing-rule layer, never a silent default.

## 23.3 The rate: linear driving force, and whose datum $k$ is

Uptake is not instantaneous. The linear driving force (LDF) model lumps the whole diffusional resistance chain into one first-order relaxation toward the local equilibrium loading $q_i^* = q_i^*(T,p)$:

$$\frac{\mathrm{d}q_i}{\mathrm{d}t} \;=\; k_i\, (q_i^* - q_i). \tag{450}$$

The coefficient $k_i$ (1/s) aggregates film, macropore and crystal diffusion *of that particle in that bed*; Glueckauf's classic estimate $k \approx 15\, D_e / r_p^2$ makes the point — $k$ depends on the particle radius, i.e. on the *equipment*, not on

the chemical identity of the pair. It therefore lives in the case dict (with declared scope and source), never in the standards catalogue, and a missing $k_i$ for an active species is a named refusal, not a default. In a closed vessel (450) couples to the gas inventory ($\mathrm{d}n_i/\mathrm{d}t = -m_{\mathrm{ads}}\,\mathrm{d}q_i/\mathrm{d}t$, $p_i = n_i RT/V_{\mathrm{gas}}$); in the Henry limit the coupled system is *linear* with

$$q(t) = q_\infty\big(1 - e^{-k_{\mathrm{eff}}t}\big), \qquad k_{\mathrm{eff}} = k\Big(1 + \frac{m_{\mathrm{ads}} HRT}{V_{\mathrm{gas}}}\Big), \tag{451}$$

the closed form the integrator must reproduce to $10^{-8}$ (`tutorials/batch/adsorber/batch11_adsorber_henry`) — the analytic gate that certifies the numerics before any nonlinear case is trusted.

## 23.4 The 1-D isothermal fixed bed

The bed balance is written the way it is solved: conservative finite volumes. For cell $j$ of volume $V_j$, species $i$:

$$\varepsilon\, V_j\, \frac{\mathrm{d}c_{ij}}{\mathrm{d}t} \;=\; F^{\mathrm{conv}}_{i,j-\frac{1}{2}} - F^{\mathrm{conv}}_{i,j+\frac{1}{2}} \;+\; F^{\mathrm{disp}}_{i,j-\frac{1}{2}} - F^{\mathrm{disp}}_{i,j+\frac{1}{2}} \;-\; \rho_b\, V_j\, \frac{\mathrm{d}q_{ij}}{\mathrm{d}t}, \tag{452}$$

with upwind convective face fluxes $u\,A\,c$ ($u$ the superficial velocity), dispersive fluxes $-\varepsilon D_{\mathrm{ax}} A\,\partial c/\partial z$ at the interior faces, and the LDF sink (450) evaluated at the local $p_{ij}$. Two boundary statements (Danckwerts): at the inlet the *flux* is imposed — the first face carries exactly $u\,c_{\mathrm{in}}$, so what enters is what you fed, dispersion included — and at the outlet $\partial c/\partial z = 0$, so nothing diffuses back in from the void beyond the bed.

Two honesty notes belong to this equation. First, the density trap: $\rho_b$ is the *bulk* (packed-bed) density, kg of solid per $\mathrm{m}^3$ of bed, so $\rho_b\, q$ is already mol per bed volume — writing $(1-\varepsilon)\rho_p$ with $\varepsilon$ and $\rho_p$ from different sources double-counts the porosity, the classic error, and is forbidden. Second, the declared closure: $u$, $P$, $T$ are held constant in this phase of the model (momentum and energy balances are later extensions), yet a bed taking up 15 % of its feed *does* slow the gas. Holding $u$ constant therefore fabricates carrier gas at exactly the net uptake rate — and the engine says so out loud, prints the fabricated moles, pins them as a KPI, and discounts them in the campaign ledger. The model's residual is *declared*, at a named magnitude, instead of being absorbed invisibly — the difference between a wrong number and an honest model.

## 23.5 The stoichiometric time: where the front must arrive

Before integrating anything, an overall mass balance already fixes where the breakthrough front must sit. Feed a clean bed at superficial velocity $u$ and inlet concentration $c_{\mathrm{in}}$; idealise the front as sharp. When it has just traversed the bed, everything fed has gone into filling the bed's two inventories — gas in the voids and adsorbate on the solid:

$$u\,A\,c_{\mathrm{in}}\,t_{\mathrm{st}} \;=\; L\,A\,\big[\varepsilon\,c_{\mathrm{in}} + \rho_b\,q^*(c_{\mathrm{in}})\big] \quad\Longrightarrow\quad t_{\mathrm{st}} \;=\; \frac{L}{u}\underbrace{\left[\varepsilon + \frac{\rho_b\,q^*(c_{\mathrm{in}})}{c_{\mathrm{in}}}\right]}_{R_f}, \tag{453}$$

where $R_f$ is the retention factor and $u_{\mathrm{sh}} = u/R_f$ the front (shock) velocity — typically hundreds of times slower than the gas. Kinetics ($k$) and dispersion ($D_{\mathrm{ax}}$) spread the front into a mass-transfer zone *around* $t_{\mathrm{st}}$, but cannot move its centroid: because (452) telescopes exactly, the integral identity

$$\int_0^\infty \left(1 - \frac{c_{\mathrm{out}}}{c_{\mathrm{in}}}\right)\mathrm{d}t \;=\; t_{\mathrm{st}} \tag{454}$$

holds for the conservative scheme on *any* mesh, while the useful capacity $t_b$ (breakthrough at, say, 5 %) converges with the mesh at the upwind scheme's first order — run the mesh study and watch both statements hold (`tutorials/batch/adsorber/batch13_breakthrough_co2`, `batch15_mesh_study`; the pure-transport twin `batch14_transport_only` pins the $k = 0$ bed against the analytic advection–dispersion solution).

## 23.6 Reading a "steady" swing unit: the twin-bed time average

A fixed bed is intrinsically transient — so what is a pressure-swing or temperature-swing unit doing in a *steady* flowsheet? It is the *time average of a pair of columns at cyclic steady state*: one bed adsorbs at high pressure (or low temperature) while its twin regenerates at low pressure (or high temperature), the valves swap, and over a full cycle the pair's averaged inlet and outlet flows are constant. The working capacity per cycle is bounded by the equilibrium swing $\Delta q = q^*(p_{\mathrm{high}}) - q^*(p_{\mathrm{low}})$ for PSA — or, through (448), by $q^*(T_{\mathrm{low}}) - q^*(T_{\mathrm{high}})$ for TSA. The case `tutorials/steady/separation/psa01_h2_psa` is exactly this abstraction for the pressure swing: competitive Langmuir loadings at $P_{\mathrm{high}}$ and $P_{\mathrm{low}}$ set the equilibrium ceiling, a declared utilisation derating and a purge sweep complete the shortcut, and the run header lists out loud what the shortcut does *not* model (breakthrough and cycle time, thermal effects, blowdown loss). The transient bed of the previous subsection is the machinery that will eventually replace the shortcut's derating with a computed one — the two levels of description are the same physics at two Pareto points.

> **Runnable case.** `tutorials/props/adsorption/` — the isotherm bench: evaluation gates (`isotherm01`), the multi-T regression with identifiability verdicts (`isotherm02`), the refusal case (`isotherm03`).
> `tutorials/batch/adsorber/` — the uptake staircase: closed vessel to inventory equilibrium (`batch09`),

extended-Langmuir competition (`batch10`), the analytic Henry gate (`batch11`), the pressure refusal (`batch12`), breakthrough with the announced $t_{st}$ and the declared model residual (`batch13`), pure transport vs the analytic solution (`batch14`), the mesh study (`batch15`).
`tutorials/steady/separation/psa01_h2_psa` — the twin-bed equilibrium shortcut in a steady flowsheet.

# 24 Rotating equipment: compressors, turbines, pumps

*What you'll learn.* A compressor, turbine or pump is specified by the *work* you put on its shaft and its efficiency — never by the outlet pressure. That is the credo for rotating gear (Chapter 1): you furnish $W_{\text{shaft}}$ and $\eta$, the pressure *emerges*. This chapter shows how the discharge state is found from an isentropic reference using the real-fluid entropy and enthalpy of Chapter 9's neighbourhood, why it needs a nested Newton for a compressible fluid, and why a pump — incompressible — needs none at all.

## 24.1 The isentropic reference and the efficiency

A reversible, adiabatic (hence *isentropic*) machine is the thermodynamic ideal. Real machines fall short by the isentropic efficiency $\eta$. The construction has three steps:

1. the ideal discharge has the *same entropy* as the inlet, $S(T_{\text{isen}}, P_{\text{out}}) = S(T_{\text{in}}, P_{\text{in}})$, which fixes $T_{\text{isen}}$ once $P_{\text{out}}$ is known;
2. the ideal work is the enthalpy change to that state, $w_{\text{isen}} = h(T_{\text{isen}}, P_{\text{out}}) - h_{\text{in}}$;
3. the real work relates to it by $\eta$ — and here compressor and turbine differ:

$$\text{compressor: } w_{\text{real}} = \frac{w_{\text{isen}}}{\eta}, \qquad \text{turbine: } w_{\text{real}} = \eta\, w_{\text{isen}}, \tag{455}$$

because a compressor needs *more* work than ideal while a turbine delivers *less*.

The real discharge temperature then follows from the first law, $h(T_{\text{out}}, P_{\text{out}}) = h_{\text{in}} + w_{\text{real}}$ — the lost work reappears as heat, so $T_{\text{out}} > T_{\text{out,isen}}$.

## 24.2 Why a nested Newton (compressor / turbine)

The catch: $P_{\text{out}}$ is unknown — it is what we furnished work to produce. We know $w_{\text{real}} = W_{\text{shaft}}/\dot{n}$, hence $w_{\text{isen}}$, and we search for the $P_{\text{out}}$ whose isentropic enthalpy rise equals it. Choupo (`IsentropicCore`) runs an *outer* Newton-1D on $P_{\text{out}}$,

$$F(P_{\text{out}}) = h\big(T_{\text{isen}}(P_{\text{out}}), P_{\text{out}}\big) - (h_{\text{in}} + w_{\text{isen}}) = 0, \tag{456}$$

wrapping an *inner* Newton that solves the entropy match for $T_{\text{isen}}(P_{\text{out}})$ at each trial pressure. $H$ and $S$ are the real-fluid $H = H^{\text{ig}} + H^{R}$, $S = S^{\text{ig}} + S^{R}$ with the residuals from the cubic EoS (SRK / PR). For an ideal gas the two levels decouple and it converges in one outer step; for a real gas they couple (4–10 steps). The expansion branch simply searches $P_{\text{out}} < P_{\text{in}}$.

**Worked example 24.1.** Air compressor, `compressor01_air`10 kW of shaft work, $\eta = 0.75$, on 1 mol/s of air: the nested solve returns $P_{\text{out}} = 8.90$ bar and $T_{\text{out}} = 633$ K, within 1 % of the hand calculation. A companion DesignSpec (`compressor02_designspec_pout`) manipulates $W_{\text{shaft}}$ to *hit* a target $P_{\text{out}} = 8$ bar — the rotating-gear analogue of sizing an evaporator's area, the "specify the pressure" that is really a disguised DesignSpec.

## 24.3 The pump: incompressible, no EoS, no Newton

A liquid's density barely changes with pressure, so a pump needs none of the above. The work raises the pressure directly,

$$\Delta P = \frac{\eta\, W_{\text{shaft}}}{\dot{n}\,\bar{v}}, \qquad P_{\text{out}} = P_{\text{in}} + \Delta P, \tag{457}$$

with $\bar{v}$ the liquid molar volume; the small inefficiency dissipates as a tiny temperature rise $\Delta T = (1-\eta) w_{\text{real}}/C_{p,L}$. A closed form — the incompressible limit needs no equation of state and no iteration.

*Intuition.* Because that relation is *linear* in the work, it inverts in closed form: writing $Q_v = \dot{n}\,\bar{v}$ for the volumetric flow (which the feed already fixes),

$$\Delta P = \frac{\eta\, W_{\text{shaft}}}{Q_v} \quad \Longleftrightarrow \quad W_{\text{shaft}} = \frac{Q_v\, \Delta P}{\eta}.$$

So the pump can be specified *either* by the motor power (and the discharge pressure is the result) *or* by the discharge pressure $P_{\text{out}}$ — or rise $\Delta P$ — it must reach (and the shaft power is the result). The second is the everyday engineering case: a pump feeds a header of *known* pressure, and you want to know the power it draws. Crucially this is **not** a DesignSpec — the target is local to the unit, so it is the same one-shot algebra solved the other way, not an outer Newton loop. A DesignSpec is only needed when the target lives *elsewhere* in the flowsheet. (Reaching for a DesignSpec on every one of a plant's dozens of pumps, just to set their discharge pressures, would be needless iteration.) The unit requires *exactly one* of $\{W_{\text{shaft}}, P_{\text{out}}, \Delta P\}$; a let-down with $P_{\text{out}} < P_{\text{in}}$ is a `valve`, not a pump.

**Runnable case.** `tutorials/steady/rotating/compressor01_air` — furnish work, the pressure emerges. `tutorials/steady/rotating/turbine01_air` — expansion, $T_{\text{out}} > T_{\text{out,isen}}$ (losses stay as heat). `tutorials/steady/rotating/pump01_water` — the incompressible closed form, head $\approx 736$ m. `tutorials/steady/rotating/pump02_pressure_spec` — the SAME pump set by its discharge pressure ($P_{\text{out}} = 30$ bar); the shaft power is the result — the exact inverse of pump01.

## 24.4 The pressure exchanger: pressure moved from one liquid to another

A seawater reverse-osmosis train rejects more than half of what it pressurises: at 45 % recovery, 55 of every 100 $\text{m}^3$ pumped to 55 bar leave as brine still carrying almost all of that pressure. Throttling it away costs the plant most of its energy. The *rotary pressure exchanger* takes that pressure back without a pump: a ceramic rotor of longitudinal ducts turns between two end covers; each duct is filled in turn by the high-pressure brine, which pushes out the seawater the duct held at nearly the same pressure, and by low-pressure seawater, which pushes the depressurised brine to drain. Liquid displaces liquid, with no piston between them, which is why the device is rotary and ceramic and why it has exactly the three imperfections its data sheet quotes.

The unit `pressureExchanger` carries no physics of its own beyond three algebraic balances on its two inlets — the brine (high pressure in) and the seawater (low pressure in) — each on a volumetric basis, $Q = \dot{n}\,\bar{v}$ with $\bar{v}$ the liquid molar volume of the stream:

$$Q_{\text{HP,out}} = (1-\lambda)\,Q_{\text{HP,in}}, \qquad Q_{\text{LP,out}} = Q_{\text{LP,in}} + \lambda\,Q_{\text{HP,in}}, \tag{458}$$

the *leakage* $\lambda$ (1–2 %) being the lubrication flow across the rotor–stator gap, from the high-pressure side to the drain;

$$n_{i,\text{HP,out}} = m\,Q_{\text{HP,out}}\,\frac{F_b z_{b,i}}{Q_b} + (1-m)\,Q_{\text{HP,out}}\,\frac{F_s z_{s,i}}{Q_s}, \qquad n_{i,\text{LP,out}} = F_b z_{b,i} + F_s z_{s,i} - n_{i,\text{HP,out}}, \tag{459}$$

the *mixing* $m$ (3–6 %) being the share of the pressurised outlet's volume that is brine left in the duct — matter closes by construction, and the salinity the feed gains is the device's hidden penalty, because the osmotic pressure the membrane must beat rises with it; and

$$\eta = \frac{Q_{\text{HP,out}}P_{\text{HP,out}} + Q_{\text{LP,out}}P_{\text{LP,out}}}{Q_{\text{HP,in}}P_{\text{HP,in}} + Q_{\text{LP,in}}P_{\text{LP,in}}}, \qquad P_{\text{HP,out}} = \frac{\eta\,(Q_{\text{HP,in}}P_{\text{HP,in}} + Q_{\text{LP,in}}P_{\text{LP,in}}) - Q_{\text{LP,out}}P_{\text{LP,out}}}{Q_{\text{HP,out}}}, \tag{460}$$

the pressure-transfer *efficiency* $\eta$ (95–98 %), Stover's definition (Desalination 203 (2007) 168–175), solved for the one unknown once the drain pressure $P_{\text{LP,out}}$ is declared. The three numbers are a maker's: none has a default, and a case that omits one is refused by name. Temperature is unchanged. The device does no work on the fluid, publishes $Q = 0$, and what it dissipates, $(1-\eta)$ of the pressure energy entering, is published beside the hydraulic duty the booster pump must then pay to lift the pressurised seawater the last 1–2 bar back to the high-pressure pump's discharge.

*Intuition.* Where did the specific energy of seawater RO go, from ∼8 to ∼3 $\text{kWh}\,\text{m}^{-3}$ in thirty years? Decompose it. The thermodynamic minimum for a feed of osmotic pressure $\pi_0$ at recovery $r$ is the work of pushing each increment of permeate against the brine's rising osmotic pressure, $\pi_0/(1-x)$:

$$w_{\min} = \frac{1}{r}\int_0^r \frac{\pi_0}{1-x}\,dx = \pi_0\,\frac{-\ln(1-r)}{r},$$

about 1 $\text{kWh}\,\text{m}^{-3}$ for seawater at 45 %. A single-pass train with no recovery device pays the high-pressure pump for the WHOLE feed, $Q_{\text{feed}}\,\Delta P/\eta_{\text{pump}}$, and throws the brine's share away — three to four times the minimum. With a pressure exchanger the high-pressure pump moves only the permeate's volume (plus the leak); the brine's volume is re-pressurised by the brine itself, at a cost of the device's $(1-\eta)$ and a small booster. What remains above the minimum is the membrane's own overpressure (the flux needs a driving force) and the pumps' efficiencies. The witness below prints every term.

**Runnable case.** `tutorials/steady/membranes/membrane27_swro_energy_recovery` — high-pressure pump + rotary exchanger + booster + one SW30HR-380 stage on membrane01's seawater: the specific energy with and without the device, beside the thermodynamic minimum, and the exchanger's specification sheet (an array of rotors from a declared rated flow, not costed).

# 25 Pipe pressure drop: Darcy–Weisbach and the friction factor

*What you'll learn.* A pump has a shaft and you furnish its work; a pipe has neither shaft nor knob. The pressure a pipe costs you is *never* a specification — it is a **computed result** of the geometry (diameter, length, roughness, elevation, fittings) and the flow (velocity, density, viscosity). This chapter derives the Darcy–Weisbach equation from the mechanical-energy balance, shows where the dimensionless friction factor $f$ comes from in each flow regime — the exact laminar 64/Re, the implicit Colebrook–White reference, and the explicit Haaland and Churchill fits Choupo offers — and adds the static-head and minor-loss terms that close a real engineering line. The watchword is the valve's: *a pipe only dissipates, exactly as much as the physics says.*

## 25.1 The mechanical-energy balance for an incompressible liquid

Write the steady-flow mechanical-energy balance (the engineering Bernoulli equation) between the inlet (1) and outlet (2) of a constant-area pipe carrying an incompressible liquid of density $\rho$:

$$\frac{P_1}{\rho} + \tfrac{1}{2}u_1^2 + g z_1 \;=\; \frac{P_2}{\rho} + \tfrac{1}{2}u_2^2 + g z_2 \;+\; \underbrace{g\,h_f}_{\text{friction}} . \tag{461}$$

For a uniform bore the velocity is unchanged ($u_1 = u_2 = u$, since the volumetric flow $Q_v = uA$ is conserved and the area $A$ is constant), so the kinetic terms cancel. Multiplying through by $\rho$ and collecting,

$$\boxed{\Delta P \equiv P_1 - P_2 \;=\; \underbrace{\rho\, g\, h_f}_{\text{dissipation}} \;+\; \underbrace{\rho\, g\, \Delta z}_{\text{static head}} , \qquad \Delta z = z_2 - z_1 .} \tag{462}$$

The first term is the irreversible loss — mechanical energy that disappears into the fluid's internal energy through viscous shear at the wall. The second is the *reversible* cost of lifting the fluid a height $\Delta z$ (a rise costs pressure; a drop returns it). Crucially $z$ is a property of the *geometry*, not the flow, so the static head appears even at zero velocity.

*Intuition.* Where does the dissipated mechanical energy go? Into heat. But Choupo holds $T$ across the pipe and does *not* re-inject the loss as a temperature rise — exactly as it does for the `valve`. The reason is honesty about scale: for a well-insulated short line the viscous heating raises $T$ negligibly ($\Delta T \sim \Delta P/(\rho C_p) \sim 10^{-2}\,\mathrm{K}$ for a 0.4 bar drop in water), and the energy that *matters* — the irreversibility — is already accounted for, in the pressure. The drop in $P$ *is* the lost work; re-injecting it as heat would double-count it.

## 25.2 Darcy–Weisbach: the friction head

The friction head $h_f$ is not arbitrary. Dimensional analysis of fully-developed pipe flow shows the wall loss scales with the velocity head $\rho u^2/2$, the length-to-diameter ratio $L/D$, and a dimensionless *Darcy friction factor* $f$:

$$\rho\, g\, h_f \;=\; f\,\frac{L}{D}\,\frac{\rho u^2}{2}, \qquad \text{so} \qquad h_f \;=\; f\,\frac{L}{D}\,\frac{u^2}{2g}. \tag{463}$$

This is the **Darcy–Weisbach equation** [97]. The velocity, area and Reynolds number follow from the feed and the bore,

$$A = \frac{\pi D^2}{4}, \quad u = \frac{Q_v}{A}, \quad Q_v = \frac{\dot m}{\rho} = \frac{\dot n\, M_w}{\rho}, \quad \mathrm{Re} = \frac{\rho\, u\, D}{\mu}, \tag{464}$$

with $\dot n$ the molar flow, $M_w$ the mixture molar mass, and $\mu$ the liquid viscosity (from the `transport` block of Chapter 2, or an explicit `operation` override). The only unknown left in Eq. (463) is $f$, and *all* of pipe hydraulics is the one question: *what is $f$?*

## 25.3 The friction factor in each regime

**Laminar ($\mathrm{Re} < 2300$).** Here the velocity profile is the exact Hagen–Poiseuille parabola, and integrating the wall shear gives a *closed form* with no adjustable constant and *no roughness dependence* (a smooth laminar sublayer buries the wall texture):

$$f_{\mathrm{lam}} = \frac{64}{\mathrm{Re}}. \tag{465}$$

**Turbulent ($\mathrm{Re} > 4000$).** Now the wall roughness matters, through the *relative roughness* $\varepsilon/D$ ($\varepsilon$ the absolute peak height, e.g. $4.6 \times 10^{-5}$ m for commercial steel). The reference correlation is the implicit **Colebrook–White** equation [98], which interpolates Nikuradse's smooth- and rough-pipe data:

$$\frac{1}{\sqrt{f}} = -2\log_{10}\!\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\mathrm{Re}\,\sqrt{f}}\right). \tag{466}$$

It is transcendental in $f$ (the unknown sits on both sides), so Choupo solves it by fixed-point iteration on $x = 1/\sqrt{f}$,

$$x_{k+1} = -2\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{2.51\,x_k}{\mathrm{Re}}\right), \tag{467}$$

seeded with the Haaland value below so it converges in a handful of steps (this is the glass-box `Colebrook` correlation you can read in `FrictionFactorCorrelation.cpp`).

**An explicit shortcut: Haaland.** To avoid the iteration, Haaland [99] gives an explicit approximation within ∼2 % of Colebrook over the whole turbulent range:

$$\frac{1}{\sqrt{f}} = -1.8\log_{10}\left[\left(\frac{\varepsilon/D}{3.7}\right)^{1.11} + \frac{6.9}{\mathrm{Re}}\right]. \tag{468}$$

Choupo's `Haaland` and `Colebrook` models both fall back to the exact 64/Re of Eq. (465) below Re = 2300.

**The transition band ($2300 \le \mathrm{Re} < 4000$): no correlation.** Between the two regimes there is no friction law to give. Laminar pipe flow is, as far as is known, linearly stable at every Re — a small enough disturbance dies away — but above Re ≈ 2000 a *large* enough one makes it turbulent, so whether a given line runs laminar, turbulent or intermittent (turbulent puffs travelling through laminar flow) depends on the disturbances it meets — the inlet, the stillness of the supply, vibration, the wall — and *not on* Re *alone*. The state also depends on the path: a flow brought up gently from rest can stay laminar far above the Re at which an already turbulent flow, slowed down, becomes laminar again. Reynolds saw all of this in 1883 [100]: with the greatest care over the stillness of his supply tank the breakdown was postponed to Re ≈ 13 000, a coil of wire in the tube brought eddies in at less than half the previous critical velocity, and the velocity below which existing eddies die out was lower again, in his ratio of 43.7 to 278 — Re close to 2000 [101]. Following puffs for very long times, the onset of *sustained* turbulence has since been placed near Re = 2040 [102], between 2020 (every flow eventually laminar) and 2060 (turbulence persists) [101]. The 2300 and 4000 that bound Choupo's band are conventions, not that critical point. So in the band $f$ is not a function of Re: at one Re the same pipe can carry laminar flow at 64/Re, turbulent flow at roughly twice that, or a mixture that changes in time. No correlation predicts it, and a design that sits in the band is one that should be moved out of it (change $D$ or the flow). A `pipe` whose Re falls there is **announced** once, at its site and in the end-of-run caveat block, as a `validity` advisory saying exactly this (`Pipe.cpp`, `pipeTransition`); its number is kept, because a solver needs a continuous $f$, and is not moved. The EduTool *Why engineering correlations are simple* (page 5) draws the band and tells the history.

**One expression for every regime: Churchill (the default).** Churchill [103] joined the laminar law and a turbulent branch into a *single* explicit formula defined for every Re and every $\varepsilon/D$ — so it needs no laminar branch and is continuous through the transition band. Through that band its value is an *interpolation* that keeps a solver continuous, not a prediction: the paragraph above is why there is no curve there for it to describe.

$$f = 8\left[\left(\frac{8}{\mathrm{Re}}\right)^{12} + \frac{1}{(A+B)^{3/2}}\right]^{1/12}, \tag{469}$$

$$A = \left\{-2.457\,\ln\left[\left(\frac{7}{\mathrm{Re}}\right)^{0.9} + 0.27\,\frac{\varepsilon}{D}\right]\right\}^{16}, \qquad B = \left(\frac{37530}{\mathrm{Re}}\right)^{16}.$$

As Re → 0 the $(8/\mathrm{Re})^{12}$ term dominates and Eq. (469) collapses to $f = 8(8/\mathrm{Re}) = 64/\mathrm{Re}$ — it reproduces the laminar law automatically. This is why Choupo defaults to `Churchill`: its continuity, not any claim about the transition. The other two are offered for pedagogy and for matching textbook hand calculations; in the band they are turbulent fits evaluated below the Re they were fitted for, and the engine flags them.

## 25.4 Minor losses and the assembled $\Delta P$

Bends, valves, expansions and tees each dissipate a fixed multiple of the velocity head, captured by a *loss coefficient* $K$. Summing the fittings and substituting Eq. (463) into Eq. (462) gives the assembled pressure drop Choupo computes:

$$\boxed{\Delta P = \underbrace{f\,\frac{L}{D}\,\frac{\rho u^2}{2}}_{\text{distributed}} + \underbrace{\left(\sum_j K_j\right)\frac{\rho u^2}{2}}_{\text{minor (fittings)}} + \underbrace{\rho\, g\, \Delta z}_{\text{elevation}}, \qquad P_{\text{out}} = P_{\text{in}} - \Delta P.} \tag{470}$$

The friction head reported as a KPI is $h_f = (\Delta P_{\text{fric}} + \Delta P_{\text{fit}})/(\rho g)$ — the dissipative part only, excluding the reversible static head. If $P_{\text{out}}$ comes out non-positive the solver warns aloud: the line is grossly undersized or too long for this flow, and you need a larger $D$ or a pump — it never silently "fixes" the geometry.

**Power.** Three lines of physics — a velocity-head scaling, a friction-factor correlation, and a static head — predict the pressure a real piping run costs, with the factor traceable to Nikuradse's data through Colebrook. The model is closed-form (or one short fixed-point loop), so it costs almost nothing inside a flowsheet pass.

**Limit.** The equations above are for a *single-phase incompressible liquid*. For a single *gas*, $\rho$ falls as $P$ drops along the pipe, so one $(\rho, u)$ evaluation is wrong — compressible flow needs the isothermal- or adiabatic-pipe integral, a documented future extension. *Gas-liquid two-phase* flow, by contrast, IS handled — it is the next subsection. There is no fitting database (you furnish each $K$), and the $K$-method itself is a lumped approximation to genuinely 3-D fitting flows.

**Worked example 25.1.** Water line, `pipe01_water_line`Water at 300 K, 5 bar, 2000 kmol/h flows through $D = 0.1$ m, $L = 50$ m commercial steel ($\varepsilon = 4.6 \times 10^{-5}$ m, so $\varepsilon/D = 4.6 \times 10^{-4}$), rising $\Delta z = 2$ m, with two elbows ($K = 0.9$ each) and an open globe valve ($K = 10$), giving $\sum K = 11.8$. The properties are $\rho = 875.5\,\mathrm{kg/m^3}$ and $\mu = 8.54 \times 10^{-4}$ Pa s. The kinematics:

$$\dot{m} = \dot{n} M_w = 0.5556\ \tfrac{\mathrm{kmol}}{\mathrm{s}} \times 18.02 = 10.0\ \mathrm{kg/s}, \quad Q_v = \frac{10.0}{875.5} = 1.14 \times 10^{-2}\ \tfrac{\mathrm{m^3}}{\mathrm{s}},$$

$$A = \frac{\pi (0.1)^2}{4} = 7.85 \times 10^{-3}\ \mathrm{m^2}, \quad u = 1.455\ \mathrm{m/s}, \quad \mathrm{Re} = \frac{875.5 \cdot 1.455 \cdot 0.1}{8.54 \times 10^{-4}} = 1.49 \times 10^5.$$

Turbulent, so Churchill (Eq. 469) returns $f = 0.01929$. The velocity head is $\rho u^2/2 = 927.3$ Pa, and the three terms are

$$\Delta P_{\mathrm{fric}} = 0.01929 \cdot \tfrac{50}{0.1} \cdot 927.3 = 8\,943\ \mathrm{Pa}, \quad \Delta P_{\mathrm{fit}} = 11.8 \cdot 927.3 = 10\,943\ \mathrm{Pa},$$

$$\Delta P_{\mathrm{elev}} = 875.5 \cdot 9.807 \cdot 2 = 17\,172\ \mathrm{Pa}, \quad \Rightarrow \quad \Delta P = 37\,058\ \mathrm{Pa} = 0.371\ \mathrm{bar},$$

so $P_{\mathrm{out}} = 4.63$ bar and $h_f = (8943 + 10943)/(875.5 \cdot 9.807) = 2.32$ m. Note the elevation term *dominates* here — a useful reminder that for a short line the static head, not the friction, often sets the pressure budget. Switching the model to `Haaland` or `Colebrook` changes $f$ (and hence $\Delta P$) by under 2 %.

**Runnable case.** `tutorials/steady/hydraulics/pipe01_water_line` — a single commercial-steel water line; $\Delta P$ is the RESULT of geometry + flow, split into its friction, fittings and elevation parts (swap `model Churchill` for `Haaland` / `Colebrook` and watch $f$ barely move).

## 25.5 Gas-liquid two-phase flow: Lockhart–Martinelli

When the feed flashes two-phase at $(T, P)$ — the vapour fraction $0 < V/F < 1$ the flowsheet already resolves (§2.4) — the pipe switches AUTOMATICALLY from the single-phase law above to a gas-liquid model. Two-phase friction is dramatically higher than either phase alone: the phases jostle, the slower one is held back, and the interface adds drag. **Layer 1** offers two $\sigma$-free models (they need only the phase densities and viscosities the package already gives).

**Homogeneous (no-slip).** Treat the mixture as one pseudo-fluid with a quality-weighted density and a McAdams viscosity,

$$\frac{1}{\rho_H} = \frac{x}{\rho_G} + \frac{1-x}{\rho_L}, \qquad \frac{1}{\mu_H} = \frac{x}{\mu_G} + \frac{1-x}{\mu_L}, \tag{471}$$

($x$ the mass quality) and apply the single-phase Darcy law to it. Simple, and adequate for finely-dispersed bubbly/mist flow; it assumes the phases travel at the *same* velocity (no slip), so it under-predicts the liquid holdup.

**Lockhart–Martinelli (the default).** Compute the pressure gradient each phase *would* have flowing alone in the pipe at its superficial velocity $j_k = \dot{m}_k/(\rho_k A)$, $(\mathrm{d}P/\mathrm{d}z)_k = f_k\, \rho_k j_k^2/(2D)$, and form the **Martinelli parameter**

$$X^2 = \frac{(\mathrm{d}P/\mathrm{d}z)_L}{(\mathrm{d}P/\mathrm{d}z)_G}. \tag{472}$$

The two-phase gradient is the liquid-alone gradient times a **multiplier** (Chisholm's algebraic form of the original Lockhart–Martinelli chart):

$$\left(\frac{\mathrm{d}P}{\mathrm{d}z}\right)_{2\phi} = \phi_L^2 \left(\frac{\mathrm{d}P}{\mathrm{d}z}\right)_L, \qquad \phi_L^2 = 1 + \frac{C}{X} + \frac{1}{X^2}, \tag{473}$$

with $C$ set by which phase is turbulent (each phase laminar if its alone-Re $< 1000$): $C = 20$ (both turbulent, *tt*), 12 (*vt*), 10 (*tv*), 5 (both viscous, *vv*). The liquid holdup uses Butterworth's [104] fit of the L–M void fraction, $\alpha = (1 + 0.28\, X^{0.71})^{-1}$, $\varepsilon_L = 1 - \alpha$, and the elevation term uses the holdup mixture density $\rho_{\mathrm{mix}} = \varepsilon_L \rho_L + \alpha \rho_G$.

*Intuition.* For the air-water line of `pipe02_airwater_twophase` ($j_G \approx 4.3$, $j_L \approx 0.58$ m/s, both phases turbulent), $X = 2.67$ and $\phi_L^2 = 8.6$ — the two-phase $\Delta P$ is **8.6 times** what the liquid alone would lose. And

the holdup is $\varepsilon_L = 0.36$ against a no-slip $j_L/(j_L + j_G) = 0.12$: the slow heavy liquid is *held up* (it occupies three times the pipe fraction its flow rate suggests), because the light gas races past it. Switch to `homogeneous` and the holdup drops to the no-slip 0.12 — a one-line way to see why slip matters.

**Layer 2: surface-tension models.** Two further models add the interfacial physics (they need $\sigma$, supplied by `transport { surfaceTension { model BrockBird; } }`). **Friedel** [105] gives a general two-phase multiplier $\phi_{LO}^2$ on the total mass flux flowing as liquid, built from density and viscosity ratios and the Froude and Weber numbers, $\phi_{LO}^2 = E + 3.24\,F\,H/(\mathrm{Fr}^{0.045}\,\mathrm{We}^{0.035})$ — the recommended general correlation. **Beggs–Brill** [106] goes further: from the no-slip holdup $\lambda_L$ and the Froude number it classifies the *flow pattern* (segregated / intermittent / distributed), reads the holdup from a pattern-specific $H_L(0) = a\,\lambda_L^b/\mathrm{Fr}^c$ (clamped $\geq \lambda_L$), corrects it for pipe *inclination* (the Payne et al. $\psi$ factor; horizontal $\Rightarrow \psi = 1$) and scales the no-slip friction factor by a two-phase ratio $e^S$ — so it is the one for inclined or hilly lines. On the `pipe02` air-water case the four models span $\Delta P = 6.0$–$9.0$ kPa: that $\sim$50 % scatter between two-phase $\Delta P$ correlations is itself the lesson — they are engineering fits, not laws.

**Scope and trust.** The correlations are general ($\rho_L, \rho_G, \mu_L, \mu_G, \sigma$), but Lockhart & Martinelli [107] developed and validated them on *isothermal air-water* pipe data, so air-water is exactly where they are trusted. All four are **adiabatic** (the inlet $V/F$ is held along the line — no flashing/condensing in the pipe). The liquid density still comes from the Rackett branch, $\sim$12 % low for water — it biases the *absolute* $\Delta P$, not the model structure.

**Runnable case.** `tutorials/steady/hydraulics/pipe02_airwater_twophase` — air ($N_2$) + water; the two-phase path activates on its own, prints $X$, $\phi_L^2$, the holdup and the regime, and lets you compare `LockhartMartinelli` against `homogeneous`.

# 26 Pneumatic conveying: carrying solids in a gas

*What you'll learn.* The pipe chapter above moves a single fluid. Now suspend SOLIDS in the gas and blow them down the line. The pressure drop is no longer a wall-friction story: most of it goes into the SOLIDS — accelerating them, dragging them against slip, lifting them, and (the punchline) re-accelerating them after every bend. This section derives the dilute-phase model Choupo uses: the additive pressure budget, the particle velocity from a force balance, the Yang solids-friction correlation, the saltation floor that decides whether the line PLUGS, and why a fitting can cost more than a hundred metres of pipe. Worked in `tutorials/steady/solids/pneumaticConveyor01_sand` and `pneumaticConveyor02_bends`.

## 26.1 The additive pressure budget

In developed dilute-phase flow the total drop is a SUM of independent momentum sinks — gas and solids treated separately, each contributing acceleration, friction and static (gravity) heads:

$$\Delta P = \underbrace{\tfrac{1}{2}\rho_g u_g^2}_{\text{gas accel}} + \underbrace{\rho_s^* u_p^2}_{\text{solids accel}} + \underbrace{2 f_g \rho_g u_g^2 \tfrac{L}{D}}_{\text{gas friction}} + \underbrace{f_p \tfrac{\rho_s^* u_p^2 L}{2D}}_{\text{solids friction}} + \underbrace{(\rho_g + \rho_s^*) g\, \Delta z}_{\text{static}} + \sum_{\text{bends}} \Delta P_{\text{bend}}. \tag{474}$$

Here $u_g$ is the superficial gas velocity, $u_p$ the particle velocity, and $\rho_s^* = \dot{m}_s/(u_p A)$ the *suspension density* — the mass of solids per unit pipe volume. The whole point of conveying is the solids terms: in a real line they dominate. Choupo PRINTS this breakdown line by line, so you SEE where the pressure goes.

## 26.2 Particle velocity: a force balance, not a fit

The particles do not travel at the gas velocity — they SLIP. In developed flow a particle no longer accelerates, so the gas DRAG balances the resistances it feels: the Coulomb friction against the pipe wall plus the gravity component along the pipe,

$$\tfrac{1}{2}\rho_g C_D \tfrac{\pi}{4} d_p^2 \,(u_g - u_p)^2 \;=\; \tfrac{\pi}{6} d_p^3 \rho_p \, g\,(\sin\theta + \mu_w \cos\theta), \tag{475}$$

solved for the slip $u_g - u_p$ (a scalar root-find; $C_D$ depends on the slip Reynolds number). Two limits make it physical: for a VERTICAL line ($\theta = 90°$) the balance is drag = gravity, so the slip equals the terminal velocity $u_t$; for a HORIZONTAL line ($\theta = 0$) it is drag = wall friction, so the slip is $u_t\sqrt{\mu_w}$. This REPLACES a fitted slip correlation — there is no black formula and no fallback; $\mu_w$, the particle–wall friction coefficient, is declared and announced.

## 26.3 Solids friction: Yang (1974)

The solids friction factor is the hard part — particles differ in size, shape and charge, and there is no universal Reynolds number. Choupo uses the correlation of Yang (1974)[108], which ties $f_p$ to the voidage $\varepsilon$ ($1-\varepsilon = \rho_s^*/\rho_p$, the solids holdup) and the terminal-to- particle Reynolds ratio $(Re)_t/(Re)_p = u_t/u_p$:

$$\text{vertical:}\quad f_p \frac{\varepsilon^3}{1-\varepsilon} = 0.0206 \left[(1-\varepsilon)\frac{(Re)_t}{(Re)_p}\right]^{-0.869}, \tag{476}$$

$$\text{horizontal:}\quad f_p \frac{\varepsilon^3}{1-\varepsilon} = 0.117 \left[(1-\varepsilon)\frac{(Re)_t}{(Re)_p}\frac{u_g}{\sqrt{gD}}\right]^{-1.15}. \tag{477}$$

The correlations are validated for voidage 0.994–0.999; outside that window (a more dilute line) $f_p$ is an EXTRAPOLATION and Choupo says so aloud. The horizontal form carries a pipe-diameter (Froude) dependence the vertical one does not, because in a horizontal pipe gravity segregates the particles radially.

## 26.4 The saltation floor — will the line plug?

Drop the gas velocity too far and the particles fall out of suspension and settle: the line SALTATES and eventually blocks. The saltation velocity $u_{\text{salt}}$ (Rizk, 1973) is the floor you must stay above, and the single most important safety number in a conveying design is the MARGIN

$$\frac{u_g}{u_{\text{salt}}} \geq 1.5\text{–}2, \tag{478}$$

which Choupo reports as a headline KPI and flags in red when it approaches 1. Under-velocity plugging — not the pressure drop — is the number-one real failure of a conveying line; a beautiful $\Delta P$ on a line that will plug is worthless.

## 26.5 Bends: the re-acceleration that dominates

A straight pipe accelerates the solids ONCE, at the inlet. A bend undoes it: the particles crash the outer wall, decelerate to $u_{p,\mathrm{bend}} = r\,u_p$, and the gas must RE-ACCELERATE them over the straight run downstream — the same momentum work as the line entry, once per bend,

$$\Delta P_{\mathrm{bend}} = \frac{\dot{m}_s\,(u_p - u_{p,\mathrm{bend}})}{A} = \frac{\dot{m}_s\,u_p\,(1-r)}{A}. \tag{479}$$

The exit ratio $r$ is set by the bend TYPE: a long-radius sweep keeps most of the velocity ($r \approx 0.75$), a blind (dead-leg) tee nearly stops the solids ($r \approx 0.25$) — three times the pressure cost, but almost no erosion, because the solids cushion on a stagnant pocket of their own kind rather than scouring the wall. In a bendy line the bends carry the majority of $\Delta P$ even though they add no length: *a fitting, not the pipe, sets the pressure budget.* (Choupo's term is the re-acceleration MOMENTUM, a mechanistic lower bound; the extra friction in the re-acceleration zone is not separately counted.)

## 26.6 Scope: dilute phase

This whole model is DILUTE phase — particles suspended and carried, loading up to ∼15 kg solids per kg gas. DENSE phase (plugs and slugs moving as a bed) is a different momentum regime with no honest general correlation; Choupo refuses to fake it and warns when the loading leaves the dilute range.

# 27 Gas-solid separation: cyclone, bag filter, ideal splitter

*What you'll learn.* Removing dust from a gas is a *size-classifying* operation, not a flow split: coarse particles are captured, fines escape, and the outlet streams carry different particle-size distributions than the feed. Choupo ships three models spanning the cost/accuracy spectrum — a spec'd ideal splitter, a cheap-physics cyclone, and a high-physics bag filter. This chapter gives the grade-efficiency curve that unifies them and the distinct physics of each.

## 27.1 Grade efficiency: the unifying idea

Every gas-solid separator is described by its *grade-efficiency* curve $\eta(d)$ — the fraction of particles of size $d$ that are captured. Apply it bin-by-bin to the feed PSD and you get the captured and escaping streams, each with its own (different) distribution. This is why a separator needs a PSD, not just a total solids flow: it does not divide the feed proportionally, it *re-sorts* it by size.

## 27.2 Cyclone: cheap centrifugal classification

A cyclone spins the gas; inertia throws particles to the wall, where they fall out, while the cleaned gas leaves up the core. The characteristic number is the *cut diameter* $d_{50}$ — the size captured with 50 % efficiency. The Lapple correlation gives

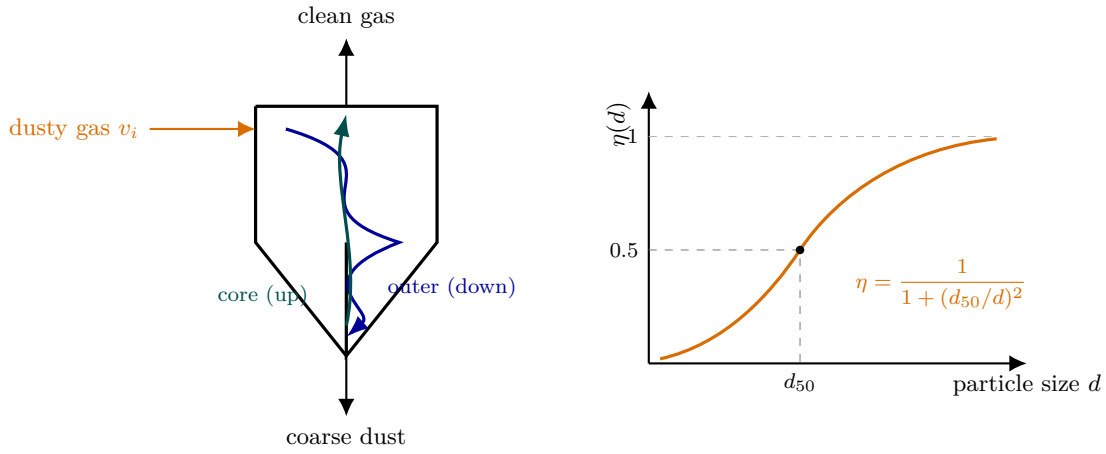



**Figure 75:** The cyclone classifies by inertia. Left: dusty gas enters tangentially and spins; the outer vortex carries particles down to the wall (blue), inertia throwing the coarse ones out, while the cleaned core swirls back up and leaves the top (teal). Right: the result is a *grade-efficiency* curve $\eta(d)$ — the fraction of size $d$ captured. The characteristic number is the **cut diameter** $d_{50}$, the size caught with 50 % efficiency (Lapple's $d_{50} = \sqrt{9\mu b/(2\pi N_e v_i \Delta\rho)}$). A separator does not divide the feed proportionally — it *re-sorts* it by size, so the two outlets carry different PSDs. This same curve, with a different shape, describes the bag filter and the ideal splitter.

$$d_{50} = \sqrt{\frac{9\,\mu\,b}{2\pi N_e\, v_i\,(\rho_p - \rho_g)}}, \tag{480}$$

($\mu$ the gas viscosity from the Chung model of Chapter 2, $b$ the inlet width, $N_e$ the effective turns, $v_i$ the inlet velocity, $\rho_p$ the particle density), and the grade-efficiency curve

$$\eta(d) = \frac{1}{1 + (d_{50}/d)^2}. \tag{481}$$

The collection efficiency follows by integrating $\eta(d)$ over the feed PSD. The pressure drop is a few velocity heads, $\Delta P = N_H \rho_g v_i^2/2$ (Shepherd–Lapple).

The cut is a *selectable sub-model* (the same explicit-factory pattern as the EoS and activity models — "all models are wrong, some useful"), chosen with `model <name>;`. There are **FIVE**, spanning the three classical approaches to the centrifugal/drag balance, and on one real cyclone they can disagree by ∼3× in $d_{50}$ (`cyclone05_dirgo_leith_validation`) — which is why you validate against data:

- **Lapple** (1951) — *timed flight*: the residence-time $d_{50}$ of Eq. (482)'s ancestor above, $d_{50} = \sqrt{9\mu b/(2\pi N_e v_i \rho_p)}$, with the S-curve $\eta = 1/(1 + (d_{50}/d)^2)$. The simple default.

- **Leith–Licht** (1972) — *back-mixing*: turbulence re-mixes uncollected dust across each plane, giving a closed-form efficiency $\eta(d) = 1 - \exp\big[-2\,(C\psi)^{1/(2n+2)}\big]$, where $\psi = \rho_p d^2 Q/(\dots)$ is the inertia parameter, $C$ a geometric constant, and $n = 1 - (1 - 0.67\,D^{0.14})(T/283)^{0.3}$ the vortex exponent (weaker spin in a hotter, wider cyclone).
- **Iozia–Leith** (1989)[109] — *flow pattern*: $d_{50}$ from the core velocity, Eq. (482) (derived just below).
- **Barth** (1956) — *equilibrium orbit*: the critical particle orbits the vortex-finder edge where its outward drift balances the inward gas, $d_{50} = \sqrt{18\mu\, v_r r_{cs}/(\rho_p v_\theta^2)}$, with $v_r = Q/(\pi D_{cs} H_{cs})$ the radial and $v_\theta = \alpha v_i$ the tangential velocity on the control surface.
- **Muschelknautz** (MMD) — the gold standard, because it adds the *solids-loading effect* the other four miss: above a limit loading $c_{\lim}$ the surplus dust is mass-separated at the wall near 100 %, $\eta = \eta_L + (1 - \eta_L)\eta_I$ with $\eta_L = 1 - c_{\lim}/c$, so a heavily-loaded cyclone is far more efficient than Lapple predicts (`cyclone04_high_loading`: 98 % vs 82 %).

### 27.2.1 The Iozia–Leith flow-pattern cut

Lapple's $d_{50}$ is a residence-time *surrogate*; Iozia & Leith (1989) instead read it off the gas FLOW PATTERN. The critical particle hangs at the core radius, where the tangential velocity peaks at $V_{t,\max}$; balancing its centrifugal throw against the inward drag of the gas draining into the vortex gives

$$d_{50} = \sqrt{\frac{9\,\mu\, Q}{\pi\, z_c\, \rho_p\, V_{t,\max}^2}}, \tag{482}$$

where $z_c$ is the core length and $Q$ the gas flow. The two flow-pattern quantities are regressed from the cyclone's OWN eight dimensions:

$$V_{t,\max} = 0.61\, V_i\, (ab/D^2)^{-0.61} (D_e/D)^{-0.74} (H/D)^{-0.33}, \tag{483}$$

$$d_c = 0.52\, D\, (ab/D^2)^{-0.25} (D_e/D)^{1.53}, \tag{484}$$

with $V_i = Q/(ab)$ the inlet velocity, $a, b$ the inlet height and width, $D_e$ the gas-outlet diameter, $H$ the total height, and $z_c = H - S$ (or, if the core $d_c$ exceeds the dust outlet $B$, the shorter distance where it meets the cone). Because it uses all eight dimensions it captures cut changes with geometry that Lapple's single $d_{50}$ cannot. (Source note: the printed 1989 prefactor on $V_{t,\max}$ is 6.1, which yields a supersonic $V_{t,\max} \sim 400$ m/s and a ten-fold-too-fine cut — contradicting the paper's own Figure 4, where $V_{t,\max} \le 40$ m/s. The factor of exactly ten is a misplaced decimal; Choupo uses the Figure-4-consistent 0.61.) Validate it against the measured fractional-efficiency curves of Dirgo & Leith (1985)[110] in `cyclone05_dirgo_leith_validation` — where the five sub-models spread over ∼3× in $d_{50}$ on one real cyclone, the reason data, not a single correlation, has the last word.

## 27.3 Bag filter: the cake is the filter

A fabric filter is almost total: the dust cake that builds on the cloth does the filtering, so the grade efficiency is high and only mildly size-dependent,

$$\eta(d) = 1 - P_0\, e^{-d/d_c}, \tag{485}$$

(fines penetrate most). Here the headline result is not efficiency but the cake-dominated *pressure drop* — Darcy through clean cloth plus cake:

$$\Delta P = \mu\, V\, (K_1 + K_2 W), \tag{486}$$

with $V = Q/A$ the face velocity (the air-to-cloth ratio), $W$ the areal dust load, $K_1$ the clean-cloth and $K_2$ the cake resistance. Same dust as the cyclone, the bag filter captures **99.93 %** (vs ∼82 %) at $\Delta P \approx 2.7$ kPa — which is why the classic train is cyclone (cheap pre-cleaner) then bag filter (polish).

## 27.4 Ideal splitter: when you just want a number

Sometimes a physics model is overkill — "assume 90 % capture". The `gasSolidSplitter` applies the split the user *states* (`solidsRecovery`, `gasCarryover`); the feed PSD passes *unchanged* into both outlets (a spec'd split does not classify by size — the very distinction from a cyclone). Use it as a placeholder, a bound, or when the physics is not worth the trouble.

**Key idea.** The three separators span a deliberate spectrum on the same PSD-aware interface: spec'd (`gasSolidSplitter`, you state the number), cheap-physics (`cyclone`, ∼82 % from a correlation), high-physics (`bagFilter`, ∼99.9 % from the cake). Pick the fidelity the problem deserves — the Pareto principle made into a menu.

**Runnable case.** `tutorials/steady/gas-solid/cyclone01_dust_removal` — Lapple, $d_{50} = 2.7\,\mu$m, 82 % collection (and cyclone02–04 for the other models + loading).
`tutorials/steady/gas-solid/bagFilter01_dust` — the same dust, 99.93 % + the cake $\Delta P$.

> `tutorials/steady/gas-solid/gasSolidSplitter01_ideal` — the spec'd split.

# 28 Drying: spray dryer and solid dryer

This section is the drying *units* as the simulator runs them; the theory they draw on — the dry-basis moisture and its equilibrium, the constant- and falling-rate periods and the drying-time integral, the three atomiser families and their correlations, the single-droplet and chamber balances — is §38.

> *What you'll learn.* Drying couples four physics at once — a mass/energy balance, droplet atomisation, the wet-bulb temperature, and the drying-rate kinetics — and ends at a residual moisture set by the air, not by wishful thinking. This chapter walks the spray dryer (atomise a liquid into a powder) and the solid dryer (finish an already-solid powder), and shows the recurring split the simulator enforces: the sorption *equilibrium* (thermodynamics, on the material's `.dat`) is kept separate from the drying *curve* (kinetics, in `constant/dryingKinetics`).

## 28.1 The four coupled pieces of a spray dryer

**Mass and energy balance.** Hot air meets atomised feed; the

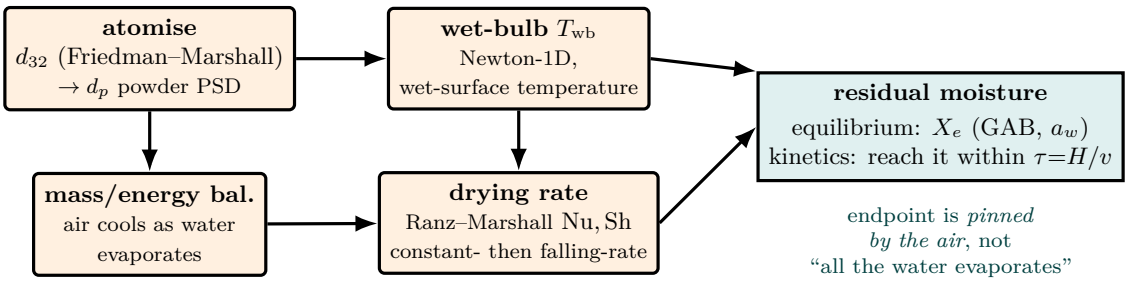



**Figure 76:** A spray dryer is four physics solved at once, ending at a moisture the air sets. **Atomisation** (a Friedman–Marshall $d_{32}$ shrinking to the powder $d_p$) gives the particle PSD; the **mass/energy balance** ties the air's cooling to the water evaporated; the **wet-bulb** temperature (a Newton-1D) is the wet-surface temperature during constant-rate drying; and the **Ranz–Marshall** transfer coefficients set the drying-rate curve. They converge on the *residual moisture* — and here is the lesson: the endpoint is *not* "everything evaporates". The GAB sorption isotherm pins the equilibrium moisture $X_e$ to the air's humidity (thermodynamics, kept on the `.dat`); the falling-rate curve says whether the powder reaches $X_e$ within the residence time $\tau = H/v$ (kinetics, kept separate). Mixing the two is a category error the simulator refuses.

solvent evaporates and the air cools. An adiabatic balance ties the air outlet temperature to the water evaporated and the powder production.

**Atomisation.** A rotary wheel shears the liquid into droplets whose Sauter mean diameter follows a Friedman–Marshall correlation in the wheel speed, diameter and liquid properties; the droplet then shrinks to the powder particle by the solid fraction,

$$d_p = d_{32}\,(x_{\text{solid}}\,\rho_L/\rho_{\text{solid}})^{1/3}, \tag{487}$$

so the powder carries a PSD.

**Wet-bulb temperature.** While the surface is wet the droplet sits at the psychrometric wet-bulb temperature (a Lewis-number-$\approx 1$ balance of convective heat in against evaporative cooling out), found by a Newton-1D — the constant-rate surface temperature.

**Drying-rate kinetics.** The droplet's Reynolds number (from its terminal velocity) feeds the Ranz–Marshall correlation $\mathrm{Nu} = \mathrm{Sh} = 2 + 0.6\,\mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}$, giving the heat- and mass-transfer coefficients $h, k_c$ and hence the constant-rate drying time.

## 28.2 The residual moisture: equilibrium vs kinetics

The "all the water evaporates" assumption is wrong: a powder equilibrates with the air's humidity at a finite moisture. Two distinct things set the endpoint, and the simulator keeps them in separate files — keeping the sorption isotherm separate from the characteristic drying curve.

*Equilibrium* (on the component `.dat`, axiom 1): the GAB sorption isotherm gives the equilibrium moisture at the air's water activity $a_w$ (= the relative humidity),

$$X_e = \frac{X_m\, C\, K\, a_w}{(1 - K a_w)(1 - K a_w + C K a_w)}. \tag{488}$$

*Kinetics* (in `constant/dryingKinetics`, axiom 3): below the critical moisture $X_c$ the rate falls; the Lewis falling-rate model $dX/dt = -k(X - X_e)$, with $k$ tied to the constant rate by continuity at $X_c$ (no new parameter), carries $X$ from $X_c$ toward $X_e$. The final moisture is the drying curve evaluated at the *residence time*, itself a RESULT of the chamber geometry, $\tau = H/v_{\text{particle}}$ (credo: the chamber diameter and height are the hardware, residence is computed). The powder leaves carrying that residual water as bound liquid *alongside* the dry solid, so the mass balance closes.

> *Intuition.* Whether a powder comes out at 1 % or 5 % moisture is not a free choice — it is pinned by the air's humidity through the sorption isotherm, and reached (or not) within the residence time through the drying curve. Mixing the critical moisture $X_c$ (a kinetics number) into the isotherm (thermodynamics) is a category error; the simulator refuses to, and so should the student.

## 28.3 The solid dryer: finishing an already-solid powder

A contact / through-circulation `solidDryer` takes an *already-solid* powder (crystals plus bound moisture) and polishes it *toward* the equilibrium moisture $X_e$ set by the inlet air's relative humidity. The air is a real stream: it brings the heat and carries the water away, the dryer is adiabatic and declares no duty, and the outlet temperature and the water removed are RESULTS. The water evaporated is the least of three, and the run announces which one binds: the *isotherm* ($X_{\text{final}} = \min(X_{\text{in}}, X_e)$), the *heat* the air can pay for with the solid leaving at the air's wet-bulb temperature (the model's floor), or the *saturation* of the exhaust at its own outlet temperature ($Y \le Y_{\text{sat}}(T_{\text{out}})$, i.e. $a_w \le 1$). So $X_{\text{final}} = \min(X_{\text{in}}, X_e)$ only when the isotherm binds; a starved or cold air leaves the powder wetter, and the run says why. It complements the spray dryer (atomise a liquid → powder); the solid dryer finishes a solid — the two ends of a drying train.

**Worked example 28.1.** Sugar spray dryer, `sprayDryer01_sugar`A 40 wt% sugar solution dried by air at 200 °C, 0.1 m disc at 20000 rpm: $d_{32} \approx 58\,\mu$m, particle $\approx 37\,\mu$m, $T_{\text{wb}} \approx 45$ °C, $T_{\text{out}} \approx 98$ °C, Stokes regime; with the GAB isotherm + a ∼7 s residence the powder reaches $X_e \approx 0.014$ (1.3 wt% — a realistic sugar-powder moisture), and the mass balance closes 100 %.

**Runnable case.** `tutorials/steady/drying/sprayDryer01_sugar` — atomise → wet-bulb → constant-rate → falling-rate → equilibrium.
`tutorials/steady/drying/sprayDryer02_residence_sweep` — the chamber-height sweep, the drying curve appearing as the chamber grows.
`tutorials/steady/drying/solidDryer01_sugar` — finishing a solid to the air-set equilibrium moisture.
`tutorials/steady/drying/solidDryer02_starved_air` — the same dryer with one eighth of the air: the exhaust saturates first, and the powder leaves wetter.

# 29 Two-stream heat exchanger: the $\varepsilon$-NTU rating method

*What you'll learn.* The single-stream `heater` (next section) takes a furnished duty $Q$; a two-stream `heatExchanger` takes only *hardware* — area $A$ and overall coefficient $U$ — and the duty, the outlet temperatures and the LMTD all emerge. That is the *rating* problem, and the effectiveness-NTU method solves it without iteration (unlike LMTD, which would need it when the outlets are unknown). The split between the two units is by *number of process streams* (1 vs 2), not by folklore.

With heat-capacity rates $C = \dot{n}\,C_p$ for each stream, $C_{\min}$, $C_{\max}$ and their ratio $C_r = C_{\min}/C_{\max}$, the number of transfer units and the effectiveness are

$$\mathrm{NTU} = \frac{UA}{C_{\min}}, \qquad \varepsilon_{\text{counter}} = \frac{1 - e^{-\mathrm{NTU}(1-C_r)}}{1 - C_r\, e^{-\mathrm{NTU}(1-C_r)}} \tag{489}$$

(and the co-current form $\varepsilon = [1 - e^{-\mathrm{NTU}(1+C_r)}]/(1 + C_r)$). The duty follows directly from the maximum possible temperature approach,

$$Q = \varepsilon\, C_{\min}\,(T_{h,\mathrm{in}} - T_{c,\mathrm{in}}), \tag{490}$$

and the outlet temperatures from each stream's energy balance. The LMTD is computed *a posteriori* and the identity $UA\,\mathrm{LMTD} = Q$ is a clean self-check. Tutorial `heatExchanger01_water_water`: $A = 10\,\mathrm{m}^2$, $U = 500\,\mathrm{W/m^2K}$ counter-current cool $360 \to 315\,\mathrm{K}$ against $300 \to 337\,\mathrm{K}$, $\varepsilon = 0.745$, $Q = 93.8\,\mathrm{kW}$.

# 30 Heat exchanger DESIGN: geometry, pressure drop, and the sizing workflow

*What you'll learn.* The $\varepsilon$–NTU section above RATES an exchanger whose $U$ and area you already know. But where do $U$ and the area come from? This section opens that box: the overall coefficient BUILT from the two film coefficients and the wall, the pressure drop both sides, the shell diameter implied by a tube count, and — the payoff — the DESIGN loop that inverts the problem: given a duty, SIZE the exchanger. This is the workflow a real project follows, and Choupo does it glass-box: every $Nu$, every resistance, every friction factor is printed and cited. Worked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty` (size it) → `hxWorkflow2_rate_designed` (rate it).

## 30.1 Two questions: rating vs design

A shell-and-tube exchanger poses two inverse questions:

- **Rating** (`model geometry;`): the geometry is GIVEN; find $U$, the area, the pressure drop, and hence the duty. "I have this exchanger — what will it do?"
- **Design** (`model design;`): the DUTY is given; find a geometry that delivers it within a pressure budget. "I need this duty — size me an exchanger."

Design is built ON rating: you cannot size without being able to rate a trial.

## 30.2 The overall coefficient, built

Heat crosses three resistances in series — the tube-side film, the tube wall, the shell-side film — referred here to the OUTSIDE area $A_o = \pi\, d_o\, L\, N$ (the convention that makes $U_o A_o$ the group in $Q = U_o A_o\, \Delta T_{\mathrm{lm}}$):

$$\frac{1}{U_o} = \underbrace{\frac{1}{h_o}}_{\text{shell film}} + \underbrace{\frac{r_o \ln(r_o/r_i)}{k_{\mathrm{wall}}}}_{\text{cylindrical wall}} + \underbrace{\frac{r_o}{r_i\, h_i}}_{\text{tube film}} . \tag{491}$$

The engine prints the three terms and names the largest the *controlling resistance* — the one worth attacking (a better shell-side baffle cut buys nothing if the tube film controls).

**Tube-side film (Gnielinski).** For turbulent in-tube flow,

$$Nu = \frac{(f/8)(Re - 1000)\, Pr}{1 + 12.7\sqrt{f/8}\,(Pr^{2/3} - 1)}, \qquad f = (0.790 \ln Re - 1.64)^{-2}, \tag{492}$$

(Gnielinski; Incropera & DeWitt)[111, 55] with $h_i = Nu\, k/d_i$, evaluated at the stream's mean temperature with real (IF97 for water) $\rho, \mu, k, c_p$.

**Shell-side film (Kern).** The shell side is a baffled crossflow; the Kern method[112] lumps it into a bundle correlation on an EQUIVALENT diameter $D_e$ and the crossflow mass flux $G_s$:

$$Nu = 0.36\, Re_s^{0.55}\, Pr^{1/3}, \quad Re_s = \frac{G_s D_e}{\mu}, \quad G_s = \frac{\dot m_s}{A_s}, \quad A_s = \frac{D_s\,(p - d_o)\,B}{p}, \tag{493}$$

with $A_s$ the crossflow area at the shell centreline, $B$ the baffle spacing, $p$ the tube pitch, and $D_e = 4(p^2 - \pi d_o^2/4)/(\pi d_o)$ (square pitch). Kern is a ±25% method — honest for teaching and a first design; Bell–Delaware (leakage/bypass corrections) is the refinement.

## 30.3 Pressure drop: the other half of design

An exchanger is not sized for area alone — it lives within a PUMPING budget. The tube side (friction over all $n_p$ passes plus the turnaround losses):

$$\Delta P_t = \left(4 f\, \frac{L\, n_p}{d_i} + 4 n_p\right) \frac{\rho u_t^2}{2}, \qquad f = 0.079\, Re^{-1/4} \text{ (Blasius[113])}. \tag{494}$$

The shell side (Kern), over the $N_b$ baffles:

$$\Delta P_s = f_s\, \frac{G_s^2\, D_s\, (N_b + 1)}{2\, \rho\, D_e}, \qquad f_s = \exp(0.576 - 0.19 \ln Re_s). \tag{495}$$

The shell drop typically EXCEEDS the tube drop — baffled crossflow dissipates more — and the duty-vs-pumping trade-off is precisely the engineering the map of $U$ against $\Delta P$ makes visible.

## 30.4 The design loop

Design inverts rating. Fix the tube choices (diameter, length, pitch, passes, material); everything else follows from ONE unknown, the tube count $N$:

1. From the duty target (an outlet $T$) and an energy balance, get $Q_{\mathrm{req}}$, both outlet temperatures, and $\Delta T_{\mathrm{lm}}$; hence the required $UA_{\mathrm{req}} = Q_{\mathrm{req}}/\Delta T_{\mathrm{lm}}$.
2. The shell diameter follows from $N$ by the Kern/Sinnott bundle correlation $D_b = d_o(N/K_1)^{1/n_1}$ + clearance (triangular pitch, one pass: $K_1 = 0.319$, $n_1 = 2.142$).
3. $U(N)\,A(N)$ increases monotonically with $N$ (area grows linearly, $U$ falls only slowly as the tube velocity drops), so a BISECTION on $N$ finds the count that meets $UA_{\mathrm{req}}$. Round UP: never undersize.
4. Rate the sized bundle (the same film coefficients) and read the pressure drop — if it busts the budget, fewer passes or a wider baffle spacing, and re-check.

## 30.5 The design workflow

The cardinal rule: *do not invent the geometry up front.* The standard practice splits the job into three:

1. a **Heater / shortcut** block establishes the DUTY while the flowsheet converges (no geometry);
2. a **detailed design** step sizes the exchanger from that duty (generates the geometry);
3. a **detailed rating** step checks the sized exchanger and probes off-design.

Choupo folds steps 1–2 into `model design;` — it computes the duty from your target, ANNOUNCES it, and sizes in one honest step — then `model geometry;` is the rating (step 3). The two-part tutorial sequence walks exactly this: Part 1 sizes 146 tubes / 0.344 m shell from a 330→315 K target; Part 2 rates that bundle (delivers 315.0 K, a touch over-surfaced from the round-up) and invites the off-design study (turn the coolant down, watch the outlet warm).

## 30.6 The deliverable: the datasheet

A design ends in a *datasheet* — the TEMA specification sheet an engineer signs off. Choupo's GUI builds one (the unit panel → "Open datasheet"): a formatted sheet with the service, the thermal and hydraulic results, the geometry, and a SCHEME of the shell-and-tube drawn from the solved numbers (the tube count, the baffles, the passes, the nozzle temperatures), printable to PDF; the raw numbers also export to a spreadsheet. Open, cited, and reproducible — the black box a commercial tool hands you, opened.

# 31 The water cooling tower: Merkel's enthalpy driving force

*What you'll learn.* Every exchanger in the last three sections moved heat from one stream to another; the cooling tower is where the plant finally *rejects* it — and it does so by evaporating a little of the very water it cools. Sensible and latent transfer happen together on the same film, and Merkel's 1925 insight is that with one approximation (the Lewis factor $\simeq 1$, an air–water near-coincidence) the TWO driving forces collapse into ONE: the difference between the enthalpy of saturated air at the local water temperature, $h^*(T)$, and the enthalpy of the bulk air, $h$. The whole tower then lives on a single diagram — $h$ against $T$ — and that diagram is exactly what the `coolingTower` unit publishes as its profile.

## 31.1 The construction

Working per kilogram of dry air (humidity $Y$, datum: dry gas and liquid water at 273.15 K), the humid enthalpy is

$$h(T,Y) = c_{p,\mathrm{dry}}\,(T - T_0) + Y\left[\lambda_0 + c_{p,\mathrm{vap}}\,(T - T_0)\right], \tag{496}$$

and $h^*(T) = h(T, Y_{\mathrm{sat}}(T))$ with $Y_{\mathrm{sat}} = (M_v/M_c)\,P^{\mathrm{sat}}/(P - P^{\mathrm{sat}})$ — the same saturation curve the psychrometric chart draws (§38). A counter-current energy balance from the bottom (air in, water out) gives the *operating line*, straight in $T$:

$$h(T) = h_{\mathrm{in}} + \frac{L\,c_{p,L}}{G}\,(T - T_{\mathrm{out}}), \tag{497}$$

and eliminating the interfacial state through $\mathrm{Le} = 1$ turns the packing balance into Merkel's integral,

$$\mathrm{Me} \;=\; \frac{KaV}{L} \;=\; \int_{T_{\mathrm{out}}}^{T_{\mathrm{in}}} \frac{c_{p,L}\,\mathrm{d}T}{h^*(T) - h(T)}. \tag{498}$$

The left side is *hardware* (the packing's transfer characteristic); the right side is pure *thermodynamics* of the two curves. Exactly one of them is declared, and the other is the result: `merkelNumber` (rating: the cold-water $T$ emerges by bisection — $\mathrm{Me}(T_{\mathrm{out}})$ is monotone, diverging at the wet-bulb floor and vanishing at $T_{\mathrm{in}}$) or `T_water_out` (design: the required Me emerges by quadrature).

## 31.2 What the model assumes, and says so

Four hypotheses ride with every solve, each announced at run time, never silent: $\mathrm{Le} = 1$ (what makes $h$ the single potential); the water loss neglected *inside* the integral ($L$ constant along the packing, the classical form) while the loss itself is reported and honoured at the boundary, so the mass balance closes exactly; the exit air assumed saturated (Merkel resolves the air's enthalpy profile, not its humidity — the exit state needs one closing assumption and this is the classical one); and $c_p$ evaluated once at mean film temperatures. The engine integrates Eq. (498) by composite Simpson on 200 intervals and publishes the classical CTI four-point Chebyshev estimate *beside* it (`merkelNumber_chebyshev4`), so the hand method's deviation from the converged quadrature is a visible number — the same pedagogy as the Kremser shortcut beside the stagewise absorber (§16).

## 31.3 The two temperatures that matter

A tower's currency is not its duty but its *approach*: the cold water can never go below the inlet air's wet-bulb temperature (the adiabatic-saturation solve of §38, at $\mathrm{Le} = 1$ the same thing), and the last few kelvin toward it are exponentially expensive in packing. Witness `coolingTower01_merkel`: water 45 °C at $L/G = 0.99$ against air at 25 °C and 62 % RH ($T_{\mathrm{wb}} = 293.0$ K), packing $\mathrm{Me} = 1.5 \to T_{\mathrm{out}} = 300.78$ K — a range of 17.4 K bought at an approach of 7.8 K, evaporating 2.40 % of the circulating water (the folklore "1 % per 7 K" emerging from the latent heat, not quoted to it). A target at or below the wet bulb refuses by name; so does an operating line that touches $h^*(T)$ (the tower analogue of a pinched exchanger — the remedy is lower $L/G$), and an air inlet that is not a permanent gas at its own conditions. Gate: `check_cooling_tower` recomputes Eq. (498) from the published diagram alone, with $c_{p,L}$ taken off the operating line's slope — the engine's internals are never consulted.

# 32 Pinch analysis: the target before the network

Every heat exchanger so far in this guide has been a *device*: two streams, an area, a duty. This section asks a question that comes *before* any of them exists.

A plant has hot streams that must be cooled and cold streams that must be heated. Some of that heat can be moved from the first group to the second; the rest has to come from steam and go to cooling water. How much is *the rest*? Not for a particular network — for the best network that could ever be built.

The answer is a *thermodynamic target*, and it can be computed from the stream data alone, without drawing a single exchanger. That is the whole of pinch analysis, and it is one of the few results in this guide that a student can hand-check on paper in ten minutes and then watch the engine reproduce line for line.

**Key idea.** The target comes first, the network second. A network is judged by how close it gets to a number that was known before it was drawn. Designing first and asking "is this good?" afterwards has no answer, because there is nothing to compare against.

## 32.1 Streams, CP, and why a duty becomes a line

A stream that changes temperature carries a *heat-capacity flowrate*

$$CP \;=\; \dot{n}\,c_p \;=\; \frac{|Q|}{|T_{\text{out}} - T_{\text{in}}|} \qquad [\text{kW/K}], \tag{499}$$

and if $c_p$ is constant the stream is a straight line on a $T$–$H$ diagram with slope $1/CP$. A *hot* stream runs from a high $T_{\text{in}}$ down to a low $T_{\text{out}}$ and releases $Q = CP\,\Delta T$; a *cold* stream is the mirror image.

Choupo does not ask you to type stream data into a separate pinch tool. It reads the *converged flowsheet*: every unit that carries a duty contributes one segment, with $CP$ from Eq. (499) between its own inlet and outlet temperatures. This is the first of the method's hypotheses and it is stated out loud, because it is the one a student will trip over:

**Common pitfall.** $CP$ **is constant per segment.** A unit whose $c_p$ varies strongly across its own temperature range is *linearised* over that range — one segment, one slope. If that matters for your case, split the unit in the flowsheet: two units in series give two segments and two slopes. The engine will not silently subdivide on your behalf, because where to subdivide is a modelling decision and not an arithmetic one.

A phase change is the limiting case: a large duty across (almost) no temperature change, so $CP \to \infty$ and Eq. (499) is useless. The standard treatment, and Choupo's, is to give the latent duty a near-isothermal slice of finite width (`latentWidth`, 1 K by default) centred on the unit's temperature. The composite curve then has a nearly flat step there, which is exactly what a condenser looks like on a $T$–$H$ plot.

## 32.2 The shift: why $\Delta T_{\min}$ moves the axes

Heat flows from hot to cold, and it needs a driving force. A designer declares the smallest approach they are prepared to pay for, $\Delta T_{\min}$ — a capital-versus-energy choice, since a smaller approach recovers more heat and needs more area.

Rather than carry $\Delta T_{\min}$ through every inequality, the method *shifts the temperature axis* once, at the start:

$$T^*_{\text{hot}} = T_{\text{hot}} - \tfrac{1}{2}\Delta T_{\min}, \qquad T^*_{\text{cold}} = T_{\text{cold}} + \tfrac{1}{2}\Delta T_{\min}. \tag{500}$$

On the shifted axis, a hot stream and a cold stream at the same $T^*$ are in fact $\Delta T_{\min}$ apart in reality. So on shifted temperatures heat may be exchanged wherever a hot stream sits at or above a cold one, with no further bookkeeping. That single change of variable is what makes the rest of the method arithmetic instead of case analysis.

**Key idea.** $\Delta T_{\min}$ is a *declaration*, not a property of the plant. Choupo makes you write it (`pinchPass { dTmin 10 K; }`) and it keeps it separate from the *utility* approach declared in the utility report — a process-to-process approach and a duty-to-utility approach are different physical quantities, and giving them one name is how they come to disagree.

## 32.3 The problem table, cascade by cascade

Now the arithmetic. Sort every shifted stream endpoint into a descending list; these are the interval boundaries. In each interval, the net enthalpy surplus is

$$\Delta H_k \;=\; \Big(\sum_{\text{hot in } k} CP \;-\; \sum_{\text{cold in } k} CP\Big)\,\big(T^*_k - T^*_{k+1}\big), \tag{501}$$

positive when the hot streams present there release more than the cold streams absorb. Cascade the surpluses downward from the top with zero entering:

$$R_0 = 0, \qquad R_k = R_{k-1} + \Delta H_k. \tag{502}$$

If every $R_k \geq 0$ the cascade is feasible as it stands. In general it is not: some $R_k$ goes negative, which would mean heat flowing *up* the temperature scale. The repair is to feed in exactly enough hot utility at the top to lift the worst deficit to zero:

$$Q_{H,\min} = -\min_k R_k, \qquad Q_{C,\min} = R_N + Q_{H,\min}, \tag{503}$$

and the interval where the corrected cascade touches zero is the **pinch**. There, and only there, the two composite curves are $\Delta T_{\min}$ apart; everywhere else there is slack.

> *Intuition.* The pinch is a bottleneck in temperature, not in equipment. Above it the process is a net heat *sink* and can only be served by hot utility; below it, a net heat *source* that can only be served by cold utility. Move a single kilowatt across the pinch and you pay for it twice — one extra kilowatt of steam *and* one extra of cooling water. That is the whole design rule, and it follows from Eq. (502) alone.

Choupo prints the table *row by row* at `verbosity 3` — the interval boundaries, the $CP$ sums, $\Delta H_k$ and the running cascade — because the table *is* the method. A student who is handed only $Q_{H,\min}$ has been handed a number to trust; a student who is handed the cascade can check it.

**Worked example 32.1.** The four-stream classic, by handTwo hot streams and two cold, with constant $c_p$ so every $CP$ is exact:

| stream | type | $CP$ [kW/K] | $T_{\text{in}}$ [°C] | $T_{\text{out}}$ [°C] | $Q$ [kW] |
|---|---|---|---|---|---|
| H1 | hot | 2.0 | 150 | 60 | −180.0 |
| H2 | hot | 8.0 | 90 | 60 | −240.0 |
| C1 | cold | 2.5 | 20 | 125 | +262.5 |
| C2 | cold | 3.0 | 25 | 100 | +225.0 |

At $\Delta T_{\min} = 20\,\text{K}$ the shifted boundaries are $140, 135, 110, 80, 50, 35, 30\,°\text{C}$, and the cascade is

| interval $T^*$ | $\sum CP_{\text{hot}}$ | $\sum CP_{\text{cold}}$ | $\Delta H$ [kW] | | cascade [kW] |
|---|---|---|---|---|---|
| 140 → 135 | 2 | 0 | +10.0 | | +10.0 |
| 135 → 110 | 2 | 2.5 | −12.5 | | −2.5 |
| 110 → 80 | 2 | 5.5 | −105.0 | −107.5 | ← minimum |
| 80 → 50 | 10 | 5.5 | +135.0 | | +27.5 |
| 50 → 35 | 0 | 5.5 | −82.5 | | −55.0 |
| 35 → 30 | 0 | 2.5 | −12.5 | | −67.5 |

So $Q_{H,\min} = 107.5\,\text{kW}$, $Q_{C,\min} = -67.5 + 107.5 = 40\,\text{kW}$, and the pinch sits at $T^* = 80\,°\text{C}$ — hot 90 °C, cold 70 °C. Unintegrated, the same four duties consume 487.5 kW of heating: the target says 380 kW of that is recoverable.

> **Runnable case.** `tutorials/steady/heat/pinch01_four_stream_classic` is that table. Its `flowsheetDict` header carries the hand-worked cascade above so you can compare it against what the run prints, line by line.

## 32.4 Composite curves: the same result, drawn

The problem table is the arithmetic; the *composite curves* are the picture. Add the hot streams' $T$–$H$ segments horizontally into one hot composite, do the same for the cold, and slide the cold curve until the closest vertical approach is $\Delta T_{\min}$. The overlap is the recoverable heat; the hot curve's overhang on the right is $Q_{C,\min}$, the cold curve's on the left is $Q_{H,\min}$, and the point of closest approach is the pinch. It is the same three numbers, seen instead of counted.

Choupo writes both curves, shifted and unshifted, to `reports/pinch/compositeCurves.csv`.

## 32.5 Candidate matches: what the physics admits

Targets say *how much*. The next question is *which stream against which* — and here the guide has to be careful about what the engine does and does not claim.

Choupo clips every segment at the pinch into an above-region and a below-region, and then enumerates *every* hot–cold pair in each region. For each pair it reports the largest duty that pair could carry, which for two constant-$CP$ segments meeting counter-currently at the declared approach is exactly

$$Q_{\max} \;=\; \min\Big(Q_{\text{hot}},\; Q_{\text{cold}},\; \min(CP_h, CP_c)\,\big(T_{h,\text{hi}} - T_{c,\text{lo}} - \Delta T_{\min}\big)\Big), \tag{504}$$

and it names which of the three terms is binding. (For straight segments the approach varies monotonically along the exchanger, so checking the two ends is sufficient — that is why Eq. (504) needs no integration.)

It also applies the classical $CP$ rule — above the pinch $CP_h \le CP_c$, below it $CP_h \ge CP_c$ — and reports, per row, whether the pair obeys it.

**Common pitfall.** **The $CP$ rule binds only *at* the pinch.** A match that violates it is not thereby forbidden: it simply cannot be the match that sits against the pinch. It may carry duty perfectly well further away from it, and Choupo says exactly that in the row's own words rather than striking it out.

**Key idea.** The candidate table is an **analysis**, never a synthesis. Choupo does not draw your network, does not rewrite your flowsheet, and — deliberately, and checked by a gate — *never uses the word "optimal"*. Every feasible row is a "thermodynamically admissible candidate". Which candidates become exchangers is the engineer's decision, and it involves plot plans, control, startup and safety that no temperature table knows about.

## 32.6 The violation diagnostics, and why they add up

The last thing the pass reports is the pair of coursework violations: *heating below the pinch* and *cooling above it*. Each is a duty that a utility is currently serving and that a recovery match could serve instead, and each is listed per unit with its duty.

They are not decoration. With the raw-duty hypothesis above they satisfy an identity worth checking by hand:

$$\sum(\text{heating below}) \;+\; \sum(\text{cooling above}) \;=\; Q_{H,\text{current}} - Q_{H,\min}, \tag{505}$$

— the excess over target *is* the sum of the violations. In the four-stream example: 120 kW of cooling above the pinch plus $125 + 135 = 260$ kW of heating below it gives 380 kW, and $487.5 - 107.5 = 380$ kW. The two sides are computed by different code and must agree; the gate that ships with the case checks that they do.

## 32.7 What this pass does not do

Stated plainly, because the boundary is a decision and not a to-do list:

- It **targets and analyses**; it never rewrites the network. The flowsheet you wrote is the flowsheet that ran.
- It reports **energy** targets. Area and capital targets — the $\sum \Delta H/(\Delta T_{\text{LM}}\,U)$ construction and the capital-versus-energy trade-off that sets $\Delta T_{\min}$ economically — are a further phase and are not implemented. A guide that described them here would be describing something the engine does not have.
- It takes the **unit's net duty**. Equipment with two duties — a column with a reboiler and a condenser — enters as its net unless you model the two as separate units. For a column that is usually the wrong picture, and splitting is the fix.

# 33 Shortcut distillation: Fenske–Underwood–Gilliland

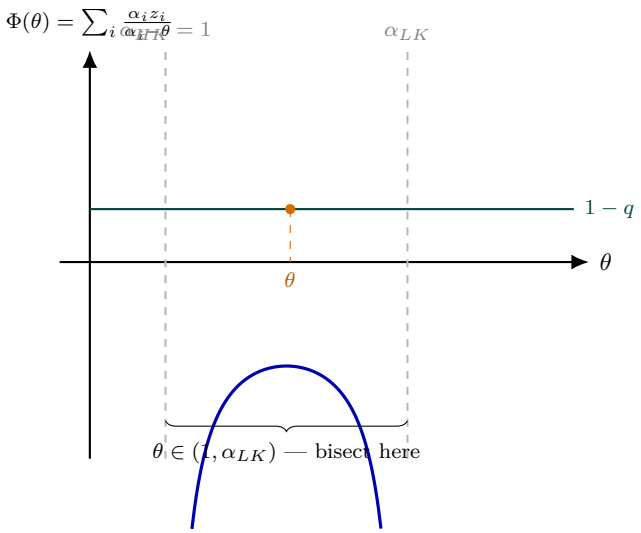



**Figure 77:** Locating the Underwood root. The function $\Phi(\theta) = \sum_i \alpha_i z_i/(\alpha_i - \theta)$ has a vertical asymptote at *every* component volatility $\alpha_i$. The physically meaningful root — the one that fixes the minimum reflux — is the value of $\theta$ that satisfies $\Phi(\theta) = 1 - q$ on the single branch trapped *between* the two key volatilities, $\theta \in (1, \alpha_{LK})$. On that interval $\Phi$ runs monotonically from $-\infty$ to $+\infty$, so it crosses the $1 - q$ level exactly once and a simple bisection brackets it with no risk of jumping to a neighbouring branch. Feeding that $\theta$ into $R_{\min} + 1 = \sum_i \alpha_i x_{D,i}/(\alpha_i - \theta)$ then returns the minimum reflux that Gilliland needs.

> *What you'll learn.* Before the rigorous Wang–Henke or MESH column (which need a built column), the FUG shortcut sizes a column from the two key components, their recoveries and the reflux — $N_{\min}$, $R_{\min}$, the actual $N$ and the feed stage all in closed form, at constant relative volatility. It is the textbook preliminary design, and the companion to the rigorous methods: FUG designs, Wang–Henke verifies.

Three classical correlations, in sequence:

- **Fenske**[114] — minimum stages at total reflux and the non-key split: $N_{\min} = \ln\big[(x_{LK}/x_{HK})_D\,(x_{HK}/x_{LK})_B\big] / \ln \alpha_{LK}$.
- **Underwood**[115] — minimum reflux from the root $\theta \in (1, \alpha_{LK})$ of $\sum_i \alpha_i z_i/(\alpha_i - \theta) = 1 - q$ (bisection), then $R_{\min} + 1 = \sum_i \alpha_i x_{D,i}/(\alpha_i - \theta)$.
- **Gilliland**[116] (Molokanov form) — the actual stage count $N$ from $N_{\min}$, the operating $R$ and $R_{\min}$; **Kirkbride** then locates the feed stage.

The relative volatilities are taken at the feed bubble point (Chapter 39 below). Tutorial `shortcut01_benzene_toluene` (50/50, LK/HK 99 %/1 %, $R = 1.3R_{\min}$): $N_{\min} = 10.07$, $R_{\min} = 1.29$, $N = 21.6$, feed stage 10.8. The method assumes constant $\alpha$ — it does *not* handle azeotropes (use the MESH column for those).

## 33.1 Reading $N(R)$: two limits and a fit between them

The three correlations are usually met as three formulae in a row. They are better read as *one curve*, because that is what a designer actually uses them for: how many stages does this separation need, as a function of how hard I am willing to reflux?

Both ends of that curve are limits with derivations behind them, and they are limits in *opposite* directions.

- **Total reflux** ($R \to \infty$, $D \to 0$): no product is withdrawn, so the operating lines collapse onto the 45° line and each stage buys a full factor of $\alpha$. Stack them and the separation factor is $\alpha^N$; asking for the separation the keys' recoveries define and solving for $N$ gives Fenske. It is the *fewest* stages the separation can ever be done in, so it is the **horizontal asymptote** $N \to N_{\min}$.
- **Minimum reflux** ($N \to \infty$): reduce the reflux and somewhere in the column a pinch appears — a zone where the operating and equilibrium lines touch and an extra stage changes nothing. The reflux at which the pinch first appears is Underwood's $R_{\min}$, and no column, however tall, separates below it. It is the **vertical asymptote** $R \to R_{\min}$.

Between them the curve is a hyperbola-like descent, and **Gilliland** is what fills it in. The correlation is written in

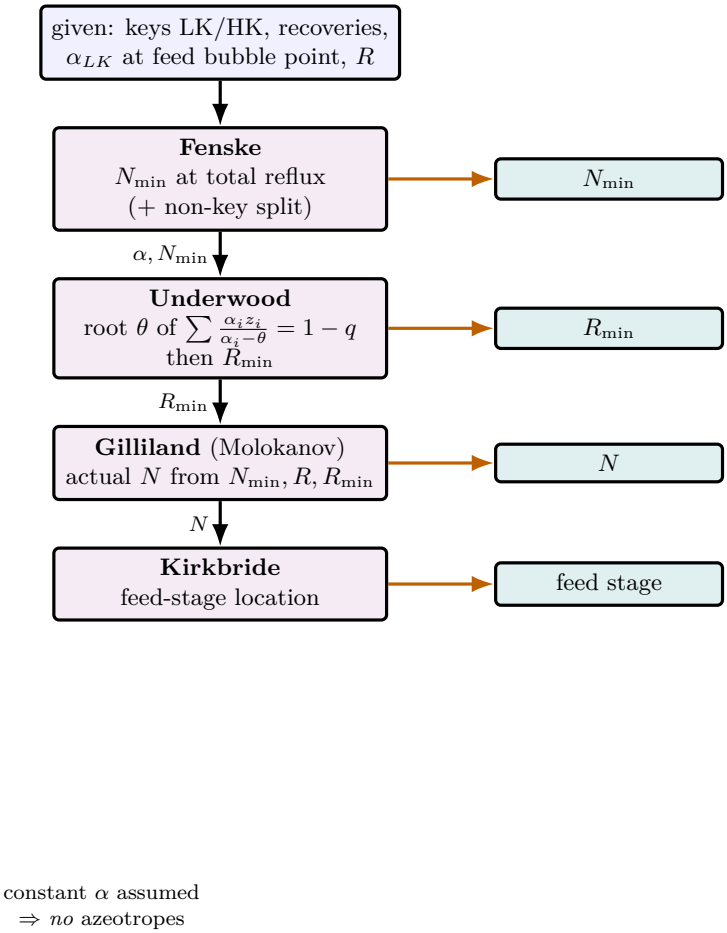



**Figure 78:** The Fenske–Underwood–Gilliland–Kirkbride shortcut as a one-pass pipeline — the textbook preliminary design, run before any rigorous column exists. **Fenske** gives the minimum stages $N_{\min}$ at total reflux (and the non-key split) from the two key recoveries and the relative volatility. **Underwood** finds the minimum reflux $R_{\min}$ via the root $\theta$ of its feed-quality equation. **Gilliland** (Molokanov form) then turns $N_{\min}$, the operating $R$ and $R_{\min}$ into the actual stage count $N$, and **Kirkbride** locates the feed stage. Each box consumes the previous result; every box also emits a headline number (orange). The whole chain rests on *constant* relative volatility, so it sizes ideal columns in closed form but cannot cross an azeotrope — there the rigorous MESH column takes over. "FUG designs, Wang–Henke verifies."

two normalised coordinates that put both limits at the origin and at $(1, 1)$:

$$X = \frac{R - R_{\min}}{R + 1}, \qquad Y = \frac{N - N_{\min}}{N + 1}, \tag{506}$$

so $X = 0$ (operating at minimum reflux) forces $Y \to 1$ and $N \to \infty$, while $X \to 1$ (total reflux) forces $Y = 0$ and $N = N_{\min}$. Choupo uses the **Molokanov** closed form of the correlation,

$$Y = 1 - \exp\left[\left(\frac{1 + 54.4\,X}{11 + 117.2\,X}\right)\frac{X - 1}{\sqrt{X}}\right], \tag{507}$$

and then inverts (506) for the stage count,

$$N = \frac{Y + N_{\min}}{1 - Y}. \tag{508}$$

Equations (506)–(508) are the whole of `ShortcutColumn`'s Gilliland step, in that order.

**Common pitfall. $N_{\min}$ and $R_{\min}$ are theorems; Gilliland is a fit.** Fenske and Underwood are derived from the stage equations under stated assumptions — they are what those assumptions *imply*. Equation (507) is not: it is an algebraic curve fitted through a scatter of plant and simulation points that Gilliland collected in 1940, tidied by Molokanov into a form a computer can evaluate. It carries that scatter with it. So the two *ends* of the $N(R)$ curve are as good as their assumptions, and everything between them is as good as a

correlation — which is a different kind of statement, and the reason the disagreement with a rigorous column is not uniform along the curve (§33.3).

## 33.2 What the shortcut returns — and what it does not

- **Constant relative volatility.** One $\alpha_{i,HK}$, for the whole column, evaluated at the feed bubble point. A real column runs 30 K colder at the top than at the bottom and its $\alpha$ varies over that span; none of that is in these numbers. The hard consequence is not inaccuracy but *expressiveness*: a constant $\alpha$ can never equal one, so **no azeotrope is representable at all**. A shortcut column applied to ethanol/water will return a stage count, and the stage count will be an answer to a system that does not exist.
- **Constant molar overflow.** Underwood's derivation assumes equal molar latent heats, so the internal $V$ and $L$ do not change from stage to stage. This is the same CMO assumption the Wang–Henke and `simultaneous` column models make (§34); `fullMESH` is where it is dropped.
- **A stage count, not a design.** FUG returns $N$, $N_{\min}$, $R_{\min}$, the feed stage and the product splits. It returns no diameter, no tray spacing, no downcomer, no flooding margin, no real trays at a Murphree efficiency — and no capital cost. Sizing is a separate pass over a column that exists.
- **An idealisation of the trays themselves.** $N$ counts *equilibrium* stages: every stage leaves its two streams in exact equilibrium. Real trays do not, which is what the Murphree efficiency exists to carry (§36).

Because all four are true and none is visible in a stream table, Choupo does not leave declaring them to the author's memory. A `shortcutColumn` emits a **declared approximation** on every run — a `problemDivergence` record naming the unit, what was requested ("distillation separation"), what was solved ("Fenske–Underwood–Gilliland shortcut; constant relative volatility at the feed bubble point; no azeotrope representable") and why ("the case declares `type shortcutColumn;`"). It is not a warning and nothing is refused: FUG is what the case asked for. It is a record that the question answered was the shortcut's question, printed above the caveats, emitted above the KPIs, and written to `converged/problemDivergence`.

## 33.3 Shortcut against rigorous: the benzene/toluene numbers

The claim "FUG designs, the rigorous column verifies" is only worth making with numbers attached, and the numbers are only worth reading if the two columns were asked the *same* question. Making them comparable takes four ties, and each one is a place where a careless comparison goes wrong.

1. **Same feed and same thermodynamics.** `shortcut01_benzene_toluene` and `column02_simultaneous` both take 100 kmol/h of a 50/50 benzene/toluene mixture at 1 atm as a saturated liquid, and both declare the same $\gamma$–$\varphi$ system (ideal liquid, ideal gas). The $K$-values are therefore the same function of $T$ and $P$ in both runs.
2. **Same specification.** A shortcut column is specified by the key *recoveries* and answers with a distillate rate; a rigorous column is specified by the distillate *rate* and answers with purities. Tying $r_{HK} = 1 - r_{LK}$ closes the gap: on a 50/50 binary feed that fixes $D = F(z_{LK}r_{LK} + z_{HK}r_{HK}) = F/2 = 50$ kmol/h — the rate the rigorous case already declares — and then $x_{D,LK} = Fz_{LK}r_{LK}/D = r_{LK}$ identically. The shortcut's recovery target and the rigorous column's distillate purity become the same number.
3. **Same reflux.** The rigorous column is run at the $R$ the shortcut returned, not at its own authored value. $R$ is an input to Gilliland and an input to MESH; leaving them different would compare two columns at two refluxes and blame the difference on the method.
4. **Same feed location.** Feed placement changes a column's purity, so the rigorous column's feed is placed by the shortcut's own Kirkbride answer, carried as a fraction of the column (feed_stage/$N$) and rounded to a tray.

One more thing has to hold before stage counts can be compared at all: the two must *count the same stages*. They do, and this was established by measurement rather than assumed. Run `column02_simultaneous` at near-total reflux ($R = 1000$) and it first reaches the 99 %/99 % products at `nStages` = 11, while Fenske's $N_{\min}$ for the same separation is 10.07 — so $\lceil N_{\min} \rceil$ and `nStages` are the same count, with no offset. That is also the cleanest single check in this section: at total reflux Gilliland is not involved at all ($X \to 1$, $Y \to 0$), so the agreement is Fenske's alone.

The comparison itself: hold the 99 % / 1 % key recoveries fixed, walk the reflux, and for each reflux ask the rigorous column how many stages it needs to reach the *same* products. Every number below is a run of one of the two cases.

| $R/R_{\min}$ | $R$ | FUG $N$ | $\lceil N \rceil$ | rigorous `nStages` |
|---|---|---|---|---|
| 1.05 | 1.359 | 29.99 | 30 | 29 |
| 1.15 | 1.488 | 25.14 | 26 | 24 |
| 1.30 | 1.682 | 21.59 | 22 | 20 |
| 1.60 | 2.071 | 17.98 | 18 | 18 |
| 2.00 | 2.588 | 15.71 | 16 | 16 |
| 3.00 | 3.883 | 13.40 | 14 | 14 |
| 5.00 | 6.471 | 11.92 | 12 | 12 |
| 10.0 | 12.94 | 10.95 | 11 | 12 |
| total reflux ($R = 1000$) | | $N_{\min} = 10.07$ | 11 | 11 |

Read the last column against the one before it. At the case's own operating point, $R = 1.3R_{\min}$, the shortcut asks for 22 trays and the rigorous column reaches the same products with 20: the shortcut is *conservative* by two stages, which is the direction a preliminary design wants to err in. Near total reflux the sign flips — the shortcut says 11 and the rigorous column needs 12 — and in the middle of the range the two agree exactly.

The disagreement is therefore not a constant, and this guide does not attribute it to one cause, because the two candidates have not been separated on this system: Gilliland's fit carries scatter, and the frozen $\alpha$ is wrong by a different amount at every reflux (the column's temperature span changes with the internal flows). What *is* established is the bracket: over a factor of ten in reflux the shortcut is never wrong by more than two stages on this separation, and it is exact at the one end where Gilliland does not act. That is what a shortcut is for — the first number, close enough to start the rigorous column from, and never the number you build.

> **Runnable case.** `tutorials/steady/distillation/shortcut01_benzene_toluene` — the FUG column.
> `tutorials/steady/distillation/column02_simultaneous` — the same separation, rigorous.
> EduTools → *Distillation shortcut (Fenske–Underwood–Gilliland)* — both of the above, run in the browser: the $N(R)$ curve with its two asymptotes, and the rigorous stage ladder beside it.

# 34 Rigorous distillation by simultaneous MESH Newton

*What you'll learn.* The *ladder* of column models — bubble-point Wang–Henke, the constant-molal-overflow simultaneous Newton, and the full energy-balance Naphtali–Sandholm MESH — which one to reach for, how the cheap model warm-starts the rigorous one, and how multiple feeds, side draws and a stage reaction are layered on top.

## 34.1 The model ladder (*all models are wrong, some useful*)

One `distillationColumn` unit carries three solver models, selected by the `model` slot, in increasing order of rigour and cost:

| `model` | method | best used when |
|---|---|---|
| `WangHenke` *(default)* | sequential bubble-point, CMO | ideal / near-ideal, ONE feed, quick screenir |
| `simultaneous` | simultaneous Newton, CMO | azeotropic / non-ideal; multi-feed, side drav |
| `fullMESH` *(= `MESH` = `NaphtaliSandholm`)* | full energy-balance MESH | wide-boiling / large $\Delta H_{\text{vap}}$ spread / when |

A naming note: `simultaneous` is the *bubble-point-reduced* solve (constant molal overflow, NO per-stage energy balance); the genuine MESH (Mass, Equilibrium, Summation, *Heat*) is `fullMESH` — aliased `MESH` and `NaphtaliSandholm` so the name never means less than it says.

## 34.2 Wang–Henke vs the simultaneous CMO Newton

The Wang–Henke bubble-point method of Chapter 15 is robust for ideal systems but converges slowly ($\mathcal{O}(100)$ outer iterations) and can step *through* an azeotrope non-physically. Selecting `model simultaneous` solves all the stage equations at once by Newton (`solver::newtonND`): the variables are the per-stage $(x_{j,0..n-2}, T_j)$, the residuals the $n-1$ component balances plus the per-stage bubble-point $\sum_i K_i x_i - 1 = 0$ that fixes $T_j$, with $K_j(T_j)$ live. It converges *quadratically* (5–6 iterations vs ∼108) and, crucially, is stable on azeotropic systems: `column03_azeotrope_mesh` (ethanol/water, NRTL) finds a physical distillate *below* the azeotrope on the very case the bubble-point method cannot solve — and refuses to invent a non-physical "past the azeotrope" answer, because none exists.

Both Wang–Henke and `simultaneous` assume **constant molal overflow** (CMO): the internal flows $L_j$, $V_j$ are fixed by a mass march down the column, and only $x_j, T_j$ are solved. CMO is exact only when all components have equal molar latent heats and sensible/mixing effects are negligible.

**The Jacobian has a shape, and the solver is told so.** Stage $j$'s residuals see only stages $j-1$, $j$ and $j+1$, so the Newton Jacobian of the simultaneous CMO model is *block-tridiagonal* with $N$ diagonal blocks of size $n_v = n$ (the $n-1$ compositions and $T_j$). The column declares that structure to the solver, and the declaration buys two classical pieces: a *colored* finite-difference Jacobian (Curtis, Powell & Reid, *J. Inst. Maths Applics* 13 (1974) 117[117] — one variable slot of every third stage is perturbed at once, so the cost is about $3\,n_v$ residual evaluations per Jacobian, *independent of the stage count*, against $Nn_v$ dense) and a block-Thomas solve on the three bands, $\mathcal{O}(Nn_v^3)$ against $\mathcal{O}((Nn_v)^3)$. A declared structure is a solver aid, and aids report aloud: on the first iteration the solver also builds the dense Jacobian, measures the largest entry outside the bands, *refuses by name* when the declaration is materially false, and prints the measured maximum with the evaluation count (`[solver] block-tridiagonal structure VERIFIED ...`). On the 48-stage reactive sour-water stripper, where every residual evaluation is 48 reactive flashes, the same 10 iterations to the same residual took 28.9 s structured against 62.7 s dense, with 18 residual evaluations per Jacobian instead of 288. The full MESH below keeps its dense solve: its structure is *bordered* tridiagonal, and declaring it clean would be exactly the lie the audit exists to catch.

## 34.3 The full MESH (Naphtali–Sandholm): the energy balance

`model fullMESH` drops the CMO assumption. It promotes $V_j$ and $L_j$ to *unknowns* (so the per-stage variable count rises from $n$ to $n+2$: $x_{j,0..n-2}, T_j, V_j, L_j$) and adds, per interior stage, the **total-mass balance** and the **energy balance** — the enthalpy residual the CMO never carried:

$$L_{j-1}\,h_L(x_{j-1},T_{j-1}) + V_{j+1}\,h_V(y_{j+1},T_{j+1}) + F_j\,h_{F,j} \;-\; L_j\,h_L(x_j,T_j) \;-\; V_j\,h_V(y_j,T_j) \;=\; 0. \tag{509}$$

The internal flows now follow the *real* latent heats instead of being assumed equimolal. What stage 1 is follows from the declared condenser type (`condenser { type total|partial; }`, default `total`): under a **total** condenser stage 1 is the top tray and carries the same energy balance, the reflux $R\,D$ entering it as the condensate of its own vapour at the bubble point of $y_1$ and $V_1 = (R+1)\,D$ leaving it to the condenser, which is not an equilibrium stage and is not counted; under a **partial** condenser stage 1 *is* the condenser, an equilibrium stage whose rows fix $V_1 = 0$ and $L_1 = R\,D$ in place of an energy residual. The reboiler fixes $L_N = B$. The condenser and reboiler duties are

*results*: $Q_{\text{cond}}$ from the condenser's own envelope, $Q_{\text{reb}}$ from the reboiler's. Validated stage-by-stage against the original Naphtali & Sandholm (1971) Problem 2 in `column07_naphtaliSandholm` ($T$ profile within $\leq 1.8$ K over 20 stages); `column06_fullMESH` shows it agreeing with CMO on benzene/toluene (close latent heats), the small gap being exactly the energy-balance correction CMO drops.

## 34.4 The warm-started ladder (switching model mid-solve)

A rigorous Newton from a cold start can fail (the linearization is only valid near the solution). Choupo therefore *switches model mid-solve*: the `fullMESH` run first converges the cheap CMO problem, then hands the CMO's converged stage profile $(x_j, T_j, V_j, L_j)$ to the full MESH as its *initial guess*. The intermediate profile is *kept*, not discarded — the rigorous model then converges in $\sim$3–4 iterations instead of from scratch. This homotopy/continuation pattern recurs throughout the simulator: non-reactive $\rightarrow$ reactive (the reaction is switched on, seeded from the non-reactive profile), and the catalyst-mass ramp in kinetic mode. It is the "cheap model bootstraps the rigorous one" principle, and it is why selecting a more rigorous model is cheap once a coarser one has run.

## 34.5 Multiple feeds and side draws

The `simultaneous` and `fullMESH` models generalise the single fixed feed into a per-stage *flow staircase*. Any stage $j$ may receive a feed (a flowsheet stream mapped by `operation.feeds ({stream; stage;}...)`) and bleed a liquid ($U_j$) or vapour ($W_j$) *side draw* (`operation.sideDraws`). Each stage balance then carries an $F_j z_{F,j}$ source and $(L_j{+}U_j), (V_j{+}W_j)$ outflows; the general form reduces *exactly* to the single-feed equations when there is one feed and no draw, so every legacy tutorial is unchanged. Feeds are FLOWSHEET STREAMS so the plant-boundary balance sees them (*information follows the streams*); a side draw is inherently less pure than an end product (it picks up the components boiling just above and below it). `column08_radfrac_multidraw` splits a C3–C6 mix four ways with two feeds and two side draws.

## 34.6 Reactive distillation

A `reaction{}` block makes named catalytic stages reactive. For a mole-conserving ($\sum_i \nu_i = 0$) reaction the CMO flows are undisturbed; the generation $\nu_i\,\xi_j$ enters each reactive stage's component balance with the molar extent $\xi_j$ fixed either by chemical `equilibrium` (an activity-based $K_a(T)$ via van't Hoff, residual $\sum_i \nu_i \ln(\gamma_i x_i) = \ln K_a$ — one extra unknown $\xi_j$ and one extra equation per stage) or by `kinetics` (a rate law $\times$ catalyst mass; $\xi_j$ then a function of $x_j, T_j$, no extra unknown). Converged by the same homotopy: the column is solved non-reactive first, then the reaction is switched on from that profile. `column05_reactive_methylacetate` (methyl acetate, validated vs Pöpken 2001) reaches the equilibrium ceiling this way.

## 34.7 A column over a chemistry: the effective stage $K$

Every column so far has stood on a *molecular* thermodynamic package: the stage asks for $K_i(T_j, P, \mathbf{x}_j)$ and the package answers with $\gamma_i P_i^{\text{sat}}/P$ or a fugacity ratio. Nothing in that exchange knows about chemistry, and for a hydrocarbon splitter nothing needs to.

A sour-water stripper breaks it. Ammonia's volatility is not a property of ammonia: it is set by how much of the ammonia in the liquid is free $NH_3$ and how much is $NH_4^+$, and *that* is set by how much $CO_2$ the liquid still carries — the carbon dioxide is what holds the ammonia as ammonium, so stripping it sets the ammonia free and the free ammonia then strips. *The separation and the speciation are the same problem*, and no molecular $K$-value can express it.

(The familiar shorthand for this is "stripping $CO_2$ raises the pH". It is a true statement about *one* solution whose composition changes, and it is *not* how the finished column reads tray by tray — which is worth a page of its own once we have the profile in hand, below.)

> **Key idea.** The stage does not stop being a stage. What changes is the *answer to one question* — what is the equilibrium ratio here? — and so that is the only place the engine changes. The MESH equations, the tridiagonal solve, the Jacobian, the flows: all untouched.

**The seam** Every stage evaluation, in every one of the three column models, goes through one call:

$$\mathbf{K}_j \;=\; \texttt{thermo.stageK}(T_j,\, P,\, \mathbf{z}_j,\, \mathbf{x}_j,\, \mathbf{y}_j).$$

For a molecular package that call forwards, unchanged, to the ordinary $K$-value routine — the twelve molecular column tutorials are bit-identical to what they were before it existed. For a package that resolves chemistry, it solves the stage's *own* equilibrium at its own $(T_j, P, \mathbf{z}_j)$ — speciation, phase transfer and electroneutrality together — and reports an **effective apparent** $K$: the ratio the apparent components would need in order to produce the split the chemistry actually gives.

The apparent components stay the state, exactly as they do in a reactive flash. The ions are internal to the stage; the Jacobian never sees one.

> **Common pitfall.** This is *not* a rigorous species-basis MESH. The chemistry is re-solved inside each residual evaluation — the same nested posture the two-liquid flash uses — and the Jacobian sees apparent components only. It also costs one reactive flash per stage per residual, and the Jacobian is numerical, so a pass costs of order (stages × unknowns) flashes. A case that runs this must say so; the tutorial does, in its header.

**A $K$-value is an incipient quantity** Here is the mistake worth making once, because it is invisible in a flash and fatal in a column, and because the first implementation made it.

The obvious definition is $K_i = y_i/x_i$, read off the converged flash. That is correct *whenever the stage is two-phase*. A column solver, however, spends most of its life asking about states that are not: on the way to the answer the Newton proposes stage liquids well below their bubble point. For those, a flash reports no vapour at all, so $y_i = 0$ and every $K_i$ comes back zero.

Follow that through. The bubble-point residual $\sum_i K_i x_i - 1$ then reads $-1$ at every stage; worse, it reads $-1$ for a *range* of $T$, so $\partial r/\partial T = 0$ and the Jacobian is singular. The column does not diverge — it does not move at all. The measured symptom is a line in the log that says `Newton iters:  0` above a profile that is still the initial guess.

The repair is to say what a $K$-value actually is. It is an **incipient** quantity: defined over a liquid whether or not a vapour phase happens to be present, because it describes the vapour that *would* form. The reactive solver already computes exactly that when it tests for saturation — the equilibrium partial pressure of each volatile over the fully speciated liquid — so

$$K_i \;=\; \frac{p_i^{\mathrm{eq}}(T, \mathbf{x})/P}{x_i}, \tag{510}$$

which is the ordinary $\gamma$–$\phi$ $K$-value, $\gamma_i P_i^{\mathrm{sat}}/P$, generalised to a liquid whose *speciation* sets the volatility. The two definitions agree where they meet: at a two-phase state the partial pressures sum to $P$ by construction, so $p_i/P$ is $y_i$ and Eq. (510) is $y_i/x_i$ again.

> *Intuition.* The molecular case hides this because $\gamma_i P_i^{\mathrm{sat}}/P$ is written down without ever mentioning a vapour phase — nobody is tempted to define it as a ratio of two things a flash returned. Under chemistry the temptation exists, because the chemistry solver returns a whole phase split, and taking $y/x$ from it feels like reading the answer rather than inventing one.

**Trial states that are not states** A second thing goes wrong, and its repair is a matter of honesty rather than of physics.

A Newton on a column takes finite differences, and it overshoots. It overshoots the *minor* components first: on the sour-water witness the solver proposed a stage liquid whose $CO_2$ mole fraction was $-8.5 \times 10^{-4}$, against a feed value of $8 \times 10^{-3}$. There is no liquid whose carbon-dioxide content is minus a thousandth. Asking a thermodynamic model to price one is not a hard question — it is not a question.

So the trial composition is projected onto the physical simplex before the chemistry is solved. Every MESH does this; what Choupo adds is that it **says so**, once, naming what it did and why (see §1: a solver aid is allowed, disguising one is not). The rule is deliberately narrow: a *negative* entry is clamped, because negative means the solver overshot a component that is genuinely present; an *exactly zero* entry stays zero, because zero means absent, and absent means absent.

The projection cannot reach the answer. At convergence every mole fraction is positive — if the clamp were still binding there, the residual could not have reached zero.

**Reading the tray's own chemistry** A profile that prints $x$, $y$ and $T$ shows the *consequence* and hides the *cause*. When the package resolves a chemistry, Choupo therefore speciates each tray liquid once after convergence — one flash per stage, not one per residual — and writes the pH, the ionic strength and every species molality beside the compositions.

For the four-stage sour-water column:

| stage | $T$ [K] | pH | $I$ [mol/kg] | free-$NH_3$ | $CO_3$ loading |
|---|---|---|---|---|---|
| 1 (top) | 354.63 | 8.485 | 0.636 | 0.7767 | 0.2120 |
| 2 | 367.34 | 8.361 | 0.130 | 0.8319 | 0.1627 |
| 3 | 368.14 | 8.328 | 0.110 | 0.8283 | 0.1670 |
| 4 (reboiler) | 371.48 | 8.532 | 0.029 | 0.9091 | 0.0872 |

where the free-ammonia fraction is $m_{\mathrm{NH_3(aq)}}/(m_{\mathrm{NH_3(aq)}} + m_{\mathrm{NH_4^+}})$ and the loading is total carbonate over total ammonia, both read off the species inventory.

**And the shorthand that does not survive contact with the table** Every account of this process, including the one at the head of this section, says: *strip the $CO_2$ and the pH rises, the rising pH frees the ammonia, and the ammonia strips.* Look at the pH column. It runs 8.485 → 8.361 → 8.328 → 8.532 — monotone in neither direction.

The shorthand is not wrong; it is a statement about *one solution whose composition changes*, and it is being read as a statement about *four different solutions compared side by side.* Those are not the same claim, and the difference is the ionic strength column: it varies 22-fold from the top tray to the reboiler.

**Key idea. A pH is an activity; a loading is an inventory.** $\mathrm{pH} = -\log_{10}(\gamma_{\mathrm{H^+}} m_{\mathrm{H^+}})$ moves with the activity coefficients as well as with the amounts, so across trays at wildly different ionic strengths it is not a clean ordering variable. The carbonate loading is a ratio of inventories and carries no $\gamma$ at all.

Order the trays by loading and the free-ammonia fraction orders *strictly* the other way, across all four. That is the mechanism stated in a form that survives: the carbon dioxide a liquid still carries is what holds its ammonia as ammonium. It is also the form Choupo gates, because a check on the monotone pH would have been a check on a coincidence — true for this feed, false for the next one.

**Common pitfall.** The same care applies one row down. $H^+$ and $OH^-$ do *not* fall monotonically down the column even though every solute does: they are not solutes being stripped but water's own dissociation products, pinned by the pH. $m_{\mathrm{OH^-}}$ rises between trays 1 and 2. An earlier version of the regression check asserted that they fall, and failed — which is a better outcome than a check that passes for the wrong reason.

**How you know the $K$ is right** The effective apparent $K$ is a *definition we made.* A wrong normalisation, a confusion between the apparent and species bases, or the speciation counted twice would each produce a column that converges to a profile which looks entirely plausible. Nothing about the column itself would give it away.

The check is an identity, and it is one a student can state in a sentence: *an equilibrium stage is a flash of its own two products.* Draw both phases off one stage, put them back together, and let the mixture separate again at the same pressure. Nothing may happen — not to the temperature, not to either composition, not to the phase ratio. Two independent pieces of code, one answer.

**Runnable case.** `tutorials/steady/distillation/column12_stage_is_a_flash` does that on benzene/toluene, where nobody doubts the answer — the control. `tutorials/steady/distillation/column13_sour_water_stage_identity` does it under the chemistry: four stages of $NH_3$ + $CO_2$ in water, converging in seven Newton iterations to $\|F\| \approx 8 \times 10^{-10}$, with ammonia climbing from 0.005 in the reboiler liquid to 0.30 in the overhead vapour. The recombined stage comes back unchanged to about $10^{-9}$ relative — *and so does its ion inventory*, which is the half of the check that a comparison of apparent compositions alone would miss.

# 35 The enthalpy-concentration diagram: putting energy inside the geometry

*What you'll learn.* McCabe–Thiele teaches a staircase between an equilibrium curve and an operating line, and it works because somebody quietly assumed the energy balance away. Here we put the energy balance back — not as an extra equation to solve, but as a second coordinate on the diagram. You will learn what the enthalpy-concentration ($h$–$x$–$y$) plane is, why the total, component and energy balances collapse into a single statement that three points are *collinear*, what the difference point $\Delta$ is and why the reflux ratio is a ratio of two lengths on the page, and exactly what constant molar overflow assumes — stated as a property of the two saturation curves, so you can look at a real pair of them and see how far from true it is.

## 35.1 Two curves and the ties between them

Fix a pressure. A saturated liquid of composition $x$ boils at its bubble temperature $T_{\mathrm{bub}}(x)$ and has, at that temperature, a molar enthalpy $h(x)$. A saturated vapour of composition $y$ condenses at its dew temperature $T_{\mathrm{dew}}(y)$ with molar enthalpy $H(y)$. Plot both against composition on one pair of axes — composition across, enthalpy up — and you have the **enthalpy-concentration diagram**: a lower curve $h(x)$, an upper curve $H(y)$, and between them the **tie lines**, each joining a liquid to the vapour that is in equilibrium *with it*.

Three things about that sentence are worth slowing down for.

First, temperature does not appear on the diagram, yet it is everywhere in it: each point sits at its own saturation temperature, and the two ends of a tie line share one. The diagram is drawn at constant $P$, and $T$ is the readout, not an axis.

Second, a tie line is *not* vertical. The vapour in equilibrium with a liquid has a different composition — that is what separation is — so a tie line leans, and it leans by exactly the amount the equilibrium says. The slope of the tie lines *is* the equilibrium data.

Third, both enthalpies must be measured from the *same* zero. This sounds like bookkeeping and is not: everything below is built from *differences* of $h$ and $H$, so a constant offset cancels in every answer — which means a diagram drawn on two inconsistent datums produces numbers that are wrong and a picture that looks exactly as convincing as a correct one. Choupo carries one enthalpy base, the elements at 298.15 K, and Section 35.4 says what that means for the file the engine writes.

## 35.2 Why three balances become one straight line

Take any envelope cut across a column — say everything above the cut between stage $n$ and stage $n+1$, with the condenser inside it. Vapour $V_{n+1}$ enters the envelope from below, liquid $L_n$ leaves it downward, distillate $D$ leaves at the top, and the condenser removes $Q_C$. Three balances:

$$V_{n+1} = L_n + D, \tag{511}$$

$$V_{n+1}\, y_{n+1} = L_n\, x_n + D\, x_D, \tag{512}$$

$$V_{n+1}\, H_{n+1} = L_n\, h_n + D\, h_D + Q_C. \tag{513}$$

Now define the **difference point**

$$\Delta_D \;\equiv\; \left( x_D,\; h_D + \frac{Q_C}{D} \right), \tag{514}$$

a point on the same plane whose composition is the distillate's and whose "enthalpy" is the distillate's plus the condenser duty spread over it. It is not a stream. It is the net flow leaving the top of the envelope, energy included, per mole — and with it the three balances above read as one statement:

$$V_{n+1} \begin{pmatrix} 1 \\ y_{n+1} \\ H_{n+1} \end{pmatrix} \;=\; L_n \begin{pmatrix} 1 \\ x_n \\ h_n \end{pmatrix} \;+\; D \begin{pmatrix} 1 \\ x_D \\ \Delta_D \end{pmatrix}. \tag{515}$$

Divide by $V_{n+1}$ and use (511): the two coefficients $L_n/V_{n+1}$ and $D/V_{n+1}$ are non-negative and sum to one. So the point $(y_{n+1}, H_{n+1})$ is a *convex combination* of $(x_n, h_n)$ and $\Delta_D$ — it lies on the segment joining them. The three points are collinear, and the two weights are their barycentric coordinates:

$$\frac{L_n}{V_{n+1}} = \frac{\overline{V\Delta}}{\overline{L\Delta}}, \qquad \frac{D}{V_{n+1}} = \frac{\overline{LV}}{\overline{L\Delta}}, \qquad R \;=\; \frac{L_n}{D} \;=\; \frac{\overline{V\Delta}}{\overline{LV}}. \tag{516}$$

**Key idea.** The lever rule here is not an analogy with a see-saw. A stream of $F$ moles at composition $x$ and enthalpy $h$ contributes the triple $F\,(1, x, h)$, and *all three* balances are linear in that same triple. A linear relation among such triples whose first components cancel is exactly the statement that the three $(x, h)$ points are collinear with the flows as barycentric weights. The energy balance is not solved on this diagram; it has been absorbed into where the points are allowed to sit.

The construction then alternates two moves. From a liquid $x_n$, the *tie line* gives the vapour $y_n$ leaving the same stage — that is equilibrium, and it comes from the diagram's own tie data. From that vapour, the *line through* $\Delta_D$ gives the liquid $x_{n+1}$ passing it — that is the three balances, and it comes from (516). Alternate, and you have counted stages. It is McCabe–Thiele's staircase with the energy balance carried along in the geometry instead of assumed away, and the flows $L_n$, $V_{n+1}$ come out *different on every stage* because (516) says so.

## 35.3 What constant molar overflow actually assumes

McCabe–Thiele's operating line is straight in $x$–$y$ because $L$ and $V$ are taken constant within a section. Stated on the enthalpy-concentration plane, the assumption is a shape claim about the two saturation curves:

$$h(x) \;=\; a + b\,x, \qquad H(y) \;=\; a + \lambda + b\,y, \tag{517}$$

two *straight* lines, *parallel*, a constant vertical distance $\lambda$ apart. Substitute (517) into (513) and use the other two balances: everything cancels except $\Delta_D = a + b\,x_D + \lambda V/D$. The energy balance stops constraining $L/V$ — it is implied by the first two — and the enthalpy coordinate becomes redundant. That is the whole of CMO, and the whole of why McCabe–Thiele can leave energy out.

Reading (517) backwards gives the physical conditions, and each one is a place where a real mixture departs:

- **Equal molar latent heats.** The vertical gap $\lambda$ between the curves is a heat of vaporisation; requiring it constant requires the two pure components to have the same one, per mole. This is why CMO is often defended on a molar basis for chemically similar pairs and quietly fails for, say, ammonia–water.
- **Negligible sensible heat.** The curves are straight only if $h$ is linear in $x$ — but each point sits at its *own* bubble temperature, and that temperature sweeps across the diagram between the two boiling points. The sensible heat of that sweep bends the lower curve.
- **No heat of mixing.** $h^E(x)$ adds curvature that no straight line can carry. It is the term most often left out of a model, and it is exactly the term this diagram would show.
- **Adiabatic, isobaric.** Heat leaking through the shell, or a tray-to-tray pressure drop, moves the saturation curves themselves from stage to stage — the diagram is drawn at one $P$.

**Common pitfall.** CMO is not a statement about the column, and it is not a small-error approximation you can bound by looking at the answer. It is a statement about the *thermophysical data*, and a McCabe–Thiele result carries no mark saying how false it was. The enthalpy-concentration diagram is the cheapest honest test: draw the two saturation curves and look at whether they are straight and parallel. If they are, use McCabe–Thiele and save yourself the work. If they are not, you now know by how much.

## 35.4 What Choupo publishes, and what it does not

The construction above needs data that no ordinary flash result contains: saturated-phase enthalpy *against composition*, on both curves, with the equilibrium correspondence between them. The `enthalpyConcentration` property operation publishes exactly that, for a binary at one declared pressure, and three of its choices matter to anyone drawing on it.

**One datum.** Both enthalpy columns come from the same two kernels every energy balance and duty in the engine is priced on — `H_liquid_formation` for the liquid and `H_stream_formation` at unit vapour fraction for the vapour — so the numbers are absolute on the elements datum at 298.15 K. A component that cannot be placed on that datum is refused *by name*, with the remedy, rather than substituted with a heat-capacity integral from a private zero. The refusal is the point: an inconsistent datum cancels out of nothing and shows up in nothing.

**Every row is one equilibrium.** A row of the table carries *both* ends of a tie line — $(x_1, h)$ and $(y_1, H)$ — and the one temperature they share. A drawing tool therefore never pairs rows by index and never interpolates one end of a tie, which are the two ways a diagram comes out smooth, closed and wrong. The two saturation curves are simply the two ends read off the same rows.

**One saturation kernel.** The dew locus of a binary is the image of its bubble locus under the equilibrium map, so a dew node is found by inverting that map rather than by solving a second residual to a second tolerance. Two independent solvers would leave the tie lines landing *near* the vapour curve instead of on it — and tie lines ending on that curve is the entire construction.

What the engine does *not* do is draw. There is no difference point, no lever arm and no staircase anywhere in the simulator: those belong to a tool reading the published locus, and keeping the split there is what stops the geometry and the thermodynamics from drifting apart.

> **Runnable case.** `tutorials/props/scan/hxy01_ethanol_water_1atm` — ethanol/water at 1 atm, NRTL from the curated DECHEMA pair, ideal vapour. The bubble temperature runs from 372.5 K at pure water to 351.5 K at pure ethanol, and the azeotrope shows as the one composition where the tie line becomes vertical. The number to look at is the tie-line gap $\lambda = H(y^*) - h(x)$: it runs from 40 680 J/mol to 45 053 J/mol across the composition range, an 11 % spread. Constant molar overflow is the assumption that those two numbers are the same one.

# 36 Tray hydraulics: can the column be built?

Everything up to here answers one question: *what separates?* The MESH equations know the stage temperatures, the compositions, the flows. They do not know the diameter of the column, and they never asked. A column of any diameter converges to the same distillate purity, because nothing in the stage equations depends on how wide the trays are.

That is a real hole. Vapour rising through a tray drags liquid with it, and above a certain velocity it carries the liquid to the tray above instead of letting it fall — *entrainment flooding*. Liquid crossing a tray must then descend through the downcomer against the tray's own pressure drop, and if the head it needs exceeds the space available it backs up to the tray above — *downcomer flooding*. Too little vapour and the liquid rains straight through the holes instead of bubbling — *weeping*. A flooded column separates nothing. It also converges.

So hydraulics is a **rating pass on the converged profile**, run after the stage equations, never inside them. That scope is a decision, not an oversight: a column whose pressure drop shifts its own stage temperatures needs the pressure profile carried back into the MESH, and Choupo does not pretend to do that. What it does do is refuse to let the student believe a stream table that no physical tray could produce.

## 36.1 First, the tray is not ideal: Murphree efficiency

Every column derived so far is built from *equilibrium stages*: the vapour leaving a tray is in equilibrium with the liquid leaving it. No real tray does that. The vapour bubbles through a froth of finite depth for a finite time and comes off somewhere between what entered it and what equilibrium would give. Murphree measured that shortfall on the vapour side:

$$y_j \;=\; E_{MV}\, y_j^{*} \;+\; (1 - E_{MV})\, y_{j+1}, \qquad y_j^{*} = K_j x_j, \tag{518}$$

so $E_{MV}$ is the *fraction of the way* the vapour travels from what rose into the tray, $y_{j+1}$, towards equilibrium with the liquid on it. $E_{MV} = 1$ is the ideal stage; $E_{MV} = 0$ is a tray that does nothing.

Two consequences, and the second is the one that matters for the rest of this section. First, purity falls: fourteen real trays do less than fourteen ideal ones, which is why a column designed on ideal stages and built on real trays comes up short. Second, and more important, **once $E_{MV} < 1$ the stage count is a count of REAL TRAYS** — physical objects with a diameter, a weir and holes in them — and it is those the hydraulics below sizes. Without an efficiency we would be dimensioning to the millimetre a tray we then assumed was perfect.

> **Key idea.** $E_{MV}$ is a *declared* number in Choupo, exactly like the orifice coefficient $C_o$ and the weep constant $K_2$ that follow. It comes from a correlation (O'Connell's, from the liquid viscosity and the relative volatility), a chart, or plant data. The engine will not invent it, and it will not silently assume 1 when you meant something else.

Numerically, Eq. (518) couples tray $j$ to the vapour rising from $j+1$, which is itself an unknown — and that would destroy the tridiagonal structure the bubble-point method depends on. Choupo folds the efficiency into an *effective* $K$,

$$K_{j,i}^{\mathrm{eff}} \;=\; E_{MV}\, K_{j,i} \;+\; (1 - E_{MV})\, \frac{y_{j+1,i}}{x_{j,i}}, \tag{519}$$

with $y_{j+1}$ taken from the *previous* outer pass. Every equation in the tridiagonal sweep is untouched: $y_j = K_j^{\mathrm{eff}} x_j$ as before. The lag vanishes at convergence, where $y^{\mathrm{old}} = y$, so the converged profile satisfies Eq. (518) exactly. The outer loop pays for the coupling, and it is visible in the iteration count. The simultaneous MESH method has no outer pass to lag against — there the relation would have to enter the residual and the Jacobian — so it *refuses* the keyword rather than quietly handing back ideal stages.

## 36.2 Entrainment flooding: Souders–Brown and Fair

Suspend a droplet in the rising vapour. It falls under its own buoyant weight and is dragged up by the gas; at the flooding velocity the two balance. Writing the drag with a constant coefficient and clearing the algebra gives the *Souders–Brown* form [118]: the flooding velocity scales as the square root of the density difference over the vapour density,

$$u_{\mathrm{flood}} \;=\; C_{\mathrm{SB}} \left(\frac{\sigma}{20}\right)^{0.2} \sqrt{\frac{\rho_L - \rho_V}{\rho_V}}, \tag{520}$$

referred to the *net area* (the tower cross-section less one downcomer, which is the area the vapour actually has). Surface tension enters because it is what holds a droplet together against the shear that would atomise it; $\sigma$ is in dyn/cm and the correlation is anchored at 20 dyn/cm.

The capacity parameter $C_{\mathrm{SB}}$ is not a constant. It falls as the tray carries more liquid relative to vapour, which is measured by the dimensionless *flow parameter*

$$F_{LV} \;=\; \frac{L}{V}\sqrt{\frac{\rho_V}{\rho_L}}, \qquad \text{(mass flows)} \tag{521}$$

and it rises with the tray spacing, because a droplet thrown off one tray has further to travel before it reaches the next. Fair [119] measured $C_{\mathrm{SB}}(F_{LV}, \mathrm{TS})$ and published it as a *chart*. A chart cannot be integrated into a solver, so Choupo uses the algebraic representation of Lygeros and Magoulas [120],

$$C_{\mathrm{SB}} \;=\; 0.0105 \;+\; 8.127\times10^{-4}\;\mathrm{TS}^{0.755}\;\exp\!\left(-1.463\,F_{LV}^{0.842}\right) \qquad [\mathrm{m/s}], \tag{522}$$

with the tray spacing TS in millimetres. This fit is *known* to misbehave below $F_{LV} \approx 0.03$: its slope turns the wrong way, where the real chart rises. Choupo says so out loud when a tray lands there rather than reporting a number it does not believe. This is the anti-black-box stance in its plainest form — a correlation is a fit to somebody's data, and when you leave the data behind, you are told.

> **Key idea.** Kister and Haas (1990) published a physically grounded, fully algebraic entrainment-flood correlation, built on the clear-liquid height at the froth-to-spray transition. It is the better model, and it is *not* implemented: two of its constants could not be verified against a primary source. A teaching tool does not ship numbers it has not read. What you get instead is Fair's chart, an honest fit to it, and a warning where the fit is known to fail.

## 36.3 Pressure drop, as heads of liquid

A tray's pressure drop is most naturally written not in pascals but as the height of clear liquid that pressure would support, in millimetres. Three contributions add [121]:

$$h_d = 51\left(\frac{u_h}{C_o}\right)^2\frac{\rho_V}{\rho_L} \qquad \text{dry tray: the orifice equation,} \tag{523}$$

$$h_{ow} = 750\left(\frac{L_w}{\rho_L\, l_w}\right)^{2/3} \qquad \text{crest over the weir (Francis),} \tag{524}$$

$$h_r = \frac{12.5\times10^3}{\rho_L} \qquad \text{residual,} \tag{525}$$

so that the total head over a tray carrying a weir of height $h_w$ is $h_t = h_d + h_w + h_{ow} + h_r$. Here $u_h$ is the vapour velocity through the holes, $L_w$ the liquid mass flow over the weir and $l_w$ the weir length. The residual term $h_r$ is Sinnott's deliberate simplification: the froth on a real tray is aerated, less dense than clear liquid, and rather than carry a charted aeration factor he absorbs the whole effect into a term that depends only on the liquid density. It is worth knowing this is a simplification and not a law.

The discharge coefficient $C_o$ is itself a chart, against the hole-area ratio and the plate-thickness-to-hole-diameter ratio [122]. Choupo will not invent it: it is a *declared* input, printed back at you with its provenance, and it is yours to own.

## 36.4 Downcomer backup

The liquid leaving a tray must fall through the downcomer and squeeze out under the apron at the bottom. That last constriction costs head,

$$h_{dc} \;=\; 166\left(\frac{L_{wd}}{\rho_L\, A_m}\right)^2, \tag{526}$$

$A_m$ being the smaller of the clearance area under the apron and the downcomer area itself. The liquid therefore stands in the downcomer at

$$h_b \;=\; (h_w + h_{ow}) \;+\; h_t \;+\; h_{dc}, \tag{527}$$

high enough to push itself through the gap *and* to overcome the pressure drop of the tray it is leaving. If $h_b$ reaches the tray above, the column floods — from below this time, for an entirely different reason. The froth in the downcomer is roughly half the density of clear liquid, which gives the criterion in its familiar form:

$$h_b \;<\; \tfrac{1}{2}\,(\mathrm{TS} + h_w)\,. \tag{528}$$

## 36.5 Weeping

At the other end of the operating window, if the vapour is too slow it cannot hold the liquid up on the plate and the liquid drains through the holes. The weep point [121] is

$$u_{h,\min} \;=\; \frac{K_2 - 0.90\,(25.4 - d_h)}{\sqrt{\rho_V}}, \tag{529}$$

with $d_h$ the hole diameter in millimetres. $K_2$ is read off a plot against $(h_w + h_{ow})$. It is optional in Choupo, and its absence is announced: give $K_2$ and the weep check runs, omit it and the column tells you the check was not performed. It never guesses.

Between the flood point above and the weep point below lies the whole operating window of a tray column, and *turndown* — the ratio of the two — is why a column designed for one throughput cannot always be run at half of it.

## 36.6 The two directions

Equation (520) contains the diameter nowhere; it is a velocity. The diameter enters only when the vapour volumetric flow $Q_V$ is turned into a velocity through the net area, $u_V = Q_V / A_{\mathrm{net}}$. So the same equation runs both ways:

| | |
|---|---|
| **Rating** | $D$ given $\Rightarrow$ how close is each tray to flood? |
| **Design** | $D$ absent $\Rightarrow$ what $D$ keeps every tray at $\phi\,u_{\mathrm{flood}}$? |

In design the widest tray sets the column, because a column has one diameter. Which tray that is, is not obvious in advance: below the feed the liquid traffic rises, which raises $F_{LV}$ and *lowers* the capacity parameter, but the vapour volumetric flow falls too. The column is sized by whichever tray loses that argument, and you find out by looking.

`column09_tray_hydraulics` sizes the benzene–toluene column of §15 at $D = 1.322$ m, set by the last stripping tray at exactly the 80 % of flood it was asked for. `column10_flooding` builds the same column 1.10 m wide. The stream table is identical to the last digit, the MESH converges just as fast, the distillate is still 98.12 % benzene — and stage 14 runs at 115.6 % of its flooding velocity. On paper it is a perfectly good column. It would not work.

# 37 Multi-stream heat exchanger (MHeatX): a checker, not a solver

*What you'll learn.* The two-stream `heatExchanger` of Chapter 29 is a *rating* model: you give it hardware ($U$, $A$) and it *computes* the outlet temperatures. The multi-stream `multiStreamHX` (alias `MHeatX`) does the *opposite*. With $N$ hot and $M$ cold streams sharing a single adiabatic shell, the detailed conductances are not knowable without a full network model, so this unit asks the student to *furnish every outlet temperature* and then *verifies* two things the first and second laws demand: that $\sum_k Q_k = 0$ (the shell is adiabatic), and that the pooled hot and cold composite curves never cross ($\Delta T_{\min} > 0$). It is a *checker*, not a *solver* — the cleanest possible expression of the no-magic credo ("information follows the streams"): the unit never back-computes an outlet the topology cannot know.

## 37.1 The degrees of freedom: why verify, not solve

A single two-stream exchanger has a clean rating problem because two hardware numbers ($U$, $A$) plus the four inlet conditions close the degrees of freedom (Chapter 29). Put $N + M$ streams in one shell and the *internal* heat-transfer network — which stream sees which, over what area, with what local $U$ — carries far more unknowns than the inlet/outlet temperatures alone can fix. Detailed-rating tools resolve this by demanding a fuller specification (zones, areas, or all-but-one outlet) and then closing the last balance. Choupo refuses to invent the missing information. Instead it adopts the spec that keeps the unit honest: *every* inlet stream carries its own outlet target temperature, and the unit's job is to *audit* the resulting energy balance, never to back-solve a duty the topology did not give it.

## 37.2 Per-stream duty and the first-law audit

For each stream $k$ the duty is the enthalpy change between the furnished end states, evaluated with the *same* enthalpy machinery the rest of Choupo uses:

$$Q_k \;=\; \dot{n}_k \left[\, h_k(T_{k,\mathrm{out}}, P_k, v_{f,k}^{\mathrm{out}}) - h_k(T_{k,\mathrm{in}}, P_k, v_{f,k}^{\mathrm{in}}) \,\right], \tag{530}$$

where $h_k$ is the formation-referenced molar enthalpy $H_{\mathrm{stream}}$ when every component carries a Gibbs-of-formation datum (so a stream that partly flashes inside the bundle is summed consistently), and the sensible $H^L$ datum otherwise. No reaction happens inside an exchanger, so the per-component reference cancels in the *difference* (530): latent heat captured by the declared inlet→outlet path (a change in $v_f$) is the only non-trivial term beyond sensible heat. Streams sort themselves by sign:

$$Q_{\mathrm{hot}} = \sum_{Q_k<0} Q_k \;(\le 0,\ \text{releasing}), \qquad Q_{\mathrm{cold}} = \sum_{Q_k>0} Q_k \;(\ge 0,\ \text{absorbing}). \tag{531}$$

The first-law residual — the duty the bundle would have to import ($>0$) or export ($<0$) to satisfy the furnished outlets — is simply the sum over *all* streams,

$$\mathcal{I} \;=\; \sum_k Q_k \;=\; Q_{\mathrm{hot}} + Q_{\mathrm{cold}}. \tag{532}$$

A truly adiabatic shell has $\mathcal{I} = 0$. The unit reports $\mathcal{I}$ *honestly* as an external duty rather than silently nudging an outlet to force closure; the test is fractional,

$$\text{PASS} \iff \frac{|\mathcal{I}|}{\max\!\big(|Q_{\mathrm{hot}}|,\, |Q_{\mathrm{cold}}|\big)} \;\le\; \tau, \tag{533}$$

with $\tau =$ `operation.tolerance` (default 1 %). A failing balance is a loud warning, not a fixed-up answer.

## 37.3 The internal pinch: pooled composite curves

The second-law check reuses the composite-curve construction of pinch analysis, applied *inside* the one shell. All hot streams are pooled into a single hot composite and all cold streams into a single cold composite — each a piecewise-linear polyline in the (cumulative-duty, temperature) plane. Over each temperature band $[T_a, T_b]$ between consecutive breakpoints, a pool's duty is the sum of the *active* streams' contributions, each linearised as its share of its own ($T_{\mathrm{in}} \to T_{\mathrm{out}}$) span:

$$\Delta Q_{\mathrm{pool}}(T_a, T_b) \;=\; \sum_{k \in \mathrm{pool}} |Q_k| \, \frac{\min(T_b, T_k^{\mathrm{hi}}) - \max(T_a, T_k^{\mathrm{lo}})}{T_k^{\mathrm{hi}} - T_k^{\mathrm{lo}}}, \tag{534}$$

accumulated from $Q = 0$ at the pool's coldest end. Each pool is built over *its own* breakpoints, anchored at zero at its cold end — never padded with the other pool's range, which would corrupt the interpolation. Adopting the counter-current convention, the two composites are aligned at $q = 0$ and the approach is sampled over the shared duty span $Q_{\mathrm{span}} = \min(Q_{\mathrm{hot,tot}}, Q_{\mathrm{cold,tot}})$:

$$\Delta T_{\min} \;=\; \min_{0 \le q \le Q_{\mathrm{span}}} \left[\, T_{\mathrm{hot}}(q) - T_{\mathrm{cold}}(q) \,\right]. \tag{535}$$

A non-positive $\Delta T_{\min}$ is a *temperature cross*: the requested outlets would require heat to flow up a temperature gradient inside one adiabatic bundle, which the second law forbids. The unit flags it as a hard warning rather than silently accepting an infeasible design.

> *Intuition.* A single bundle "passes" only if the hot composite lies entirely above the cold composite over their shared duty range. $\mathcal{I} = 0$ (first law) says the two curves carry the *same total* duty; $\Delta T_{\min} > 0$ (second law) says they do not *touch or cross* while doing so. Both can fail independently: balanced-but-crossing (an infeasible internal arrangement), or feasible-but-unbalanced (the bundle needs an outside utility). The student sees, in one report, exactly which law is in trouble.

## 37.4 Worked mini-example

Three streams in one shell: a hot stream H1 releasing $Q_{\mathrm{H1}} = -100\,\mathrm{kW}$ cooling $400 \to 320\,\mathrm{K}$, and two cold streams absorbing it — C1 at $+60\,\mathrm{kW}$ ($300 \to 360\,\mathrm{K}$) and C2 at $+40\,\mathrm{kW}$ ($310 \to 350\,\mathrm{K}$). The first-law residual (532) is $\mathcal{I} = -100 + 60 + 40 = 0\,\mathrm{kW}$, so the shell is adiabatic and the balance PASSES. The hot composite runs $320 \to 400\,\mathrm{K}$ carrying $100\,\mathrm{kW}$; the pooled cold composite runs $300 \to 360\,\mathrm{K}$ carrying the same $100\,\mathrm{kW}$. Sampling (535) over the shared duty gives a positive minimum approach at the cold end ($320 - 300 = 20\,\mathrm{K}$), so $\Delta T_{\min} > 0$ and the design is thermodynamically feasible. Had the analyst instead asked C1 to leave at $410\,\mathrm{K}$ (above H1's inlet), the composites would cross, $\Delta T_{\min} < 0$, and the unit would refuse the design with a temperature-cross warning — even though the first law could still be made to close.

## 37.5 Power and limit

**Power:** the MHeatX is the honest multi-stream audit — it verifies first- and second-law feasibility of a heat-integration proposal *without* pretending to know an internal conductance network it cannot, and it surfaces the in-bundle pinch ($\Delta T_{\min}$, $Q_{\mathrm{hot}}$, $Q_{\mathrm{cold}}$, $\mathcal{I}$, plus a per-stream $Q_k$, all as KPIs) so the result feeds straight into the flowsheet-level pinch analysis and into sensitivity / sizing. **Limit:** precisely because it is a checker, it will *not* hand back an unknown outlet temperature — every inlet must carry a target; and the linearised two-point composites (535) assume each stream's $Q$–$T$ relation is monotone and roughly straight between its declared end states (a stream crossing its dew or bubble dome inside the bundle should be split into segments so the latent kink is resolved). It models a single adiabatic shell — any genuine loss to the surroundings appears, correctly, as a non-zero $\mathcal{I}$ rather than being absorbed.

# 38 Drying of solids and spray drying

This section is the *theory*; the units that run it — `sprayDryer` and `solidDryer`, the four coupled pieces the spray dryer solves at once, and the tutorials — are §28.

Drying removes a liquid (usually water) from a wet solid by evaporation. It competes with distillation as the most energy-intensive unit operation — the latent heat must be supplied and the (usually air) drying medium is thermodynamically inefficient — so the model must close both a mass and an energy balance. Choupo models drying at the "all models are wrong, but some are useful" altitude: measured drying curves and algebraic transfer correlations, no pore-scale or CFD detail.

## 38.1 Moisture content and the equilibrium

Moisture is reported on a *dry basis* $X$ = kg water / kg dry solid (so $0 \le X < \infty$), related to the wet basis $W$ by $X = W/(1-W)$. A wet solid held in air of a given relative humidity reaches an *equilibrium moisture* $X_e(a_w)$ — its *sorption isotherm* — below which that air cannot dry it. The *free moisture* $X - X_e$ is what can be removed; the bound fraction (held in capillaries and by sorption, its vapour pressure depressed) dries only under a lower humidity. Choupo reads $X_e$ from a GAB isotherm on the solid component's `sorption` block, and no dryer takes a solid below it: the solid dryer's outlet is $X_{\text{final}} = \min(X_{\text{in}}, X_e)$ when nothing else binds first — the air may be unable to pay for the evaporation, or its exhaust unable to carry the water without supersaturating, and then the powder leaves wetter (`SolidDryer.cpp`); the spray dryer's is the drying curve evaluated at the residence time, never below $X_e$.

## 38.2 The drying-rate curve: constant and falling rate

Plot the drying rate $N = -\frac{m_s}{A}\frac{dX}{dt}$ against $X$ and two regimes appear. In the **constant-rate period** ($X > X_c$) the surface stays wet: the particle surface sits at the *wet-bulb temperature* $T_w$ and the rate is limited by external gas-film heat/mass transfer, so $N = N_c$ is flat. Below the **critical moisture** $X_c$ the surface dries out and a receding front / crust governs; the **falling-rate period** ramps $N$ down toward zero at $X_e$. The critical moisture $X_c$ is a KINETIC datum (it depends on the drying intensity), so it lives in `constant/dryingKinetics`, not on the equilibrium isotherm. The drying *time* is the integral

$$t \;=\; \frac{m_s}{A}\int_{X_2}^{X_1}\frac{dX}{N(X)} \;=\; \underbrace{\frac{m_s(X_1 - X_c)}{A\,N_c}}_{\text{constant rate}} + \underbrace{\frac{m_s(X_c - X_e)}{A\,N_c}\,\ln\frac{X_c - X_e}{X_2 - X_e}}_{\text{falling rate (linear model)}},$$

the falling-rate branch using the linear (Lewis) closure $N \propto (X - X_e)$ matched to $N_c$ at $X_c$ — an honest first approximation that does *not* resolve crust diffusion; calibrate $X_c$/the curve against a measured drying curve of the real material.

## 38.3 Spray drying

A spray dryer atomises a solution into a cloud of droplets and dries them in hot gas to a powder of controlled particle size — seconds of residence, because the droplets are tiny. Three pieces compose.

## 38.4 Atomisation — the drop-size correlations

Atomisation sets the whole dryer. The spray is a *distribution* of drops, and its Sauter mean diameter $d_{32} = \sum n_i d_i^3 / \sum n_i d_i^2$ (the volume-to-surface mean) is the number that matters: the drying rate scales as $1/d$ (Ranz–Marshall, below), the powder inherits it ($d_p = d_{32}(x_{\text{solid}}\rho_L/\rho_{\text{solid}})^{1/3}$), and the chamber must be tall enough for the *largest* drops to dry. Choupo exposes the three families of Figure 79 as the selectable `atomiser` sub-model; each returns $d_{32}$ and a Rosin–Rammler width $n$ for the spread.

**The physics, in one line.** A drop forms when the atomising energy overcomes the surface tension holding the liquid sheet/ligament together, against the viscosity resisting its deformation. So every correlation shares one skeleton — $d_{32}$ grows with the cohesive properties ($\sigma$, $\mu_L$, $\rho_L$) and shrinks as you spend energy ($N$, $\Delta P$, air velocity). Because a concentrated, viscous feed therefore sprays *coarser*, $\sigma$ and $\mu_L$ are drawn from the property layer — the *solution* viscosity via its mixing rule, not a fixed water value.

**Rotary wheel — three regimes, one lever.** Liquid fed to the centre of a disc (or cup) spinning at $N$ rev/s is driven to the rim by centrifugal force. On a *flat* disc the liquid slips — it leaves below the rim speed, so the drop size is unpredictable — and industrial wheels therefore carry radial *vanes*: the liquid is trapped against a vane and discharged at the wheel's peripheral (tip) speed $U = \pi D N$, the true scaling velocity (100–200 m/s in practice). So the honest lever is not $N$ alone but $U$: a small fast wheel and a large slow one that share $U$ atomise alike.

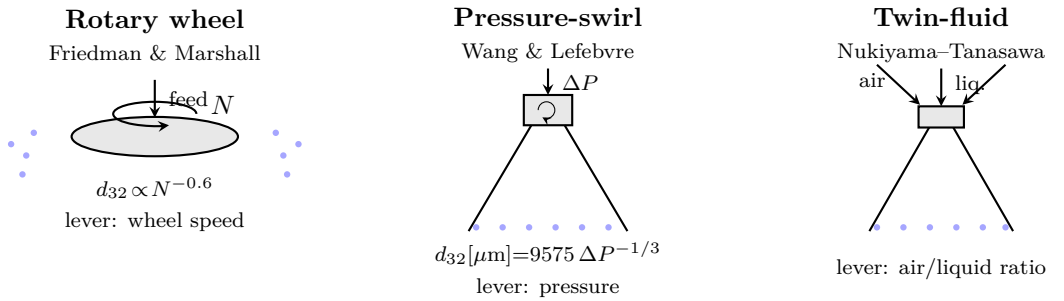



**Figure 79:** The three atomiser families. A drop is born when the atomising energy overcomes the surface tension holding the sheet together, against the liquid's viscous resistance: $d_{32} \sim$ (cohesion $\sigma, \mu_L, \rho_L$)/(energy $N, \Delta P, U_{\text{air}}$). The wheel sizes by speed, the pressure nozzle by pressure, the twin-fluid by the air/liquid ratio; the twin-fluid gives the finest drops (best for viscous feeds).

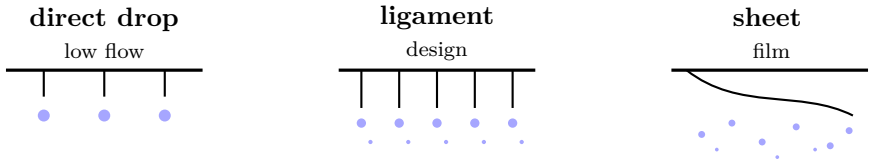



**Figure 80:** The three break-up regimes at a rotary wheel's rim as the feed rate rises (after Christensen & Steely; Lefebvre & McDonell, *Atomization and Sprays*). *Direct drop*: discrete, near-monodisperse drops shed one at a time. *Ligament*: radial jets break by Rayleigh instability into main drops + satellites — the industrial regime. *Sheet/film*: a continuous sheet overhangs the rim and breaks irregularly — a broad, coarse spray. Turning the feed up to raise throughput can silently cross ligament→sheet and *widen* the powder.

As the feed rate rises the rim passes through the three regimes of Figure 80; Tanasawa's critical flow rates (cgs: $q$ [cm$^3$/s], $D$ [cm], $n$ [rpm]) fix the transitions,

$$q_{\text{direct}} = 2.8\left(\tfrac{D}{n}\right)^{2/3}\tfrac{\sigma}{\rho_L}\left[1+10\left(\tfrac{\mu_L}{(\rho_L\sigma D)^{1/2}}\right)^{1/3}\right]^{-1}, \qquad q_{\text{film}} \geq 5.3\left(\tfrac{D}{n}\right)^{2/3}\tfrac{\sigma}{\rho_L}\left(\tfrac{\rho_L}{\mu_L}\right)^{1/3},$$

the ligament (design) regime lying between. In it the Sauter mean follows the Friedman–Gluckert–Marshall correlation ($\Gamma = \dot{m}/\pi D$, the loading per wetted perimeter),

$$d_{32} = 0.4\,D\left(\frac{\Gamma}{\rho_L N D^2}\right)^{0.6}\left(\frac{\mu_L}{\Gamma}\right)^{0.2}\left(\frac{\sigma\rho_L D}{\Gamma^2}\right)^{0.1} \propto N^{-0.6},$$

a comparatively narrow spray ($n \approx 2.2$) — though the Rayleigh break-up leaves *satellite* drops, so the true distribution is mildly bimodal. Serrating the rim (a *toothed* wheel) delays the ligament→sheet transition; Christensen & Steely give the teeth needed to hold ligament formation, $z = 0.215(\rho_L\omega^2 D^3/\sigma)^{5/12}(\rho_L\sigma D/\mu_L^2)^{1/6}$ (e.g. $\approx 95$ teeth on a 5 cm wheel at 3000 rpm). The wheel's asset is operational: since $d_{32} \propto \dot{m}^{0.2}$ only, drop size is *decoupled from throughput* — set size by speed, rate independently, with no small orifice to clog. This is why large food/dairy towers atomise by wheel, where a pressure nozzle would tie size and flow to one $\Delta P$ through one hole.

**Pressure-swirl nozzle.** Liquid forced through a swirl chamber leaves as a thin conical sheet that breaks up. The class-average design curve $d_{32}[\mu\text{m}] = 9575\,\Delta P[\text{Pa}]^{-1/3}$ is the quick estimate; the two-term SMD of Wang & Lefebvre resolves the properties and the sheet geometry,

$$d_{32} = 4.52\left(\frac{\sigma\mu_L^2}{\rho_G\Delta P^2}\right)^{0.25}(t\cos\theta)^{0.25} + 0.39\left(\frac{\sigma\rho_L}{\rho_G\Delta P}\right)^{0.25}(t\cos\theta)^{0.75},$$

$t$ the film thickness and $\theta$ the half-cone angle. Pressure is the lever ($d_{32} \propto \Delta P^{-1/3}$); wider spray than a wheel ($n \approx 1.8$).

**Twin-fluid (airblast) nozzle.** A high-velocity air stream shreds the liquid — the finest drops of the three, the choice for viscous feeds. Nukiyama & Tanasawa (empirical, cgs; $U_{\text{rel}}$ the air–liquid slip velocity, $Q_L/Q_A$ the liquid-to-air volume ratio):

$$d_{32}\,[\mu\text{m}] = \frac{585}{U_{\text{rel}}}\sqrt{\frac{\sigma}{\rho_L}} + 597\left(\frac{\mu_L}{\sqrt{\sigma\rho_L}}\right)^{0.45}\left(1000\,\frac{Q_L}{Q_A}\right)^{1.5}.$$

A higher air/liquid ratio gives finer drops; broad spray ($n \approx 1.6$). This form is dimensional (cgs) — extrapolate with care.

**The atomising energy — pressure follows the stream, power is a result.** The atomising pressure of a pressure nozzle is *not* a free number: it is the feed's pressure above the chamber, $\Delta P = P_{\text{feed}} - P_{\text{chamber}}$, imposed by the feed pump on the *stream* (a bare-atomiser study may override it with `deltaP`). Each atomiser then carries a **shaft-power** estimate. For the pressure nozzle it is the hydraulic pumping power $\dot{W} = Q\,\Delta P/\eta_{\text{pump}}$. For the rotary wheel it splits in two: the kinetic energy imparted to the liquid, accelerated centre-to-rim to the tip speed, $\frac{1}{2}\dot{m}U^2$ (exact), *plus* **windage** — the wheel dragging the surrounding gas. The spray-drying literature gives windage only qualitatively ("rotary needs more power"), so Choupo anchors it on the classic rotating-disk drag (von Kármán 1921; Schlichting, *Boundary Layer Theory*),

$$\dot{W}_{\text{wind}} = \tfrac{1}{2}\,C_M\,\rho_g\,\omega^3 R^5, \qquad C_M = 0.146\,\mathrm{Re}_\omega^{-1/5}\ \text{(turbulent)}, \quad \mathrm{Re}_\omega = \frac{\omega R^2}{\nu_g},$$

flagged honestly as the *smooth free disk* — a vaned wheel pumps air like a fan and draws *more*, so this is a lower bound. The split (liquid KE, windage, drive efficiency) is printed in the atomiser note; the power feeds sizing and costing.

**References.** Marshall (1954); Masters (1991); Lefebvre & McDonell (2017, DOI 10.1201/9781315120911); Wang & Lefebvre, *J. Propulsion & Power* 3 (1987) 11 (DOI 10.2514/3.22946); Nukiyama & Tanasawa, *Trans. Soc. Mech. Eng. Japan* 5 (1939) 68.

**Single-droplet drying.** A droplet decelerates to its terminal velocity in milliseconds, after which the slip Reynolds number is small and the Ranz–Marshall correlation $\mathrm{Nu} = 2 + 0.6\,\mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}$ collapses to its conduction limit $\mathrm{Nu} = 2$, i.e. $h = 2k_g/d$. The constant-rate time to reach the critical diameter $D_c$ is then

$$t_c \;=\; \frac{\rho_L\,\Delta H_{\text{vap}}}{8\,k_g\,\Delta T_{\ln}}\left(D_0^2 - D_c^2\right),$$

with the log-mean driving force $\Delta T_{\ln}$ over the constant-rate span (the gas cools as it dries). The droplet then shrinks to the powder particle by mass, $d_f = d_0\left[(1+X_f)/(1+X_0)\right]^{1/3}$. The $\mathrm{Nu} = 2$ short-cut holds once the droplet has slowed; it *underpredicts* early-flight and large-droplet drying where the convective $\mathrm{Re}\,\mathrm{Pr}$ term matters.

**Chamber: co- vs. counter-current.** The chamber closes a lumped adiabatic mass + energy balance on the gas (fixing the dry-air rate $G$, the outlet humidity and temperature, and the air state at the critical point), with the residence time a RESULT of the hardware ($V \approx \dot{Q}_{\text{gas}}\,t,\ t \geq t_c$). *Co-current* (hot gas meets the wettest droplet) keeps the droplet near $T_w$ — gentle, for heat-sensitive foods (milk, coffee); *counter-current* (hot gas meets the driest, crusted particle) is thermally efficient but harsh — for robust materials (detergents).

**References.** Marshall, *Atomization and Spray Drying*, AIChE Monograph 2 (1954); Masters, *Spray Drying Handbook*, 5th ed. (1991); Mujumdar (ed.), *Handbook of Industrial Drying*, 4th ed., CRC Press 2015 (DOI 10.1201/b17208). Atomisation: Lefebvre & McDonell, *Atomization and Sprays*, 2nd ed. 2017 (DOI 10.1201/9781315120911); Wang & Lefebvre, *J. Propulsion & Power* 3 (1987) 11 (DOI 10.2514/3.22946). Droplet drying and axial chamber profiles: Ranz & Marshall, *Chem. Eng. Prog.* 48 (1952) 141, 173; Langrish & Kockel, *Chem. Eng. J.* 84 (2001) 69 (DOI 10.1016/S1385-8947(00)00384-3, the characteristic drying curve); Mezhericher, Levy & Borde, *Chem. Eng. & Processing* 49 (2010) 1205 (DOI 10.1016/j.cep.2010.09.002). Falling-rate kinetics — the two non-CFD schools: the *characteristic drying curve* $N/N_c = f(\Phi)$ above (Langrish & Kockel), and the *Reaction Engineering Approach*, Chen & Xie, *Drying Technology* 15 (1997) 1063 (DOI 10.1080/07373939708917282) and Chen, *Drying Technology* 26 (2008) 627 (DOI 10.1080/07373930802045922), which replaces the critical-moisture break with a smooth apparent activation energy $\Delta E_v(X)$.

# 39 The simple unit operations

Not every unit needs a chapter. Four are one-liners, listed here for completeness — and because their very simplicity is the point: the flowsheet's plumbing (mixing, splitting, a furnished duty, a phase specification) is not where the modelling effort goes.

**Mixer**
Sums the component molar flows of its inlets and closes an adiabatic energy balance for the outlet temperature (a Newton-1D in $T$ on $\sum H_{\text{in}} = H_{\text{out}}$).

**Splitter**
Divides one inlet into outlets by furnished split fractions — same $T$, $P$, composition in every branch (a flow split, *not* a separation: contrast the `gasSolidSplitter`, which carries solids but likewise does not classify).

**Heater**
A single process stream with a furnished duty $Q$ (W, the only hardware); $T_{\text{out}}$ is a RESULT from a Newton-1D on the enthalpy balance $H(T_{\text{out}}) = H_{\text{in}} + Q$. Phase-aware: a gas stream integrates $H^{\text{ig}}$, a liquid uses $H^{L}$ — the same datum on both sides, so a combustor reaching $> 1000\,\text{K}$ is fine. A target outlet temperature is a DesignSpec on $Q$, not a spec.

**Saturation & adiabatic flash**
`bubbleT` solves $\sum_i K_i(T)\, x_i = 1$ for the bubble temperature; `dewT` solves $\sum_i y_i / K_i(T) = 1$ for the dew temperature (Newton-1D in $T$). The `adiabaticFlash` wraps the isothermal flash (Chapter 9's neighbourhood) in an outer Newton on $T$ that drives the energy balance $H_{\text{out}} = H_{\text{in}}$ to zero — the Joule–Thomson / valve flash.

**Valve**
An isenthalpic pressure let-down: the outlet pressure is furnished (the only hardware) and the outlet state is the `adiabaticFlash` of the inlet at that lower $P$, holding $H_{\text{out}} = H_{\text{in}}$. A throttle that may flash part of the stream to vapour — the Joule–Thomson valve of the item above, exposed as a unit of its own.

**ElectricLoad**
A passive shaft sink: the electric generator (or any mechanical brake) wired to a turbine's shaft. It carries no process stream — the only thing crossing its boundary is the shaft work $W$ delivered through an energy wire, which it absorbs. Its rating is not furnished; it is the $W$ it actually receives, so it closes a turbine–generator shaft balance (Chapter 24) without itself setting a pressure.

# Part VII — Equipment design and economics

After the simulator pass converges, the case may declare a `system/postDict` chain. Two post-processors are currently shipped: `sizing` turns each named unit op into a set of physical dimensions; `costing` turns those dimensions into a purchased + bare-module + total-module cost in EUR. Neither feeds back into the simulator — the contract is unidirectional, augmenting the converged `SimulationResult`.

> *Intuition.* Post-processing is what makes a process simulator more than a material-balance calculator. The same converged flowsheet, run with different `postDict` chains, can produce: a stream table for a PFD, a vessel list with weights and wall thicknesses, an EPC budget, a HAZOP-feed reactor inventory list, or a utility-summary report. The simulator does NOT know which use case; it just delivers a converged **F** and KPI table. The post-processor decides the rest.

# 1 Vessel sizing: the stirred tank

*What you'll learn.* How the simulator turns a CSTR's $V_R$ KPI into a cylindrical drum with diameter $D$, length $L$, wall thickness $t$, weight, and a selected construction material.

## 1.1 Volume to geometry

For a vertical cylindrical drum with an $L/D$ ratio specified by `designRules.L_over_D` (default 2.5), the diameter and length follow from the requested reactor volume:

$$V_R \;=\; \frac{\pi D^2}{4}\,L, \qquad L \;=\; (\mathrm{L/D})\,D \;\Longrightarrow\; D \;=\; \left(\frac{4V_R}{\pi\,(L/D)}\right)^{1/3}. \tag{536}$$

## 1.2 Wall thickness: ASME stress design

The vessel must withstand the design pressure $P_{\mathrm{design}} =$ `designRules.pressureDesign`. ASME Section VIII Div. 1 gives, for a thin cylindrical shell (transcribed, as `src/postProcessing/sizing/VesselMechanics.H` records, from [123], App. A; the standard itself was not read here):

$$t \;=\; \frac{P_{\mathrm{design}}\,D/2}{S\,E - 0.6\,P_{\mathrm{design}}} \;+\; t_{\mathrm{corr}}, \tag{537}$$

which the engine writes in the equivalent form $P_{\mathrm{design}}D/(2SE \;-\; 1.2\,P_{\mathrm{design}}) \;+\; t_{\mathrm{corr}}$ in `src/postProcessing/sizing/VesselMechanics.cpp`, reading $P_{\mathrm{design}}$ as a GAUGE pressure. $E$ is the joint efficiency (`jointEfficiency`, default 1.0), and $t_{\mathrm{corr}}$ is the corrosion allowance (`corrosionAllow`, default 3 mm). $S$ is the allowable stress, and the engine applies NO safety factor to obtain it: $S$ is the material record's field `sigma_y`, used as it stands. The field's NAME is misleading — it reads as a yield strength, but `src/materials/Material.H` declares it to hold the ALLOWABLE stress, and the carbon-steel record labels its 140 MPa as an allowable at 250 °C. A record that stored a true yield strength there would be sized on it with no margin at all. A record holds ONE such value for every temperature: the wall of a 700 K shell is sized on the same $S$ as a 300 K one.

All material properties (`sigma_y`, $\rho$, $F_M$, $T_{\max}$, $P_{\max}$) come from `data/standards/assets/<name>.dat` — carbon steel, SS304, SS316, aluminium are shipped, and their own headers call their numbers unsourced teaching assertions. The two limits are COMPARED and ANNOUNCED, never enforced (`src/postProcessing/sizing/MaterialLimits.H`): a design pressure above the record's $P_{\max}$, and a service temperature above its $T_{\max}$, each print a `[rating]` warning that the end-of-run caveat block repeats, and the item is sized and costed anyway. Choupo has no design temperature, so the temperature compared is the hottest state on the item's ports — a lower bound on what the metal sees (a hot spot inside a bed or a fire case is not in it). Neither limit is a code allowable at temperature; each is the record's own service envelope.

## 1.3 Weight

With the wall thickness, the shell metal mass is that of the cylinder alone — no heads, no nozzles, no end-closure correction (`src/postProcessing/sizing/VesselMechanics.cpp`):

$$m \;=\; \rho\,A_{\mathrm{shell}}\,t, \qquad A_{\mathrm{shell}} \;=\; \pi D L. \tag{538}$$

This $m$ is published on the item's design sheet, and it is NOT what the cost reads. The Turton vessel correlation is sized on the VOLUME $V_R$ (`src/postProcessing/costing/Turton.cpp`, the vessel coefficients' size key), so the weight — and the density with it — moves no euro. The material reaches the cost through $F_M$, and through `sigma_y` once more inside the pressure factor $F_P$.

# 2 Heat-exchanger sizing: shell-and-tube

For a single-pass shell-and-tube HX the area follows from the

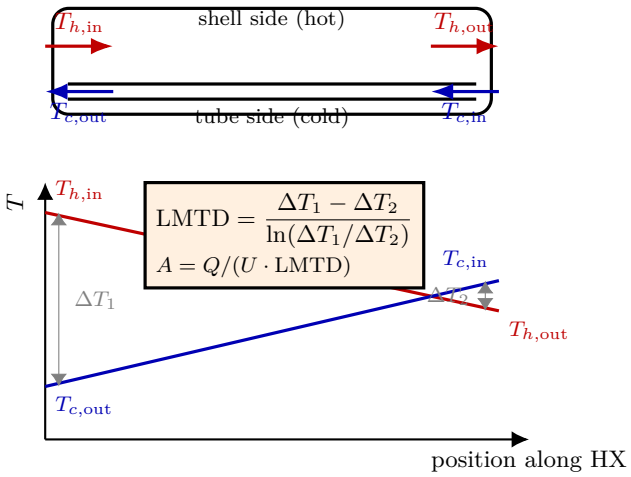



**Figure 81:** The shell-and-tube sizing picture. Top: a single-pass counter-current exchanger — hot stream in the shell, cold stream in the tubes, flowing in opposite directions. Bottom: the temperature profiles down the unit. The approach $\Delta T$ is *not* constant — it is $\Delta T_1$ at one end and $\Delta T_2$ at the other — so the correct mean driving force is the *log-mean* LMTD, not the arithmetic average. Once the duty $Q$ is known (a unit-op KPI) and $U$ is supplied, the area is simply $A = Q/(U \cdot \text{LMTD})$. Counter-current flow (cold rising against hot falling) keeps the approach open along the whole length — why it out-performs co-current for the same area.

duty $Q$ (a unit-op KPI) and the user-supplied overall coefficient $U$ and log-mean temperature difference LMTD:

$$A = \frac{Q}{U \cdot \text{LMTD}}. \tag{539}$$

Both $U$ and LMTD are read from `designRules`; the simulator does not yet auto-compute LMTD from the streams (planned for). The pressure-design wall thickness uses Eq. (537) on the shell diameter.

# 3 Discounted cash flow: from a cost to a decision

Costing (§4) ends with a capital figure. A capital figure is not a decision: money now and money later are different quantities, and the appraisal that compares them is the discounted cash flow Choupo implements in `EconomicsPass`.

**The ledger, year by year.** Each operating year carries a revenue $R$, a cash operating cost $C$, a depreciation charge $D_k$ that is not a cash flow but reduces the taxable income, a tax at rate $t$, and the resulting after-tax cash flow

$$\mathrm{CF}_k = (R - C - D_k)(1 - t) + D_k. \tag{540}$$

Depreciation is added back because it left the ledger only to compute the tax. Discounting at rate $i$ and summing over the project life $N$, against the capital placed at the start,

$$\mathrm{NPV} = \sum_{k=1}^{N} \frac{\mathrm{CF}_k}{(1+i)^k} - (\mathrm{FCI} + \mathrm{WC}), \tag{541}$$

and the internal rate of return is the $i$ that makes it vanish.

**The start-up years.** Equation (541) is the one-year construction: the whole capital at $t = 0$ and full production from year 1. A case may declare more (all four keys optional). With `constructionPeriod` $N_c$ and a `capitalSchedule` $(f_1, \ldots, f_{N_c})$ summing to one, the fraction $f_j$ of the FCI is spent at the END of construction year $j$; Choupo's $t = 0$ stays the end of construction (the start-up instant), so year $j$ sits at $t = j - N_c \le 0$ and its outflow is compounded forward by the same factor $(1+i)^{-t}$. The working capital is invested in full at $t = 0$ and recovered at $t = N$. A `rampUp` $(p_1, p_2, \ldots)$ sets the production fraction of the first operating years: the revenue becomes $p_k R$ and, of the cost of manufacture $\mathrm{COM_d} = 0.180\,\mathrm{FCI} + 2.73\,C_{OL} + 1.23\,(C_{RM} + C_{UT} + C_{WT})$, only the last group scales with $p_k$ — the equipment and its crew are there from year 1. A `startupCostFraction` $s$ adds an expense $s\,\mathrm{FCI}$ to year 1, deducted from its taxable income. So

$$\mathrm{NPV} = \sum_{j=1}^{N_c} \frac{-f_j\,\mathrm{FCI}}{(1+i)^{\,j-N_c}} - \mathrm{WC} + \sum_{k=1}^{N} \frac{\mathrm{CF}_k}{(1+i)^k} + \frac{\mathrm{WC}}{(1+i)^N}, \tag{542}$$

with $\mathrm{CF}_k$ of Eq. (540) evaluated at $p_k R$, the year's own cost of manufacture and, in year 1, the start-up cost. The placement follows CAPCOST 2017, the costing program distributed with Turton et al. [123], which counts its year 0 from the START of the project; the IRR and the paybacks do not depend on where $t = 0$ is, and the run prints the NPV referred to the start of construction beside the one at start-up. A construction period above one with no schedule is refused: how the capital is spent moves the NPV, and an even split is a policy nobody chose. Worked case: `economics03_startup_years`, whose `postDict` redoes every row by hand.

**Three cautions the engine states aloud.** First, the IRR need not be unique: when the cumulative cash flow changes sign more than once the equation has more than one root, and the run says so and tells the reader to use the NPV. Second, the discount rate is a *declaration about risk and the cost of capital*, not a property of the plant — two firms appraising the same flowsheet correctly disagree. Third, and most important pedagogically: **a discounted cash flow inherits every uncertainty upstream of it**. The correlations behind the FCI are study-grade, so the NPV is a study-grade number wearing however many digits the arithmetic produced.

**What is NOT modelled, named rather than implied.** A ramp-up that degrades yield or quality (the production fraction scales products and raw materials alike), a utility bill not proportional to throughput, a working capital built up with the ramp-up (it is invested in full at start-up, and the run says so), and any tax treatment of the start-up cost other than an expense of the first operating year.

# 4 Turton bare-module costing

*What you'll learn.* How the sized equipment list becomes a bare-module and total-module cost, anchored on Turton-style polynomial purchased-cost correlations, CEPCI year scaling, and a USD→EUR conversion.

The model is registered as `Turton` (`costing { method Turton; }`): every coefficient it holds is Turton et al.'s [123], App. A, set in the bare-module form Guthrie (1969) introduced. The word `Guthrie` named this model until 2026-10-05 and is now refused by name, naming `Turton`.

## 4.1 Purchased cost from size

For each piece of equipment the model evaluates a polynomial fit of $\log_{10} C_p$ vs $\log_{10}(\text{size})$, with the size attribute equipment-specific (vessel → mass; HX → area):

$$\log_{10} C_p^{2001\,\text{USD}} \;=\; K_1 \;+\; K_2\,\log_{10}(\text{size}) \;+\; K_3\,\big[\log_{10}(\text{size})\big]^2. \tag{543}$$

The coefficients $(K_1, K_2, K_3)$ come from Turton 4th ed. [123] and live inside `src/postProcessing/costing/Turton.cpp` — one set per equipment type.

## 4.2 Bare-module factor

The base purchased cost assumes carbon steel and atmospheric operation. Corrections for actual material and design pressure are folded into the bare-module factor:

$$C_{\text{BM}} \;=\; C_p\,\big(B_1 + B_2\,F_M\,F_P\big), \tag{544}$$

with $B_1, B_2$ from Turton (e.g. 2.25 and 1.82 for a vertical vessel, the set the code applies to every process vessel; 1.49 and 1.52 are the horizontal one's), $F_M$ the material factor read from `data/standards/assets/<name>.dat`, and $F_P$ the pressure factor from a Turton polynomial in $\log_{10} P_{\text{design}}$.

## 4.3 Total-module cost

Adding contingency and fees, with the factor of the same Turton correlation set the code transcribes ([123], App. A; `src/postProcessing/costing/Turton.H`):

$$C_{\text{TM}} \;=\; 1.18\,C_{\text{BM}}. \tag{545}$$

## 4.4 Tray stacks: bought by the tray

A tray is priced per tray, not per item, so a column's tray stack has its own shape. With $A$ the tower cross-section and $N$ the number of trays,

$$C_p \;=\; N \cdot 10^{\,K_1 + K_2\log_{10} A + K_3(\log_{10} A)^2}, \qquad C_{\text{BM}} \;=\; C_p\,F_{\text{BM}}\,F_q, \tag{546}$$

$$\log_{10} F_q \;=\; 0.4771 + 0.08516\log_{10} N - 0.3473(\log_{10} N)^2 \quad (N < 20), \qquad F_q = 1 \quad (N \geq 20), \tag{547}$$

with no pressure factor and no $F_M$: the trays' material enters through $F_{\text{BM}}$, chosen by the material class the case declares (`trayMaterial carbonSteel | stainlessSteel | nickelAlloy;` in the column's `designRules`). For sieve trays $K_1 = 2.9949$, $K_2 = 0.4465$, $K_3 = 0.3961$ on $0.07 \leq A \leq 12.3$ m$^2$ and $F_{\text{BM}} = 1.00/1.83/5.58$. These were read from CAPCOST 2017, the costing program the authors of [123] distribute with its 5th edition (sheet *Equipment Cost Data*, blocks *Tower Trays* and *FBM*, and its macro `calculateTrayCosts`); the book's printed table was not consulted. $F_q > 1$ for a short run: at fourteen trays the bare-module cost carries $F_q = 1.31$ (`column09_tray_hydraulics`). The tray cost is then escalated and converted exactly as below, and $C_{\text{TM}} = 1.18\,C_{\text{BM}}$.

## 4.5 Year and currency scaling

Costs are escalated from the Turton reference year (2001) to the user's target year via the Chemical Engineering Plant Cost Index (CEPCI), and converted from USD to EUR:

$$C(\text{year}, \text{EUR}) \;=\; C_{\text{TM}}^{2001\,\text{USD}} \cdot \frac{\text{CEPCI}_{\text{year}}}{\text{CEPCI}_{2001}} \cdot \texttt{usdToEur}. \tag{548}$$

The user supplies `year`, `cepci`, `cepci2001` (default 397), and `usdToEur` in `postDict`.

**Common pitfall.** Turton's correlations are accurate to about ±30% for order-of-magnitude work — they are *not* engineering-grade estimates. They are pedagogically valuable because the entire chain from $V_R$ to EUR is exposed (no proprietary database, no black-box adjustment), so a student can perturb any link and see the downstream effect. They are *not* a substitute for a vendor quote.

**Runnable case.** `tutorials/steady/flowsheets/process02_with_design` — the canonical end-to-end: two-unit flowsheet (→ CSTR + flash), `postDict` declares sizing for both, Turton costing in $\mathrm{EUR}_{2026}$ at the user-chosen CEPCI. The output table shows $F_M$, $F_P$, $C_{\mathrm{purchased}}$, $C_{\mathrm{BM}}$, $C_{\mathrm{TM}}$ per unit and the grand total.

# Appendix A — Symbol glossary

| Symbol | Meaning |
|---|---|
| $T$ | Temperature, K |
| $P$ | Pressure, bar |
| $F, L, V$ | Total molar flow: feed, liquid, vapour, kmol/h |
| $z_i, x_i, y_i$ | Mole fraction of $i$ in feed, liquid, vapour |
| $K_i$ | Equilibrium $K$-value, $K_i = y_i/x_i$ |
| $\beta$ | Vapour fraction, $V/F$ |
| $\gamma_i$ | Activity coefficient of $i$ in the liquid |
| $\varphi_i^V$ | Fugacity coefficient of $i$ in the vapour |
| $P_i^{\mathrm{sat}}(T)$ | Saturation pressure of pure $i$, bar |
| $\Delta H_{\mathrm{vap}}(T)$ | Molar heat of vaporisation, J/mol |
| $R_{\mathrm{g}}$ | Universal gas constant, $8.314\,\mathrm{J/(mol\,K)}$ |
| $\mu_i$ | Chemical potential of $i$ |
| $\hat{f}_i$ | Fugacity of $i$ in the mixture |
| $G^E$ | Excess Gibbs energy of the liquid mixture, J/mol |
| $\xi$ | Reaction extent, mol/h or mol/s |
| $\boldsymbol{\nu}$ | Stoichiometric vector of a reaction |
| $r(\mathbf{x}, T)$ | Reaction rate, $\mathrm{mol/(m^3\,s)}$ |
| Da | Damköhler number, $k\,\tau$ |
| $\mathbf{J}$ | Jacobian matrix (residual w.r.t. unknowns or w.r.t. parameters) |
| $q$ | Wegstein acceleration factor (Eq. 219) |
| $\lambda$ | Damping factor: Newton line search or Marquardt damping (Eq. 209) |
| $\mathbf{p}$ | Adjustable parameter vector in regression (Ch. 4) |
| $\mathbf{r}$ | Residual vector, $r_i = T_i^{\mathrm{pred}}(\mathbf{p}) - T_i^{\mathrm{exp}}$ |
| $\chi^2(\mathbf{p})$ | Sum of squared residuals (Eq. 200) |
| $\tau, G, \alpha$ | NRTL binary parameters |
| $\Lambda$ | Wilson binary parameter |

# Appendix B — References

# References


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[2] CAPE-OPEN Laboratories Network, *Unit Operation Interface Specification*, current version, `https://www.colan.org/specifications/unit-operation-interface-specification/`. The vendor-neutral standard (since 1999) that codifies the ports-vs-parameters separation, the material / energy / information stream taxonomy, and the contract that any third-party unit operation must satisfy to be droppable into a CAPE-OPEN-compliant flowsheeter. Choupo's `UnitOperation` interface is a deliberate subset of this standard; v1.0 will add adapter wrappers so Choupo modules can load into a CAPE-OPEN host and vice versa.

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# Appendix C — Historical context: Choupo and FLOWTRAN

Steady-state sequential-modular chemical process simulation as we know it begins with FLOWTRAN, developed at Monsanto by Robert Cavett, Edward Rosen, Allen Pauls and colleagues between 1964 and 1966 [124]. FLOWTRAN was the first system to package the abstractions used in this manual — streams as information records, unit operations as "blocks" acting on streams, an outer recycle convergence loop on tear streams — into a coherent, redistributable simulator. Every modern chemical-process simulator inherits this architecture. Choupo does too, by design: when you write a block that inherits from `UnitOperation` and registers itself with the factory, you are continuing a lineage that runs through Cavett's 1964 "building-block architecture" memo, the 1966 internal release, the 1969 commercial product, and the 1974 distribution through CACHE.

## FLOWTRAN's scope, as documented by Rosen & Pauls (1977)

| Component | What it was |
|---|---|
| FLOWTRAN simulator | A FORTRAN program translating a user's flow-sheet description into a series of `CALL` statements; compiled and run against a library of pre-built blocks. Sequential-modular, with recycle convergence by bounded Wegstein. |
| PROPTY | Companion program: fits Antoine, ideal-gas $C_p$ and other property correlations from raw experimental data, writing parameters into the property file. |
| VLE | Companion program: fits liquid-phase activity-coefficient parameters (van Laar and others) from VLE data. |
| INF | Information-retrieval program; reads/writes the property file accessed by FLOWTRAN. |

The unit-operation block library, as published in Table 3 of [124], contained approximately fifty-eight blocks across eleven categories:

| Category | Blocks | Count |
|---|---|---|
| Flash | `IFLSH`, `AFLSH`, `BFLSH`, `KFLSH`, `FLSH3` | 5 |
| Absorption / stripping | `ABSBR` (rigorous) | 1 |
| Extraction | `EXTRC` (rigorous LLE) | 1 |
| Heat exchange | `EXCH1/2/3`, `HEATR`, `CLCN1`, `DESUP`, `HTR3`, `BOILR` | 8 |
| Miscellaneous unit ops | `ADD`, `MIX`, `SPLIT` | 3 |
| Distillation | `FRAKB` (matrix), `DISTL`, `DSTWU`, `SEPR`, `AFRAC` | 5 |
| Reaction | `REACT`, `AREAC`, `XTNT` | 3 |
| Pump / compressor | `PUMP`, `GCOMP`, `MULPY` | 3 |
| Convergence | Bounded Wegstein (tear streams) | 1 |
| Control | `CNTRL`, `PCVB`, `DSPLT`, `RCNTL` | 4 |
| Cost & sizing | `CAFLH`, `CIFLH`, `CKFLH`, `CFLH3`, `CFRKB`, `CAFRC`, `CDSTL`, `CABSB`, `CTABS`, `CEXC1/2/3`, `CHETR`, `CCLN1`, `CPUMP`, `CCOMP` | 16 |
| Report writer | `SUMRY` et al. | 4 |
| Profit analysis | `RAWMT`, `BPROD`, `PRODT`, `PRBFT` | 4 |
| **Total** | | ∼58 |

The public physical-property database carried data for 180 chemical species [124] — twenty times the Choupo inventory of nine. By 1975 FLOWTRAN ran on the CDC 6000 series, the GE/Honeywell 600/6080 and the IBM 360/370 series; approximately seventy companies held licences; twenty-one universities were using the CACHE distribution in coursework.

## An honest LOC estimate

Rosen & Pauls (1977) does not publish a lines-of-code figure for FLOWTRAN, and no quantitative size measurement appears in any indexed source I could locate. The figure is presumably extractable from the FLOWTRAN System Collection at the Science History Institute (donated by Rosen in two accessions, 2006 and 2016), but that is a physical archive and not yet digitised in a form suitable for keyword search.

A defensible *estimate* can nevertheless be made from the information that Rosen & Pauls do publish. A typical FORTRAN-66 unit-operation subroutine of the era, with its property-table interface and convergence guards, runs to 200–400 lines. Taking the midpoint:

| Component | LOC (estimated) |
|---|---|
| 58 unit-operation blocks × 300 LOC mean | ∼17 000 |
| System shell (preprocessor, executor, recycle, I/O, error handling) | ∼ 8 000 |
| PROPTY + VLE + INF (three auxiliary FORTRAN programs) | ∼10 000 |
| **Total (estimated)** | **∼35 000 LOC FORTRAN-66** |

The figure above is the author's calibrated guess from the published architecture, not a primary citation. If the Science History Institute archive eventually yields a primary number, this appendix should be updated.

## Comparison with the current state

| Metric | FLOWTRAN (1977) | Choupo (2 |
|---|---|---|
| Unit-operation block types | ∼58 (incl. costing/profit) | |
| Components in the public database | 180 | |
| Activity-coefficient models | van Laar, NRTL, others | ideal, NRTL, W |
| Distillation methods | matrix (rigorous) + shortcuts | Wang-Henke bubble- |
| Recycle convergence | bounded Wegstein | bounded Weg |
| Parameter estimation | companion program (VLE fitting) | in-simulator Levenberg-Marquardt (`fitParameter` |
| Equipment sizing + costing | 20+ blocks built in | two equipment + Turton co |
| LOC (source) | ∼35 000 FORTRAN-66 (est.) | ∼10 000 C- |
| Equivalent C++ LOC | (FORTRAN-66 → modern C++ compresses ∼ 3×) ⇒ ∼12 000 | 10 000 (d |
| External dependencies | none (FORTRAN library) | none (C++ 17 standard lib |
| Licence | commercial (Monsanto), then CACHE-academic | GPL-3.0-or-later (open so |

In feature density per source line, Choupo is in the same ballpark as FLOWTRAN of fifty years ago. What FLOWTRAN had and Choupo does not (yet): a rigorous matrix-method distillation solver (`FRAKB`); rigorous absorption (`ABSBR`) and extraction (`EXTRC`) blocks; dedicated pump and compressor sizing/cost blocks; a profit-

analysis pass. Each of these is a candidate item for the roadmap.

What Choupo has and FLOWTRAN did not: an embedded modern parameter-estimation engine (`fitParameters`) inside the simulator rather than as a companion program; a Web-based GUI with flowsheet canvas, plots and a WASM solver; an open-source licence that lets a student modify any layer; auditable transparency in the manuals and the source.

## What survives

The Rosen donation (Science History Institute, accessions 2006 and 2016) preserves the FLOWTRAN source as paper listings — specifically Box 1 Folder 17 (PROPTY, 1974–78), Box 1 Folder 18 (simulator core, 1986 and 1993), Box 2 Folder 4 (VLE, undated) — and as physical FORTRAN punch cards in Series V. The collection is meticulously curated for research access but has not been digitised: there is no machine-readable copy of FLOWTRAN in circulation. Copyright passed from Monsanto to Solutia (1997), to Pharmacia/Pfizer (2002), to the new Monsanto Agriculture, and on to Bayer's acquisition in 2018. Reviving FLOWTRAN as runnable code would require systematic OCR of the listings, transcription against the punch-card decks, a port to modern GFortran, and a copyright clarification with Bayer. Choupo does not attempt this revival — it is an independent re-implementation of the same architectural pattern in modern C++.

## The fate is not unique

FLOWTRAN's outcome is the rule, not the exception, for chemical process simulators of the mainframe era. PACER (P. T. Shannon, Dartmouth, mid-1960s), CHESS (R. L. Motard, 1968 — eventually re-implemented commercially), PROCESS (Simulation Sciences of Los Angeles, 1966), GEMCS and CONCEPT (both reviewed alongside FLOWTRAN by Peters & Barker, 1974), and HYSIM (Hyprotech, Calgary, 1991): every foundational chemical-process simulator of the mainframe era either disappeared or was absorbed into a commercial product whose source remained closed. None was open-sourced before the original mainframe was decommissioned. The one notable exception is ASCEND (Carnegie Mellon, 1978–), an equation-oriented modelling environment that was open source from inception and is still actively developed half a century later, although with limited adoption outside its research community.

The historical lesson is sobering and motivating in equal measure: the architectural primitives that make modern process simulation work — streams, blocks, the outer recycle loop, the physical property file — were already in place in 1966; the lineage spans the entire C++ era and the FORTRAN-66 era both; and yet the machine-readable code that implemented those primitives in production has been disappearing for fifty years, one decommissioned mainframe at a time. Choupo, by being on git under GPL-3.0-or-later from day one, takes the bet that machine-readable persistence is itself a pedagogical goal — not a side effect.

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*Colophon.* This manual was typeset in Latin Modern with the Choupo shared LaTeX preamble. Sources live in `docs/`; rebuild with `cd docs && make`. The current edition covers, in eight parts, the foundations of vapour–liquid equilibrium, the activity-coefficient models (ideal, NRTL, Wilson), the numerical methods (Newton 1-D and $n$-D, Levenberg-Marquardt, Wegstein, Nelder-Mead, RK4), the selected unit operations (isothermal flash with VL / LL / VLLE branches, CSTR, PFR, Gibbs reactor by free-energy minimisation, Wang-Henke distillation), the equipment-sizing + Turton-costing chain that closes the loop from a converged flowsheet to a EUR figure, and — — the dynamic simulation classes: batch reactors (isothermal + adiabatic + multi-reaction), Rayleigh batch distillation, continuous dynamic CSTR, and the feedback control layer with the ideal PID controller. Five binaries (`choupoSolve`, `choupoBatch`, `choupoSemiContinuous`, `choupoCtrl`, `choupoProps`) ship with this edition, one per problem class. Appendix C places Choupo in its historical lineage, comparing it block-by-block with Monsanto's FLOWTRAN (1966–1993) following the description by Rosen & Pauls [124].

# Part VIII — Dynamic simulation and process control

Parts I–VII solved steady problems: find the state $\mathbf{x}^\star$ such that $\mathbf{F}(\mathbf{x}^\star) = \mathbf{0}$. A whole other class of chemical-engineering questions asks instead: starting from a known state $\mathbf{Y}(0)$, what is $\mathbf{Y}(t)$ for $t > 0$? Batch reactors, simple batch stills, continuous reactors with finite hold-up and control loops — none of these have a steady state to find; they tell a story over time. Choupo ships three extra binaries for this class, `choupoBatch` (closed vessels with recipe events), `choupoSemiContinuous` (continuous flow with no control loop – start-up, disturbance, fed-batch, feed & bleed) and `choupoCtrl` (the same continuous flow with controllers; the two share one driver), all built on the same shared library used by `choupoSolve`.

The integration scheme is the same explicit RK4 derived in Section 9 — only the state vector and the right-hand side change.

# 1 Time-stepping with a packed state vector

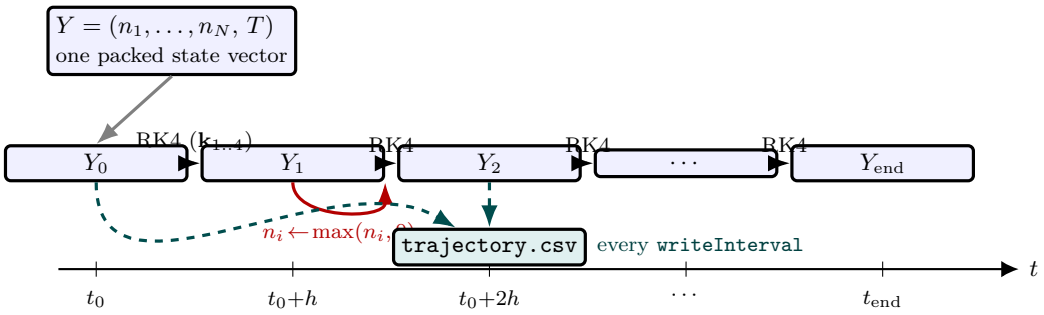



**Figure 82:** Time-marching with a packed state vector (Eq. 550). Every state variable of the vessel — the $N$ component mole numbers *and* the temperature — is packed into one vector $Y$, and the generic RK4 stencil (Figure 46) advances it one fixed step $h =$ `deltaT` at a time along the instant ladder $Y_0 \to Y_1 \to \ldots$ After each step the mole numbers are projected onto the non-negative orthant ($n_i \leftarrow \max(n_i, 0)$, red) to absorb the small negative drifts RK4 produces near depletion. The solver does not store every instant: it *samples* the trajectory to `trajectory.csv` every `writeInterval` seconds (green). The same machinery serves the isothermal batch ($Y = \mathbf{n}$), the adiabatic batch ($Y = (\mathbf{n}, T)$, Section 2) and the dynamic CSTR (Section 4) — only $f(Y)$ changes.

> *What you'll learn.* The RK4 chapter wrote the PFR as $d\mathbf{x}/dV = f(\mathbf{x}, T)$ with $V$ playing the role of the independent variable. In a closed batch vessel, the same RK4 integrates $dY/dt = f(Y, t)$ where $Y$ packs *every state variable* of the vessel. The dimension of $Y$ grows with the number of components and the energy balance, but the RK4 stencil is component-by-component identical.

For a single closed vessel with $N$ components reacting under $R$ reactions at isothermal conditions, the state is just the component mole numbers $\mathbf{n} = (n_1, \ldots, n_N)$ and the right-hand side is the standard rate-balance

$$\frac{dn_i}{dt} \;=\; V \sum_{r=1}^{R} \nu_{i,r}\, r_r(T, \mathbf{n}), \qquad i = 1, \ldots, N, \tag{549}$$

with $r_r$ in kmol/(m$^3 \cdot$ s) and $V$ the constant-density liquid volume. In adiabatic mode the temperature joins as state variable $N{+}1$ and the right-hand side gains an energy balance (Section 2). In a continuous dynamic CSTR (Section 4) the right-hand side gains inflow and outflow terms. In every case Choupo builds a single "packed state vector" $Y = (n_1, \ldots, n_N, T)$ (or just $\mathbf{n}$ in isothermal mode) and lets a generic RK4 step

$$\begin{aligned} \mathbf{k}_1 &= f(Y_n) \\ \mathbf{k}_2 &= f(Y_n + \tfrac{h}{2}\,\mathbf{k}_1) \\ \mathbf{k}_3 &= f(Y_n + \tfrac{h}{2}\,\mathbf{k}_2) \\ \mathbf{k}_4 &= f(Y_n + h\,\mathbf{k}_3) \\ Y_{n+1} &= Y_n + \tfrac{h}{6}\big(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\big) \end{aligned} \tag{550}$$

do the work. The step size $h$ is `deltaT` in `controlDict`; output is sampled to `trajectory.csv` every `writeInterval` seconds.

The mole numbers are projected onto the non-negative orthant after each step ($n_i \leftarrow \max(n_i, 0)$) to absorb the small negative drifts that RK4 produces near depletion; an early-stop on $|\mathbf{n}| \to 0$ keeps the still chapter (Section 3) from dividing by zero in its bubble-T call.

# 2 Adiabatic batch reactor: coupled mass + energy balance

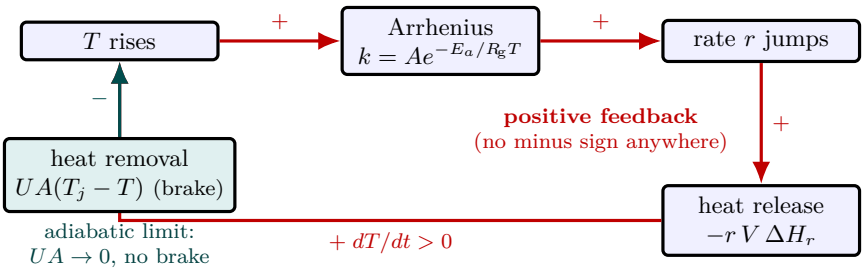



**Figure 83:** The thermal-runaway feedback loop of an adiabatic batch reactor. A rise in $T$ lifts the Arrhenius rate constant $k$ exponentially, which raises the reaction rate $r$, which releases more heat, which raises $T$ further — every arrow carries a "+", so the loop is *positive* feedback with no restoring term. The only brake is heat removal $UA(T_j - T)$ (green); in the adiabatic limit $UA \to 0$ it vanishes and, once the "thermal Jacobian" $J_{T,T} \propto E_a/(R_g T^2 \sum_i n_i C_{p,i})$ exceeds the (here zero) loss term, the system has no stable steady state and *must* run away. The case `batch02_adiabatic` sits on this side: $\sim 200\,\mathrm{s}$ of quiescence while $r$ is tiny, then the Arrhenius factor catches and $T$ climbs $345 \to 482\,\mathrm{K}$ in $\sim 100\,\mathrm{s}$.

> *What you'll learn.* Adding the energy balance to a batch reactor is what gives the classical "thermal runaway" picture: a mild exotherm that lifts $T$ at $\sim 1\,\mathrm{K/s}$ in the early phase, kinetics that double every $\sim 10\,\mathrm{K}$, and a self-amplifying ascent until the reactor empties of reactants or hits a heat-removal limit. Choupo integrates the balance trivially because RK4 already supports a packed state vector — only the right-hand side $f(Y)$ has to know about $T$.

## The energy balance

For a perfectly mixed, constant-volume, well-insulated vessel (adiabatic limit, no jacket), the per-mole heat capacity of the liquid mixture is $\sum_i n_i\, C_{p,i}^{L}(T)$ in kJ/K (with $n_i$ in kmol and $C_p$ in J/(mol K) — the factor of 1000 cancels because of the kmol $\leftrightarrow$ mol ratio). An exothermic reaction releases $-\Delta H_r$ per mole of limiting reactant consumed. The energy balance becomes:

$$\Big(\sum_{i=1}^{N} n_i\, C_{p,i}^{L}(T)\Big)\,\frac{dT}{dt} \;=\; -\sum_{r=1}^{R} r_r(T,\mathbf{n})\,V\,\Delta H_r. \tag{551}$$

The sign convention is the standard one: $\Delta H_r < 0$ for an exothermic reaction, so the RHS is positive and $dT/dt > 0$. The heat-of-reaction is the elements-datum sum $\Delta H_r(T) = \sum_i \nu_i\, h_i(T)$, each $h_i$ carrying its $\Delta h_{f,i}^{\circ}$ (§6.3), resolved through the same shared `reactionHeat()` every steady reactor uses — evaluated at the live $T$, so the Kirchhoff temperature dependence rides in automatically through the $c_p$ integrals. A `dH_rxn` entry in `constant/reactions` is *not* a primary input: it is an announced, cross-checked override honoured only when a species deliberately lacks formation data, and the engine warns when both exist and disagree.

The full state derivative is, packed:

$$Y = (n_1, \ldots, n_N, T), \qquad f_i(Y) = V \sum_r \nu_{i,r}\, r_r(T,\mathbf{n}), \qquad f_{N+1}(Y) = \frac{-\sum_r r_r\, V\,\Delta H_r}{\sum_i n_i\, C_{p,i}^{L}(T)},$$

with $C_p$ evaluated from the polynomial `liquidHeatCapacity` block in each component file.

## The runaway signature

A linear stability analysis of (549) + (551) around a stationary point $(\mathbf{n}^*, T^*)$ gives a Jacobian with the block

$$J_{T,T} \;=\; \frac{\partial}{\partial T}\Big[\frac{-r(T,\mathbf{n}^*)\,V\,\Delta H}{\sum_i n_i^*\, C_{p,i}}\Big] \;\propto\; \frac{E_a/RT^{*2}}{\sum_i n_i^*\, C_{p,i}^{L}(T^*)}$$

for first-order Arrhenius kinetics. When this positive "thermal Jacobian" exceeds the heat-loss term, the system has no stable steady state and *must* runaway. The pedagogical case `tutorials/batch/reactor/batch02_adiabatic` sits on the runaway side: $T_0 = 320\,\mathrm{K}$, $k(T_0) \approx 3.7 \times 10^{-4}\,\mathrm{s}^{-1}$, $\tau_0 \approx 2700\,\mathrm{s}$. The first $\sim 200\,\mathrm{s}$ look quiescent

(the small $r$ feeds back a $\sim 0.01\,\mathrm{K/s}$ heat rate), then the Arrhenius factor catches and the reactor runs from 345 to 482 K between $t = 230\,\mathrm{s}$ and $t = 330\,\mathrm{s}$.

The maximum adiabatic temperature rise from full conversion is $\Delta T_{\mathrm{ad}} = |\Delta H|/\bar{C}_p$ where $\bar{C}_p$ is a molar-weighted average; for the 15 kJ/mol exotherm of `batch02_adiabatic` and an average Cp of $\sim 92\,\mathrm{J/(mol\,K)}$ (the harmonic of pure ethanol's 112 and water's 75), this gives $\Delta T_{\mathrm{ad}} \approx 163\,\mathrm{K}$ — which is what the simulator reports to within 1 K.

## Multi-reaction case

When $R > 1$ reactions share the vessel, both (549) and (551) simply sum over $r$. The matrix $(\nu_{i,r})$ of stoichiometric coefficients does the species bookkeeping; each reaction's own $r_r$, $E_a$, $A$ and $\Delta H_r$ feed in independently. Choupo's `BatchReactor` holds a `std::vector<ReactionSpec>` and walks it inside `derivatives_`.

> **Runnable case.** batch/batch03_consecutive Consecutive A → B → C at 350 K with $k_1 = 10\,k_2$. The intermediate B accumulates and then decays as the second reaction overtakes the first; the maximum $n_B/n_{A,0} = 0.669$ at $t_{\max} = 200\,\mathrm{s}$ matches the closed-form $(k_1/k_2)^{k_2/(k_2-k_1)}$ and $\ln(k_1/k_2)/(k_1 - k_2)$ to 0.1 %.

# 3 Rayleigh batch distillation

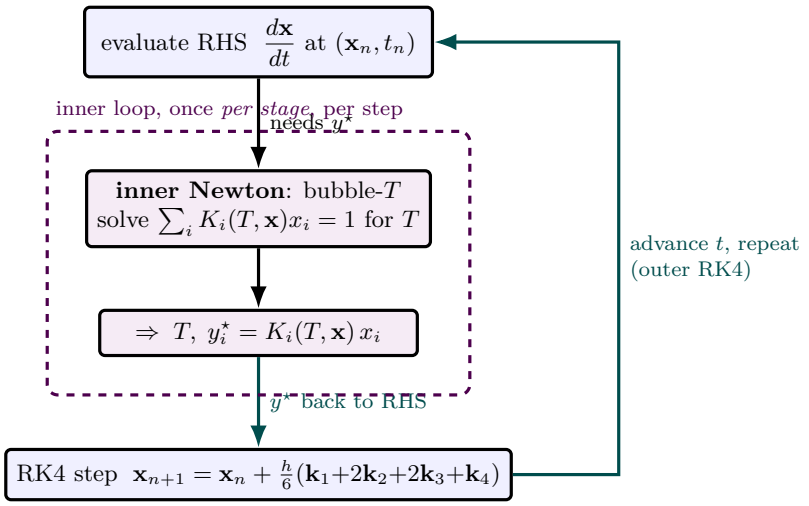



**Figure 84:** The nested-Newton pattern that recurs in every dynamic VLE problem, shown for Rayleigh batch distillation. The *outer* loop is RK4 marching the residue composition $\mathbf{x}$ in time (blue). But each RK4 stage needs the equilibrium vapour $y_i^\star = K_i(T,\mathbf{x})x_i$, and that requires the boiling temperature — so a full *inner* Newton iteration (violet) solves the bubble-point condition $\sum_i K_i(T,\mathbf{x})x_i = 1$ for $T$ before the right-hand side can even be evaluated (`BubblePoint::compute` used as a black box). The inner solve runs once *per stage*, four times per RK4 step; its result feeds the outer step, time advances, and the whole thing repeats. The same Newton-inside-RK4 structure drives the dynamic flash and every moving-VLE unit op.

> *What you'll learn.* The classical Rayleigh distillation strips a vessel of its more volatile components at a rate $F_{\text{vap}}$ [kmol/s] of vapour removed; mass balance fixes the residue composition over time as a function of the vapour-liquid equilibrium $y^\star = K(T,\mathbf{x})\,\mathbf{x}$. Choupo integrates the corresponding ODE with RK4 using `BubblePoint::compute` as a black-box bubble-T solver inside the right-hand side — a nested-Newton pattern that recurs in every dynamic VLE problem.

## The mass balance

For a closed pot at constant pressure $P$ from which vapour is removed at molar rate $F_{\text{vap}}$, component $i$ obeys

$$\frac{dn_i}{dt} \;=\; -\,F_{\text{vap}}\,y_i^\star(T,\mathbf{x}), \qquad \mathbf{x} = \mathbf{n}/\sum_j n_j, \qquad y_i^\star = K_i(T,\mathbf{x})\,x_i. \tag{552}$$

The temperature $T$ is determined algebraically each instant by the bubble-point condition $\sum_i K_i(T,\mathbf{x})\,x_i = 1$; the RHS of (552) therefore depends on $\mathbf{x}$ through both the direct $x_i$ in $y_i^\star$ and the implicit $T(\mathbf{x})$ in $K_i$. The classical *Rayleigh equation* is obtained by dividing through to eliminate $t$:

$$\frac{d\mathbf{x}}{dL} \;=\; \frac{\mathbf{x} - \mathbf{y}^\star(T,\mathbf{x})}{L}, \qquad L = \sum_i n_i,$$

which integrates by quadrature for ideal binaries (constant relative volatility $\alpha$) to

$$\ln\frac{L}{L_0} \;=\; \frac{1}{\alpha-1}\left[\ln\frac{x}{x_0} \;+\; \alpha\,\ln\frac{1-x_0}{1-x}\right].$$

Choupo integrates (552) directly with RK4 rather than reaching for the closed form — the same code runs ideal and NRTL cases without change.

## The warm-seeded bubble-T inside the integrator

Each RK4 stage evaluation has to call `BubblePoint::compute` on the current $\mathbf{x}$. Choupo caches the previous step's converged $T_{\text{bub}}$ and feeds it back as the initial guess for the next call. Because $\mathbf{x}$ changes slowly (the per-step composition jump is $O(F_{\text{vap}}\,h\,/\,L)$, typically $< 10^{-4}$) the warm-seed converges in 2–3

Newton iterations. This is the same warm-start pattern used by every multi-step ODE integrator that embeds an algebraic solver.

The pattern surfaced a long-standing bug in `solver::newton1D`: when the warm seed is already within tolerance the routine used to take one zero-length Newton step, find $\Delta x \approx 0$, and then *bisect* into the centre of the bracket because of an `x_new == x_hi` edge condition. The fix — check $|f| < \text{tol}$ before any state mutation — ships from.

## Azeotropes from below: the Rayleigh barrier

The `still02_ethanol_water_azeo` tutorial starts at $x_{\text{eth}} = 0.40$ (below the 0.894 minimum-boiling azeotrope at 1 atm). In this regime $y^{\star}_{\text{eth}} > x_{\text{eth}}$, so ethanol leaves faster than water, and the pot composition drifts *monotonically toward pure water*, ending at $x_{\text{eth}} \approx 3 \times 10^{-11}$ with $T \to 372.4\,\text{K}$ (within 1 K of pure water's $T_b$). A Rayleigh still simply cannot cross the azeotrope from below; it can only walk further away from it. Crossing the azeotrope requires a column with reflux, an entrainer (extractive distillation), or a pressure swing.

> **Runnable case.** batch/still01_benzene_toluene 50/50 benzene/toluene at 1 atm, $F_{\text{vap}} = 1.5 \times 10^{-6}$ kmol/s for 600 s. Residue at 90 % distilled: $x_{\text{benzene}} = 0.082$. Closed-form Rayleigh with the mean Antoine-based $\alpha \approx 2.4$ predicts 0.080 — agreement at the 2 % level.

## Batch rectification: the pot grows a column

The Rayleigh still wastes what the vapour knows: $y^{\star}$ is richer than the pot, but every mole of it leaves. The classical batch rectifier (`model rectifier;` on the same `batchStill`) inserts $N$ ideal equilibrium stages between the pot and a *total* condenser, and returns a fraction of the condensate as reflux. With stage and condenser hold-up neglected — the standard textbook idealisation — the column is *quasi-steady*: at every instant it is an algebraic system riding on the slow pot ODE.

Let $V$ be the boilup leaving the pot (the dict's `F_vap` in this model), $R$ the reflux ratio. A total condenser splits $V$ into distillate product $D = V/(R+1)$ and reflux $L = R\,D$. The pot then loses *product only*:

$$\frac{\mathrm{d}n_i}{\mathrm{d}t} \;=\; -\,D\,x_{D,i}(t), \tag{553}$$

with $x_D(t)$ solved at each instant from the cascade

$$\begin{aligned} y_0 &= K(T_{\text{bub}}(x_B))\;x_B && \text{(pot vapour)} \\ V\,y_{s-1} &= L\,x_s + D\,x_D, && s = 1\ldots N \quad \text{(section balance)} \\ y_s &= K(T_{\text{bub}}(x_s))\;x_s && \text{(stage equilibrium)} \\ x_D &= y_N. && \text{(total condenser)} \end{aligned} \tag{554}$$

The unknown is the *vector* $x_D$; Choupo iterates the closure $x_D \leftarrow y_N$ and declares the single residual $\max_i |y_{N,i} - x_{D,i}|$, with the section balances holding algebraically at every iterate (so $\sum_i x_{s,i} = 1$ follows from $\sum_i y_i = \sum_i x_{D,i} = 1$ — nothing is ever renormalised into looking converged). Two numerical honesty notes, both learnt the hard way: the closure tolerance sits at $10^{-7}$, just *above* the $10^{-8}$ Newton floor of each stage's bubble-$T$ (a $10^{-10}$ demand can never close — it chases the stages' round-off); and an infeasible iterate (a negative stage composition, which the natural first guess $x_D = y_0$ produces by over-enriching) is projected onto the cascade's own last vapour, which corrects toward the reachable set from either side.

Two limits anchor the model. With $R = 0$ or $N = 0$ no liquid runs down the column, $x_D = y_0$ and eq. (553) degenerates to the Rayleigh still exactly — the regression suite pins the equivalence to the last digit. As $R \to \infty$ at fixed $N$ the cascade approaches total reflux, where the Fenske relation [114] for an ideal binary gives

$$\frac{x_D}{1-x_D} \;=\; \alpha^{\,N+1}\,\frac{x_B}{1-x_B} \tag{555}$$

($N$ stages + the pot are $N{+}1$ equilibrium contacts). For benzene/toluene at $x_B = 0.5$ with $\alpha = 2.35$–$2.6$ the band is $x_D = 0.968$–$0.979$; the solver at $R = 200$, $N = 3$ lands at 0.9765 — inside, computed by a route (per-stage Antoine bubble-$T$) that shares nothing with eq. (555).

At *finite* reflux the constant-$R$ policy trades purity for time: as the pot strips, $y_0$ leans and the whole cascade leans with it, so $x_D(t)$ drifts downward while the receiver accumulates the run average — two different compositions, deliberately both reported (the trajectory carries the instantaneous $x_{D,i}$, $R$ and $D$; the

receiver holds the mean). The complementary policy (`refluxPolicy constantComposition`) *holds* $x_D$ of a declared light key at a target by solving $R(t)$ each step: a *bracketed* solve of $f(R) = x_{D,\mathrm{LK}}(R) - x_D^{\mathrm{target}}$ over $[0, R_{\max}]$ on a feasibility map — never a bare Newton, because $x_D(R)$ can lose monotonicity (an azeotrope) and the solve must refuse an ambiguous branch rather than jump one silently. Three infeasibilities get three distinct diagnostics: a target *below* the $R = 0$ floor (even zero reflux over-purifies); a target unreachable from the charge at $R_{\max}$ (an azeotrope caps the cascade at *any* reflux — ethanol/water refuses a 0.95 target against its 0.894 barrier by computed VLE, not by a message); and the mid-campaign exhaustion, where the leaning pot drags the reachable set below the target and the run announces `onRefluxLimit stopDistillation`: $D = 0$, the pot inventory freezes, and the campaign idles *visibly* to `endTime` with the event in the trajectory and the KPIs — never a magic termination. $R(t)$ is a control input updated once per step at the committed state (first order in $\Delta t$, announced as such). No reboiler duty is reported by this model: $V$ alone does not prove an energy balance, and a $V \cdot \lambda$ shortcut would be a decorative number.

> **Runnable case.** batch/still04_rectifier_benzene_toluene The still01 pot grown a column: $N = 3$, $R = 3$, $V = 1.5 \times 10^{-6}$ kmol/s at 1 atm. Initial distillate $x_D = 0.949$ (vs. 0.71 for the raw pot vapour) drifting to 0.906 over 600 s; the receiver accumulates 0.931; the pot loses exactly $D\,t = 0.225$ mmol — three quarters of the boilup refluxes.

> **Runnable case.** batch/still05_rectifier_constant_xD The same column under the second policy: $x_D$ held at $0.9300 \pm 10^{-4}$ while $R$ climbs $2.0 \to 7.9$; at $t = 863$ s the ceiling $R_{\max} = 8$ can no longer reach the target and distillation stops — the receiver ends with the WHOLE cut on-spec at 0.930, the policy's selling point over still04's drifting purity.

## Campaign accounting: the material ledger and the mass balance

Batch vessels are materially CONNECTED: a recipe `transfer` is a discrete material edge, a `dischargeTo` route a continuous one. Choupo records every such edge as a structured *ledger* entry — $t_{\mathrm{start}}$, $t_{\mathrm{end}}$, source, destination, the per-component amount moved, and the transported enthalpy

$$H_{\mathrm{moved}} \;=\; \sum_{\text{packages } p} n_p\, h(T_p, x_p)\,, \tag{556}$$

integrated *package by package at each package's own temperature* (elements datum). A single end-temperature stamped on an accumulated amount would destroy exactly the information an energy balance needs — a still's condensate leaves at a $T$ that climbs all campaign. The timeline and the Gantt view are *projections* of this ledger, never independent records. A vessel that sheds material with no declared receiver is an EXTERNAL outlet — an unrouted product stream, recorded and named, never a hidden loss.

The transported enthalpy carries an explicit *validity*: a package containing a species with no formation datum cannot be priced, and the record then withholds $H_{\mathrm{moved}}$ entirely and names the species. Validity is *monotonic in time* — once one package was unpriceable the integral has lost a parcel forever, and a later priceable package must never resurrect it; an emitted enthalpy that silently dropped part of its own history would be worse than no number at all. The campaign reports the census (how many ledger records carry a valid $H$) as a first-class verdict.

> **Runnable case.** batch/still06_ledger_mixed_validity Two stills poison their distillate records in opposite temporal orders (unpriceable → priceable and the reverse); both records must end invalid, naming the datum-less species, with no partial integral emitted.

The material ledger has an energy twin. Each vessel contributes *energy records*: one per segment of constant physics — a recipe `setParameter` or a material charge closes the running segment, because the record's $h_i(T)$ belong to one temperature and the state difference it prices must be purely reactive. The isothermal reactor's jacket duty is the flagship: for a closed vessel at constant pressure the first law gives $Q = \Delta H$ over the segment, and on the elements datum Hess's law collapses the per-reaction story into a state difference,

$$Q \;=\; \sum_j \Delta H_{\mathrm{rxn},j}\,\Delta\xi_j \;=\; \sum_i \Delta n_i\, h_i(T), \tag{557}$$

so the integral is *exact* — computed from committed mole numbers, never a quadrature riding on the integrator's step size. When a lumped species carries no formation data but the reaction declares its heat, the duty falls back to the announced-override branch: extents are solved from the stoichiometry

($N\boldsymbol{\xi} = \Delta\mathbf{n}$, refused aloud if the reaction set is linearly dependent) and priced per reaction. Neither datum nor declaration — the record is invalid and names the reaction.

Material transfers respect the same first law: a recipe charge solves its mix temperature from *enthalpy equality* on the elements datum,

$$H(\mathbf{n}_1 + \mathbf{n}_2,\, T_{\mathrm{mix}}) \;=\; H(\mathbf{n}_1, T_1) + H(\mathbf{n}_2, T_2), \tag{558}$$

by a 1-D Newton warm-started at the molar average — so charging is a material act, never a thermal one, and internal transfers cancel *exactly* in the campaign sum. Heat-capacity differences matter: a hot ethanol charge into cold water lands ∼4 K above the equal-$C_p$ molar average. When a species cannot be priced the charge falls back to the molar average, *announced* — and that named occasion is enough to withhold the balance verdict below.

The campaign *energy balance* follows the same honesty rule as the mass balance, with one addition: it only claims a verdict when *every* piece is ledgered. A vessel type whose duty physics has not landed yet, an instantaneous `setParameter` temperature jump, a charge that fell back to the molar-$T$ average, or an external outlet whose transported enthalpy could not be priced makes the balance report UNAVAILABLE, naming each missing piece — never a decorative closure. When it does close, it closes over everything that moved: $\sum_{\mathrm{vessels}} \Delta H = Q_{\mathrm{ledger}} - H_{\mathrm{external\,out}}$, with the closure scaled by the total enthalpy that MOVED ($\sum |\Delta H_{\mathrm{vessel}}|$), so a pure transfer campaign whose signed sum cancels to zero still reports its closure honestly instead of a 0/0.

> **Runnable case.** batch/batch08_isothermal_duty The override branch end-to-end: lumped $A \to B$ with a declared $\Delta H_{\mathrm{rxn}} = -50\,\mathrm{kJ/mol}$ prices the jacket duty $E = \Delta H\,\Delta\xi = -49.88\,\mathrm{kJ}$ against the analytic first-order decay, while the balance stays unavailable naming the datum-less species.

> **Runnable case.** batch/recipe04_charge_enthalpy The H-equality charge: hot ethanol into cold water mixes to 317.06 K (the enthalpy-true temperature, ∼4 K above the molar average), and the campaign balance CLOSES across the transfer at $2 \times 10^{-12}$ — the first law surviving a recipe action.

The still's duties follow the same discipline: no invented $V\lambda$, only phase-enthalpy differences of the *actual* packages. For the Rayleigh pot the first law gives the reboiler directly,

$$Q_{\mathrm{reb}} \;=\; \Delta H_{\mathrm{pot}} + \sum_p n_p\, h^{\mathrm{vap}}(T_p, \mathbf{y}_p), \qquad Q_{\mathrm{cond}} \;=\; -\sum_p n_p \left[h^{\mathrm{vap}} - h^{\mathrm{liq}}\right](T_p), \tag{559}$$

each shed package priced at its own hand-off temperature — so the reboiler integral is a state difference plus a package sum, exact. The rectifier's condenser is the physical one — it condenses the whole boilup $V$, not the product $D$, at the top-stage temperature — and its reboiler closes the box first law as the exact residual. With that, a full react-then-distil campaign closes its energy balance end to end: reactor duty, charge, distillation and the unrouted product all on one elements-datum ledger.

> **Runnable case.** batch/recipe01_react_then_distill The showcase, now energy-closed: esterification duty, the reactor-to- still charge, reboiler/condenser duties and the external distillate balance to $6 \times 10^{-16}$.

An instantaneous `setParameter` temperature change is an energy *intervention*, and the ledger prices it as such: an `impulse` record with $E = H(\mathbf{n}, T_{\mathrm{new}}) - H(\mathbf{n}, T_{\mathrm{old}})$ on the same elements datum — a state difference including any phase leg, never a $\sum nC_p\Delta T$ shortcut when enthalpy data exist. A staged temperature programme thus reads off the ledger as its full story: reaction segments at each held temperature, with the jump energies in between, and the balance closes across all of them.

The ledger closes into *utilities*: every valid duty record carries the process temperature its service must beat — the segment's *worst-case* end (the hottest pot state for a reboiler, the coldest condensing temperature for a coolant) — and the campaign allocates each record to the lowest-grade catalogue utility that still serves it, the same tightest-margin rule the steady allocation uses, through one shared picker. Mass consumption uses the catalogue's phase-aware duty per kilogram (condensing steam is latent heat, never a $c_p\Delta T$), and the campaign cost is simply $\sum |E| \times$ price. A record no utility can serve is *listed*, never silently dropped.

The batch crystalliser's latent duty is another exact state difference: $Q = -\Delta H_{\mathrm{cryst}}\,\Delta n_{\mathrm{cryst}}$ with $n_{\mathrm{cryst}} = V\rho_c k_v \mu_3 / M$ read off the third moment, and $\Delta H_{\mathrm{cryst}}$ resolved through the *same* shared resolver the steady

crystalliser uses (ion-derived `dissolutionEnthalpy`, the molecular solution tier, or a declared constant that confesses itself) — the source travels in the record's basis, and steady and batch can never diverge on the number. The vessel-enthalpy half is an honestly *named* gap: a magma is dissolved solute plus crystal, and until inventories price on the solid rung the campaign balance reports it unavailable, verbatim.

The campaign closes on inventories. Each vessel reports its ALL-phase per-component inventory (a crystalliser adds its crystal magma $V\rho_c k_v \mu_3/M$ to the dissolved holdup — without it, crystallisation reads as mass loss), and

$$m_0 \;=\; m_{\text{final}} \;+\; m_{\text{external}}, \tag{560}$$

in kilograms via the component molar masses, because *reactions conserve mass, not moles.* For reacting campaigns the sharper invariant is the ELEMENTS: formulas are parsed against the periodic table (a lumped teaching species whose "formula" is `A` makes no elemental claim) and each element must close independently — a chemically wrong stoichiometry can hide inside a closed total mass, but not inside a closed oxygen balance. Two honest findings shipped with this machinery: catalogue molar masses from mixed sources drift from the stoichiometry at the $10^{-5}$ level ($\sum_j \nu_j M_j \neq 0$ exactly), reported as a curation note; and a genuine violation reports the run NON-CONVERGED — a leak is a failure, not a log line.

> **Runnable case.** batch/recipe01_react_then_distill The react-then-still campaign: one discrete transfer at $t = 400\,\text{s}$, the still's unrouted distillate recorded as an external outlet (0.889 kg), and the campaign closing to $10^{-15}$ with every element independently balanced.

# 4 Continuous dynamic CSTR with finite hold-up

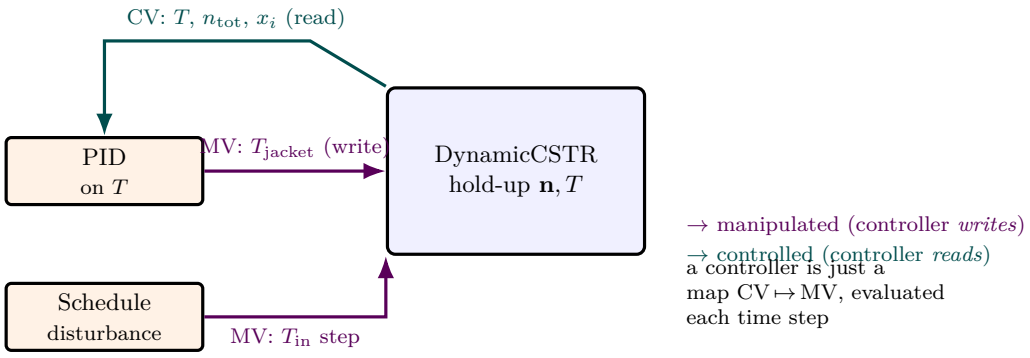



**Figure 85:** How a controller couples to a dynamic unit op, the Choupo `Controller` contract. The unit (`DynamicCSTR`) exposes a set of *controlled variables* the controller may read — here the reactor temperature $T$, total holdup, and composition $x_i$ (green, "measurements") — and a set of *manipulated variables* a controller may write each step ($T_{\text{jacket}}, T_{\text{in}}, F_{\text{in}}$; violet). A `PIDController` reads $T$ and writes $T_{\text{jacket}}$ to hold setpoint; a degenerate `ScheduleController` ignores the measurement entirely and writes a piecewise-constant MV (e.g. a $\pm 15\,\text{K}$ step in $T_{\text{in}}$) to inject scripted disturbances. Both plug into the *same* infrastructure because a controller is nothing more than a map (controlled variables) ↦ (manipulated variables) evaluated once per time step, between RK4 steps.

> *What you'll learn.* A continuous stirred-tank reactor with finite liquid hold-up (*not* the steady CSTR of Section 6) has the same mass balance as a batch reactor plus convective in/out terms. Closed-form steady-state results recover for $dn_i/dt = 0$; the transient between two steady states is governed by the residence time $\tau = n_{\text{total}}/F_{\text{in}}$. This unit is the basic vehicle for studying control: it is non-trivial enough to have interesting closed-loop behaviour, simple enough that students can hand-derive the open-loop response.

## Mass and energy balance

For a constant-volume liquid reactor with continuous inlet at $(F_{\text{in}}, T_{\text{in}}, \mathbf{z}_{\text{in}})$ and outlet at composition matching the tank ($x_i = n_i / \sum_j n_j$), with $R$ reactions and a jacket exchanging heat at rate $Q_{\text{j}} = UA\,(T_{\text{jacket}} - T)$ in W (so $Q_{\text{j}}/1000$ in kJ/s for the units used in Section 2):

$$\frac{dn_i}{dt} = F_{\text{in}}\, z_{\text{in},\,i} \;-\; F_{\text{out}}\, x_i \;+\; V \sum_r \nu_{i,r}\, r_r(T, \mathbf{n}), \tag{561}$$

$$\Big(\sum_i n_i\, C_{p,i}^{L}\Big)\frac{dT}{dt} = F_{\text{in}}\, \bar{C}_{p,\text{in}}\,(T_{\text{in}} - T) \;+\; \frac{UA}{1000}\,(T_{\text{jacket}} - T) \;-\; \sum_r r_r\, V\, \Delta H_r. \tag{562}$$

With constant volume (incompressible liquid) the overall mass balance fixes $F_{\text{out}} = F_{\text{in}}$ — there is no accumulation of mass — but the *component* balance does accumulate because composition $x_i$ tracks $n_i$.

The state is again packed as $Y = (n_1, \ldots, n_N, T)$ and RK4 (550) integrates it directly. Choupo's `DynamicCSTR` exposes $T_{\text{jacket}}$, $T_{\text{in}}$ and $F_{\text{in}}$ as *manipulated variables*: a controller or a recipe event can write to any of them at run time. The current $T$, $n_{\text{total}}$ and $x_i$ are exposed as *controlled variables*.

## Open-loop steady state

Setting (561)–(562) to zero gives the steady state $(\mathbf{n}^*, T^*)$ at the operating point. For first-order kinetics in the limiting reactant $A$, the component balance simplifies to

$$F_{\text{in}}\,(z_{\text{in},A} - x_A^*) \;=\; k(T^*)\, x_A^*\, n_{\text{total}} \;\Longrightarrow\; x_A^* \;=\; \frac{z_{\text{in},A}}{1 + k(T^*)\,\tau}, \qquad \tau = n_{\text{total}}/F_{\text{in}}.$$

The classical CSTR formula recovers exactly — the dynamic simulator is the steady CSTR *plus* a transient envelope.

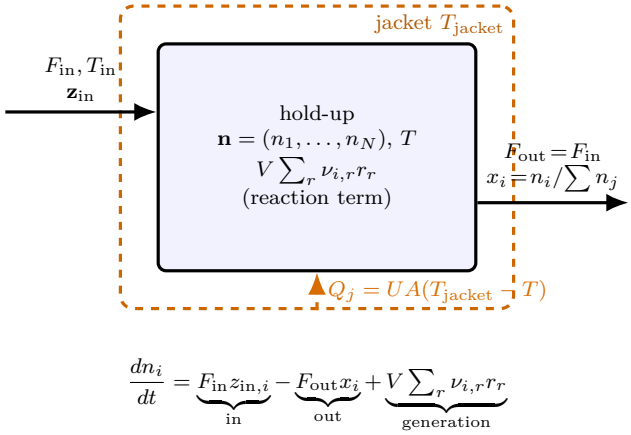



**Figure 86:** The continuous dynamic CSTR as a single 0-D well-mixed hold-up cell (`DynamicCSTR`). Unlike the *steady* CSTR of Section 6, the contents *accumulate*: the state $\mathbf{Y} = (\mathbf{n}, T)$ evolves in time. Three currents cross the boundary of each component balance — convective inflow $F_{\text{in}} z_{\text{in},i}$, convective outflow $F_{\text{out}} x_i$ at the *tank* composition (perfect mixing), and chemical generation $V \sum_r \nu_{i,r} r_r$ — and the jacket adds a heat current $Q_j = UA(T_{\text{jacket}} - T)$ to the energy balance. Constant liquid volume forces $F_{\text{out}} = F_{\text{in}}$, so total mass does not accumulate, yet *composition* does because $x_i$ tracks $n_i$. The manipulated variables a controller may write $(T_{\text{jacket}}, T_{\text{in}}, F_{\text{in}})$ are exactly the labelled inlet and jacket quantities.

## Time constants and what control can do

The residence-time constant of the mass balance is $\tau$; the thermal time constant is $\tau_T \approx \sum_i n_i C_p / (F_{\text{in}} \bar{C}_p + UA/1000)$. When the jacket coupling $UA$ is comparable to $F_{\text{in}} \bar{C}_p$, the thermal response is order-$\tau$ — fast enough that a well-tuned PID can hold $T$ to within a few mK of setpoint over disturbances of a few K (see Section 6).

> **Runnable case.** ctrl/ctrl01_cstr_temp_control Mildly exothermic compA → compB at 350 K setpoint. Open-loop the reactor settles at the natural steady state 320.8 K (just above the cool inlet at 320 K because of the mild exotherm). A PID on $T_{\text{jacket}}$ with $K_p = 4$, $K_i = 0.04$, $K_d = 0$ drives $T$ to 350 K and holds it to within $10^{-4}$ K at $t = 1500$ s.

# 5 The liquid tank with a level: integrating and self-regulating inventory

The vessel of the previous section holds its volume fixed. The most common control loop of a plant does not: it holds a *level*, and a level exists only if the liquid volume is free to move. `dynamicTank` is the minimal such vessel — a vertical cylinder of declared diameter $D$ and height $H_v$, a liquid, one feed, one outflow, no reaction. Its state is the inventory $n_i$ and the stored enthalpy $H$ (the temperature $T$ is a readout of $H(\mathbf{n}, T)$ on the package's mixture liquid surface, the same "mixture-$H$" route as `DynamicCSTR`):

$$\frac{dn_i}{dt} = \dot{n}_{\mathrm{in},i} - F_{\mathrm{out}}\, x_i, \tag{563}$$

$$\frac{dH}{dt} = F_{\mathrm{in}}\, h(\mathbf{z}_{\mathrm{in}}, T_{\mathrm{in}}) - F_{\mathrm{out}}\, h(\mathbf{x}, T) + \frac{UA}{1000}\,(T_j - T). \tag{564}$$

The volume is not declared but derived from the package, $V = \sum_i n_i M_i / \rho(T, P, \mathbf{x})$ (the liquid density of the case's package; a package that cannot price one refuses by name), and the level is

$$h \;=\; \frac{V}{\pi D^2/4}. \tag{565}$$

A tank that runs dry ($V \le 0$) or overflows ($h > H_v$) is *refused* at that instant, never clamped: the liquid that would spill is neither in the vessel nor in any stream.

**Two outflows, two behaviours.** The outflow is declared, and the two declarations are two different processes.

*Manipulated flow* (`mode flow`, a flow-controlled pump). $F_{\mathrm{out}}$ is the manipulated variable; with one component,

$$\frac{dh}{dt} \;=\; \frac{M}{\rho A}\,\big(F_{\mathrm{in}} - F_{\mathrm{out}}\big),$$

a pure *integrator*: nothing brings the level back, and a constant imbalance runs the tank dry or over the top. A level controller on $F_{\mathrm{out}}$ must be *reverse acting* (a level above the setpoint raises the outflow), which with the direct-acting $u = u_0 + K_p e + K_i \int e$, $e = \mathrm{SP} - h$, means negative gains. The closed loop is second order, $\omega_n^2 = a|K_i|$, $2\zeta\omega_n = a|K_p|$ with $a = M/(\rho A)$; a P-only controller settles with an offset $h - \mathrm{SP} = (F_{\mathrm{in}} - u_0)/|K_p|$ that the integral term removes.

*Valve on the outlet* (`mode valve`, manipulated variable the opening $x \in [0, 1]$):

$$Q = C_v\, x\, \sqrt{\Delta P/\rho}, \qquad \Delta P = P + \rho g h - P_{\mathrm{down}}, \qquad F_{\mathrm{out}} = Q\rho/M, \tag{566}$$

where $C_v$ is the valve's effective flow *area* in $\mathrm{m}^2$ (the orifice equation's $C_d A$) and the outlet sits at the bottom. Now the outflow grows with the level and the tank *regulates itself*: at steady state $F_{\mathrm{out}} = F_{\mathrm{in}}$ and

$$h^\star \;=\; \frac{(Q/C_v x)^2}{g} \;-\; \frac{P - P_{\mathrm{down}}}{\rho g}, \qquad Q = F_{\mathrm{in}} M/\rho, \tag{567}$$

approached with the time constant $\tau = A/(dQ/dh) = 2Ah^\star/Q$ for $P = P_{\mathrm{down}}$. Because $Q \propto \sqrt{h}$, the steady level grows with the *square* of the inflow. The arithmetic of (565)–(567) (volume, level, valve, the dry/overflow refusals) has one home, `VariableVolumeHoldup`, shared with the variable-volume vessels to come.

> **Runnable case.** unsteady/unsteady05_tank_valve_drain Water tank drained through a valve, no controller. A 50 % step in the inflow (0.20 to 0.30 kmol/s) takes the level from 0.998 m to the closed-form $h^\star = 2.245$ m of (567); the run reaches it by itself.

> **Runnable case.** ctrl/ctrl21_tank_level_control The same tank with a manipulated outflow and a PI level controller ($K_p = -2$, $K_i = -0.01$). The same inflow step is rejected and the level returns to the 2.0 m setpoint; with $K_i = 0$ it settles 0.050 m high.

# 6 Feedback control: the ideal PID, derivative-on-PV, anti-windup

*What you'll learn.* The PID controller is one of the oldest objects in process control, the workhorse for over 90 % of single-input single-output loops in industry. The mathematics is two paragraphs. But the *implementation* choices — whether the derivative term sees the error or the process variable, what to do when the actuator saturates, how to discretise the integral — decide whether the closed loop behaves nicely or oscillates pathologically. This section derives the ideal form and explains the two implementation choices Choupo makes (`derivative-on-PV` and `clamping anti-windup`).

## The ideal form

Let SP be the set point, $\mathrm{PV}(t)$ the measured process variable, $e(t) = \mathrm{SP} - \mathrm{PV}(t)$ the error, and $u(t)$ the controller output (the manipulated variable). The *ideal-form* PID is

$$u(t) \;=\; u_{\mathrm{bias}} \;+\; K_p\, e(t) \;+\; K_i \int_0^t e(\tau)\, d\tau \;-\; K_d\, \frac{d\,\mathrm{PV}(t)}{dt}. \tag{568}$$

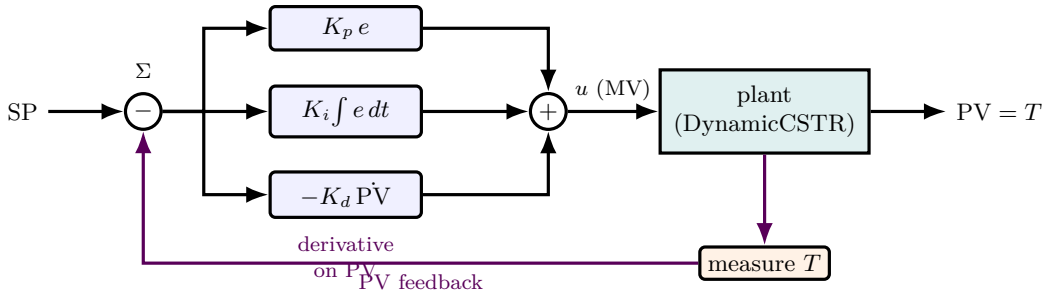



**Figure 87:** The ideal-form PID loop Choupo's `PIDController` implements (`src/control/PIDController`). The summing junction forms the error $e = \mathrm{SP} - \mathrm{PV}$; three parallel terms act on it — proportional ($K_p e$, the present), integral ($K_i \int e\, dt$, the accumulated past, killing offset) and derivative (the anticipated future) — and their sum is the manipulated variable $u$ driving the plant (here $T_{\mathrm{jacket}}$ of the dynamic CSTR). The measured $T$ is fed back (violet) to close the loop. Two Choupo choices are visible: the derivative tap reads PV directly, *not* $e$, so an operator's set-point step produces no derivative "kick"; and the integral term is governed by the anti-windup gate of Figure 88.

The *bias* $u_{\mathrm{bias}}$ is the controller's idle output when $e = 0$ — typically chosen to be the operating-point MV. Each of the three terms has a clear physical meaning:

- $K_p\, e$ — **proportional**: react to the present error. Larger $K_p$ means faster response and tighter tracking but more aggressive transients and a tendency to overshoot.
- $K_i \int e\, dt$ — **integral**: eliminate any residual offset. In a P-only loop, a non-zero $u_{\mathrm{bias}}$ mismatch yields a permanent steady-state error (called *offset*); the integral keeps accumulating error until offset is removed. Larger $K_i$ removes offset faster but adds phase lag, destabilising fast loops.
- $-K_d\, d\mathrm{PV}/dt$ — **derivative**: anticipate where the process is heading. Larger $K_d$ damps oscillation but amplifies measurement noise.

## Why derivative-on-PV, not derivative-on-error

A naive form puts $de/dt = d\mathrm{SP}/dt - d\mathrm{PV}/dt$ in the derivative term. When the operator changes SP, $d\mathrm{SP}/dt$ is a Dirac impulse, and the controller fires a derivative "kick" of the same shape onto $u$ — a momentary spike that the actuator sees as a step change of theoretical magnitude $-K_d/0^+$. For a fast valve or heater this can be a real disturbance.

The *derivative-on-PV* form drops the SP contribution:

$$u_{\mathrm{D}}(t) \;=\; -\,K_d\, \frac{d\,\mathrm{PV}(t)}{dt},$$

which still damps the *response* to disturbances and load changes (because they all show up as PV changes) but makes set-point changes show up only through the proportional and integral terms. No spike, no SP-kick. Choupo's `PIDController` implements this form.

## Anti-windup

The integral term is unbounded in (568). When the actuator saturates — the output $u$ hits a physical limit, e.g. the cooling-jacket cannot go below $T_{\text{jacket,min}}$ — the controller *commands* an even more extreme $u$, but the process responds as if it were saturated. The error stays large, the integral keeps accumulating, and when the constraint clears the controller has wound up so much integral that the recovery overshoot is enormous.

Choupo uses the simplest robust scheme, *clamping anti-windup*. At each step, compute the unclamped output $u_{\text{unc}}$ from (568), then clamp it:

$$u \;=\; \text{clip}\big(u_{\text{unc}},\; u_{\min},\; u_{\max}\big),$$

and **advance the integral only when the unclamped output is inside the bounds**:

$$I_{n+1} \;=\; \begin{cases} I_n + e_n\,h, & u_{\min} \le u_{\text{unc},n} \le u_{\max} \\ I_n, & \text{otherwise.} \end{cases} \tag{569}$$

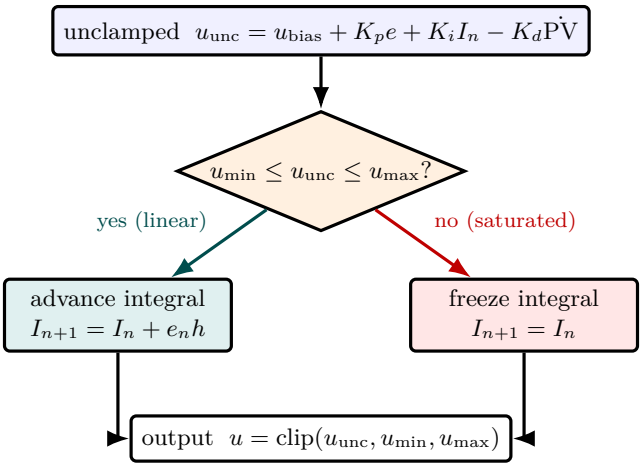



**Figure 88:** Clamping anti-windup, the integral gate inside Choupo's PID. Each step computes the *unclamped* output from the full ideal-form law. If it lands within the actuator limits the loop is operating linearly and the integral is advanced normally ($I_{n+1} = I_n + e_n h$, green). But if it has saturated — the cooling jacket cannot go below $T_{\text{jacket,min}}$, say — the integral is *frozen* (red) so it cannot "wind up" a huge reserve during the constrained phase that would then drive an enormous recovery overshoot when the constraint clears. Either way the delivered $u$ is the clipped value. This is the simplest robust scheme; back-calculation is the more elaborate alternative deferred in the text.

This prevents the integral from growing during saturated phases. More sophisticated alternatives (back-calculation, conditional integration with a track-back constant) exist; for the tutorials (569) suffices.

## Discretisation

For a fixed-step explicit integrator (Choupo's RK4 wrapper around `DynamicCSTR`) the simplest discretisation is rectangular for the integral and backward finite difference for the derivative:

$$I_{n+1} = I_n + e_n\,h, \qquad \dot{\text{PV}}_n \approx (\text{PV}_n - \text{PV}_{n-1})/h.$$

On the very first step the derivative cannot be evaluated — there is no previous PV — so Choupo zeros it (`primed` flag). This avoids an artificial spike at $t = 0$. The controller's output is held *constant* across the RK4 stages within the step (zero-order hold), which is acceptable when $h \ll \tau$.

> **Runnable case.** ctrl/ctrl02_disturbance_rejection Same plant + PID as `ctrl01`, but a `Schedule` controller injects step disturbances of $\pm 15\,\text{K}$ in $T_{\text{in}}$ at $t = 700\,\text{s}$ and $t = 1300\,\text{s}$. The PID has no

foreknowledge — it only sees $T_{\text{reactor}}$ drift away from setpoint. Peak excursion after each step is $\approx 0.7\,\text{K}$, settling to within $10^{-3}\,\text{K}$ of setpoint in $\approx 200\,\text{s}$.

## Open-loop schedules: the `ScheduleController`

A degenerate "controller" that ignores the measurement and writes a piecewise-constant MV from a list of (time, value) pairs slots into the same `Controller` infrastructure. Choupo uses it for two purposes: (1) injecting scripted disturbances on $T_{\text{in}}$ to test closed-loop disturbance rejection (above); (2) running open-loop ramps (e.g. linearly increasing feed rate over time) without writing a new unit op. The user-side dict syntax (`type Schedule;`, `schedule ( time; value; ...);`) is documented in the User Guide §8.2.

The forcing vocabulary culminates in the *PRBS* — the classic system-identification input. A maximal-length Fibonacci LFSR of $n$ registers generates a binary sequence of period exactly $2^n - 1$ bits; its autocorrelation approximates an impulse, so one record excites the plant across the band and carries the information for a full low-order model fit, where a sinusoid probes a single frequency. Choupo's `type prbs;` is deterministic by construction — the seed and the feedback taps are *declared* in the dict (taps may come from the versioned canonical table of primitive polynomials), the initialiser *walks* the register and refuses any tap set whose cycle is not exactly $2^n - 1$, and the announce line prints the leading bits so the student can derive the same sequence by hand. Randomness at run time would make the experiment unrepeatable; a declared LFSR makes it exact.

**Runnable case.** ctrl/ctrl08_prbs_ident $T_{\text{in}} = 320 \pm 10\,\text{K}$ by an $n = 4$ LFSR (period 15 bits $\times$ 120 s), the hand-derived sequence `000100110101111` in the case header matching the announce; seed 0 and non-maximal taps refuse aloud.

# 7 The process reaction curve: identifying an FOPDT model, and tuning from it

*What you'll learn.* Before a loop can be tuned it has to be *measured*. The classical measurement is the *process reaction curve*: open the loop, step the manipulated variable once, and record what the plant does. From that one curve a student reads three numbers — a gain $K$, a time constant $\tau$, an apparent dead time $\theta$ — and three published rules turn them into controller settings. This section derives the model those three numbers belong to, gives both the hand construction and the regression Choupo runs, and then spends most of its length on the three questions that decide whether the answer means anything: *has the response finished?*, *is the dead time resolved by the record?*, and *how much of the gain is a property of the plant rather than of the step you happened to ask with?* The tuning rules themselves are two lines of algebra each; the disagreement between them — a factor of eight on the worked case — is the lesson.

## The experiment

The loop is *open*. If a controller is acting while the step is applied, the recorded response is the closed loop's, and the model identified from it already contains the controller you were about to design. So: bring the plant to a steady state, hold it there long enough to be sure, change one manipulated variable by a declared amount, and record until the plant is steady again.

Everything in this section rests on those two steady states. The gain is the ratio of the change in the measured variable to the change in the manipulated one,

$$K \;=\; \frac{\Delta y_\infty}{\Delta u} \;=\; \frac{y(\infty) - y(0^-)}{\Delta u}, \tag{570}$$

and both ends of that fraction are steady-state quantities. A curve that has not finished has no $y(\infty)$ to put in the numerator, and a step applied to a plant that was still moving has no $y(0^-)$ to subtract.

## The model, and what it assumes

*First-order plus dead time* (FOPDT) is the smallest model that can carry the two features a controller designer needs: a lag, and a delay. In the Laplace domain

$$G(s) \;=\; \frac{K\, e^{-\theta s}}{\tau s + 1}, \tag{571}$$

and in the time domain, for a step $\Delta u$ applied at $t = 0$ from a steady state $y_0$,

$$y(t) \;=\; y_0 \;+\; K\,\Delta u \left[1 - e^{-(t-\theta)/\tau}\right] \mathbf{1}(t \ge \theta). \tag{572}$$

**Key idea.** An FOPDT model is a *surrogate*, not a description. Almost no real plant is first order, and most plants with an apparent dead time have no transport delay at all — what the fit calls $\theta$ is the curvature of the leading edge, absorbed into the only parameter that can hold it. The three numbers describe *the response you showed the fit*, over *the step you asked with*, at *that operating point*.

That is not a reason to distrust the method; it is the reason the method works. A controller does not need to know the plant, only enough of it to be tuned. But it is a reason to publish the *residual* beside the three parameters, which Choupo does and this section insists on: a fitted parameter with no residual is a number the reader cannot judge.

Where does an apparent dead time come from, if not from transport? Put two first-order lags in series, $\tau_1 \ll \tau_2$. The response starts flat — the second lag sees nothing until the first has moved — and then curves up. An FOPDT fit reads that initial flatness as delay, and the delay it reports grows with $\tau_1/\tau_2$. This is exactly the situation in the worked case below: the jacket heats the liquid (fast), and the warmer liquid reacts faster (slow), so the composition responds through two lags in series and carries a $\theta$ of about 25 s, while the *temperature* in the same experiment responds through one lag and carries no resolvable delay at all.

## Reading the curve by hand

Two constructions, both doable with a ruler on a printed curve, and both worth doing once before trusting any regression.

**The tangent construction (Ziegler and Nichols, 1942).** Draw the tangent to the reaction curve at its steepest point. Call its slope $R$ (response per unit time per unit change in controller output) and call $L$ the time at which that tangent crosses the initial value. For an FOPDT model the steepest point is the instant the response begins, where the slope is $K/\tau$ per unit $\Delta u$ and the tangent crosses at $t = \theta$; so

$$R = \frac{K}{\tau}, \qquad L = \theta. \tag{573}$$

This is the construction Ziegler and Nichols wrote their rules in terms of, which is why their settings are quoted below in both forms. Its weakness is that a tangent slope is a *derivative* read off a drawing: on a noisy record it is the least reliable quantity on the page.

**The two-point construction (Sundaresan and Krishnaswamy, 1978).** Instead of a slope, use two *crossings* [125]. Let $t_{35.3}$ and $t_{85.3}$ be the times at which the response has completed 35.3 % and 85.3 % of its total change. Then

$$\tau = 0.67\,(t_{85.3} - t_{35.3}), \qquad \theta = 1.3\,t_{35.3} - 0.29\,t_{85.3}. \tag{574}$$

Two readings, no derivative. Choupo publishes this construction beside its regression on every segment, for two reasons: it is arithmetic a student can repeat, and a regression nobody can check by hand is a black box with a residual attached.

## Reading the curve by regression

The engine's answer is a least-squares fit of Equation (572) to every sample in the segment:

$$\min_{K,\,\tau,\,\theta \ge 0} \; S(K,\tau,\theta) \;=\; \sum_i \Big[\, y_i - y_0 - K\,\Delta u\, g(t_i;\tau,\theta) \,\Big]^2, \qquad g = \big[1 - e^{-(t-\theta)/\tau}\big]\mathbf{1}(t \ge \theta). \tag{575}$$

Three parameters, but only two of them are awkward: for any trial $(\tau, \theta)$ the amplitude $a = K\,\Delta u$ enters *linearly*, so it is eliminated in closed form,

$$a^\star(\tau,\theta) \;=\; \frac{\sum_i g_i\,(y_i - y_0)}{\sum_i g_i^2}, \tag{576}$$

and the search runs over $(\tau, \theta)$ only. Choupo drives it with the project's shared Nelder-Mead simplex, seeded from Equation (574).

> **Common pitfall.** $\theta$ is bounded below at zero because a negative dead time is not causal, and that bound *binds* on plants with no delay. When it does, Choupo says so — and then refuses every tuning rule that divides by $\theta$, by name. A bound that silently absorbs a parameter is the crutch this project refuses to build; a bound that reports when it binds is a measurement.

> **Common pitfall.** The objective in Equation (575) is a sum of squares in whatever units $y$ carries, so its absolute size is arbitrary — on the worked case it is around $10^{-11}$, because $y$ is a holdup in kmol. A solver whose convergence test compares that number against a fixed floor will stop far from the minimum and report success. This is not hypothetical: it happened here, and the fit was 2.6 % off in sum of squares while announcing itself converged. Normalise the objective by something of its own size before handing it to a general-purpose optimiser.

## Three questions the number has to survive

**Has it settled?** Both $K$ and $\tau$ are read off the end of the curve. If the response is still moving when the record stops, $K$ is the change *so far* rather than the change, and $\tau$ is whichever exponential happens to pass through an unfinished transient — a number with a plausible magnitude and no meaning. Choupo measures the largest excursion from the final value over the closing fraction of each segment, as a fraction of the step change, and *refuses* to report a time constant when it exceeds a declared tolerance.

**Is $\theta$ resolved?** The record is sampled. A step that occurs between two samples is bracketed, not located, so the identified dead time is an upper bound within one sampling interval and cannot be narrowed by any amount of arithmetic — only by sampling faster. A $\theta$ smaller than the interval it was measured on is not a small dead time; it is no measurement at all, and Choupo treats it as such.

**How much of $K$ belongs to the plant?** This is the question a single step cannot answer, and the one most often skipped. A gain identified from one step and then quoted as *the* gain of the plant is a remembered measurement dressed as a property. Step the same plant by a different amount, or in the other direction, and a nonlinear plant answers differently. The honest procedure is to run several steps in one experiment and publish the *spread*; if only one step was run, the honest report says the sensitivity was not measured.

> *Intuition.* Normalise each segment's response by its own total change and plot them together. A linear plant gives one curve. Where the normalised curves fail to superpose is exactly where the plant is nonlinear, and by how much.

## The tuning rules

Three rules, all closed-form algebra on $(K, \tau, \theta)$, all published, and all different. Write $r = \theta/\tau$.

| Rule | $K_c$ | $T_i$ | $T_d$ |
|---|---|---|---|
| Z–N PI | $0.9\,\tau/(K\theta)$ | $3.33\,\theta$ | — |
| Z–N PID | $1.2\,\tau/(K\theta)$ | $2\theta$ | $0.5\theta$ |
| C–C PI | $\dfrac{\tau}{K\theta}\left(0.9 + \dfrac{r}{12}\right)$ | $\theta\,\dfrac{30+3r}{9+20r}$ | — |
| C–C PID | $\dfrac{\tau}{K\theta}\left(\dfrac{4}{3} + \dfrac{r}{4}\right)$ | $\theta\,\dfrac{32+6r}{13+8r}$ | $\theta\,\dfrac{4}{11+2r}$ |
| IMC PI | $\dfrac{1}{K}\,\dfrac{\tau+\theta/2}{\lambda+\theta/2}$ | $\tau+\theta/2$ | — |
| IMC PID | $\dfrac{1}{K}\,\dfrac{\tau+\theta/2}{\lambda+\theta/2}$ | $\tau+\theta/2$ | $\dfrac{\tau\theta}{2\tau+\theta}$ |

The sources, and they travel with the numbers in Choupo's published tuning table rather than living only here:

- **Ziegler, J. G. and Nichols, N. B.**, "Optimum Settings for Automatic Controllers", *Trans. ASME* **64** (1942) 759–768. Written in terms of the reaction curve's $R$ and $L$ of Equation (573): the PID gain is $1.2/(RL)$, which is the table's $1.2\,\tau/(K\theta)$.
- **Cohen, G. H. and Coon, G. A.**, "Theoretical Consideration of Retarded Control", *Trans. ASME* **75** (1953) 827–834. The same family of objectives, corrected for the ratio $r = \theta/\tau$ so that the settings remain usable when the delay is a large fraction of the lag.
- **Rivera, D. E., Morari, M. and Skogestad, S.**, "Internal Model Control: 4. PID Controller Design", *Ind. Eng. Chem. Process Des. Dev.* **25** (1986) 252–265. The FOPDT entry with a first-order Padé approximation of the delay.
- **Sundaresan, K. R. and Krishnaswamy, P. R.**, "Estimation of time delay, time constant parameters in time, frequency, and Laplace domains", *Can. J. Chem. Eng.* **56** (1978) 257–262, for the two-point construction of Equation (574).

## Why they disagree, and why that is the point

They disagree because they are answering different questions.

Ziegler and Nichols aimed at a *quarter-decay* response to a load disturbance: each swing a quarter of the last. That is a fast loop and a poorly damped one, and the settings are correspondingly aggressive. Cohen and Coon stay in the same family but carry the ratio $r$ through the algebra, so their gains hold up where Ziegler–Nichols' assumption of a small delay does not.

IMC asks something else entirely: *how fast do you want the closed loop to be?* The answer is the parameter $\lambda$, and it is not in the data. No open-loop record can tell you how fast a loop should respond; that is a

statement about the process's economics, its actuator, and how much model error you are willing to be wrong by. Choupo therefore *refuses* to run the IMC rule until $\lambda$ is declared. A small $\lambda$ buys speed and spends robustness; a large one does the reverse.

Two structural consequences follow, and both are visible in the worked case. First, the two quarter-decay rules put $\theta$ in a *denominator*, so they blow up as the delay vanishes — and are undefined, not merely large, when it is unresolved. The IMC gain does not: at $\theta \to 0$ it becomes $\tau/(K\lambda)$, perfectly finite. That is why IMC is the rule of choice for a lag-dominant loop, and it is not a matter of taste. Second, when all three *are* defined they still differ by a large factor, and the reader has to choose. Choupo publishes all three with their sources and recommends none.

**Key idea.** None of these settings has been tested by closing the loop. Choupo identifies and publishes; it does not simulate the tuned controller and does not claim any of the three rules works. A number obtained from an open-loop record and a published formula is a *proposal*.

## The worked case

**Runnable case.** ctrl/ctrl17_step_reaction_curve One jacketed CSTR, open loop, and one staircase on the jacket temperature: +5, back, +20, back, each held 3000 s. Run `choupoCtrl` for the experiment, then `choupoProps` on the same directory for the identification.
Read as a *composition* loop (jacket → product holdup) the four steps identify $K = 1.240 \times 10^{-5}$ to $1.834 \times 10^{-5}$ kmol K$^{-1}$, $\tau \approx 233$ to 240 s and $\theta \approx 19$ to 31 s, each with an r.m.s. residual under 0.35 % of its own step change. The gain at the hotter operating point is *48 % larger* than at the base — Arrhenius, measured — which is the honest error bar on any single-step identification and is published as `K_spread_relative`. The +5 and −5 steps connect the same two steady states, so their steady-state gains must agree by construction; they do, to $3.5 \times 10^{-5}$ on the ±5 pair and $1.9 \times 10^{-4}$ on the ±20 pair, which is a free check on the whole chain — integration, sampling and regression together. Ziegler–Nichols PI asks for $K_c = 6.68 \times 10^5$ where IMC at $\lambda = \tau$ asks for $8.07 \times 10^4$: a factor of eight, from one curve.
Read as a *temperature* loop (jacket → reactor temperature), the same experiment has one dominant lag and no resolvable delay: $\theta$ comes back below the 5 s sampling interval on every segment, and one of the four hits its lower bound of zero exactly. Ziegler–Nichols and Cohen–Coon are refused by name; IMC still publishes. One plant, one experiment, two honest answers.

# 8 Choosing a column control structure, and where the instruments go

*What you'll learn.* A two-product distillation column has five valves and five things to hold, and the arithmetic of that sentence is the whole subject: pressure takes one valve and each of the two levels takes another, so exactly two are left for composition — and the structure is *named after those two.* This section derives that count, names the candidate structures as the literature names them, states why a composition loop almost always measures a *tray temperature* instead of a composition, and then answers the sharpest question a student asks: *on which tray?* Two classical criteria are given, both computable from a profile the rigorous MESH column of Section 34 already writes, and on a worked column they select *different trays* — which is the lesson, not a defect. What this section does not do is recommend one structure: the published authorities disagree, their conditions differ, and the disagreement is the material.

## Five valves, five inventories, and the naming convention

Around a two-product column with a total condenser there are five manipulable flows:

- the condenser duty (equivalently the coolant valve),
- the distillate $D$,
- the reflux $L$,
- the boilup $V$ (equivalently the reboiler steam valve),
- the bottoms $B$.

and five variables that must be held: the column pressure, the reflux-drum level, the sump (reboiler) level, and the two product compositions. Pressure is settled first, on the condenser duty or a vent. Each level must then be held by a stream *leaving* the vessel it belongs to — a level controlled by an inflow that also sets production is not self-regulating — and that consumes two more valves.

Two valves remain, and the convention is to name the structure after them. Writing $\{\cdot\cdot\}$ for the pair left free for composition:

| name | free for composition | drum level held by | sump level held by |
|---|---|---|---|
| LV | $L$, $V$ | $D$ | $B$ |
| LB | $L$, $B$ | $D$ | $V$ |
| DV | $D$, $V$ | $L$ | $B$ |
| DB | $D$, $B$ | $L$ | $V$ |
| $(L/D)(V/B)$ | $L$, $V$ as *ratios* | $D$ | $B$ |

LV is the conventional arrangement — each level held by the product stream leaving it — and it is the one Skogestad recommends for most columns, in combination with a fast inner temperature loop (*The dos and don'ts of distillation column control*, *Chem. Eng. Res. Des.* **85**(A1), 13–23, 2007). DV moves the drum level onto the reflux, which is what a large reflux ratio invites: the rule of thumb is to hold a level with the *larger* stream leaving that drum, and when $R = L/D$ is large that stream is the reflux, which frees $D$ for composition. The $(L/D)(V/B)$ scheme keeps LV's levels but drives each composition loop through a ratio station, so a throughput change moves both members of a pair together.

DB looks impossible on a steady-state material balance — both products set independently, with no valve left to absorb the difference — and this is the classic trap. Skogestad, Lundström and Jacobsen (*AIChE J.* **36**(5), 753–764, 1990) report that DB has an *infinite* steady-state relative gain and can nonetheless be controlled well, because its high-frequency relative gain is close to one. The lesson generalises far beyond distillation: a criterion can return a definite, dramatic number and still not settle the question, because it asked at the wrong frequency.

## Why the drawing carries the lesson

None of the above needs a transient. Which measurement drives which valve is a *drawing* question — it is exactly what a piping and instrumentation diagram records — and switching the structure moves lines on that drawing rather than changing any equation. This is why Choupo's EduTool for this material is a

*cross-section* of the column with the instrument bubbles and the signal lines on it, over a steady solution: the reader changes the structure and watches the wiring move.

What the drawing cannot honestly show is a level rising and falling. A level that moves is an accumulation term; it needs a dynamic column, and Choupo has none — the dynamic tier is the continuous stirred tank of Section 4 and one reference plant, and no column anywhere in the corpus is under control. A plausible-looking animation drawn from nothing would be worse than the static picture, because a moving image is more persuasive and would teach a specific wrong thing with confidence.

## The composition loop measures a temperature

An on-line composition analyser is slow and expensive, so a column's "composition" loop almost always measures a *sensitive tray temperature* instead. For a binary at fixed pressure the substitution is exact: bubble-point temperature and liquid composition are in one-to-one correspondence (Section 13), so

$$T_n = T^{\mathrm{bub}}(x_n, P) \qquad \Longleftrightarrow \qquad x_n = x(T_n, P). \tag{577}$$

With more than two components the correspondence fails — many compositions share a bubble point — and if the column pressure moves, the temperature moves with no composition change at all. Both are limits of the substitution, not of the instrument.

That substitution is what creates the question this section exists for. If we are going to measure a temperature, *which tray*?

## Criterion 1: the slope

Read the converged profile and take the tray where it is steepest:

$$n^{\star}_{\mathrm{slope}} = \arg\max_{n} \left| T_{n+1} - T_n \right|, \tag{578}$$

the difference between *adjacent* trays. A steep profile means a large temperature change for a small composition change, so the measurement carries information; a flat region means the thermocouple is nearly blind.

It costs one solve — the profile is already on the screen — and a student can apply it with a ruler. What it cannot say is whether that tray's temperature actually *moves* when the manipulated variable moves.

## Criterion 2: the sensitivity

Perturb the manipulated variable the loop will use, solve again, and take the tray that responded most:

$$n^{\star}_{\mathrm{sens}} = \arg\max_{n} \left| \frac{\partial T_n}{\partial u} \right| \approx \arg\max_{n} \left| \frac{T_n(u + \Delta u) - T_n(u)}{\Delta u} \right|. \tag{579}$$

This is a finite difference over two rigorous solutions — no new physics, one extra solve — and it is the criterion a simulator can offer that a table in a book cannot, because the number belongs to *this* column at *these* settings with *this* handle.

Note what $u$ smuggles in. Equation (579) has no answer until the manipulated variable is named, and the answer moves with it. The sign matters too: the derivative changes sign somewhere between the two ends of the column, and a sensor at the crossing is nearly deaf to the handle however steep the profile looks there — something Equation (578) cannot warn about, because it never asks the handle a question.

## Worked: the two criteria disagree

Take the benzene/toluene column of `tutorials/steady/distillation/column02_simultaneous` — 15 equilibrium stages (stage 1 the top tray below a total condenser, which is not counted; stage 15 the reboiler), feed on stage 8, $R = 2.0$, $D = 50\,\mathrm{kmol/h}$, rigorous MESH. Its solved profile runs from 354.21 K at the top to 382.89 K in the reboiler. Restricting attention to the stages that are trays (the reboiler cannot carry a tray thermocouple):

| criterion | handle | tray selected |
|---|---|---|
| slope, Eq. (578) | — | **10** (3.140 K to stage 11) |
| sensitivity, Eq. (579) | reflux $L$ | **11** (+0.0910 K per 1 % of $L$) |
| sensitivity, Eq. (579) | distillate $D$ | **5** (+1.336 K per 1 % of $D$) |

Three answers, one column. The slope and the reflux sensitivity differ by one tray; the *distillate* sensitivity moves the sensor into the rectifying section entirely — five stages and a section boundary away from where the ruler pointed — and it is fifteen times the size, because at fixed reflux the distillate rate is the strong handle on this split. The reflux sensitivity also changes sign between stages 6 and 7: raising the reflux *cools* everything above stage 7 and *heats* everything below it, and a thermocouple on either side of that crossing reads a gain near zero.

Both numbers are differences of temperatures the engine published; nobody tuned anything to produce the disagreement.

Two things follow, and they are the practical content of this section. First, a tray number quoted without naming the manipulated variable is incomplete. Second, the pairing rule Skogestad states — use a flow at the *same end* of the column as the sensor — is not a nicety: a sensor at stage 5 driven by a valve at the bottom of the column would be reading one section and acting on the other, with the whole cascade's lag in between.

## What is not settled, and by whom

Luyben (*J. Process Control* **16**, 115–134, 2006) compares five criteria for this choice — slope, sensitivity, singular value decomposition, invariant temperature, and minimum product variability — over systems from ideal binary to azeotropic multicomponent, and concludes that singular value decomposition is a simple and effective method. Choupo computes the first two and *not* the one that comparison prefers: the SVD criterion (due to Moore, in *Practical Distillation Control*, W. L. Luyben, ed., Van Nostrand Reinhold, 1992) needs a steady-state gain matrix over several perturbations, and nothing in the engine publishes one. That is a stated gap, not an oversight.

On the structure itself the disagreement is sharper, and it is a disagreement about *conditions* rather than about facts. Selecting a configuration on its steady-state relative gain rejects LV at high purity, because the two composition loops interact strongly; on a frequency-dependent analysis with optimal PI designs, Skogestad, Lundström and Jacobsen (1990) find $(L/D)(V/B)$ the best of four candidates for two-point composition control. Seventeen years later Skogestad (2007) recommends plain LV for most columns — and the reason the two are not in contradiction is that closing a fast temperature loop first collapses the steady-state relative gain of what remains, in his example from about 35 to below 2. The 1990 result answers "which configuration for dual composition control, judged in the frequency domain"; the 2007 recommendation answers "what should most columns do", with an inner loop that has already changed the question.

A student who leaves knowing which rules exist, who stated them, and under what conditions each was asserted has gained the thing they came for. A student handed one confident recommendation has gained a rule they will misapply the first time the conditions differ.

**Common pitfall. Relative gain is not computed anywhere in Choupo.** The paragraphs above quote published relative gains; the engine publishes no gain matrix, at any frequency, and the EduTool says so on its own face. Do not read a number here as something this simulator checked.

# 9 Numerical integration of stiff systems

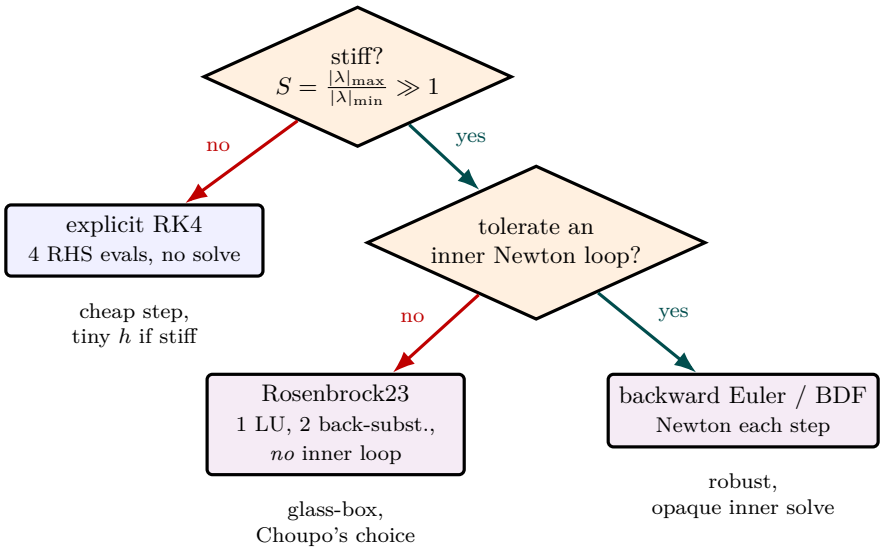



**Figure 89:** The integrator decision tree. If the eigenvalue spread $S$ is $\mathcal{O}(1)$ the system is non-stiff and explicit RK4 wins — four right-hand-side evaluations, nothing solved. Once $S \gg 1$ an explicit method is hostage to the dead fast mode (Figure 91), so an implicit method is required, and the only remaining question is whether to pay for an inner Newton iteration. Choupo takes the left branch: the *linearly-implicit* Rosenbrock23 keeps full implicit stability but replaces the inner loop with one LU factorisation reused by two back-substitutions — the trait that keeps the stiff solver as readable as the non-stiff one.

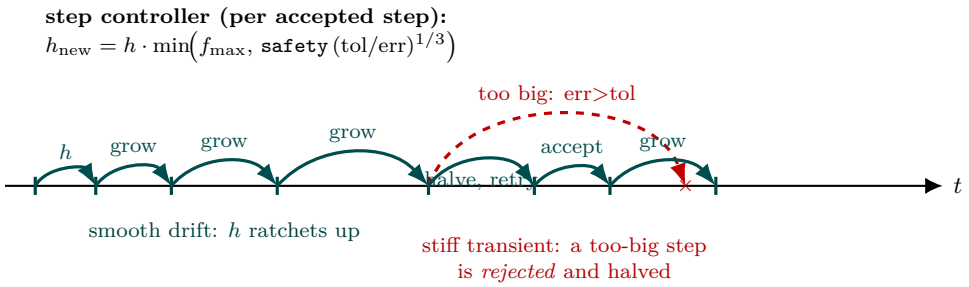



**Figure 90:** The adaptive step-size trajectory of Rosenbrock23 over several steps (the per-step internals are in Figure 93). On the smooth drift the embedded error stays well below tolerance, so after each accepted step the controller *grows* $h$ (green arcs ratchet wider) — $h_{\text{new}} = h \cdot \min(f_{\max}, \texttt{safety}\,(\text{tol}/\text{err})^{1/3})$ for the order-2/3 pair. When the solution hits a stiff transient a trial step overshoots tolerance: it is *rejected* ($\times$), $h$ is halved, the matrix $\mathbf{W}$ refactorised, and the step retried until it is accepted (red dashed $\rightarrow$ short green arcs). The integrator thus spends large steps where nothing happens and small steps exactly where the action is — automatically, from the embedded error alone.

> *What you'll learn.* The RK4 chapters (Sections 9 and 1) integrate $d\mathbf{Y}/dt = f(\mathbf{Y})$ with a fixed explicit step. That works while every state variable changes on a comparable timescale. A reacting mixture violates this: a radical can be born and consumed a million times faster than the bulk heats up. The system is *stiff*, and an explicit method must either crawl or explode. This chapter explains stiffness, why RK4 fails on it, and the linearly-implicit *Rosenbrock* cure Choupo uses.

## 9.1 The initial-value problem and explicit stepping

A closed reactor's state obeys an initial-value problem

$$\frac{d\mathbf{Y}}{dt} = f(\mathbf{Y}), \qquad \mathbf{Y}(0) = \mathbf{Y}_0, \tag{580}$$

with $\mathbf{Y}$ the packed mole numbers (and temperature) of Section 1. The simplest integrator is explicit (forward) Euler, $\mathbf{Y}_{n+1} = \mathbf{Y}_n + h\, f(\mathbf{Y}_n)$; RK4 (550) is the same idea with four stages and a fourth-order local error. Both are *explicit*: the new state is an arithmetic combination of right-hand sides already evaluated at known states. Nothing is solved; the step is cheap.

The price of that cheapness is a stability limit. Linearise (580) about a point, $\delta\mathbf{Y}' = \mathbf{J}\,\delta\mathbf{Y}$ with $\mathbf{J} = \partial f/\partial\mathbf{Y}$ the Jacobian. An explicit step multiplies each eigenmode $\lambda$ of $\mathbf{J}$ by an amplification factor; for forward Euler that factor is $1 + h\lambda$. Stability ($|1 + h\lambda| \le 1$) requires

$$h \lesssim \frac{2}{|\lambda|_{\max}}, \tag{581}$$

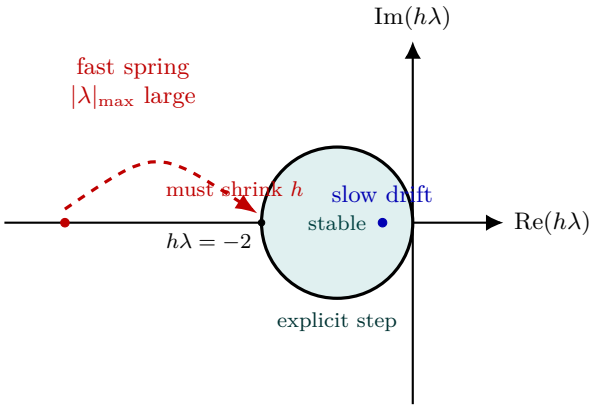



**Figure 91:** The stability region of an explicit step (forward Euler shown: the disc $|1 + h\lambda| \le 1$, centre $-1$, radius 1). Every eigenmode $\lambda$ of the Jacobian, multiplied by the step $h$, must land *inside* the disc or that mode is amplified and the integration blows up. A slow drift mode (blue) sits comfortably inside for almost any $h$; but the fast spring (red, large $|\lambda|_{\max}$) forces $h \lesssim 2/|\lambda|_{\max}$ to drag it back across the boundary at $h\lambda = -2$ — even after that mode has died and stopped contributing to the answer. That is stiffness: the dead fast mode, not accuracy, sets the step. The implicit cure (Rosenbrock23, Figure 93) has *no* such boundary in the left half-plane.

*regardless of accuracy.* The fastest-decaying mode dictates the step even after it has died out and no longer contributes anything to the answer.

## 9.2 What stiffness is

Stiffness is a separation of timescales. Define the stiffness ratio

$$S = \frac{|\lambda|_{\max}}{|\lambda|_{\min}}, \tag{582}$$

the spread of the Jacobian's eigenvalues (with negative real parts). When $S \sim 1$ the system is non-stiff and RK4 is ideal. When $S \gg 1$ it is stiff.

Chemical kinetics is the textbook offender. In a hydrogen flame the radicals H, O, OH reach a quasi-steady balance on a $\sim 10^{-8}$ s timescale, while the heat release that actually moves the temperature runs on $\sim 10^{-3}$ s. The ratio $S \sim 10^5$–$10^6$ is built into the physics, not an artefact. By (581) an explicit method must take steps of order $10^{-8}$ s to stay stable, yet the interesting evolution lasts milliseconds: tens of millions of steps, each limited by a mode that contributes nothing to the answer. Push the step any larger and the amplification factor exceeds one — the trace radical, driven negative, feeds a rate law $\prod c_j^{m_j}$ that overflows to a non-finite number within a few steps.

> *Intuition.* Stiffness is not "the problem is hard"; it is "the problem hides a fast spring inside a slow drift." An explicit integrator has to resolve the spring even when you only care about the drift. An implicit one lets the spring relax to its equilibrium each step and follows the drift.

## 9.3 The implicit cure

Backward (implicit) Euler, $\mathbf{Y}_{n+1} = \mathbf{Y}_n + h\, f(\mathbf{Y}_{n+1})$, has amplification factor $1/(1 - h\lambda)$, which is below one for *every* $h$ when $\mathrm{Re}\,\lambda < 0$. It is unconditionally stable — the step is set by accuracy, not by the dead

fast mode. The cost is that $\mathbf{Y}_{n+1}$ now appears on both sides: each step solves a nonlinear system, normally by an inner Newton iteration.

A *linearly-implicit* method keeps the stability and removes the inner loop by linearising $f$ once. The semi-implicit Euler step,

$$(\,\mathbf{I} - h\,\mathbf{J}\,)\,\Delta\mathbf{Y} = h\,f(\mathbf{Y}_n), \qquad \mathbf{Y}_{n+1} = \mathbf{Y}_n + \Delta\mathbf{Y}, \tag{583}$$

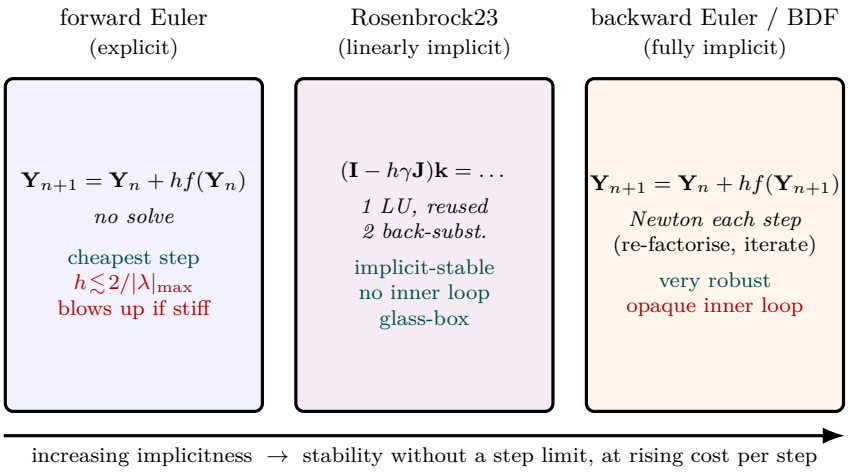



**Figure 92:** Three rungs of implicitness for the stiff IVP $d\mathbf{Y}/dt = f(\mathbf{Y})$. Forward Euler (left) solves nothing per step but is hostage to the stability limit $h \lesssim 2/|\lambda|_{\max}$ and overflows on a stiff system. Backward Euler/BDF (right) is unconditionally stable but pays an *inner Newton iteration* every step — robust, but the iteration is the opaque box Choupo wants to avoid. Rosenbrock23 (centre) is the *linearly-implicit* compromise: it linearises $f$ once, assembles $\mathbf{W} = \mathbf{I} - h\gamma\mathbf{J}$, factorises it *once*, and reuses the factorisation across both stages — keeping full implicit stability *without* an inner loop. That is why Choupo's stiff solver (Figure 93) reads almost as plainly as its RK4.

is one matrix assembly and one linear solve — no Newton iteration. It is the readable "hello, stiffness" stepping-stone, first order and without error control.

## 9.4 Rosenbrock23: the scheme Choupo uses

Choupo's stiff integrator is a Rosenbrock–Wanner method of order two with an embedded third-order estimate (the "ode23s" / Rosenbrock23 family of [126]). It generalises (583) to several stages that all reuse the *same* factorised matrix $\mathbf{W} = \mathbf{I} - h\gamma\mathbf{J}$:

$$\begin{aligned} \mathbf{W}\,\mathbf{k}_1 &= f(\mathbf{Y}_n), \\ \mathbf{W}\,\mathbf{k}_2 &= f(\mathbf{Y}_n + a\,h\,\mathbf{k}_1) - c\,\mathbf{k}_1, \\ \mathbf{Y}_{n+1} &= \mathbf{Y}_n + h\,(b_1\mathbf{k}_1 + b_2\mathbf{k}_2), \end{aligned} \tag{584}$$

with the scheme constants $\gamma, a, c, b_i$ fixed by the order conditions. The properties that matter:

- **L-stable** — not merely stable but *strongly* damping of the fast modes, so a stiff transient is absorbed rather than rung.
- **One Jacobian, one LU factorisation per step.** The factorisation of $\mathbf{W}$ is reused across the stages; each stage is a cheap back-substitution. Choupo's hand-rolled `luFactor`/`luSolve` (split out of the Gauss solver) does this.
- **No inner Newton loop** — unlike backward Euler or a BDF method, the stages are linear in the unknowns. This is what keeps the code short and glass-box.
- **Embedded error estimate** — the difference between the order-2 and order-3 combinations gives a local error per step that drives an adaptive $h$: shrink when the error exceeds the tolerance, grow when it is comfortably below.

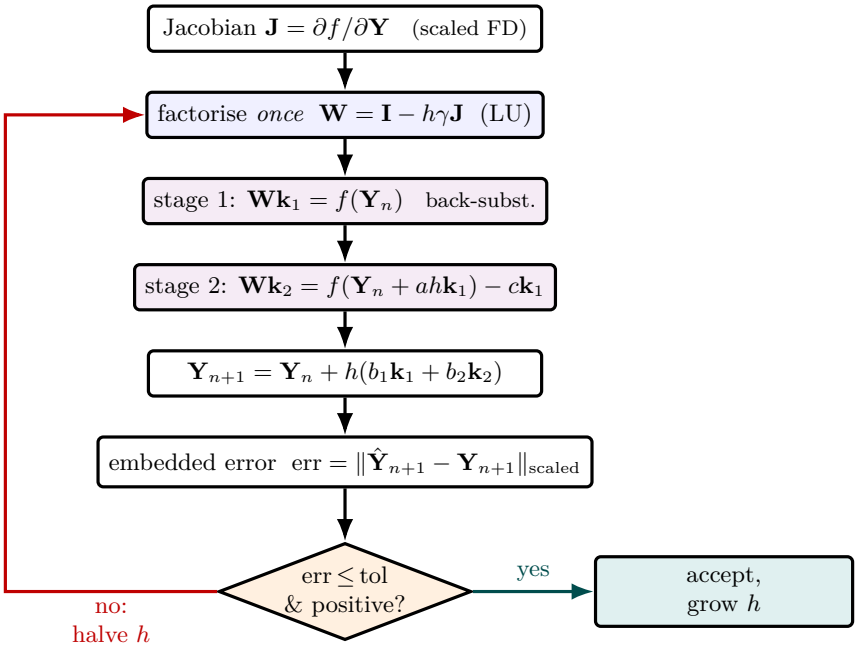



**Figure 93:** One adaptive step of Rosenbrock23, the stiff integrator (`src/solver/ODE/Rosenbrock23`). The key economy is the blue box: the matrix $\mathbf{W} = \mathbf{I} - h\gamma\mathbf{J}$ is LU-factorised *once* per step and then *reused* by both linear stages (violet) — each stage is just a back-substitution, and because the stages are *linear* in the unknowns there is no inner Newton loop (the trait that keeps the code glass-box, unlike backward-Euler/BDF). The order-2 update and an embedded order-3 estimate $\hat{\mathbf{Y}}$ differ by the local error, which drives the controller: if the (component-scaled) error is within tolerance *and* no concentration went negative, the step is accepted and $h$ grows; otherwise $h$ is halved, $\mathbf{W}$ refactorised, and the step retried (red). L-stability means a stiff transient is *absorbed*, not rung — where fixed-step RK4 would overflow on the Robertson benchmark.

## 9.5 The Jacobian, and two correctness traps

The method needs $\mathbf{J} = \partial f/\partial \mathbf{Y}$. Choupo forms it by finite differences (a perturbation of each state in turn) with an analytic option; either is acceptable, but two details are not optional in a chemical system:

- **Per-component scaling.** The state mixes mole numbers that span orders of magnitude (a bath gas at $\mathcal{O}(1)$, a radical at $\mathcal{O}(10^{-10})$) with a temperature row in kelvin. A single scalar tolerance lets the abundant species drive the step controller and *under-resolve the trace radical that is the whole event.* The error norm $\|\mathrm{err}/(\mathrm{atol} + \mathrm{rtol}\,|\mathbf{Y}|)\|$ and the finite-difference perturbation must both be scaled component by component.
- **Positivity rejection.** A step that drives a concentration negative is rejected and halved, because the next rate evaluation would raise a negative number to a fractional order. The embedded controller does not give this for free.

**Worked example 9.1.** The Robertson problemThe standard stiff benchmark is Robertson's autocatalytic system

$$\dot{y}_1 = -0.04\,y_1 + 10^4\,y_2y_3, \quad \dot{y}_2 = 0.04\,y_1 - 10^4\,y_2y_3 - 3\times10^7 y_2^2, \quad \dot{y}_3 = 3\times10^7 y_2^2,$$

with $\mathbf{y}(0) = (1, 0, 0)$, integrated to $t = 10^5$. The intermediate $y_2$ shoots up on a microsecond scale then decays across five decades of time — stiffness ratio $\sim 10^6$. Rosenbrock23 integrates it in a couple of thousand adaptive steps, conserving $y_1 + y_2 + y_3 = 1$ to round-off and matching a reference solution to four figures. Explicit RK4 at any practical fixed step overflows to a non-finite value: the lesson that the implicit machinery earns its complexity.

*Intuition.* RK4 is still the right default for a mild, non-stiff trajectory: it is cheaper per step and needs no Jacobian. Choupo keeps it as a selectable integrator and announces, at run time, the stiffness it measures — so a student *sees* why a particular problem needed the Rosenbrock solver rather than being told.

The classic references are Hairer and Wanner [127] and Sandu et al. [128].

# 10 Chemical kinetics in the gas phase

*What you'll learn.* The reactor chapters took a reaction rate as given. This chapter is the rate itself: the Arrhenius law and its modified form, the pressure-dependent corrections (third body, fall-off), and — the piece most often got wrong — how a reversible reaction's reverse rate is fixed by thermodynamics, with the concentration-basis factor that an isobaric mole-changing reaction must carry.

## 10.1 Mass action and the Arrhenius law

For an elementary reaction the rate is the law of mass action: a product of reactant concentrations raised to their stoichiometric orders, times a temperature-dependent rate coefficient,

$$r \;=\; k(T)\,\prod_{j} c_j^{\,m_j}. \tag{585}$$

The coefficient follows the Arrhenius law

$$k(T) \;=\; A\,\exp\!\Big(-\frac{E_a}{R_{\mathrm{g}}T}\Big), \tag{586}$$

with $A$ the pre-exponential factor and $E_a$ the activation energy. Collision theory and transition-state theory both predict a weak extra temperature dependence in the prefactor, captured by the *modified Arrhenius* form

$$k(T) \;=\; A\,T^{\,b}\,\exp\!\Big(-\frac{E_a}{R_{\mathrm{g}}T}\Big), \tag{587}$$

where the exponent $b$ is typically between $-1$ and $+3$. Choupo reads $A$, $b$, $E_a$ from the `kinetics` block; published mechanisms quote them in CHEMKIN units (cm–mol–s, cal/mol), which the parser converts to SI on load and announces, so the conversion is a visible step rather than a hidden one.

## 10.2 Third-body and pressure fall-off reactions

Some reactions need a collision partner M to carry away energy — a recombination such as $\mathrm{H} + \mathrm{O_2} + \mathrm{M} \rightarrow \mathrm{HO_2} + \mathrm{M}$. The rate scales with the total third-body concentration, weighted by per-species collision efficiencies $\varepsilon_j$ (unlisted species take $\varepsilon = 1$):

$$[\mathrm{M}] \;=\; \sum_{j} \varepsilon_j\, c_j. \tag{588}$$

At high pressure the partner is always available and the reaction saturates to a pressure-independent (high-pressure-limit) rate; at low pressure it is starved and the rate is proportional to [M]. The transition between the two limits is the *fall-off* region. The simplest description (Lindemann) blends the low- and high-pressure coefficients $k_0$ and $k_\infty$ through the reduced pressure $P_r = k_0[\mathrm{M}]/k_\infty$:

$$k \;=\; k_\infty\,\frac{P_r}{1+P_r}\,F, \tag{589}$$

with $F = 1$ for Lindemann. The Troe form supplies a more accurate broadening factor $F(P_r, T)$ from three or four tabulated parameters. This is why a gas-phase rate can depend on pressure even though (586) contains only temperature.

## 10.3 Reversible reactions: the reverse rate from thermodynamics

A reversible reaction must relax to chemical equilibrium, so its forward and reverse coefficients are not independent — their ratio is the equilibrium constant. Kinetics gives the forward $k_f$; thermodynamics gives the reverse,

$$k_r \;=\; \frac{k_f}{K_c}, \tag{590}$$

where $K_c$ is the equilibrium constant *on a concentration basis*. The subtlety — and a standing trap — is that thermodynamics naturally delivers the constant on a *pressure* basis. From the standard-state Gibbs energies of reaction,

$$K_p \;=\; \exp\!\Big(-\frac{\sum_i \nu_i\, g_i^{\circ}}{R_{\mathrm{g}}T}\Big), \tag{591}$$

with $\nu_i$ the (signed) stoichiometric coefficients and $g_i^\circ$ the standard-state molar Gibbs energies the property layer already integrates from $C_p$, $\Delta H_f$ and $S^\circ$. Converting to the concentration basis uses $c_i = P_i/(R_\mathrm{g}T)$ for an ideal gas, which introduces a factor for every net mole produced:

$$\boxed{K_c \;=\; K_p \left(\frac{P^\circ}{R_\mathrm{g}T}\right)^{\Delta\nu}, \qquad \Delta\nu = \sum_i \nu_i.} \tag{592}$$

**Common pitfall.** The $(P^\circ/R_\mathrm{g}T)^{\Delta\nu}$ factor is *mandatory* whenever $\Delta\nu \neq 0$. Applying $K_p$ directly to a concentration product — omitting the factor — silently corrupts the reverse rate of every mole-changing reversible gas reaction (a recombination, $\Delta\nu = -1$; a dissociation, $\Delta\nu = +1$). It is invisible on a mole-conserving reaction such as the water–gas shift ($\mathrm{CO + H_2O \rightleftharpoons CO_2 + H_2}$, $\Delta\nu = 0$, factor $= 1$), which is exactly why the bug can hide for a long time. Choupo prints $K_p$, $\Delta\nu$ and $K_c$ at verbosity 3 so the conversion is auditable.

**Worked example 10.1.** A chain-branching stepThe single most important reaction in hydrogen and hydrocarbon ignition is the chain-branching step $\mathrm{H + O_2 \rightleftharpoons O + OH}$. It is mole-conserving ($\Delta\nu = 0$), so $K_c = K_p$, and its forward coefficient has a strongly non-Arrhenius shape captured by the modified form (587) with a negative $b$. Its reverse, $\mathrm{O + OH \to H + O_2}$, is not transcribed independently: Choupo computes it from (590) using the cited thermodynamics of H, $\mathrm{O_2}$, O, OH. One forward triple plus the species' Gibbs data fixes both directions consistently — the reverse can never drift out of thermodynamic agreement.

A note on data and provenance. The numerical rate constants of a detailed mechanism are measured facts, each traceable to a primary kinetics study, and a value cited to its primary source is the honest unit of reuse. Choupo therefore authors a mechanism reaction by reaction from the primary literature rather than vendoring a compiled mechanism file, whose particular arrangement may carry its compiler's licence. Standard graduate texts for this chapter are C. K. Law, *Combustion Physics* (Cambridge, 2006), and S. R. Turns, *An Introduction to Combustion* (McGraw-Hill).

# 11 Polymerisation: chain statistics from conversion

*What you'll learn.* A polymerisation reactor's interesting outputs are not concentrations but *chain statistics*: how long the chains are on average, and how broad the spread is. For step-growth polymerisation these follow in closed form from a single number, the conversion $p$ — a clean glass-box lesson in reaction engineering, where the whole molar-mass distribution is an analytic curve rather than the output of an engine.

## 11.1 Carothers: degree of polymerisation from conversion

In step-growth polymerisation (a diacid and a diol esterifying to a polyester, releasing water) any two functional groups can react. Let $p$ be the fraction of functional groups that have reacted — the conversion. Carothers' counting argument, in the closed form Choupo takes from [129], gives the number-average degree of polymerisation, the mean number of repeat units per chain,

$$\bar{X}_n \;=\; \frac{1}{1-p}. \tag{593}$$

The divergence as $p \to 1$ is the central fact of step-growth chemistry: a high polymer demands near-complete conversion. At $p = 0.99$, $\bar{X}_n = 100$; reaching $\bar{X}_n = 200$ needs $p = 0.995$. The number-average molar mass follows from the repeat-unit mass $M_0$,

$$M_n \;=\; M_0\,\bar{X}_n \;=\; \frac{M_0}{1-p}. \tag{594}$$

## 11.2 The distribution and the polydispersity index

A mean is not the whole story — chains of every length coexist. For the equal-reactivity, uniform-time case the probability that a chain is exactly $x$ units long is the *Flory–Schulz* (most-probable) distribution [129],

$$n(x) \;=\; (1-p)\,p^{\,x-1}, \tag{595}$$

the mole fraction, monotonically decaying in $x$; the corresponding weight fraction

$$w(x) \;=\; x\,(1-p)^2\,p^{\,x-1} \tag{596}$$

is peaked near $x \approx \bar{X}_n$ and broadens as $p$ rises. The weight-average degree of polymerisation and molar mass are [129]

$$\bar{X}_w \;=\; \frac{1+p}{1-p}, \qquad M_w \;=\; M_0\,\frac{1+p}{1-p}. \tag{597}$$

The ratio of the two averages is the polydispersity index, the single number that reports the breadth of the distribution,

$$\boxed{\mathrm{PDI} \;=\; \frac{M_w}{M_n} \;=\; 1+p \;\xrightarrow[p\to 1]{}\; 2.} \tag{598}$$

A step-growth polymer taken to high conversion always approaches a polydispersity of two. Watching $w(x)$ broaden while PDI climbs toward two is what makes the number mean something.

**Common pitfall.** Equations (595)–(598) assume every chain has seen the *same reaction time* — true in a batch vessel or an ideal plug-flow reactor. A continuous stirred tank imposes a residence-time distribution: some chains leave young, some old, and the molar-mass distribution is broader than Flory–Schulz, with $\mathrm{PDI} > 1 + p$. Printing $\mathrm{PDI} = 1 + p$ next to a CSTR result would teach a visibly wrong equation. Choupo therefore reports the bare conversion-only averages ($p$, $\bar{X}_n$, $M_n$, which are residence-time independent) on any reactor, but gates the distribution and $\mathrm{PDI} = 1 + p$ to the batch / plug-flow case.

**Worked example 11.1.** Polyesterification to high conversionA diacid and diol esterify in a plug-flow reactor to $p = 0.99$ with a repeat-unit mass $M_0 = 0.111\,\mathrm{kg/mol}$. Then $\bar{X}_n = 1/(1-0.99) = 100$,

$M_n = 11.1\,\mathrm{kg/mol}$, $\bar{X}_w = 1.99/0.01 = 199$, $M_w = 22.1\,\mathrm{kg/mol}$, and PDI $= 1.99$. The Flory–Schulz weight curve peaks near 100 repeat units with a long tail. Sweeping the reactor volume traces PDI climbing from near 1 toward 2 as $p$ rises — the entire Carothers/Flory story in one run.

These results are derived from a small set of closed-form algebraic relations layered on the existing reactor conversion, not from a new engine, which is why they cost almost nothing and stay transparent. The primary references are P. J. Flory, *Principles of Polymer Chemistry* (Cornell University Press, 1953), Chs. VIII–IX; G. Odian, *Principles of Polymerization* (Wiley, 4th ed.), Ch. 2; and H. S. Fogler, *Elements of Chemical Reaction Engineering* (Pearson, 5th ed.), §9.

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*The information you have is not the information you want.*
*The information you want is not the information you need.*
*The information you need is not the information you can obtain.*
John Gall, Inaccessibility Theorem, *Systemantics* (1975)

This manual ends here rather than on a summary, because the sentence above is the one it has been earning throughout. Every refusal in this simulator — a component with no critical constants, an activity model with no pair parameter, a phase envelope that will not close its own nose — is an instance of it. A tool that answers anyway is not more capable than one that declines; it is only quieter about the same gap.